

CONVEX COUNTEREXAMPLES TO THE SCHIFFER AND POMPEIU CONJECTURES IN DIMENSIONS TWO TO EIGHTEEN

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ABSTRACT. We construct the first convex planar counterexample to the Schiffer and Pompeiu conjectures: a bounded strictly convex non-disc domain with real-analytic boundary carrying a nonconstant solution of $\Delta u + u = 0$ with $u = 1$ and $\partial_\nu u = 0$ on the boundary. Together with the higher-dimensional constructions, this gives convex non-ball counterexamples in every dimension from two to eighteen, and in dimensions twenty and twenty-one, with real-analytic boundaries diffeomorphic to spheres and indicator Fourier transforms vanishing on the unit sphere. In dimensions 4, 6, 8, 10 and 14, planar reductions through compact Lie algebras of rank two use Harish-Chandra's radial part formula and Kostant's convexity theorem. In dimensions 3, 5 and 7, an axisymmetric formulation on the unit ball gives a quartic equation in weighted coefficient spaces. The exact two-sided inverse of the residual-corrected linearisation combines a Dirichlet Helmholtz inverse, a holomorphic boundary real-part problem and the material-derivative identity; it gives the planar Schiffer domain and, by one dimension-parametrised argument, the domains in dimensions 5, 9, 11 to 13, 15 to 18, 20 and 21. We also construct the first convex planar counterexample to Berenstein's conjecture: a strictly convex non-disc domain with real-analytic boundary carrying a sign-changing solution of $\Delta u + u = 0$ with $u = 0$ and $\partial_\nu u = 1$ on the boundary. Planar existence for this problem is due to Colbrook, Sadeghi and Stepaniants. All existence proofs in this paper are computer-assisted.

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1. INTRODUCTION

A bounded domain has the *Pompeiu property* if a continuous function whose integral over every rigid motion of the domain is zero must vanish identically. Schiffer's conjecture asserts that a bounded smooth domain with spherical boundary carrying a nonconstant Helmholtz eigenfunction with constant Dirichlet data and zero Neumann data is a ball. We construct convex counterexamples, beginning with the plane: in every dimension from two to eighteen, and in dimensions twenty and twenty-one, we find a bounded convex domain $\Omega \subset \mathbb{R}^n$, not a ball, and a nonconstant function u such that

$$(1.1) \quad \Delta u + u = 0 \quad \text{in } \Omega, \quad u = 1 \quad \text{and} \quad \nabla u = 0 \quad \text{on } \partial\Omega.$$

Let D_m be the group of order $2m$ generated by the rotation through $2\pi/m$ and the reflection in the real axis.

Write $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^{n-1}$ for $n \in \{3, 4, 7\} \cup \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\}$, and $x = (x', x'') \in \mathbb{R}^3 \times \mathbb{R}^3$ for points of $\mathbb{R}^6$.

For $n \in \{8, 10, 14\}$ let $(\mathfrak{k}, G)$ be $(\mathfrak{su}(3), SU(3))$, $(\mathfrak{so}(5), SO(5))$ and $(\mathfrak{g}_2, G_2)$, respectively. We identify $\mathfrak{k}$ with $\mathbb{R}^n$ by a linear isometry for an $\text{Ad}(G)$ -invariant inner product.

Theorem 1.1 (Main theorem: convex counterexamples). *In every dimension from two to eighteen, and in dimensions twenty and twenty-one, there exist a bounded convex domain $\Omega \subset \mathbb{R}^n$ and a nonconstant real-valued function u , smooth on $\overline{\Omega}$, such that (1.1) holds. For $n \geq 3$, u is real analytic on an open neighbourhood of $\overline{\Omega}$. Moreover:*

- (i) Ω is not a ball;
- (ii) Ω is strictly star-shaped with respect to the origin, and $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^{n-1} ;
- (iii) for $n = 2$, Ω is strictly convex and invariant under D_{182} ; for $n \in \{3, 7\} \cup \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\}$, Ω is invariant under $O(n-1)$ on x' and under $x_1 \mapsto -x_1$; for $n = 4$, Ω is invariant under the action of $O(3)$ on x' and under $x_1 \mapsto -x_1$; for $n = 6$, Ω is invariant under the action of $O(3) \times O(3)$ on (x', x'') and under the exchange $(x', x'') \mapsto (x'', x')$; for $n \in \{8, 10, 14\}$, $\Omega \subset \mathfrak{k}$ is invariant under the adjoint action of G and under $x \mapsto -x$.

In particular, Schiffer's conjecture (Conjecture 1 below) fails in every dimension $2 \leq n \leq 18$ and in dimensions 20 and 21, even within the class of convex domains.

Corollary 1.2 (Pompeiu). *Let n satisfy $2 \leq n \leq 18$ or $n \in \{20, 21\}$ and let $\Omega \subset \mathbb{R}^n$ be the corresponding domain of Theorem 1.1. The Fourier transform of $\mathbf{1}_\Omega$ vanishes on the unit sphere of $\mathbb{R}^n$. Consequently $\int_{\sigma(\Omega)} \cos x_1 dx = 0$ for every rigid motion σ of $\mathbb{R}^n$, the domain Ω fails the Pompeiu property, and the Pompeiu conjecture fails in $\mathbb{R}^n$, even within the class of convex domains.*

1.1. The planar domain. Write $\mathbb{D} = \{w \in \mathbb{C} : |w| < 1\}$. Our normalisation fixes the physical eigenvalue to be one; dilating a domain changes the eigenvalue by the inverse square of the dilation factor. Here ν denotes the outward unit normal.

Theorem 1.3. *There exist a bounded strictly convex domain $\Omega \subset \mathbb{R}^2$, strictly star-shaped with respect to the origin and different from a disc, with real-analytic boundary, and a nonconstant real function u , smooth on $\overline{\Omega}$, such that*

$$(1.2) \quad \Delta u + u = 0 \quad \text{in } \Omega, \quad u = 1, \quad \partial_\nu u = 0 \quad \text{on } \partial\Omega.$$

The domain is invariant under D_{182} . It is the image of $\mathbb{D}$ under a conformal map ψ normalised by $\psi(0) = 0$, $\psi'(0) > 0$, with

$$(1.3) \quad 814.45 < \psi'(0) < 814.46,$$

$$(1.4) \quad \operatorname{Re} \left(1 + \frac{w\psi''(w)}{\psi'(w)} \right) > 0.2998879 \quad (|w| = 1).$$

More precisely, ψ is holomorphic and univalent on $|w| < R$, where $R = 2^{1/182} > 1.0038$, and satisfies $\operatorname{Re} \psi'(w) > 808$ there. On the closed unit disc $|\psi'(w)| > 811$, and the coefficient of w^{183} in ψ has absolute value greater than 0.01645. The Fourier transform of $\mathbf{1}_\Omega$ vanishes on $|\xi| = 1$, so Ω fails the Pompeiu property.

The exact solution is specified by a contraction ball rather than a finite Taylor polynomial. With the spaces of Subsection 2.1 and the exactly clamped centre (g_0, p_0) of Subsection 3.1, there is a unique zero of

$$(1.5) \quad F(g, p) = g + |p|^2(1 + Kg)$$

(Here F denotes the cubic map of the plane; from Subsection 1.6 on, F is the planar field on a Cartan section.) The zero lies in

$$\|g - g_0\|_Y + \|p - p_0\|_W \leq 10^{-32}, \quad \psi(w) = \int_0^w p(z) dz.$$

This description, together with the decimal input in the ancillary files, determines the domain of Theorem 1.3.

The domain lies within relative Hausdorff distance about $2 \cdot 10^{-5}$ of a centred disc, but its curvature varies substantially. The centre has conformal radius approximately 814.457 and 182 small ripples. A binary64 evaluation of the centre gives a peak-to-trough radial variation of about $4 \cdot 10^{-5}$ relative to the conformal radius, and curvature between about 0.35 and 1.69 times the curvature of the circle of that radius; see Figure 1. These values describe the centre and are not interval enclosures. The strict lower bound (1.4) comes from the existence-ball estimates of Section 4.

1.2. A convex domain for Berenstein’s problem. Berenstein’s Dirichlet overdetermined problem prescribes $u = 0$ and $\partial_\nu u = \text{const} \neq 0$ for a Helmholtz eigenfunction on a bounded domain; it is conjectured to be solvable only on balls. Colbrook, Sadeghi and Stepaniants [15] constructed a planar counterexample. Theorem 5.1 gives the first convex planar counterexample: a bounded strictly convex non-disc domain with real-analytic boundary and a sign-changing solution with eigenvalue and outward flux both equal to one. Equivalently, the Fourier transform of its boundary arc-length measure vanishes on the unit circle. Part 2 proves the result by a Robin inverse and an exact two-sided inverse of the residual-corrected fixed-disc linearisation.

1.3. The Pompeiu problem and Schiffer’s conjecture. Let $n \geq 2$. A bounded domain $\Omega \subset \mathbb{R}^n$ has the *Pompeiu property* if the only continuous function $f: \mathbb{R}^n \rightarrow \mathbb{C}$ such that

$$\int_{\sigma(\Omega)} f(x) dx = 0 \quad \text{for every rigid motion } \sigma \text{ of } \mathbb{R}^n$$

is $f \equiv 0$. The question of which domains have this property goes back to Pompeiu [53–55]. Balls do not have it: if $J_{n/2}(kR) = 0$ for some $k > 0$, then the Fourier transform of $\mathbf{1}_{B_R}$ vanishes on the sphere $\{|\xi| = k\}$, and $f(x) = e^{ikx_1}$ integrates to zero over every ball of radius R . Brown, Schreiber and Taylor [7] characterised the failure of the Pompeiu property through the zero set of the Fourier–Laplace transform of the indicator function; in particular,

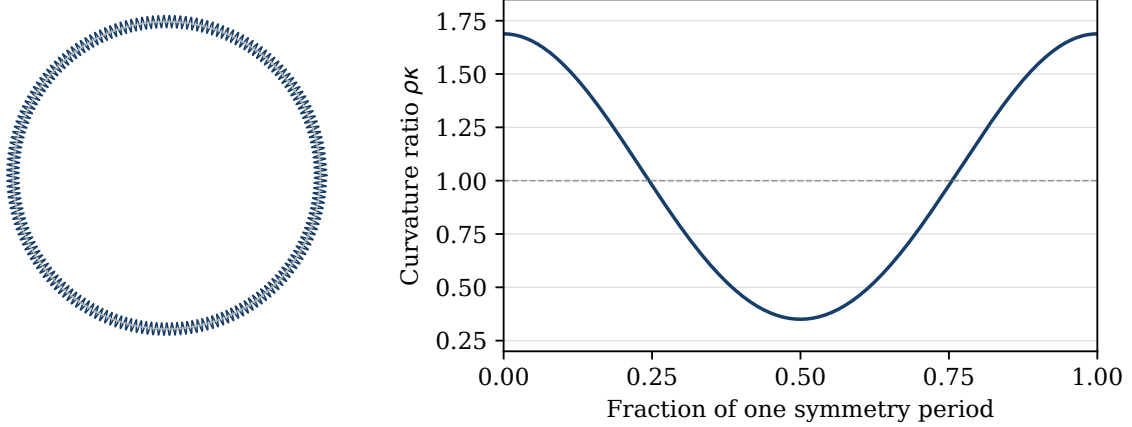
182 ripples; radial deviation $\times 2000$ 

FIGURE 1. The forty-mode centre ψ_0 of Subsection 3.1, with conformal radius $\rho = \psi'_0(0)$. Left: Schematic boundary, in which the radial deviation from the circle of radius ρ (grey) is multiplied by 2000; this exaggerated curve is not the domain of Theorem 1.3. Right: the curvature of the centre, multiplied by ρ , over one symmetry period of the conformal angle. Both panels are binary64 evaluations of the centre, not interval enclosures.

a bounded domain fails the property whenever $\widehat{1}_\Omega$ vanishes identically on a sphere $\{|\xi| = k\}$ with $k > 0$ (this direction is elementary, see Section 33). The *Pompeiu conjecture* asserts that a bounded domain with Lipschitz boundary homeomorphic to S^{n-1} fails the Pompeiu property only if it is a ball. Some topological assumption is necessary: for every $k > 0$ the function $R \mapsto R^{n/2}J_{n/2}(kR)$ takes equal values at suitable distinct radii $R_1 < R_2$, so the spherical shell $\{R_1 < |x| < R_2\}$ fails the Pompeiu property, and such shells are simply connected when $n \geq 3$.

Conjecture 1 (Schiffer). *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with smooth boundary homeomorphic to S^{n-1} . If, for some $\mu > 0$, there is a nonconstant solution of*

$$(1.6) \quad \Delta u + \mu u = 0 \quad \text{in } \Omega, \quad u = \text{const} \quad \text{and} \quad \partial_\nu u = 0 \quad \text{on } \partial\Omega,$$

then Ω is a ball.

On a ball B_R , the radial Neumann eigenfunctions $u = r^{1-n/2}J_{n/2-1}(\sqrt{\mu}r)$ solve (1.6) precisely when $J_{n/2}(\sqrt{\mu}R) = 0$. Formulations of the conjecture in the literature differ in their regularity and topological hypotheses (smooth boundary, simply connected domain, connected boundary); the domains constructed below satisfy all of them. Williams [67] showed that, for bounded domains with Lipschitz boundary homeomorphic to a sphere, the failure of the Pompeiu property is equivalent to the solvability of (1.6) for some $\mu > 0$, building on [7]; the direction from (1.6) to the failure of the Pompeiu property is a consequence of Green's identity. In the planar case Schiffer's problem appears as Problem 80 in Yau's list [68]. We refer to Zalcman's survey [69] for the classical bibliography.

A large literature proves the conjectured rigidity under additional hypotheses: for domains carrying solutions for infinitely many eigenvalues [9], for low eigenvalues [20], for perturbations of balls [2, 10, 46], for planar convex domains [6] and centrally symmetric planar domains [52], in particular at high frequency [25], and under further conditions [17, 44]. Section 36 compares our examples with the results of [6, 9, 16, 20, 25, 46, 52] and with the statements of [4, 5, 23] quoted there from secondary sources.

1.4. Previous constructions. In August 2026 the planar conjectures were disproved independently by two groups. Colbrook and Stepaniants [14] constructed a single explicit ten-fold symmetric noncircular domain $\Omega \subset \mathbb{R}^2$ with real-analytic boundary for which (1.6) holds with $\sqrt{\mu} \in (31.967007261, 31.967007293)$. Their proof pulls the problem back to the unit disc by a conformal map, which turns it into a cubic equation on a space of disk-polynomial coefficients, and establishes the existence of an exact solution near an explicit approximation by a computer-assisted contraction argument. Cao-Labora and de Dios Pont [12] constructed infinitely many N -fold symmetric planar counterexamples, for N sufficiently large, by a bifurcation argument for a relaxed problem in which N is a real parameter, with a branch length that is uniform in N . Colbrook, Sadeghi and Stepaniants [15] adapted the conformal method to disprove the planar Berenstein conjecture. Dolha [24] gave a further computer-assisted planar counterexample, with seven-fold dihedral symmetry, in a public software repository. None of [14], [12] and [15] asserts convexity; the published centres of the examples of [14, 24] have negative boundary curvature at some points. Singh [63] used the counterexample of [14] to obtain noncircular homogeneous dielectric inclusions that are exactly non-scattering for an explicit incident wave in the two-dimensional scalar Maxwell reduction. Earlier constructions of Schiffer-type domains in related settings include domains in the flat cylinder $\mathbb{R}^N \times \mathbb{R}/2\pi\mathbb{Z}$ and on S^2 [31], contractible domains on the half-sphere [13], doubly connected planar domains on whose two boundary components the eigenfunction takes different constants [29], and non-circular planar solutions of the related problem with nonzero constant Neumann data [66]. Schiffer-type domains in round spheres bounded by isoparametric hypersurfaces were found earlier [56, 61]; their boundaries are in general not topological spheres. Apart from our earlier preprint [36] and version 1 of this paper (see the end of this section), to the best of our knowledge no counterexample to either conjecture was previously known for bounded domains in $\mathbb{R}^n$, $n \geq 3$, with boundary homeomorphic to a sphere.

Rigidity for convex planar domains is known under several additional hypotheses, including width and central-symmetry conditions, and at high frequency above a domain-dependent threshold [6, 16, 25, 52]. To our knowledge, no convex planar counterexample was known. Theorem 1.3 supplies such an example.

1.5. The higher-dimensional domains.

Theorem 1.4 (Convex example in dimension three). *There exist a bounded convex domain $\Omega \subset \mathbb{R}^3$ and a nonconstant real-valued function u , real analytic on an open neighbourhood of $\bar{\Omega}$, such that (1.1) holds. The domain is not a ball, is strictly star-shaped with respect to the origin, and has compact real-analytic boundary diffeomorphic to S^2 . It is invariant under $O(2)$ acting on x' and the reflection $x_1 \mapsto -x_1$.*

The case $n = 2$ is Theorem 1.3, the case $n = 3$ is Theorem 1.4, the cases $n = 5, 7$ are proved in Sections 22 and 23, the cases $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\} \setminus \{5\}$ and a second proof for $n = 5$ are given in Part 5, and the cases $n = 4, 6, 8, 10, 14$ are proved in Part 3. Table 1 lists for each dimension the symmetry reduction, the method, the certified radius, linear defect and geometric bound, and the place of the proof.

Rescaling Ω and u gives, for every $\mu > 0$, a counterexample with eigenvalue μ in (1.6).

The convex three-dimensional example is distinct from the nonconvex example of Proposition 21.27, which is retained below. The comparison with rigidity results and other claims is in Section 36.

Remark 1.5 (The domains). In dimensions 4, 6, 8, 10 and 14 the domains are obtained from a planar domain $D = \psi(\mathbb{D}) \subset \mathbb{R}^2 \cong \mathbb{C}$, the image of the unit disc under a map ψ with real

n	Symmetry reduction	Method	Certified bounds	Proof
2	D_{182}	Exact coupled inverse	$r = 10^{-32}$, $Z < 2.122 \cdot 10^{-38}$; convexity factor > 0.2998879	§§2–4
3	$O(2) \times \mathbb{Z}_2$	Unit-ball pull-back	$r = 10^{-7}$, $Z < 0.776701$; convexity factor > 0.7878	§21
4	$\mathfrak{u}(2)$	Cartan section	$r = 1/50000$, $Z < 0.360262$; convexity factor > 0.414074	§§10–18
5	$O(4) \times \mathbb{Z}_2$	Degree-weighted pull-back	$r = 10^{-10}$, $Z < 0.887235$; convexity factor > 0.839460	§22; §§24–32
6	$\mathfrak{so}(4)$	Cartan section	$r = 1/5000$, $Z < 0.464977$; convexity factor > 0.71569200	§19
7	$O(6) \times \mathbb{Z}_2$	Squared-degree- weighted pull-back	$r = 10^{-9}$, $Z_0 < 0.898934481$; convexity factor > 0.951916263887	§23
8	$\mathfrak{su}(3)$	Cartan section	$r = 1/30000$, $Z < 0.418572$; convexity factor > 0.83153153	§20
9	$O(8) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-43}$, $Z < 1.64019 \cdot 10^{-83}$; curvature numerator $> 9.93044 \cdot 10^{-1}$	§§24–32
10	$\mathfrak{so}(5)$	Cartan section	$r = 10^{-7}$, $Z < 0.552897$; convexity factor > 0.97305365	§20
11	$O(10) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-38}$, $Z < 6.87444 \cdot 10^{-70}$; curvature numerator $> 9.10902 \cdot 10^{-1}$	§§24–32
12	$O(11) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-50}$, $Z < 4.65561 \cdot 10^{-98}$; curvature numerator $> 9.75281 \cdot 10^{-1}$	§§24–32
13	$O(12) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-50}$, $Z < 4.3153 \cdot 10^{-98}$; curvature numerator $> 9.37264 \cdot 10^{-1}$	§§24–32
14	$\mathfrak{g}_2$	Cartan section	$r = 1/6000$, $Z < 0.604948$; convexity factor > 0.89406714	§20
15	$O(14) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-49}$, $Z < 5.15043 \cdot 10^{-80}$; curvature numerator $> 8.32915 \cdot 10^{-1}$	§§24–32
16	$O(15) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-72}$, $Z < 2.91643 \cdot 10^{-129}$; curvature numerator $> 9.98507 \cdot 10^{-1}$	§§24–32
17	$O(16) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-55}$, $Z < 6.23668 \cdot 10^{-104}$; curvature numerator $> 9.94623 \cdot 10^{-1}$	§§24–32
18	$O(17) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-52}$, $Z < 3.45744 \cdot 10^{-96}$; curvature numerator $> 8.25454 \cdot 10^{-1}$	§§24–32
20	$O(19) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-71}$, $Z < 6.35578 \cdot 10^{-132}$; curvature numerator $> 9.64444 \cdot 10^{-1}$	§§24–32
21	$O(20) \times \mathbb{Z}_2$	Meridian pull-back; exact inverse	$r = 10^{-82}$, $Z < 8.05097 \cdot 10^{-142}$; curvature numerator $> 9.483 \cdot 10^{-1}$	§§24–32

TABLE 1. The convex examples. The radius and linear defect use the norms of the indicated proof; the row $n = 5$ records the proof of Section 22, and the second proof in Part 5 has the bounds of Tables 18–23; for the rows $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\} \setminus \{5\}$, the curvature numerator is χ_* of (31.1), which controls both meridian and transverse curvatures; for the other entries, the convexity factor is a lower bound for $\operatorname{Re}(1 + w\psi''/\psi')$ on the unit circle of the planar or meridian map. The equation in Part 5 is quartic. The earlier equations are cubic for $n = 2, 4$, quartic for $n = 3, 5, 6, 7$, and of degree $m + 2$ for the Cartan reductions $n = 2m + 2$.

Taylor coefficients; the axisymmetric domains of Part 4 and Theorem 24.1 are images of unit balls under axisymmetric extensions of such maps.

Dimension three. Here $\Omega = \Theta(B^3)$, where B^3 is the unit ball and

$$\Theta(x_1, x') = \left(\operatorname{Re} \psi(x_1 + i|x'|), \frac{\operatorname{Im} \psi(x_1 + i|x'|)}{|x'|} x' \right), \quad \psi(w) = \sum_{j \text{ odd}} c_j w^j.$$

The exact coefficient vector lies in the ball

$$\sum_{j \text{ odd}} 8b j \left(\frac{21}{20} \right)^j |c_j - c_j^\circ| \leq 10^{-7}, \quad b = 7626062690497999 \cdot 2^{-46},$$

about the dyadic centre in `anc/r3_convex/candidate.json`. Its linear coefficient is approximately 108.372869; the nearby radial Schiffer ball has radius $j_{3/2,34} \approx 108.37572$, the thirty-fourth positive root of $\tan x = x$. The non-ball witness is $c_{19} \neq 0$. An independent example near $j_{3/2,15} \approx 48.67414$ is strictly star-shaped but nonconvex (Proposition 21.27).

Dimension five. Here $\Omega = \rho \Theta(B^5)$, where B^5 is the unit ball, $\rho = 7951044098899425 \cdot 2^{-47} \approx 56.4955663$, and, for $x' \in \mathbb{R}^4$,

$$\Theta(x_1, x') = \left(\operatorname{Re} \psi(x_1 + i|x'|), \frac{\operatorname{Im} \psi(x_1 + i|x'|)}{|x'|} x' \right), \quad \psi(w) = \sum_{j \text{ odd}} c_j w^j.$$

The exact coefficient vector lies in the ball

$$\sum_{j \text{ odd}} 30\rho^2 j \left(\frac{21}{20} \right)^j \left(1 + \frac{j-1}{10^6} \right) |c_j - c_j^\circ| \leq 10^{-10}$$

about the dyadic centre embedded in `anc/r5/verify_r5.py`; one has $c_1^\circ \approx 1.0002252$ and $c_3^\circ \approx 4.457 \cdot 10^{-4}$, and the non-ball witness is $c_3 \neq 0$. Numerically, the distance of $\partial\Omega$ from the origin lies between 56.4763 and 56.6242. The ball of $\mathbb{R}^5$ that carries a radial solution of (1.1) with the closest radius has radius $j_{5/2,17} \approx 56.495566$, the seventeenth positive zero of $J_{5/2}$, and the relative deviation of $\partial\Omega$ from that sphere lies between -0.034% and $+0.228\%$. Expanding the boundary radius in zonal spherical harmonics on S^4 , the dominant nonconstant term has degree twelve, with relative amplitude about $2.5 \cdot 10^{-3}$.

Dimension seven. Here $\Omega = \rho \Theta(B^7)$ with the same lift, $x' \in \mathbb{R}^6$ and $\rho = 731659610854495 \cdot 2^{-43} \approx 83.1800674$. The shape coefficients satisfy

$$\sum_{j \text{ odd}} 25\rho^2 j \left(\frac{6}{5} \right)^j \left(1 + \frac{j-1}{10^6} \right)^2 |c_j - c_j^\circ| \leq 10^{-9},$$

about the exact dyadic centre embedded in `anc/r7/verify_r7.py`, with $c_1^\circ \approx 1.0000971$ and $c_3^\circ \approx 1.899137 \cdot 10^{-4}$. The non-ball witness is $c_3 \neq 0$. Numerically, the unscaled meridian radius lies between 0.9999409 and 1.0009305; the nearest radial Schiffer ball has radius $j_{7/2,25} \approx 83.1800674$, and the dominant nonconstant zonal harmonic has degree fourteen. Section 23 certifies strict convexity on the whole existence ball.

Dimension four. Here $\Omega = \{(x_1, x') : (x_1, |x'|) \in D\}$ and $\psi(w) = \sum_{j \text{ odd}} c_j w^j$. The coefficients satisfy

$$\sum_{j \text{ odd}} (c_1^\circ)^2 (j+1) \left(\frac{21}{20} \right)^j |c_j - c_j^\circ| \leq \frac{1}{50000},$$

where $c_1^\circ, c_3^\circ, \dots, c_{41}^\circ$ are explicit dyadic rationals listed in the verification script (Section 34) and $c_j^\circ = 0$ for $j > 41$; one has $c_1^\circ \approx 14.840043$ and $c_3^\circ \approx 0.0912$, $c_5^\circ \approx 0.0942$, $c_7^\circ \approx 0.0964$. For orientation we record some numerical observations about the centre ψ° , which are not

used in the proofs. The boundary of D is a radial graph whose radius lies between 14.7213 and 15.1343. The ball of $\mathbb{R}^4$ that carries a radial solution of (1.1) with the same eigenvalue and closest radius has radius $j_{2,4} \approx 14.79595$, the fourth positive zero of J_2 ; thus the relative deviation of $\partial\Omega$ from that sphere lies between -0.5% and $+2.3\%$. Expanding the boundary radius of Ω in zonal spherical harmonics on S^3 , the dominant nonconstant term is the zonal harmonic of degree six, with relative amplitude about 2.2×10^{-2} at the poles.

Dimension six. Here $\Omega = \{(x', x'') : (|x'|, |x''|) \in D\}$ and $\psi(w) = \sum_{j \equiv 1 \pmod{4}} c_j w^j$. The coefficients satisfy

$$\sum_{j \equiv 1 \pmod{4}} \frac{(c_1^\circ)^3}{2} (j+2) \left(\frac{11}{10}\right)^{j+1} |c_j - c_j^\circ| \leq \frac{1}{5000},$$

where $c_1^\circ, c_5^\circ, \dots, c_{57}^\circ$ are explicit dyadic rationals listed in the ancillary data file `center_r6.json` (Section 34.5) and $c_j^\circ = 0$ for $j > 57$; one has $c_1^\circ \approx 28.864751$, $c_5^\circ \approx -0.0852970$ and $c_9^\circ \approx -0.0837734$, while $|c_j^\circ| < 4 \cdot 10^{-4}$ for $j \geq 13$. Numerically, the boundary of D is a radial graph whose radius lies between 28.6958 and 28.9596, and the ball of $\mathbb{R}^6$ with the same volume as Ω has radius 28.908339. The ball of $\mathbb{R}^6$ that carries a radial solution of (1.1) with the same eigenvalue and closest radius has radius $j_{3,8} \approx 28.908351$, the eighth positive zero of J_3 ; the relative deviation of $\partial\Omega$ from that sphere lies between -0.74% and $+0.18\%$, so that the radius of $\partial\Omega$ varies by about 0.92% . Expanding the boundary radius in the $O(3) \times O(3)$ -invariant spherical harmonics on S^5 , which are the functions $U_k(|\omega'|^2 - |\omega''|^2)$ of degree $2k$ (Section 35.2), the dominant nonconstant term has degree eight, with relative amplitude about 7.2×10^{-3} .

Dimensions eight, ten and fourteen. Let $m = 3, 4, 6$, respectively. Here $\Omega = \text{Ad}(G)D$, where $D = \Omega \cap \mathfrak{t}$ is the section of Ω by a Cartan subalgebra $\mathfrak{t} \cong \mathbb{C}$ (Section 1.6), and $\psi(w) = \sum_{j \equiv 1 \pmod{2m}} c_j w^j$ lies in an explicit weighted ℓ^1 ball around explicit dyadic coefficients c_j° (Section 20). One has $c_1^\circ \approx 20.787029$, 38.152926 , 88.497305 and $c_{2m+1}^\circ \approx -0.0774840$, -0.0138325 , 0.0463087 , respectively. Numerically, the boundary of D is a radial graph whose radius lies between 20.7086 and 20.8635, between 38.1390 and 38.1667, and between 88.4477 and 88.5402. The balls that carry a radial solution of (1.1) with the closest radii have radii $j_{4,5} \approx 20.826933$, $j_{5,10} \approx 38.159869$ and $j_{7,25} \approx 88.474363$, and the relative deviations of $\partial\Omega$ from these spheres lie between -0.57% and $+0.18\%$, between -0.055% and $+0.018\%$, and between -0.030% and $+0.074\%$. In each case the dominant nonconstant term of the radius of D is a multiple of $\cos(2m\theta)$, which corresponds to the $\text{Ad}(G)$ -invariant spherical harmonic of degree $2m$; its restriction to the unit circle of $\mathfrak{t}$ is a multiple of $1 + 2\cos(2m\theta)$.

Dimension nine. For the centre of Part 5 (Table 18), $\rho \approx 169.58703$, $c_1^\circ \approx 0.9999926$ and $c_3^\circ \approx -1.4533 \cdot 10^{-5}$. Numerically, the boundary distance from the origin is approximately between 169.5727 and 169.5875. The nearest radial Schiffer ball has radius $j_{9/2,52} \approx 169.587034$, with relative boundary deviations approximately -0.00846% to $+0.00027\%$. The dominant nonconstant zonal harmonic has degree 20, with relative polar amplitude about $8.5 \cdot 10^{-5}$.

Dimension eleven. For the centre of Part 5 (Table 18), $\rho \approx 57.86006$, $c_1^\circ \approx 1.0005497$ and $c_3^\circ \approx 1.0243 \cdot 10^{-3}$. Numerically, the boundary distance from the origin is approximately between 57.8560 and 58.0511. The nearest radial Schiffer ball has radius $j_{11/2,16} \approx 57.860063$, with relative boundary deviations approximately -0.00701% to $+0.33013\%$. The dominant nonconstant zonal harmonic has degree 10, with relative polar amplitude about $3.4 \cdot 10^{-3}$.

Dimension twelve. For the centre of Part 5 (Table 18), $\rho \approx 42.77848$, $c_1^\circ \approx 1.0002524$ and $c_3^\circ \approx 4.5391 \cdot 10^{-4}$. Numerically, the boundary distance from the origin is approximately between 42.7775 and 42.8315. The nearest radial Schiffer ball has radius $j_{6,11} \approx 42.778482$, with relative boundary deviations approximately -0.00241% to $+0.12399\%$. The dominant nonconstant zonal harmonic has degree 8, with relative polar amplitude about $1.3 \cdot 10^{-3}$.

Dimension thirteen. For the centre of Part 5 (Table 18), $\rho \approx 84.57460$, $c_1^\circ \approx 1.0003023$ and $c_3^\circ \approx 5.6777 \cdot 10^{-4}$. Numerically, the boundary distance from the origin is approximately between 84.5726 and 84.7411. The nearest radial Schiffer ball has radius $j_{13/2,24} \approx 84.574596$, with relative boundary deviations approximately -0.00241% to $+0.19682\%$. The dominant nonconstant zonal harmonic has degree 12, with relative polar amplitude about $2.0 \cdot 10^{-3}$.

Dimension fifteen. For the centre of Part 5 (Table 18), $\rho \approx 32.11120$, $c_1^\circ \approx 0.9973022$ and $c_3^\circ \approx -4.3725 \cdot 10^{-3}$. Numerically, the boundary distance from the origin is approximately between 31.8317 and 32.1169. The nearest radial Schiffer ball has radius $j_{15/2,7} \approx 32.111196$, with relative boundary deviations approximately -0.87031% to $+0.01780\%$. The dominant nonconstant zonal harmonic has degree 6, with relative polar amplitude about $1.3 \cdot 10^{-2}$.

Dimension sixteen. For the centre of Part 5 (Table 18), $\rho \approx 369.83635$, $c_1^\circ \approx 0.9999986$ and $c_3^\circ \approx -2.6652 \cdot 10^{-6}$. Numerically, the boundary distance from the origin is approximately between 369.8305 and 369.8364. The nearest radial Schiffer ball has radius $j_{8,114} \approx 369.836345$, with relative boundary deviations approximately -0.00159% to $+0.00001\%$. The dominant nonconstant zonal harmonic has degree 28, with relative polar amplitude about $1.6 \cdot 10^{-5}$. The centre is extremely close to a ball: its smallest and largest boundary distances $r_{\min}, r_{\max}$ give the relative radial range $(r_{\max} - r_{\min})/(r_{\max} + r_{\min}) \approx 8.0 \cdot 10^{-6}$.

Dimension seventeen. For the centre of Part 5 (Table 18), $\rho \approx 74.91722$, $c_1^\circ \approx 1.0000441$ and $c_3^\circ \approx 7.9917 \cdot 10^{-5}$. Numerically, the boundary distance from the origin is approximately between 74.9171 and 74.9348. The nearest radial Schiffer ball has radius $j_{17/2,20} \approx 74.917221$, with relative boundary deviations approximately -0.00014% to $+0.02340\%$. The dominant nonconstant zonal harmonic has degree 10, with relative polar amplitude about $2.3 \cdot 10^{-4}$.

Dimension eighteen. For the centre of Part 5 (Table 18), $\rho \approx 20.80705$, $c_1^\circ \approx 0.9944513$ and $c_3^\circ \approx -7.7897 \cdot 10^{-3}$. Numerically, the boundary distance from the origin is approximately between 20.5106 and 20.8138. The nearest radial Schiffer ball has radius $j_{9,3} \approx 20.807048$, with relative boundary deviations approximately -1.42481% to $+0.03229\%$. The dominant nonconstant zonal harmonic has degree 4, with relative polar amplitude about $2.0 \cdot 10^{-2}$.

Dimension twenty. For the centre of Part 5 (Table 18), $\rho \approx 83.43919$, $c_1^\circ \approx 0.9996765$ and $c_3^\circ \approx -5.7956 \cdot 10^{-4}$. Numerically, the boundary distance from the origin is approximately between 83.3038 and 83.4397. The nearest radial Schiffer ball has radius $j_{10,22} \approx 83.439190$, with relative boundary deviations approximately -0.16231% to $+0.00061\%$. The dominant nonconstant zonal harmonic has degree 10, with relative polar amplitude about $1.7 \cdot 10^{-3}$.

Dimension twenty-one. For the centre of Part 5 (Table 18), $\rho \approx 112.60858$, $c_1^\circ \approx 1.0003139$ and $c_3^\circ \approx 5.7641 \cdot 10^{-4}$. Numerically, the boundary distance from the origin is approximately between 112.6081 and 112.8144. The nearest radial Schiffer ball has radius $j_{21/2,31} \approx 112.608584$, with relative boundary deviations approximately -0.00045% to $+0.18275\%$. The dominant nonconstant zonal harmonic has degree 12, with relative polar amplitude about $1.8 \cdot 10^{-3}$.

1.6. Compact Lie algebras of rank two. The domains in dimensions 4, 6, 8, 10 and 14 are instances of a single construction. Let $\mathfrak{k}$ be a compact Lie algebra of rank two, with an Ad-invariant inner product, let $\mathfrak{t} \subset \mathfrak{k}$ be a Cartan subalgebra, identified with $\mathbb{C}$ by a linear isometry, and let W be the Weyl group. If $\mathfrak{k}$ is not abelian, then W is the dihedral group of order $2m$ with $m \in \{1, 2, 3, 4, 6\}$, there are m positive roots, $\dim \mathfrak{k} = 2m + 2$, and, up to isomorphism, $\mathfrak{k}$ is $\mathfrak{u}(2)$, $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, $\mathfrak{su}(3)$, $\mathfrak{so}(5) \cong \mathfrak{sp}(2)$ or $\mathfrak{g}_2$. The product ϖ of the positive roots is a W -anti-invariant harmonic polynomial of degree m ; in suitable coordinates $\varpi(z)$ is a nonzero multiple of $\operatorname{Im} z^m$. For an Ad-invariant C^2 function u with restriction $f = u|_{\mathfrak{t}}$, a classical formula of Harish-Chandra [37] (see [38, Ch. II]) states that

$$(1.7) \quad (\Delta_{\mathfrak{k}} u)|_{\mathfrak{t}} = \varpi^{-1} \Delta_{\mathfrak{t}}(\varpi f)$$

at the regular points of $\mathfrak{t}$. Consequently, Ad-invariant solutions of (1.1) on $\Omega \subset \mathfrak{k}$ correspond to W -anti-invariant solutions $F = \varpi f$ of the planar problem

$$(1.8) \quad \Delta F + F = 0 \text{ in } D, \quad F = \varpi \text{ and } \nabla F = \nabla \varpi \text{ on } \partial D,$$

on the Cartan section $D = \Omega \cap \mathfrak{t}$. There is no potential term, so the conformal method of [14] applies to (1.8). Conversely, if D is W -invariant, contains 0 and has a positive real-analytic radial function, then $\Omega = \text{Ad}(G)D$ is a bounded domain whose boundary is a real-analytic hypersurface diffeomorphic to S^{2m+1} , and an analytic solution of (1.8) lifts to a solution of (1.1) on Ω (Section 20); some hypothesis on D is needed here, since for an annulus D the lifted boundary has two components. By Kostant's convexity theorem [48], in the form of [49, Theorems 2.2 and 3.6], Ω is convex if and only if D is. For $\mathfrak{k} = \mathfrak{u}(2) \cong \mathbb{R} \times \mathbb{R}^3$ the adjoint group acts as $SO(3)$ on the second factor and ϖ is a multiple of y ; for $\mathfrak{k} = \mathfrak{so}(4) \cong \mathbb{R}^3 \times \mathbb{R}^3$ it acts as $SO(3) \times SO(3)$ and ϖ is a multiple of $y_1 y_2$. These are the reductions of Sections 10 and 19, which we present there in elementary form. For the abelian algebra $\mathbb{R}^2$ one has $\varpi = 1$, and (1.8) is the planar Schiffer problem itself. Among splittings $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with symmetry group $O(d_1) \times O(d_2)$, the meridian problem has no potential term exactly for $n \in \{2, 4, 6\}$ (Remark 10.5), which are the cases $m \in \{0, 1, 2\}$ above; in dimension three no such reduction is available, and Theorem 1.4 is proved by a different formulation.

The ball $B_\rho \subset \mathfrak{k}$ carries the radial solution $u = r^{-m} J_m(r)$ of (1.1) exactly when $J_{m+1}(\rho) = 0$; in $\mathfrak{t}$ it corresponds to $F = J_m(r) \sin m\theta$. An Ad-invariant spherical harmonic of degree ℓ , which exists for $\ell \in m\mathbb{Z}_{\geq 0}$, corresponds to the planar mode $J_{m+\ell}(r) \sin(m+\ell)\theta$, and the linearisation of the Schiffer problem at B_ρ in this direction is degenerate exactly when $J_{m+\ell}(\rho) = 0$. By Siegel's theorem [62] this never happens for $\ell \geq 2$; the case $\ell = 1$, which occurs only for $m = 1$, corresponds to translations and is excluded by the symmetry $x_1 \mapsto -x_1$. So there is no bifurcation from balls in these symmetry classes; our solutions are found instead at near-resonances, where $J_{m+\ell}(\rho)$ is small (Section 35.2 and Table 2).

1.7. Strategy of the proof. Near-resonance. Numerically, the centres lie close to Schiffer balls at near-resonant eigenvalues. Let ρ be a zero of $J_{n/2}$ and $\ell \geq 2$ the degree of a harmonic in the symmetry class. The linearisation in that direction is degenerate exactly when $J_{\ell+n/2-1}(\rho) = 0$, which never happens. In the higher-dimensional classes considered here, a formal reduced equation has a nonzero quadratic coefficient; the amplitude is then of the order of the small divisor. In the plane the quadratic coefficient vanishes, and the cubic term gives an amplitude of the order of the square root of the small divisor. A mode of angular frequency m changes curvature on the scale m^2 times its relative amplitude, so planar near-resonance competes directly with convexity. The planar proof uses the full equation, independently of this formal balance; Section 35.2 describes the higher-dimensional mechanism.

The plane: an exact inverse. In the fixed-disc formulation of Colbrook and Stepaniants [14], the equation is $F(g, p) = g + |p|^2(1 + Kg) = 0$. We use angularly weighted radial supremum norms. The additional ingredient is an exact two-sided inverse of a residual-corrected linearisation on the entire space: a symmetric-sector Dirichlet Helmholtz inverse gives a normal trace, a holomorphic boundary real-part problem gives the shape variation, and the material-derivative identity gives the field variation. A complete spectral window and the Gram identity for its full complement bound the Dirichlet inverse; analytic boundary tails bound the residual of the exactly clamped centre. The shape-derivative identity is the classical Hadamard mechanism [40, 65], which is not specific to two dimensions; conformal coordinates make the coupled inverse explicit here. In [14] the inverse is approximate, with a finite matrix completed by an explicit map on the tail coefficients.

n	symmetry	ρ	ℓ	$J_{\ell+n/2-1}(\rho)$
2	D_{182}	$j_{1,259} \approx 814.45744$	182	$J_{182}(\rho) \approx 1.6 \cdot 10^{-6}$
3	$O(2) \times \mathbb{Z}_2$	$j_{3/2,34} \approx 108.37572$	18	$J_{37/2}(\rho) \approx 1.25 \cdot 10^{-4}$
4	$\mathfrak{u}(2)$	$j_{2,4} \approx 14.79595$	6	$J_7(\rho) \approx -4.9 \cdot 10^{-3}$
5	$O(4) \times \mathbb{Z}_2$	$j_{5/2,17} \approx 56.49557$	12	$J_{27/2}(\rho) \approx -6.1 \cdot 10^{-4}$
6	$\mathfrak{so}(4)$	$j_{3,8} \approx 28.90835$	8	$J_{10}(\rho) \approx -3.0 \cdot 10^{-3}$
7	$O(6) \times \mathbb{Z}_2$	$j_{7/2,25} \approx 83.18007$	14	$J_{33/2}(\rho) \approx 2.3 \cdot 10^{-4}$
8	$\mathfrak{su}(3)$	$j_{4,5} \approx 20.82693$	6	$J_9(\rho) \approx -3.3 \cdot 10^{-3}$
9	$O(8) \times \mathbb{Z}_2$	$j_{9/2,52} \approx 169.58703$	20	$J_{47/2}(\rho) \approx -1.9 \cdot 10^{-5}$
10	$\mathfrak{so}(5)$	$j_{5,10} \approx 38.15987$	8	$J_{12}(\rho) \approx -4.4 \cdot 10^{-4}$
11	$O(10) \times \mathbb{Z}_2$	$j_{11/2,16} \approx 57.86006$	10	$J_{29/2}(\rho) \approx -6.5 \cdot 10^{-4}$
12	$O(11) \times \mathbb{Z}_2$	$j_{6,11} \approx 42.77848$	8	$J_{13}(\rho) \approx -2.3 \cdot 10^{-4}$
13	$O(12) \times \mathbb{Z}_2$	$j_{13/2,24} \approx 84.57460$	12	$J_{35/2}(\rho) \approx 3.3 \cdot 10^{-4}$
14	$\mathfrak{g}_2$	$j_{7,25} \approx 88.47436$	12	$J_{18}(\rho) \approx -9.0 \cdot 10^{-4}$
15	$O(14) \times \mathbb{Z}_2$	$j_{15/2,7} \approx 32.11120$	6	$J_{25/2}(\rho) \approx -2.0 \cdot 10^{-3}$
16	$O(15) \times \mathbb{Z}_2$	$j_{8,114} \approx 369.83635$	28	$J_{35}(\rho) \approx -1.9 \cdot 10^{-6}$
17	$O(16) \times \mathbb{Z}_2$	$j_{17/2,20} \approx 74.91722$	10	$J_{35/2}(\rho) \approx -2.9 \cdot 10^{-5}$
18	$O(17) \times \mathbb{Z}_2$	$j_{9,3} \approx 20.80705$	4	$J_{12}(\rho) \approx 2.7 \cdot 10^{-3}$
20	$O(19) \times \mathbb{Z}_2$	$j_{10,22} \approx 83.43919$	10	$J_{19}(\rho) \approx 1.6 \cdot 10^{-4}$
21	$O(20) \times \mathbb{Z}_2$	$j_{21/2,31} \approx 112.60858$	12	$J_{43/2}(\rho) \approx -1.5 \cdot 10^{-4}$

TABLE 2. Near-resonances: the Schiffer ball B_ρ of $\mathbb{R}^n$ at eigenvalue one and the degree ℓ of the dominant spherical harmonic in the symmetry class. Here ρ is the Schiffer-ball radius; for $n = 2$ it exceeds the conformal radius of Theorem 1.3 by approximately $3 \cdot 10^{-5}$. The Bessel values are numerical evaluations.

Dimensions four, six, eight, ten and fourteen: conformal fixed-disc formulation. Following [14] we write $D = \psi(\mathbb{D})$ and $V = F \circ \psi$, so that $\Delta V + |\psi'|^2 V = 0$ in $\mathbb{D}$, and $V - \varpi \circ \psi$ has vanishing value and normal derivative on $\partial\mathbb{D}$. We write $V = Kg + \varpi \circ \psi$, where K is an explicit inverse of the Laplacian with vanishing Cauchy data on its natural range. For $n = 4$, where $\varpi = y$, this gives the cubic equation

$$(1.9) \quad \mathcal{F}(g, \psi) = g + |\psi'|^2 (Kg + \text{Im } \psi) = 0,$$

and in general the polynomial equation

$$(1.10) \quad g + |\psi'|^2 (Kg + \varpi \circ \psi) = 0$$

of degree $m + 2$ in the unknowns. Compared with the equation $g + |p|^2(1 + Kg) = 0$ of [14], the constant boundary datum is replaced by $\varpi \circ \psi$, so that the unknown map enters the data as well; the eigenvalue is absorbed into the scale of ψ . The symmetry class consists of maps $\psi(w) = \sum_{j \equiv 1 \pmod{2m}} c_j w^j$ with real coefficients, which is slightly more symmetric than the Weyl group requires, and g lies in the sine sector of angular frequencies $\equiv m \pmod{2m}$. The part of the linearisation acting on high shape modes has a harmonic part which is a difference operator of step $2m$ (upper bidiagonal for $m = 1$), and we invert it exactly. We prove that (1.10) has an exact zero in an explicit ball around a numerically computed centre, using a Newton–Kantorovich (radii polynomial) argument in weighted ℓ^1 spaces of disk-polynomial coefficients. The approximate inverse consists of a finite matrix, of size 525, 465, 200, 259 and 610 for $n = 4, 6, 8, 10, 14$, inverted and checked in ball arithmetic, together with an exact

inverse on the tail. All infinitely many columns of the defect operator are covered: finitely many are evaluated without truncation in interval arithmetic, and the remaining ones are bounded by analytic estimates. From the zero we obtain a univalent map ψ , a strictly convex planar domain D (by a lower bound for $\operatorname{Re}(1 + w\psi''/\psi')$ on the closed disc, uniform over the existence ball), a solution of (1.8) that extends analytically across ∂D , and finally the domain Ω and the function u , by the rotations of Sections 10 and 19 for $n = 4, 6$ and by the adjoint lift of Section 20 for $n = 8, 10, 14$.

Dimension three. Here there is no reduction to the planar Laplacian (Remark 10.5), and we pull the three-dimensional problem back to the unit ball B^3 . Let ψ be a map as in Remark 1.5, with odd powers and real coefficients, and Θ its axisymmetric extension. The differential of Θ is a similarity with factor $|\psi'|$ in the meridian directions and multiplication by $q = \operatorname{Im} \psi(w)/\operatorname{Im} w$ in the azimuthal direction, where $w = x_1 + i|x'|$. Hence, for an axisymmetric function $U = u \circ \Theta$ on B^3 , problem (1.1) becomes

$$(1.11) \quad \operatorname{div}(q\nabla U) + q|\psi'|^2 U = 0 \quad \text{in } B^3, \quad U = 1 \quad \text{and} \quad \nabla U = 0 \quad \text{on } \partial B^3.$$

We write $U = 1 + K_3 g$, where K_3 is an explicit inverse of the Laplacian of $\mathbb{R}^3$ with vanishing Cauchy data on ∂B^3 , acting on the axisymmetric ball polynomials $r^\ell P_\ell(x_1/r) P_s^{(0, \ell+1/2)}(2r^2 - 1)$ with ℓ even and $s \geq 1$. Since q is a real-analytic function of x_1 and $|x'|^2$ that depends linearly on the coefficients of ψ , (1.11) becomes a quartic polynomial equation in the unknown field coefficients and shape coefficients, with no division by the unknown q . The disk-polynomial algebra of [14] is replaced by weighted ℓ^1 norms adapted to the Legendre–Chebyshev connection coefficients; multiplication estimates follow from the positivity of the Gaunt coefficients and of Jacobi connection coefficients. The approximate inverse consists of a finite block of dimension 7371, with 7280 field and 91 shape unknowns, and an explicit inverse of the principal boundary part on high shape modes, given by a convergent Neumann series. The Newton–Kantorovich argument again covers all columns, finitely many by interval evaluation and all others by analytic estimates. Univalence and positivity of q follow from a lower bound for $\operatorname{Re} \psi'$ on a disc of radius 101/100, uniform over the existence ball, which also shows that Θ is a real-analytic diffeomorphism near $\overline{B^3}$; strict star-shapedness follows from a lower bound for $\operatorname{Re}(w\psi'/\psi)$ on the unit circle, and strict convexity from a positive lower bound for $\operatorname{Re}(1 + w\psi''/\psi')$ on the whole closed disc and throughout the projected shape ball. The field $u = U \circ \Theta^{-1}$ extends analytically across the axis and across $\partial\Omega$ by elliptic regularity.

Dimension five. The same pull-back works in $\mathbb{R}^5$ with q^3 in place of q : for an axisymmetric function $U = u \circ \Theta$ on B^5 , problem (1.1) becomes $\operatorname{div}(q^3 \nabla U) + q^3 |\psi'|^2 U = 0$ with $U = 1$ and $\nabla U = 0$ on ∂B^5 , and after removal of the positive factor q^2 the equation is again a quartic polynomial equation in the field and shape coefficients. The new difficulty is functional-analytic: for $n \geq 5$ the Jacobi coefficient norm used for $n = 3$ does not control $K_n g$ in C^1 , since the gradients of K_n applied to normalised ball polynomials of radial degree s grow like $s^{(n-4)/2}$. We therefore grade all coefficient norms by the weight $1 + D/10^6$ of the polynomial degree D . In these norms K_5 is bounded into C^1 with a large but finite constant, which is used only qualitatively; the multiplication constants are unchanged, and the stencil, inverse and tail bounds acquire factors of at most 1.001. The approximate inverse has a finite block of dimension 4698, and a new analytic bound covers all shape columns beyond $j = 599$. The resulting domain is convex, and its non-ball witness is the coefficient c_3 (Section 22).

Dimension seven. The same axisymmetric construction uses q^5 in the divergence equation and the squared weight $(1 + D/10^6)^2$. The finite block has dimension 9010; Section 23 gives the seven-dimensional stencils and all field and shape tail estimates.

The exact inverse in general dimension. Part 5 applies the material-derivative construction of the planar proof to the axisymmetric quartic equation on B^n . The field norm sums analytically weighted Gegenbauer angular coefficients with a supremum norm in the radial variable, replacing the graded Jacobi coefficient norms of Sections 22 and 23. A complete Bessel spectral window and the Gram identity for its full complement certify a symmetric-sector Dirichlet inverse; heat-kernel smoothing of the low angular degrees and the radial maximum principle on the high degrees transfer it to the working norm. The Poisson normal trace gains the derivative needed by the holomorphic boundary real-part problem, and the material identity gives an exact two-sided inverse of the residual-corrected linearisation, including its regular axis term. An exactly clamped centre, quartic remainder bounds and a radii polynomial then give the domains of Theorem 24.1, with univalence, both principal-curvature estimates and a nonzero shape coefficient holding on the entire existence ball.

Relation with earlier versions. Version 1 of this paper proved Theorem 1.1 in dimensions 3, 4, 6, 8, 10, 14. This version adds convex counterexamples in dimensions 2, 5, 7, 9, 11–13, 15–18, 20 and 21, a second proof of the five-dimensional case by the exact-inverse method, and the first convex planar counterexample to Berenstein’s conjecture (Theorem 5.1). The earlier preprint and its verification files are archived in [36].

Organisation. Part 1 treats the plane first: Section 2 gives the spaces and Dirichlet inverse, Section 3 constructs the exactly clamped centre and the coupled inverse, and Section 4 proves existence and geometry. Part 2 treats Berenstein’s problem: Sections 5–7 give the fixed-disc equation and Robin inverse, Section 8 constructs the residual-corrected coupled inverse, and Section 9 proves existence, geometry and sign change. Part 3 gives the Cartan-plane reductions: Sections 10–18 prove the four-dimensional case, Section 19 the six-dimensional case, and Section 20 the cases $n = 8, 10, 14$. Part 4 proves the axisymmetric cases $n = 3, 5, 7$ in Sections 21, 22 and 23, retaining the additional nonconvex three-dimensional example. Part 5 proves Theorem 24.1 by one dimension-parametrised exact-inverse argument. Part 6 contains the Pompeiu implication, the computer-assisted verification, the mechanism and further examples, the comparison with rigidity results, and the open problems.

Part 1. The plane

2. FUNCTION SPACES AND THE DIRICHLET INVERSE

In this part $m = 182$, $p = \psi'$, and W is the holomorphic shape space; Y is the real invariant field space and K is the zero-Dirichlet Poisson inverse, restricted to its clamped range when acting on X_g . These symbols are local: in later parts W may denote a Weyl group, Y a residual bound, and K a compatible inverse in a coefficient space. In the spectral subsection ρ is the radius of the reference shape; thereafter it is the radius of the forty-mode centre, rather than the angular coefficient weight used in the Cartan reductions.

2.1. Conformal formulation and Banach spaces.

2.1.1. *Pull-back to the disc.* Identify $\mathbb{R}^2$ with $\mathbb{C}$. For a holomorphic univalent map ψ with derivative p , conformal covariance of the Laplacian gives

$$(2.1) \quad \Delta_w(u \circ \psi) = |p|^2(\Delta_z u) \circ \psi.$$

Along the unit circle the vector $wp(w)$ points in the outward normal direction to the image boundary and has length $|p(w)|$. Hence, for $U = u \circ \psi$,

$$U = 1, \quad \partial_r U = 0 \quad \text{on } \partial\mathbb{D} \quad \Longleftrightarrow \quad u = 1, \quad \partial_\nu u = 0 \quad \text{on } \partial\Omega.$$

The pulled-back equation is $\Delta U + |p|^2 U = 0$. We encode both boundary conditions in the field space before solving the equation.

Throughout the construction set

$$(2.2) \quad m = 182, \quad R = 2^{1/m}, \quad R^{jm} = 2^j.$$

For a continuous function on $\overline{\mathbb{D}}$ with angular coefficients $f_n(r)$, define

$$(2.3) \quad f(r, \theta) = \sum_{n \in \mathbb{Z}} f_n(r) e^{in\theta}, \quad \|f\|_{\mathcal{Y}} = \sum_{n \in \mathbb{Z}} R^{|n|} \sup_{0 \leq r \leq 1} |f_n(r)|.$$

The coefficients must have the origin compatibility of a continuous Cartesian function. In particular, $f_n(0) = 0$ for $n \neq 0$. The space $\mathcal{Y}$ consists of those functions for which (2.3) is finite. Let Y be its real D_m -invariant subspace. Its only frequencies are $n \in m\mathbb{Z}$, and its coefficients satisfy $f_{-n} = f_n \in \mathbb{R}$.

Lemma 2.1. *The spaces $\mathcal{Y}$ and Y are Banach spaces, and*

$$(2.4) \quad \|fh\|_{\mathcal{Y}} \leq \|f\|_{\mathcal{Y}} \|h\|_{\mathcal{Y}}.$$

The boundary trace has norm at most one into the Laurent coefficient space with norm $\sum_n R^{|n|} |f_n(1)|$.

Proof. The weighted sum in (2.3) gives uniform convergence of the angular series. A Cauchy sequence converges in each radial coefficient in $C([0, 1])$ and in the weighted sum of these coefficient norms. The limit is uniform on the closed disc, so it preserves Cartesian continuity and origin compatibility. The reality and symmetry conditions are closed. For products, absolute convergence gives

$$(fh)_n = \sum_k f_k h_{n-k}, \quad R^{|n|} \leq R^{|k|} R^{|n-k|}.$$

Taking radial suprema and summing proves (2.4). The trace estimate follows coefficient by coefficient. $\square$

The shape derivative belongs to the real-coefficient holomorphic space

$$(2.5) \quad W = \left\{ p(w) = \sum_{j \geq 0} p_j w^{jm} : p_j \in \mathbb{R}, \quad \|p\|_W = \sum_{j \geq 0} 2^j |p_j| < \infty \right\}.$$

Every $p \in W$ is holomorphic on $|w| < R$ and continuous on its closure. As an element of $\mathcal{Y}$ on the unit disc it has norm exactly $\|p\|_W$. We normalise its primitive by

$$(2.6) \quad \psi(w) = \int_0^w p(z) dz = \sum_{j \geq 0} c_j w^{jm+1}, \quad c_j = \frac{p_j}{jm+1}.$$

The shape space fixes the origin, the orientation, and the reflection symmetry.

We also need one angular derivative of boundary data. For a Laurent series set

$$(2.7) \quad \|h\|_{W_1} = \sum_{n \in \mathbb{Z}} R^{|n|} (1 + |n|) |h_n|.$$

The holomorphic version uses only nonnegative indices. Both retain the analytic weight. If W_0 denotes the same coefficient norm without $1 + |n|$, then

$$\|h\|_{W_1} = \|h\|_{W_0} + \|\partial_\theta h\|_{W_0}.$$

The inequalities $R^{|n+k|} \leq R^{|n|}R^{|k|}$ and $1 + |n+k| \leq (1 + |n|)(1 + |k|)$ show that W_1 is a Banach algebra. We use the same notation for a holomorphic series and its boundary values when the norm is identical.

2.1.2. The clamped range of the Laplacian. Let K denote the zero-Dirichlet inverse of Δ on the disc. Define the closed real subspace

$$(2.8) \quad X_g = \left\{ g \in Y : \int_0^1 r^{|n|+1} g_n(r) dr = 0 \quad (n \in m\mathbb{Z}) \right\}.$$

Each moment is a bounded functional, so this common kernel is closed.

Lemma 2.2. *For $g \in Y$, the zero-Dirichlet solution $v = Kg$ satisfies*

$$(2.9) \quad \|Kg\|_Y \leq \frac{1}{4} \|g\|_Y, \quad \partial_r v_n(1) = \int_0^1 r^{|n|+1} g_n(r) dr.$$

Consequently $g \in X_g$ if and only if Kg has zero Neumann trace.

Proof. For $n \geq 0$, the positive Green kernel of $-\Delta_n$, with radial measure $s ds$, is

$$G_n(r, s) = \frac{(a/b)^n - (ab)^n}{2n} \quad (n > 0), \quad G_0(r, s) = -\log b, \quad a = \min(r, s), \quad b = \max(r, s).$$

The formula at $r = 0$ is understood by continuity. Since

$$0 \leq G_n(r, s) \leq \frac{1 - b^{2n}}{2n} \leq -\log b, \quad \int_0^1 G_0(r, s) s ds = \frac{1 - r^2}{4},$$

the supremum norm of each radial inverse is at most $1/4$. Summing with the angular weights proves the norm bound. The n th radial equation is

$$v_n'' + \frac{1}{r} v_n' - \frac{n^2}{r^2} v_n = g_n.$$

After multiplication by $r^{|n|+1}$ the left side is the derivative of $r^{|n|+1} v_n' - |n| r^{|n|} v_n$. This function tends to zero as $r \rightarrow 0$ for the regular solution, and $v_n(1) = 0$. Integration over $[0, 1]$ therefore gives

$$\int_0^1 r^{|n|+1} g_n(r) dr = v_n'(1) - |n| v_n(1) = v_n'(1).$$

Continuous right sides give $v \in W^{2,s}(\mathbb{D})$ for every finite s , hence $v \in C^1(\overline{\mathbb{D}})$ by taking $s > 2$. Consequently $v_n'(1)$ is the n th Fourier coefficient of the Neumann trace $\partial_r v|_{r=1}$, and the identity just proved gives its Fourier expansion. Since $|v_n'(1)| \leq \sup |g_n|/(|n| + 2)$, the trace series is absolutely convergent. The Neumann trace therefore vanishes if and only if every moment in (2.8) vanishes, which is the last assertion. $\square$

2.1.3. The cubic equation. We work on real Banach spaces

$$(2.10) \quad X = X_g \times W, \quad \|(h, k)\|_X = \|h\|_Y + \|k\|_W, \quad F: X \rightarrow Y, \quad F(g, p) = g + |p|^2(1 + Kg).$$

The algebra estimate makes F a continuous cubic polynomial map. For any zero set $U = 1 + Kg$. Lemma 2.2 gives its exact boundary data, while $F = 0$ gives

$$(2.11) \quad \Delta U + |p|^2 U = 0 \quad \text{in } \mathbb{D}, \quad U = 1, \quad \partial_r U = 0 \quad \text{on } \partial\mathbb{D}.$$

Univalence and strict convexity are proved for every shape in the final ball, after existence has been established in X .

We distinguish two fixed shapes. The eight-mode reference p_{ref} is used only to compute the Dirichlet spectral inverse in Subsection 2.2. The forty-mode shape p_0 defines the clamped

centre and every subsequent nonlinear constant. Their proximity transfers the reference inverse to p_0 at the end of Subsection 2.2. The exact shape p supplied by the contraction is a third object. The finite arrays specify the first two objects, while the contraction specifies the last.

2.2. The Dirichlet operator in the symmetric sector. This section constructs an inverse of $L_{\text{ref}} = \Delta + |p_{\text{ref}}|^2$ in the real even D_m sector and transfers it to $L = \Delta + |p_0|^2$. In the spectral construction only, abbreviate

$$p = p_{\text{ref}}, \quad \rho = p_{\text{ref},0}, \quad q = |p|^2, \quad V = q - \rho^2.$$

The L^2 measure is $r dr d\theta / (2\pi)$. All projections and operators in this section act inside that invariant sector.

2.2.1. The spectral window. Let $a_{j,k}$ denote the positive zeros of J_{jm} in increasing order. An orthonormal Dirichlet basis is

$$(2.12) \quad b_{j,k}(r)\chi_j(\theta), \quad b_{j,k}(r) = \frac{\sqrt{2}J_{jm}(a_{j,k}r)}{|J_{jm+1}(a_{j,k})|}, \quad \chi_0 = 1, \quad \chi_j = \sqrt{2}\cos(jm\theta) \quad (j > 0).$$

Indeed, the angular factors are orthonormal, and integration of the Bessel equation and its parameter derivative gives

$$\int_0^1 r J_n(ar)^2 dr = \frac{J_{n+1}(a)^2}{2} \quad (J_n(a) = 0).$$

The usual Dirichlet spectral decomposition on the disc establishes completeness. The potential V preserves this sector.

Let P be the projection onto all basis states with

$$(2.13) \quad |a_{j,k} - \rho| < 160,$$

and set $Q = I - P$. Complete zero enumeration gives the counts in Table 3; their sum is 408.

Angular order	0	182	364	546	728	910	1092
States in the window	102	99	91	74	37	5	0

TABLE 3. The complete symmetric-sector spectral window. All higher angular orders have no state in the window.

Lemma 2.3. *The interval sign and spacing tests described below enumerate every state of (2.13). The independent quarter-step sign grid gives the same enumeration without using the root proposals.*

Proof. For integer $n \geq 0$, put $y(x) = \sqrt{x}J_n(x)$. The Bessel equation becomes

$$(2.14) \quad y'' + Q_n(x)y = 0, \quad Q_n(x) = 1 - \frac{n^2 - 1/4}{x^2}.$$

If y vanishes at $a < b$ and $Q_n \leq Q_{\max}$ on $[a, b]$, integration by parts and Wirtinger's inequality imply

$$\frac{\pi^2}{(b-a)^2} \int_a^b y^2 \leq \int_a^b (y')^2 = \int_a^b Q_n y^2 \leq Q_{\max} \int_a^b y^2.$$

Hence two distinct zeros of y in an interval on which $Q_n \leq Q_{\max}$ are separated by at least $\pi/\sqrt{Q_{\max}}$. Since $Q_n(x) \leq 5/4$ for $x \geq 1$ and every integer $n \geq 0$, zeros in $x \geq 1$ are separated by more than 2.8. Each proposed root is enclosed by opposite signs at exact offsets $\pm 2^{-220}$ from its exact midpoint. Its left endpoint is checked to exceed 1 and its width to be less than

1/10, so the bracket contains exactly one root. For successive brackets, the distance between the outer endpoints is checked to be less than $2\pi/\sqrt{Q_{\max}}$, where $Q_{\max}$ is the maximum of Q_n between those endpoints. A further root between the two enclosed roots would require the opposite inequality. Guard roots on both sides cover the endpoints of the window.

There is a separate initial check when the angular order lies above the lower endpoint of the window. For $n > 0$, the positive leading power of J_n gives $J_n, J'_n > 0$ near zero. A first zero of J'_n in $(0, n)$ while $J_n > 0$ would satisfy

$$J''_n = (n^2/x^2 - 1)J_n > 0,$$

which is incompatible with a first crossing from positive derivative. A first zero of J_n is then also impossible. The limiting values at n give $y(n), y'(n) > 0$. Comparison in (2.14) with the constant coefficient $Q_{\max}$ places the first subsequent zero at distance at least $\pi/(2\sqrt{Q_{\max}})$ from n . A hidden first zero before the proposed one would place the latter at distance at least $3\pi/(2\sqrt{Q_{\max}})$; the reverse strict inequality is checked. For $n \geq 1092 > \rho + 160$, positivity up to n excludes the entire window.

For the second enumeration, use again the universal bound

$$Q_n(x) \leq \frac{5}{4} \quad (x \geq 1), \quad \frac{\pi}{\sqrt{5/4}} > 2.8.$$

Every cell of length 1/4 in $x \geq 1$ therefore contains at most one zero. Positive zeros are simple by uniqueness for the Bessel ODE. Nonzero endpoint signs determine whether the cell contains one zero or none. Evaluate the signs at the 1281 exact points

$$(2.15) \quad x_k = \rho - 160 + k/4, \quad 0 \leq k \leq 1280,$$

for $n = 0, 182, 364, 546, 728, 910, 1092$. Increasing precision until the sign is resolved yields Table 3. The positivity argument excludes all remaining orders. Finally, each root bracket from the first method must lie in a distinct sign-changing cell of the second method. Equality of the reconstructed state lists establishes agreement at the level of individual roots, as well as total and sector counts. $\square$

2.2.2. Matrix entries and certified quadrature. On the 408-dimensional range of P , form

$$(2.16) \quad B = PVP, \quad B_2 = PV^2P, \quad H = \text{diag}(a_{j,k}^2 - \rho^2) - B.$$

These are exact matrices, evaluated through enclosures. If $p(w) = \sum p_j w^{jm}$ is the eight-mode reference, write

$$\begin{aligned} q_h(r) &= \sum_{k \geq 0} p_{k+h} p_k r^{(2k+h)m} \quad (h > 0), \\ q_0(r) &= \sum_{k > 0} p_k^2 r^{2km}. \end{aligned}$$

Then $V = q_0 + \sum_{h>0} 2q_h \cos(hm\theta)$; absent coefficients are zero. The Fourier coefficients of V^2 are the full finite convolution over both positive and negative indices. With the angular normalisation in (2.12), the angular integrals of multiplication by V are

$$q_0 \quad (0, 0), \quad \sqrt{2}q_j \quad (0, j), \quad q_{|i-j|} + q_{i+j} \quad (i, j > 0).$$

The same formulas applied to the convolved coefficients give the entries of B_2 . Thus the remaining integrations are one-dimensional radial integrals.

Use $N_G = 1536$ Gauss–Legendre nodes on $[-1, 1]$. Each is enclosed as a sign-changing bracket of the Legendre polynomial P_{N_G} ; the disjoint brackets and the degree prove that all nodes are covered. At a true node x_i , the weight is

$$w_i = \frac{2}{(1 - x_i^2)P'_{N_G}(x_i)^2}.$$

The identities

$$(x^2 - 1)P'_{N_G} = N_G(xP_{N_G} - P_{N_G-1}), \quad \max_{[-1,1]} |P''_{N_G}| = \frac{(N_G - 1)N_G(N_G + 1)(N_G + 2)}{8}$$

bound evaluation at the unknown true root from evaluation at its enclosing midpoint. The transformed radial node is $r_i = (1 + x_i)/2$, with radial measure factor $w_i r_i/2$.

The Bessel values at exact midpoint arguments are enclosed at 2048 bits. For real arguments and integer order, the integral representation implies $|J'_n| \leq 1$. It therefore bounds the additional uncertainty in arguments arising from enclosed roots and nodes. A binary64 value exported for matrix assembly is interpreted as an exact dyadic number, accompanied by a verified row error; the resulting matrix enclosure includes that error and the quadrature error.

For completeness, the quadrature majorant is explicit. On the Bernstein ellipse of parameter 2 in $x = 2r - 1$,

$$|r| \leq 9/8, \quad |\operatorname{Im} r| \leq 3/8, \quad |J_n(ar)| \leq e^{a|\operatorname{Im} r|}.$$

Set

$$P_e = \sum_j |p_j|(9/8)^{jm}, \quad V_e = P_e^2 + \rho^2, \quad b_* = \min |J_{jm+1}(a_{j,k})|, \quad a_{\max} = \max a_{j,k},$$

where the minimum and the maximum are taken over the 408 states of the window (2.13); in particular $a_{\max} < \rho + 160$. The common bound

$$(2.17) \quad M_e = 8(1 + 9/8)(1 + V_e^2)e^{3a_{\max}/4}/b_*^2$$

dominates every integrand for B and B_2 , including the radial measure, both basis normalisations and angular factors. Its analytic Chebyshev coefficients have absolute value at most $2M_e 2^{-k}$. Truncate at degree $2N_G - 1$. The Gauss rule is exact on that polynomial, and both the integral and the positive quadrature rule have total mass two. The error is therefore at most

$$(2.18) \quad 4 \sum_{k \geq 2N_G} 2M_e 2^{-k} = 16M_e 2^{-2N_G}.$$

This error is added to every matrix entry, before any inverse estimate is made.

2.2.3. The entire complement. Let $s = \sum_{j>0} |p_j|$ and define

$$(2.19) \quad v_* = 2\rho s + s^2, \quad g_* = 160(2\rho - 160), \quad \gamma = g_* - v_*.$$

Here $\|V\|_\infty \leq v_*$. On the complementary Dirichlet operator domain set

$$D_Q = Q(-\Delta - \rho^2)Q, \quad M_Q = D_Q - QVQ.$$

The operator D_Q is generally indefinite. Its spectrum nonetheless has a two-sided gap of size g_* . For a root $a \leq \rho - 160$, put $t = \rho - a$; then $t \in [160, \rho]$ and

$$|a^2 - \rho^2| = t(2\rho - t) \geq 160(2\rho - 160).$$

For $a \geq \rho + 160$ the corresponding bound is $160(2\rho + 160)$, which is larger. Thus D_Q^{-1} is bounded with norm at most $1/g_*$. If $\gamma > 0$, the factorization

$$(2.20) \quad M_Q = (I - QVQD_Q^{-1})D_Q$$

on the domain of D_Q has an invertible bounded first factor, by its Neumann series. Consequently

$$(2.21) \quad \|M_Q^{-1}\| \leq 1/\gamma.$$

This argument uses a gap in absolute spectral value, not positivity of M_Q .

The following identity is the reason no further complementary radial enumeration is required:

$$(2.22) \quad G := B_2 - B^2 = PV(P + Q)VP - (PVP)^2 = (QVP)^*(QVP) \succeq 0.$$

It holds for the full orthogonal complement Q . In particular, positivity is an analytic property of the exact matrix G , even when its numerical enclosure has small entrywise uncertainty.

Proposition 2.4 (Schur complement estimate). *Assume that $\gamma > 0$ in (2.19). Let C be a real 408×408 matrix and D a real diagonal matrix of the same size with nonzero diagonal entries D_i , and suppose that the numbers ε_C , ε and η satisfy*

$$(2.23) \quad \|C^T C - I\|_\infty \leq \varepsilon_C < 1,$$

$$(2.24) \quad \| |D|^{-1/2}(C^T H C - D) |D|^{-1/2} \|_\infty \leq \varepsilon,$$

$$(2.25) \quad \frac{\text{tr}(|D|^{-1} C^T G C)}{\gamma} \leq \eta, \quad \varepsilon + \eta < 1.$$

Then L_{ref} , with its Dirichlet domain in the symmetric sector, has a bounded inverse on L^2 , and

$$(2.26) \quad \|L_{\text{ref}}^{-1}\|_{L^2 \rightarrow L^2} \leq B_{\text{ref}} := (1 + (v_*/\gamma)^2) S_* + 1/\gamma,$$

$$(2.27) \quad S_* := \frac{1 + \varepsilon_C}{\min_i |D_i|(1 - \varepsilon - \eta)}.$$

Proof. The block form. The range of P is spanned by finitely many Dirichlet eigenfunctions, so it lies in the operator domain of L_{ref} , and P commutes with Δ on that domain. Consequently Q maps the operator domain into itself, and the operator domain is the direct sum of the range of P and the domain of D_Q . For $u = u_P + u_Q$ in this decomposition, $P(-\Delta - \rho^2)u_Q = 0$ and $Q(-\Delta - \rho^2)u_P = 0$. Hence

$$-L_{\text{ref}} = \begin{pmatrix} H & b^* \\ b & M_Q \end{pmatrix}, \quad b = -QVP, \quad b^* = -PVQ,$$

where $\|b\| \leq \|V\|_\infty \leq v_*$. The operator D_Q is self-adjoint and diagonal in the basis (2.12), and QVQ is bounded and symmetric. Thus M_Q is self-adjoint on the domain of D_Q , and its inverse, which exists by (2.20), is bounded and self-adjoint. The Schur complement

$$S = H - b^* M_Q^{-1} b$$

is a real symmetric matrix on the range of P .

The Schur complement. Both defect matrices in (2.23) and (2.24) are symmetric. Their ℓ^2 operator norms are therefore bounded by their maximum row sums. The first defect makes $C^T C$, and hence the square matrix C , invertible, and gives $\|C\|^2 \leq 1 + \varepsilon_C$. Set $X_Q = bC|D|^{-1/2}$.

By (2.22), $X_Q^* X_Q = |D|^{-1/2} C^T G C |D|^{-1/2}$ is positive semidefinite, so its norm is at most its trace $\text{tr}(|D|^{-1} C^T G C)$. With (2.21) and (2.25),

$$\|X_Q^* M_Q^{-1} X_Q\| \leq \gamma^{-1} \|X_Q^* X_Q\| \leq \eta.$$

The normalised Schur matrix is

$$|D|^{-1/2} C^T S C |D|^{-1/2} = \text{sgn}(D) + |D|^{-1/2} (C^T H C - D) |D|^{-1/2} - X_Q^* M_Q^{-1} X_Q,$$

that is, $\text{sgn}(D)$ plus a perturbation of norm at most $\varepsilon + \eta < 1$. Since $\text{sgn}(D)$ is an isometry, the normalised Schur matrix is invertible and its inverse has norm at most $(1 - \varepsilon - \eta)^{-1}$. Undoing the congruence,

$$S^{-1} = C |D|^{-1/2} (|D|^{-1/2} C^T S C |D|^{-1/2})^{-1} |D|^{-1/2} C^T, \quad \|S^{-1}\| \leq S_*,$$

with S_* as in (2.27). The signs of D are retained throughout; no definiteness of H or of S is used.

The block inverse. For $f = f_P + f_Q$ in L^2 , put

$$u_P = S^{-1}(f_P - b^* M_Q^{-1} f_Q), \quad u_Q = M_Q^{-1}(f_Q - b u_P).$$

Then u_Q lies in the domain of D_Q , so $u = u_P + u_Q$ lies in the operator domain, and

$$H u_P + b^* u_Q = S u_P + b^* M_Q^{-1} f_Q = f_P, \quad b u_P + M_Q u_Q = f_Q.$$

Thus $-L_{\text{ref}} u = f$. Conversely, if $-L_{\text{ref}} u = 0$ for some u in the operator domain, the second block row gives $u_Q = -M_Q^{-1} b u_P$, the first gives $S u_P = 0$, and therefore $u = 0$. Hence $-L_{\text{ref}}$ is a bijection of its domain onto L^2 , with the bounded inverse

$$(-L_{\text{ref}})^{-1} = \begin{pmatrix} I \\ -M_Q^{-1} b \end{pmatrix} S^{-1} \begin{pmatrix} I & -b^* M_Q^{-1} \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & M_Q^{-1} \end{pmatrix}.$$

Since M_Q^{-1} is self-adjoint, the row operator is the adjoint of the column. The squared norm of the column is at most $1 + (v_*/\gamma)^2$, because $\|M_Q^{-1} b\| \leq \|b\|/\gamma \leq v_*/\gamma$. Together with $\|S^{-1}\| \leq S_*$ and (2.21), this gives (2.26). $\square$

For the eight-mode reference, C and D are binary64 proposals read as exact dyadic matrices, and the enclosed calculation gives

$$(2.28) \quad \begin{aligned} v_* &< 5004.03, & \gamma &> 230022.34, & \min_i |D_i| &> 0.1953, \\ \varepsilon_C &< 10^{-12}, & \varepsilon &< 10^{-9}, & \eta &< 0.240915, \end{aligned}$$

and the quadrature error (2.18) is below $1.35 \cdot 10^{-360}$. The hypotheses of Proposition 2.4 therefore hold, and $B_{\text{ref}} < 6.748$. Floating-point root and congruence proposals enter this conclusion only through the root coverage of Lemma 2.3, the enclosed matrix entries and the defects (2.23)–(2.25).

2.2.4. Transfer to the forty-mode centre. Return now to p_0 . The pointwise difference on $\overline{\mathbb{D}}$ satisfies

$$(2.29) \quad \| |p_0|^2 - |p_{\text{ref}}|^2 \|_\infty \leq \delta_q, \quad \delta_q = (\|p_0\|_\infty + \|p_{\text{ref}}\|_\infty) \|p_0 - p_{\text{ref}}\|_\infty.$$

Bounding the three supremum norms by unweighted sums of coefficients gives $\delta_q < 8.376 \cdot 10^{-14}$. On the Dirichlet domain,

$$\Delta + |p_0|^2 = \left(I + (|p_0|^2 - |p_{\text{ref}}|^2) L_{\text{ref}}^{-1} \right) L_{\text{ref}},$$

and the first factor is invertible by its Neumann series because $B_{\text{ref}}\delta_q < 1$. Hence

$$(2.30) \quad \|(\Delta + |p_0|^2)^{-1}\|_{L^2 \rightarrow L^2} \leq B_{L^2} := \frac{B_{\text{ref}}}{1 - B_{\text{ref}}\delta_q} < 6.748.$$

All constants from Subsection 2.3 onward use p_0 and its conformal radius $\rho = p_0(0)$.

Remark 2.5. The inverse in (2.30) is confined to the symmetric sector. For a genuine Schiffer solution, physical derivatives of u solve a homogeneous Dirichlet problem at the same eigenvalue. Their angular sectors are different from the invariant even sector, so they do not contradict the inverse proved here.

2.3. From the spectral inverse to the analytic norm. In this subsection and Section 3, put $p = p_0$, $q = |p_0|^2$, $\rho = p_0(0)$, and $L = \Delta + q$; write $p(w) = \sum_{j \geq 0} p_j w^{jm}$. Define

$$(2.31) \quad P = \sum_{j \geq 0} 2^j |p_j|, \quad P_1 = \sum_{j \geq 0} 2^j j m |p_j|, \quad V_* = P^2 - \rho^2.$$

The finite centre makes these sums finite. The product estimate shows

$$(2.32) \quad \|q\|_Y \leq P^2, \quad \|q - \rho^2\|_Y \leq V_*.$$

For the second bound, expand $p\bar{p}$ and remove the $(j, k) = (0, 0)$ term; $2^{|j-k|} \leq 2^{j+k}$ bounds the remaining terms.

Proposition 2.6. *Suppose the symmetric-sector Dirichlet inverse has L^2 bound B_{L^2} . Let h be the least positive integer with $hm > \rho$, and set*

$$(2.33) \quad d_h = (hm)^2 - \rho^2, \quad \beta = V_*/d_h, \quad W_\ell = 1 + 2 \sum_{j=1}^{h-1} 2^j.$$

If $d_h > 0$ and $0 \leq \beta < 1$, then the zero-Dirichlet solution of $Lv = f$, for $f \in Y$, belongs to Y and satisfies

$$(2.34) \quad \|v\|_Y \leq B_Y \|f\|_Y, \quad B_Y = \frac{W_\ell(1 + P^2 B_{L^2})/(2\sqrt{2}) + 1/d_h}{1 - \beta}.$$

Its normal trace $N = \partial_r v|_{r=1}$ belongs to W_1 , with

$$(2.35) \quad \|N\|_{W_1} \leq B_N \|f\|_Y, \quad B_N = 1 + P^2 B_Y.$$

Proof. Parseval's identity and the radial measure give

$$\|f\|_{L^2}^2 = \sum_n \int_0^1 |f_n(r)|^2 r \, dr \leq \frac{1}{2} \sum_n \sup_r |f_n(r)|^2 \leq \frac{1}{2} \|f\|_Y^2.$$

The spectral inverse therefore gives $\|v\|_{L^2} \leq B_{L^2} \|f\|_Y / \sqrt{2}$. For every angular coefficient, apply the radial Poisson inverse to $(f - qv)_n$. Its kernel is dominated by $-\log \max(r, s) \leq -\log s$, and

$$\int_0^1 (\log s)^2 s \, ds = \frac{1}{4}.$$

Cauchy–Schwarz yields

$$(2.36) \quad \sup_r |v_n(r)| \leq \frac{1}{2} \|(f - qv)_n\|_{L^2(r \, dr)} \leq \frac{1 + P^2 B_{L^2}}{2\sqrt{2}} \|f\|_Y.$$

This estimates all coefficients, but we use it only for $|n| < hm$. Their total weighted norm is bounded by W_ℓ times the right side of (2.36).

For $|n| \geq hm$, the positive potential of $-\Delta_n - \rho^2$ is

$$\frac{n^2}{r^2} - \rho^2 \geq n^2 - \rho^2 \geq d_h > 0.$$

The radial maximum principle bounds its zero-Dirichlet inverse in supremum norm by $1/(n^2 - \rho^2)$. The direct sum of these inverses is consequently bounded by $1/d_h$ in the weighted radial supremum norm on the high sector $|n| \geq hm$ of Y . Let Π_H be the angular projection onto that sector; it has norm one on Y , and multiplication by $q - \rho^2$ has norm at most V_* by (2.4) and (2.32). Let $v_\ell = v - \Pi_H v$ be the low part already estimated, which belongs to Y , and solve the high equations

$$(\Delta + \rho^2)\tilde{v}_H = \Pi_H(f - (q - \rho^2)(v_\ell + \tilde{v}_H)), \quad \tilde{v}_H = 0 \text{ on } \partial\mathbb{D},$$

by the contraction mapping theorem in the high sector of Y . The contraction constant is at most $\beta < 1$, so there is a unique solution $\tilde{v}_H$, and its weighted norm is finite. This construction establishes finiteness of the weighted high norm before any inequality involving that norm is rearranged.

We identify $\tilde{v}_H$ with the high part $v_H = \Pi_H v$ of the original L^2 solution. Put

$$F_H = \Pi_H(f - (q - \rho^2)(v_\ell + \tilde{v}_H)) - \rho^2 \tilde{v}_H \in Y.$$

In every angular mode, $\tilde{v}_{H,n}$ is the regular solution of the radial equation $\Delta_n \tilde{v}_{H,n} = F_{H,n}$ with $\tilde{v}_{H,n}(1) = 0$; hence $\tilde{v}_H = KF_H$. Since F_H is continuous, $\tilde{v}_H \in W^{2,s}(\mathbb{D})$ for every finite s , as in the proof of Lemma 2.2. In particular $\tilde{v}_H \in H^2 \cap H_0^1$, and the high equation holds almost everywhere. Applying Π_H to $(\Delta + \rho^2)v = f - (q - \rho^2)v$ shows that $v_H \in H^2 \cap H_0^1$ satisfies the same equation with $\tilde{v}_H$ replaced by v_H . The difference $z = \tilde{v}_H - v_H$ therefore lies in the high sector and solves

$$(\Delta + \rho^2)z = -\Pi_H((q - \rho^2)z).$$

On this sector, $r \leq 1$ and angular orthogonality give

$$\int_{\mathbb{D}} |\nabla z|^2 \geq \int_{\mathbb{D}} \frac{|\partial_\theta z|^2}{r^2} \geq (hm)^2 \int_{\mathbb{D}} |z|^2.$$

Multiplying the equation for z by $-z$ and integrating over $\mathbb{D}$, and using that Π_H is self-adjoint in L^2 with $\Pi_H z = z$, gives

$$d_h \int_{\mathbb{D}} |z|^2 \leq \int_{\mathbb{D}} |\nabla z|^2 - \rho^2 \int_{\mathbb{D}} |z|^2 = \int_{\mathbb{D}} (q - \rho^2)|z|^2 \leq V_* \int_{\mathbb{D}} |z|^2,$$

because $\|q - \rho^2\|_\infty \leq \|q - \rho^2\|_Y \leq V_*$. Since $d_h > V_*$, $z = 0$. Thus $v_H = \tilde{v}_H$ belongs to Y .

The high fixed-point equation now gives

$$\|v_H\|_Y \leq d_h^{-1} \|f\|_Y + \beta(\|v_\ell\|_Y + \|v_H\|_Y).$$

Add the low norm and solve for the total norm $\|v\|_Y = \|v_\ell\|_Y + \|v_H\|_Y$ to obtain (2.34). Finally apply the trace formula from Lemma 2.2 to $\Delta v = f - qv$:

$$(2.37) \quad N_n = \int_0^1 r^{|n|+1} (f - qv)_n(r) dr.$$

It follows that

$$R^{|n|}(1 + |n|)|N_n| \leq R^{|n|} \frac{1 + |n|}{2 + |n|} \sup_r |(f - qv)_n(r)|.$$

Summation and (2.32) prove (2.35). $\square$

For the centre in this paper, $h = 5$ and $W_\ell = 61$. The numerical hypotheses and consequences are

$$(2.38) \quad \begin{aligned} d_h &> 164759.13, & \beta &< 0.061958, \\ B_Y &< 1.045 \cdot 10^8, & B_N &< 7.039 \cdot 10^{13}. \end{aligned}$$

These are bounds in the same angular weight used for the shape space. The normal trace gains the factor $(|n| + 2)^{-1}$, which pays for one derivative when the shape variation is reconstructed in Subsection 3.2. No reduction of the analytic radius is needed.

3. THE CENTRE AND THE EXACT LINEARISATION INVERSE

3.1. The exactly clamped approximate solution.

3.1.1. *The finite input and the analytic lift.* The frozen centre contains 41 exact decimal entries in each of the arrays c_j , a_j and s_j , indexed by $0 \leq j \leq 40$. Every s_j is positive. The file key $\mathbf{p}$ stores c_j , not the derivative coefficient $p_j = (jm + 1)c_j$. Define

$$(3.1) \quad \psi_0(w) = \sum_{j=0}^{40} c_j w^{jm+1}, \quad p_0 = \psi'_0, \quad \rho = c_0.$$

The exact decimal value of ρ lies in $(814.45, 814.46)$. The finite physical field is

$$(3.2) \quad u_*(z) = \sum_{j=0}^{40} \frac{a_j}{s_j} J_{jm}(|z|) \cos(jm \arg z), \quad U_*(w) = u_*(\psi_0(w)).$$

The normalisation divisors s_j are part of the exact input; no identity with a Bessel value is assumed. Each summand extends smoothly through $z = 0$ and is an entire real-analytic solution of $(\Delta + 1)u = 0$. Thus $\Delta U_* + qU_* = 0$, where $q = |p_0|^2$.

Let D_j and N_j be the full cosine coefficients in

$$U_*(1, \theta) - 1 = \sum_{j \geq 0} D_j \cos(jm\theta), \quad \partial_r U_*(1, \theta) = \sum_{j \geq 0} N_j \cos(jm\theta).$$

For $n = jm$ introduce

$$(3.3) \quad h_n^D(r) = r^n \left(1 - \frac{n}{2}(r^2 - 1)\right), \quad h_n^N(r) = \frac{1}{2}r^n(r^2 - 1).$$

Their Dirichlet and radial derivative traces at $r = 1$ are $(1, 0)$ and $(0, 1)$, respectively. Set

$$(3.4) \quad \begin{aligned} \ell(r, \theta) &= \sum_{j \geq 0} (D_j h_{jm}^D(r) + N_j h_{jm}^N(r)) \cos(jm\theta), \\ U_0 &= U_* - \ell, \quad g_0 = \Delta(U_0 - 1). \end{aligned}$$

The bounds below imply convergence of this series with all derivatives needed in the proof, on a neighbourhood of the closed disc. Consequently

$$(3.5) \quad U_0 = 1, \quad \partial_r U_0 = 0 \quad \text{on } \partial\mathbb{D}, \quad g_0 \in X_g, \quad Kg_0 = U_0 - 1.$$

The inclusion follows by Green's identity against the harmonic monomials; the last equality follows from Dirichlet uniqueness for Δ . Thus (g_0, p_0) is an exactly specified point of X , even though g_0 is defined by an infinite analytic lift.

3.1.2. *Uniform strip bounds for the field.* Complexify $\phi = m\theta$ to $|\operatorname{Im} \phi| \leq \log 4$. Define

$$(3.6) \quad \begin{aligned} S &= \sum_{j>0} 4^j |c_j|, & d &= \frac{2\rho S + S^2}{\rho}, & \varepsilon_s &= \frac{1}{2} \log \frac{\rho + S}{\rho - S}, \\ P_4 &= \sum_j 4^j (jm + 1) |c_j|. \end{aligned}$$

For fixed $0 \leq r \leq 1$, write

$$A_{\pm} = \rho + \sum_{j>0} c_j r^{jm} e^{\pm i j \phi}, \quad B_s = (A_+ A_-)^{1/2}.$$

Both $A_{\pm}$ lie within S of ρ . For the centre $S < 0.07$, so $\rho > 2S$ and each of $A_{\pm}$ has argument of modulus less than $\pi/6$; this places $A_+ A_-$ in the right half-plane and selects the square root B_s with positive real part. Since $|B_s^2 - \rho^2| = |A_+ A_- - \rho^2| \leq 2\rho S + S^2$ and $|B_s + \rho| \geq \rho$, and since $|A_{\pm}/B_s|^2 = |A_{\pm}/A_{\mp}| \leq (\rho + S)/(\rho - S)$,

$$(3.7) \quad |B_s - \rho| \leq d, \quad |A_+/B_s| \leq e^{\varepsilon_s}, \quad |A_-/B_s| \leq e^{\varepsilon_s}.$$

The continuation of the positive angular branch in (3.2) is

$$J_n(r B_s) e^{in\theta} (A_+/B_s)^n, \quad n = jm,$$

and the negative branch uses A_- . This representation also follows from the entire power series in the two independent variables $z, \bar{z}$, so the apparent polar coordinates introduce no singularity at the origin.

For integer $n \geq 0$, shift the contour in the Bessel integral by $i\alpha$, $\alpha \geq 0$. The integrand is entire and periodic, so

$$(3.8) \quad |J_n(z)| \leq \exp(-n\alpha + |\operatorname{Re} z| \sinh \alpha + |\operatorname{Im} z| \cosh \alpha).$$

Choose $\alpha_j = 0$ if $jm \leq \rho + d$, and otherwise

$$\alpha_j = \operatorname{acosh} \frac{jm}{\rho + d}.$$

Combining (3.7) and (3.8), define

$$(3.9) \quad b_j = \frac{|a_j|}{s_j} 4^j \exp(jm\varepsilon_s - jm\alpha_j + (\rho + d) \sinh \alpha_j + d \cosh \alpha_j).$$

Then $M_U = \sum_j b_j$ bounds $|U_*|$ on the complex strip, uniformly in r . Cauchy's estimate in the periodic variable gives a Fourier coefficient bound $M_U 4^{-|k|}$; summing the weight $2^{|k|}$ yields

$$(3.10) \quad \|U_*\|_Y \leq U_b := 3M_U.$$

We need a bound for a Cartesian derivative in the same weight, rather than only a bound for an angular derivative. The Wirtinger identities are

$$\begin{aligned} \partial_z (J_n(|z|) e^{in \arg z}) &= \frac{1}{2} J_{n-1}(|z|) e^{i(n-1) \arg z}, \\ \partial_{\bar{z}} (J_n(|z|) e^{-in \arg z}) &= -\frac{1}{2} J_{n+1}(|z|) e^{-i(n+1) \arg z}. \end{aligned}$$

Hence, for $n = jm$,

$$\partial_z u_* = \sum_{j=0}^{40} \frac{a_j}{4s_j} \left(J_{n-1}(|z|) e^{i(n-1) \arg z} - J_{n+1}(|z|) e^{-i(n+1) \arg z} \right),$$

and $\partial_w U_* = p_0 (\partial_z u_*) \circ \psi_0$. We bound the continuation of each summand to the strip as before; the two angular factors continue to $e^{i(n-1)\theta} (A_+/B_s)^{n-1}$ and $e^{-i(n+1)\theta} (A_-/B_s)^{n+1}$. Let $j \geq 1$.

With the same α_j , the contour bound (3.8) for the orders $n \pm 1$ exceeds the bound for the order n by at most the factor e^{α_j} . On the strip $|e^{\pm i\theta}| \leq 4^{1/m}$ and $|A_{\pm}/B_s| \leq e^{\varepsilon_s}$, so the angular factor of order $n + 1$ is at most $4^{1/m}e^{\varepsilon_s}$ times the bound $4^j e^{n\varepsilon_s}$ used in (3.9), while the factor of order $n - 1$ does not exceed that bound. For $j = 0$ the summand is $-(a_0/2s_0)J_1(|z|)e^{-i\arg z}$, and $\alpha_0 = 0$. Finally $|p_0| \leq P_4$ on the strip. The j th summand of $\partial_w U_*$ is therefore at most $\frac{1}{2} 4^{1/m} P_4 e^{\varepsilon_s} b_j e^{\alpha_j}$ for every $j \geq 0$, and the same holds for $\partial_{\bar{w}} U_*$ by reality. Dropping the factor $1/2$, both Wirtinger derivatives satisfy, on the strip and uniformly in r ,

$$(3.11) \quad |\partial_w U_*|, |\partial_{\bar{w}} U_*| \leq M_{\partial} := 4^{1/m} P_4 e^{\varepsilon_s} \sum_j b_j e^{\alpha_j}.$$

The derivative $\partial_w U_*$ has angular frequencies $km - 1$, $k \in \mathbb{Z}$. The function $e^{i\theta} \partial_w U_*$ is therefore periodic in ϕ , and it is bounded by $4^{1/m} M_{\partial}$ on the strip. Cauchy's estimate bounds its Fourier coefficients by $4^{1/m} M_{\partial} 4^{-|k|}$, while $R^{|km-1|} \leq R 2^{|k|}$. Summation over k gives

$$(3.12) \quad \|\partial_w U_*\|_Y \leq D_b := 3R 4^{1/m} M_{\partial} = 3R 4^{2/m} P_4 e^{\varepsilon_s} \sum_j b_j e^{\alpha_j}.$$

Reality gives the same bound for $\partial_{\bar{w}} U_*$. The resulting enclosures are $U_b < 131.610$ and $D_b < 110996$.

Both boundary defects have a common strip bound. Since $2M_U \geq 1$ for the centre (indeed $M_U = U_b/3 > 43$),

$$|U_* - 1| \leq M_U + 1 \leq 3M_U = U_b.$$

The radial derivative is $\partial_r U_* = e^{i\theta} \partial_w U_* + e^{-i\theta} \partial_{\bar{w}} U_*$, and $|e^{\pm i\theta}| \leq 4^{1/m}$ on the strip. By (3.11),

$$|\partial_r U_*| \leq 2 \cdot 4^{1/m} M_{\partial} = \frac{2D_b}{3R}.$$

We can therefore use

$$(3.13) \quad M = U_b + 2D_b$$

as a common majorant for the two boundary functions.

3.1.3. All boundary modes and the residual. The boundary computation encloses 1024 equispaced values at 4096 bits and their cosine coefficients through $j = 400$. For a periodic strip-analytic function bounded by M , the discrete coefficient differs from the continuum coefficient by the sum of all aliased coefficients. Their geometric bound gives, uniformly for $0 \leq j \leq 400$,

$$(3.14) \quad e_Q = \frac{8M 4^{-(1024-400)}}{1 - 4^{-1024}}.$$

For $j > 400$, Cauchy's estimate gives $|D_j|, |N_j| \leq 2M 4^{-j}$. The constant mode is included with its own coefficient convention; in particular no defect is discarded because it belongs to a finite equation used to choose the centre.

For $0 \leq k \leq 3$, form the full moments

$$(3.15) \quad D^{[k]} = \sum_{j \geq 0} 2^j (jm + 1)^k |D_j|, \quad N^{[k]} = \sum_{j \geq 0} 2^j (jm + 1)^k |N_j|.$$

The tail begins at $j = 401$. Expanding $(m(401 + t) + 1)^k$ by the binomial theorem reduces its bound to

$$(3.16) \quad \sum_{t \geq 0} t^{\ell} 2^{-t} = 2, \quad 2, \quad 6, \quad 26 \quad \text{for } \ell = 0, 1, 2, 3 \text{ respectively,}$$

where $t^0 = 1$, also at $t = 0$. Thus every moment in (3.15) includes the infinite tail. The stronger coefficient decay 4^{-j} also proves smooth convergence of the lift on any closed disc strictly inside radius $4^{1/m}$.

Direct differentiation gives

$$\Delta(h_n^D \cos n\theta) = -2n(n+1)r^n \cos n\theta, \quad \Delta(h_n^N \cos n\theta) = 2(n+1)r^n \cos n\theta.$$

On $0 \leq r \leq 1$, $|h_n^D| \leq 1$ and $|h_n^N| \leq 1/(n+2)$. For the first inequality, h_n^D increases to one, since its derivative is $n(n+2)r^{n-1}(1-r^2)/2$ for $n > 0$; the $n = 0$ case is constant. For the second, maximize $r^n(1-r^2)/2$. The following enlarged constants then suffice:

$$(3.17) \quad \begin{aligned} L_0 &= D^{[0]} + N^{[0]}/2, & L_1 &= 2R(D^{[2]} + N^{[1]}), \\ L_\Delta &= 2(D^{[2]} + N^{[1]}), & L_{\Delta,1} &= 2R(D^{[3]} + N^{[2]}). \end{aligned}$$

They bound ℓ , $\partial_w \ell$, $\Delta \ell$ and $\partial_w \Delta \ell$, respectively, in $\mathcal{Y}$. To see the derivative bounds, express $r^n \cos n\theta = (w^n + \bar{w}^n)/2$ and multiply by $w\bar{w}$ for the second term of each lift. A Wirtinger derivative costs at most the degree and one factor R in the angular weight. The coefficients in (3.17) dominate all resulting terms, including $n = 0$.

The residual at the clamped centre is

$$(3.18) \quad E = F(g_0, p_0) = \Delta U_0 + qU_0 = -(\Delta + q)\ell.$$

Since $\|\partial_w q\|_{\mathcal{Y}} \leq RPP_1$, the residual and its derivative have bounds

$$(3.19) \quad \|E\|_{\mathcal{Y}} \leq E_0 = L_\Delta + P^2 L_0, \quad \|\partial_w E\|_{\mathcal{Y}} \leq E_1 = L_{\Delta,1} + RPP_1 L_0 + P^2 L_1.$$

Since $\Delta U_0 = -qU_* - \Delta \ell$, we also have the fixed-centre derivative bounds

$$(3.20) \quad \begin{aligned} \|\Delta U_0\|_{\mathcal{Y}} &\leq \Lambda_0 := P^2 U_b + L_\Delta, \\ \|\partial_w \Delta U_0\|_{\mathcal{Y}} &\leq \Lambda_1 := RPP_1 U_b + P^2 D_b + L_{\Delta,1}. \end{aligned}$$

The resulting enclosures are

$$(3.21) \quad E_0 < 9.643 \cdot 10^{-61}, \quad E_1 < 5.418 \cdot 10^{-56}, \quad \Lambda_0 < 8.865 \cdot 10^7, \quad \Lambda_1 < 7.489 \cdot 10^{10}.$$

The small residual is a bound for the clamped centre in the Banach-space equation, including all modes. The existence argument uses this residual and the inverse proved next.

3.2. An exact inverse of the corrected linearisation. All quantities are evaluated at (g_0, p_0) ; abbreviate $p = p_0$, $q = |p|^2$. The centre U_0 is smooth on a neighbourhood of $\mathbb{D}$ and has the exact Cauchy data (3.5). Its residual is $E = \Delta U_0 + qU_0$. For a variation $(\delta g, \delta p) \in X$, set

$$(3.22) \quad \eta(w) = \int_0^w \delta p(z) dz, \quad v = \eta/p, \quad T = v\partial_w U_0 + \bar{v}\partial_{\bar{w}} U_0.$$

Here p has no zero on $|w| \leq R$ as soon as $j_* > 0$ in (3.24) below, because $|p(w) - \rho| \leq P - \rho$ there; v is then holomorphic on $|w| < R$. The real vector field represented by the complex number v acts on U_0 as T . Define the bounded correction

$$(3.23) \quad \mathcal{E}(\delta p) = v\partial_w E + \bar{v}\partial_{\bar{w}} E + (v' + \bar{v}')E, \quad \tilde{D} = DF(g_0, p_0) - \mathcal{E},$$

where $\mathcal{E}$ on X acts only on the second component.

Proposition 3.1 (The coupled inverse). *Suppose $L = \Delta + q$ has the symmetric-sector Dirichlet inverse and trace bound of Proposition 2.6, and the centre satisfies*

$$(3.24) \quad d_* := \rho^2 - (P^2 - \rho^2) - E_0 > 0, \quad j_* := \rho - (P - \rho) > 0.$$

Then $\tilde{D}: X \rightarrow Y$ has a bounded two-sided inverse $A: Y \rightarrow X$. Define

$$(3.25) \quad I_0 = d_*^{-1}, \quad I_1 = I_0 + I_0^2(2PP_1 + 2RE_1),$$

$$(3.26) \quad \mathcal{J}_0 = j_*^{-1}, \quad \mathcal{J}_1 = \mathcal{J}_0 + P_1\mathcal{J}_0^2,$$

$$(3.27) \quad B_p = (P + P_1)I_1B_N, \quad B_g = B_N + 2(R\Lambda_1 + \Lambda_0)I_1B_N.$$

Then, with $A_* = B_g + B_p$,

$$(3.28) \quad \begin{aligned} \|A\|_{Y \rightarrow X} &\leq A_*, & \|AF(g_0, p_0)\|_X &\leq Y_0 := A_*E_0, \\ \|I_X - ADF(g_0, p_0)\|_{X \rightarrow X} &\leq Z := 2A_*(RE_1 + E_0)\mathcal{J}_1. \end{aligned}$$

Proof. The material-derivative identity. Use $\Delta = 4\partial_w\partial_{\bar{w}}$ and holomorphicity of v . Direct differentiation gives

$$(3.29) \quad \Delta T = v\partial_w\Delta U_0 + \bar{v}\partial_{\bar{w}}\Delta U_0 + (v' + \bar{v}')\Delta U_0.$$

There is no second derivative of v in this formula. For example,

$$4\partial_w\partial_{\bar{w}}(v\partial_w U_0) = v'\Delta U_0 + v\partial_w\Delta U_0;$$

the conjugate term has the analogous expression. Since $\eta = pv$,

$$(3.30) \quad \begin{aligned} \delta q &= \bar{p}\eta' + p\bar{\eta}' \\ &= v\partial_w q + \bar{v}\partial_{\bar{w}} q + q(v' + \bar{v}'). \end{aligned}$$

Insert $\Delta U_0 = E - qU_0$ into (3.29), apply the product rule, and use (3.30). The terms involving derivatives of q combine with δqU_0 , and the terms involving derivatives of U_0 combine with qT . The result is the exact identity

$$(3.31) \quad (\Delta + q)T + \delta qU_0 = v\partial_w E + \bar{v}\partial_{\bar{w}} E + (v' + \bar{v}')E = \mathcal{E}(\delta p).$$

For $\delta U = K\delta g$, differentiation of F gives

$$(3.32) \quad DF(g_0, p_0)(\delta g, \delta p) = (\Delta + q)\delta U + \delta qU_0.$$

Thus $\tilde{D}(\delta g, \delta p) = L(\delta U - T)$.

The boundary Hessian. The constant Dirichlet data and zero normal derivative imply $\nabla U_0 = 0$ on the circle. Differentiate this vector identity along the circle: the Hessian annihilates its tangent vector. Since the Hessian is symmetric, it has only a normal–normal component, and its trace is that component. Therefore

$$\text{Hess } U_0 = (\Delta U_0)\nu \otimes \nu \quad \text{on } \partial\mathbb{D}.$$

The derivative of the vector field v in the normal derivative of T multiplies ∇U_0 and vanishes. With $\nu = e^{i\theta}$ and $U_0 = 1$ this proves

$$(3.33) \quad T = 0, \quad \partial_r T = (E - q)\text{Re}(ve^{-i\theta}) \quad \text{on } \partial\mathbb{D}.$$

The boundary reciprocal. Put $d_\partial = (q - E)|_{\partial\mathbb{D}}$. In the weighted boundary algebra W_0 ,

$$\|d_\partial - \rho^2\|_{W_0} \leq P^2 - \rho^2 + E_0 < \rho^2.$$

A Neumann series in W_0 gives $d_\partial^{-1} \in W_0$ and norm at most I_0 . The angular derivative satisfies

$$\|\partial_\theta q\|_{W_0} \leq 2PP_1, \quad \|\partial_\theta E\|_{W_0} \leq 2RE_1.$$

For the second inequality use $\partial_\theta E = i(w\partial_w E - \bar{w}\partial_{\bar{w}} E)$ and reality. Differentiate the reciprocal to obtain

$$\|d_\partial^{-1}\|_{W_1} \leq I_0 + I_0^2\|\partial_\theta d_\partial\|_{W_0} \leq I_1.$$

This argument uses a smallness condition only in W_0 , and the derivative identity in W_1 ; it does not require the stronger W_1 perturbation to be small.

Construction of the right inverse. For $f \in Y$, solve

$$(3.34) \quad L\Phi = f \quad \text{in } \mathbb{D}, \quad \Phi = 0 \quad \text{on } \partial\mathbb{D}, \quad N = \partial_r \Phi|_{r=1}.$$

Set $H_\partial = N/d_\partial$. Proposition 2.6 and the reciprocal bound give $\|H_\partial\|_{W_1} \leq I_1 B_N \|f\|_Y$. This function is real and even, with frequencies in $m\mathbb{Z}$. If its complex Fourier coefficients are H_n , define

$$(3.35) \quad a(w) = H_0 + 2 \sum_{j>0} H_{jm} w^{jm}.$$

Then a has real coefficients, $\text{Re } a = H_\partial$ on the circle, and

$$(3.36) \quad \|a\|_{W_1} = \|H_\partial\|_{W_1} \leq I_1 B_N \|f\|_Y.$$

The equality uses $H_{-n} = H_n$ and the factor two on positive frequencies. This real-part problem has a unique solution in the real-coefficient class: the usual imaginary constant is excluded by that class. The constant mode H_0 is allowed and determines the dilation component of the shape.

Define

$$(3.37) \quad v = wa, \quad \eta = pv, \quad \delta p = (pv)', \quad \delta U = \Phi + T, \quad \delta g = \Delta \delta U.$$

Now $\text{Re}(ve^{-i\theta}) = \text{Re } a = N/d_\partial$. Equations (3.33) and (3.34) give zero Dirichlet and Neumann traces of δU . Consequently $\delta g \in X_g$ and $K\delta g = \delta U$. Its shape variation lies in W , with the required real coefficients and frequencies. Formula (3.31) then gives $\tilde{D}(\delta g, \delta p) = L\Phi = f$. The construction is linear in f ; denote it by Af .

Bounds and regularity of the construction. Multiplication and differentiation of the holomorphic series give

$$(3.38) \quad \|v\|_Y \leq R\|a\|_{W_1}, \quad \|v'\|_Y \leq \|a\|_{W_1}, \quad \|\delta p\|_W = \|pa\|_{W_1} \leq (P + P_1)\|a\|_{W_1}.$$

In the middle inequality, writing $a(w) = \sum_{j \geq 0} \hat{a}_j w^{jm}$, one has $v' = \sum_j (jm + 1) \hat{a}_j w^{jm}$, so equality actually holds. The last identity follows from differentiating wpa . For the field, $\Delta\Phi = f - q\Phi$ and (3.29) imply

$$\|\delta g\|_Y \leq B_N \|f\|_Y + 2(R\Lambda_1 + \Lambda_0)\|a\|_{W_1}.$$

Together with (3.36), these are B_p and B_g in (3.27) and prove $\|A\| \leq A_*$. They also prove that the recipe maps all of Y into X .

There is no loss of angular radius in these estimates. Although a general input field is only continuous in Y , its Poisson potential and the Dirichlet solution are in $W^{2,s}$ for finite s and have C^1 traces. The function v is holomorphic on the larger disc $|w| < R$; hence T is smooth near $\overline{\mathbb{D}}$. The identities hold distributionally for these general inputs, and the boundary identities are classical for T . Thus no density assertion in a stronger differentiability norm is required.

The left inverse. Take any $(\delta g, \delta p) \in X$ and define η, v, T as in (3.22). For $a = \eta/(wp)$, the apparent singularity at zero is removable. Moreover,

$$(3.39) \quad \|\eta/w\|_{W_1} = \|\delta p\|_W.$$

The W_0 Neumann series for $1/p$, followed by differentiation of the reciprocal, gives $\|1/p\|_{W_1} \leq \mathcal{J}_1$ from (3.26). Thus $a \in W_1$, with real coefficients and frequencies jm .

Put $\Phi = K\delta g - T$. It has zero Dirichlet trace and satisfies $L\Phi = \tilde{D}(\delta g, \delta p)$. Dirichlet uniqueness in the symmetric sector recovers exactly the function in (3.34) when the recipe is applied to this right side. Its normal trace is, by (3.33),

$$\partial_r \Phi = -\partial_r T = d_\partial \text{Re } a.$$

Dividing by d_∂ and solving the real-part problem recovers a uniquely. The recipe therefore recovers v , η , δp and δg . Hence $A\tilde{D} = I_X$, as well as $\tilde{D}A = I_Y$. In particular A is injective.

The Newton defect. Equation (3.39) gives $\|a\|_{W_1} \leq \mathcal{J}_1 \|\delta p\|_W$. From (3.23) and (3.38),

$$\|\mathcal{E}(\delta p)\|_Y \leq 2(RE_1 + E_0)\mathcal{J}_1 \|\delta p\|_W.$$

Since $A\tilde{D} = I_X$,

$$I_X - ADF(g_0, p_0) = -A\mathcal{E}.$$

This proves the left defect estimate in (3.28). The bound for $AF(g_0, p_0)$ follows from $\|E\|_Y \leq E_0$. $\square$

Remark 3.2 (Shape differentiation). The material derivative $T = v \cdot \nabla U_0$ and the use of the boundary normal displacement are the classical Hadamard shape-derivative mechanism; see, for example, [40, 65]. This mechanism is not specific to two dimensions. The planar conformal coordinates and holomorphic real-part problem make the resulting inverse explicit here. The residual term in (3.31) records that the centre solves the equation only approximately.

For the forty-mode centre, exact rational arithmetic gives $P < 820.7003$, $P_1 < 1176.5538$ and $V_* = P^2 - \rho^2 < 10208.097$. With the bound for E_0 in (3.21),

$$(3.40) \quad d_* > 6.5313 \cdot 10^5, \quad j_* > 808.21,$$

so (3.24) holds. The enclosed constants are $I_1 < 6.059 \cdot 10^{-6}$, $\mathcal{J}_1 < 3.039 \cdot 10^{-3}$, $B_p < 8.517 \cdot 10^{11}$ and $B_g < 6.419 \cdot 10^{19}$, and the inverse bound is $A_* < 6.419 \cdot 10^{19}$. Its size is compatible with the contraction because the residual bounds in (3.21) are much smaller. The two-sided construction is needed because the Newton map acts on X : its contraction estimate uses $I_X - ADF(g_0, p_0)$, a left-inverse defect on X .

4. EXISTENCE, GEOMETRY AND THE POMPEIU PROPERTY

4.1. The radii polynomial. Let $x_0 = (g_0, p_0)$ and use the fixed bounded inverse A of Proposition 3.1. Write $\kappa_K = 1/4$ and $U_c = U_b + L_0$. For $(h, k) \in X$, expansion of the cubic map gives

$$(4.1) \quad \begin{aligned} F(g_0 + h, p_0 + k) &= E + DF(g_0, p_0)(h, k) \\ &\quad + (p_0 \bar{k} + \bar{p}_0 k)Kh + |k|^2 U_0 + |k|^2 Kh. \end{aligned}$$

For $s = \|h\|_Y + \|k\|_W$, the algebra estimate and Lemma 2.2 bound the quadratic terms by $(2P\kappa_K + U_c)s^2$ and the cubic term by $\kappa_K s^3$. Define

$$(4.2) \quad C_2 = A_*(2P\kappa_K + U_c), \quad C_3 = A_*\kappa_K.$$

These constants also bound the corresponding polarised bilinear and trilinear maps. For example, polarising the first quadratic term gives the average of $2P\kappa_K \|k_1\| \|h_2\|$ and $2P\kappa_K \|k_2\| \|h_1\|$; each is bounded by the product of the two X norms. Polarizing $|k|^2 U_0$ uses $\text{Re}(k_1 \bar{k}_2) U_0$ with the same bound. The symmetric trilinear form for $|k|^2 Kh$ has bound κ_K in the product of three X norms. Consequently the derivative majorants are $2C_2 s$ and $3C_3 s^2$.

Lemma 4.1. *If a radius $r_* > 0$ satisfies*

$$(4.3) \quad Y_0 + (Z - 1)r_* + C_2 r_*^2 + C_3 r_*^3 < 0, \quad Z + 2C_2 r_* + 3C_3 r_*^2 < 1,$$

then F has a unique zero in the closed X ball $\|x - x_0\|_X \leq r_$.*

Proof. For the Newton map $\mathcal{T}(x) = x - AF(x)$, Proposition 3.1 and (4.1) imply

$$\|\mathcal{T}(x_0 + (h, k)) - x_0\|_X \leq Y_0 + Zs + C_2s^2 + C_3s^3.$$

The first inequality of (4.3) makes this ball invariant. The polarised derivative bounds give

$$\|D\mathcal{T}(x)\|_{X \rightarrow X} \leq Z + 2C_2r_* + 3C_3r_*^2 < 1$$

on the ball. Completeness gives a unique fixed point. At that point $AF(x) = 0$, and injectivity of A gives $F(x) = 0$. Every zero in the ball is a fixed point, so the uniqueness is uniqueness of a zero of F . $\square$

For the forty-mode centre take $r_* = 10^{-32}$. The constants obtained from the exact interval chain are recorded in Table 4. Upper bounds are rounded upward; negative upper bounds are rounded toward zero. The two inequalities of Lemma 4.1 are strict.

Quantity	Enclosed bound
B_{L^2}	< 6.748
B_Y	$< 1.045 \cdot 10^8$
B_N	$< 7.039 \cdot 10^{13}$
E_0	$< 9.643 \cdot 10^{-61}$
E_1	$< 5.418 \cdot 10^{-56}$
A_*	$< 6.419 \cdot 10^{19}$
Y_0	$< 6.189 \cdot 10^{-41}$
Z	$< 2.122 \cdot 10^{-38}$
C_2	$< 3.479 \cdot 10^{22}$
C_3	$< 1.605 \cdot 10^{19}$
$Y_0 + (Z - 1)r_* + C_2r_*^2 + C_3r_*^3$	$< -9.99999993 \cdot 10^{-33}$
$Z + 2C_2r_* + 3C_3r_*^2$	$< 6.957 \cdot 10^{-10}$
$4C_2Y_0$	$< 8.611 \cdot 10^{-18}$

TABLE 4. The forty-mode certificate, with $r_* = 10^{-32}$. Every row is rounded from its own interval enclosure. In particular the row for Y_0 and the last three rows are not recomputed from the rounded rows above; for example, the rounded bounds for A_* and E_0 alone give only $Y_0 < 6.190 \cdot 10^{-41}$.

The inequalities (4.3) remain valid at $r_* = 10^{-32}$ when A_* , and hence Y_0 , Z , C_2 and C_3 , are multiplied by 10^8 : the first expression is then less than $-3.46 \cdot 10^{-33}$ and the second less than 0.0696 (ancillary file `ROBUSTNESS_CHECK.json`).

4.2. Geometry on the entire ball. Let (g, p) be the zero obtained above, and write $p(w) = \sum_j p_j w^{jm}$ and $\psi(w) = \sum_j c_j w^{jm+1}$ as in (2.6). For the centre write $p_0(w) = \sum_j p_j^{(0)} w^{jm}$ and $\psi_0(w) = \sum_j c_j^{(0)} w^{jm+1}$, so that $p_0^{(0)} = c_0^{(0)} = \rho$ and $P = \sum_j 2^j |p_j^{(0)}|$. The estimates of this subsection use only

$$\|p - p_0\|_W = \sum_{j \geq 0} 2^j |p_j - p_j^{(0)}| \leq r_*,$$

so they hold for every shape in the ball. On $|w| \leq R$ the weighted coefficient bound gives

$$(4.4) \quad \operatorname{Re} p(w) \geq \rho - (P - \rho) - r_* > 808.$$

If $w_1 \neq w_2$ are in the disc $|w| < R$, then

$$\frac{\psi(w_2) - \psi(w_1)}{w_2 - w_1} = \int_0^1 p(w_1 + t(w_2 - w_1)) dt$$

has positive real part. Thus ψ is univalent on that larger disc. The same estimate makes p nonzero there, and ψ is analytic on a neighbourhood of $\overline{\mathbb{D}}$.

For unit-disc convexity, use the sharper unweighted centre sums

$$(4.5) \quad L_* = \rho - \sum_{j>0} |p_j^{(0)}| - r_*, \quad \Theta_* = \sum_{j>0} jm |p_j^{(0)}| + \frac{m+1}{2} r_*.$$

Here $\sum_j |p_j - p_j^{(0)}| \leq r_*$, and $\sum_j jm |p_j - p_j^{(0)}| \leq \frac{m+1}{2} r_*$ because $jm/2^j \leq (m+1)/2$ for every $j \geq 1$. Throughout the closed unit disc therefore

$$|p| \geq L_* > 811, \quad |wp'| \leq \Theta_*.$$

On the boundary,

$$(4.6) \quad \operatorname{Re} \left(1 + \frac{wp'}{p} \right) \geq 1 - \frac{\Theta_*}{L_*} > 0.2998879.$$

For $z(\theta) = \psi(e^{i\theta})$, the derivative of the tangent angle is $\operatorname{Re}(1 + wp'/p)$ and the speed is $|p|$. Thus the outwardly oriented curvature is

$$(4.7) \quad \kappa(\theta) = \frac{\operatorname{Re}(1 + wp'/p)}{|p|} > 0.$$

Univalence makes the boundary a simple closed curve. Its positive curvature and total turning 2π make its enclosed domain strictly convex. Equivalently, this is the analytic convexity criterion for conformal maps [26, §2.5]. The same coefficient estimates give the convexity inequality in additive form: for every shape in the ball and every $|w| \leq 1$,

$$(4.8) \quad |p(w)| - |wp'(w)| \geq \rho - \sum_{j>0} (jm+1) |p_j^{(0)}| - \frac{m+1}{2} r_* > 243.3264,$$

because $jm+1 \leq \frac{m+1}{2} 2^j$ for every $j \geq 0$.

Since $\psi(0) = 0$, the origin lies in Ω . Every tangent line to the boundary supports this strictly convex domain and therefore does not pass through the origin. Thus every ray from the origin meets the boundary exactly once and transversally. The coefficient bounds also give, for $|w| = 1$,

$$\left| p(w) - \frac{\psi(w)}{w} \right| \leq P - \rho + r_*, \quad \left| \frac{\psi(w)}{w} \right| \geq \operatorname{Re} \int_0^1 p(tw) dt > 808,$$

and hence

$$\operatorname{Re} \frac{w\psi'(w)}{\psi(w)} \geq 1 - \frac{P - \rho + r_*}{808} > 0,$$

because $P < 820.7003$, $\rho > 814.457$ and $r_* = 10^{-32}$. This proves strict star-shapedness with respect to the origin. The real-analytic parametrisation $\theta \mapsto \psi(e^{i\theta})$, with nonvanishing derivative, identifies the boundary diffeomorphically with S^1 .

The first nonlinear coefficient of ψ obeys

$$(4.9) \quad |c_1| \geq |c_1^{(0)}| - \frac{r_*}{2(m+1)} > 0.01645,$$

because $2|p_1 - p_1^{(0)}| \leq \|p - p_0\|_W \leq r_*$ and $c_1 = p_1/(m+1)$. The coefficient restrictions give

$$\psi(e^{2\pi i/m}w) = e^{2\pi i/m}\psi(w), \quad \psi(\bar{w}) = \overline{\psi(w)}.$$

The domain is therefore D_m -invariant. If it were a disc, its centre would be fixed by rotation through $2\pi/m$ and hence would be zero. Its normalised conformal map would then be linear, contrary to (4.9). Hence Ω is not a disc. Since $\psi'(0) = p(0)$ and $|p(0) - \rho| \leq r_*$, while the exact decimal ρ satisfies $814.457 < \rho < 814.458$, the bound (1.3) follows. Analyticity and nonvanishing of ψ' near the circle give a real-analytic boundary.

4.3. Transfer of the field. At the zero of F , $U = 1 + Kg$ satisfies (2.11) initially in the distributional sense, with its C^1 boundary traces. The coefficient $|p|^2$ is smooth near $\overline{\mathbb{D}}$. Dirichlet elliptic regularity, applied successively to $\Delta U = -|p|^2 U$ with smooth constant Dirichlet data, gives smoothness up to the boundary. The previously established zero normal trace is therefore the classical normal condition.

Set $\Omega = \psi(\mathbb{D})$ and $u = U \circ \psi^{-1}$. Formula (2.1) proves $(\Delta + 1)u = 0$. The normal transformation gives $\partial_\nu u = |p|^{-1}\partial_r U = 0$, while $u = 1$ on the boundary. The function cannot be constant: its boundary value would force that constant to be one, and the constant one does not solve $(\Delta + 1)u = 0$. This establishes the differential-equation and geometric assertions of Theorem 1.3.

The Green's identity argument in Section 33 proves that $\widehat{1}_\Omega$ vanishes on the unit circle and that Ω fails the Pompeiu property, completing Theorem 1.3.

Part 2. A convex planar domain for Berenstein's problem

5. THE DOMAIN AND THE FIXED-DISC EQUATION

The Dirichlet counterpart of Schiffer's problem prescribes zero boundary value and a nonzero constant normal derivative. Colbrook, Sadeghi and Stepaniants [15] constructed a planar counterexample to Berenstein's conjecture. The construction below gives the first convex planar counterexample.

Theorem 5.1. *There exist a bounded strictly convex domain $\Omega \subset \mathbb{R}^2$, different from a disc, with real-analytic boundary and D_{77} symmetry, and a real-valued sign-changing function $u \in C^\infty(\overline{\Omega})$ such that*

$$(5.1) \quad \Delta u + u = 0 \quad \text{in } \Omega, \quad u = 0, \quad \partial_\nu u = 1 \quad \text{on } \partial\Omega.$$

Here ν is the outward unit normal. The domain is $\psi(\mathbb{D})$, with $\psi(0) = 0$, $\psi'(0) > 0$, and

$$(5.2) \quad 237.97 < \psi'(0) < 237.98, \quad \operatorname{Re} \left(1 + \frac{w\psi''(w)}{\psi'(w)} \right) > 0.64407 \quad (|w| = 1).$$

The map is holomorphic and univalent on $|w| < R = 2^{1/77} > 1.00904$, with $\operatorname{Re} \psi'(w) > 235.7747$ there. On $\mathbb{D}$, $|\psi'| > 237.2785$. Writing $\psi(w) = \sum_{j \geq 0} c_j w^{jm+1}$, $m = 77$, one has $|c_1| > 0.0038627$ and $u(0) > 19.3341$. There is a unique zero of the fixed-disc equation (5.12) in

$$(5.3) \quad \|g - g_0\|_Y + \|p - p_0\|_{W_1} \leq 10^{-36}, \quad p = \psi'.$$

Here Y , W_1 and X are the spaces of Subsection 5.2, and (g_0, p_0) is the clamped centre of Subsection 5.3.

The second quantity in (5.2) is the conformal convexity factor. The physical curvature is this factor divided by $|\psi'|$. For the finite centre, a floating-point evaluation gives physical curvature in the approximate range $[2.719 \cdot 10^{-3}, 5.283 \cdot 10^{-3}]$.

5.1. The arc-length Fourier transform. Let $\sigma_{\partial\Omega}$ denote arc-length measure and use the convention $\widehat{\sigma}(\xi) = \int_{\partial\Omega} e^{-i\xi \cdot x} ds$. For a smooth bounded domain with connected Jordan boundary, the existence of a real solution u of (5.1) is equivalent to

$$(5.4) \quad \widehat{\sigma}_{\partial\Omega}(\xi) = 0 \quad (|\xi| = 1).$$

Indeed, if $v = e^{-i\xi \cdot x}$, Green's identity gives $0 = \int_{\Omega} (u\Delta v - v\Delta u) = \int_{\partial\Omega} (u\partial_{\nu}v - v\partial_{\nu}u) = -\widehat{\sigma}(\xi)$. Conversely, let Φ be the outgoing fundamental solution with $(\Delta + 1)\Phi = -\delta$ and put $S(x) = \int_{\partial\Omega} \Phi(x - y) ds_y$. Its far-field amplitude is a nonzero constant multiple of $\widehat{\sigma}(\widehat{x})$, so (5.4) and Rellich uniqueness give $S = 0$ in the connected exterior. The single layer is continuous across the boundary, and its normal derivatives obey $\partial_{\nu}S|_{\text{ext}} - \partial_{\nu}S|_{\text{int}} = -1$. Thus $u = \text{Re } S|_{\Omega}$ solves (5.1); smooth boundary regularity gives the classical traces. This is the Fourier-circle equivalence of [15, Proposition 2.1]. In particular, the theorem produces a strictly convex noncircular curve whose arc-length Fourier transform vanishes on the unit circle.

5.2. Spaces and lifts. In this part $m = 77$, $R = 2^{1/m}$, and the symbols $Y, \mathcal{Y}, K, X_g$ have the meanings of Subsection 2.1, with this value of m . The field is real, reflection-even and invariant under rotation through $2\pi/m$. In particular,

$$(5.5) \quad \|f\|_Y = \sum_{n \in m\mathbb{Z}} R^{|n|} \sup_{0 \leq r \leq 1} |f_n(r)|, \quad \|b\|_{W_s} = \sum_{n \in m\mathbb{Z}} R^{|n|} (1 + |n|)^s |b_n|.$$

The full angular space $\mathcal{Y}$ permits all integer frequencies. Continuity is Cartesian, so nonconstant angular coefficients vanish at the origin. Holomorphic versions of W_s use nonnegative frequencies and real coefficients. The algebra proof of Lemma 2.1 applies for every $R > 1$ and every m ; the additional weight satisfies $1 + |n + k| \leq (1 + |n|)(1 + |k|)$, making W_s a Banach algebra for $s = 0, 1, 2$. A holomorphic real-coefficient series h satisfies

$$(5.6) \quad \|\text{Re } h\|_{W_s} = \|h\|_{W_s}.$$

For each positive frequency, taking the real part puts half the Taylor coefficient at each of the two opposite Fourier frequencies; their weights are equal. The constant coefficient is real and retains its weight.

The zero-Dirichlet Poisson inverse K and its clamped range are

$$(5.7) \quad X_g = \left\{ g \in Y : \int_0^1 r^{|n|+1} g_n(r) dr = 0 \ (n \in m\mathbb{Z}) \right\}, \quad \|K\|_{Y \rightarrow Y} \leq \frac{1}{4}.$$

Lemma 2.2 uses neither the Schiffer boundary value nor the symmetry order: its radial Green formula and moment identity apply verbatim. Thus Kg has zero Cauchy data precisely when $g \in X_g$. For a boundary series introduce the Neumann lift

$$(5.8) \quad \mathcal{B}b = \frac{1}{2} \sum_n b_n r^{|n|} (r^2 - 1) e^{in\theta}.$$

Its Cauchy traces are $(0, b)$, and

$$(5.9) \quad \|\mathcal{B}b\|_Y \leq \frac{1}{2} \|b\|_{W_0}, \quad \|\Delta \mathcal{B}b\|_Y \leq 2 \|b\|_{W_1}.$$

Indeed $\Delta(r^{|n|}(r^2 - 1)e^{in\theta}/2) = 2(|n| + 1)r^{|n|}e^{in\theta}$, and the radial factor has modulus at most $1/2$ on $[0, 1]$.

Write $p = \psi' = \sum_j p_j w^{jm}$ and $\psi(w) = \int_0^w p(z) dz$, so $c_j = p_j/(jm + 1)$. Conformal covariance (2.1) and the normal transformation give the following fixed-disc problem for $U = u \circ \psi$:

$$(5.10) \quad \Delta U + |p|^2 U = 0 \quad \text{in } \mathbb{D}, \quad U = 0, \quad U_r = |p| \quad \text{on } \partial\mathbb{D}.$$

The boundary vector $wp(w)$ is outward normal to the image and has length $|p(w)|$, so the normal derivative in physical coordinates is $|p|^{-1}U_r$. We work on the real Banach space

$$(5.11) \quad X = X_g \times W_1, \quad \|(g, p)\|_X = \|g\|_Y + \|p\|_{W_1}, \quad U(g, p) = \mathcal{B}(|p|) + Kg.$$

The shape component p of X is a holomorphic real-coefficient series in W_1 : $p = \sum_{j \geq 0} p_j w^{jm}$ with $p_j \in \mathbb{R}$. Its reflection symmetry $p(\bar{w}) = \overline{p(w)}$ is equivariance; it does not make p real on $\partial\mathbb{D}$. The field is real-valued and reflection-even in Y . On the ball (5.3), the boundary function $|p|$ lies in the full two-sided space W_1 , and every shape satisfies $\operatorname{Re} p \geq 2\rho - P - 10^{-36} > 0$ on the closed weighted disc $|w| \leq R$, with P of (7.1); see (9.7). Thus p is nonzero throughout that disc. We solve

$$(5.12) \quad F(g, p) = g + \Delta \mathcal{B}(|p|) + |p|^2 (\mathcal{B}(|p|) + Kg) = 0.$$

On the ball $\|p - p_0\|_{W_1} < 1/J_1$, with J_1 as in (8.24), write $p = p_0(1 + k/p_0)$. The binomial series for the square roots of $1 + k/p_0$ and its reflection converge in W_1 , and $|p_0| \in W_1$ by (8.25). Thus F is a real-analytic map from this neighbourhood of the centre in X to Y . The term $\Delta \mathcal{B}(|p|)$ requires the W_1 shape norm. For every $(g, p) \in X$ in this neighbourhood, $U(g, p)$ has the exact Cauchy data $(0, |p|)$, and $F = \Delta U + |p|^2 U$ distributionally. The Schiffer cubic equation and its Dirichlet linear inverse are therefore replaced by (5.12) and a Robin inverse.

5.3. The exactly clamped centre. The input consists of exact decimal coefficients $c_j^{(0)}$, a_j and positive divisors s_j , $0 \leq j \leq 72$. The divisors are specified positive numbers; no identity with a Bessel value is assumed. The decimal strings are the entries $0 \leq j \leq 72$ of `p`, `a` and `scales` in `anc/berenstein/centre_frozen.json`; the array `p` stores $c_j^{(0)}$, while $p_j^{(0)} = (jm + 1)c_j^{(0)}$. Put

$$(5.13) \quad \begin{aligned} \psi_0(w) &= \sum_{j=0}^{72} c_j^{(0)} w^{jm+1}, \quad p_0 = \psi'_0, \quad \rho = c_0^{(0)}, \quad q = |p_0|^2, \\ u_*(z) &= \sum_{j=0}^{72} \frac{a_j}{s_j} J_{jm}(|z|) \cos(jm \arg z), \quad U_* = u_* \circ \psi_0. \end{aligned}$$

Each physical summand extends through the origin as an entire real-analytic Helmholtz solution, so $(\Delta + q)U_* = 0$ identically. The finite conformal polynomial and finite field specify an infinite-dimensional clamped centre as follows. Let D_j and N_j be the complete cosine coefficients of

$$(5.14) \quad U_*(1, \theta) = \sum_{j \geq 0} D_j \cos(jm\theta), \quad \partial_r U_*(1, \theta) - |p_0(e^{i\theta})| = \sum_{j \geq 0} N_j \cos(jm\theta).$$

For $n \geq 0$ use the polynomials of (3.3),

$$h_n^D = r^n(1 - n(r^2 - 1)/2), \quad h_n^N = r^n(r^2 - 1)/2,$$

whose traces $(1, 0)$ and $(0, 1)$ do not depend on the prescribed Cauchy data. Define

$$(5.15) \quad \begin{aligned} \ell &= \sum_{j \geq 0} (D_j h_{jm}^D + N_j h_{jm}^N) \cos(jm\theta), \\ U_0 &= U_* - \ell, \quad g_0 = \Delta(U_0 - \mathcal{B}(|p_0|)). \end{aligned}$$

The strip estimates of Subsection 8.1 give convergence with all derivatives needed near $\overline{\mathbb{D}}$. Thus U_0 has traces $(0, |p_0|)$, the quantity inside the Laplacian has zero Cauchy data, $g_0 \in X_g$, and

Poisson uniqueness gives $Kg_0 = U_0 - \mathcal{B}(|p_0|)$. In particular $U(g_0, p_0) = U_0$ and

$$(5.16) \quad E := F(g_0, p_0) = -(\Delta + q)\ell.$$

All boundary harmonics enter this definition. The exact zero enclosed in (5.3) need not have a finite conformal or field series.

6. THE ROBIN INVERSE IN THE SYMMETRIC SECTOR

At the centre define

$$(6.1) \quad LV = (\Delta + q)V = f, \quad V_r + \beta V = 0 \quad \text{on } \partial\mathbb{D}, \quad \beta = 1 + d, \quad d = \operatorname{Re} \frac{wp'_0}{p_0}.$$

All operators in this section act in the real even D_m sector, with interior measure $r dr d\theta/(2\pi)$ and boundary measure $d\theta/(2\pi)$. By (4.7), $\beta = |p_0|\kappa_0$, where κ_0 is the curvature of $\partial\psi_0(\mathbb{D})$. The symmetric form associated with $-L$ is

$$(6.2) \quad \mathfrak{a}(z, y) = \int_{\mathbb{D}} \nabla z \cdot \nabla y + \int_{\partial\mathbb{D}} \beta zy - \int_{\mathbb{D}} qzy, \quad z, y \in H^1(\mathbb{D}).$$

The form is closed and bounded below after addition of a sufficiently large multiple of the L^2 inner product. Its self-adjoint operator is the Robin realisation of $-L$. Boundary multiplication is treated as a form on H^1 .

6.1. The Dini basis and complete spectral window. Let $\alpha_{j,k} > 0$ run over the roots

$$(6.3) \quad \alpha J'_n(\alpha) + J_n(\alpha) = 0, \quad n = jm.$$

The Robin-one Laplacian $-\Delta_{R,1}$ has the complete orthonormal Dini basis

$$(6.4) \quad e_{j,k} = b_{j,k}(r)\chi_j(\theta), \quad b_{j,k}(r) = \frac{\sqrt{2}J_n(\alpha_{j,k}r)}{|J_n(\alpha_{j,k})|\sqrt{1 + (1 - n^2)/\alpha_{j,k}^2}},$$

where $\chi_0 = 1$ and $\chi_j = \sqrt{2}\cos(jm\theta)$ for $j > 0$. The Dini parameter is 1, with $n + 1 > 0$, so no exceptional nonpositive radial eigenvalue is present. Orthogonality and completeness also follow from the separated self-adjoint Robin Sturm–Liouville problems and the complete angular cosine basis. The identity

$$\int_0^1 r J_n(\alpha r)^2 dr = \frac{1}{2} (J'_n(\alpha)^2 + (1 - n^2/\alpha^2) J_n(\alpha)^2)$$

follows by integrating the Bessel equation and its parameter derivative. At (6.3) it gives $\frac{1}{2} J_n(\alpha)^2 (1 + (1 - n^2)/\alpha^2)$, which is positive: every root exceeds n for $n \geq 1$, and the $n = 0$ expression is positive. In particular the basis functions have nonzero boundary values.

Let $\mathbf{P}$ select every root satisfying

$$(6.5) \quad |\alpha_{j,k} - \rho| < \omega, \quad \omega = 160,$$

and put $\mathbf{Q} = I - \mathbf{P}$. The window contains 300 states, with the sector counts of Table 5. Enumerate these states by $i \leftrightarrow (j, k)$ and write $n_i = jm$, $\alpha_i = \alpha_{j,k}$, $\chi_i = \chi_j$ and $\tau_i = b_{j,k}(1)$.

Angular order	0	77	154	231	308	385	462
States	102	91	60	33	13	1	0

TABLE 5. Complete Robin-one spectral window. All higher angular orders have no state in the window.

Lemma 6.1 (Two coverage arguments). *The Dirichlet interlacing and the quarter-step Robin sign grid described below enumerate all states of (6.5).*

Proof. For $y(x) = \sqrt{x}J_n(x)$ the Bessel equation gives

$$y'' + Q_n(x)y = 0, \quad Q_n(x) = 1 - (n^2 - 1/4)/x^2.$$

On an interval with $Q_n \leq Q_{\max}$, integration by parts between two zeros and Wirtinger's inequality give their spacing at least $\pi/\sqrt{Q_{\max}}$, exactly as in Lemma 2.3. Enclose proposed Dirichlet zeros by opposite signs in disjoint brackets of width less than $1/10$. For successive brackets the distance between their outer endpoints is less than $2\pi/\sqrt{Q_{\max}}$, where $Q_{\max}$ is the maximum between those endpoints. Three zeros there would require the reverse inequality, so each bracket contains exactly one zero and none lies between them. Guard zeros beyond both endpoints of the window complete this argument whenever a lower Dirichlet zero is present.

For every positive angular order whose first Dirichlet zero lies above the lower window endpoint, the initial interval requires a separate guard. This includes $n = m, 2m, \dots, 5m$ in the present window. When $n > 0$, J_n and J'_n are positive on $(0, n]$: at a first derivative zero below n the Bessel ODE would give $J''_n = (n^2/x^2 - 1)J_n > 0$, which cannot be a first downward crossing. This also excludes a first zero of J_n there. The identity $(xJ'_n)' = (n^2/x - x)J_n > 0$ on $(0, n)$ gives $J'_n(n) > 0$, and hence $y(n), y'(n) > 0$. Constant-coefficient comparison puts the first zero after n at distance at least $\pi/(2\sqrt{Q_{\max}})$. A hidden first zero before the proposed bracket would put the bracketed zero at distance at least $3\pi/(2\sqrt{Q_{\max}})$. The strict opposite inequality is imposed on its upper endpoint, with $Q_{\max}$ taken on the entire interval from n to that endpoint. This proves the initial Dirichlet guard.

Between consecutive Dirichlet zeros consider $z = xJ'_n/J_n$. It satisfies $xz' = n^2 - x^2 - z^2$. Thus $1+z$ runs from $+\infty$ to $-\infty$ and, at any zero, has derivative $-(x^2+1-n^2)/x < 0$. There is exactly one Robin root in each such gap. In the initial gap it is likewise unique between n and the first Dirichlet zero, since $1+z > 0$ at n . Each proposed Robin bracket lies strictly inside its corresponding certified gap. The location relative to both window endpoints is resolved by strict inequalities. For all remaining angular orders $n > \rho + \omega$, positivity up to n excludes any root in the window.

For the second argument put $\mathfrak{g}_n = J_n + xJ'_n$ and $\mathfrak{d}_n = x^2 + 1 - n^2$. On $x > \max(n, 1)$, direct differentiation gives

$$(6.6) \quad \left(\mathfrak{g}_n \sqrt{x/\mathfrak{d}_n}\right)'' + \left(1 + \frac{1/4 - n^2}{x^2} + \frac{4}{\mathfrak{d}_n} - \frac{3x^2}{\mathfrak{d}_n^2}\right) \mathfrak{g}_n \sqrt{x/\mathfrak{d}_n} = 0.$$

The coefficient is at most $21/4$: $\mathfrak{d}_n > 1$, the second term is at most $1/4$, and the last term is nonpositive. The same spacing argument gives separation greater than $\pi/\sqrt{21/4} > 1$. Roots are simple because at a root $\mathfrak{g}'_n = -(\mathfrak{d}_n/x)J_n \neq 0$; simultaneous vanishing of $\mathfrak{g}_n$ and J_n would also force $J'_n = 0$. Below n the sign of $\mathfrak{g}_n$ is positive. The nonzero endpoint signs at $x_k = \rho - \omega + k/4$, $0 \leq k \leq 8\omega$, therefore determine the number of roots in every quarter-step cell: one for opposite signs, zero for equal signs. In a cell crossing n the portion below n contains no root and the portion above it contains at most one. Each bracket from the first argument belongs to a distinct sign-changing cell, and all such cells are represented. This establishes a bijection of the state lists, including the empty next angular sector. $\square$

6.2. Matrices and integration. Write $\mathcal{V} = q - \rho^2$ and form the exact matrices on $\text{Ran } \mathbf{P}$

$$(6.7) \quad \begin{aligned} \mathbf{B} &= \mathbf{P}\mathcal{V}\mathbf{P}, & \mathbf{B}_2 &= \mathbf{P}\mathcal{V}^2\mathbf{P}, \\ \mathbf{D}_{ij} &= \int_{\partial\mathbb{D}} d\tau_i \chi_i \tau_j \chi_j, & \mathbf{D}_{2,ij} &= \int_{\partial\mathbb{D}} d^2\tau_i \chi_i \tau_j \chi_j, \\ \mathbf{H} &= \text{diag}(\alpha_i^2 - \rho^2) - \mathbf{B} + \mathbf{D}. \end{aligned}$$

The plus sign in $\mathbf{D}$ follows from (6.2). The two Gram identities are

$$(6.8) \quad \mathbf{G} := \mathbf{B}_2 - \mathbf{B}^2 = (\mathbf{Q}\mathcal{V}\mathbf{P})^*(\mathbf{Q}\mathcal{V}\mathbf{P}), \quad x^T \mathbf{D}_2 x = \|d \text{Tr}(\mathbf{P}x)\|_2^2.$$

Both include all complementary modes; $\mathbf{D}_2$ is not a truncated square of $\mathbf{D}$.

Writing $p_0(w) = \sum_j p_j^{(0)} w^{jm}$, the complex Fourier coefficients of q are

$$q_h(r) = \sum_k p_{k+h}^{(0)} p_k^{(0)} r^{(2k+h)m} \quad (h \geq 0).$$

Subtract ρ^2 from q_0 to obtain those of $\mathcal{V}$, and take the full two-sided convolution for $\mathcal{V}^2$. For complex coefficients v_h , the angular factors in the cosine basis are v_0 in position $(0, 0)$, $\sqrt{2}v_j$ in $(0, j)$, and $v_{|i-j|} + v_{i+j}$ for $i, j > 0$. The same rule applies to d and d^2 .

The radial integrations use $N_G = 1536$ Gauss–Legendre nodes. Disjoint sign brackets of the Legendre polynomial in $(-1, 1)$, equal in number to its degree, enclose all its roots. The weights $2/[(1 - x_i^2)\text{Leg}'_{N_G}(x_i)^2]$ include root uncertainty using

$$\max_{[-1,1]} |\text{Leg}''_{N_G}| = (N_G - 1)N_G(N_G + 1)(N_G + 2)/8.$$

Real Bessel arguments are evaluated at exact midpoint arguments and enlarged by their argument radii, since the integer-order integral representation gives $|J'_n| \leq 1$ on the real axis. An exported dyadic basis value carries its interval evaluation error into every matrix entry.

For the analytic quadrature error use the Bernstein ellipse of parameter two in $x = 2r - 1$. It has $|r| \leq 9/8$ and $|\text{Im } r| \leq 3/8$. Set

$$(6.9) \quad \begin{aligned} P_e &= \sum_j |p_j^{(0)}| (9/8)^{jm}, & V_e &= P_e^2 + \rho^2, \\ M_e &= 8(1 + 9/8)(1 + V_e^2) e^{3\alpha_{\max}/4} / b_*^2, \end{aligned}$$

where b_* is the smallest denominator in (6.4) and $\alpha_{\max}$ is the largest root in the window. The Bessel integral gives $|J_n(z)| \leq e^{|\text{Im } z|}$, so M_e bounds every integrand, including the angular factors and normalisations. The Chebyshev argument of (2.18) depends only on this majorant and Gauss exactness; hence the error in each entry is at most

$$(6.10) \quad 16M_e 2^{-2N_G} < 9.734 \cdot 10^{-51}.$$

For the boundary entries complexify $\phi = m\theta$. On $|\text{Im } \phi| \leq \log 4$ define $S_4 = \sum_{j>0} 4^j |p_j^{(0)}|$ and $P_{1,4} = \sum_j 4^j j m |p_j^{(0)}|$. The Laurent continuation of d is the half-sum of wp'_0/p_0 and its reflection. For the centre, $S_4 < 7.762 < \rho$; this excludes its poles and gives strip bound $M_d = P_{1,4}/(\rho - S_4)$. A trapezoidal sum with $N_d = 512$ nodes adds to every required Fourier coefficient of d and d^2 the error

$$(6.11) \quad \frac{4(1 + M_d^2)4^{-(N_d - h_{\max})}}{1 - 4^{-N_d}}.$$

Here $h_{\max}$ is the largest Fourier index used by the matrices. This follows by summing the aliased coefficients, each bounded by its strip supremum times $4^{-|h|}$.

6.3. The complement as a quadratic form. Let

$$(6.12) \quad \gamma_0 = \omega(2\rho - \omega), \quad t_Q = \frac{2(1 + \sqrt{\gamma_0 + \rho^2})}{\gamma_0}, \quad v_* = 2\rho\sigma_0 + \sigma_0^2, \quad d_* = \sigma_1/(\rho - \sigma_0).$$

Here $\sigma_0 = \sum_{j>0} |p_j^{(0)}|$ and $\sigma_1 = \sum_{j>0} jm|p_j^{(0)}|$. On $\overline{\mathbb{D}}$, $\rho - \sigma_0 \leq |p_0| \leq \rho + \sigma_0$ and $|wp'_0| \leq \sigma_1$, so $\|\mathcal{V}\|_\infty \leq v_*$ and $\|d\|_\infty \leq d_*$. On $\text{Ran } Q$ set $D_Q = Q(-\Delta_{R,1} - \rho^2)Q$. Every omitted state obeys $|\alpha^2 - \rho^2| \geq \gamma_0$: below the window this is $(\rho - \alpha)(\rho + \alpha) \geq \omega(2\rho - \omega)$, and above the window the bound is larger. Thus $|D_Q| \geq \gamma_0$. For $z \in H^1(\mathbb{D})$ the divergence theorem applied to $x|z|^2$ gives

$$(6.13) \quad \|\text{Tr } z\|_2^2 \leq 2\|z\|_2^2 + 2\|z\|_2 \|\nabla z\|_2.$$

The estimate extends from smooth functions by H^1 approximation. For $z \in QH^1$, the reference Robin-one form and $\alpha^2 \leq |\alpha^2 - \rho^2| + \rho^2$ give

$$\|\nabla z\|_2^2 \leq (1 + \rho^2/\gamma_0) \langle |D_Q|z, z \rangle.$$

Dropping its positive boundary term and inserting this inequality in (6.13) proves

$$(6.14) \quad \|\text{Tr } Q|D_Q|^{-1/2}\|^2 \leq t_Q.$$

Consequently the complementary form, conjugated by $|D_Q|^{-1/2}$, is $\text{sgn } D_Q$ plus a bounded self-adjoint perturbation of norm at most

$$(6.15) \quad \mu_Q = v_*/\gamma_0 + d_*t_Q.$$

If $\mu_Q < 1$, its inverse in this energy pairing has norm at most $(1 - \mu_Q)^{-1}$. The sign operator is a self-adjoint isometry; no positivity of D_Q is required.

Proposition 6.2 (Robin form Schur bound). *Let C be a real square matrix of dimension 300 and Θ a real diagonal matrix with nonzero entries. Suppose*

$$(6.16) \quad \begin{aligned} \|C^T C - I\|_{\text{row}} &\leq \epsilon_C < 1, \\ \| |\Theta|^{-1/2} (C^T H C - \Theta) | \Theta|^{-1/2} \|_{\text{row}} &\leq \epsilon. \end{aligned}$$

Define

$$(6.17) \quad \begin{aligned} g_V &= \text{tr}(|\Theta|^{-1} C^T G C), & g_d &= \text{tr}(|\Theta|^{-1} C^T D_2 C), \\ \eta_Q &= \frac{(\sqrt{g_V/\gamma_0} + \sqrt{t_Q g_d})^2}{1 - \mu_Q}, \\ S_* &= \frac{1 + \epsilon_C}{\min_i |\Theta_i| (1 - \epsilon - \eta_Q)}, \\ t_P &= \max_j \sum_{i:n_i=jm} \tau_i^2, & k_* &= v_*/\sqrt{\gamma_0} + d_*\sqrt{t_Q t_P}, \\ \delta_Q &= \frac{k_*}{\sqrt{\gamma_0}(1 - \mu_Q)}. \end{aligned}$$

If $\mu_Q < 1$ and $\epsilon + \eta_Q < 1$, the problem (6.1) has an L^2 inverse with

$$(6.18) \quad \|L^{-1}\|_{2 \rightarrow 2} \leq B_{L^2} := (1 + \delta_Q^2) S_* + \frac{1}{\gamma_0(1 - \mu_Q)}.$$

Proof. The finite spectral projection preserves H^1 , and the energy space of $|D_Q|^{1/2}$ is QH^1 with an equivalent norm. Thus the block decomposition of (6.2) is legitimate on $\text{Ran } P \oplus QH^1$. Its finite diagonal block is H and its complementary block is the form just estimated. Denote the coupling from the finite block into the complementary dual space by $\mathfrak{b}$. Its volume part is $-QVP$; its boundary part is the pairing with $d \text{Tr } P$. By (6.8) and (6.14), after multiplication on the right by $C|\Theta|^{-1/2}$, their energy Hilbert–Schmidt norms are at most $\sqrt{g_V/\gamma_0}$ and $\sqrt{t_Q g_d}$. The triangle inequality bounds their sum. Pairing twice with the complementary inverse yields the relative Schur correction at most η_Q .

The matrices in (6.16) are symmetric, so their maximum row sums bound their Euclidean operator norms. The first makes C invertible and gives $\|C\|^2 \leq 1 + \epsilon_C$. If S is the finite Schur complement, then $|\Theta|^{-1/2} C^T S C |\Theta|^{-1/2}$ is $\text{sgn } \Theta$ plus a symmetric perturbation of norm at most $\epsilon + \eta_Q < 1$. Its inverse norm is at most $(1 - \epsilon - \eta_Q)^{-1}$, and undoing the congruence gives $\|S^{-1}\| \leq S_*$.

The finite trace map has squared norm at most t_P : angular orthogonality gives $\|\text{Tr } Px\|_2^2 = \sum_j |\sum_{i:n_i=jm} \tau_i x_i|^2 \leq t_P \|x\|_2^2$. The unweighted coupling into the complementary energy space therefore has norm at most k_* . The complementary inverse, followed by inclusion into L^2 , bounds the off-diagonal solve by δ_Q and its L^2 diagonal solve by $1/[\gamma_0(1 - \mu_Q)]$. The symmetric block inverse of $-L$ has the form

$$\begin{pmatrix} I \\ -X_Q \end{pmatrix} S^{-1} \begin{pmatrix} I & -X_Q^* \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & \mathfrak{M}_Q^{-1} \end{pmatrix}, \quad \|X_Q\| \leq \delta_Q,$$

where $\mathfrak{M}_Q^{-1}$ is the complementary form inverse restricted to L^2 . This proves (6.18). The formula constructs a weak solution for every L^2 right side; the invertible complementary and Schur blocks also prove uniqueness. Smooth coefficients and boundary give the usual Robin H^2 regularity, so this form inverse is the operator inverse asserted in the proposition. $\square$

For the centre (5.13), the enclosures give

$$(6.19) \quad \begin{aligned} d_* &< 0.355923 < 1, & \mu_Q &< 0.011202, & \epsilon_C &< 5.276 \cdot 10^{-13}, & \epsilon &< 2.580 \cdot 10^{-11}, \\ \min_i |\Theta_i| &> 0.5561235, & \eta_Q &< 0.013407, & B_{L^2} &< 1.823. \end{aligned}$$

In particular $\beta \geq 1 - d_* > 0$.

7. TRANSFER TO THE WORKING NORMS

Define the centre coefficient sums

$$(7.1) \quad \begin{aligned} P &= \sum_j 2^j |p_j^{(0)}|, & P_1 &= \sum_j 2^j j m |p_j^{(0)}|, & P_2 &= \sum_j 2^j (j m)^2 |p_j^{(0)}|, \\ J_0 &= (2\rho - P)^{-1}, & d_0 &= J_0 P_1, & d_1 &= J_0 (P_1 + P_2) + J_0^2 P_1^2. \end{aligned}$$

The reciprocal is well defined because $P < 240.1776 < 2\rho$. The Neumann series for $1/p_0$ in W_0 gives norm at most J_0 . For $Q_0 = wp'_0/p_0$, (5.6) and differentiation give

$$(7.2) \quad \|d\|_{W_0} \leq d_0, \quad \|d\|_{W_1} \leq d_1.$$

Indeed $\|wp'_0\|_{W_0} = P_1$, while $\|\partial_\theta Q_0\|_{W_0} \leq J_0 P_2 + J_0^2 P_1^2$. Also multiplication by q on Y has norm at most P^2 , and

$$(7.3) \quad \|q - \rho^2\|_Y \leq P^2 - \rho^2.$$

For the latter, remove the constant–constant product in $p_0 \overline{p_0}$ and bound $2^{|j-k|}$ by 2^{j+k} on every remaining term.

Proposition 7.1 (Robin inverse and two trace derivatives). *Let j_h be the least positive integer with $j_h m > \rho$. Put*

$$(7.4) \quad \gamma_h = (j_h m)^2 - \rho^2, \quad b_h = \frac{P^2 - \rho^2}{\gamma_h} + \frac{d_0}{1 + \sqrt{\gamma_h}}.$$

Suppose that (6.1) has an inverse on L^2 of the symmetric sector with norm at most B_{L^2} , and that $\beta > 0$ and $b_h < 1$. For $f \in Y$ the Robin solution of (6.1) belongs to Y , its boundary trace belongs to W_2 , and

$$(7.5) \quad \|V\|_Y \leq B_Y \|f\|_Y, \quad \|\text{Tr } V\|_{W_s} \leq B_{T_s} \|f\|_Y \quad (s = 1, 2),$$

where, with $P_{\text{un}} = \sum_j |p_j^{(0)}|$,

$$(7.6) \quad \begin{aligned} \tau_*^2 &= 2B_{L^2}^2 + 2B_{L^2} \sqrt{P_{\text{un}}^2 B_{L^2}^2 + B_{L^2}}, \\ b_{\text{low}} &= [(1 + P^2 B_{L^2})/2 + \tau_*] / \sqrt{2}, \\ B_Y &= \frac{(1 + 2 \sum_{j=1}^{j_h-1} 2^j) b_{\text{low}} + 1/\gamma_h}{1 - b_h}, \\ B_{T1} &= 1 + P^2 B_Y + d_0 B_Y, \\ B_{T2} &= 1 + P^2 B_Y + d_1 B_{T1}. \end{aligned}$$

Proof. Energy and low modes. The weak equation (6.2), tested with V , gives

$$\|\nabla V\|_2^2 + \int_{\partial \mathbb{D}} \beta V^2 = \int_{\mathbb{D}} q V^2 - \int_{\mathbb{D}} f V.$$

Dropping the positive boundary term and using the L^2 inverse yields

$$\|\nabla V\|_2^2 \leq (P_{\text{un}}^2 B_{L^2}^2 + B_{L^2}) \|f\|_2^2.$$

The divergence estimate (6.13) then gives $\|\text{Tr } V\|_2 \leq \tau_* \|f\|_2$. For each angular coefficient write $V_n = K_n(f - qV)_n + t_n r^{|n|}$, where K_n is the zero-Dirichlet inverse of the radial operator in Lemma 2.2 and t_n is its boundary trace. The radial Green kernel in Lemma 2.2 is bounded by $-\log \max(r, s)$ and $\int_0^1 (\log s)^2 s ds = 1/4$. Therefore

$$\sup_r |K_n(f - qV)_n| \leq \frac{1}{2} \|(f - qV)_n\|_{L^2(r dr)}.$$

The three comparisons completing the estimate are

$$\|f\|_2 \leq \|f\|_Y / \sqrt{2}, \quad \|(f - qV)_n\|_{L^2(r dr)} \leq \|f - qV\|_2, \quad |t_n| \leq \|\text{Tr } V\|_2.$$

The first follows from Parseval and $\int_0^1 r dr = 1/2$, the second from orthogonality, and the third from the boundary Fourier coefficient formula. Thus every coefficient has supremum at most $b_{\text{low}} \|f\|_Y$. The low part V_{low} , consisting of $|n| < j_h m$, has weighted norm at most $(1 + 2 \sum_{j=1}^{j_h-1} 2^j) b_{\text{low}} \|f\|_Y$.

Construction of the high modes. For $|n| \geq j_h m$, the operator $-\Delta_n - \rho^2$ has positive radial potential $n^2/r^2 - \rho^2 \geq \gamma_h$. The maximum principle with Robin-one boundary data gives interior-forcing norm at most $1/\gamma_h$. This may also be seen by comparing with the constant supersolution $\|f\|_\infty / \gamma_h$ for the interior forcing f of (7.7). The supersolution has positive Robin boundary datum; apply the comparison to both signs of the real solution. For a pure boundary source the regular homogeneous solution is a multiple of $J_{|n|}(\rho r)$, which is positive and

increasing because $|n| > \rho$. Its logarithmic derivative $z_n = r\rho J'_{|n|}(\rho r)/J_{|n|}(\rho r)$ obeys

$$rz'_n = n^2 - \rho^2 r^2 - z_n^2.$$

The power series at zero gives $z_n = |n| - \rho^2 r^2/[2(|n| + 1)] + O(r^4)$, greater than $\sqrt{n^2 - \rho^2 r^2}$ for small positive r . At a first contact with this decreasing comparison function, $z'_n = 0$ and the difference has positive derivative, excluding the contact. Consequently the solution with boundary source ς_n has supremum at most $|\varsigma_n|/(1 + \sqrt{\gamma_h})$.

Let Π_H be the angular high projection. In the high subspace of Y solve by contraction

$$(7.7) \quad \begin{aligned} (\Delta + \rho^2)\tilde{V}_H &= \Pi_H(f - (q - \rho^2)(V_{\text{low}} + \tilde{V}_H)), \\ (\tilde{V}_H)_r + \tilde{V}_H &= -\Pi_H(d \operatorname{Tr}(V_{\text{low}} + \tilde{V}_H)). \end{aligned}$$

The interior and boundary estimates, the trace norm $Y \rightarrow W_0$ at most one, and (7.2) give Lipschitz constant $b_h < 1$. This constructs $\tilde{V}_H$ with finite Y norm before any total analytic norm is assumed.

For its identification with $\Pi_H V$ we first justify the energy equation. In each mode the constructed solution is regular at zero and solves its radial ODE with Robin-one boundary source. If $\mathfrak{f}_n$ and ς_n are the right sides of (7.7), the mode energy identity is

$$\int_0^1 (|\tilde{V}_{H,n}'|^2 + (n^2/r^2 - \rho^2)|\tilde{V}_{H,n}|^2)r dr + |\tilde{V}_{H,n}(1)|^2 = -\operatorname{Re} \int_0^1 \mathfrak{f}_n \overline{\tilde{V}_{H,n}} r dr + \operatorname{Re}(\varsigma_n \overline{\tilde{V}_{H,n}(1)}).$$

The weighted supremum sums for the solution and $\mathfrak{f}_n$, and the weighted boundary sum for ς_n , make the sum of the absolute right sides finite. Adding the finite $\rho^2 L^2$ term shows convergence of the mode series in H^1 . The mode equations, first tested against finite angular sums and then by H^1 density, give the weak high-subspace equation and its boundary pairing. In particular the boundary source is an actual weak Robin datum.

The original L^2 inverse gives $V \in H^2$; testing its weak equation against a radial test function times $\cos(n\theta)$ gives

$$(7.8) \quad V'_n(1) = -(\beta \operatorname{Tr} V)_n.$$

Here V_n is C^2 on $(0, 1]$ since $(f - qV)_n$ is continuous. Thus $V_H = \Pi_H V$ satisfies the same weak projected equation as $\tilde{V}_H$, with its own occurrence on the right side. Their difference $z \in \Pi_H H^1$ satisfies

$$\|\nabla z\|_2^2 + \int_{\partial \mathbb{D}} \beta z^2 = \int_{\mathbb{D}} q z^2.$$

Angular orthogonality gives $\|\nabla z\|_2^2 \geq (j_h m)^2 \|z\|_2^2$. The inequality $b_h < 1$ implies $(j_h m)^2 > P^2 \geq \|q\|_\infty$. Since $\beta > 0$, $z = 0$. This identifies the analytic high solution with the original Robin solution.

The high fixed-point equation now gives $\|V_H\|_Y \leq \|f\|_Y/\gamma_h + b_h \|V\|_Y$. Add the low bound and rearrange to obtain B_Y .

Two boundary derivatives. Integrate the radial equation $\Delta_n V_n = (f - qV)_n$ against $r^{|n|+1}$. The moment identity underlying Lemma 2.2, now with nonzero Dirichlet trace, reads

$$\int_0^1 r^{|n|+1} (f - qV)_n dr = V'_n(1) - |n|t_n.$$

Using (7.8) gives

$$(7.9) \quad (|n| + 1)t_n = -\int_0^1 r^{|n|+1} (f - qV)_n dr - (dt)_n.$$

Since $t \in W_0$ already, summing this formula with weights $R^{|n|}$ first proves $t \in W_1$. Its norm is at most $\frac{1}{2}\|f - qV\|_Y + d_0\|t\|_{W_0}$, bounded by the stated B_{T1} . Multiply the same identity by $|n| + 1$ and use $(|n| + 1)/(|n| + 2) \leq 1$. The W_1 algebra bound on dt then proves $t \in W_2$ and the stated B_{T2} . $\square$

The centre enclosures give $b_h < 0.033901 < 1$ and

$$(7.10) \quad B_Y < 1.117 \cdot 10^6, \quad B_{T1} < 6.443 \cdot 10^{10}, \quad B_{T2} < 1.187 \cdot 10^{13}.$$

The second boundary derivative is needed because reconstruction of p from its primitive consumes one derivative while the unknown shape belongs to W_1 .

8. ALL-MODE RESIDUAL AND THE EXACT LINEARISATION INVERSE

8.1. Strip bounds and the clamped residual. Complexify $\phi = m\theta$ to $|\operatorname{Im} \phi| \leq \log 4$. Set

$$(8.1) \quad \begin{aligned} S_c &= \sum_{j>0} 4^j |c_j^{(0)}|, & \delta_s &= (2\rho S_c + S_c^2)/\rho, & \varepsilon_s &= \frac{1}{2} \log \frac{\rho + S_c}{\rho - S_c}, \\ P_4 &= \sum_j 4^j (jm + 1) |c_j^{(0)}|. \end{aligned}$$

For $0 \leq r \leq 1$ put $A_\pm = \rho + \sum_{j>0} c_j^{(0)} r^{jm} e^{\pm i j \phi}$. For the centre, $S_c < 0.0571$, so $\rho > 2S_c$; this places each $A_\pm$ in the disc $|A - \rho| \leq S_c < \rho/2$. Define

$$(8.2) \quad B_s = \sqrt{A_+} \sqrt{A_-}$$

using the principal roots of the individual factors. Both roots are analytic, and $|\arg B_s| < \pi/6$, hence $\operatorname{Re} B_s > 0$ and $|B_s + \rho| \geq \rho$. Therefore

$$(8.3) \quad |B_s - \rho| = \frac{|A_+ A_- - \rho^2|}{|B_s + \rho|} \leq \delta_s, \quad |A_\pm / B_s| \leq e^{\varepsilon_s}.$$

The complexified physical radius is rB_s . The positive angular branch of order $n = jm$ is $J_n(rB_s) e^{ij\phi} (A_+/B_s)^n$, with the reflected expression for the negative branch. The entire Cartesian Bessel power series interprets this formula also at $r = 0$. For integer order the contour shift in (3.8) uses only the Bessel integral and gives

$$|J_n(z)| \leq \exp(-n\vartheta + |\operatorname{Re} z| \sinh \vartheta + |\operatorname{Im} z| \cosh \vartheta) \quad (\vartheta \geq 0).$$

Choose $\vartheta_j = 0$ for $jm \leq \rho + \delta_s$ and otherwise $\vartheta_j = \operatorname{acosh}(jm/(\rho + \delta_s))$. Define

$$(8.4) \quad \mathbf{u}_j = \frac{|a_j|}{s_j} 4^j \exp(jm\varepsilon_s - jm\vartheta_j + (\rho + \delta_s) \sinh \vartheta_j + \delta_s \cosh \vartheta_j).$$

Since $|\operatorname{Re}(rB_s)| \leq \rho + \delta_s$ and $|\operatorname{Im}(rB_s)| \leq \delta_s$, $\sum_j \mathbf{u}_j$ bounds U_* uniformly on the strip. The coefficient bound $M4^{-|j|}$ yields the weighted sum factor $1 + 2\sum_{j \geq 1} 2^{-j} = 3$. Consequently

$$(8.5) \quad \begin{aligned} \|U_*\|_Y &\leq U_b := 3 \sum_j \mathbf{u}_j, \\ \|\partial_w U_*\|_Y, \|\partial_{\bar{w}} U_*\|_Y &\leq \mathfrak{D}_U := 3R4^{2/m} P_4 e^{\varepsilon_s} \sum_j \mathbf{u}_j e^{\vartheta_j}. \end{aligned}$$

For the derivative bound use the Wirtinger recurrences displayed in Subsection 3.1: a physical derivative changes Bessel order by one, costs at most e^{ϑ_j} in the contour bound, and changes the angular factor by at most $4^{1/m} e^{\varepsilon_s}$. The chain rule contributes P_4 . The resulting angular frequencies are $km \pm 1$; multiplying by $e^{\mp i\theta}$ makes them periodic in ϕ , and $R^{|km \pm 1|} \leq R2^{|k|}$

costs the further factors $R4^{1/m}$. These estimates give the enlarged $\mathfrak{D}_U$ in (8.5), including the order-zero term via the Bessel identity $J_{-1}(z) = -J_1(z)$.

The common strip majorant for both defects in (5.14) is

$$(8.6) \quad M_{\text{tr}} = U_b + 2\mathfrak{D}_U + P_4.$$

Indeed the radial derivative is $e^{i\theta}U_w + e^{-i\theta}U_{\bar{w}}$, whose strip bound is at most $2\mathfrak{D}_U$ by the estimates just derived. The complexification of $|p_0|$ is the product of the principal square roots of $p_0(e^{i\theta})$ and its reflected series. The inequality $S_4 < \rho$ of Subsection 6.2 makes this product analytic on the strip, with modulus at most P_4 . This accounts for the additional term in (8.6).

Use $N_b = 1024$ interval trapezoidal nodes and retain the cosine coefficients through $L_b = 400 < N_b/2$. Summing the aliases gives the common retained-coefficient error

$$(8.7) \quad e_b = \frac{8M_{\text{tr}}4^{-(N_b-L_b)}}{1 - 4^{-N_b}}.$$

For $j > L_b$, $|D_j|, |N_j| \leq 2M_{\text{tr}}4^{-j}$. Cauchy's estimate gives the same bound for every $j \geq 1$, and the cosine coefficients of $|p_0|$ are bounded by $2P_44^{-j}$. Thus the series (5.15) for ℓ and the series for $\mathcal{B}(|p_0|)$ converge with all derivatives on every closed disc of radius less than $4^{1/m}$. The complete moments are

$$(8.8) \quad D^{[k]} = \sum_{j \geq 0} 2^j(jm+1)^k |D_j|, \quad N^{[k]} = \sum_{j \geq 0} 2^j(jm+1)^k |N_j| \quad (0 \leq k \leq 3).$$

Complex Fourier coefficients and one-sided cosine coefficients give the same norm here, since the two half-coefficients at opposite frequencies sum to the full cosine coefficient. With $J = L_b + 1$, the tail added to each moment is at most

$$(8.9) \quad 2M_{\text{tr}}2^{-J} \sum_{i=0}^k \binom{k}{i} (mJ+1)^{k-i} m^i \mathfrak{m}_i, \quad (\mathfrak{m}_0, \mathfrak{m}_1, \mathfrak{m}_2, \mathfrak{m}_3) = (2, 2, 6, 26).$$

These four integers are the exact sums $\sum_{t \geq 0} t^i 2^{-t}$, as in (3.16). Expanding $(m(J+t)+1)^k$ proves (8.9) and includes every uncomputed harmonic.

The lift polynomials are identical to those of the Schiffer centre; their norm estimates depend only on the defect coefficients. In the present notation they give

$$(8.10) \quad \begin{aligned} L_0 &= D^{[0]} + N^{[0]}/2, & L_1 &= 2R(D^{[2]} + N^{[1]}), \\ L_\Delta &= 2(D^{[2]} + N^{[1]}), & L_{\Delta,1} &= 2R(D^{[3]} + N^{[2]}). \end{aligned}$$

They bound ℓ , each Wirtinger derivative of ℓ , $\Delta\ell$, and each Wirtinger derivative of $\Delta\ell$, respectively. For example,

$$\begin{aligned} \Delta(h_n^D \cos n\theta) &= -2n(n+1)r^n \cos n\theta, \\ \Delta(h_n^N \cos n\theta) &= 2(n+1)r^n \cos n\theta. \end{aligned}$$

Writing the lifts in $w, \bar{w}$ proves the derivative bounds with one factor R for the shifted angular frequency; the displayed enlargements include $n = 0$. Since $\|\partial_w q\|_{\mathcal{Y}} \leq RPP_1$ and multiplication by q has norm at most P^2 , equation (5.16) gives

$$(8.11) \quad \begin{aligned} \|E\|_{\mathcal{Y}} &\leq E_0 := L_\Delta + P^2 L_0, \\ \|\partial_w E\|_{\mathcal{Y}}, \|\partial_{\bar{w}} E\|_{\mathcal{Y}} &\leq E_1 := L_{\Delta,1} + RPP_1 L_0 + P^2 L_1, \\ \|\text{Tr } E\|_{W_1} &\leq E_{W1} := E_0 + 2RE_1. \end{aligned}$$

The last bound uses $\partial_\theta E = i(wE_w - \bar{w}E_{\bar{w}})$. Likewise $\Delta U_0 = -qU_* - \Delta\ell$ implies

$$(8.12) \quad \|\Delta U_0\|_Y \leq \Lambda_0 := P^2 U_b + L_\Delta, \quad \|\partial_w \Delta U_0\|_Y \leq \Lambda_1 := RPP_1 U_b + P^2 \mathfrak{D}_U + L_{\Delta,1},$$

with the same bound for the conjugate derivative. In particular,

$$(8.13) \quad U_b < 68.459, \quad \mathfrak{D}_U < 17609.075, \quad E_0 < 9.699 \cdot 10^{-67}, \quad E_1 < 2.728 \cdot 10^{-62}.$$

8.2. Material derivative and boundary identities. All quantities below are evaluated at $x_0 = (g_0, p_0)$. For a variation $(\delta g, k) \in X$ define

$$(8.14) \quad \eta(w) = \int_0^w k(z) dz, \quad v = \eta/p_0 = wa, \quad T = v\partial_w U_0 + \bar{v}\partial_{\bar{w}} U_0.$$

The symbol a denotes only the holomorphic real-coefficient shape function in this formula; the a_j in (5.13) are the fixed field coefficients. Put $b = |p_0|$ on the circle and denote its real derivative by $Db[k] = b \operatorname{Re}(k/p_0)$. Differentiation of (5.12) yields

$$(8.15) \quad DF(x_0)(\delta g, k) = LH + \delta q U_0, \quad H = K\delta g + \mathcal{B}(Db[k]), \quad \delta q = 2 \operatorname{Re}(\bar{p}_0 k).$$

The material identity of the Schiffer part uses only holomorphicity and the equation $\Delta U_0 + qU_0 = E$, so it gives here

$$(8.16) \quad LT + \delta q U_0 = \mathcal{E}(k), \quad \mathcal{E}(k) = vE_w + \bar{v}E_{\bar{w}} + (v' + \bar{v}')E.$$

To see all cancellations, $\Delta(vU_w) = v\partial_w \Delta U + v'\Delta U$, and the conjugate formula gives

$$(8.17) \quad \Delta T = v\partial_w \Delta U_0 + \bar{v}\partial_{\bar{w}} \Delta U_0 + (v' + \bar{v}')\Delta U_0.$$

The identity $k = (p_0 v)'$ gives $\delta q = vq_w + \bar{v}q_{\bar{w}} + (v' + \bar{v}')q$. Substitute $\Delta U_0 = E - qU_0$ in (8.17); the terms involving derivatives of q cancel with $\delta q U_0$, and the other terms involving U_0 cancel with qT . This proves (8.16).

The boundary Hessian calculation differs from the Schiffer case. Exact clamping gives $U_0 = 0$, $(U_0)_r = b$, $(U_0)_{r\theta} = b_\theta$, and $(U_0)_{rr} = E - b$ on the circle. Since $T = r \operatorname{Re} a (U_0)_r + \operatorname{Im} a (U_0)_\theta$, radial differentiation gives

$$(8.18) \quad T = b \operatorname{Re} a, \quad T_r = b \operatorname{Re}(a + wa') + (E - b) \operatorname{Re} a + b_\theta \operatorname{Im} a.$$

For $\mathcal{Q}_0 = wp'_0/p_0$ one has

$$Db[k] = b \operatorname{Re}((1 + \mathcal{Q}_0)a + wa'), \quad b_\theta = -b \operatorname{Im} \mathcal{Q}_0.$$

The tangential terms cancel in the difference, leaving

$$(8.19) \quad Db[k] - T_r = (\beta - E/b)T, \quad \beta = 1 + \operatorname{Re} \mathcal{Q}_0.$$

Thus the residual introduces a boundary term as well as the interior term $\mathcal{E}$. Define its Neumann lift and the corrected linearisation by

$$(8.20) \quad \mathcal{C}(k) = \mathcal{B}(E \operatorname{Re} a), \quad \tilde{D} = DF(x_0) - \mathcal{E} + L\mathcal{C}.$$

Equations (8.15) and (8.16) then give $\tilde{D}(\delta g, k) = L(H - T + \mathcal{C}(k))$.

8.3. The exact two-sided inverse.

Proposition 8.1. *Suppose that (6.1) has the inverse and trace bounds of Proposition 7.1, and that $P < 2\rho$. Then $\tilde{D}: X \rightarrow Y$ of (8.20) has a bounded two-sided inverse $A: Y \rightarrow X$ with $\|A\| \leq A_*$, where A_* is defined in (8.27).*

Proof. Right inverse. Given $f \in Y$, solve (6.1) by the Robin inverse and let $t = \text{Tr } V \in W_2$. Since b does not vanish, prescribe

$$(8.21) \quad \text{Re } a = -t/b.$$

The reciprocal estimates in Subsection 8.4 below give $1/b \in W_2$. The right side is real, even and has frequencies in $m\mathbb{Z}$. If its cosine coefficients are $\hat{a}_j$, define $a(w) = \sum_{j \geq 0} \hat{a}_j w^{jm}$. Thus the cosine coefficients of $-t/b$ are the Taylor coefficients of a . Equivalently retain the constant complex Fourier coefficient and double every positive one. This is the unique solution in the holomorphic real-coefficient class; the imaginary constant of the general real-part problem is excluded. Equation (5.6) gives equality of the W_2 norms. Define, in this order,

$$(8.22) \quad v = wa, \quad k = (p_0 wa)', \quad H = V + T - \mathcal{C}(k), \quad \delta g = \Delta(H - \mathcal{B}(\text{Db}[k])).$$

The boundary value of H is $t + b \text{Re } a = 0$. Using $V_r = -\beta t = \beta T$ and (8.19),

$$H_r = \beta T + T_r - E \text{Re } a = \text{Db}[k].$$

These equalities hold mode by mode by (7.8). The function $H - \mathcal{B}(\text{Db}[k])$ has zero Cauchy data and a Laplacian in Y . The radial moment identity therefore gives $\delta g \in X_g$, and Poisson uniqueness gives $K\delta g = H - \mathcal{B}(\text{Db}[k])$. The interior identity now gives $\tilde{D}(\delta g, k) = LV = f$. The recipe is linear in f .

Derivative count and regularity. For an arbitrary $k \in W_1$,

$$(8.23) \quad \|\eta/w\|_{W_2} = \|k\|_{W_1}, \quad \|(p_0 wa)'\|_{W_1} = \|p_0 a\|_{W_2}.$$

Both identities follow coefficient by coefficient from multiplication or division by $jm + 1$. The Robin solve gains two boundary derivatives, division by b preserves W_2 , and reconstruction of k spends one. The Laplacian of the Neumann lift spends one boundary derivative, which is available from $k \in W_1$. All holomorphic shape functions are smooth near $\mathbb{D}$ because $R > 1$. The Robin solution is a weak H^2 solution, with continuous right side; standard elliptic regularity gives $W^{2,s}$ for finite s and C^1 boundary traces. The mode-wise boundary identity proved above justifies the zero Neumann trace of $H - \mathcal{B}(\text{Db}[k])$. The estimates below show that its Laplacian belongs to Y , and hence that the recipe maps Y to X .

Left inverse. Start with any $(\delta g, k) \in X$. Form η, v, a, T by (8.14), H by (8.15), and put $V = H - T + \mathcal{C}(k)$. The apparent singularity in $a = \eta/(wp_0)$ at zero is removable, and (8.23) gives $a \in W_2$. Its Dirichlet trace is $-b \text{Re } a$, and (8.19) gives $V_r + \beta V = 0$. Also $LV = \tilde{D}(\delta g, k)$ distributionally. The terms involving the fixed smooth centre and the holomorphic shape are smooth near the circle; $K\delta g$ has C^1 traces. Thus this function is the weak Robin solution. Uniqueness recovers the same V when the right-inverse recipe is applied to $\tilde{D}(\delta g, k)$. Its Dirichlet trace divided by $-b$ recovers $\text{Re } a$, and the unique real-coefficient real-part solution recovers a . Then $k, T, \mathcal{C}(k), H$ and δg are recovered in turn. Consequently $A\tilde{D} = I_X$ and $\tilde{D}A = I_Y$. $\square$

8.4. Inverse norm and left Newton defect. The reciprocal bounds required above are

$$(8.24) \quad J_1 = J_0 + P_1 J_0^2, \quad J_2 = J_0 + (2P_1 + P_2) J_0^2 + 2P_1^2 J_0^3.$$

They bound $1/p_0$ in W_1 and W_2 , respectively; J_2 also bounds $1/b$ in W_2 . To prove the holomorphic bounds differentiate the reciprocal once and twice in the angular variable and use

$\|h\|_{W_2} = \|h\|_{W_0} + 2\|\partial_\theta h\|_{W_0} + \|\partial_\theta^2 h\|_{W_0}$. For the boundary reciprocal, $\log b = (\log p_0 + \log \overline{p_0})/2$ has first derivative norm at most $J_0 P_1$ and second derivative norm at most $J_0 P_2 + J_0^2 P_1^2$. The product of the individual square-root reciprocal series gives $\|1/b\|_{W_0} \leq J_0$; differentiating $\exp(-\log b)$ gives exactly the stated J_2 enlargement.

With $\delta_p = (P - \rho)/\rho < 1$, set

$$(8.25) \quad B_{01} = \rho(1 + \delta_p)^2 + \frac{1 + \delta_p}{\sqrt{1 - \delta_p}} P_1.$$

This bounds $\|b\|_{W_1}$. Indeed the sum of absolute nonconstant coefficients in the series for $\sqrt{1+z}$ at $|z| = \delta_p$ is $1 - \sqrt{1 - \delta_p} \leq \delta_p$. Apply this to $\sqrt{p_0/\rho}$ and its reflection and multiply. Differentiating a square root adds at most $P_1/[2\sqrt{\rho}\sqrt{1 - \delta_p}]$; the two product derivatives give the second term of (8.25). In particular,

$$(8.26) \quad \|Db[k]\|_{W_1} \leq B_{01} J_1 \|k\|_{W_1},$$

by (5.6) applied to k/p_0 .

Define

$$(8.27) \quad \begin{aligned} B_a &= J_2 B_{T2}, & B_p &= (P + 2P_1 + P_2) B_a, \\ B_g &= 1 + P^2 B_Y + 2(R\Lambda_1 + \Lambda_0 + E_{W1}) B_a + 2B_{01} J_1 B_p, \\ A_* &= B_g + B_p. \end{aligned}$$

For $Af = (\delta g, k)$, equation (8.21) bounds $\|a\|_{W_2} \leq B_a \|f\|_Y$, and the exact second identity of (8.23) gives $\|k\|_{W_1} \leq B_p \|f\|_Y$ because $\|p_0\|_{W_2} = P + 2P_1 + P_2$. As in (3.38), $\|v\|_Y \leq R\|a\|_{W_1}$ and $\|v'\|_Y = \|a\|_{W_1}$. The product identity (8.17) therefore gives

$$\|\Delta T\|_Y \leq 2(R\Lambda_1 + \Lambda_0) \|a\|_{W_1}.$$

Also $\|\Delta \mathcal{C}(k)\|_Y \leq 2E_{W1} \|a\|_{W_1}$. Applying these estimates and (8.26) to (8.22), with $\Delta V = f - qV$, proves $\|\delta g\|_Y \leq B_g \|f\|_Y$ and $\|A\| \leq A_*$.

For an arbitrary input shape, $\|a\|_{W_2} \leq J_2 \|k\|_{W_1}$ by the first identity of (8.23). Hence

$$(8.28) \quad \begin{aligned} \|AF(x_0)\|_X &\leq Y_0 := A_* E_0, \\ \|I_X - ADF(x_0)\|_{X \rightarrow X} &\leq Z, \end{aligned}$$

$$Z := A_* J_2 [2(RE_1 + E_0) + 2E_{W1} + P^2 E_0/2].$$

Indeed $I_X - ADF = A(-\mathcal{E} + LC)$, the material correction has norm at most $2(RE_1 + E_0) \|a\|_{W_1}$, and $\|LC\|_Y \leq (2E_{W1} + P^2 E_0/2) \|a\|_{W_1}$. The last term includes $q\mathcal{C}$ as well as its Laplacian. Both identities of Proposition 8.1 are used: the left identity gives this defect on X , and the right identity makes A injective, so a fixed point of the Newton map is a zero of F .

8.5. Symmetry directions and the fixed parameter. At a true solution, the Eulerian derivative $\dot{u}$ associated with normal displacement h_ν has boundary data $\dot{u} = -h_\nu$ and $\partial_\nu \dot{u} = \kappa h_\nu$. Thus $\partial_\nu \dot{u} + \kappa \dot{u} = 0$, agreeing with (6.1). Translations have angular character ± 1 and are absent from the D_m invariant sector. The rotation derivative is reflection-odd and is absent from its real even subspace. The real conformal coefficients and $\psi(0) = 0$, $\psi'(0) > 0$ fix the conformal normalisation. The constant coefficient $\hat{a}_0$ of a generates the radial reparametrisation $w \mapsto (1 + \varepsilon \hat{a}_0)w$ of the parameter disc. It is retained in the inverse and equals the mean of $-t/b$ in (8.21). Euclidean dilation is not a homogeneous kernel with eigenvalue and flux fixed. The flux-preserving family $u_s(x) = s u(x/s)$ has parameter s^{-2} ; its derivative $u - x \cdot \nabla u$ satisfies $(\Delta + 1)(u - x \cdot \nabla u) = 2u$. Multiplying the field by a constant changes its prescribed flux. The absence of a kernel within the working sector follows from the Robin inverse and the two-sided construction above.

9. EXISTENCE, GEOMETRY AND EARLIER RESULTS

9.1. The full nonlinear bound. Fix $r_* > 0$ with $J_1 r_* < 1/4$ and consider the closed X ball about x_0 of that radius. Put

$$(9.1) \quad \begin{aligned} P_b &= P + r_*, & B_b &= 4B_{01}, & J_b &= 2J_1, & U_c &= U_b + L_0, \\ U_{\text{ball}} &= U_c + r_*/4 + B_b J_b r_*/2. \end{aligned}$$

On the ball, $\|p\|_{W_0} \leq P_b$, $\|p\|_{W_1} \leq B_b$, $\|1/p\|_{W_1} \leq J_b$, and $\|U(g, p)\|_Y \leq U_{\text{ball}}$. For the reciprocal and square root write $p = p_0(1 + k/p_0)$. The W_1 algebra gives $\|k/p_0\|_{W_1} < 1/4$; its Neumann series bounds the reciprocal by $(4/3)J_1 \leq J_b$. Multiplying the square-root series of $1 + k/p_0$ and its reflection bounds the modulus by $4B_{01} = B_b$. Along the segment from p_0 to p , the formula $D|p|[k] = |p| \operatorname{Re}(k/p)$ bounds its change in W_1 by $B_b J_b \|k\|_{W_1}$. The lift bound and $\|K\| \leq 1/4$ give the last bound in (9.1).

Lemma 9.1. *Throughout the ball, F has second derivative bound*

$$(9.2) \quad \|F''\|_{X \times X \rightarrow Y} \leq M_2 := (4 + P_b^2)B_b J_b^2 + 2P_b B_b J_b + P_b + 2U_{\text{ball}}.$$

Proof. Differentiating the modulus derivative once more and applying the W_1 product inequality gives $\|D^2|p|[k_1, k_2]\|_{W_1} \leq 2B_b J_b^2 \|k_1\|_{W_1} \|k_2\|_{W_1}$. Let $q_p = |p|^2$ for the varying shape; q continues to denote the fixed centre potential $|p_0|^2$. For $x_i = (g_i, k_i)$ the full second derivative is

$$(9.3) \quad \begin{aligned} F''[x_1, x_2] &= (\Delta + |p|^2)\mathcal{B}(D^2|p|[k_1, k_2]) \\ &+ Dq_p[k_1](\mathcal{B}(D|p|[k_2]) + Kg_2) \\ &+ Dq_p[k_2](\mathcal{B}(D|p|[k_1]) + Kg_1) + D^2q_p[k_1, k_2]U. \end{aligned}$$

Here $\|Dq_p[k]\|_Y \leq 2P_b \|k\|_{W_1}$ and $\|D^2q_p[k_1, k_2]\|_Y \leq 2\|k_1\|_{W_1} \|k_2\|_{W_1}$. The first line of (9.3) is bounded by $(4 + P_b^2)B_b J_b^2$, using both bounds in (5.9). The two shape/lift products give $2P_b B_b J_b$, and the two shape/Poisson products are bounded by P_b using $\|K\| \leq 1/4$. The final product gives $2U_{\text{ball}}$. For unit X directions these enlarged bounds sum to (9.2). $\square$

9.2. The radii polynomial. Set $C_2 = A_* M_2/2$. The Newton map $\mathcal{T}(x) = x - AF(x)$ satisfies, for $\|x - x_0\|_X \leq r_*$,

$$(9.4) \quad \|\mathcal{T}(x) - x_0\|_X \leq Y_0 + Zr_* + C_2 r_*^2, \quad \|D\mathcal{T}(x)\| \leq Z + 2C_2 r_*.$$

These follow from Taylor's integral remainder, (8.28) and Lemma 9.1. Thus

$$(9.5) \quad Y_0 + (Z - 1)r_* + C_2 r_*^2 < 0, \quad Z + 2C_2 r_* < 1$$

make the ball invariant and the map a contraction. The fixed-point argument in Lemma 4.1 applies with the quadratic Taylor majorant (9.4) in place of the cubic expansion. The contraction has a unique fixed point; there $AF(x) = 0$. The identity $\tilde{D}A = I_Y$ implies $F(x) = 0$. Every zero in the ball is a fixed point, so uniqueness is uniqueness of the zero of (5.12).

Take $r_* = 10^{-36}$. The bounds in Table 6 give both strict inequalities in (9.5).

9.3. Geometry on the whole ball. Let σ_0, σ_1 be the sums defined after (6.12). For every shape in the ball, the W_1 variation norm bounds both $\sum_j |p_j - p_j^{(0)}|$ and $\sum_j jm|p_j - p_j^{(0)}|$ by r_* . Consequently, on $\mathbb{D}$,

$$(9.6) \quad |p| \geq \rho - \sigma_0 - r_* > 237.2785, \quad \operatorname{Re}(1 + wp'/p) \geq 1 - \frac{\sigma_1 + r_*}{\rho - \sigma_0 - r_*} > 0.64407.$$

Quantity	Enclosed bound
B_{L^2}	< 1.823
B_Y	$< 1.117 \cdot 10^6$
B_{T1}	$< 6.443 \cdot 10^{10}$
B_{T2}	$< 1.187 \cdot 10^{13}$
E_0	$< 9.699 \cdot 10^{-67}$
E_1	$< 2.728 \cdot 10^{-62}$
A_*	$< 1.947 \cdot 10^{22}$
Y_0	$< 1.888 \cdot 10^{-44}$
Z	$< 2.981 \cdot 10^{-39}$
M_2	< 65827.163
C_2	$< 6.406 \cdot 10^{26}$
$Y_0 + (Z - 1)r_* + C_2r_*^2$	$< -9.99999980 \cdot 10^{-37}$
$Z + 2C_2r_*$	$< 1.282 \cdot 10^{-9}$
$4C_2Y_0$	$< 4.836 \cdot 10^{-17}$

TABLE 6. The certificate at $r_* = 10^{-36}$. The last row gives $4C_2Y_0 < (1 - Z)^2$, the discriminant condition for the radii polynomial.

On the larger disc $|w| \leq R$,

$$(9.7) \quad \operatorname{Re} p(w) \geq 2\rho - P - r_* > 235.7747 > 0.$$

For distinct w_1, w_2 inside that disc, the difference quotient

$$\frac{\psi(w_2) - \psi(w_1)}{w_2 - w_1} = \int_0^1 p(w_1 + t(w_2 - w_1)) dt$$

has positive real part, proving injectivity. Thus ψ is univalent and p nonzero on a neighbourhood of $\overline{\mathbb{D}}$; $R > 1.00904$. The boundary is a real-analytic Jordan curve. The tangent-angle calculation (4.7) is a purely conformal identity and applies to the present boundary:

$$(9.8) \quad \kappa(\theta) = \frac{\operatorname{Re}(1 + wp'/p)}{|p|} > 0.$$

Its positive curvature and total turning 2π make the bounded image strictly convex.

For its first nonlinear coefficient,

$$(9.9) \quad |c_1| \geq |c_1^{(0)}| - \frac{r_*}{2(m+1)^2} > 0.0038627.$$

Indeed the $j = 1$ term in $\|p - p_0\|_{W_1}$ is $2(m+1)|p_1 - p_1^{(0)}|$, and $c_1 = p_1/(m+1)$. The real coefficient and frequency restrictions give

$$\psi(e^{2\pi i/m}w) = e^{2\pi i/m}\psi(w), \quad \psi(\bar{w}) = \overline{\psi(w)}.$$

Thus Ω has D_m symmetry. A disc invariant under this rotation would be centred at zero, and its normalised Riemann map would be linear, contradicting (9.9). Finally $|\psi'(0) - \rho| \leq r_*$ and the exact centre coefficient give $237.97 < \psi'(0) < 237.98$.

9.4. The field and its signs. At the zero, $U = \mathcal{B}(|p|) + Kg$ solves (5.10) distributionally. Initially U is continuous, with the Cauchy traces given by the lift and Lemma 2.2. The coefficient $|p|^2$ is real analytic near $\overline{\mathbb{D}}$. Dirichlet elliptic regularity upgrades U to a smooth

solution on the closed disc, and the Cauchy traces are classical. With $u = U \circ \psi^{-1}$, conformal covariance and the normal transformation give exactly (5.1).

At the origin the finite field has value a_0/s_0 . Here s_0 is the positive field divisor from (5.13), while the shape sums are σ_0, σ_1 . The lift and ball bounds give

$$(9.10) \quad U(0) \geq \frac{a_0}{s_0} - L_0 - r_*/4 - B_b J_b r_*/2 > 19.3341 > 0.$$

For any boundary point x , smoothness and the outward derivative one give $u(x - t\nu(x)) = -t + O(t^2) < 0$ for small positive t . Thus u changes sign. This completes the proof of Theorem 5.1 and, by the Green identity of Subsection 5.1, its arc-length Fourier formulation.

9.5. Relation with earlier work.

9.5.1. Sign, spectral sequences and low eigenvalues. Serrin's symmetry theorem [60], in the form recorded in [19, Theorem 2], concerns a bounded connected open set Ω with C^2 boundary and $u \in C^2(\overline{\Omega})$ satisfying $\Delta u + \lambda u + \varepsilon = 0$, $\lambda > 0$, $\varepsilon \in \{0, \pm 1\}$, and $u = 0$, $\partial_\nu u = c$ on $\partial\Omega$: if $u > 0$ in Ω , then Ω is a ball. Changing the sign of the solution gives the negative case. The field in (9.10) has both signs. Berenstein's planar spectral-sequence theorem [5], as stated in [18, Proposition 1.1] for simply connected bounded planar domains with $C^{2,\varepsilon}$ boundary and in [9, §0], forces a disc when the Dirichlet overdetermined problem is solvable for infinitely many distinct spectral parameters. Berenstein and Yang [9, Theorem 1] give the corresponding result for bounded Lipschitz domains with connected boundary in all dimensions, for either the Neumann or Dirichlet overdetermined problem. The present construction supplies one parameter. These sequence results imply that the domain of Theorem 5.1 has only finitely many overdetermined Dirichlet eigenvalues.

Dalmasso [19, Theorem 1] proves that a bounded connected domain with C^1 boundary is a ball if $\Delta u + \lambda u = 0$ has a nonconstant solution $u \in C^2(\overline{\Omega})$ with constant Dirichlet and Neumann data and $\lambda \leq \lambda_1$. If the domain is also convex and symmetric about a hyperplane, the same conclusion holds for $\lambda \leq \lambda_2$. The geometric and symmetry hypotheses of the latter statement hold here. Its spectral inequality does not: the trial functions $1 - r^2$ and $r^n(1 - r^2)\cos n\theta$, $r^n(1 - r^2)\sin n\theta$, $1 \leq n \leq 6$, span a thirteen-dimensional zero-Dirichlet space on the disc. As in (36.5), their Rayleigh quotients are $2(n+1)(n+3) \leq 126$. Conformal invariance of energy and $|p| > 237.2785$ therefore give

$$(9.11) \quad \lambda_2(\Omega) \leq \lambda_{13}(\Omega) \leq 126/(237.2785)^2 < 1.$$

For the finite centre, comparison with the inscribed and circumscribed origin-centred discs, using binary64 Bessel zeros, shows that between 14037 and 14040 Dirichlet eigenvalues do not exceed one. The second-Dirichlet-eigenvalue case of the Schiffer problem is recorded in [9, §0] as a consequence of the Payne–Weinberger isoperimetric inequality and attributed to Berenstein [5] in [19, Remark 1]. The corresponding application of Dai, Liu and Sun [23] is quoted in [25, §1]. These statements concern the Schiffer boundary data.

Dalmasso's finiteness theorem [18, Theorem 1.1] concerns $\Delta u + \lambda u + \mu = 0$ with $u = 0$, $\partial_\nu u = c \neq 0$. For a bounded convex planar domain with smooth positive-curvature boundary, the Schiffer property and nonconstant width, it allows at most finitely many pairs (λ, μ) at fixed c . It is a finiteness statement and permits the single pair $(1, 0)$. Liu's criterion [50], as recorded in [15, 25], involves constancy of the second interior normal derivative. Here the boundary decomposition of the Laplacian gives directly

$$(9.12) \quad u_{\nu\nu} = -\kappa u_\nu = -\kappa \quad \text{on } \partial\Omega :$$

the tangential second derivative of the zero boundary trace vanishes, and $\Delta u = -u = 0$ there. Constant curvature would make the boundary a circle, so (9.9) also excludes constancy of $u_{\nu\nu}$.

9.5.2. Dai–Sun–Wei–Zhang and the numerical remainder comparison. Dai, Sun, Wei and Zhang [25, Theorem 1.1, version 2] prove Dirichlet rigidity above a threshold $\alpha_0(\Omega)$ for bounded uniformly convex planar domains with connected $C^{2,\varepsilon}$ boundary, $0 < \varepsilon < 1$. The geometric hypotheses hold here. The threshold depends on the full geometry of the fixed domain. In support coordinates in Steiner position, let $h(\theta)$ be the support function and $\varrho = h + h''$ the radius of curvature. Their amplitude defect is

$$(9.13) \quad \mathfrak{B}(\Omega) = \max_{\theta, v \in \{-1, 1\}} \left| \sqrt{\varrho(\theta)} e^{vh'(\theta)} - \sqrt{\varrho(\theta + \pi)} e^{-vh'(\theta + \pi)} \right|.$$

This is the quantity $\mathcal{B}(\Omega)$ of (36.1).

For a controlled C^5 class, put

$$I_D(\lambda, v, \theta) = \int_{\partial\Omega} e^{v\eta_\perp \cdot x + i\lambda\xi \cdot x} ds, \quad \xi = (\cos \theta, \sin \theta), \quad \eta_\perp \perp \xi.$$

Its two-point leading term is

$$(9.14) \quad \sqrt{\frac{2\pi}{\lambda}} \left[\sqrt{\varrho(\theta)} e^{vh'(\theta)} e^{i\lambda h(\theta) - i\pi/4} + \sqrt{\varrho(\theta + \pi)} e^{-vh'(\theta + \pi)} e^{-i\lambda h(\theta + \pi) + i\pi/4} \right].$$

Writing R_D for the difference, their expansion gives $|R_D| \leq C_* \lambda^{-3/2}$ for $\lambda \geq 3$, $|v| \leq 1$. The constant C_* depends on the class and is not supplied numerically. Their exclusion lemma, for $\mathfrak{B} \geq \delta > 0$, has threshold

$$(9.15) \quad \alpha > \left[\max \left\{ 3, \frac{C_*}{\sqrt{2\pi\delta}} \right\} \right]^2 - 1.$$

The qualitative theorem chooses $\delta = \mathfrak{B}/2$.

The following comparison uses floating-point quadrature for the finite centre. Let Ω_1 be its image rescaled to conformal radius one; its parameter is $\alpha_u = \rho^2 \approx 56632.65$. Any constant admissible for its stationary-phase expansion must dominate every evaluated remainder. In the unit scale define

$$G_u(\zeta) = \zeta^{3/2} \max_{\theta} |R_D^{\Omega_1}(\zeta, 0, \theta)|.$$

The computed values in Table 7 use trapezoidal integration in the conformal angle and the support coordinates of the two stationary points.

ζ	Sampled $G_u(\zeta)$
5000	3953
23309.55191881634	7096
100000	8467

TABLE 7. Numerical stationary-phase remainders for the normalised Berenstein centre in binary64 arithmetic. Each sampled angular maximum estimates a lower bound for $G_u(\zeta)$.

The comparison also includes dilation. For $s\Omega_1$ the parameter is α_u/s^2 , and

$$R_D^{s\Omega_1}(\lambda, 0, \theta) = s R_D^{\Omega_1}(s\lambda, 0, \theta), \quad \mathfrak{B}(s\Omega_1) = \sqrt{s} \mathfrak{B}_u(s),$$

where $\mathfrak{B}_u(s) = \max_{\theta, v=\pm 1} |\sqrt{\varrho} e^{vsh'} - \sqrt{\varrho(\theta + \pi)} e^{-vsh'(\theta + \pi)}|$ uses the unit-scale support data. An admissible C_* for a class containing $s\Omega_1$ satisfies

$$C_* \geq \lambda^{3/2} |R_D^{s\Omega_1}(\lambda', 0, \theta)|, \quad \lambda' \geq 3.$$

Taking the angular maximum gives $C_* \geq s^{-1/2} G_u(s\lambda')$. With the most favourable choice $\delta = \mathfrak{B}(s\Omega_1)$, exclusion of α_u/s^2 by (9.15) requires $\lambda_s = \sqrt{\alpha_u/s^2 + 1} > 3$ and

$$C_* < \sqrt{2\pi} \lambda_s \mathfrak{B}(s\Omega_1) = \sqrt{2\pi(\alpha_u + s^2)} s^{-1/2} \mathfrak{B}_u(s).$$

Hence exclusion requires

$$(9.16) \quad G_u(\zeta) < T_D(s) := \sqrt{2\pi(\alpha_u + s^2)} \mathfrak{B}_u(s) \quad \text{for every } \zeta \geq 3s.$$

For $s > \sqrt{\alpha_u/8}$ the parameter is below eight and cannot exceed (9.15). On the remaining scales $0 < s \leq \sqrt{\alpha_u/8} \approx 84.137$, the numerical calculation gives $T_D(s)$ approximately in the range 92.6–228.8. The sampled remainder at the largest frequency in Table 7 is approximately 8467, about 37 times the largest computed threshold. In this computation the condition (9.16) fails at every admissible scale, so the stationary-phase exclusion does not remove the finite centre.

Version 1 of [25] required central symmetry. The present domain is not centrally symmetric. Any centre of symmetry would be fixed by its nontrivial rotational symmetry and hence be the origin. Then normalised Riemann-map uniqueness would imply $\psi(-w) = -\psi(w)$. Since $m = 77$ is odd, its nonzero coefficient c_1 multiplies the even power w^{m+1} , which contradicts that identity.

9.5.3. The planar counterexample of Colbrook–Sadeghi–Stepanians. Colbrook, Sadeghi and Stepanians [15] proved existence on a bounded simply connected non-disc domain with real-analytic Jordan boundary, D_{13} symmetry and a sign-changing eigenfunction. For their numerical centre $\varphi^\circ(w) = w + \sum_{j=1}^{20} \varphi_j^\circ w^{13j+1}$, the binary64 evaluation of its appendix coefficients in the present paper gives

$$(9.17) \quad \min_{|w|=1} \operatorname{Re}(1 + w(\varphi^\circ)''/(\varphi^\circ)') \approx -2.5506, \quad \min \kappa^\circ \approx -2.968.$$

Here κ° is the curvature of $\varphi^\circ(\partial\mathbb{D})$ in the normalisation $(\varphi^\circ)'(0) = 1$. The centre curve is not convex: one of its points lies at distance about $1.0 \cdot 10^{-2}$ inside the boundary of its convex hull. The exact boundary of [15] differs from the centre curve pointwise by less than $4.67 \cdot 10^{-10}$ [15, Subsection 5.1], so their domain is not convex. The interval triangle witness is given in `anc/diagnostics/berenstein/csb_nonconvex.json`; its construction is described in Subsection 34.2. Theorem 5.1 supplies a convex domain. The conformal fixed-disc method is due to [14], and [15] adapted it to the Dirichlet problem. The coupled inverse here uses the Dirichlet trace of a Robin solution to recover the shape, together with the explicit residual Neumann lift (8.20).

Part 3. Planar reductions: convex counterexamples in dimensions four, six, eight, ten and fourteen

10. REDUCTION TO A PLANAR PROBLEM

Sections 10–18 contain the proof for $n = 4$; the proof for $n = 6$ is given in Section 19. Write points of the meridian plane as $(x_1, y) \in \mathbb{R}^2$, identified with $z = x_1 + iy \in \mathbb{C}$, and let $\Pi: \mathbb{R}^4 \rightarrow \mathbb{R}^2$, $\Pi(x_1, x') = (x_1, |x'|)$. The map Π is continuous and its image is the closed half-plane $\{y \geq 0\}$.

Lemma 10.1. *Let f be a C^2 function on an open subset of $\{y > 0\}$ and $F(x_1, y) = yf(x_1, y)$. Then*

$$F_{x_1x_1} + F_{yy} = y \left(f_{x_1x_1} + f_{yy} + \frac{2}{y} f_y \right).$$

The expression in parentheses, evaluated at $(x_1, |x'|)$, is the Laplacian of the function $x \mapsto f(x_1, |x'|)$ at every point with $x' \neq 0$.

Proof. $(yf)_{yy} = yf_{yy} + 2f_y$. The second statement is the formula $\partial_y^2 + \frac{2}{y}\partial_y$ for the Laplacian of $\mathbb{R}^3$ acting on radial functions. $\square$

Lemma 10.2. *Let $N \subset \mathbb{R}^2$ be open and symmetric under $(x_1, y) \mapsto (x_1, -y)$, and let F be real analytic on N and odd in y . Then there is a real-analytic function G , defined on an open set containing $\{(x_1, y^2) : (x_1, y) \in N\}$, such that $F(x_1, y) = yG(x_1, y^2)$ on N . Consequently $x \mapsto G(x_1, |x'|^2)$ is real analytic on the open set $\Pi^{-1}(N)$, and it equals $F(x_1, |x'|)/|x'|$ where $x' \neq 0$.*

Proof. For $t > 0$ with $(x_1, \sqrt{t}) \in N$ put $G(x_1, t) = F(x_1, \sqrt{t})/\sqrt{t}$; this is real analytic. Let $(a, 0) \in N$. For some $\delta > 0$, $F(x_1, y) = \sum_{i,k \geq 0} c_{ik}(x_1 - a)^i y^k$ converges absolutely for $|x_1 - a| < \delta$, $|y| < \delta$, and $c_{ik} = 0$ for even k by oddness. Then $G(x_1, t) = \sum_{i,j} c_{i,2j+1}(x_1 - a)^i t^j$ converges for $|x_1 - a| < \delta$, $|t| < \delta^2$ and agrees with the previous definition where $0 < t < \delta^2$. Two such local definitions agree on their overlap, which is a connected product of intervals containing points with $t > 0$. This defines G . The last assertion follows because $x \mapsto (x_1, |x'|^2)$ is polynomial. $\square$

Proposition 10.3 (From the plane to $\mathbb{R}^4$). *Let $D \subset \mathbb{R}^2$ be a bounded domain, symmetric under $y \mapsto -y$, whose boundary is a Jordan curve meeting the axis $\{y = 0\}$ in exactly two points. Let F be real analytic on an open set $N \supset \overline{D}$ symmetric under $y \mapsto -y$, odd in y , and such that*

$$(10.1) \quad \Delta F + F = 0 \text{ in } D, \quad F = y \text{ and } \nabla F = e_y \text{ on } \partial D.$$

Put $\Omega = \Pi^{-1}(D)$ and let u be the function $x \mapsto G(x_1, |x'|^2)$ of Lemma 10.2. Then Ω is a bounded open set, u is real analytic on the neighbourhood $\Pi^{-1}(N)$ of $\overline{\Omega}$, u is nonconstant, and (1.1) holds.

Proof. Ω is open and bounded because Π is continuous and proper. By continuity $\overline{\Omega} \subset \Pi^{-1}(\overline{D}) \subset \Pi^{-1}(N)$, and $\partial\Omega \subset \Pi^{-1}(\partial D)$, since a point $x \in \overline{\Omega}$ with $\Pi(x) \in D$ lies in the open set Ω .

Off the axis $\{x' = 0\}$, $u(x) = f(x_1, |x'|)$ with $f = F/y$, so Lemma 10.1 gives $|x'|(\Delta u + u)(x) = (\Delta F + F)(\Pi(x)) = 0$ in $\Omega \setminus \{x' = 0\}$. By continuity of $\Delta u + u$ the equation holds in Ω .

Let $x \in \partial\Omega$ with $y = |x'| > 0$. Then $\Pi(x) \in \partial D$, so $f = F/y = 1$ there. Since $\nabla F = e_y$ on ∂D , we have $f_{x_1} = F_{x_1}/y = 0$ and $f_y = (F_y - f)/y = 0$, hence $\nabla u(x) = (f_{x_1}, f_y x'/|x'|) = 0$. The set $\partial\Omega \cap \{x' \neq 0\} = \Pi^{-1}(\partial D \cap \{y > 0\})$ is dense in $\partial\Omega$, since ∂D meets the axis in only two points and each of them is a limit of points of $\partial D \cap \{y > 0\}$. By continuity $u = 1$ and $\nabla u = 0$ on all of $\partial\Omega$. Finally, a constant solution of $\Delta u + u = 0$ vanishes identically, which contradicts $u = 1$ on $\partial\Omega$. $\square$

The reduction is reversible in the analytic class, which explains why nothing is lost by working in the plane.

Proposition 10.4 (From $\mathbb{R}^4$ to the plane). *Let $\Omega \subset \mathbb{R}^4$ be a bounded domain invariant under $O(3)$ acting on x' , and let u be real analytic on a neighbourhood of $\overline{\Omega}$ and solve (1.1). Put $D = \{(x_1, y) : (x_1, |y|, 0, 0) \in \Omega\}$ and $\tilde{u}(x) = \int_{O(3)} u(x_1, Qx') dQ$ (Haar probability measure).*

Then $\tilde{u}$ also solves (1.1), it has the form $\tilde{u}(x) = f(x_1, |x'|)$ with f real analytic and even in y , and $F = yf$ solves (10.1).

Proof. The Laplacian commutes with rotations, and the boundary conditions are rotation invariant; hence $\tilde{u}$ solves (1.1). The function $f(x_1, y) = \tilde{u}(x_1, y, 0, 0)$ is real analytic and even in y (use the rotation $x' \mapsto -x'$), and $\tilde{u}(x) = f(x_1, |x'|)$ by invariance. Lemma 10.1 gives $\Delta F + F = 0$ in $D \cap \{y \neq 0\}$, and by analyticity in D . On $\partial D \cap \{y > 0\}$ we have $f = 1$ and $\nabla f = 0$, hence $F = y$ and $\nabla F = fe_y + y\nabla f = e_y$; continuity extends this to the points of ∂D on the axis. $\square$

Remark 10.5 (Splittings without potential). For a function on $\mathbb{R}^d$ that depends only on $y = |\xi|$, $\xi \in \mathbb{R}^d$, the Laplacian is $\partial_y^2 + (d-1)y^{-1}\partial_y$, and for $F = y^{(d-1)/2}f$ one has

$$\partial_y^2 F = y^{(d-1)/2} \left(f_{yy} + \frac{d-1}{y} f_y \right) + \frac{(d-1)(d-3)}{4} \frac{F}{y^2}.$$

Let $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ and let $u(x) = f(y_1, y_2)$ be invariant under $O(d_1) \times O(d_2)$, where y_i is the modulus of the i -th component of x (for $d_i = 1$ one may use the coordinate itself). Then $F = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2} f$ satisfies

$$F_{y_1 y_1} + F_{y_2 y_2} = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2} \Delta u + \left(\frac{(d_1-1)(d_1-3)}{4y_1^2} + \frac{(d_2-1)(d_2-3)}{4y_2^2} \right) F.$$

The inverse-square potential vanishes exactly when $d_1, d_2 \in \{1, 3\}$, that is, for $n = d_1 + d_2 \in \{2, 4, 6\}$. In these cases the weight $\varphi = y_1^{(d_1-1)/2} y_2^{(d_2-1)/2}$ is one of the harmonic polynomials $1, y, y_1 y_2$, and the boundary conditions $u = 1, \nabla u = 0$ become $F = \varphi, \nabla F = \nabla \varphi$. For $n = 4$ ($d_1 = 1, d_2 = 3$) this is Lemma 10.1, and for $n = 6$ ($d_1 = d_2 = 3$) it is Lemma 19.1 below. The conformal method uses in an essential way that the planar operator is the Laplacian, whose conformal pull-back is a multiple of the Laplacian. For $n = 3, 5, 7$ every splitting into two blocks contains a block of dimension different from 1 and 3, and the meridian problem contains a first-order drift term, or equivalently an inverse-square potential; a different fixed-domain formulation is needed, and it is given in Section 21 for $n = 3$, in Section 22 for $n = 5$ and in Section 23 for $n = 7$. Symmetry classes of adjoint type, which give in addition the dimensions 8, 10 and 14, are treated in Section 20.

Let $x_1 + iy = re^{i\phi}$ be planar polar coordinates, and let U_ℓ denote the Chebyshev polynomial of the second kind, so that $U_\ell(\cos \phi)$ is the zonal spherical harmonic of degree ℓ on S^3 . Under $u = F/y$, the planar Helmholtz solution $F = J_m(r) \sin m\phi$, with m odd, corresponds to the four-dimensional Helmholtz solution

$$u = r^{-1} J_m(r) U_{m-1}(\cos \phi).$$

For instance, $F = J_1(r) \sin \phi$ gives the radial solution $r^{-1} J_1(r)$, whose derivative is $-r^{-1} J_2(r)$; this is why the Schiffer balls of $\mathbb{R}^4$ at eigenvalue one have radii $j_{2,k}$, the positive zeros of J_2 .

11. THE CONFORMAL FIXED-DISC FORMULATION

Let $\mathbb{D} = \{w \in \mathbb{C} : |w| < 1\}$, $w = re^{i\theta}$. The following lemma pulls (10.1) back to the disc; it is the analogue of the reformulation in [14].

Lemma 11.1. *Let ψ be holomorphic and injective on a neighbourhood of $\overline{\mathbb{D}}$, with $\psi' \neq 0$ there, and let $D = \psi(\mathbb{D})$. Let $V \in C^1(\overline{\mathbb{D}})$ be real valued and satisfy*

$$(11.1) \quad \Delta V + |\psi'|^2 V = 0 \text{ in } \mathcal{D}'(\mathbb{D}), \quad V - \operatorname{Im} \psi = 0 \text{ and } \nabla(V - \operatorname{Im} \psi) = 0 \text{ on } \partial \mathbb{D}.$$

Then $F = V \circ \psi^{-1} \in C^1(\overline{D})$ satisfies $\int_D (\nabla F \cdot \nabla \varphi - F \varphi) dA = 0$ for all $\varphi \in C_c^\infty(D)$, and $F = y$, $\nabla F = e_y$ on ∂D . If moreover ψ is odd with real Taylor coefficients, $V(\overline{w}) = -V(w)$ and $V(-\overline{w}) = V(w)$, then D is symmetric under both coordinate reflections, F is odd in y and even in x_1 .

Proof. For $\varphi \in C_c^\infty(D)$, the function $\varphi \circ \psi$ belongs to $C_c^\infty(\mathbb{D})$. Conformal invariance of the Dirichlet integral and the Jacobian $|\psi'|^2$ give

$$\int_D \nabla F \cdot \nabla \varphi dA = \int_{\mathbb{D}} \nabla V \cdot \nabla (\varphi \circ \psi) dA = \int_{\mathbb{D}} |\psi'|^2 V(\varphi \circ \psi) dA = \int_D F \varphi dA,$$

where the middle equality is the distributional equation for $V \in C^1$. The function $V - \text{Im } \psi = (F - y) \circ \psi$ has zero value and gradient on $\partial \mathbb{D}$; since the Jacobian matrix of ψ is invertible on $\partial \mathbb{D}$, the chain rule gives $F - y = 0$ and $\nabla(F - y) = 0$ on ∂D . If ψ is odd with real coefficients, then $\psi(\overline{w}) = \overline{\psi(w)}$ and $\psi(-\overline{w}) = -\overline{\psi(w)}$, which gives the symmetries of D and, with the stated parities of V , those of F . $\square$

We shall look for V of the form $V = Kg + \text{Im } \psi$, where g is an unknown function on $\mathbb{D}$ and K is an inverse of the Laplacian with zero Cauchy data on $\partial \mathbb{D}$ (Section 13). Since $\text{Im } \psi$ is harmonic, (11.1) becomes $g + |\psi'|^2(Kg + \text{Im } \psi) = 0$, which is (1.9). We normalise the eigenvalue to one; the size of D is then an unknown, encoded in the coefficient c_1 of ψ . The symmetry class removes the degeneracies of the problem: the scaling is fixed by the eigenvalue, translations along the axis are excluded by the symmetry $x_1 \mapsto -x_1$, and the Möbius automorphisms of the disc are excluded by requiring $\psi(0) = 0$ with real Taylor coefficients.

12. DISK POLYNOMIALS AND COEFFICIENT SPACES

12.1. Disk polynomials. Write $w = re^{i\theta}$ and $t = r^2$. For $m \in \mathbb{Z}$ and an integer $s \geq 0$ let

$$(12.1) \quad \Phi_{m,s}(w) = r^{|m|} P_s^{(0,|m|)}(2r^2 - 1) e^{im\theta},$$

where $P_s^{(\alpha,\beta)}$ is the Jacobi polynomial in the standard normalisation, so that $P_s^{(0,n)}(1) = 1$ and $\Phi_{m,s} = e^{im\theta}$ on $\partial \mathbb{D}$. Explicitly,

$$(12.2) \quad P_s^{(0,n)}(2t - 1) = \sum_{k=0}^s a_{s,k}^{(n)} t^k, \quad a_{s,k}^{(n)} = (-1)^{s-k} \frac{(n+s+k)!}{k!(s-k)!(n+k)!},$$

and, equivalently, $P_s^{(0,n)}(2t - 1) = \frac{1}{s!} t^{-n} \frac{d^s}{dt^s} \{t^{n+s}(t-1)^s\}$. We set $a_{s,k}^{(n)} = 0$ for $k \notin \{0, \dots, s\}$ and $\Phi_{m,-1} = 0$. Each $\Phi_{m,s}$ is a polynomial in $w, \overline{w}$ of degree $|m| + 2s$; these are the disk (Zernike) polynomials [47]. Complex conjugation gives $\overline{\Phi_{m,s}} = \Phi_{-m,s}$.

Lemma 12.1. *The disk polynomials have the following properties.*

- (a) For $n, s \geq 0$, the polynomial $P_s^{(0,n)}(2t - 1)$ is orthogonal in $L^2([0, 1], t^n dt)$ to all polynomials of degree less than s .
- (b) The functions $\Phi_{m,s}$ are pairwise orthogonal in $L^2(\mathbb{D})$, and $\int_{\mathbb{D}} |\Phi_{m,s}|^2 dA = \pi/(|m| + 2s + 1)$.
- (c) $|\Phi_{m,s}(w)| \leq 1$ for $|w| \leq 1$.

Proof. (a) Integrate the Rodrigues formula by parts s times against $t^n \cdot t^i$, $i < s$; all boundary terms vanish because the derivatives of order less than s of $t^{n+s}(t-1)^s$ vanish at $t = 0$ and $t = 1$.

(b) Angular integration separates different m . For equal m with $n = |m|$, $\int_{\mathbb{D}} \Phi_{m,s} \overline{\Phi_{m,s'}} dA = \pi \int_0^1 t^n P_s^{(0,n)} P_{s'}^{(0,n)} dt$, which vanishes for $s \neq s'$ by (a). For $s = s'$, by (a) only the leading

coefficient $a_{s,s}^{(n)} = \frac{(n+2s)!}{s!(n+s)!}$ of one factor contributes, and s integrations by parts in the Rodrigues formula give $\int_0^1 t^{n+s} P_s^{(0,n)} dt = \int_0^1 t^{n+s} (1-t)^s dt = \frac{(n+s)! s!}{(n+2s+1)!}$. The product is $1/(n+2s+1)$.

(c) Let $|w| \leq 1$, $h = \sqrt{1-|w|^2}$, and consider the unitary matrix $U = \begin{pmatrix} w & -h \\ h & \bar{w} \end{pmatrix}$, acting on the space of homogeneous polynomials of degree $N = n+2s$ in two variables e_1, e_2 , with the $U(2)$ -invariant inner product in which the monomials are orthogonal. The induced operator is unitary, so its diagonal matrix coefficients with respect to the normalised monomials have modulus at most one. The coefficient of $e_1^{n+s} e_2^s$ in $(we_1 + he_2)^{n+s} (-he_1 + \bar{w}e_2)^s$ is

$$\sum_{k=0}^s \binom{n+s}{k} \binom{s}{k} w^{n+s-k} \bar{w}^{s-k} (-h^2)^k = e^{in\theta} r^n \sum_{k=0}^s \binom{n+s}{k} \binom{s}{k} t^{s-k} (t-1)^k.$$

By Leibniz's rule applied to the Rodrigues formula, the last sum equals $P_s^{(0,n)}(2t-1)$. This proves (c) for $m = n \geq 0$; conjugation gives $m < 0$. $\square$

Fix $\rho > 1$ (we shall take $\rho = 21/20$). Let $\mathcal{B}_\rho$ be the space of complex sequences $f = (f_{m,s})_{m \in \mathbb{Z}, s \geq 0}$ with

$$\|f\|_\rho = \sum_{m,s} \rho^{|m|} |f_{m,s}| < \infty.$$

By Lemma 12.1(c), $\sum f_{m,s} \Phi_{m,s}$ converges absolutely and uniformly on $\overline{\mathbb{D}}$, with $\sup_{\overline{\mathbb{D}}} |f| \leq \|f\|_\rho$; by Lemma 12.1(b) the coefficients are determined by the function. We identify f with this continuous function. $\mathcal{B}_\rho$ is a Banach space, and every bounded linear operator on it (or between such weighted ℓ^1 spaces) is determined by its values on the basis vectors.

Lemma 12.2 (Multiplication by w and $\bar{w}$). *Let $n = |m|$. Then*

$$(12.3) \quad w \Phi_{m,s} = \begin{cases} \frac{n+s+1}{n+2s+1} \Phi_{m+1,s} + \frac{s}{n+2s+1} \Phi_{m+1,s-1}, & m \geq 0, \\ \frac{n+s}{n+2s+1} \Phi_{m+1,s} + \frac{s+1}{n+2s+1} \Phi_{m+1,s+1}, & m < 0, \end{cases}$$

and $\bar{w} \Phi_{m,s} = \overline{w \Phi_{-m,s}}$. In both cases the coefficients are nonnegative and sum to one. Consequently $\|wf\|_\rho \leq \rho \|f\|_\rho$ and $\|\bar{w}f\|_\rho \leq \rho \|f\|_\rho$, and for every holomorphic $p(w) = \sum_{a \geq 0} p_a w^a$ with $N_\rho(p) := \sum_a \rho^a |p_a| < \infty$,

$$(12.4) \quad \|pf\|_\rho \leq N_\rho(p) \|f\|_\rho, \quad \|\bar{p}f\|_\rho \leq N_\rho(p) \|f\|_\rho.$$

Moreover, if $m \geq 0$ then $w^N \Phi_{m,s}$ is a convex combination of $\Phi_{m+N,s'}$ with $s' \leq s$.

Proof. For $m = n \geq 0$ both sides of the first formula are $r^{n+1} e^{i(n+1)\theta}$ times a polynomial in t , and the claim is the identity $a_{s,k}^{(n)} = \frac{n+s+1}{n+2s+1} a_{s,k}^{(n+1)} + \frac{s}{n+2s+1} a_{s-1,k}^{(n+1)}$. Dividing by $a_{s,k}^{(n+1)}$ (which is nonzero for $0 \leq k \leq s$) and using (12.2), it reduces to $(n+s+1)(n+s+k+1) - s(s-k) = (n+k+1)(n+2s+1)$. For $m = -n < 0$ the claim is $a_{s,k-1}^{(n)} = \frac{n+s}{n+2s+1} a_{s,k}^{(n-1)} + \frac{s+1}{n+2s+1} a_{s+1,k}^{(n-1)}$; dividing by $a_{s,k}^{(n-1)}$ for $0 \leq k \leq s$ it reduces to $(n+s)(s+1-k) - (s+1)(n+s+k) = -k(n+2s+1)$, and the case $k = s+1$ is checked directly. The formula for $\bar{w}$ follows from $\overline{\Phi_{m,s}} = \Phi_{-m,s}$. Since $|m \pm 1| \leq |m| + 1$, the weight grows by at most a factor ρ , which gives the bounds for w and $\bar{w}$; (12.4) follows by iteration and absolute summation. The last assertion follows from the case $m \geq 0$ of (12.3). $\square$

Lemma 12.3 (Radial index under monomials). *Let $\alpha, \beta \geq 0$, $m \in \mathbb{Z}$, $s \geq 0$. Then $w^\alpha \bar{w}^\beta \Phi_{m,s}$ is a finite linear combination of $\Phi_{m+\alpha-\beta,s'}$ with $s' \geq s - \max(\alpha, \beta)$.*

Proof. Put $m' = m + \alpha - \beta$. The product equals $r^{\alpha+\beta+|m|} P_s^{(0,|m|)}(2t-1) e^{im'\theta}$, so, by Lemma 12.1(b) its coefficient on $\Phi_{m',s'}$ is a multiple of

$$\int_0^1 t^{|m|} P_s^{(0,|m|)}(2t-1) t^e P_{s'}^{(0,|m'|)}(2t-1) dt, \quad e = \frac{\alpha + \beta + |m'| - |m|}{2}.$$

Since $|m'| - |m| \equiv \alpha + \beta \pmod{2}$ and $||m'| - |m|| \leq |\alpha - \beta|$, e is an integer with $0 \leq \min(\alpha, \beta) \leq e \leq \max(\alpha, \beta)$. The second factor is a polynomial of degree $e + s'$, so the integral vanishes if $e + s' < s$ by Lemma 12.1(a). $\square$

Lemma 12.4 (Monomials). *Let $\alpha \geq \beta \geq 0$. Then $w^\alpha \bar{w}^\beta = \sum_{s=0}^\beta \kappa_s \Phi_{\alpha-\beta,s}$ with $\kappa_s \geq 0$, $\sum_s \kappa_s = 1$ and $\kappa_0 = \frac{\alpha - \beta + 1}{\alpha + 1}$.*

Proof. $\bar{w}^\beta = \Phi_{-\beta,0}$. Applying (12.3) α times preserves nonnegativity and total mass one, raises the radial index by at most one in each of the first β steps and never afterwards. The harmonic coefficient is the L^2 projection onto $r^{\alpha-\beta} e^{i(\alpha-\beta)\theta}$, namely $\int_0^1 r^{2\alpha+1} dr / \int_0^1 r^{2(\alpha-\beta)+1} dr$. $\square$

12.2. The sine sector. For odd $n \geq 1$ and $s \geq 0$ let

$$S_{n,s} = \text{Im } \Phi_{n,s} = r^n P_s^{(0,n)}(2r^2 - 1) \sin n\theta = \frac{\Phi_{n,s} - \Phi_{-n,s}}{2i}.$$

Let $\mathcal{S}$ be the real Banach space of $f = \sum_{n \text{ odd}, s \geq 0} f_{n,s} S_{n,s}$ with $\|f\|_\rho = \sum \rho^n |f_{n,s}| < \infty$; this is also its norm as an element of $\mathcal{B}_\rho$. The elements of $\mathcal{S}$ are real-valued, odd under $w \mapsto \bar{w}$ and even under $w \mapsto -\bar{w}$. Conversely, since $\Phi_{m,s}(\bar{w}) = \Phi_{-m,s}(w)$ and $\Phi_{m,s}(-\bar{w}) = (-1)^m \Phi_{-m,s}(w)$, every real-valued $f \in \mathcal{B}_\rho$ with these two symmetries belongs to $\mathcal{S}$. The functions $S_{n,0} = r^n \sin n\theta$ are harmonic; we call $f_{n,0}$ the harmonic coefficients and put

$$\mathcal{G} = \{g \in \mathcal{S} : g_{n,0} = 0 \text{ for all } n\}.$$

13. THE COMPATIBLE INVERSE OF THE LAPLACIAN

For odd $n \geq 1$ and $s \geq 1$ put $q = n + 2s$ and define

$$(13.1) \quad KS_{n,s} = \frac{S_{n,s-1}}{4q(q+1)} - \frac{S_{n,s}}{2q(q+2)} + \frac{S_{n,s+1}}{4(q+1)(q+2)}.$$

Lemma 13.1. *For odd $n \geq 1$ and $s \geq 1$:*

- (a) $\Delta KS_{n,s} = S_{n,s}$;
- (b) $KS_{n,s} = 0$ and $\nabla KS_{n,s} = 0$ on $\partial\mathbb{D}$;
- (c) $KS_{n,s} = \frac{(1-r^2)^2}{4s(s+1)} r^n P_{s-1}^{(2,n)}(2r^2 - 1) \sin n\theta$;
- (d) *the absolute sum of the coefficients in (13.1) is $1/(q(q+2))$. Hence K extends to a bounded operator $K: \mathcal{G} \rightarrow \mathcal{S}$ with $\|K\| = 1/15$.*

Proof. Write A, B, C for the three coefficients in (13.1) and $a_{s,k} = a_{s,k}^{(n)}$.

(a) Since $\Delta(r^{n+2k} \sin n\theta) = 4k(n+k)r^{n+2k-2} \sin n\theta$, the claim is equivalent to $Aa_{s-1,k+1} + Ba_{s,k+1} + Ca_{s+1,k+1} = a_{s,k}/(4(k+1)(n+k+1))$ for $0 \leq k \leq s$. By (12.2) the three ratios $a_{s-1,k+1}/a_{s,k}$, $a_{s,k+1}/a_{s,k}$, $a_{s+1,k+1}/a_{s,k}$ equal $(s-k)(s-k-1)$, $-(n+s+k+1)(s-k)$ and $(n+s+k+2)(n+s+k+1)$, each divided by $(k+1)(n+k+1)$, and a direct computation gives $A(s-k)(s-k-1) - B(n+s+k+1)(s-k) + C(n+s+k+2)(n+s+k+1) = \frac{1}{4}$.

(b) At $r = 1$ each $r^n P_t^{(0,n)}(2r^2 - 1)$ equals one, and $A + B + C = 0$. Its radial derivative at $r = 1$ is $\beta_t = n + 2t(t+n+1)$, because $\frac{d}{dx} P_t^{(0,n)}(1) = t(t+n+1)/2$. Since $\beta_s - \beta_{s-1} = 2q$ and

$\beta_{s+1} - \beta_s = 2(q+2)$, we get $A\beta_{s-1} + B\beta_s + C\beta_{s+1} = -2qA + 2(q+2)C = 0$. The tangential derivative vanishes because the boundary value vanishes identically.

(c) By (b), $KS_{n,s} = r^n R(t) \sin n\theta$ where R is a polynomial of degree $s+1$ with $R(1) = 0$ and $nR(1) + 2R'(1) = 0$, so $R(t) = (1-t)^2 Q(t)$ with $\deg Q = s-1$. As a combination of $P_{s-1}^{(0,n)}, P_s^{(0,n)}, P_{s+1}^{(0,n)}$, the polynomial R is orthogonal in $L^2(t^n dt)$ to polynomials of degree less than $s-1$ (Lemma 12.1(a)); hence Q is orthogonal in $L^2(t^n(1-t)^2 dt)$ to them and is a multiple of $P_{s-1}^{(2,n)}(2t-1)$. Comparing leading coefficients, $Ca_{s+1,s+1}$ for R and $\frac{(n+2s)!}{(s-1)!(n+s+1)!}$ for $P_{s-1}^{(2,n)}(2t-1)$, gives the factor $1/(4s(s+1))$.

(d) $A - B + C = \frac{(q+2)+2(q+1)+q}{4q(q+1)(q+2)} = \frac{1}{q(q+2)}$. The operator K does not change the angular index, hence not the weight. Since $q \geq 3$, the supremum of $1/(q(q+2))$ is $1/15$, attained at $(n, s) = (1, 1)$. $\square$

Lemma 13.2. *For $g \in \mathcal{G}$, the function $W = Kg$ belongs to $C^1(\overline{\mathbb{D}})$, satisfies $\Delta W = g$ in $\mathcal{D}'(\mathbb{D})$, and $W = 0, \nabla W = 0$ on $\partial\mathbb{D}$.*

Proof. Let g_N be the truncations of g to finitely many coefficients and $W_N = Kg_N$, a polynomial with $\Delta W_N = g_N$ and zero Cauchy data by Lemma 13.1. The Dirichlet Green function of the disc is $G(w, \zeta) = \frac{1}{2\pi} \log \left| \frac{w-\zeta}{1-w\bar{\zeta}} \right|$, and $W_N(w) = \int_{\mathbb{D}} G(w, \zeta) g_N(\zeta) dA(\zeta)$. We have $|\nabla_w G(w, \zeta)| \leq \frac{1}{2\pi} (|w-\zeta|^{-1} + |\zeta| |1-w\bar{\zeta}|^{-1})$. For $|w| \leq 1$, $\int_{\mathbb{D}} |w-\zeta|^{-1} dA(\zeta) \leq 4\pi$, since $\mathbb{D}$ lies in the disc of radius 2 about w . If $|w| < 1/2$ then $|\zeta|/|1-w\bar{\zeta}| \leq 2$; if $|w| \geq 1/2$ then $|\zeta|/|1-w\bar{\zeta}| \leq 2/|\zeta - 1/\bar{w}|$ and $\mathbb{D}$ lies in the disc of radius 3 about $1/\bar{w}$; in both cases the integral of this term is at most 12π . Hence

$$\sup_{\mathbb{D}} |\nabla(W_N - W_L)| \leq 8 \sup_{\mathbb{D}} |g_N - g_L| \leq 8 \|g_N - g_L\|_{\rho}, \quad \sup_{\mathbb{D}} |W_N - W_L| \leq \frac{1}{15} \|g_N - g_L\|_{\rho}.$$

Thus (W_N) is Cauchy in $C^1(\overline{\mathbb{D}})$; its limit is Kg (the coefficients converge in $\mathcal{S}$), the zero Cauchy data pass to the limit, and so does the distributional equation. $\square$

Remark 13.3. Conversely, if $W \in C^1(\overline{\mathbb{D}})$ has $\Delta W = g \in \mathcal{S}$ in $\mathcal{D}'(\mathbb{D})$ and zero Cauchy data, then $g \in \mathcal{G}$ and $W = Kg$: the extension of W by zero is C^1 and satisfies $\Delta W = g \mathbf{1}_{\mathbb{D}}$ in $\mathcal{D}'(\mathbb{R}^2)$, and pairing with the harmonic polynomials $r^n \sin n\theta$ gives $g_{n,0} = 0$; then $W - Kg$ is harmonic with zero boundary values. This is why the equation is posed on $\mathcal{G}$.

14. THE CUBIC EQUATION

From now on $\rho = 21/20$, $M = 41$ and $S = 24$. The centre $x^\circ = (g^\circ, c^\circ)$ consists of 504 coefficients $g_{n,s}^\circ$, $n \in \{1, 3, \dots, 41\}$, $1 \leq s \leq 24$, and 21 coefficients c_j° , $j \in \{1, 3, \dots, 41\}$; all other coefficients of x° vanish. Each of these 525 numbers is an explicitly given dyadic rational (a binary64 number interpreted exactly). We write $b = c_1^\circ \approx 14.840043$ and

$$\omega_j = b^2(j+1)\rho^j \quad (j \text{ odd}).$$

Let X be the real Banach space of pairs $x = (g, c)$, with $g \in \mathcal{G}$ and $c = (c_j)_{j \text{ odd} \geq 1}$ real, normed by

$$\|(g, c)\|_X = \|g\|_{\rho} + \sum_{j \text{ odd}} \omega_j |c_j|.$$

To c we associate $\psi_c(w) = \sum_j c_j w^j$, which is holomorphic in $|w| < \rho$ and continuous up to $|w| = \rho$ because $|c_j| \rho^j \leq \omega_j |c_j| / (2b^2)$. Note that $\text{Im } \psi_c = \sum_j c_j S_{j,0} \in \mathcal{S}$. Define

$$(14.1) \quad \mathcal{F}(g, c) = g + |\psi'_c|^2 (Kg + \text{Im } \psi_c).$$

Lemma 14.1. $\mathcal{F}$ is a continuous cubic polynomial map $X \rightarrow \mathcal{S}$. If $\mathcal{F}(g, c) = 0$, then $V = Kg + \text{Im } \psi_c$ belongs to $C^1(\overline{\mathbb{D}})$, is odd under $w \mapsto \bar{w}$ and even under $w \mapsto -\bar{w}$, and satisfies (11.1) with $\psi = \psi_c$.

Proof. $\psi'_c = \sum_j j c_j w^{j-1}$ has real coefficients and even powers, and $N_\rho(\psi'_c) = \sum_j j \rho^{j-1} |c_j| \leq \|c\|/(b^2 \rho)$, where $\|c\| = \sum \omega_j |c_j|$. Thus $|\psi'_c|^2$ is real and invariant under $w \mapsto \bar{w}$ and $w \mapsto -\bar{w}$, and multiplication by it maps $\mathcal{S}$ into $\mathcal{S}$ with norm at most $N_\rho(\psi'_c)^2$ by (12.4). Together with $\|Kg\|_\rho \leq \|g\|_\rho/15$ and $\|\text{Im } \psi_c\|_\rho = \sum \rho^j |c_j|$, this shows that (14.1) is a sum of bounded multilinear maps of degree at most three. If $\mathcal{F}(g, c) = 0$, then Lemma 13.2 gives $V \in C^1(\overline{\mathbb{D}})$, $V - \text{Im } \psi_c = Kg$ with zero Cauchy data, and $\Delta V = g = -|\psi'_c|^2 V$ in $\mathcal{D}'(\mathbb{D})$; the last identity holds pointwise because both sides are uniformly convergent disk-polynomial series with the same coefficients. The parities hold because $V \in \mathcal{S}$. $\square$

14.1. Expansion at the centre. Write $p = \psi'_{c^\circ} = \sum_{a=0,2,\dots,40} p_a w^a$ with $p_a = (a+1)c_{a+1}^\circ$, so $p_0 = b$, and put

$$P = N_\rho(p), \quad V^\circ = Kg^\circ + \text{Im } \psi_{c^\circ}, \quad V_0 = \|V^\circ\|_\rho.$$

For $h = (\delta g, \eta) \in X$ let $\delta V = K\delta g + \text{Im } \psi_\eta$ and, for shape perturbations η, η_1, η_2 ,

$$L(\eta) = \bar{p} \psi'_\eta + p \overline{\psi'_\eta} = 2 \text{Re}(\bar{p} \psi'_\eta), \quad Q(\eta_1, \eta_2) = \text{Re}(\psi'_{\eta_1} \overline{\psi'_{\eta_2}}).$$

Since $|p + \psi'_\eta|^2 = |p|^2 + L(\eta) + Q(\eta, \eta)$, expanding (14.1) gives

$$(14.2) \quad \mathcal{F}(x^\circ + h) = \mathcal{F}(x^\circ) + D\mathcal{F}(x^\circ)h + B_2(h, h) + B_3(h, h, h),$$

where

$$(14.3) \quad \begin{aligned} D\mathcal{F}(x^\circ)h &= \delta g + |p|^2 K\delta g + L(\eta)V^\circ + |p|^2 \text{Im } \psi_\eta, \\ B_2(h_1, h_2) &= \frac{1}{2} \{L(\eta_1) \delta V_2 + L(\eta_2) \delta V_1\} + Q(\eta_1, \eta_2)V^\circ, \\ B_3(h_1, h_2, h_3) &= \frac{1}{3} \{Q(\eta_1, \eta_2) \delta V_3 + Q(\eta_1, \eta_3) \delta V_2 + Q(\eta_2, \eta_3) \delta V_1\}. \end{aligned}$$

There are no terms of higher order.

Lemma 14.2. Let $\kappa = 1/15$, $\vartheta_1 = 1/(b^2 \rho)$ and $\vartheta_0 = 1/(2b^2)$. Then for all $h_i \in X$,

$$\|B_2(h_1, h_2)\|_\rho \leq c_2 \|h_1\|_X \|h_2\|_X, \quad \|B_3(h_1, h_2, h_3)\|_\rho \leq c_3 \|h_1\|_X \|h_2\|_X \|h_3\|_X,$$

with $c_2 = \max\{P\vartheta_1\kappa, 2P\vartheta_1\vartheta_0 + \vartheta_1^2 V_0\}$ and $c_3 = \vartheta_1^2 \max\{\kappa, \vartheta_0\}$.

Proof. Write $a_i = \|\delta g_i\|_\rho$ and $e_i = \sum_j \omega_j |\eta_{i,j}|$. Termwise, $N_\rho(\psi'_{\eta_i}) \leq \vartheta_1 e_i$ (as $j/(j+1) \leq 1$), $\|\text{Im } \psi_{\eta_i}\|_\rho \leq \vartheta_0 e_i$ (as $j+1 \geq 2$), and $\|K\delta g_i\|_\rho \leq \kappa a_i$, so $\|\delta V_i\|_\rho \leq \kappa a_i + \vartheta_0 e_i$. By (12.4), $\|L(\eta_i)f\|_\rho \leq 2P\vartheta_1 e_i \|f\|_\rho$ and $\|Q(\eta_i, \eta_k)f\|_\rho \leq \vartheta_1^2 e_i e_k \|f\|_\rho$. Hence $\|B_2(h_1, h_2)\|_\rho \leq P\vartheta_1 \kappa (e_1 a_2 + e_2 a_1) + (2P\vartheta_1 \vartheta_0 + \vartheta_1^2 V_0) e_1 e_2 \leq c_2 (a_1 + e_1)(a_2 + e_2)$. Similarly every monomial $e_i e_k a_l$ in the bound for B_3 has coefficient $\vartheta_1^2 \kappa/3$ and $e_1 e_2 e_3$ has coefficient $\vartheta_1^2 \vartheta_0$; each is at most its coefficient in $c_3 \prod_i (a_i + e_i)$. $\square$

15. THE CONTRACTION ARGUMENT AND THE APPROXIMATE INVERSE

15.1. A Newton–Kantorovich theorem.

Proposition 15.1. Let $\mathcal{F}: X \rightarrow \mathcal{S}$ have the form (14.2) with symmetric bounded B_2 and B_3 , and let $A: \mathcal{S} \rightarrow X$ be bounded, linear and injective. Suppose that

$$\|A\mathcal{F}(x^\circ)\|_X \leq Y, \quad \|I - A D\mathcal{F}(x^\circ)\|_{X \rightarrow X} \leq Z, \quad \|A\| c_2 \leq C_2, \quad \|A\| c_3 \leq C_3,$$

and that $r > 0$ satisfies

$$(15.1) \quad Y + (Z - 1)r + C_2 r^2 + C_3 r^3 < 0, \quad Z - 1 + 2C_2 r + 3C_3 r^2 < 0.$$

Then $\mathcal{F}$ has exactly one zero in the closed ball $\overline{B}_X(x^\circ, r)$.

Proof. Let $\mathcal{N}(x) = x - A\mathcal{F}(x)$. For $\|h\|_X \leq r$, (14.2) gives $\mathcal{N}(x^\circ + h) - x^\circ = -A\mathcal{F}(x^\circ) + (I - A\mathcal{D}\mathcal{F}(x^\circ))h - AB_2(h, h) - AB_3(h, h, h)$, whose norm is at most $Y + Zr + C_2r^2 + C_3r^3 < r$. For h_1, h_2 in the ball, $B_2(h_1, h_1) - B_2(h_2, h_2) = B_2(h_1 - h_2, h_1 + h_2)$ and $B_3(h_1, h_1, h_1) - B_3(h_2, h_2, h_2) = B_3(h_1 - h_2, h_1, h_1) + B_3(h_2, h_1 - h_2, h_1) + B_3(h_2, h_2, h_1 - h_2)$, so $\mathcal{N}$ is Lipschitz on the ball with constant $Z + 2C_2r + 3C_3r^2 < 1$. By the contraction mapping principle $\mathcal{N}$ has a unique fixed point in the ball, and since A is injective the fixed points of $\mathcal{N}$ are exactly the zeros of $\mathcal{F}$. $\square$

In the weighted ℓ^1 spaces X and $\mathcal{S}$, operator norms are computed column by column.

Lemma 15.2. *Let E and E' be weighted ℓ^1 spaces with bases (e_i) and (e'_i) and weights (λ_i) and (λ'_i) , and let $T: E \rightarrow E'$ be bounded. Then $\|T\| = \sup_i \|Te_i\|_{E'}/\lambda_i$.*

Proof. The inequality $\geq$ follows by testing on e_i/λ_i , and $\leq$ by the triangle inequality and absolute summation, using continuity of T and density of finitely supported vectors. $\square$

We shall apply Lemma 15.2 with the bases $\{e_{(n,s)} : s \geq 1\} \cup \{e_j\}$ of X , of weights ρ^n and ω_j , and $\{S_{n,s} : s \geq 0\}$ of $\mathcal{S}$, of weights ρ^n . We call the basis vectors of X *inputs* and those of $\mathcal{S}$ *rows*.

15.2. Finite block and tail. The *finite inputs* are the 504 coordinates $g_{n,s}$ with $n \leq M$, $1 \leq s \leq S$, and the 21 shape coordinates c_j with $j \leq M$. The *finite rows* are the 504 rows (n, s) with $n \leq M$, $1 \leq s \leq S$, and the 21 harmonic rows $(n, 0)$ with $n \leq M$. All other inputs and rows are *tail* inputs and rows. We write $X = X_f \oplus X_t$ and $\mathcal{S} = \mathcal{S}_f \oplus \mathcal{S}_t$ accordingly; $\dim X_f = \dim \mathcal{S}_f = 525$. Let M_f be the 525×525 matrix of the restriction of $D\mathcal{F}(x^\circ)$ to finite inputs and finite rows.

On tail inputs we use the following explicit model of $D\mathcal{F}(x^\circ)$, obtained by replacing ψ° by bw and V° by $b \operatorname{Im} w$:

$$(15.2) \quad T(\delta g, \eta) = \delta g + b^2 \sum_{j > M} \eta_j \{(j+1)S_{j,0} - (j-1)S_{j-2,0} - S_{j-2,1}\}.$$

Lemma 15.3. *For odd $j \geq 3$, $L_b(w^j) b \operatorname{Im} w + b^2 \operatorname{Im} w^j = b^2 \{(j+1)S_{j,0} - (j-1)S_{j-2,0} - S_{j-2,1}\}$, where $L_b(\eta) = 2 \operatorname{Re}(b\psi'_\eta)$.*

Proof. $L_b(w^j) b \operatorname{Im} w = 2b^2 j r^j \cos((j-1)\theta) \sin \theta = b^2 j r^j \{\sin j\theta - \sin(j-2)\theta\}$. Moreover $r^2 \Phi_{j-2,0} = \frac{1}{j} \Phi_{j-2,1} + \frac{j-1}{j} \Phi_{j-2,0}$, because $P_1^{(0,n)}(2t-1) = (n+2)t - (n+1)$ by (12.2); hence $j r^j \sin(j-2)\theta = S_{j-2,1} + (j-1)S_{j-2,0}$. $\square$

Write $Te = (Be, T_{tt}e)$ for tail inputs e , with B the part in finite rows and T_{tt} the part in tail rows. Only the shape input $j = M+2$ reaches finite rows:

$$(15.3) \quad B(\delta g, \eta) = -b^2 \eta_{M+2} ((M+1)e_{(M,0)} + e_{(M,1)}).$$

Put $\tau = (1 - \rho^{-2})^{-1}$.

Lemma 15.4. *$T_{tt}: X_t \rightarrow \mathcal{S}_t$ is a bounded bijection. For a tail residual $y = \sum y_{n,s} S_{n,s} \in \mathcal{S}_t$ its inverse is*

$$(15.4) \quad \eta_n = \sum_{\substack{m \geq n \\ m \text{ odd}}} \frac{y_{m,0}}{b^2(m+1)} \quad (n > M), \quad \delta g_{n,s} = y_{n,s} + \mathbf{1}_{s=1} b^2 \eta_{M+2} \quad ((n, s) \text{ a tail } g\text{-input}).$$

Moreover $\|T_{tt}\| \leq 1 + \rho^{-2}$ and $\|T_{tt}^{-1}\| \leq \max\{1, \tau + \tau/((M+3)\rho^2)\}$.

Proof. By (15.2), the tail harmonic rows read $b^2(n+1)(\eta_n - \eta_{n+2}) = y_{n,0}$ for $n \geq M+2$, the tail rows $(n, 1)$ read $\delta g_{n,1} - b^2\eta_{n+2} = y_{n,1}$, and all other tail rows are $\delta g_{n,s} = y_{n,s}$. Since $\sum \omega_n |\eta_n| < \infty$ forces $\eta_n \rightarrow 0$, summation of the harmonic equations gives (15.4), and a solution with $y = 0$ vanishes; conversely (15.4) solves the equations whenever the series converge. For the bounds we use Lemma 15.2. The shape input e_j has three nonzero entries, $b^2(j+1)$ in row $(j, 0)$, $-b^2(j-1)$ in row $(j-2, 0)$ and $-b^2$ in row $(j-2, 1)$; its weighted norm divided by ω_j is $1 + j\rho^{-2}/(j+1) \leq 1 + \rho^{-2}$. Tail g -inputs are mapped to the identical rows. For T_{tt}^{-1} , a nonharmonic tail row is mapped to the identical input. The normalised harmonic row $\rho^{-m}S_{m,0}$ is mapped to $\eta_n = \rho^{-m}/(b^2(m+1))$ for $M < n \leq m$ and $\delta g_{n-2,1} = b^2\eta_n$ for $M+4 \leq n \leq m$, with norm

$$\sum_{M < n \leq m} \frac{(n+1)\rho^n}{(m+1)\rho^m} + \sum_{M+4 \leq n \leq m} \frac{\rho^{n-2}}{(m+1)\rho^m} \leq \tau + \frac{\tau}{(M+3)\rho^2}. \quad \square$$

15.3. The approximate inverse. Let $\hat{A}$ be a 525×525 matrix with dyadic rational entries (in the computation, the midpoints of an interval enclosure of M_f^{-1}), regarded as an exact linear map $\mathcal{S}_f \rightarrow X_f$, and suppose that

$$(15.5) \quad \|I - \hat{A}M_f\|_{X_f \rightarrow X_f} < 1.$$

Define $A: \mathcal{S} = \mathcal{S}_f \oplus \mathcal{S}_t \rightarrow X = X_f \oplus X_t$ by

$$(15.6) \quad A = \begin{pmatrix} \hat{A} & -\hat{A}BT_{tt}^{-1} \\ 0 & T_{tt}^{-1} \end{pmatrix}.$$

Lemma 15.5. *Under (15.5), A is bounded and injective, and $ATe = e$ for every tail input e .*

Proof. By (15.5) and the Neumann series, $\hat{A}M_f$ is invertible, so $\hat{A}$ is invertible. The operator $\begin{pmatrix} \hat{A}^{-1} & B \\ 0 & T_{tt} \end{pmatrix}$ is a left inverse of A . For a tail input e , $ATe = (\hat{A}Be - \hat{A}BT_{tt}^{-1}T_{tt}e, T_{tt}^{-1}T_{tt}e) = (0, e)$. $\square$

Let $e_{(n,s)}$ denote the finite rows as elements of $\mathcal{S}_f$, and define

$$\|R\| = \max_{(n,s) \text{ finite row}} \frac{\|\hat{A}e_{(n,s)}\|_{X_f}}{\rho^n}, \quad H = \|\hat{A}(e_{(M,1)} + (M+1)e_{(M,0)})\|_{X_f},$$

and, for odd $n \geq M+2$,

$$(15.7) \quad h_n = \frac{H + \sum_{j=M+2, M+4, \dots, n} (j+1)\rho^j + \sum_{j=M+4, M+6, \dots, n} \rho^{j-2}}{(n+1)\rho^n}.$$

Lemma 15.6. *Let $N_H \geq M+2$ be odd. Then A maps every residual supported in tail rows with norm at most*

$$A_t := \max \left\{ 1, \max_{M+2 \leq n \leq N_H} h_n, \tau + \frac{\tau}{(N_H+3)\rho^2} + \frac{H}{(N_H+3)\rho^{N_H+2}} \right\},$$

it is the identity on residuals supported in nonharmonic tail rows, and $\|A\| \leq \max\{\|R\|, A_t\}$.

Proof. We use Lemma 15.2. A finite row $e_{(n,s)}$ is mapped to $\hat{A}e_{(n,s)}$. A nonharmonic tail row is mapped to the identical tail input by (15.4), and B vanishes on it. The normalised harmonic tail row $\rho^{-n}S_{n,0}$ is mapped to the shape coordinates $\eta_j = \rho^{-n}/(b^2(n+1))$, $M+2 \leq j \leq n$, to $\delta g_{j-2,1} = b^2\eta_j$, $M+4 \leq j \leq n$, and, by (15.3), to the finite part $b^2\eta_{M+2}\hat{A}((M+1)e_{(M,0)} + e_{(M,1)})$; the sum of the three norms is exactly h_n . For $n \geq N_H+2$ we bound $(j+1)/(n+1) \leq 1$ in the first sum, sum the geometric series $\sum_{i \geq 0} \rho^{-2i} = \tau$, and use $n+1 \geq N_H+3$ in the other two terms. $\square$

16. THE DEFECT BOUNDS

In this section we explain how the bounds Y and Z of Proposition 15.1 are obtained for the operator A of (15.6). By Lemma 15.2, $\|I - ADF(x^\circ)\|$ is the supremum over all inputs e of $\|(I - ADF(x^\circ))e\|_X$ divided by the weight of e . We partition the inputs into five classes and bound each class separately; every input belongs to exactly one class, and no column is estimated by sampling. Throughout, $d = \deg p = 40$, and $\Lambda = M + d = 81$.

16.1. The residual and the finite columns. The residual $\mathcal{F}(x^\circ)$ is a polynomial in $w, \bar{w}$. Using (12.3) and (13.1), its expansion in the basis $S_{n,s}$ is computed without truncation (it has 2286 coefficients) in interval arithmetic. Only finitely many tail rows occur, so $A\mathcal{F}(x^\circ)$ is computed without truncation from (15.6), (15.3) and (15.4), and Y is an upper bound of its norm.

For each of the 525 finite inputs e , $DF(x^\circ)e$ is again a polynomial by (14.3), and $(I - ADF(x^\circ))e$ is computed in the same way. We denote by Z_f an upper bound for the maximum of the normalised norms of these columns.

16.2. Tail g -columns. Let $n_{\max} = M + 2d + 2 = 123$ and $s_{\max} = S + d + 3 = 67$.

Lemma 16.1. *Let $e = e_{(n,s)}$ be a tail g -input and $q = n + 2s$. Then $(I - ADF(x^\circ))e = -A(|p|^2 K S_{n,s})$. Moreover, writing $p = p_{\leq 16} + p_{> 16}$ for the splitting of p into terms of degree ≤ 16 and > 16 , and $P_s = N_\rho(p_{\leq 16})$, $P_t = N_\rho(p_{> 16})$,*

$$(16.1) \quad \frac{\|(I - ADF(x^\circ))e\|_X}{\rho^n} \leq \frac{\|A(|p_{\leq 16}|^2 K S_{n,s})\|_X}{\rho^n} + \frac{\|A\| (2P_s P_t + P_t^2)}{q(q+2)}.$$

Proof. By (14.3), $DF(x^\circ)e = S_{n,s} + |p|^2 K S_{n,s}$, and $Te = S_{n,s}$, so $AS_{n,s} = ATe = e$ by Lemma 15.5. Since $|p|^2 - |p_{\leq 16}|^2 = \overline{p_{\leq 16}} p_{> 16} + \overline{p_{> 16}} p_{\leq 16} + |p_{> 16}|^2$, (12.4) and Lemma 13.1(d) give $\|(|p|^2 - |p_{\leq 16}|^2) K S_{n,s}\|_\rho \leq (2P_s P_t + P_t^2) \rho^n / (q(q+2))$. $\square$

For the $62 \cdot 67 - 21 \cdot 24 = 3650$ tail g -inputs with $n \leq n_{\max}$ and $s \leq s_{\max}$, the first term on the right of (16.1) is computed without truncation; let $Z_{g,\partial}$ bound the maximum of the right-hand side over these inputs.

Lemma 16.2. *For every tail g -input $e_{(n,s)}$ with $n > n_{\max}$ or $s > s_{\max}$,*

$$\frac{\|(I - ADF(x^\circ))e_{(n,s)}\|_X}{\rho^n} \leq Z_{g,\infty} := \max \left\{ \frac{P^2}{137 \cdot 139}, \frac{A_t P^2}{127 \cdot 129} \right\}.$$

Proof. We have $|p|^2 = \sum_{a,a'} p_a p_{a'} w^a \bar{w}^{a'}$ with $a, a' \leq d$. By (13.1) and Lemma 12.3, every component of $|p|^2 K S_{n,s}$ has radial index at least $s - 1 - d$ and angular frequency of modulus at least $n - d$. Also $\||p|^2 K S_{n,s}\|_\rho \leq P^2 \rho^n / (q(q+2))$ with $q = n + 2s$.

If $s > s_{\max}$, all radial indices are at least $s - 1 - d \geq S + 3$, so all components lie in nonharmonic tail rows, on which A is the identity (Lemma 15.6); and $q \geq 1 + 2(s_{\max} + 1) = 137$.

If $s \leq s_{\max}$ and $n > n_{\max}$, all frequencies are at least $n - d \geq n_{\max} + 2 - d > M$, so all components lie in tail rows, on which A has norm at most A_t ; and $q \geq n_{\max} + 2 + 2 = 127$. $\square$

16.3. Shape columns. For the 480 shape inputs e_j with $M < j \leq j_* = 1001$, $(I - ADF(x^\circ))e_j$ is computed without truncation as in Section 16.1; let $Z_{\text{sh},\partial}$ bound their normalised norms. For $j > j_*$ we prove a uniform bound.

Let $W^\circ = Kg^\circ$, so that $V^\circ = W^\circ + \text{Im } \psi_{c^\circ}$. By (14.3) and Lemma 15.3, for odd $j > M$,

$$DF(x^\circ)e_j = 2 \operatorname{Re} \left[w^{j-1} \left(j \bar{p} V^\circ + \frac{|p|^2 w}{2i} \right) \right], \quad Te_j = 2 \operatorname{Re} \left[w^{j-1} \left(j b^2 \operatorname{Im} w + \frac{b^2 w}{2i} \right) \right].$$

Since $ATe_j = e_j$, we get $(I - AD\mathcal{F}(x^\circ))e_j = -A\Delta_j$ with

$$(16.2) \quad \Delta_j = 2 \operatorname{Re} [w^{j-1}(j\Gamma_1 + \Gamma_0)] + 2 \operatorname{Re} [jw^{j-1}\bar{p}W^\circ],$$

where $\Gamma_1 = \bar{p} \operatorname{Im} \psi_{c^\circ} - b^2 \operatorname{Im} w$ and $\Gamma_0 = (|p|^2 - b^2)w/(2i)$. Writing $\operatorname{Im} \psi_{c^\circ} = (\psi_{c^\circ} - \overline{\psi_{c^\circ}})/(2i)$ and $|p|^2 = \sum p_a p_{a'} w^a \bar{w}^{a'}$, and using $p_0 = c_1^\circ = b$,

$$2i\Gamma_1 = \sum_{(a,l) \neq (0,1)} p_a c_l^\circ (w^l \bar{w}^a - \bar{w}^{a+l}), \quad 2i\Gamma_0 = \sum_{(a,a') \neq (0,0)} p_a p_{a'} w^{a+1} \bar{w}^{a'},$$

with $a, a' \in \{0, 2, \dots, 40\}$ and $l \in \{1, 3, \dots, 41\}$. After combining equal monomials, $2i\Gamma_1 = \sum_{\alpha, \beta} q_{\alpha\beta}^1 w^\alpha \bar{w}^\beta$ and $2i\Gamma_0 = \sum_{\alpha, \beta} q_{\alpha\beta}^0 w^\alpha \bar{w}^\beta$ with finitely many real coefficients. The differences $k = \alpha - \beta$ that occur are odd and lie in $[k_-, k_+] = [-81, 41]$. For odd $k \in [k_-, k_+]$ define

$$\begin{aligned} \bar{q}_k &= \sum_{\alpha-\beta=k} q_{\alpha\beta}^1, & \varepsilon_k &= \frac{1}{j_* + 1} \sum_{\alpha-\beta=k} ((1+\beta)|q_{\alpha\beta}^1| + |q_{\alpha\beta}^0|), & \bar{\eta}_k &= \frac{1}{b^2} \sum_{l \geq k} \bar{q}_l, \\ \delta_k &= \frac{1}{b^2} \sum_{l \geq k} \left[\varepsilon_l \left(1 + \frac{|l-1|}{j_* + k_-} \right) + \frac{|\bar{q}_l| |l-1|}{j_* + k_-} \right], & E_k &= |\bar{\eta}_k| + \delta_k, \end{aligned}$$

where the sums over l run over odd $l \leq k_+$. By Lemma 13.1(c), $W^\circ = (1 - |w|^2)^2 Q^\circ$ with

$$Q^\circ = \sum_{n,s} \frac{g_{n,s}^\circ}{4s(s+1)} r^n P_{s-1}^{(2,n)} (2r^2 - 1) \sin n\theta,$$

and Q° has a finite disk-polynomial expansion, which we compute from the connection formulas (they follow from [22, Eq. 18.9.5] and the symmetry $P_k^{(\alpha,\beta)}(-x) = (-1)^k P_k^{(\beta,\alpha)}(x)$, [22, Table 18.6.1])

$$P_k^{(a+1,n)} = \frac{2k+a+n+1}{k+a+n+1} P_k^{(a,n)} + \frac{k+n}{k+a+n+1} P_{k-1}^{(a+1,n)} \quad (a = 0, 1).$$

The function $w^\Lambda \bar{p}Q^\circ$ has only nonnegative frequencies, since the frequencies of $\bar{p}Q^\circ$ are at least $-M - d$; write $w^\Lambda \bar{p}Q^\circ = \sum_{m \geq 0, t \geq 0} \gamma_{m,t}^W \Phi_{m,t}$ (a finite sum, computed in interval arithmetic). Finally let

$$\begin{aligned} U_1 &= \sum_k \left(1 + \frac{\max(0, k-1)}{j_* + 1} \right) \rho^{k-1} E_k, & U_2 &= E_{k_-} \rho^{k_- - 3} \tau, \\ U_3 &= \frac{1}{j_* + 1} \left(\sum_k \rho^{k-3} E_k + E_{k_-} \rho^{k_- - 5} \tau \right), & U_4 &= \frac{H E_{k_-}}{(j_* + 1) \rho^{j_*}}, \\ U_5 &= \sum_{\alpha, \beta} \frac{(|q_{\alpha\beta}^1| + |q_{\alpha\beta}^0|/j_*) \beta \rho^{\alpha-\beta-1}}{b^2(j_* + \alpha)}, \\ U_W &= \frac{8}{b^2(j_* - \Lambda)^2} \sum_{m,t} |\gamma_{m,t}^W| (2t+1)(2t+3) \rho^{m-1-\Lambda}, \end{aligned}$$

where k runs over odd integers in $[k_-, k_+]$.

Lemma 16.3. *For integers $m, t, N \geq 0$,*

$$\|(1 - |w|^2)^2 w^N \Phi_{m,t}\|_\rho \leq \frac{4(2t+1)(2t+3)}{(m+N+1)^2} \rho^{m+N}.$$

Proof. Applying (12.3) for w and then for $\bar{w}$ gives $|w|^2\Phi_{m,t} = \lambda_t^+\Phi_{m,t+1} + (1 - \lambda_t^+ - \lambda_t^-)\Phi_{m,t} + \lambda_t^-\Phi_{m,t-1}$ with

$$\lambda_t^+ = \frac{(t+1)(m+t+1)}{(m+2t+1)(m+2t+2)} \leq \frac{t+1}{m+1}, \quad \lambda_t^- = \frac{t(m+t)}{(m+2t)(m+2t+1)} \leq \frac{t}{m+1}.$$

Hence $(1 - |w|^2)\Phi_{m,t}$ has the same frequency and absolute coefficient sum $2(\lambda_t^+ + \lambda_t^-) \leq 2(2t+1)/(m+1)$, and its radial indices are at most $t+1$. Applying this twice, $\|(1 - |w|^2)^2\Phi_{m,t}\|_\rho \leq 4(2t+1)(2t+3)\rho^m/(m+1)^2$. By Lemma 12.2, $w^N\Phi_{m,t}$ is a convex combination of $\Phi_{m+N,t'}$ with $t' \leq t$; apply the previous bound to each term. $\square$

Proposition 16.4. *Assume $j_* + k_- - 1 > M + 2$ and $j_* > 2M + d + 2$. Then for every odd $j > j_*$,*

$$\frac{\|(I - AD\mathcal{F}(x^\circ))e_j\|_X}{\omega_j} \leq Z_{\text{sh},\infty} := \mathbf{U}_1 + \mathbf{U}_2 + \mathbf{U}_3 + \mathbf{U}_4 + \mathbf{U}_5 + A_t \mathbf{U}_W.$$

Proof. Fix an odd $j > j_*$. For real γ , $2\operatorname{Re}[w^{j-1}\gamma w^\alpha \bar{w}^\beta/(2i)] = \gamma \operatorname{Im}(w^{j-1+\alpha}\bar{w}^\beta)$. For every monomial occurring, $k = \alpha - \beta \geq k_-$ and $j - 1 + k > M + 2 \geq 1$, so by Lemma 12.4 $\operatorname{Im}(w^{j-1+\alpha}\bar{w}^\beta) = \sum_s \kappa_s S_{j-1+k,s}$ with nonnegative coefficients summing to one and harmonic coefficient $\kappa_0 = 1 - \beta/(j + \alpha)$. Accordingly we split $\Delta_j = \Delta^H + \Delta^N + \Delta^W$ into the harmonic part and the nonharmonic part of the first term of (16.2), and the second term Δ^W . All rows occurring in Δ^H and Δ^N have frequency at least $j - 1 + k_- > M + 2$, so they are tail rows.

The harmonic part. The coefficient of Δ^H in row $(j - 1 + k, 0)$ is $y_k = \sum_{\alpha-\beta=k} (jq_{\alpha\beta}^1 + q_{\alpha\beta}^0)(1 - \frac{\beta}{j+\alpha})$. Since $\frac{j}{j+1}(1 - \frac{\beta}{j+\alpha}) - 1 = -\frac{1}{j+1} - \frac{j\beta}{(j+1)(j+\alpha)}$ and $0 \leq 1 - \frac{\beta}{j+\alpha} \leq 1$, we have $|y_k/(j+1) - \bar{q}_k| \leq \varepsilon_k$. By (15.4), the shape part of $A\Delta^H$ is $\eta_{j-1+k} = b^{-2} \sum_{l \geq k} y_l/(j+l)$ in the window $k_- \leq k \leq k_+$, and $\eta_n = \eta_{j-1+k_-}$ for $M < n < j - 1 + k_-$. Writing $\frac{y_l}{j+l} = \frac{y_l}{j+1}(1 + \frac{1-l}{j+1})$ and using $|\frac{1-l}{j+1}| \leq \frac{|l-1|}{j_*+k_-}$, we get $|\eta_{j-1+k} - \bar{\eta}_k| \leq \delta_k$, hence $|\eta_{j-1+k}| \leq E_k$, and $|\eta_n| \leq E_{k_-}$ below the window. We now bound the parts of $A\Delta^H$, divided by $\omega_j = b^2(j+1)\rho^j$.

- Shape coordinates in the window: $\omega_{j-1+k}/\omega_j = (1 + \frac{k-1}{j+1})\rho^{k-1}$, which gives at most $\mathbf{U}_1$.
- Shape coordinates below the window: $\sum_{n \leq j-3+k_-} (n+1)\rho^n/((j+1)\rho^j) \leq \rho^{k_- - 3}\tau$, which gives at most $\mathbf{U}_2$.
- The coupled coordinates $\delta g_{n-2,1} = b^2\eta_n$, of weight ρ^{n-2} : at most $\mathbf{U}_3$, by the same two computations and $j+1 > j_* + 1$.
- The finite part $b^2\eta_{M+2}\hat{A}((M+1)e_{(M,0)} + e_{(M,1)})$: its norm is at most $b^2HE_{k_-}$, which gives at most $\mathbf{U}_4$ since $j > j_*$.

The nonharmonic part. Δ^N lies in nonharmonic tail rows, on which A is the identity, and its normalised norm is at most

$$\sum_{\alpha,\beta} \frac{|jq_{\alpha\beta}^1 + q_{\alpha\beta}^0|}{j+1} \cdot \frac{\beta}{j+\alpha} \cdot \frac{\rho^{j-1+\alpha-\beta}}{b^2\rho^j} \leq \mathbf{U}_5.$$

The part Δ^W . We have $jw^{j-1}\bar{p}W^\circ = j(1 - |w|^2)^2w^{j-1-\Lambda}(w^\Lambda\bar{p}Q^\circ)$ with $j - 1 - \Lambda \geq 0$. Since conjugation preserves $\|\cdot\|_\rho$, $\|2\operatorname{Re} f\|_\rho \leq 2\|f\|_\rho$, and Lemma 16.3 with $N = j - 1 - \Lambda$ gives

$$\frac{\|\Delta^W\|_\rho}{\omega_j} \leq \frac{2j}{b^2(j+1)\rho^j} \sum_{m,t} |\gamma_{m,t}^W| \frac{4(2t+1)(2t+3)\rho^{m+j-1-\Lambda}}{(m+j-\Lambda)^2} \leq \mathbf{U}_W.$$

All frequencies in Δ^W are at least $j - 1 - \Lambda > M$ because $j_* > 2M + d + 2$, so $\|A\Delta^W\|_X \leq A_t\|\Delta^W\|_\rho$ by Lemma 15.6.

Adding the three contributions proves the claim. $\square$

Remark 16.5. Each of ε_k , δ_k , U_3 , U_5 is $O(1/j_*)$, $U_4 = O(\rho^{-j_*}/j_*)$ and $U_W = O(j_*^{-2})$. Hence, as $j_* \rightarrow \infty$, the majorant $Z_{\text{sh},\infty}$ tends to $\sum_k \rho^{k-1} |\bar{\eta}_k| + |\bar{\eta}_{k-}| \rho^{k-3} \tau \approx 0.19723$; for comparison, the directly computed normalised column at $j = 1003$ is about 0.19752.

Collecting the five classes, the number

$$Z = \max\{Z_f, Z_{g,\partial}, Z_{g,\infty}, Z_{\text{sh},\partial}, Z_{\text{sh},\infty}\}$$

bounds $\|I - \text{ADF}(x^\circ)\|_{X \rightarrow X}$.

17. CERTIFIED BOUNDS AND THE EXACT ZERO

The quantities defined in Sections 14–16 are finite expressions in the 525 dyadic centre coefficients and in $\rho = 21/20$, apart from the entries of $\hat{A}$, which are dyadic numbers produced by the computation and are then treated as exact data. They were evaluated in ball arithmetic as described in Section 34. Table 8 lists the results, rounded outward.

Quantity	Where defined	Certified bound
$b = c_1^\circ$	Section 14	14.8400430096918... (exact dyadic)
$P = N_\rho(p)$	Section 14	< 16.833325
$V_0 = \ V^\circ\ _\rho$	Section 14	< 134.261058
$\ I - \hat{A}M_f\ $	(15.5)	$< 1.343 \cdot 10^{-34}$
$\ R\ $	Lemma 15.6	< 675.571877
H	Lemma 15.6	< 1966.719238
A_t ($N_H = 443$)	Lemma 15.6	< 10.777973
$\ A\ $	Lemma 15.6	< 675.571877
Y	Section 16.1	$< 6.297631 \cdot 10^{-6}$
Z_f (525 columns)	Section 16.1	< 0.180814
$2P_s P_t + P_t^2$	Lemma 16.1	< 0.056966
$Z_{g,\partial}$ (3650 columns, maximum at (43, 1))	Lemma 16.1	< 0.313479
$Z_{g,\infty}$	Lemma 16.2	< 0.186417
$Z_{\text{sh},\partial}$ (480 columns, maximum at $j = 43$)	Section 16	< 0.360262
U_1	Proposition 16.4	< 0.217542
U_2	Proposition 16.4	$< 4.45704 \cdot 10^{-4}$
U_3	Proposition 16.4	$< 1.97149 \cdot 10^{-4}$
U_4	Proposition 16.4	$< 3.018 \cdot 10^{-24}$
U_5	Proposition 16.4	$< 9.23594 \cdot 10^{-4}$
$\sum_{m,t} \gamma_{m,t}^W (2t+1)(2t+3) \rho^{m-1-\Lambda}$	Proposition 16.4	< 19930.219512
$U_W, A_t U_W$	Proposition 16.4	$< 8.55375 \cdot 10^{-4}, < 9.219199 \cdot 10^{-3}$
$Z_{\text{sh},\infty}$ (all $j > 1001$)	Proposition 16.4	< 0.228328
Z	Section 16	< 0.360262
$C_2 = \ A\ c_2$	Lemma 14.2	< 3.278617
$C_3 = \ A\ c_3$	Lemma 14.2	$< 8.42289 \cdot 10^{-4}$

TABLE 8. Certified bounds, rounded outward, for $M = 41$, $S = 24$, $\rho = 21/20$, $j_* = 1001$. The complete interval enclosures are recorded in the certificate file (Section 34).

Theorem 17.1. *Let $r = 1/50000$. There is exactly one $x^* = (g^*, c^*) \in X$ with $\|x^* - x^\circ\|_X \leq r$ and $\mathcal{F}(x^*) = 0$.*

Proof. The conditions $j_* + k_- - 1 = 919 > M + 2$ and $j_* = 1001 > 2M + d + 2 = 124$ of Proposition 16.4 hold, and (15.5) holds by Table 8; hence A is bounded and injective (Lemma 15.5). By Table 8, Proposition 15.1 applies with the rational majorants $Y = 7 \cdot 10^{-6}$, $Z = 37/100$, $C_2 = 4$ and $C_3 = 10^{-3}$. For these values and $r = 1/50000$, exact rational arithmetic gives

$$\begin{aligned} Y + (Z - 1)r + C_2 r^2 + C_3 r^3 &= -5.5983999999920 \cdot 10^{-6} < 0, \\ Z - 1 + 2C_2 r + 3C_3 r^2 &= -0.6298399999988 < 0. \end{aligned}$$

□

Remark 17.2 (The centre). The centre x° was obtained numerically, and its provenance plays no role in the proof. A spectral collocation computation in spherical coordinates, started from the ball of radius $j_{2,4}$ perturbed by the zonal harmonic of degree six (see Section 35), produced an approximate solution of (1.1); its meridian domain was mapped conformally to the disc, and Newton's method was applied to the Galerkin truncation of (14.1) to the 21 shape coefficients $c_1, c_3, \dots, c_{41}$ and the 504 coefficients of g described above. The resulting binary64 numbers were then frozen and are interpreted as exact dyadic rationals.

18. RECONSTRUCTION OF THE DOMAIN FOR $n = 4$

Let $x^* = (g^*, c^*)$ be the zero of Theorem 17.1, $\psi^* = \psi_{c^*}$, and $\delta c = c^* - c^\circ$. Then $\sum_j \omega_j |\delta c_j| \leq r$, and consequently

$$(18.1) \quad \begin{aligned} |c_1^* - b| &\leq \frac{r}{2b^2\rho}, \quad \sum_j j\rho^{j-1} |\delta c_j| \leq \frac{r}{b^2\rho}, \\ \sum_{j \geq 3} |\delta c_j| &\leq \frac{r}{4b^2\rho^3}, \quad \sum_j j(j-1) |\delta c_j| \leq \frac{r\rho}{b^2(\rho-1)^2}, \end{aligned}$$

because $j\rho^{j-1}/\omega_j \leq 1/(b^2\rho)$, $1/\omega_j \leq 1/(4b^2\rho^3)$ for $j \geq 3$, and $j(j-1)/\omega_j \leq j\rho^{-j}/b^2 \leq \sum_{i \geq 1} i\rho^{-i}/b^2 = \rho/(b^2(\rho-1)^2)$. In particular ψ^* is holomorphic in $|w| < \rho$.

18.1. Univalence, star-shapedness and the non-disc property.

Proposition 18.1. *The map ψ^* is injective on the disc $\{|w| < 41/40\}$, and $\psi^{*'} has no zeros there. The domain $D = \psi^*(\mathbb{D})$ is bounded by the real-analytic Jordan curve $\psi^*(\partial\mathbb{D})$, it is strictly star-shaped with respect to 0, symmetric under $x_1 \mapsto -x_1$ and $y \mapsto -y$, and it is not a disc.$*

Proof. Let $R = 41/40 < \rho$, $b_- = b - r/(2b^2\rho) \leq c_1^*$, and

$$s_1 = \sum_{j \geq 3} j|c_j^\circ| + \frac{r}{b^2\rho}, \quad s_0 = \sum_{j \geq 3} |c_j^\circ| + \frac{r}{4b^2\rho^3}.$$

By (18.1), $\sum_{j \geq 3} j|c_j^*| \leq s_1$ and $\sum_{j \geq 3} |c_j^*| \leq s_0$. The computation (Section 34) certifies

$$(18.2) \quad \operatorname{Re} \psi^{*'}(w) \geq b_- - \sum_{j \geq 3} j|c_j^*| R^{j-1} - \frac{r}{b^2\rho} > 13.081358 \quad (|w| \leq R),$$

$$(18.3) \quad \frac{s_1}{b_-} < 0.104547, \quad \frac{s_1 + s_0}{b_- - s_0} < 0.126890, \quad |c_3^*| \geq |c_3^\circ| - \frac{r}{4b^2\rho^3} > 0.0911958,$$

where in (18.2) we used $j|\delta c_j| R^{j-1} \leq j|\delta c_j| \rho^{j-1}$ and (18.1).

Univalence. For $w_1 \neq w_2$ in the convex disc $|w| < R$, $\frac{\psi^*(w_2) - \psi^*(w_1)}{w_2 - w_1} = \int_0^1 \psi^{*'}(w_1 + \sigma(w_2 - w_1)) d\sigma$ has positive real part by (18.2).

Star-shapedness. For $|w| = 1$ we have $|\psi^*(w) - c_1^*w| \leq s_0$, $|w\psi^{*'}(w) - \psi^*(w)| \leq \sum_{j \geq 3} (j-1)|c_j^*| \leq s_1$ and $|\psi^*(w)| \geq b_- - s_0 > 0$, hence $|w\psi^{*'}(w)/\psi^*(w) - 1| < 0.127$ by (18.3). Therefore $\frac{d}{d\theta} \arg \psi^*(e^{i\theta}) = \operatorname{Re}(w\psi^{*'}(w)/\psi^*(w)) > 0$. The winding number of $\psi^*(e^{i\theta})$ about 0 equals that of $c_1^*e^{i\theta}$, namely one, since $|\psi^*(w) - c_1^*w| \leq s_0 < c_1^*$ on $\partial\mathbb{D}$. Hence the argument increases by exactly 2π , and ∂D is a radial graph $\{R_D(\phi)e^{i\phi}\}$ with R_D positive and real analytic; D is strictly star-shaped.

Symmetry. ψ^* is odd with real coefficients, so $\psi^*(\bar{w}) = \overline{\psi^*(w)}$ and $\psi^*(-w) = -\psi^*(w)$.

Not a disc. If D were a disc, the two reflection symmetries would force its centre to be 0, and $w \mapsto \psi^*(w)/\psi^{*'}(0)$ would be a conformal map of $\mathbb{D}$ onto a disc centred at 0 fixing 0 with derivative one there; by the Schwarz lemma applied to it and to its inverse, it is the identity, so ψ^* is linear. This contradicts $c_3^* \neq 0$. $\square$

18.2. Convexity.

Proposition 18.2. *For $|w| \leq 1$,*

$$\operatorname{Re}\left(1 + \frac{w\psi^{*''}(w)}{\psi^{*'}(w)}\right) > 0.414074.$$

Consequently D is strictly convex, and $\Omega = \Pi^{-1}(D)$ is convex.

Proof. Let $A_1 = \sum_{j \geq 3} j|c_j^*|$, $A_2 = \sum_{j \geq 3} j(j-1)|c_j^*|$, $E_1 = r/(b^2\rho)$ and $E_2 = r\rho/(b^2(\rho-1)^2)$. By (18.1), for $|w| \leq 1$,

$$|\psi^{*'}(w)| \geq c_1^* - \sum_{j \geq 3} j|c_j^*| \geq b - A_1 - E_1 =: L, \quad |\psi^{*''}(w)| \leq \sum_{j \geq 3} j(j-1)|c_j^*| \leq A_2 + E_2 =: U.$$

The computation certifies $A_1 < 1.5514770$, $A_2 < 7.7860764$, $E_1 < 8.6491 \cdot 10^{-8}$, $E_2 < 3.81425 \cdot 10^{-5}$, hence $L > 13.288565$, $U < 7.786115$ and $\operatorname{Re}(1 + w\psi^{*''}/\psi^{*'}) \geq 1 - U/L > 0.414074$.

Let $z(\theta) = \psi^*(e^{i\theta})$. Then $z'(\theta) = ie^{i\theta}\psi^{*'}(e^{i\theta}) \neq 0$ and

$$\frac{d}{d\theta} \arg z'(\theta) = \operatorname{Re}\left(1 + \frac{w\psi^{*''}(w)}{\psi^{*'}(w)}\right)\Big|_{w=e^{i\theta}} > 0.$$

Thus the tangent direction of the simple closed analytic curve ∂D turns strictly monotonically, through a total angle 2π (the winding number of $\psi^{*'}(e^{i\theta})$ about 0 is zero, as $\psi^{*'}$ has no zeros in $\mathbb{D}$), and ∂D bounds a strictly convex domain; this is the classical analytic convexity criterion for conformal maps of the disc, cf. [26, Chapter 2].

Now let $(x_1, x'), (z_1, z') \in \Omega$, $\sigma \in [0, 1]$, and put $y_\sigma = \sigma|x'| + (1-\sigma)|z'|$. By convexity of D , $(\sigma x_1 + (1-\sigma)z_1, \pm y_\sigma) \in D$, and by convexity again the vertical segment joining these two points lies in D . Since $|\sigma x' + (1-\sigma)z'| \leq y_\sigma$, the point $\sigma(x_1, x') + (1-\sigma)(z_1, z')$ lies in Ω . $\square$

18.3. Analytic regularity and continuation.

Lemma 18.3. *Let $U \subset \mathbb{R}^2$ be open and $F \in C(U)$ with $\int_U F(\Delta\varphi + \varphi) = 0$ for all $\varphi \in C_c^\infty(U)$. Then F is real analytic in U .*

Proof. Let $\mathcal{H}(x_1, y, \zeta) = e^\zeta F(x_1, y)$ on $U \times \mathbb{R}$. For $\Phi \in C_c^\infty(U \times \mathbb{R})$, integration by parts in ζ gives $\int \mathcal{H} \Delta_3 \Phi = \int_U F(\Delta\varphi + \varphi) = 0$ with $\varphi = \int e^\zeta \Phi(\cdot, \zeta) d\zeta \in C_c^\infty(U)$. By Weyl's lemma the continuous function $\mathcal{H}$ is harmonic, hence real analytic, and so is $F = \mathcal{H}(\cdot, \cdot, 0)$. $\square$

Lemma 18.4. *Let $D \subset \mathbb{R}^2$ be a bounded domain whose boundary is a real-analytic regular Jordan curve, and let $F \in C^1(\bar{D})$ satisfy $\int_D (\nabla F \cdot \nabla \varphi - F\varphi) = 0$ for all $\varphi \in C_c^\infty(D)$, together with $F = y$ and $\nabla F = e_y$ on ∂D . Then F extends to a real-analytic solution of $\Delta F + F = 0$ on an open neighbourhood of $\bar{D}$.*

Proof. Local Cauchy problem. Fix $z_0 \in \partial D$ and a real-analytic regular parametrisation γ of ∂D near z_0 with $\gamma(0) = z_0$; extend γ holomorphically to a disc $\{|a| < a_0\}$ on which $\gamma' \neq 0$ and γ is injective, and put $\gamma^\dagger(b) = \overline{\gamma(\bar{b})}$. In the complex coordinates $\zeta = x_1 + iy$, $\chi = x_1 - iy$ the Laplacian is $4\partial_\zeta\partial_\chi$, and the substitution $\zeta = \gamma(a)$, $\chi = \gamma^\dagger(b)$ turns $4\partial_\zeta\partial_\chi F + F = 0$ into $\partial_a\partial_b F = -kF$ with $k(a, b) = \gamma'(a)\gamma'^\dagger(b)/4$. The real plane corresponds to $b = \bar{a}$, and ∂D to the real diagonal $a = b \in \mathbb{R}$. The function $F_0(a, b) = (\gamma(a) - \gamma^\dagger(b))/(2i)$ is the holomorphic extension of y and satisfies $\partial_a\partial_b F_0 = 0$. On the closed bidisc $\{|a|, |b| \leq \varepsilon\}$ consider

$$(\mathcal{T}W)(a, b) = - \int_{[b, a]} \int_{[t, b]} k(t, v) (W(t, v) + F_0(t, v)) dv dt$$

along straight segments, which stay in the bidisc by convexity. $\mathcal{T}$ maps the Banach space of functions continuous on the closed bidisc and holomorphic in its interior into itself, and $\|\mathcal{T}W - \mathcal{T}\tilde{W}\|_\infty \leq 2\varepsilon^2 K_0 \|W - \tilde{W}\|_\infty$ with $K_0 = \sup |k|$, since $\int_{[b, a]} |t - b| |dt| = |a - b|^2/2 \leq 2\varepsilon^2$. For $2\varepsilon^2 K_0 < 1$ the contraction mapping principle gives a fixed point W , and differentiation of the iterated integral gives $W_{ab} = -k(W + F_0)$ and $W = W_a = W_b = 0$ on the diagonal $a = b$. Hence $\tilde{F}(x_1, y) = (F_0 + W)(\gamma^{-1}(x_1 + iy), (\gamma^\dagger)^{-1}(x_1 - iy))$ is a real-analytic (possibly complex-valued) solution of $\Delta\tilde{F} + \tilde{F} = 0$ on a disc U about z_0 , with $\tilde{F} = y$ and $\nabla\tilde{F} = e_y$ on $\partial D \cap U$.

Matching. Shrink U so that ∂D divides it into two connected pieces. The function $Z = F - \tilde{F}$ on $D \cap U$, extended by zero to $U \setminus D$, is C^1 on U , because its value and gradient vanish on $\partial D \cap U$. It is a weak solution of $\Delta Z + Z = 0$ on U . Indeed, let $\varphi \in C_c^\infty(U)$, and let $\chi_\delta \in C_c^\infty(D)$ be equal to one at distance at least 2δ from ∂D , with $0 \leq \chi_\delta \leq 1$ and $|\nabla\chi_\delta| \leq C/\delta$. Since F and $\tilde{F}$ are weak solutions in $D \cap U$ and $\chi_\delta\varphi \in C_c^\infty(D \cap U)$,

$$\int_U (\nabla Z \cdot \nabla\varphi - Z\varphi) = \int_{D \cap U} (1 - \chi_\delta)(\nabla Z \cdot \nabla\varphi - Z\varphi) - \int_{D \cap U} \varphi \nabla Z \cdot \nabla\chi_\delta.$$

The integrands are supported in the layer $\Sigma_\delta = \{\text{dist}(\cdot, \partial D) < 2\delta\} \cap D \cap \text{supp } \varphi$, whose area is $O(\delta)$. The first integral is $O(\delta)$, and the second is bounded by $C' \sup_{\Sigma_\delta} |\nabla Z|$, which tends to zero because ∇Z is continuous and vanishes on $\partial D \cap U$. By Lemma 18.3 (applied to the real and imaginary parts) Z is real analytic on U ; it vanishes on the open set $U \setminus \bar{D}$, hence on U . Thus $\tilde{F} = F$ on $D \cap U$.

Gluing. Cover ∂D by finitely many discs $B(z_i, r_i/2)$ such that on $B(z_i, r_i)$ a local extension $\tilde{F}_i$ as above exists, and let $0 < \varepsilon \leq \min_i r_i/6$. For z with $\text{dist}(z, \partial D) < \varepsilon$ choose $q \in \partial D$ with $|z - q| < \varepsilon$ and i with $q \in B(z_i, r_i/2)$, and put $\tilde{F}(z) = \tilde{F}_i(z)$. If another choice q', i' is made, then $|q - q'| < 2\varepsilon$, so $B(q', \varepsilon) \subset B(z_i, r_i) \cap B(z_{i'}, r_{i'})$; both $\tilde{F}_i$ and $\tilde{F}_{i'}$ coincide with F on the nonempty open set $B(q', \varepsilon) \cap D$, hence on the disc $B(q', \varepsilon) \ni z$. Thus $\tilde{F}$ is well defined and real analytic on $\{\text{dist}(\cdot, \partial D) < \varepsilon\}$, and together with F it gives the required extension. It is real valued because it is real on D . $\square$

18.4. Proof of Theorem 1.1 for $n = 4$. Let $V^* = Kg^* + \text{Im } \psi^*$. By Lemma 14.1, $V^* \in C^1(\bar{\mathbb{D}})$ satisfies (11.1) with $\psi = \psi^*$ and has the parities required in Lemma 11.1. By Proposition 18.1, ψ^* is injective with nonvanishing derivative on a neighbourhood of $\bar{\mathbb{D}}$. Lemma 11.1 shows that $F = V^* \circ (\psi^*)^{-1} \in C^1(\bar{D})$ is a weak solution of $\Delta F + F = 0$ in D with $F = y$ and $\nabla F = e_y$ on ∂D , odd in y and even in x_1 . By Lemmas 18.3 and 18.4, F extends to a real-analytic solution on an open neighbourhood of $\bar{D}$; replacing this neighbourhood by the connected component containing $\bar{D}$ of its intersection with its images under the two reflections, we obtain a symmetric connected neighbourhood N on which, by the identity theorem, F is still odd in y and even in x_1 .

Since D is strictly star-shaped and symmetric, ∂D meets the axis in exactly two points. Proposition 10.3 now shows that $\Omega = \Pi^{-1}(D)$ and $u = F(x_1, |x'|)/|x'|$ (extended analytically to the axis) satisfy (1.1), that u is real analytic on the neighbourhood $\Pi^{-1}(N)$ of $\bar{\Omega}$, and that u is nonconstant.

Geometry of Ω . Let $R_D(\phi)$ be the radial function of D from Proposition 18.1. It is positive, real analytic, 2π -periodic and even. Its Fourier series $R_D(\phi) = \sum_{k \geq 0} \alpha_k \cos k\phi$ has exponentially decaying coefficients, so $\tilde{R}(\tau) = \sum_k \alpha_k T_k(\tau)$, with T_k the Chebyshev polynomials, converges on a complex neighbourhood of $[-1, 1]$ (since $|T_k| \leq \varrho^k$ on the Bernstein ellipse of parameter ϱ) and defines a real-analytic function with $\tilde{R}(\cos \phi) = R_D(\phi)$, positive near $[-1, 1]$. For $x \neq 0$, $\Pi(x)$ has polar angle $\phi = \arccos(x_1/|x|) \in [0, \pi]$ and modulus $|x|$. Hence

$$\Omega = \{0\} \cup \{x \neq 0 : |x| < \tilde{R}(x_1/|x|)\}, \quad \partial\Omega = \{\tilde{R}(\omega_1)\omega : \omega \in S^3\}.$$

Thus Ω is strictly star-shaped with respect to 0, in particular connected. The map $\omega \mapsto \tilde{R}(\omega_1)\omega$ is real analytic and injective on S^3 (its inverse is $x \mapsto x/|x|$), and its differential maps a tangent vector $v \perp \omega$ to $\tilde{R}'(\omega_1)v_1\omega + \tilde{R}(\omega_1)v$, whose component tangent to S^3 is $\tilde{R}(\omega_1)v \neq 0$. Hence it is a real-analytic embedding, and $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^3 . The symmetries of Ω follow from $\Omega = \Pi^{-1}(D)$ and the symmetry of D under $x_1 \mapsto -x_1$.

Ω is not a ball. If it were, invariance under $x' \mapsto -x'$ and $x_1 \mapsto -x_1$ would put its centre at 0; then $\Pi(\Omega) = D \cap \{y \geq 0\}$ would be the upper half of a disc centred at 0, and by the symmetry $y \mapsto -y$, D would be that disc, contradicting Proposition 18.1. Finally, Ω is convex by Proposition 18.2. This completes the proof of Theorem 1.1 for $n = 4$. $\square$

19. DIMENSION SIX

In this section we prove Theorem 1.1 for $n = 6$. The argument follows Sections 10–18 step by step. We state explicitly which results of those sections are used verbatim, and why, and we give complete proofs of everything that changes. Points of $\mathbb{R}^6$ are written $x = (x', x'') \in \mathbb{R}^3 \times \mathbb{R}^3$, points of the meridian plane $(y_1, y_2) \in \mathbb{R}^2$, identified with $z = y_1 + iy_2 \in \mathbb{C}$, and

$$\Pi_6 : \mathbb{R}^6 \rightarrow \mathbb{R}^2, \quad \Pi_6(x', x'') = (|x'|, |x''|).$$

The map Π_6 is continuous, its image is the closed quadrant $[0, \infty)^2$, and $|\Pi_6(x)| = |x|$.

19.1. Reduction to a planar problem.

Lemma 19.1. *Let f be a C^2 function on an open subset of $\{y_1 > 0, y_2 > 0\}$ and $F = y_1 y_2 f$. Then*

$$F_{y_1 y_1} + F_{y_2 y_2} = y_1 y_2 \left(f_{y_1 y_1} + f_{y_2 y_2} + \frac{2}{y_1} f_{y_1} + \frac{2}{y_2} f_{y_2} \right).$$

The expression in parentheses, evaluated at $\Pi_6(x)$, is the Laplacian of the function $x \mapsto f(|x'|, |x''|)$ at every point with $x' \neq 0$ and $x'' \neq 0$.

Proof. $(y_1 y_2 f)_{y_1 y_1} = y_2 (y_1 f_{y_1 y_1} + 2f_{y_1})$, and similarly for y_2 . The Laplacian of $\mathbb{R}^3 \times \mathbb{R}^3$ is the sum of the Laplacians of the two factors, each of which acts on functions of the modulus y of its variable as $\partial_y^2 + \frac{2}{y} \partial_y$. $\square$

Analyticity across the two axes is obtained by applying the division argument of Lemma 10.2 in each variable.

Lemma 19.2. *Let $N \subset \mathbb{R}^2$ be open and invariant under $(y_1, y_2) \mapsto (-y_1, y_2)$ and $(y_1, y_2) \mapsto (y_1, -y_2)$, and let F be real analytic on N , odd in y_1 and odd in y_2 . Then there is a real-analytic function G , defined on an open set containing $\{(y_1^2, y_2^2) : (y_1, y_2) \in N\}$, such that*

$F(y_1, y_2) = y_1 y_2 G(y_1^2, y_2^2)$ on N . Consequently $x \mapsto G(|x'|^2, |x''|^2)$ is real analytic on the open set $\Pi_6^{-1}(N)$, and it equals $F(\Pi_6(x))/(|x'| |x''|)$ where $x' \neq 0$ and $x'' \neq 0$.

Proof. Let $a = (a_1, a_2) \in N$. Choose $\delta > 0$, with $\delta < |a_i|$ whenever $a_i \neq 0$, such that $F(y) = \sum_{i,k \geq 0} c_{ik} (y_1 - a_1)^i (y_2 - a_2)^k$ converges absolutely for $|y_1 - a_1|, |y_2 - a_2| < \delta$. For $i = 1, 2$ let $I_i = (-\delta^2, \delta^2)$ if $a_i = 0$, and $I_i = (|a_i| - \delta)^2, (|a_i| + \delta)^2) \subset (0, \infty)$ if $a_i \neq 0$, and let σ_i be the sign of a_i . We define G on the box $I_1 \times I_2$.

- If $a_1 = a_2 = 0$, oddness in both variables gives $c_{ik} = 0$ unless i and k are odd, and we put $G(t_1, t_2) = \sum_{i,k} c_{2i+1, 2k+1} t_1^i t_2^k$.
- If $a_1 = 0 \neq a_2$, oddness in y_1 gives $c_{ik} = 0$ for even i , so $F = y_1 H(y_1^2, y_2)$ with $H(t_1, y_2) = \sum_{i,k} c_{2i+1, k} t_1^i (y_2 - a_2)^k$, and we put $G(t_1, t_2) = H(t_1, \sigma_2 \sqrt{t_2})/(\sigma_2 \sqrt{t_2})$. The case $a_2 = 0 \neq a_1$ is symmetric.
- If $a_1 a_2 \neq 0$, we put $G(t_1, t_2) = F(\sigma_1 \sqrt{t_1}, \sigma_2 \sqrt{t_2})/(\sigma_1 \sigma_2 \sqrt{t_1 t_2})$.

In each case G is real analytic on $I_1 \times I_2$, because $\sqrt{t_i}$ is real analytic and positive on I_i when $a_i \neq 0$. By the oddness of F in each variable and the invariance of N , $G(t_1, t_2) = F(\sqrt{t_1}, \sqrt{t_2})/\sqrt{t_1 t_2}$ at the points of the box with $t_1, t_2 > 0$. The intersection of two such boxes is a product of two intervals, each of which is contained in $(0, \infty)$ or symmetric about 0; if it is nonempty, it is connected and contains points with $t_1, t_2 > 0$, at which the two definitions agree. By the identity theorem they agree on the whole intersection, and G is well defined. It satisfies $F = y_1 y_2 G(y_1^2, y_2^2)$ near every point of N . The last assertion follows because $x \mapsto (|x'|^2, |x''|^2)$ is polynomial. $\square$

Proposition 19.3 (From the plane to $\mathbb{R}^6$). *Let $D \subset \mathbb{R}^2$ be a bounded domain, invariant under $(y_1, y_2) \mapsto (-y_1, y_2)$ and $(y_1, y_2) \mapsto (y_1, -y_2)$, whose boundary is a Jordan curve meeting each coordinate axis in exactly two points. Let F be real analytic on an open set $N \supset \overline{D}$ invariant under the same two reflections, odd in y_1 and in y_2 , and such that*

$$(19.1) \quad \Delta F + F = 0 \text{ in } D, \quad F = y_1 y_2 \text{ and } \nabla F = (y_2, y_1) \text{ on } \partial D.$$

Put $\Omega = \Pi_6^{-1}(D)$ and let u be the function $x \mapsto G(|x'|^2, |x''|^2)$ of Lemma 19.2. Then Ω is a bounded open set, u is real analytic on the neighbourhood $\Pi_6^{-1}(N)$ of $\overline{\Omega}$, u is nonconstant, and (1.1) holds.

Proof. We follow the proof of Proposition 10.3. The set Ω is open and bounded because Π_6 is continuous and proper; $\overline{\Omega} \subset \Pi_6^{-1}(\overline{D}) \subset \Pi_6^{-1}(N)$ and $\partial\Omega \subset \Pi_6^{-1}(\partial D)$. Let $E = \{x : x' \neq 0, x'' \neq 0\}$; its complement is the union of two three-dimensional subspaces. On $\Omega \cap E$ we have $u = f \circ \Pi_6$ with $f = F/(y_1 y_2)$, and Lemma 19.1 gives $|x'| |x''| (\Delta u + u)(x) = (\Delta F + F)(\Pi_6(x)) = 0$. Since $\Omega \cap E$ is dense in Ω and $\Delta u + u$ is continuous, $\Delta u + u = 0$ in Ω .

Let $x \in \partial\Omega \cap E$. Then $\Pi_6(x) \in \partial D$ lies in the open quadrant, and $f = F/(y_1 y_2) = 1$ there. From $F_{y_1} = y_2 f + y_1 y_2 f_{y_1} = y_2$ and the analogous identity for y_2 we get $f_{y_1} = f_{y_2} = 0$, hence $\nabla u(x) = (f_{y_1} x'/|x'|, f_{y_2} x''/|x''|) = 0$.

It remains to show that $\partial\Omega \cap E$ is dense in $\partial\Omega$. By the reflection symmetries, the two points of ∂D on the axis $\{y_1 = 0\}$ are $(0, \pm a)$ with $a > 0$. The reflection $y_1 \mapsto -y_1$ fixes them and maps $\{y_1 > 0\}$ to $\{y_1 < 0\}$, so it exchanges the two arcs of ∂D between them, and one of these arcs lies in $\{y_1 > 0\}$. Hence $(0, a)$ is a limit of points of ∂D in the open quadrant; the same holds for the points of ∂D on the other axis. Every point of $\Pi_6^{-1}(\partial D)$ lies outside Ω and is a limit of points of Ω , so $\Pi_6^{-1}(\partial D) = \partial\Omega$, and lifting the above sequences by Π_6 shows that $\partial\Omega \cap E$ is dense in $\partial\Omega$. By continuity $u = 1$ and $\nabla u = 0$ on $\partial\Omega$. Finally, a constant solution of $\Delta u + u = 0$ vanishes identically. $\square$

As in Proposition 10.4, averaging over $O(3) \times O(3)$ shows that the reduction is reversible in the analytic class. Let $y_1 + iy_2 = re^{i\phi}$. Since $y_1 y_2 = \frac{1}{2} r^2 \sin 2\phi$ and $\sin m\phi = \sin 2\phi U_{m/2-1}(\cos 2\phi)$ for even m , the planar Helmholtz solution $F = J_m(r) \sin m\phi$, with $m \equiv 2 \pmod{4}$, corresponds to the six-dimensional Helmholtz solution

$$u = 2r^{-2} J_m(r) U_{m/2-1}(\cos 2\phi), \quad \cos 2\phi = \frac{|x'|^2 - |x''|^2}{|x|^2}.$$

For $m = 2$ this is the radial solution $2r^{-2} J_2(r)$, whose derivative is $-2r^{-2} J_3(r)$; the Schiffer balls of $\mathbb{R}^6$ at eigenvalue one have radii $j_{3,k}$, the positive zeros of J_3 .

19.2. Symmetry sector and conformal formulation. For a real sequence $c = (c_j)_{j \equiv 1 \pmod{4}}$ put $\psi_c(w) = \sum_{j \equiv 1 \pmod{4}} c_j w^j$. Since $i^j = i$ for $j \equiv 1 \pmod{4}$,

$$\psi_c(\bar{w}) = \overline{\psi_c(w)}, \quad \psi_c(-\bar{w}) = -\overline{\psi_c(w)}, \quad \psi_c(i\bar{w}) = i \overline{\psi_c(w)}.$$

Hence, if ψ_c is injective on $\mathbb{D}$, then $D = \psi_c(\mathbb{D})$ is invariant under $z \mapsto \bar{z}$, $z \mapsto -\bar{z}$ and $z \mapsto i\bar{z}$, that is, under $y_2 \mapsto -y_2$, $y_1 \mapsto -y_1$ and the exchange $(y_1, y_2) \mapsto (y_2, y_1)$.

Lemma 19.4. *Let ψ be holomorphic and injective on a neighbourhood of $\bar{\mathbb{D}}$, with $\psi' \neq 0$ there, let $D = \psi(\mathbb{D})$ and $h_\psi = \text{Im}(\psi^2)/2$. Let $V \in C^1(\bar{\mathbb{D}})$ be real valued and satisfy*

$$(19.2) \quad \Delta V + |\psi'|^2 V = 0 \text{ in } \mathcal{D}'(\mathbb{D}), \quad V - h_\psi = 0 \text{ and } \nabla(V - h_\psi) = 0 \text{ on } \partial\mathbb{D}.$$

Then $F = V \circ \psi^{-1} \in C^1(\bar{D})$ satisfies $\int_D (\nabla F \cdot \nabla \varphi - F \varphi) dA = 0$ for all $\varphi \in C_c^\infty(D)$, and $F = y_1 y_2$, $\nabla F = (y_2, y_1)$ on ∂D . If moreover $\psi = \psi_c$ as above and $V(\bar{w}) = -V(w)$, $V(-\bar{w}) = -V(w)$, $V(i\bar{w}) = V(w)$, then F is odd in y_1 and in y_2 and invariant under $(y_1, y_2) \mapsto (y_2, y_1)$.

Proof. Since $y_1 y_2 = \text{Im}(z^2)/2$, we have $h_\psi = (y_1 y_2) \circ \psi$ and $V - h_\psi = (F - y_1 y_2) \circ \psi$. With this change, the proof of Lemma 11.1 applies verbatim, and the parities of F follow from those of V and the displayed symmetries of ψ_c . $\square$

For $n \equiv 2 \pmod{4}$, $n \geq 2$, and $s \geq 0$ let $S_{n,s} = \text{Im } \Phi_{n,s} = r^n P_s^{(0,n)}(2r^2 - 1) \sin n\theta$. Fix $\rho = 11/10$ in this section, and let $\mathcal{S}_6$ be the real Banach space of $f = \sum_{n \equiv 2 \pmod{4}, s \geq 0} f_{n,s} S_{n,s}$ with $\|f\|_\rho = \sum \rho^n |f_{n,s}| < \infty$, which is also its norm in $\mathcal{B}_\rho$. Since $\Phi_{m,s}(\bar{w}) = \Phi_{-m,s}(w)$, $\Phi_{m,s}(-\bar{w}) = (-1)^m \Phi_{-m,s}(w)$ and $\Phi_{m,s}(i\bar{w}) = i^m \Phi_{-m,s}(w)$, the real-valued elements of $\mathcal{B}_\rho$ with the three parities of Lemma 19.4 are exactly the elements of $\mathcal{S}_6$: the first two parities force m to be even, and the third one forces $i^m = -1$. Let $\mathcal{G}_6 = \{g \in \mathcal{S}_6 : g_{n,0} = 0 \text{ for all } n\}$, and define K on $\mathcal{G}_6$ by (13.1), now for $n \equiv 2 \pmod{4}$.

Lemma 19.5 (Results used verbatim). *The following statements hold, with the proofs given in the previous sections.*

- (a) *Lemmas 12.1, 12.2, 12.3 and 12.4, and the bounds (12.4). They are stated for all $m \in \mathbb{Z}$ and $s \geq 0$.*
- (b) *Lemma 13.1(a)–(c), for every $n \geq 1$ and $s \geq 1$. Its proof uses only the explicit coefficients $a_{s,k}^{(n)}$ of (12.2) and Lemma 12.1(a), and never the parity of n . The absolute column sum of (13.1) is $1/(q(q+2))$ with $q = n+2s$, as in Lemma 13.1(d); in the present sector $q \geq 4$, so $K: \mathcal{G}_6 \rightarrow \mathcal{S}_6$ is bounded with $\|K\| = 1/24$, attained at $(n, s) = (2, 1)$.*
- (c) *Lemma 13.2 and Remark 13.3, with $\mathcal{G}_6$, $\mathcal{S}_6$ and the harmonic polynomials $r^n \sin n\theta$, $n \equiv 2 \pmod{4}$ (and with $1/24$ in place of $1/15$ in the uniform bound).*
- (d) *Lemma 15.2, Proposition 15.1 in the quartic form of Proposition 19.8 below, and Lemma 15.5 for the operator A defined in (19.13) below; the proof of Lemma 15.5 uses only the block structure of A .*

- (e) Lemma 16.3, which is stated for all $m, t, N \geq 0$, and the connection formulas of Section 16 for the Jacobi polynomials $P_k^{(1,n)}$ and $P_k^{(2,n)}$, which hold for every $n \geq 0$.
- (f) Lemma 18.3, and Lemma 18.4 with the Cauchy data (y, e_y) replaced by $(y_1 y_2, (y_2, y_1))$. The only property of the datum used in the proof of Lemma 18.4 is that it is a harmonic polynomial: in the coordinates $\zeta = y_1 + iy_2$, $\chi = y_1 - iy_2$ one has $y_1 y_2 = (\zeta^2 - \chi^2)/(4i)$, whose holomorphic extension $F_0(a, b) = (\gamma(a)^2 - \gamma^\dagger(b)^2)/(4i)$ again satisfies $\partial_a \partial_b F_0 = 0$, and the rest of the proof is unchanged.

19.3. The quartic equation. In the rest of this section $\rho = 11/10$, $M = 58$ and $S = 30$. The centre $x^\circ = (g^\circ, c^\circ)$ consists of 450 coefficients $g_{n,s}^\circ$, $n \in \{2, 6, \dots, 58\}$, $1 \leq s \leq 30$, and 15 coefficients c_j° , $j \in \{1, 5, \dots, 57\}$; all other coefficients of x° vanish. Each of these 465 numbers is an explicitly given dyadic rational (a binary64 number interpreted exactly), listed in the ancillary file `center_r6.json`. We write $b = c_1^\circ \approx 28.864751$, $\alpha = b^3/2$ and

$$\omega_j = \alpha(j+2)\rho^{j+1} \quad (j \equiv 1 \pmod{4}).$$

Let X_6 be the real Banach space of pairs $x = (g, c)$, with $g \in \mathcal{G}_6$ and $c = (c_j)_{j \equiv 1 \pmod{4}}$ real, normed by $\|(g, c)\|_X = \|g\|_\rho + \|c\|$, $\|c\| = \sum_j \omega_j |c_j|$. With

$$\theta_0 = \frac{1}{3\alpha\rho}, \quad \theta_1 = \frac{1}{\alpha\rho^2},$$

we have $N_\rho(\psi_c) \leq \theta_0 \|c\|$ and $N_\rho(\psi'_c) \leq \theta_1 \|c\|$, because $\rho^j/\omega_j = 1/(\alpha(j+2)\rho)$ and $j\rho^{j-1}/\omega_j = j/(\alpha(j+2)\rho^2)$. In particular ψ_c is holomorphic in $|w| < \rho$ and continuous up to $|w| = \rho$. Put

$$h_c = \frac{1}{2} \operatorname{Im}(\psi_c^2) = \frac{1}{2} \sum_{j,k} c_j c_k S_{j+k,0} \in \mathcal{S}_6$$

(note that $j+k \equiv 2 \pmod{4}$), and define

$$(19.3) \quad \mathcal{F}_6(g, c) = g + |\psi'_c|^2 (Kg + h_c).$$

Lemma 19.6. $\mathcal{F}_6$ is a continuous polynomial map of degree four from X_6 to $\mathcal{S}_6$. If $\mathcal{F}_6(g, c) = 0$, then $V = Kg + h_c$ belongs to $C^1(\overline{\mathbb{D}})$, has the three parities of Lemma 19.4, and satisfies (19.2) with $\psi = \psi_c$.

Proof. $\psi'_c = \sum_j j c_j w^{j-1}$ has real coefficients and only powers divisible by four. Hence $|\psi'_c|^2$ is real and invariant under $w \mapsto \bar{w}$ and $w \mapsto iw$, multiplication by it preserves the three parities, and by (12.4) it maps $\mathcal{S}_6$ into $\mathcal{S}_6$ with norm at most $N_\rho(\psi'_c)^2$. Since $w^k = \Phi_{k,0}$ has norm ρ^k and $\|\operatorname{Im} f\|_\rho \leq \|f\|_\rho$, we have $\|h_c\|_\rho \leq N_\rho(\psi_c)^2/2$. Together with $\|K\| = 1/24$, this shows that (19.3) is a sum of bounded multilinear maps of degree at most four. The rest of the proof is that of Lemma 14.1, using Lemma 19.5(c) and the fact that h_c is harmonic. $\square$

19.4. Expansion at the centre. Write $p = \psi'_{c^\circ} = \sum_{a \in \{0,4,\dots,56\}} p_a w^a$ with $p_a = (a+1)c_{a+1}^\circ$, so $p_0 = b$ and $d = \deg p = 56$, and put

$$P = N_\rho(p), \quad C = N_\rho(\psi_{c^\circ}), \quad V^\circ = Kg^\circ + h_{c^\circ}, \quad V_0 = \|V^\circ\|_\rho.$$

For $h = (\delta g, \eta) \in X_6$ let

$$L(\eta) = 2 \operatorname{Re}(\bar{p} \psi'_\eta), \quad Q(\eta_1, \eta_2) = \operatorname{Re}(\psi'_{\eta_1} \overline{\psi'_{\eta_2}}), \quad \ell(\eta) = \operatorname{Im}(\psi_{c^\circ} \psi_\eta), \quad I(\eta_1, \eta_2) = \operatorname{Im}(\psi_{\eta_1} \psi_{\eta_2}),$$

and $\delta V = K\delta g + \ell(\eta)$. Since $|p + \psi'_\eta|^2 = |p|^2 + L(\eta) + Q(\eta, \eta)$ and $h_{c^\circ + \eta} = h_{c^\circ} + \ell(\eta) + \frac{1}{2}I(\eta, \eta)$, expanding (19.3) gives

$$(19.4) \quad \mathcal{F}_6(x^\circ + h) = \mathcal{F}_6(x^\circ) + D\mathcal{F}_6(x^\circ)h + B_2(h, h) + B_3(h, h, h) + B_4(h, h, h, h),$$

where

$$(19.5) \quad \begin{aligned} D\mathcal{F}_6(x^\circ)h &= \delta g + |p|^2(K\delta g + \ell(\eta)) + L(\eta)V^\circ, \\ B_2(h, h) &= \frac{1}{2}|p|^2I(\eta, \eta) + L(\eta)\delta V + Q(\eta, \eta)V^\circ, \\ B_3(h, h, h) &= \frac{1}{2}L(\eta)I(\eta, \eta) + Q(\eta, \eta)\delta V, \quad B_4(h, h, h, h) = \frac{1}{2}Q(\eta, \eta)I(\eta, \eta), \end{aligned}$$

and B_2, B_3, B_4 denote the symmetric multilinear maps with these diagonals.

Lemma 19.7. *Let $\kappa = 1/24$. For all $h_i \in X_6$ and $k = 2, 3, 4$, $\|B_k(h_1, \dots, h_k)\|_\rho \leq c_k\|h_1\|_X \cdots \|h_k\|_X$, where*

$$c_2 = \max\{2P\theta_1\kappa, 2P\theta_1C\theta_0 + \frac{1}{2}P^2\theta_0^2 + \theta_1^2V_0\}, \quad c_3 = \max\{\theta_1^2\kappa, P\theta_1\theta_0^2 + \theta_1^2C\theta_0\}, \quad c_4 = \frac{1}{2}\theta_1^2\theta_0^2.$$

Proof. Write $a_i = \|\delta g_i\|_\rho$ and $e_i = \|\eta_i\|$. As in the proof of Lemma 14.2, $N_\rho(\psi'_{\eta_i}) \leq \theta_1 e_i$, $N_\rho(\psi_{\eta_i}) \leq \theta_0 e_i$, $\|\ell(\eta_i)\|_\rho \leq C\theta_0 e_i$, $\|I(\eta_i, \eta_k)\|_\rho \leq \theta_0^2 e_i e_k$, $\|\delta V_i\|_\rho \leq \kappa a_i + C\theta_0 e_i$, $\|L(\eta_i)f\|_\rho \leq 2P\theta_1 e_i \|f\|_\rho$ and $\|Q(\eta_i, \eta_k)f\|_\rho \leq \theta_1^2 e_i e_k \|f\|_\rho$. Hence

$$\|B_2(h_1, h_2)\|_\rho \leq P\theta_1\kappa(e_1 a_2 + e_2 a_1) + (\frac{1}{2}P^2\theta_0^2 + 2P\theta_1C\theta_0 + \theta_1^2V_0)e_1 e_2 \leq c_2(a_1 + e_1)(a_2 + e_2).$$

In the bound for B_3 , every monomial $e_i e_k a_l$ has coefficient $\theta_1^2\kappa/3$ and $e_1 e_2 e_3$ has coefficient $P\theta_1\theta_0^2 + \theta_1^2C\theta_0$; in the bound for B_4 , $e_1 e_2 e_3 e_4$ has coefficient $\frac{1}{2}\theta_1^2\theta_0^2$. Each is at most its coefficient in $c_k \prod_i (a_i + e_i)$. $\square$

Proposition 19.8. *Let $\mathcal{F}_6: X_6 \rightarrow \mathcal{S}_6$ have the form (19.4) with symmetric bounded B_2, B_3, B_4 , and let $A: \mathcal{S}_6 \rightarrow X_6$ be bounded, linear and injective. Suppose that $\|A\mathcal{F}_6(x^\circ)\|_X \leq Y$, $\|I - A D\mathcal{F}_6(x^\circ)\|_{X \rightarrow X} \leq Z$ and $\|A\| c_k \leq C_k$ for $k = 2, 3, 4$, and that $r > 0$ satisfies*

$$(19.6) \quad Y + (Z - 1)r + C_2 r^2 + C_3 r^3 + C_4 r^4 < 0, \quad Z - 1 + 2C_2 r + 3C_3 r^2 + 4C_4 r^3 < 0.$$

Then $\mathcal{F}_6$ has exactly one zero in the closed ball $\overline{B}_{X_6}(x^\circ, r)$.

Proof. This is the proof of Proposition 15.1, with one more term: the difference $B_4(h_1, \dots, h_1) - B_4(h_2, \dots, h_2)$ is a sum of four terms, each of which contains $h_1 - h_2$ once and three factors h_1 or h_2 , so that the norm of its image under A is at most $4C_4 r^3 \|h_1 - h_2\|_X$. $\square$

19.5. The principal part of the linearisation. In the model on high shape modes the map is replaced by the disc $\psi = bw$ and g by 0. We need the following radial expansion; here $S_{m,s} = r^m P_s^{(0,m)}(2r^2 - 1) \sin m\theta$ for any integer $m \geq 0$.

Lemma 19.9. *For every integer $m \geq 0$,*

$$r^{m+4} \sin m\theta = \frac{m+1}{m+3} S_{m,0} + \frac{2}{m+4} S_{m,1} + \frac{2}{(m+3)(m+4)} S_{m,2}.$$

Proof. Dividing by $r^m \sin m\theta$ and writing $t = r^2 = 1 + u$, the claim is $t^2 = \frac{m+1}{m+3} P_0 + \frac{2}{m+4} P_1 + \frac{2}{(m+3)(m+4)} P_2$ with $P_s = P_s^{(0,m)}(2t - 1)$. By (12.2), $P_0 = 1$, $P_1 = 1 + (m+2)u$ and $P_2 = 1 + 2(m+3)u + \frac{1}{2}(m+3)(m+4)u^2$. The right-hand side is therefore $c_0 + c_1 u + c_2 u^2$ with $c_2 = 1$, $c_1 = \frac{2(m+2)}{m+4} + \frac{4}{m+4} = 2$ and $c_0 = \frac{(m+1)(m+4)+2(m+3)+2}{(m+3)(m+4)} = 1$, which is $(1+u)^2 = t^2$. $\square$

Proposition 19.10. *For every $x = (g, c) \in X_6$ and every $j \equiv 1 \pmod{4}$,*

$$(19.7) \quad D\mathcal{F}_6(x)e_j = 2 \operatorname{Re}(jw^{j-1}\overline{\psi'_c})(Kg + h_c) + |\psi'_c|^2 \operatorname{Im}(\psi_c w^j).$$

Let $x_b = (0, b e_1)$, so that $\psi_{x_b}(w) = bw$. For every $j \equiv 1 \pmod{4}$ with $j \geq 5$,

$$(19.8) \quad T_j := D\mathcal{F}_6(x_b)e_j = b^3 \left\{ \frac{j+2}{2} S_{j+1,0} - \frac{j-2}{2} S_{j-3,0} - \frac{j}{j+1} S_{j-3,1} - \frac{1}{j+1} S_{j-3,2} \right\}.$$

Proof. Replacing c by $c + \varepsilon e_j$ changes ψ_c by εw^j ; since $\partial_\varepsilon |\psi'_c + \varepsilon j w^{j-1}|^2 = 2 \operatorname{Re}(j w^{j-1} \overline{\psi'_c})$ and $\partial_\varepsilon \frac{1}{2} \operatorname{Im}((\psi_c + \varepsilon w^j)^2) = \operatorname{Im}(\psi_c w^j)$ at $\varepsilon = 0$, we obtain (19.7). At x_b we have $g = 0$, $\psi'_c = b$ and $h_c = \frac{1}{2} b^2 \operatorname{Im}(w^2)$, so

$$T_j = b^3 [j \operatorname{Re}(w^{j-1}) \operatorname{Im}(w^2) + \operatorname{Im}(w^{j+1})] = b^3 [j r^{j+1} \cos((j-1)\theta) \sin 2\theta + r^{j+1} \sin((j+1)\theta)].$$

Since $2 \cos((j-1)\theta) \sin 2\theta = \sin((j+1)\theta) - \sin((j-3)\theta)$,

$$T_j = b^3 \left[\frac{j+2}{2} r^{j+1} \sin((j+1)\theta) - \frac{j}{2} r^{j+1} \sin((j-3)\theta) \right].$$

The first term is $\frac{j+2}{2} S_{j+1,0}$. Lemma 19.9 with $m = j-3 \geq 2$ gives

$$-\frac{j}{2} r^{j+1} \sin((j-3)\theta) = -\frac{j}{2} \left[\frac{j-2}{j} S_{j-3,0} + \frac{2}{j+1} S_{j-3,1} + \frac{2}{j(j+1)} S_{j-3,2} \right],$$

which is the rest of (19.8). $\square$

Both frequencies $j+1$ and $j-3$ in (19.8) are $\equiv 2 \pmod{4}$, so $T_j \in \mathcal{S}_6$. The harmonic part of T_j is $\alpha(j+2)$ in row $(j+1, 0)$ and $-\alpha(j-2)$ in row $(j-3, 0)$; this replaces the upper bidiagonal structure of Lemma 15.3 by a difference operator of step four.

19.6. Tail model and approximate inverse. The *finite inputs* are the 450 coordinates $g_{n,s}$ with $n \leq M$, $1 \leq s \leq S$, and the 15 shape coordinates c_j with $j \leq M-1$. The *finite rows* are the 450 rows (n, s) with $n \leq M$, $1 \leq s \leq S$, and the 15 harmonic rows $(n, 0)$ with $n \leq M$. All other inputs and rows are *tail* inputs and rows, and we write $X_6 = X_f \oplus X_t$, $\mathcal{S}_6 = \mathcal{S}_f \oplus \mathcal{S}_t$; $\dim X_f = \dim \mathcal{S}_f = 465$. Let M_f be the 465×465 matrix of the restriction of $D\mathcal{F}_6(x^\circ)$ to finite inputs and finite rows. On tail inputs we use the model

$$(19.9) \quad T(\delta g, \eta) = \delta g + \sum_{j \geq M+3} \eta_j T_j.$$

Write $Te = (Be, T_{tt}e)$ for tail inputs e , with B the part in finite rows. By (19.8) with $j = M+3 = 61$, only the shape input $j = M+3$ reaches finite rows:

$$(19.10) \quad B(\delta g, \eta) = -\eta_{M+3} \left(\alpha(M+1) e_{(M,0)} + \frac{b^3(M+3)}{M+4} e_{(M,1)} + \frac{b^3}{M+4} e_{(M,2)} \right).$$

Put $\tau_4 = (1 - \rho^{-4})^{-1}$.

Lemma 19.11. $T_{tt}: X_t \rightarrow \mathcal{S}_t$ is a bounded bijection with $\|T_{tt}\| \leq 1 + \rho^{-4}$. For a tail residual $y = \sum y_{n,s} S_{n,s} \in \mathcal{S}_t$ its inverse is given by

$$(19.11) \quad \eta_{n-1} = \sum_{k \geq 0} \frac{y_{n+4k,0}}{\alpha(n+4k+1)} \quad (n \geq M+4, n \equiv 2 \pmod{4}),$$

$$\delta g_{n,1} = y_{n,1} + \frac{b^3(n+3)}{n+4} \eta_{n+3}, \quad \delta g_{n,2} = y_{n,2} + \frac{b^3}{n+4} \eta_{n+3} \quad (n \geq M+4),$$

and $\delta g_{n,s} = y_{n,s}$ for all other tail g -inputs.

Proof. By (19.9) and (19.8), the tail harmonic rows read $\alpha(n+1)(\eta_{n-1} - \eta_{n+3}) = y_{n,0}$ for $n \geq M+4$, the tail rows $(n, 1)$ and $(n, 2)$ with $n \geq M+4$ read $\delta g_{n,1} - \frac{b^3(n+3)}{n+4} \eta_{n+3} = y_{n,1}$ and $\delta g_{n,2} - \frac{b^3}{n+4} \eta_{n+3} = y_{n,2}$, and all other tail rows read $\delta g_{n,s} = y_{n,s}$. Since $\sum \omega_j |\eta_j| < \infty$ forces $\eta_j \rightarrow 0$, summation of the harmonic equations gives (19.11), and a solution with $y = 0$ vanishes; conversely (19.11) solves the equations whenever the series converge, which is the case for every $y \in \mathcal{S}_t$ by the proof of Lemma 19.12 below (with $H = 0$). For $j \geq M+7$ the column $T_{tt}e_j$ has the four entries $\alpha(j+2)$, $-\alpha(j-2)$, $-b^3 j/(j+1)$ and $-b^3/(j+1)$ in

the rows $(j+1, 0)$, $(j-3, 0)$, $(j-3, 1)$ and $(j-3, 2)$; its weighted norm divided by ω_j is $1 + j\rho^{-4}/(j+2) \leq 1 + \rho^{-4}$. For $j = M+3$ only the first entry lies in a tail row, and tail g -inputs are mapped to the identical rows. Lemma 15.2 gives the bound for $\|T_{tt}\|$. $\square$

Let $\hat{A}$ be a 465×465 matrix with dyadic rational entries, regarded as an exact linear map $\mathcal{S}_f \rightarrow X_f$, such that

$$(19.12) \quad \|I - \hat{A}M_f\|_{X_f \rightarrow X_f} < 1,$$

and define

$$(19.13) \quad A = \begin{pmatrix} \hat{A} & -\hat{A}BT_{tt}^{-1} \\ 0 & T_{tt}^{-1} \end{pmatrix} : \mathcal{S}_f \oplus \mathcal{S}_t \rightarrow X_f \oplus X_t.$$

By Lemma 15.5 (Lemma 19.5(d)), A is bounded and injective, and $ATe = e$ for every tail input e . In the computation, $\hat{A}$ is a fixed binary64 approximation of M_f^{-1} , stored in the ancillary file `inverse_r6_M58_S30.npy`, whose entries are interpreted as exact dyadic rationals; only (19.12) is used.

Lemma 19.12. *Let*

$$\|R\| = \max_{(n,s) \text{ finite row}} \frac{\|\hat{A}e_{(n,s)}\|_X}{\rho^n},$$

$$H = \left\| \hat{A} \left(\alpha(M+1)e_{(M,0)} + \frac{b^3(M+3)}{M+4}e_{(M,1)} + \frac{b^3}{M+4}e_{(M,2)} \right) \right\|_X,$$

and, for $n \geq M+4$ with $n \equiv 2 \pmod{4}$,

$$(19.14) \quad h_n = \frac{1}{(n+1)\rho^n} \left(\sum_{\substack{M+4 \leq k \leq n \\ k \equiv 2(4)}} (k+1)\rho^k + 2 \sum_{\substack{M+4 \leq k \leq n-4 \\ k \equiv 2(4)}} \rho^k + \frac{H}{\alpha} \right).$$

Let $N_H \geq M+4$ with $N_H \equiv 2 \pmod{4}$. Then A maps every residual supported in tail rows with norm at most

$$A_t := \max \left\{ 1, \max_{M+4 \leq n \leq N_H} h_n, \tau_4 + \frac{2\tau_4\rho^{-4}}{N_H+5} + \frac{H}{\alpha(N_H+5)\rho^{N_H+4}} \right\},$$

it is the identity on residuals supported in nonharmonic tail rows, and $\|A\| \leq \max\{\|R\|, A_t\}$.

Proof. We use Lemma 15.2. A finite row $e_{(n,s)}$ is mapped to $\hat{A}e_{(n,s)}$. A nonharmonic tail row is mapped to the identical tail input by Lemma 19.11, and B vanishes on it. The normalised harmonic tail row $\rho^{-n}S_{n,0}$ is mapped to the shape coordinates $\eta_{k-1} = \rho^{-n}/(\alpha(n+1))$, $M+4 \leq k \leq n$, of weights $\omega_{k-1} = \alpha(k+1)\rho^k$; to the coordinates $\delta g_{k,1}$ and $\delta g_{k,2}$, $M+4 \leq k \leq n-4$, of weight ρ^k and total absolute value $b^3\rho^{-n}/(\alpha(n+1)) = 2\rho^{-n}/(n+1)$; and, by (19.10) and (19.13), to the finite part $\eta_{M+3}\hat{A}(\alpha(M+1)e_{(M,0)} + \dots)$, of norm $H\rho^{-n}/(\alpha(n+1))$. The sum of the three norms is exactly h_n . For $n \geq N_H+4$ we bound $(k+1)/(n+1) \leq 1$ in the first sum and sum the geometric series $\sum_{i \geq 0} \rho^{-4i} = \tau_4$; the second term is at most $2\tau_4\rho^{-4}/(n+1)$, and the third one decreases in n . $\square$

19.7. The defect bounds. We bound $\|I - A\mathcal{F}_6(x^\circ)\|$ as in Section 16, by partitioning the inputs into five classes; every input belongs to exactly one class, and no column is estimated by sampling.

The residual and the finite columns. $\mathcal{F}_6(x^\circ)$ is a polynomial in $w, \bar{w}$; its expansion in the basis $S_{n,s}$ (it has 2601 coefficients) is computed without truncation in interval arithmetic, using (12.3) and (13.1). Only finitely many tail rows occur, so $A\mathcal{F}_6(x^\circ)$ is computed without

truncation from (19.13), (19.10) and Lemma 19.11, and Y is an upper bound of its norm. For each of the 465 finite inputs e , $(I - AD\mathcal{F}_6(x^\circ))e$ is computed in the same way, and Z_f bounds the maximum of the normalised norms of these columns.

Tail g -columns.

Lemma 19.13. *Let $e = e_{(n,s)}$ be a tail g -input and $q = n + 2s$. Then $(I - AD\mathcal{F}_6(x^\circ))e = -A(|p|^2 K S_{n,s})$. Let $p = p_{\leq 24} + p_{> 24}$ be the splitting of p into terms of degree ≤ 24 and > 24 , and $P_s = N_\rho(p_{\leq 24})$, $P_t = N_\rho(p_{> 24})$, $\delta_a = 2P_s P_t + P_t^2$. Then*

$$(19.15) \quad \frac{\|(I - AD\mathcal{F}_6(x^\circ))e\|_X}{\rho^n} \leq \frac{\|A(|p_{\leq 24}|^2 K S_{n,s})\|_X}{\rho^n} + \frac{\|A\| \delta_a}{q(q+2)}.$$

Moreover, the right-hand side is at most $(A_t P_s^2 + \|A\| \delta_a)/(88 \cdot 90)$ if $n \geq M + 28 = 86$, and at most $(P_s^2 + \|A\| \delta_a)/(114 \cdot 116)$ if $s \geq S + 26 = 56$.

Proof. By (19.5), $D\mathcal{F}_6(x^\circ)e = S_{n,s} + |p|^2 K S_{n,s}$, and $Te = S_{n,s}$, so $AS_{n,s} = e$. The splitting is estimated as in Lemma 16.1. The function $|p_{\leq 24}|^2 = \sum_{a,a' \leq 24} p_a p_{a'} w^a \bar{w}^{a'}$ changes angular frequencies by at most 24 in modulus, and $\||p_{\leq 24}|^2 K S_{n,s}\|_\rho \leq P_s^2 \rho^n / (q(q+2))$. If $n \geq 86$, all components of $|p_{\leq 24}|^2 K S_{n,s}$ have frequencies of modulus at least $n - 24 \geq 62 > M$, so they lie in tail rows, on which A has norm at most A_t (Lemma 19.12); and $q \geq 88$. If $s \geq 56$, then by (13.1) and Lemma 12.3 all components have radial index at least $s - 1 - 24 \geq 31 > S$, so they lie in nonharmonic tail rows, on which A is the identity; and $q \geq 2 + 2 \cdot 56 = 114$. $\square$

The tail g -inputs with $n \leq 82$ and $s \leq 55$ are the $21 \cdot 55 - 450 = 705$ *near* ones; for them the first term of (19.15) is computed without truncation, and $Z_{g,\partial}$ bounds the maximum of the right-hand side. The remaining tail g -inputs are covered by the last statement of Lemma 19.13; let $Z_{g,\infty}$ be the larger of the two bounds.

Near shape columns. For the 85 shape inputs e_j with $M + 3 = 61 \leq j \leq 397$, we have $(I - AD\mathcal{F}_6(x^\circ))e_j = -A\Delta_j$ with $\Delta_j = D\mathcal{F}_6(x^\circ)e_j - T_j$, because $ATe_j = e_j$; Δ_j is a polynomial, and these columns are computed without truncation. Let $Z_{\text{sh},\partial}$ bound their normalised norms.

Far shape columns. Let $W^\circ = Kg^\circ$ and $h^\circ = h_{c^\circ}$. By (19.7) at x° and at x_b , for every $j \equiv 1 \pmod{4}$, $j \geq 5$,

$$(19.16) \quad \Delta_j = 2 \operatorname{Re}(jw^{j-1}\Gamma_1) + \operatorname{Im}(w^j\Gamma_0) + 2 \operatorname{Re}(jw^{j-1}\bar{p}W^\circ),$$

where

$$\Gamma_1 = \bar{p}h^\circ - \frac{1}{2}b^3 \operatorname{Im}(w^2), \quad \Gamma_0 = |p|^2 \psi_{c^\circ} - b^3 w.$$

By Lemma 13.1(c) (Lemma 19.5(b)), $W^\circ = (1 - |w|^2)^2 Q^\circ$ with

$$Q^\circ = \sum_{n,s} \frac{g_{n,s}^\circ}{4s(s+1)} r^n P_{s-1}^{(2,n)} (2r^2 - 1) \sin n\theta,$$

and the connection formulas of Section 16 give the finite expansion of Q° in the basis $S_{n,t}$; its connection coefficients are positive rational numbers. Let $\Lambda = M + d = 114$. The function $w^\Lambda \bar{p}Q^\circ$ has only nonnegative frequencies; write $w^\Lambda \bar{p}Q^\circ = \sum_{m \geq 0, t \geq 0} \gamma_{m,t}^W \Phi_{m,t}$ (a finite sum, computed in interval arithmetic). Let $j_* = 401$ and

$$U_\Gamma = \frac{1}{\alpha} \left(\frac{2\|\Gamma_1\|_\rho}{\rho^2} + \frac{\|\Gamma_0\|_\rho}{\rho(j_* + 2)} \right), \quad U_W = \frac{8}{\alpha(j_* - \Lambda)^2} \sum_{m,t} |\gamma_{m,t}^W| (2t+1)(2t+3) \rho^{m-\Lambda-2},$$

where $\|\cdot\|_\rho$ is the norm of $\mathcal{B}_\rho$.

Proposition 19.14. *For every $j \equiv 1 \pmod{4}$ with $j \geq j_*$,*

$$\frac{\|(I - AD\mathcal{F}_6(x^\circ))e_j\|_X}{\omega_j} \leq Z_{\text{sh},\infty} := A_t(\mathbf{U}_\Gamma + \mathbf{U}_W).$$

Proof. The frequencies of $\bar{p}$ lie in $[-d, 0] = [-56, 0]$, those of h° in $[-114, 114]$, since h° is a combination of $S_{j+k,0}$ with $j, k \leq 57$, and those of W° and Q° in $[-M, M]$. Hence all frequencies of $w^{j-1}\Gamma_1$, $w^j\Gamma_0$ and $w^{j-1}\bar{p}W^\circ$ are at least $j-1-170 \geq 230 > M$. Taking real and imaginary parts only changes the signs of frequencies, so all components of $\Delta_j \in \mathcal{S}_6$ lie in tail rows, and $\|A\Delta_j\|_X \leq A_t\|\Delta_j\|_\rho$ by Lemma 19.12. By Lemma 12.2 and $\|\operatorname{Re} f\|_\rho, \|\operatorname{Im} f\|_\rho \leq \|f\|_\rho$,

$$\frac{\|2\operatorname{Re}(jw^{j-1}\Gamma_1)\|_\rho}{\omega_j} \leq \frac{2j\rho^{j-1}\|\Gamma_1\|_\rho}{\alpha(j+2)\rho^{j+1}} \leq \frac{2\|\Gamma_1\|_\rho}{\alpha\rho^2}, \quad \frac{\|\operatorname{Im}(w^j\Gamma_0)\|_\rho}{\omega_j} \leq \frac{\|\Gamma_0\|_\rho}{\alpha\rho(j_*+2)}.$$

Finally $jw^{j-1}\bar{p}W^\circ = j(1-|w|^2)^2w^{j-1-\Lambda}(w^\Lambda\bar{p}Q^\circ)$ with $j-1-\Lambda \geq 0$, and Lemma 16.3 with $N = j-1-\Lambda$ gives

$$\frac{\|2\operatorname{Re}(jw^{j-1}\bar{p}W^\circ)\|_\rho}{\omega_j} \leq \frac{2j}{\alpha(j+2)\rho^{j+1}} \sum_{m,t} |\gamma_{m,t}^W| \frac{4(2t+1)(2t+3)\rho^{m+j-1-\Lambda}}{(m+j-\Lambda)^2} \leq \mathbf{U}_W.$$

Adding the three contributions proves the claim. $\square$

Remark 19.15. This far-shape bound is simpler and cruder than Proposition 16.4: it does not exploit the cancellation between $D\mathcal{F}_6(x^\circ)e_j$ and T_j in the harmonic rows, and $\mathbf{U}_\Gamma$ does not tend to zero as $j_* \rightarrow \infty$. It suffices here because Γ_1 and Γ_0 measure the deviation of the centre from the disc, which is small compared with $\alpha = b^3/2 \approx 1.2 \cdot 10^4$; the directly computed normalised column at $j = 397$ is about 0.082, while $Z_{\text{sh},\infty} < 0.465$ (Table 9).

Collecting the five classes, the number

$$Z = \max\{Z_f, Z_{g,\partial}, Z_{g,\infty}, Z_{\text{sh},\partial}, Z_{\text{sh},\infty}\}$$

bounds $\|I - AD\mathcal{F}_6(x^\circ)\|_{X \rightarrow X}$.

19.8. Certified bounds and the exact zero. The quantities defined above are finite expressions in the 465 dyadic centre coefficients, the entries of $\hat{A}$ and $\rho = 11/10$. They were evaluated in ball arithmetic at 128 bits as described in Section 34.5. Table 9 lists the results, rounded outward.

Theorem 19.16. *Let $r = 1/5000$. There is exactly one $x^* = (g^*, c^*) \in X_6$ with $\|x^* - x^\circ\|_X \leq r$ and $\mathcal{F}_6(x^*) = 0$.*

Proof. The support condition of Proposition 19.14, $j_* - 1 - 170 = 230 > M$, holds, and (19.12) holds by Table 9; hence A is bounded and injective. By Table 9, Proposition 19.8 applies with the rational majorants $Y = 53 \cdot 10^{-6}$, $Z = 47/100$, $C_2 = 251/1000$, $C_3 = 3 \cdot 10^{-7}$ and $C_4 = 3 \cdot 10^{-15}$. For these values and $r = 1/5000$, exact rational arithmetic gives

$$Y + (Z-1)r + C_2r^2 + C_3r^3 + C_4r^4 = -5.29899599999975999999999952 \cdot 10^{-5} < 0,$$

$$Z - 1 + 2C_2r + 3C_3r^2 + 4C_4r^3 = -0.5298995999999639999999999904 < 0. \quad \square$$

Remark 19.17 (The centre for $n = 6$). The centre x° was obtained numerically, and its provenance plays no role in the proof. A Fourier–Bessel collocation in $\mathbb{R}^6$, with u expanded in the regular Helmholtz solutions $r^{-2}J_{\ell+2}(\rho_*r)U_{\ell/2}(\cos 2\phi)$, $\ell \equiv 0 \pmod{4}$, at the ball wavenumber $\rho_* = j_{3,8}$ and seeded with the second-order amplitude predicted by the mechanism of Section 35.2, produced a numerical solution of (1.1) with Cauchy defects of order 10^{-13} . Its

Quantity	Where defined	Certified bound
$b = c_1^\circ$	Section 19	28.864750643300045... (exact dyadic)
P, C	Lemma 19.7	$< 31.143514, < 32.088747$
V_0	Lemma 19.7	< 7841.683303
$\ I - \hat{A}M_f\ $	(19.12)	$< 6.714 \cdot 10^{-11}$
$\ R\ $	Lemma 19.12	< 1402.313800
H	Lemma 19.12	$< 5.848182 \cdot 10^8$
A_t ($N_H = 458$)	Lemma 19.12	< 3.164016
$\ A\ $	Lemma 19.12	< 1402.314
Y	Section 19	$< 5.243310 \cdot 10^{-5}$
Z_f (465 columns)	Section 19	< 0.073227
P_s, P_t	Lemma 19.13	$< 31.143246, < 2.675771 \cdot 10^{-4}$
$\delta_a = 2P_s P_t + P_t^2$	Lemma 19.13	< 0.016667
$Z_{g,\partial}$ (705 columns, max. at (62, 1))	Lemma 19.13	< 0.313291
$Z_{g,\infty}$	Lemma 19.13	< 0.390424
$Z_{\text{sh},\partial}$ (85 columns, max. at $j = 61$)	Section 19	< 0.387180
$\ \Gamma_1\ _\rho, \ \Gamma_0\ _\rho$	Proposition 19.14	$< 844.817923, < 4110.324378$
$\sum_{m,t} \gamma_{m,t}^W (2t+1)(2t+3)\rho^{m-\Lambda-2}$	Proposition 19.14	< 3721574.315769
$\mathbf{U}_\Gamma, \mathbf{U}_W$	Proposition 19.14	$< 0.11689839, < 0.03005935$
$Z_{\text{sh},\infty}$ (all $j \geq 401$)	Proposition 19.14	< 0.464977
Z	Section 19	< 0.464977
$C_2 = \ A\ c_2$	Lemma 19.7	< 0.250135
$C_3 = \ A\ c_3$	Lemma 19.7	$< 2.7601 \cdot 10^{-7}$
$C_4 = \ A\ c_4$	Lemma 19.7	$< 2.1035 \cdot 10^{-15}$

TABLE 9. Certified bounds for $n = 6$, rounded outward, for $M = 58$, $S = 30$, $\rho = 11/10$, $j_* = 401$. The complete interval enclosures are recorded in the certificate `certificate_r6.json` and in the seven interval receipts (Section 34.5); P, C and V_0 are intermediate values of the final verification script, which uses $\|A\| \leq 1402.314$ in C_2, C_3, C_4 .

meridian domain was mapped conformally to the disc by Theodorsen's method, and Newton's method was applied to the Galerkin truncation of (19.3) to the 15 shape coefficients $c_1, c_5, \dots, c_{57}$ and the 450 coefficients of g described above, with residual about 10^{-10} in double precision. The resulting binary64 numbers were then frozen and are interpreted as exact dyadic rationals. (The wavenumber ρ_* used to locate the centre is unrelated to the weight $\rho = 11/10$ of the proof.)

19.9. Reconstruction, geometry and convexity. Let $x^* = (g^*, c^*)$ be the zero of Theorem 19.16, $\psi^* = \psi_{c^*}$ and $\delta c = c^* - c^\circ$. Then $\sum_j \omega_j |\delta c_j| \leq r$, and consequently

$$(19.17) \quad |c_1^* - b| \leq \frac{r}{\omega_1}, \quad \sum_j j \rho^{j-1} |\delta c_j| \leq \theta_1 r, \quad \sum_j \rho^j |\delta c_j| \leq \theta_0 r.$$

Proposition 19.18. *The map ψ^* is injective on the disc $\{|w| < 21/20\}$, and $\psi^{*'} has no zeros there. The domain $D = \psi^*(\mathbb{D})$ is bounded by the real-analytic Jordan curve $\psi^*(\partial\mathbb{D})$, it is strictly star-shaped with respect to 0, invariant under $z \mapsto \bar{z}$, $z \mapsto -\bar{z}$ and $z \mapsto i\bar{z}$, and it is not a disc.$*

Proof. Let $R = 21/20 < \rho$, $b_- = b - r/\omega_1 \leq c_1^*$, and

$$s_1 = \sum_{j \geq 5} j |c_j^\circ| R^{j-1} + \theta_1 r, \quad s_0 = \sum_{j \geq 5} |c_j^\circ| + \theta_0 r.$$

By (19.17) and $R < \rho$, $\sum_{j \geq 5} j |c_j^*| R^{j-1} \leq s_1$ and $\sum_{j \geq 5} |c_j^*| \leq s_0$. The computation (Section 34.5) certifies

$$(19.18) \quad b_- - s_1 > 27.2138856, \quad \frac{s_1 + s_0}{b_- - s_0} < 0.0634442, \quad |c_5^*| \geq |c_5^\circ| - \frac{r}{\omega_5} > 0.08529699.$$

Thus $\operatorname{Re} \psi^{*'}(w) \geq c_1^* - \sum_{j \geq 5} j |c_j^*| R^{j-1} > 27.21$ for $|w| \leq R$, which gives univalence on the convex disc $|w| < R$ exactly as in Proposition 18.1. For $|w| = 1$ we have $|\psi^*(w) - c_1^* w| \leq s_0$, $|w \psi^{*'}(w) - \psi^*(w)| \leq s_1$ and $|\psi^*(w)| \geq b_- - s_0 > 0$, hence $|w \psi^{*'} / \psi^* - 1| < 0.0635$; the argument of Proposition 18.1 shows that D is strictly star-shaped and that ∂D is a radial graph with positive real-analytic radius. The symmetries were shown at the beginning of this section. If D were a disc, its symmetries would force its centre to be 0, and the Schwarz lemma argument of Proposition 18.1 would show that ψ^* is linear, contradicting $c_5^* \neq 0$. $\square$

Proposition 19.19. *Let c be any real sequence indexed by $j \equiv 1 \pmod{4}$ with $\sum_j \omega_j |c_j - c_j^\circ| \leq r$, for instance $c = c^*$. Then for $|w| \leq 1$,*

$$|\psi'_c(w)| > 27.67533225, \quad |\psi''_c(w)| < 7.86831783, \quad \operatorname{Re} \left(1 + \frac{w \psi''_c(w)}{\psi'_c(w)} \right) > 0.71569200.$$

Consequently $D = \psi^*(\mathbb{D})$ is strictly convex, and $\Omega = \Pi_6^{-1}(D)$ is convex.

Proof. Let $A_1 = \sum_{j \geq 5} j |c_j^\circ|$, $A_2 = \sum_{j \geq 5} j(j-1) |c_j^\circ|$, $\delta = c - c^\circ$, and

$$E_1 = r \max \left\{ \frac{1}{\omega_1}, \frac{1}{\alpha \rho^6} \right\}, \quad E_2 = \frac{r}{\alpha \rho e \log \rho}.$$

We have $|\delta_1| + \sum_{j \geq 5} j |\delta_j| = \sum_j \kappa_j \omega_j |\delta_j| \leq r \sup_j \kappa_j$ with $\kappa_1 = 1/\omega_1$ and $\kappa_j = j/\omega_j = j/((j+2)\alpha\rho^{j+1}) < 1/(\alpha\rho^6)$ for $j \geq 5$; hence this sum is at most E_1 . For $j \geq 5$, $j(j-1)/\omega_j < j\rho^{-j}/(\alpha\rho) \leq 1/(\alpha\rho e \log \rho)$, since $x \mapsto x\rho^{-x}$ attains its maximum $1/(e \log \rho)$ on $(0, \infty)$ at $x = 1/\log \rho$; hence $\sum_{j \geq 5} j(j-1) |\delta_j| \leq E_2$. Both suprema are over all j , so the infinite tail of c is covered. For $|w| \leq 1$ it follows that

$$|\psi'_c(w)| \geq c_1 - \sum_{j \geq 5} j |c_j| \geq b - A_1 - E_1 =: L, \quad |\psi''_c(w)| \leq \sum_{j \geq 5} j(j-1) |c_j| \leq A_2 + E_2 =: U.$$

The computation certifies $A_1 < 1.18941839$, $A_2 < 7.86831777$, $E_1 < 9.39 \cdot 10^{-9}$ and $E_2 < 5.84 \cdot 10^{-8}$, hence $L > 27.67533225$, $U < 7.86831783$ and $\operatorname{Re}(1 + w \psi''_c / \psi'_c) \geq 1 - U/L > 0.71569200$.

For $c = c^*$, the argument of Proposition 18.2 (the analytic convexity criterion) shows that D is strictly convex. Now let $x = (x', x'')$, $z = (z', z'') \in \Omega$ and $\sigma \in [0, 1]$, and put $m = (m_1, m_2) = \sigma \Pi_6(x) + (1 - \sigma) \Pi_6(z)$. By convexity $m \in D$, and $m_1, m_2 \geq 0$. Since D is invariant under both reflections, it contains the four points $(\pm m_1, \pm m_2)$, and by convexity their convex hull, the rectangle $[-m_1, m_1] \times [-m_2, m_2]$. By the triangle inequality in each factor $\mathbb{R}^3$, $|\sigma x' + (1 - \sigma) z'| \leq m_1$ and $|\sigma x'' + (1 - \sigma) z''| \leq m_2$, so $\Pi_6(\sigma x + (1 - \sigma) z) \in D$, that is, $\sigma x + (1 - \sigma) z \in \Omega$. $\square$

19.10. Proof of Theorem 1.1 for $n = 6$. Let $V^* = Kg^* + h_{c^*}$. By Lemma 19.6, $V^* \in C^1(\overline{\mathbb{D}})$ satisfies (19.2) with $\psi = \psi^*$ and has the parities required in Lemma 19.4. By Proposition 19.18, ψ^* is injective with nonvanishing derivative on a neighbourhood of $\overline{\mathbb{D}}$. Lemma 19.4 shows that $F = V^* \circ (\psi^*)^{-1} \in C^1(\overline{D})$ is a weak solution of $\Delta F + F = 0$ in D with $F = y_1 y_2$ and $\nabla F = (y_2, y_1)$ on ∂D , odd in y_1 and in y_2 , and invariant under the exchange of y_1 and y_2 . By Lemma 18.3 and Lemma 18.4 in the form of Lemma 19.5(f), F extends to a real-analytic solution on an open neighbourhood of $\overline{D}$; replacing this neighbourhood by the connected component containing $\overline{D}$ of its intersection with its images under the two reflections and the exchange, we obtain a neighbourhood N with these symmetries on which, by the identity theorem, F keeps its parities.

Since D is strictly star-shaped and symmetric, ∂D meets each axis in exactly two points. Proposition 19.3 shows that $\Omega = \Pi_6^{-1}(D)$ and $u = F(|x'|, |x''|)/(|x'| |x''|)$, extended analytically to $\{x' = 0\} \cup \{x'' = 0\}$, satisfy (1.1), that u is real analytic on the neighbourhood $\Pi_6^{-1}(N)$ of $\overline{\Omega}$, and that u is nonconstant.

Geometry of Ω . Let $R_D(\phi)$ be the radial function of D . It is positive and real analytic, even (by $z \mapsto \bar{z}$) and invariant under $\phi \mapsto \pi - \phi$ (by $z \mapsto -\bar{z}$), hence π -periodic, and its Fourier series has the form $R_D(\phi) = \sum_{k \geq 0} \alpha_k \cos 2k\phi$, with exponentially decaying coefficients (by the exchange symmetry $\alpha_k = 0$ for odd k , which we do not need). As in the case $n = 4$, $\tilde{R}(\tau) = \sum_k \alpha_k T_k(\tau)$ converges on a complex neighbourhood of $[-1, 1]$ and defines a real-analytic function with $\tilde{R}(\cos 2\phi) = R_D(\phi)$, positive near $[-1, 1]$. For $x \neq 0$ we have $\Pi_6(x) = |x|(\cos \phi, \sin \phi)$ with $\phi \in [0, \pi/2]$ and $\cos 2\phi = (|x'|^2 - |x''|^2)/|x|^2$. Hence

$$\Omega = \{0\} \cup \left\{ x \neq 0 : |x| < \tilde{R}\left(\frac{|x'|^2 - |x''|^2}{|x|^2}\right) \right\},$$

$$\partial\Omega = \left\{ \tilde{R}(|\omega'|^2 - |\omega''|^2) \omega : \omega = (\omega', \omega'') \in S^5 \right\}.$$

Thus Ω is strictly star-shaped with respect to 0, in particular connected. The map $\omega \mapsto \tilde{R}(|\omega'|^2 - |\omega''|^2) \omega$ is real analytic and injective on S^5 , and, exactly as for $n = 4$, its differential maps a tangent vector $v \perp \omega$ to a vector whose component tangent to S^5 is $\tilde{R}(|\omega'|^2 - |\omega''|^2) v \neq 0$. Hence $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^5 . The symmetries of Ω follow from $\Omega = \Pi_6^{-1}(D)$ and the invariance of D under the exchange of coordinates.

Ω is not a ball. If it were, its centre would be fixed by every isometry of $\mathbb{R}^6$ preserving Ω , in particular by $O(3) \times O(3)$, whose only fixed point is 0. Then $\Pi_6(\Omega) = D \cap [0, \infty)^2$ would be a quarter of a disc centred at 0, and by the reflection symmetries D would be that disc, contradicting Proposition 19.18. Finally, Ω is convex by Proposition 19.19. This completes the proof of Theorem 1.1 for $n = 6$. $\square$

20. RANK-TWO LIE ALGEBRAS

The symmetry classes used for $n = 4$ and $n = 6$ are, up to finite index and the additional reflections, the adjoint representations of the compact Lie algebras $\mathfrak{u}(2)$ and $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, and the weights y and $y_1 y_2$ are the products of their positive roots. In this section we carry out the reduction for an arbitrary compact Lie algebra of rank two, and we use it to prove the following theorem for the three simple ones.

Theorem 20.1. *Let $(\mathfrak{k}, G)$ be one of $(\mathfrak{su}(3), SU(3))$, $(\mathfrak{so}(5), SO(5))$ and $(\mathfrak{g}_2, G_2)$, so that $n = \dim \mathfrak{k}$ equals 8, 10 and 14, respectively, and equip $\mathfrak{k}$ with an $\text{Ad}(G)$ -invariant inner product, so that $\mathfrak{k} \cong \mathbb{R}^n$ as a Euclidean space. There exist a bounded convex domain $\Omega \subset \mathfrak{k}$ and a nonconstant real-valued function u , real analytic on an open neighbourhood of $\overline{\Omega}$, such that (1.1) holds. Moreover, Ω is not a ball, it is strictly star-shaped with respect to 0, its boundary*

is a compact real-analytic hypersurface diffeomorphic to S^{n-1} , and Ω is invariant under $\text{Ad}(G)$ and under $X \mapsto -X$. Consequently the Fourier transform of $\mathbf{1}_\Omega$ vanishes on the unit sphere, Ω fails the Pompeiu property, and the Schiffer and Pompeiu conjectures fail in $\mathbb{R}^8$, $\mathbb{R}^{10}$ and $\mathbb{R}^{14}$ within the class of bounded convex domains with real-analytic boundary.

Theorem 20.1 contains the cases $n = 8, 10, 14$ of Theorem 1.1 and of Corollary 1.2. Only the adjoint action of G enters, so G may be replaced by any compact connected group with Lie algebra $\mathfrak{k}$, for instance $\text{Spin}(5)$ in place of $SO(5)$. For a simple $\mathfrak{k}$ the invariant inner product is unique up to a positive factor; changing it amounts to a dilation, which is absorbed by rescaling Ω and u . For $\mathfrak{k} = \mathfrak{su}(3)$, identified with the traceless Hermitian 3×3 matrices via $X \mapsto iX$, the domain is a spectral set: Ω consists of the matrices whose eigenvalue vector, a point of the plane $\{\lambda_1 + \lambda_2 + \lambda_3 = 0\} \cong \mathbb{R}^2$, lies in a fixed planar domain.

The proof follows the pattern of the cases $n = 4, 6$. Sections 20.1–20.4 reduce the problem, for every compact Lie algebra of rank two, to a planar Helmholtz problem on a Cartan section D with the Cauchy data of $\text{Im } \tau^m$, and show that the lifted domain is analytic, spherical and convex whenever D is. Sections 20.5–20.8 solve the planar problems for $m = 3, 4, 6$ by the conformal fixed-disc method, with a computer-assisted Newton–Kantorovich argument in a symmetry class that contains the cases $m = 1, 2$ of Sections 11–19 as special cases. Section 20.9 proves Theorem 20.1, Section 34.7 describes the computations, and Section 20.10 explains why the dimensions 4, 6, 8, 10, 14 are the only ones that this reduction can reach.

20.1. The radial part of the Laplacian. Let G be a compact connected Lie group with Lie algebra $\mathfrak{k}$ and an $\text{Ad}(G)$ -invariant inner product $\langle \cdot, \cdot \rangle$ on $\mathfrak{k}$. We identify $\mathfrak{k}$ with $\mathbb{R}^n$, $n = \dim \mathfrak{k}$, by an orthonormal basis, so that $\Delta_{\mathfrak{k}}$ is the Euclidean Laplacian. Let $\mathbb{T} \subset G$ be a maximal torus with Lie algebra $\mathfrak{t}$, and let $W = N_G(\mathbb{T})/\mathbb{T}$ be the Weyl group, which acts on $\mathfrak{t}$ by orthogonal maps. There is an orthogonal decomposition

$$\mathfrak{k} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi^+} \mathfrak{k}_\alpha, \quad \dim \mathfrak{k}_\alpha = 2,$$

indexed by a set Φ^+ of positive real roots, which are linear forms on $\mathfrak{t}$, such that $\text{ad}(\tau)$ acts on $\mathfrak{k}_\alpha$ as $\alpha(\tau)J_\alpha$ for $\tau \in \mathfrak{t}$, where J_α is a rotation through a right angle. We use the following standard facts [8, Ch. IV–V], [41, Ch. 1]: every $X \in \mathfrak{k}$ is $\text{Ad}(G)$ -conjugate to an element of $\mathfrak{t}$; two elements of $\mathfrak{t}$ are $\text{Ad}(G)$ -conjugate if and only if they are W -conjugate; the roots in Φ^+ are pairwise non-proportional; W is generated by the orthogonal reflections s_α in the walls $\ker \alpha$, $\alpha \in \Phi^+$; and the function

$$\varpi(\tau) = \prod_{\alpha \in \Phi^+} \alpha(\tau)$$

satisfies $\varpi \circ w = \det(w) \varpi$ for $w \in W$. Let $\mathfrak{t}_{\text{reg}} = \{\tau \in \mathfrak{t} : \varpi(\tau) \neq 0\}$. For $\tau \in \mathfrak{t}_{\text{reg}}$ the kernel of the skew-symmetric map $\text{ad } \tau$ is $\mathfrak{t}$, hence $[\mathfrak{k}, \tau] = \mathfrak{t}^\perp$.

Proposition 20.2 (Harish-Chandra). *The polynomial ϖ is harmonic on $\mathfrak{t}$. Let $U \subset \mathfrak{k}$ be open and $\text{Ad}(G)$ -invariant, let $u \in C^2(U)$ be $\text{Ad}(G)$ -invariant, and let $f = u|_{U \cap \mathfrak{t}}$. Then for every $\tau \in U \cap \mathfrak{t}_{\text{reg}}$,*

$$(20.1) \quad \nabla u(\tau) = \nabla_{\mathfrak{t}} f(\tau) \in \mathfrak{t}, \quad (\Delta_{\mathfrak{k}} u)(\tau) = \frac{\text{div}_{\mathfrak{t}}(\varpi^2 \nabla_{\mathfrak{t}} f)}{\varpi^2}(\tau) = \frac{\Delta_{\mathfrak{t}}(\varpi f)}{\varpi}(\tau).$$

The identity (20.1) is the flat case of Harish-Chandra’s radial-part formula [37]; see also [38, Ch. II]. We include the short proof.

Proof. The Laplacian commutes with orthogonal maps, so $\Delta\varpi$ is a W -anti-invariant polynomial of degree $|\Phi^+| - 2$. An anti-invariant polynomial vanishes on each wall $\ker \alpha$, since s_α fixes it pointwise, and is therefore divisible by α ; the roots being pairwise non-proportional, it is divisible by ϖ . Hence $\Delta\varpi = 0$.

For $Y \in \mathfrak{k}$ the function $s \mapsto u(\text{Ad}(\exp sY)\tau)$ is constant, so $\nabla u(\tau) \perp [Y, \tau]$; since $[\mathfrak{k}, \tau] = \mathfrak{t}^\perp$, $\nabla u(\tau) \in \mathfrak{t}$, and it equals $\nabla_{\mathfrak{t}} f(\tau)$. Let $\mathfrak{m} = \mathfrak{t}^\perp$ and $\Psi: G/\mathbb{T} \times \mathfrak{t}_{\text{reg}} \rightarrow \mathfrak{k}$, $\Psi(g\mathbb{T}, \tau) = \text{Ad}(g)\tau$. Its differential maps the tangent vector $(\frac{d}{ds}g \exp(sY)\mathbb{T}|_{s=0}, \tau')$, $Y \in \mathfrak{m}$, $\tau' \in \mathfrak{t}$, to $\text{Ad}(g)([Y, \tau] + \tau')$. As $[Y, \tau] \in \mathfrak{m} \perp \mathfrak{t}$ and $|[Y, \tau]|^2 = \sum_\alpha \alpha(\tau)^2 |Y_\alpha|^2$, where Y_α is the component of Y in $\mathfrak{k}_\alpha$, the map Ψ is a local diffeomorphism, and the pull-back of the Euclidean metric is the orthogonal sum of $|d\tau|^2$ and a metric on $G/\mathbb{T}$ whose volume density is $\prod_\alpha \alpha(\tau)^2 = \varpi(\tau)^2$ times a fixed smooth density. Since $u \circ \Psi(g\mathbb{T}, \tau) = f(\tau)$, the coordinate formula $\Delta = |\gamma|^{-1/2} \partial_i (|\gamma|^{1/2} \gamma^{ij} \partial_j)$ for this block-diagonal metric γ gives $(\Delta_{\mathfrak{k}} u)(\text{Ad}(g)\tau) = \varpi^{-2} \text{div}_{\mathfrak{t}}(\varpi^2 \nabla_{\mathfrak{t}} f)(\tau)$. Finally $\varpi^{-2} \text{div}(\varpi^2 \nabla f) = \Delta f + 2\varpi^{-1} \nabla \varpi \cdot \nabla f = \varpi^{-1}(\Delta(\varpi f) - f \Delta \varpi) = \varpi^{-1} \Delta(\varpi f)$. $\square$

Because ϖ is harmonic, (20.1) conjugates the Laplacian of $\mathfrak{k}$, acting on invariant functions, to the flat Laplacian of $\mathfrak{t}$ without any potential term; the normalisation of ϖ cancels.

Rank two. From now on $\mathfrak{k}$ has rank two, $\dim \mathfrak{t} = 2$. Let $m = |\Phi^+| \geq 1$. Then W is a finite group generated by the reflections in the m distinct lines $\ker \alpha$, that is, a dihedral group of order $2m$, which we denote by $I_2(m)$ (so $I_2(1)$ is generated by a single reflection). Since W preserves the lattice $\ker(\exp|_{\mathfrak{t}})$, its rotations have order 1, 2, 3, 4 or 6, so $m \in \{1, 2, 3, 4, 6\}$ [41], and $n = \dim \mathfrak{k} = 2 + 2m$. We identify $\mathfrak{t}$ with $\mathbb{C}$, $\tau = x + iy = |\tau|e^{i\theta}$, by an orthonormal basis such that the real axis is a wall. The walls are then the lines $e^{ik\pi/m}\mathbb{R}$, and

$$W = \{\tau \mapsto e^{2\pi ik/m} \tau, \tau \mapsto e^{2\pi ik/m} \bar{\tau} : 0 \leq k < m\}.$$

Since $\prod_{k=0}^{m-1} \sin(\theta - k\pi/m) = 2^{1-m} \sin m\theta$, the product of the positive roots is a nonzero multiple of $\text{Im } \tau^m$; as the normalisation of ϖ plays no role, we put

$$\varpi(\tau) = \text{Im } \tau^m = |\tau|^m \sin m\theta.$$

The W -invariant polynomials $p_1(\tau) = |\tau|^2$ and $p_2(\tau) = \text{Re } \tau^m$ separate W -orbits: $p(\tau) = p(\tau')$ for $p = (p_1, p_2)$ if and only if $\tau' \in W\tau$. By Chevalley's restriction theorem [38, Ch. II] there are $\text{Ad}(G)$ -invariant polynomials $\mathbf{P}_1(X) = |X|^2$ and $\mathbf{P}_2$ on $\mathfrak{k}$ with $\mathbf{P}_j|_{\mathfrak{t}} = p_j$; we write $\mathbf{P} = (\mathbf{P}_1, \mathbf{P}_2)$. A direct computation, using $\partial_x p_2 = m \text{Re } \tau^{m-1}$ and $\partial_y p_2 = -m \text{Im } \tau^{m-1}$, gives

$$(20.2) \quad \det Dp(\tau) = -2m \varpi(\tau).$$

The compact Lie algebras of rank two with $m \geq 1$ are listed in Table 10.

$\mathfrak{k}$	n	roots	W	m	treated in
$\mathfrak{u}(2) \cong \mathbb{R} \oplus \mathfrak{su}(2)$	4	A_1	$I_2(1) \cong \mathbb{Z}_2$	1	Sections 10–18
$\mathfrak{su}(2) \oplus \mathfrak{su}(2) \cong \mathfrak{so}(4)$	6	$A_1 \times A_1$	$I_2(2) \cong \mathbb{Z}_2^2$	2	Section 19
$\mathfrak{su}(3)$	8	A_2	$I_2(3) \cong S_3$	3	this section
$\mathfrak{so}(5) \cong \mathfrak{sp}(2)$	10	$B_2 = C_2$	$I_2(4)$	4	this section
$\mathfrak{g}_2$	14	G_2	$I_2(6)$	6	this section

TABLE 10. The compact Lie algebras of rank two with at least one root. In each case $n = 2 + 2m$, and in suitable orthonormal coordinates $\tau \in \mathfrak{t} \cong \mathbb{C}$ the product of the positive roots is a multiple of $\text{Im } \tau^m$.

Remark 20.3 (Dimensions four and six). For $\mathfrak{k} = \mathfrak{u}(2)$, the group $\text{Ad}(U(2))$ acts on $\mathfrak{su}(2) \cong \mathbb{R}^3$ as $SO(3)$ and trivially on the centre; $\mathfrak{t}$ is spanned by the centre and a line in $\mathfrak{su}(2)$, and in the coordinates $\tau = x_1 + iy$ of Section 10 one has $W = \{1, y \mapsto -y\}$, $\varpi = y$ and $p = (x_1^2 + y^2, x_1)$. For $\mathfrak{k} = \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, $\text{Ad}(G) = SO(3) \times SO(3)$ acts on $\mathbb{R}^3 \times \mathbb{R}^3$, W is generated by the reflections in the two axes of $\tau = y_1 + iy_2$, $\varpi = y_1 y_2 = \frac{1}{2} \text{Im } \tau^2$ and $p_2 = y_1^2 - y_2^2$. Accordingly, the maps Π and Π_6 of Sections 10 and 19 send X to its conjugate in the closed Weyl chamber; Lemmas 10.1 and 19.1 are the cases $m = 1, 2$ of Proposition 20.2; Lemmas 10.2 and 19.2 combine the cases $m = 1, 2$ of Lemmas 20.4 and 20.5 below; the groups $O(3) \times \mathbb{Z}_2$ and $(O(3) \times O(3)) \rtimes \mathbb{Z}_2$ used there contain $\text{Ad}(G)$ and realise, in addition, the enlarged symmetry $I_2(2m)$ of Section 20.4; and (14.1) and (19.3) are the cases $m = 1, B = 1$ and $m = 2, B = 2$ of the equation (20.7) below, with the same weights, the same principal parts (Lemma 15.3, Proposition 19.10) and the same tail inverses (Lemmas 15.4 and 19.11). For $m \geq 3$ the direct arguments of Sections 18 and 19 for the geometry and the convexity of the lifted domain are replaced by Lemmas 20.6 and 20.8, and the non-ball argument uses that $\mathfrak{k}$ is simple.

20.2. Lifting from the Cartan plane. In this subsection $\mathfrak{k}$ has rank two, with the notation introduced above. A function F on a W -invariant subset of $\mathfrak{t}$ is called *anti-invariant* if $F \circ w = \det(w)F$ for all $w \in W$.

Lemma 20.4 (Anti-invariant division). *Let $N \subset \mathfrak{t}$ be open and W -invariant, and let F be real analytic and anti-invariant on N . Then $F = \varpi q$ with q real analytic and W -invariant on N .*

Proof. Let $\ell_1, \dots, \ell_m$ be the walls and λ_k a unit linear form with $\ell_k = \ker \lambda_k$, so that $\varpi = c\lambda_1 \cdots \lambda_m$ with $c \neq 0$. Since the reflection in ℓ_k lies in W and has determinant -1 , F vanishes on $N \cap \ell_k$. Let $U \subset N$ be an open disc centred at a point τ_0 . If τ_0 lies on ℓ_k , write points of U in orthonormal coordinates (λ, μ) with $\lambda = \lambda_k$; since U contains the point $(0, \mu)$ together with (λ, μ) , Hadamard's formula

$$f(\lambda, \mu) = \lambda \int_0^1 \partial_\lambda f(\sigma \lambda, \mu) d\sigma$$

shows that every real-analytic f on U vanishing on $\ell_k \cap U$ is λ_k times a real-analytic function on U . If $\tau_0 \neq 0$, then τ_0 lies on at most one wall, and dividing F by that λ_k (if any) and by the other factors of ϖ , which do not vanish near τ_0 , shows that F/ϖ extends analytically to a neighbourhood of τ_0 . If $\tau_0 = 0$, then every wall passes through the centre of U . We divide F successively by $\lambda_1, \dots, \lambda_m$: after dividing by $\lambda_1, \dots, \lambda_{k-1}$, the quotient vanishes on $(\ell_k \cap U) \setminus \{0\}$, where the divisors are nonzero, hence on $\ell_k \cap U$ by continuity, and the next division is possible. Hence $q = F/\varpi$, defined on $N \cap \mathfrak{t}_{\text{reg}}$, extends to a real-analytic function on N . It is W -invariant, because F and ϖ are both anti-invariant. $\square$

For a W -invariant open set $N \subset \mathfrak{t}$ let $\text{Ad}(G)N = \{\text{Ad}(g)\tau : g \in G, \tau \in N\}$. For a W -invariant function q on N we define $q^\sharp$ on $\text{Ad}(G)N$ by $q^\sharp(X) = q(\tau)$ for any $\tau \in N$ conjugate to X ; this is well defined, since the elements of $\mathfrak{t}$ conjugate to X form one W -orbit.

Lemma 20.5 (Invariant lift). *Let $N \subset \mathfrak{t}$ be open and W -invariant, and let q be a W -invariant function on N .*

- (a) $\text{Ad}(G)N$ is open in $\mathfrak{k}$, $\text{Ad}(G)N \cap \mathfrak{t} = N$, and if q is continuous, so is $q^\sharp$.
- (b) If q is real analytic, then $q^\sharp$ is real analytic on $\text{Ad}(G)N \setminus \{0\}$. In particular, for $\tau \in N \cap \mathfrak{t}_{\text{reg}}$, (20.1) holds for $u = q^\sharp$ and $f = q$.

Proof. (a) If $\tau' \in \mathfrak{t}$ is conjugate to some $\tau \in N$, then $\tau' \in W\tau \subset N$; this gives $\text{Ad}(G)N \cap \mathfrak{t} = N$. Let $X_j \rightarrow X$ with $X = \text{Ad}(g)\tau$, $\tau \in N$, and write $X_j = \text{Ad}(g_j)\tau_j$ with $\tau_j \in \mathfrak{t}$. Given any subsequence, a further subsequence has $g_j \rightarrow g'$ and $\tau_j \rightarrow \tau'$ (as $|\tau_j| = |X_j|$ is bounded), and

then τ' is conjugate to X , so $\tau' \in N$. Hence $\tau_j \in N$ for large j , and $q^\sharp(X_j) = q(\tau_j) \rightarrow q(\tau') = q^\sharp(X)$ if q is continuous. Both assertions follow because the subsequence was arbitrary.

(b) Let $X_0 \in \text{Ad}(G)N \setminus \{0\}$, and let $\tau_0 \in N$ be conjugate to X_0 , so $\tau_0 \neq 0$. We first find an open disc $U \ni \tau_0$ in N and a real-analytic function Q on an open set containing $p(U)$ with $q = Q \circ p$ on U . If $\varpi(\tau_0) \neq 0$, then p is a local analytic diffeomorphism at τ_0 by (20.2), and $Q = q \circ (p|_U)^{-1}$ for a small U . Otherwise τ_0 lies on exactly one wall ℓ . In orthonormal coordinates (λ, μ) with $\ell = \{\lambda = 0\}$, the functions p and q are even in λ near τ_0 , because the reflection in ℓ belongs to W . As in the proof of Lemma 10.2, $p = \tilde{P}(\lambda^2, \mu)$ and $q = \tilde{q}(\lambda^2, \mu)$ near τ_0 with $\tilde{P}$ and $\tilde{q}$ real analytic near $(0, \mu_0)$. Then $\det Dp = 2\lambda \det D\tilde{P}(\lambda^2, \mu)$ in these coordinates, while $\det Dp = \pm 2m\varpi$ by (20.2), and near τ_0 one has $\varpi = \lambda\varpi_1$ with $\varpi_1(\tau_0) \neq 0$, since only one factor of ϖ vanishes at τ_0 . Hence $\det D\tilde{P}(0, \mu_0) \neq 0$, $\tilde{P}$ is a local analytic diffeomorphism, and $Q = \tilde{q} \circ \tilde{P}^{-1}$ has the required property.

By the argument in (a), every X close to X_0 is conjugate to some $\tau \in U$ (apply an element of W to a limit point conjugate to X_0). For such X , $q^\sharp(X) = q(\tau) = Q(p(\tau)) = Q(\mathbf{P}(X))$. Since $\mathbf{P}$ is polynomial, $q^\sharp$ is real analytic near X_0 . The last assertion follows from Proposition 20.2. $\square$

Lemma 20.6 (Lifting). *Let $\mathfrak{k}$ have rank two. Let $R: \mathbb{R} \rightarrow (0, \infty)$ be real analytic with $R(\theta) = R(-\theta) = R(\theta + 2\pi/m)$, and let*

$$D = \{re^{i\theta} : 0 \leq r < R(\theta)\} \subset \mathfrak{t}.$$

Let $N \supset \overline{D}$ be a W -invariant open subset of $\mathfrak{t}$, and let F be real analytic and anti-invariant on N and satisfy

$$(20.3) \quad \Delta F + F = 0 \text{ in } D, \quad F = \varpi \text{ and } \nabla F = \nabla \varpi \text{ on } \partial D.$$

Let q be the function of Lemma 20.4, so that $F = \varpi q$, let $\Omega = \text{Ad}(G)D$ and $u = q^\sharp$. Then Ω is a bounded domain, strictly star-shaped with respect to 0, with $\Omega \cap \mathfrak{t} = D$ and $\partial\Omega = \text{Ad}(G)\partial D$, and $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^{n-1} . The function u is real analytic on the open neighbourhood $\text{Ad}(G)N$ of $\overline{\Omega}$, it is nonconstant, and (1.1) holds.

Proof. The domain. The function R is even and $2\pi/m$ -periodic, so its Fourier series has the form $R(\theta) = \sum_{k \geq 0} a_k \cos(km\theta)$, with $|a_k| \leq C\varrho^{-k}$ for some $\varrho > 1$ because R is real analytic. As $|T_k| \leq \varrho_1^k$ on the Bernstein ellipse of parameter ϱ_1 , the series $R_T(s) = \sum_k a_k T_k(s)$, with T_k the Chebyshev polynomials, converges on a complex neighbourhood of $[-1, 1]$, and $R_T(\cos m\theta) = R(\theta)$. If $X \neq 0$ is conjugate to $\tau = |\tau|e^{i\theta}$, then $\mathbf{P}_2(X)/|X|^m = \text{Re } \tau^m/|\tau|^m = \cos m\theta \in [-1, 1]$. Hence

$$\mathbf{R}(X) = R_T(\mathbf{P}_2(X)/|X|^m), \quad X \in \mathfrak{k} \setminus \{0\},$$

is real analytic, positive, $\text{Ad}(G)$ -invariant and homogeneous of degree zero, and $\mathbf{R}(\tau) = R(\theta)$ for $\tau = |\tau|e^{i\theta}$. Consequently

$$\Omega = \{0\} \cup \{X \neq 0 : |X| < \mathbf{R}(X)\}, \quad \partial\Omega = \{\mathbf{R}(\omega)\omega : \omega \in S^{n-1}\} = \text{Ad}(G)\partial D,$$

and $\Omega \cap \mathfrak{t} = D$. Thus Ω is open, bounded and strictly star-shaped with respect to 0, hence connected. Exactly as for $n = 4$ (Section 18), the map $\omega \mapsto \mathbf{R}(\omega)\omega$ is an injective real-analytic immersion of S^{n-1} with inverse $X \mapsto X/|X|$, so $\partial\Omega$ is a compact real-analytic hypersurface diffeomorphic to S^{n-1} . Moreover $\overline{\Omega} = \text{Ad}(G)\overline{D} \subset \text{Ad}(G)N$, and $\text{Ad}(G)N$ is open by Lemma 20.5(a).

The equation. By Lemma 20.5, u is continuous on $\text{Ad}(G)N$ and real analytic on $\text{Ad}(G)N \setminus \{0\}$. For $\tau \in D \cap \mathfrak{t}_{\text{reg}}$, (20.1) gives $\Delta u(\tau) = \varpi^{-1} \Delta(\varpi q)(\tau) = \varpi^{-1} \Delta F(\tau) = -u(\tau)$; since Δ commutes with the orthogonal maps $\text{Ad}(g)$, $\Delta u + u = 0$ on $\Omega \cap \mathfrak{t}_{\text{reg}}$, where $\mathfrak{t}_{\text{reg}} = \text{Ad}(G)\mathfrak{t}_{\text{reg}}$. The polynomial ϖ^2 is W -invariant, so by Chevalley's theorem $\mathfrak{t}_{\text{reg}}$ is the complement of the

zero set of an $\text{Ad}(G)$ -invariant polynomial, which is not identically zero; hence $\mathfrak{k}_{\text{reg}}$ is dense, and by continuity $\Delta u + u = 0$ on $\Omega \setminus \{0\}$. The point 0 lies in Ω , because $R > 0$. Let $U_0 \subset \Omega$ be a ball about 0, $\varphi \in C_c^\infty(U_0)$, and $\chi_\varepsilon(X) = \chi(|X|/\varepsilon)$ with $\chi \in C^\infty(\mathbb{R})$, $\chi = 0$ on $(-\infty, 1]$ and $\chi = 1$ on $[2, \infty)$. Then $\int u(\Delta + 1)(\chi_\varepsilon \varphi) = 0$, and $(\Delta + 1)(\chi_\varepsilon \varphi) - \chi_\varepsilon(\Delta + 1)\varphi = 2\nabla \chi_\varepsilon \cdot \nabla \varphi + \varphi \Delta \chi_\varepsilon$ is supported in $\{|X| \leq 2\varepsilon\}$ and bounded by $C\varepsilon^{-2}$. Since u is bounded near 0, letting $\varepsilon \rightarrow 0$ gives $\int_{U_0} u(\Delta + 1)\varphi = 0$ ($n \geq 3$). The proof of Lemma 18.3 applies verbatim in $\mathbb{R}^n$ and shows that the continuous weak solution u is real analytic on U_0 . Hence u is real analytic on $\text{Ad}(G)N$ and $\Delta u + u = 0$ in Ω .

The boundary conditions. Let $\tau \in \partial D$ with $\varpi(\tau) \neq 0$. Then $u(\tau) = q(\tau) = F(\tau)/\varpi(\tau) = 1$, and by (20.1) $\nabla u(\tau) = \nabla q(\tau) = \varpi^{-1}(\nabla F - q\nabla \varpi)(\tau) = 0$. By invariance, $u = 1$ and $\nabla u = 0$ on $\text{Ad}(G)(\partial D \cap \mathfrak{k}_{\text{reg}})$. Since ∂D meets the m walls in $2m$ points, $\partial D \cap \mathfrak{k}_{\text{reg}}$ is dense in ∂D , and since $(g, \tau) \mapsto \text{Ad}(g)\tau$ maps $G \times \partial D$ continuously onto $\partial \Omega$, the set $\text{Ad}(G)(\partial D \cap \mathfrak{k}_{\text{reg}})$ is dense in $\partial \Omega$. By continuity $u = 1$ and $\nabla u = 0$ on $\partial \Omega$. A constant solution of $\Delta u + u = 0$ vanishes, which contradicts $u = 1$ on $\partial \Omega$. $\square$

Remark 20.7 (The radial-graph hypothesis). Some hypothesis on D beyond W -invariance and analyticity of ∂D is necessary. For the annulus $D = \{a < |\tau| < b\}$, which is W -invariant with real-analytic boundary, $\text{Ad}(G)D = \{a < |X| < b\}$ is a spherical shell, whose boundary consists of two spheres. Spherical shells do fail the Pompeiu property (Section 1), so a topological hypothesis of this kind cannot be dispensed with in any statement of this type. The positive analytic radial function in Lemma 20.6 is what makes $\partial \Omega$ a sphere, including at the points where ∂D meets the walls. In our applications it is supplied by strict convexity together with $0 \in D$ (Proposition 20.21).

20.3. Convexity. Kostant's convexity theorem [48] is usually stated for a real reductive Lie algebra $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{p}$ with Cartan decomposition, the group H acting on $\mathfrak{p}$, and a maximal abelian subspace $\mathfrak{a} \subset \mathfrak{p}$; in this form it reads: for $a \in \mathfrak{a}$, the orthogonal projection of $H \cdot a$ onto $\mathfrak{a}$ is the convex hull of the orbit of a under the Weyl group of $(\mathfrak{g}, \mathfrak{a})$ [49, Theorem 2.2]. The adjoint representation of a compact $\mathfrak{k}$ is of this type. Let $\mathfrak{g} = \mathfrak{k}_{\mathbb{C}} = \mathfrak{k} \oplus i\mathfrak{k}$, regarded as a real Lie algebra, with the Cartan involution $X + iY \mapsto X - iY$, so that $\mathfrak{h} = \mathfrak{k}$ and $\mathfrak{p} = i\mathfrak{k}$. The map $X \mapsto iX$ is a G -equivariant linear isometry of $\mathfrak{k}$ onto $\mathfrak{p}$ (with $\langle iX, iY \rangle := \langle X, Y \rangle$), because $\text{Ad}(g)$ is complex linear on $\mathfrak{k}_{\mathbb{C}}$. It maps $\mathfrak{k}$ onto the maximal abelian subspace $\mathfrak{a} = i\mathfrak{t}$, and the Weyl group $N_G(\mathfrak{a})/Z_G(\mathfrak{a}) = N_G(\mathbb{T})/\mathbb{T}$ of $(\mathfrak{g}, \mathfrak{a})$ acts on $\mathfrak{a} \cong \mathfrak{t}$ as W [45, Ch. VI]. Hence, for a compact semisimple $\mathfrak{k}$ and $\tau \in \mathfrak{k}$, the orthogonal projection $\text{pr}: \mathfrak{k} \rightarrow \mathfrak{t}$ satisfies

$$(20.4) \quad \text{pr}(\text{Ad}(G)\tau) = \text{conv}(W\tau).$$

Lemma 20.8 (Convexity transfer). *Let $\mathfrak{k}$ be compact and semisimple, of any rank, let $D \subset \mathfrak{t}$ be W -invariant, and $\Omega = \text{Ad}(G)D$. Then $\Omega \cap \mathfrak{t} = D$, and Ω is convex if and only if D is convex.*

Proof. If $\tau' \in \mathfrak{t}$ is conjugate to $\tau \in D$, then $\tau' \in W\tau \subset D$; hence $\Omega \cap \mathfrak{t} = D$, and if Ω is convex, so is D . Conversely, let D be convex. By (20.4), $\text{pr}(\Omega) = \bigcup_{\tau \in D} \text{conv}(W\tau) = D$, since $W\tau \subset D$ and $\tau \in \text{conv}(W\tau)$. The convex hull $\text{conv} \Omega$ is $\text{Ad}(G)$ -invariant, as each $\text{Ad}(g)$ is linear, and $\text{pr}(\text{conv} \Omega) = \text{conv}(\text{pr} \Omega) = D$. As pr is the identity on $\mathfrak{t}$, $\text{conv} \Omega \cap \mathfrak{t} \subset D$. Every element of $\text{conv} \Omega$ is conjugate to an element of $\text{conv} \Omega \cap \mathfrak{t}$, hence $\text{conv} \Omega \subset \text{Ad}(G)D = \Omega$, and Ω is convex. $\square$

This is the case of adjoint representations of [49, Theorem 3.6], which holds for arbitrary invariant sets. No closedness or openness of D is needed.

20.4. The reduced problem and the balls. By Lemmas 20.6 and 20.8, Theorem 20.1 follows once we find a W -invariant domain D with a positive real-analytic radial function, which is convex and not a disc, and an anti-invariant real-analytic F on a W -invariant neighbourhood of $\bar{D}$ that satisfies (20.3). The reduction is reversible in the analytic class: if $\Omega = \text{Ad}(G)D$ and an $\text{Ad}(G)$ -invariant u , real analytic on a neighbourhood of $\bar{\Omega}$, satisfy (1.1), then $F = \varpi \cdot u|_{\mathfrak{t}}$ is anti-invariant and satisfies (20.3), by (20.1); any solution can be made invariant by averaging over G , as in Proposition 10.4.

Balls. Let $D = \{|\tau| < R\}$, so that Ω is the ball of radius R . The function $F = J_m(|\tau|) \sin m\theta$ solves $\Delta F + F = 0$ and is anti-invariant, and $F/\varpi = |\tau|^{-m} J_m(|\tau|)$ is the regular radial Helmholtz solution in $\mathbb{R}^{2m+2}$. Since $J'_m(r) = \frac{m}{r} J_m(r) - J_{m+1}(r)$, a multiple of F satisfies (20.3) if and only if $J_{m+1}(R) = 0$ (then $J_m(R) \neq 0$, and the multiple is $R^m F/J_m(R)$), in agreement with the Schiffer balls of $\mathbb{R}^n$ at eigenvalue one, whose radii are the zeros of $J_{n/2} = J_{m+1}$.

The enlarged symmetry. We shall look for D invariant under the dihedral group

$$I_2(2m) = \langle \tau \mapsto \bar{\tau}, \tau \mapsto e^{i\pi/m} \tau \rangle \supset W,$$

and for F with $F(\bar{\tau}) = -F(\tau)$ and $F(e^{i\pi/m} \tau) = -F(\tau)$; then $F \circ w = \det(w)F$ for $w \in W$, and ϖ transforms in the same way. The regular Helmholtz solutions in this class are the series $\sum_{k \geq 0} a_k J_{(2k+1)m}(r) \sin((2k+1)m\theta)$. Since $-1 \in I_2(2m)$, the lifted domain is centrally symmetric; for $m = 3$ the rotation through $\pi/3$ is the composition of -1 with an element of W , while for $m = 4, 6$ one has $-1 \in W$ already. For $m = 1, 2$ the enlarged symmetry is that of Sections 10 and 19.

Near-resonances. By Proposition 20.2, an $\text{Ad}(G)$ -invariant harmonic polynomial h of degree ℓ on $\mathfrak{k}$ corresponds to the anti-invariant harmonic polynomial $\varpi \cdot h|_{\mathfrak{t}}$ of degree $\ell + m$. Anti-invariance under $\tau \mapsto \bar{\tau}$ and invariance under the rotation through $2\pi/m$ force this polynomial to be a multiple of $\text{Im } \tau^{\ell+m}$ with $m \mid \ell$. Thus the invariant spherical harmonics restrict on $\mathfrak{t}$ to multiples of $U_{\ell/m}(\cos m\theta)$, where U_k is the Chebyshev polynomial of the second kind, and the planar mode $J_{m+\ell}(r) \sin((m+\ell)\theta)$ corresponds to the Helmholtz solution $r^{-m} J_{m+\ell}(r) U_{\ell/m}(\cos m\theta)$ in $\mathbb{R}^{2m+2}$; the enlarged symmetry allows $\ell \in 2m\mathbb{Z}$. The linearisation of the Schiffer problem at the ball of radius R , $J_{m+1}(R) = 0$, in the direction of such a harmonic is degenerate if and only if $J_{m+\ell}(R) = 0$, which is impossible for $\ell \geq 2$ by Siegel's theorem [62], as in Section 35.2. (The case $\ell = 1$ occurs only for $m = 1$, where it corresponds to translations along the centre of $\mathfrak{u}(2)$.) The centres used below were located at the near-resonances

$$(m, R, \ell) = (3, j_{4,5}, 6), \quad (4, j_{5,10}, 8), \quad (6, j_{7,25}, 12),$$

with $j_{4,5} \approx 20.826933$, $j_{5,10} \approx 38.159869$, $j_{7,25} \approx 88.474363$ and small divisors $J_9(j_{4,5}) \approx -3.3 \cdot 10^{-3}$, $J_{12}(j_{5,10}) \approx -4.4 \cdot 10^{-4}$, $J_{18}(j_{7,25}) \approx -9.0 \cdot 10^{-4}$ (Remark 20.20).

20.5. The conformal formulation for general m . Fix $m \geq 1$ and $\rho > 1$. For a real sequence $c = (c_j)_{j \equiv 1 \pmod{2m}}$, $j \geq 1$, put $\psi_c(w) = \sum_j c_j w^j$. Since $e^{i\pi j/m} = e^{i\pi/m}$ for $j \equiv 1 \pmod{2m}$,

$$(20.5) \quad \psi_c(\bar{w}) = \overline{\psi_c(w)}, \quad \psi_c(e^{i\pi/m} w) = e^{i\pi/m} \psi_c(w);$$

hence, if ψ_c is injective on $\bar{\mathbb{D}}$, then $D = \psi_c(\mathbb{D})$ is invariant under $I_2(2m)$.

For $n \equiv m \pmod{2m}$, $n \geq m$, and $s \geq 0$ let $S_{n,s} = \text{Im } \Phi_{n,s} = r^n P_s^{(0,n)}(2r^2 - 1) \sin n\theta$, and let $\mathcal{S}_m$ be the real Banach space of $f = \sum_{n \equiv m \pmod{2m}, s \geq 0} f_{n,s} S_{n,s}$ with $\|f\|_\rho = \sum \rho^n |f_{n,s}| < \infty$, which is also its norm in $\mathcal{B}_\rho$. Since $\Phi_{k,s}(\bar{w}) = \Phi_{-k,s}(w)$ and $\Phi_{k,s}(e^{i\pi/m} w) = e^{ik\pi/m} \Phi_{k,s}(w)$, and $e^{ik\pi/m} = -1$ exactly when $k \equiv m \pmod{2m}$, the real-valued elements $f \in \mathcal{B}_\rho$ with $f(\bar{w}) = -f(w)$ and $f(e^{i\pi/m} w) = -f(w)$ are exactly the elements of $\mathcal{S}_m$. Let $\mathcal{G}_m = \{g \in \mathcal{S}_m :$

$g_{n,0} = 0$ for all n }, and define K on $\mathcal{G}_m$ by (13.1). For $m = 1$ and $m = 2$ these are the spaces $\mathcal{S}, \mathcal{G}$ of Section 12 and $\mathcal{S}_6, \mathcal{G}_6$ of Section 19. Put

$$\kappa_m = \frac{1}{(m+2)(m+4)}.$$

Lemma 20.9 (Results used verbatim). *The statements of Lemma 19.5(a)–(f) hold with $\mathcal{S}_6, \mathcal{G}_6$ and the condition $n \equiv 2 \pmod{4}$ replaced by $\mathcal{S}_m, \mathcal{G}_m$ and $n \equiv m \pmod{2m}$, with $1/24$ replaced by κ_m , and with the Cauchy datum $(y_1 y_2, (y_2, y_1))$ in (f) replaced by $(\varpi, \nabla \varpi)$. In particular $K: \mathcal{G}_m \rightarrow \mathcal{S}_m$ is bounded with $\|K\| = \kappa_m$, the supremum of $1/(q(q+2))$ over $q = n+2s \geq m+2$ being attained at $(n, s) = (m, 1)$.*

Proof. Only (f) needs a comment. In the coordinates $\zeta = x + iy$, $\chi = x - iy$ one has $\varpi = (\zeta^m - \chi^m)/(2i)$, whose holomorphic extension $F_0(a, b) = (\gamma(a)^m - \gamma^\dagger(b)^m)/(2i)$ in the proof of Lemma 18.4 satisfies $\partial_a \partial_b F_0 = 0$; the rest of that proof is unchanged. $\square$

Let $B > 0$ (a fixed *field scale*) and $b > 0$ (in the application, $b = c_1^\circ$), and put

$$\alpha = \frac{b^{m+1}}{B}, \quad \omega_j = \alpha(j+m)\rho^{j+m-1}, \quad \theta_0 = \frac{1}{\alpha(m+1)\rho^{m-1}}, \quad \theta_1 = \frac{1}{\alpha\rho^m}.$$

Let X_m be the real Banach space of pairs $x = (g, c)$, with $g \in \mathcal{G}_m$ and $c = (c_j)_{j \equiv 1 \pmod{2m}}$ real, normed by $\|(g, c)\|_X = \|g\|_\rho + \|c\|$, $\|c\| = \sum_j \omega_j |c_j|$. Since $\rho^j/\omega_j = 1/(\alpha(j+m)\rho^{m-1})$ and $j\rho^{j-1}/\omega_j = j/(\alpha(j+m)\rho^m)$,

$$(20.6) \quad N_\rho(\psi_c) \leq \theta_0 \|c\|, \quad N_\rho(\psi'_c) \leq \theta_1 \|c\|.$$

The powers of w in ψ_c^m are $\equiv m \pmod{2m}$, so $h_c = \text{Im}(\psi_c^m)/B$ is a harmonic element of $\mathcal{S}_m$. Define

$$(20.7) \quad \mathcal{H}_m(g, c) = g + |\psi'_c|^2(Kg + h_c), \quad h_c = \frac{\text{Im}(\psi_c^m)}{B}.$$

For $m = 1$, $B = 1$ this is (14.1), with $\omega_j = b^2(j+1)\rho^j$, $\theta_0 = \vartheta_0$ and $\theta_1 = \vartheta_1$; for $m = 2$, $B = 2$ it is (19.3), with the weights of Section 19.

Lemma 20.10. *$\mathcal{H}_m$ is a continuous polynomial map of degree $m+2$ from X_m to $\mathcal{S}_m$. If $\mathcal{H}_m(g, c) = 0$, then $V = Kg + h_c$ belongs to $C^1(\mathbb{D}) \cap \mathcal{S}_m$, and it satisfies $\Delta V + |\psi'_c|^2 V = 0$ in $\mathcal{D}'(\mathbb{D})$ and $V - h_c = 0$, $\nabla(V - h_c) = 0$ on $\partial\mathbb{D}$.*

Proof. ψ'_c has real coefficients and only powers divisible by $2m$, so $|\psi'_c|^2$ is real, invariant under $w \mapsto \bar{w}$ and $w \mapsto e^{i\pi/m}w$, and multiplication by it maps $\mathcal{S}_m$ into itself with norm at most $N_\rho(\psi'_c)^2$ by (12.4). Moreover $\|h_c\|_\rho \leq N_\rho(\psi_c)^m/B$, since $w^k = \Phi_{k,0}$ has norm ρ^k and N_ρ is submultiplicative. With (20.6) and $\|K\| = \kappa_m$, (20.7) is a sum of bounded multilinear maps of degree at most $m+2$. The rest is the proof of Lemma 14.1, using Lemma 20.9. $\square$

Lemma 20.11. *Let ψ be holomorphic and injective on a neighbourhood of $\mathbb{D}$ with $\psi' \neq 0$ there, $D = \psi(\mathbb{D})$, and let $V \in C^1(\mathbb{D})$ be real valued with $\Delta V + |\psi'|^2 V = 0$ in $\mathcal{D}'(\mathbb{D})$ and $V - h = 0$, $\nabla(V - h) = 0$ on $\partial\mathbb{D}$, where $h = \text{Im}(\psi^m)/B$. Then $F = BV \circ \psi^{-1} \in C^1(\bar{D})$ satisfies $\int_D (\nabla F \cdot \nabla \varphi - F \varphi) dA = 0$ for all $\varphi \in C_c^\infty(D)$, and $F = \varpi$, $\nabla F = \nabla \varpi$ on ∂D . If moreover $\psi = \psi_c$ and $V \in \mathcal{S}_m$, then $F(\bar{\tau}) = -F(\tau)$ and $F(e^{i\pi/m}\tau) = -F(\tau)$.*

Proof. Since $\varpi \circ \psi = \text{Im} \psi^m = Bh$, we have $B(V - h) = (F - \varpi) \circ \psi$, and the proof of Lemma 11.1 applies. The parities follow from (20.5) and the characterisation of $\mathcal{S}_m$. $\square$

Expansion at a centre. Let $x^\circ = (g^\circ, c^\circ) \in X_m$ have finitely many nonzero coefficients, $b = c_1^\circ$, and put

$$p = \psi'_{c^\circ}, \quad P = N_\rho(p), \quad C = N_\rho(\psi_{c^\circ}), \quad V^\circ = Kg^\circ + h_{c^\circ}, \quad V_0 = \|V^\circ\|_\rho.$$

For $h = (\delta g, \eta) \in X_m$, $|\psi'_{c^\circ + \eta}|^2 = |p|^2 + L(\eta) + Q(\eta, \eta)$ with $L(\eta) = 2 \operatorname{Re}(\bar{p} \psi'_\eta)$ and $Q(\eta, \eta') = \operatorname{Re}(\psi'_\eta \overline{\psi'_{\eta'}})$, and $h_{c^\circ + \eta} = h_{c^\circ} + \sum_{k=1}^m H_k(\eta)$ with $H_k(\eta) = \binom{m}{k} \operatorname{Im}(\psi_{c^\circ}^{m-k} \psi_\eta^k) / B$. Hence

$$(20.8) \quad D\mathcal{H}_m(x^\circ)h = \delta g + |p|^2(K\delta g + H_1(\eta)) + L(\eta)V^\circ,$$

and $\mathcal{H}_m(x^\circ + h) = \mathcal{H}_m(x^\circ) + D\mathcal{H}_m(x^\circ)h + \sum_{q=2}^{m+2} B_q(h, \dots, h)$, where the degree- q part consists of the terms $|p|^2 H_q(\eta)$ ($q \leq m$), $L(\eta)H_{q-1}(\eta)$ ($2 \leq q \leq m+1$), $Q(\eta, \eta)H_{q-2}(\eta)$ ($3 \leq q \leq m+2$), $L(\eta)K\delta g + Q(\eta, \eta)V^\circ$ ($q = 2$) and $Q(\eta, \eta)K\delta g$ ($q = 3$), and B_q is the symmetric q -linear map with this diagonal.

Lemma 20.12. *Let $h_k = \binom{m}{k} C^{m-k} \theta_0^k / B$ for $1 \leq k \leq m$, and $h_k = 0$ otherwise. For $2 \leq q \leq m+2$ and all $h_i \in X_m$, $\|B_q(h_1, \dots, h_q)\|_\rho \leq c_q \|h_1\|_X \cdots \|h_q\|_X$, where*

$$(20.9) \quad c_q = P^2 h_q + 2P\theta_1 h_{q-1} + \theta_1^2 h_{q-2} + \mathbf{1}_{q=2}(2P\theta_1 \kappa_m + \theta_1^2 V_0) + \mathbf{1}_{q=3} \theta_1^2 \kappa_m.$$

Proof. By (20.6), (12.4), $\|\operatorname{Im} f\|_\rho \leq \|f\|_\rho$ and submultiplicativity of N_ρ ,

$$\|L(\eta)f\|_\rho \leq 2P\theta_1 \|\eta\| \|f\|_\rho, \quad \|Q(\eta, \eta')f\|_\rho \leq \theta_1^2 \|\eta\| \|\eta'\| \|f\|_\rho, \quad \|K\delta g\|_\rho \leq \kappa_m \|\delta g\|_\rho,$$

and the symmetric k -linear map with diagonal H_k is bounded by h_k , because it is a symmetrised product of k multiplications by ψ_η , each bounded by θ_0 , and of the fixed factor $\binom{m}{k} \psi_{c^\circ}^{m-k} / B$. Each degree- q term listed above is a product of q such linear factors, and the symmetrisation of a product of linear maps is bounded by the product of their norms. Adding the bounds of the terms of degree q gives c_q . $\square$

Proposition 20.13. *Let $A: \mathcal{S}_m \rightarrow X_m$ be bounded, linear and injective. Suppose that $\|A\mathcal{H}_m(x^\circ)\|_X \leq Y$, $\|I - AD\mathcal{H}_m(x^\circ)\|_{X \rightarrow X} \leq Z$ and $\|A\| c_q \leq C_q$ for $2 \leq q \leq m+2$, and that $r > 0$ satisfies*

$$(20.10) \quad Y + (Z - 1)r + \sum_{q=2}^{m+2} C_q r^q < 0, \quad Z - 1 + \sum_{q=2}^{m+2} q C_q r^{q-1} < 0.$$

Then $\mathcal{H}_m$ has exactly one zero in the closed ball $\overline{B}_{X_m}(x^\circ, r)$.

Proof. This is the proof of Proposition 19.8: $B_q(h_1, \dots, h_1) - B_q(h_2, \dots, h_2)$ is a sum of q terms, each containing $h_1 - h_2$ once, so its image under A has norm at most $q C_q r^{q-1} \|h_1 - h_2\|_X$ on the ball. $\square$

20.6. The principal part and the approximate inverse.

Proposition 20.14. *For every $x = (g, c) \in X_m$ and $j \equiv 1 \pmod{2m}$,*

$$D\mathcal{H}_m(x)e_j = 2 \operatorname{Re}(jw^{j-1} \overline{\psi'_c})(Kg + h_c) + \frac{m}{B} |\psi'_c|^2 \operatorname{Im}(\psi_c^{m-1} w^j).$$

Let $x_b = (0, b e_1)$, so that $\psi_{x_b}(w) = bw$. For $n \geq 0$ write

$$(20.11) \quad r^{n+2m} \sin n\theta = \sum_{s=0}^m q_s(n) S_{n,s};$$

then $q_s(n) \geq 0$, $\sum_s q_s(n) = 1$ and $q_0(n) = (n+1)/(n+m+1)$. For every $j \equiv 1 \pmod{2m}$ with $j \geq 2m+1$,

$$(20.12) \quad T_j := D\mathcal{H}_m(x_b)e_j = \alpha \left\{ (j+m)S_{j+m-1,0} - j \sum_{s=0}^m q_s(j-m-1)S_{j-m-1,s} \right\}.$$

Proof. The first formula follows by differentiating $|\psi'_c + \varepsilon j w^{j-1}|^2$ and $\text{Im}((\psi_c + \varepsilon w^j)^m)/B$ at $\varepsilon = 0$. The statements on q_s are Lemma 12.4 for $w^{n+m}\bar{w}^m = r^{n+2m}e^{in\theta}$, after taking imaginary parts. At x_b we have $Kg + h_c = b^m \text{Im}(w^m)/B$ and $\psi'_c = b$, so

$$\begin{aligned} T_j &= \alpha [2j \text{Re}(w^{j-1}) \text{Im}(w^m) + m \text{Im}(w^{j+m-1})] \\ &= \alpha r^{j+m-1} [(j+m) \sin((j+m-1)\theta) - j \sin((j-m-1)\theta)], \end{aligned}$$

using $2 \cos((j-1)\theta) \sin(m\theta) = \sin((j+m-1)\theta) - \sin((j-m-1)\theta)$. With $n = j-m-1 \geq m$, $n \equiv m \pmod{2m}$, (20.11) gives (20.12). $\square$

The harmonic part of T_j is $\alpha(j+m)$ in row $(j+m-1, 0)$ and $-\alpha(j-m)$ in row $(j-m-1, 0)$, a difference operator of step $2m$; the nonharmonic part lies in the rows $(j-m-1, s)$, $1 \leq s \leq m$, with total mass $\alpha j \sum_{s \geq 1} q_s(j-m-1) = \alpha m$. For $m = 1, 2$, (20.12) reduces to Lemma 15.3 and to (19.8) (Lemma 19.9 gives q_s for $m = 2$).

Finite block and tail. Let $M \equiv m \pmod{2m}$ and $S \geq m$. The *finite inputs* are the coordinates $g_{n,s}$ with $n \leq M$, $1 \leq s \leq S$, and c_j with $j \leq M-m+1$; the *finite rows* are (n, s) with $n \leq M$, $1 \leq s \leq S$, and $(n, 0)$ with $n \leq M$. All others are *tail inputs* and rows, $X_m = X_f \oplus X_t$, $\mathcal{S}_m = \mathcal{S}_f \oplus \mathcal{S}_t$, and $\dim X_f = \dim \mathcal{S}_f = (S+1)(M+m)/(2m)$. Let M_f be the matrix of $D\mathcal{H}_m(x^\circ)$ on finite inputs and finite rows. On tail inputs we use the model $T(\delta g, \eta) = \delta g + \sum_{j \geq M+m+1} \eta_j T_j$. By (20.12), only η_{M+m+1} reaches finite rows, since $S \geq m$:

$$(20.13) \quad B_f(\delta g, \eta) = -\eta_{M+m+1} \alpha \left\{ (M+1)e_{(M,0)} + (M+m+1) \sum_{s=1}^m q_s(M)e_{(M,s)} \right\}.$$

Write $Te = (B_f e, T_{tt} e)$ for tail inputs e , and put $\tau_m = (1 - \rho^{-2m})^{-1}$.

Lemma 20.15. $T_{tt}: X_t \rightarrow \mathcal{S}_t$ is a bounded bijection with $\|T_{tt}\| \leq 1 + \rho^{-2m}$. For a tail residual $y = \sum y_{n,s} S_{n,s} \in \mathcal{S}_t$ its inverse is

$$(20.14) \quad \eta_{n-m+1} = \sum_{k \geq 0} \frac{y_{n+2mk,0}}{\alpha(n+2mk+1)}, \quad \delta g_{n,s} = y_{n,s} + \alpha(n+m+1)q_s(n)\eta_{n+m+1} \quad (1 \leq s \leq m),$$

for $n \geq M+2m$, $n \equiv m \pmod{2m}$, and $\delta g_{n,s} = y_{n,s}$ for all other tail g -inputs.

Proof. By (20.12), the tail harmonic rows read $\alpha(n+1)(\eta_{n-m+1} - \eta_{n+m+1}) = y_{n,0}$ for $n \geq M+2m$, the rows (n, s) , $1 \leq s \leq m$, $n \geq M+2m$, read $\delta g_{n,s} - \alpha(n+m+1)q_s(n)\eta_{n+m+1} = y_{n,s}$, and all other tail rows read $\delta g_{n,s} = y_{n,s}$. The rest is the proof of Lemma 19.11: $\eta_j \rightarrow 0$ gives uniqueness and (20.14), and the column $T_{tt}e_j$, $j \geq M+3m+1$, has weighted norm $\omega_j(1 + j\rho^{-2m}/(j+m)) \leq (1 + \rho^{-2m})\omega_j$. $\square$

Let $\hat{A}$ be a matrix with dyadic entries, regarded as an exact linear map $\mathcal{S}_f \rightarrow X_f$, with

$$(20.15) \quad \|I - \hat{A}M_f\|_{X_f \rightarrow X_f} < 1,$$

and define $A = \begin{pmatrix} \hat{A} & -\hat{A}B_f T_{tt}^{-1} \\ 0 & T_{tt}^{-1} \end{pmatrix} : \mathcal{S}_f \oplus \mathcal{S}_t \rightarrow X_f \oplus X_t$. By Lemma 15.5 (Lemma 20.9), A is bounded and injective, and $ATe = e$ for every tail input e .

Lemma 20.16. *Let $\|R\| = \max_{(n,s) \text{ finite row}} \|\hat{A}e_{(n,s)}\|_X / \rho^n$,*

$$H = \left\| \hat{A} \alpha \left((M+1)e_{(M,0)} + (M+m+1) \sum_{s=1}^m q_s(M) e_{(M,s)} \right) \right\|_X,$$

and, for $n \geq M+2m$, $n \equiv m \pmod{2m}$,

$$(20.16) \quad h_n = \frac{1}{(n+1)\rho^n} \left(\sum_{\substack{M+2m \leq k \leq n \\ k \equiv m \pmod{2m}}} (k+1)\rho^k + m \sum_{\substack{M+2m \leq k \leq n-2m \\ k \equiv m \pmod{2m}}} \rho^k + \frac{H}{\alpha} \right).$$

Let $N_H \geq M+2m$, $N_H \equiv m \pmod{2m}$. Then A maps every residual supported in tail rows with norm at most

$$A_t := \max \left\{ 1, \max_{M+2m \leq n \leq N_H} h_n, \tau_m + \frac{m\tau_m\rho^{-2m}}{N_H+2m+1} + \frac{H}{\alpha(N_H+2m+1)\rho^{N_H+2m}} \right\},$$

it is the identity on residuals supported in nonharmonic tail rows, and $\|A\| \leq \max\{\|R\|, A_t\}$.

Proof. As in Lemma 19.12: the normalised harmonic tail row $\rho^{-n}S_{n,0}$ is mapped to $\eta_{k-m+1} = \rho^{-n}/(\alpha(n+1))$ for $M+2m \leq k \leq n$, of weights $\omega_{k-m+1} = \alpha(k+1)\rho^k$; to the coordinates $\delta g_{k,s}$, $1 \leq s \leq m$, $M+2m \leq k \leq n-2m$, of weight ρ^k and total absolute value $\alpha m \rho^{-n}/(\alpha(n+1))$, because $\sum_{s \geq 1} (k+m+1)q_s(k) = m$; and to the finite part $\eta_{M+m+1}\hat{A}\alpha(\dots)$, of norm $H\rho^{-n}/(\alpha(n+1))$. The sum is h_n . For $n \geq N_H+2m$ we use $(k+1)/(n+1) \leq 1$ and sum the geometric series. $\square$

20.7. The defect bounds. We bound $\|I - AD\mathcal{H}_m(x^\circ)\|$ by Lemma 15.2, splitting the inputs into five classes; every input belongs to exactly one class, and no column is estimated by sampling. Let $d = \deg p = M - m$ and $L = M + d$.

Residual and finite columns. $\mathcal{H}_m(x^\circ)$ and the columns $D\mathcal{H}_m(x^\circ)e$, e a finite input, are polynomials in $w, \bar{w}$; they are expanded in the basis $S_{n,s}$ without truncation, A is applied exactly using (20.13) and (20.14), and Y and Z_f bound $\|A\mathcal{H}_m(x^\circ)\|_X$ and the normalised norms of the finite columns of $I - AD\mathcal{H}_m(x^\circ)$.

Lemma 20.17. *Let $e = e_{(n,s)}$ be a tail g -input and $q = n + 2s$. Then $(I - AD\mathcal{H}_m(x^\circ))e = -A(|p|^2 K S_{n,s})$. Let $p = p_{\leq d_s} + p_{> d_s}$ be the splitting of p into the terms of degree $\leq d_s$ and $> d_s$, where $d_s \equiv 0 \pmod{2m}$ is fixed, $P_s = N_\rho(p_{\leq d_s})$, $P_t = N_\rho(p_{> d_s})$ and $\delta_a = 2P_s P_t + P_t^2$. Then*

$$(20.17) \quad \frac{\|(I - AD\mathcal{H}_m(x^\circ))e\|_X}{\rho^n} \leq \frac{\|A(|p_{\leq d_s}|^2 K S_{n,s})\|_X}{\rho^n} + \frac{\|A\| \delta_a}{q(q+2)}.$$

Moreover, the right-hand side is at most $(A_t P_s^2 + \|A\| \delta_a)/((n_f + 2)(n_f + 4))$ if $n \geq n_f := M + d_s + 2m$, and at most $(P_s^2 + \|A\| \delta_a)/((m + 2s_f)(m + 2s_f + 2))$ if $s \geq s_f := S + d_s + 2$.

Proof. This is the proof of Lemma 19.13. Multiplication by $|p_{\leq d_s}|^2$ changes frequencies by at most d_s , so for $n \geq n_f$ all components lie in rows of frequency at least $M + 2m$, which are tail rows; and by (13.1) and Lemma 12.3, for $s \geq s_f$ all radial indices are at least $s - 1 - d_s > S$, so all components lie in nonharmonic tail rows, on which A is the identity. Finally $q \geq n_f + 2$, respectively $q \geq m + 2s_f$. $\square$

The tail g -inputs with $n \leq M + d_s$ and $s \leq S + d_s + 1$ are the *near* ones; for them the first term of (20.17) is computed without truncation, and $Z_{g,\partial}$ bounds the maximum of the right-hand side. The remaining tail g -inputs are covered by the last statement of Lemma 20.17, and $Z_{g,\infty}$ is the larger of the two bounds.

Shape columns. For a tail shape input e_j , $j \geq M+m+1$, we have $(I-AD\mathcal{H}_m(x^\circ))e_j = -A\Delta_j$ with $\Delta_j = D\mathcal{H}_m(x^\circ)e_j - T_j$. For the *near* shape inputs $M+m+1 \leq j \leq j_* - 2m$, Δ_j is a polynomial, the columns are computed without truncation, and $Z_{\text{sh},\partial}$ bounds their normalised norms. Let $W^\circ = Kg^\circ$, $h^\circ = h_{c^\circ}$ and $h_{be_1} = b^m \text{Im}(w^m)/B$. By Proposition 20.14,

$$(20.18) \quad \Delta_j = 2 \text{Re}(jw^{j-1}\Gamma_1) + m \text{Im}(w^j\Gamma_0) + 2 \text{Re}(jw^{j-1}\bar{p}W^\circ),$$

where

$$\Gamma_1 = \bar{p}h^\circ - bh_{be_1}, \quad \Gamma_0 = \frac{|p|^2\psi_{c^\circ}^{m-1}}{B} - \alpha w^{m-1}.$$

By Lemma 13.1(c), $W^\circ = (1-|w|^2)^2Q^\circ$ with $Q^\circ = \sum_{n,s} \frac{g_{n,s}^\circ}{4s(s+1)} r^n P_{s-1}^{(2,n)}(2r^2-1) \sin n\theta$, whose finite expansion in the basis $S_{n,t}$ is given by the connection formulas of Section 16, with positive rational coefficients. The function $w^L\bar{p}Q^\circ$ has only nonnegative frequencies; write $w^L\bar{p}Q^\circ = \sum_{k,t \geq 0} \gamma_{k,t}^W \Phi_{k,t}$, $\Gamma_1 = \sum \gamma_{k,t}^1 \Phi_{k,t}$ and $\Gamma_0 = \sum \gamma_{k,t}^0 \Phi_{k,t}$ (finite sums, computed in interval arithmetic). For $j_* \equiv 1 \pmod{2m}$ with $j_* > L$ let

$$\begin{aligned} N_1^+ &= \sum_{k+j_*-1 > M} \rho^{|k|} |\gamma_{k,t}^1|, & N_0^+ &= \sum_{k+j_* > M} \rho^{|k|} |\gamma_{k,t}^0|, \\ \Sigma_W^+ &= \sum_{k+j_*-1-L > M} \frac{|\gamma_{k,t}^W| (2t+1)(2t+3) \rho^{k-L-m}}{(k+j_*-L)^2}, \end{aligned}$$

and let N_1^-, N_0^-, Σ_W^- be the corresponding sums over the remaining indices.

Proposition 20.18. *For every $j \equiv 1 \pmod{2m}$ with $j \geq j_* > L$,*

$$\frac{\|(I-AD\mathcal{H}_m(x^\circ))e_j\|_X}{\omega_j} \leq Z_{\text{sh},\infty} := U'_\Gamma + U'_W,$$

where

$$U'_\Gamma = \frac{1}{\alpha} \left(\frac{2(A_t N_1^+ + \|A\| N_1^-)}{\rho^m} + \frac{m(A_t N_0^+ + \|A\| N_0^-)}{\rho^{m-1}(j_*+m)} \right), \quad U'_W = \frac{8}{\alpha} (A_t \Sigma_W^+ + \|A\| \Sigma_W^-).$$

If moreover $j_*-1-((m+1)M-m^2) > M$, then $N_1^- = N_0^- = \Sigma_W^- = 0$ and $Z_{\text{sh},\infty} \leq A_t(U_\Gamma + U_W)$ with

$$U_\Gamma = \frac{1}{\alpha} \left(\frac{2\|\Gamma_1\|_\rho}{\rho^m} + \frac{m\|\Gamma_0\|_\rho}{\rho^{m-1}(j_*+m)} \right), \quad U_W = \frac{8}{\alpha(j_*-L)^2} \sum_{k,t} |\gamma_{k,t}^W| (2t+1)(2t+3) \rho^{k-L-m}.$$

Proof. Fix $j \geq j_*$. By Lemma 12.2, $w^{j-1}\Phi_{k,t}$ is a combination of $\Phi_{k+j-1,t'}$ of norm at most $\rho^{j-1+|k|}$, and taking real or imaginary parts only changes the signs of frequencies. Hence the contribution of $\gamma_{k,t}^1$ to $2 \text{Re}(jw^{j-1}\Gamma_1)$ has norm at most $2j\rho^{j-1}\rho^{|k|}|\gamma_{k,t}^1|$ and lies in rows of frequency $|k+j-1|$; if $k+j-1 > M$ these are tail rows, on which A has norm at most A_t (Lemma 20.16), and otherwise we use $\|A\| \geq A_t$. Dividing by $\omega_j = \alpha(j+m)\rho^{j+m-1}$ and using $j/(j+m) \leq 1$ gives the first term of U'_Γ . The term $m \text{Im}(w^j\Gamma_0)$ is treated in the same way, with $m\rho^j/\omega_j = m/(\alpha(j+m)\rho^{m-1}) \leq m/(\alpha(j_*+m)\rho^{m-1})$. Finally $jw^{j-1}\bar{p}W^\circ = j(1-|w|^2)^2w^{j-1-L}(w^L\bar{p}Q^\circ)$ with $j-1-L \geq 0$, and by Lemma 16.3 with $N = j-1-L$ the contribution of $\gamma_{k,t}^W$ has frequency $k+j-1-L$ and normalised norm at most

$$\frac{2j}{\alpha(j+m)\rho^{j+m-1}} |\gamma_{k,t}^W| \frac{4(2t+1)(2t+3)\rho^{k+j-1-L}}{(k+j-L)^2} \leq \frac{8|\gamma_{k,t}^W|(2t+1)(2t+3)\rho^{k-L-m}}{\alpha(k+j-L)^2}.$$

As j increases, every output frequency increases, so the sets of indices counted with A_t can only grow, and every factor above decreases; hence the bound at j_* covers all $j \geq j_*$. For the

last statement: the frequencies of $\bar{p}$, h° , ψ_{c° and Q° lie in $[-d, 0]$, $[-m(M - m + 1), m(M - m + 1)]$, $[0, M - m + 1]$ and $[-M, M]$, so the frequencies of Γ_1 , Γ_0 and $\bar{p}Q^\circ$ are all at least $-(m(M - m + 1) + d) = -((m + 1)M - m^2)$, and $L = 2M - m \leq (m + 1)M - m^2$. Under the stated condition all output frequencies therefore exceed M , so that $N_1^- = N_0^- = \Sigma_W^- = 0$, and $(k + j_* - L)^2 \geq (j_* - L)^2$ since $k \geq 0$. $\square$

Collecting the five classes, the number

$$Z = \max\{Z_f, Z_{g,\partial}, Z_{g,\infty}, Z_{\text{sh},\partial}, Z_{\text{sh},\infty}\}$$

bounds $\|I - AD\mathcal{H}_m(x^\circ)\|_{X \rightarrow X}$.

20.8. Certified bounds and the exact zeros. For $m = 3, 4, 6$ we take $\rho = 11/10$ and the data of Table 11. Each centre x° consists of the finite inputs listed there, given as binary64 numbers interpreted as exact dyadic rationals. The field scale B is such a number as well: $B = 1$ for $m = 3$, and $B \approx j_{m+1,k}^m$ is the binary64 number with hexadecimal representation `0x1.02d7fa4b541bbp+21` for $m = 4$ and `0x1.beb0441320216p+38` for $m = 6$. The finite inverse $\hat{A}$ is a fixed binary64 approximation of M_f^{-1} whose entries are interpreted as exact dyadic rationals; only (20.15) is used. All quantities of Sections 20.5–20.7 were evaluated in ball arithmetic at 128 bits (Section 34.7). Table 12 lists the results, rounded outward.

	$n = 8$	$n = 10$	$n = 14$
$(\mathfrak{k}, m)$	$(\mathfrak{su}(3), 3)$	$(\mathfrak{so}(5), 4)$	$(\mathfrak{g}_2, 6)$
(M, S)	$(45, 24)$	$(52, 36)$	$(114, 60)$
finite $g_{n,s}$	8×24	7×36	10×60
finite c_j	$c_1, c_7, \dots, c_{43}$	$c_1, c_9, \dots, c_{49}$	$c_1, c_{13}, \dots, c_{109}$
$\dim X_f$	200	259	610
B	1	≈ 2120447.2868	$\approx 4.7962817 \cdot 10^{11}$
$b = c_1^\circ$	20.78702921089377...	38.15292567121707...	88.49730472771703...
$d_s; n_f, s_f$	24; 75, 50	24; 84, 62	60; 186, 122
$j_*; L$	223; 87	305; 100	241; 222
N_H	645	852	1314
r	1/30000	10^{-7}	1/6000

TABLE 11. Parameters of the three computer-assisted proofs ($\rho = 11/10$). The finite shape inputs are c_j , $j \equiv 1 \pmod{2m}$, $j \leq M - m + 1$; the finite g -inputs are $g_{n,s}$, $n \equiv m \pmod{2m}$, $n \leq M$, $1 \leq s \leq S$. B and b are exact dyadic rationals.

Theorem 20.19. *Let $m \in \{3, 4, 6\}$, and let M, S, B, x° and r be as in Table 11. There is exactly one $x^* = (g^*, c^*) \in X_m$ with $\|x^* - x^\circ\|_X \leq r$ and $\mathcal{H}_m(x^*) = 0$.*

Proof. The condition (20.15) holds by Table 12, so A is bounded and injective. The hypotheses of Proposition 20.18 hold: for $m = 3$ and $m = 4$, $j_* - 1 - ((m + 1)M - m^2)$ equals $222 - 171 = 51 > 45$ and $304 - 244 = 60 > 52$, and for $m = 6$, $j_* = 241 > L = 222$. By Table 12, Proposition 20.13 applies with the following rational majorants:

- $m = 3$: $Y = 153 \cdot 10^{-7}$, $Z = 43/100$, $C_2 = 1/125$, $C_3 = 5 \cdot 10^{-10}$, $C_4 = 2 \cdot 10^{-18}$, $C_5 = 3 \cdot 10^{-26}$;
- $m = 4$: $Y = 10^{-8}$, $Z = 56/100$, $C_2 = 396$, $C_3 = 72/1000$, $C_4 = 4 \cdot 10^{-7}$, $C_5 = 2 \cdot 10^{-11}$, $C_6 = 4 \cdot 10^{-16}$;

Quantity	$n = 8$	$n = 10$	$n = 14$
P	< 21.788460	< 38.428146	< 91.291342
C	< 23.020116	< 42.001369	< 97.545293
V_0	< 131547.591	< 28.746248	< 79.018365
$\ I - \hat{A}M_f\ $	$< 2.2858 \cdot 10^{-14}$	$< 5.7259 \cdot 10^{-12}$	$< 5.0709 \cdot 10^{-9}$
$\ R\ $	< 963.7779	< 10194.7512	< 11766.8786
H	$< 1.4540263 \cdot 10^9$	< 718348.086	$< 1.2223739 \cdot 10^9$
A_t	< 2.302038	< 2.014454	< 1.661175
$\ A\ $ (ceiling)	964	10195	11767
Y	$< 1.519444 \cdot 10^{-5}$	$< 9.591861 \cdot 10^{-9}$	$< 4.592200 \cdot 10^{-5}$
Z_f	< 0.052262	< 0.060605	< 0.120542
P_s	< 21.788460	< 38.428143	< 91.291236
P_t	$< 1.4978 \cdot 10^{-7}$	$< 2.6609 \cdot 10^{-6}$	$< 1.05815 \cdot 10^{-4}$
δ_a	$< 6.5267 \cdot 10^{-6}$	$< 2.04502 \cdot 10^{-4}$	< 0.0193200
near g -columns; max. at	396; (51, 1)	358; (60, 1)	1215; (126, 1)
$Z_{g,\partial}$	< 0.194306	< 0.462762	< 0.576826
far bound, $n \geq n_f$	< 0.179735	< 0.393351	< 0.393946
far bound, $s \geq s_f$	< 0.043897	< 0.088871	< 0.135896
near shape columns; max. at	29; $j = 49$	31; $j = 57$	10; $j = 121$
$Z_{sh,\partial}$	< 0.284628	< 0.552897	< 0.600896
$\ \Gamma_1\ _\rho$ resp. N_1^+	< 14093.696	< 0.466786	< 5.147419
N_1^-	—	—	$< 3.3948 \cdot 10^{-8}$
$\ \Gamma_0\ _\rho$ resp. N_0^+	< 21422.774	< 0.692754	< 7.696736
N_0^-	—	—	0
Σ_W^+	—	—	< 3.293822
Σ_W^-	—	—	$< 1.0394 \cdot 10^{-17}$
U_Γ resp. U'_Γ	< 0.114684	< 0.016902	< 0.111093
U_W resp. U'_W	< 0.067068	< 0.088733	< 0.493856
$Z_{sh,\infty}$ (all $j \geq j_*$)	< 0.418572	< 0.212795	< 0.604948
Z	< 0.418572	< 0.552897	< 0.604948
C_2	< 0.0072192	< 395.40418	< 210.53260
C_3	$< 4.8836 \cdot 10^{-10}$	< 0.071133	< 0.0060610
C_4	$< 1.5491 \cdot 10^{-18}$	$< 3.2353 \cdot 10^{-7}$	$< 1.9135 \cdot 10^{-9}$
C_5	$< 2.1152 \cdot 10^{-26}$	$< 1.7470 \cdot 10^{-11}$	$< 2.2718 \cdot 10^{-14}$
C_6	—	$< 3.7236 \cdot 10^{-16}$	$< 1.5942 \cdot 10^{-19}$
C_7	—	—	$< 6.1326 \cdot 10^{-25}$
C_8	—	—	$< 9.9963 \cdot 10^{-31}$

TABLE 12. Certified bounds, rounded outward. The row $\|A\|$ gives the rational ceiling for $\max\{\|R\|, A_t\}$ used in $C_q = \|A\| c_q$; the tail estimates use this ceiling ($n = 8$) or the certified enclosure of $\|R\|$ ($n = 10, 14$). For $n = 8, 10$ the far shape bound is the second one of Proposition 20.18, $Z_{sh,\infty} = A_t(U_\Gamma + U_W)$ (for $n = 8$ with the ceiling $A_t \leq 2.303$); for $n = 14$ it is the first one, $Z_{sh,\infty} = U'_\Gamma + U'_W$. The maximum defining A_t is attained by the remainder term for $n = 8$, and at $n = M + 2m$ for $n = 10, 14$. P , C and V_0 are intermediate values of the final verification scripts; the complete enclosures of the other quantities are recorded in the certificates and receipts (Section 34.7).

- $m = 6$: $Y = 46 \cdot 10^{-6}$, $Z = 61/100$, $C_2 = 211$, $C_3 = 61 \cdot 10^{-4}$, $C_4 = 2 \cdot 10^{-9}$, $C_5 = 3 \cdot 10^{-14}$, $C_6 = 2 \cdot 10^{-19}$, $C_7 = 7 \cdot 10^{-25}$, $C_8 = 2 \cdot 10^{-30}$.

For these values and $r = 1/30000$, 10^{-7} , $1/6000$, respectively, exact rational arithmetic gives for the two expressions in (20.10)

$$\begin{aligned} & -3.699991111111111 \dots \cdot 10^{-6} \quad \text{and} \quad -0.5699994666666666 \dots \quad (m = 3), \\ & -3.399603999999992 \dots \cdot 10^{-8} \quad \text{and} \quad -0.4399207999999978 \dots \quad (m = 4), \\ & -1.31388886064814 \dots \cdot 10^{-5} \quad \text{and} \quad -0.319666661583333 \dots \quad (m = 6), \end{aligned}$$

all negative. $\square$

Remark 20.20 (The centres). The centres were obtained numerically, and their provenance plays no role in the proof. A Fourier–Bessel collocation of (20.3) in the enlarged symmetry class, at the ball wavenumber $j_{m+1,k}$ of Section 20.4 and seeded with the near-resonant mode $\ell = 2m$, produced numerical solutions with Cauchy defects of order 10^{-12} or smaller. Their domains were mapped conformally to the disc by Theodorsen’s method, and Newton’s method was applied to the Galerkin truncation of (20.7). For $m = 4, 6$ the unscaled field has boundary data of size about $j_{m+1,k}^m$, and double-precision Newton iterations stagnated at residuals of about 10^{-6} and 2; dividing the field by the frozen constant B , which does not change the Cauchy data of $F = BV \circ \psi^{-1}$, removed this obstruction. For $m = 6$ a first centre with $M = 54$ gave $Y \approx 1.7$, the defect being concentrated in the omitted angular frequency 66; this led to $M = 114$. For orientation we record some numerical observations about the centres, not used in the proofs. The boundary radius of D lies between 20.7086 and 20.8635 ($m = 3$), 38.1390 and 38.1667 ($m = 4$), 88.4477 and 88.5402 ($m = 6$), so that $\partial\Omega$ deviates from the sphere of radius $j_{m+1,k}$ by between -0.57% and $+0.18\%$, -0.055% and $+0.018\%$, -0.030% and $+0.074\%$, respectively; the minimum of $\operatorname{Re}(1 + w\psi''/\psi')$ on $\partial\mathbb{D}$ is about 0.832, 0.973 and 0.896, close to the certified lower bounds of Table 13.

20.9. Geometry and proof of Theorem 20.1.

Proposition 20.21. *Let $m \in \{3, 4, 6\}$, let c be a real sequence indexed by $j \equiv 1 \pmod{2m}$ with $\sum_j \omega_j |c_j - c_j^\circ| \leq r$ (for instance $c = c^*$), and put $R_e = 21/20$, $E_1 = r\theta_1$, $E_2 = r/(\alpha\rho^{m-1}e \log \rho)$,*

$$\Lambda_e = b - \sum_{j>1} j |c_j^\circ| R_e^{j-1} - E_1, \quad L_1 = b - \sum_{j>1} j |c_j^\circ| - E_1, \quad U_2 = \sum_{j>1} j(j-1) |c_j^\circ| + E_2.$$

Then $\operatorname{Re} \psi'_c \geq \Lambda_e$ on $\{|w| \leq R_e\}$, $|\psi'_c| \geq L_1$ and $|\psi''_c| \leq U_2$ on $\overline{\mathbb{D}}$, and $|c_{2m+1}| \geq |c_{2m+1}^\circ| - r/\omega_{2m+1}$. The computation certifies the bounds of Table 13; in particular, for $|w| \leq 1$,

$$\operatorname{Re} \left(1 + \frac{w\psi''_c(w)}{\psi'_c(w)} \right) \geq 1 - \frac{U_2}{L_1} > 0.$$

Consequently $\psi^ = \psi_{c^*}$ is injective with nonvanishing derivative on $\{|w| < 21/20\}$, and $D = \psi^*(\mathbb{D})$ is a strictly convex domain bounded by a real-analytic Jordan curve, invariant under $I_2(2m)$, not a disc, with $0 \in D$ and $\partial D = \{R(\theta)e^{i\theta}\}$ for a positive, real-analytic, W -invariant function R .*

Proof. Let $\delta = c - c^\circ$. Since $j\rho^{j-1}/\omega_j \leq \theta_1$ for all j , we have $\sum_j j\rho^{j-1}|\delta_j| \leq E_1$; since, for $j \geq 2$, $j(j-1)/\omega_j \leq j\rho^{-j}/(\alpha\rho^{m-1}) \leq 1/(\alpha\rho^{m-1}e \log \rho)$, as $x \mapsto x\rho^{-x}$ has maximum $1/(e \log \rho)$ on $(0, \infty)$, we have $\sum_{j>1} j(j-1)|\delta_j| \leq E_2$; and $|\delta_{2m+1}| \leq r/\omega_{2m+1}$. Both suprema are over all j , so the infinite tail of c is covered. For $|w| \leq R_e < \rho$ this gives $\operatorname{Re} \psi'_c(w) \geq c_1 - \sum_{j>1} j |c_j| R_e^{j-1} \geq \Lambda_e$, and similarly $|\psi'_c| \geq L_1$ and $|\psi''_c| \leq U_2$ on $\overline{\mathbb{D}}$, so $\operatorname{Re}(1 + w\psi''_c/\psi'_c) \geq 1 - |\psi''_c|/|\psi'_c| \geq 1 - U_2/L_1$.

For $c = c^*$, univalence on the convex disc $\{|w| < R_e\}$ follows from $\operatorname{Re} \psi^{*'} > 0$ as in Proposition 18.1, and strict convexity of D from the analytic convexity criterion, as in Proposition 18.2.

Quantity	$n = 8$	$n = 10$	$n = 14$
Λ_e	> 20.037195	> 37.965074	> 87.134069
L_1	> 20.231942	> 38.026654	> 87.809995
U_2	< 3.408445	< 1.024680	< 9.301964
$1 - U_2/L_1$	> 0.83153153	> 0.97305365	> 0.89406714
$ c_{2m+1} $	$ c_7 > 0.07748402$	$ c_9 > 0.01383247$	$ c_{13} > 0.04630866$

TABLE 13. Certified geometric bounds of Proposition 20.21, valid for every c in the ball of Theorem 20.19.

The symmetries follow from (20.5), and $0 = \psi^*(0) \in D$. A line tangent to ∂D at some point is a supporting line of the convex set D , so it cannot pass through the interior point 0; hence $\frac{d}{d\phi} \arg \psi^*(e^{i\phi}) \neq 0$, every ray from 0 meets ∂D exactly once and transversally, and ∂D is the graph of a positive real-analytic radial function R , which is W -invariant because D is. If D were a disc, its centre would be fixed by the rotation through π/m , hence equal to 0, and the Schwarz lemma argument of Proposition 18.1 would show that ψ^* is linear, contradicting $c_{2m+1}^* \neq 0$. $\square$

Proof of Theorem 20.1. Let $m = 3, 4, 6$ for $\mathfrak{k} = \mathfrak{su}(3), \mathfrak{so}(5), \mathfrak{g}_2$, respectively, let $x^* = (g^*, c^*)$ be the zero of Theorem 20.19, $\psi^* = \psi_{c^*}$ and $V^* = Kg^* + h_{c^*}$. By Lemma 20.10, V^* satisfies the hypotheses of Lemma 20.11 with $\psi = \psi^*$, and by Proposition 20.21 ψ^* is injective with nonvanishing derivative on a neighbourhood of $\overline{\mathbb{D}}$. Hence $F = BV^* \circ (\psi^*)^{-1} \in C^1(\overline{D})$ is a weak solution of $\Delta F + F = 0$ in D with $F = \varpi$ and $\nabla F = \nabla \varpi$ on ∂D , and $F(\bar{\tau}) = -F(\tau)$, $F(e^{i\pi/m}\tau) = -F(\tau)$. By Lemma 18.3 and Lemma 18.4 in the form of Lemma 20.9, F extends to a real-analytic solution on an open neighbourhood of $\overline{D}$. Replacing this neighbourhood by the connected component containing $\overline{D}$ of its intersection with its images under $I_2(2m)$, we obtain an $I_2(2m)$ -invariant neighbourhood N on which, by the identity theorem, F keeps its parities; in particular F is anti-invariant under W .

By Proposition 20.21, $D = \{re^{i\theta} : r < R(\theta)\}$ with R positive, real analytic and W -invariant. Lemma 20.6 shows that $\Omega = \text{Ad}(G)D$ is a bounded domain, strictly star-shaped with respect to 0, whose boundary is a compact real-analytic hypersurface diffeomorphic to S^{n-1} , and that $u = (F/\varpi)^\sharp$ is a nonconstant real-analytic function on the neighbourhood $\text{Ad}(G)N$ of $\overline{\Omega}$ satisfying (1.1). Since D is convex, Ω is convex by Lemma 20.8. It is $\text{Ad}(G)$ -invariant by construction, and $-\Omega = \text{Ad}(G)(-D) = \Omega$ because $-1 \in I_2(2m)$.

If Ω were a ball, its centre would be fixed by $\text{Ad}(G)$. As $\mathfrak{k}$ is simple, the only $\text{Ad}(G)$ -fixed vector is 0, so Ω would be a ball centred at 0 and $D = \Omega \cap \mathfrak{t}$ a disc, contradicting Proposition 20.21. Finally, the proof of Corollary 1.2 in Section 33 uses only (1.1) and the regularity of u and $\partial\Omega$, and applies verbatim: $\widehat{1}_\Omega$ vanishes on the unit sphere, and $\int_{\sigma(\Omega)} \cos x_1 dx = 0$ for every rigid motion σ of $\mathfrak{k} \cong \mathbb{R}^n$. $\square$

20.10. Which dimensions arise.

Remark 20.22. The reduction of Proposition 20.2 lowers the dimension to the rank of $\mathfrak{k}$ and never produces a potential term, because ϖ is harmonic; it is planar exactly when $\mathfrak{k}$ has rank two. Since $\mathfrak{k} = \mathfrak{z}(\mathfrak{k}) \oplus [\mathfrak{k}, \mathfrak{k}]$ with $[\mathfrak{k}, \mathfrak{k}]$ semisimple, a compact Lie algebra of rank two is isomorphic to $\mathbb{R}^2$, to $\mathbb{R} \oplus \mathfrak{su}(2) \cong \mathfrak{u}(2)$, to $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$, or to one of the simple algebras $\mathfrak{su}(3)$, $\mathfrak{so}(5)$, $\mathfrak{g}_2$ of types A_2 , $B_2 = C_2$, G_2 [8, 45]. Leaving aside the abelian case, in which there is nothing to reduce, the dimensions $n = 2 + 2m$ are 4, 6, 8, 10, 14, corresponding to $m \in \{1, 2, 3, 4, 6\}$, the orders of rotations allowed by the crystallographic restriction. Thus $\{4, 6, 8, 10, 14\}$ is the

complete list of dimensions obtainable by exact planar, potential-free reductions of adjoint type, and Theorems 1.1 and 20.1 cover all of them. The classes $O(d_1) \times O(d_2)$ of Remark 10.5 give no further dimensions: the admissible ones, $(d_1, d_2) = (1, 3)$ and $(3, 3)$, are the adjoint classes of $\mathfrak{u}(2)$ and $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. For $m = 5$ or $m \geq 7$, the planar problem (20.3) with $\varpi = \operatorname{Im} \tau^m$ still makes sense and could be treated by the method of this section, but the dihedral group $I_2(m)$ is not crystallographic, and the problem is not the reduction of a problem in a Euclidean space of this type. More generally, for the isotropy representation of a Riemannian symmetric space of rank two whose restricted roots all have the same multiplicity k , the radial part of the Laplacian is $f \mapsto J^{-1} \operatorname{div}(J \nabla f)$ with $J = |\varpi|^k$ [38, Ch. II], and conjugation by $J^{1/2}$ leaves the potential $\frac{k}{2}(\frac{k}{2} - 1)|\nabla \varpi|^2/\varpi^2$, which vanishes only for $k = 2$, the adjoint case.

Part 4. Axisymmetric counterexamples in dimensions three, five and seven

21. DIMENSION THREE

In this section we prove Theorem 1.4. Points of the physical space $\mathbb{R}^3$ are written $X = (X_1, X') \in \mathbb{R} \times \mathbb{R}^2$, so that X_1 and X' are the coordinates x_1 and x' of Theorem 1.4; the letter x is used below for points of a reference ball. The convex domain and its dyadic centre are described in Remark 1.5. The independent earlier example is stated at the end of this section.

Strategy. For $n = 3$ there is no reduction to the planar Laplacian of the kind used for $n = 4$ and $n = 6$ (Remark 10.5): for $O(2)$ -invariant functions the meridian problem contains the drift term $y^{-1} \partial_y$. We therefore keep the three-dimensional Laplacian and pull back only the meridian plane by a conformal map ψ of the unit disc, which we lift to a diffeomorphism Θ of the unit ball $B \subset \mathbb{R}^3$ (Section 21.1). With $p = \psi'$ and $q = \operatorname{Im} \psi/y$, the pulled-back problem is

$$\operatorname{div}(q \nabla U) + q|p|^2 U = 0 \text{ in } B, \quad U = 1 \text{ and } \nabla U = 0 \text{ on } \partial B.$$

We write $U = 1 + K_3 g$, where K_3 is an explicit inverse of the Laplacian of $\mathbb{R}^3$ on axisymmetric functions with vanishing Cauchy data on ∂B (Section 21.4), expand g in zonal solid harmonics times Jacobi polynomials (Section 21.2), and obtain a quartic equation $\mathcal{F}(g, c) = 0$ for g and the Taylor coefficients c of ψ (Section 21.5). The unknown coefficient q is not divided out, so that the equation is polynomial. The approximate inverse of the linearisation consists of a 7371×7371 matrix, the identity on high field modes, and the exact inverse of a Toeplitz operator that describes the linearisation on high shape modes (Section 21.6). The defect bounds (Section 21.7) cover all columns of the linearisation: the finite parts of 7280 columns and the complete columns of $457 + 310$ further inputs are evaluated in interval arithmetic, while the output tails of the former and all remaining columns are covered by explicit envelopes (91 radial classes, fourteen near angular classes, one uniform angular class and one far shape class), whose validity is proved there. Compared with $n = 4$ and $n = 6$, the equation has the variable coefficient q in its principal part, so that the model of the linearisation on high shape modes is a Toeplitz operator whose symbol is built from q and ψ' rather than a bidiagonal operator; the output space measures harmonic components through their boundary traces; and the computation is much larger, taking about 1.8 hours with five processes (Section 34.8).

21.1. Axisymmetric functions and the pull-back. Throughout this section $R = 21/20$; the notation is independent of that of Sections 10–20. Let $B \subset \mathbb{R}^3$ be the open unit ball and $B_R = \{|x| < R\}$. We write $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^2$ for points of the reference space, X for points of the physical space, and

$$y = |x'|, \quad w = x_1 + iy, \quad r = |x| = |w|, \quad t = r^2, \quad \xi = x_1/r.$$

A function on a rotation-invariant subset of $\mathbb{R}^3$ is *axisymmetric* if it is invariant under the action of $O(2)$ on x' ; it is then a function of (x_1, y) , and we use both descriptions. Polynomials in x_1 and y^2 are polynomials on $\mathbb{R}^3$. We write T_n , U_n , P_ℓ and $C_n^{(2)}$ for the Chebyshev polynomials of the first and second kind, the Legendre polynomials and the Gegenbauer polynomials of index two, with $U_{-1} = 0$.

Let $\mathcal{E}$ be the Banach space of real sequences $c = (c_j)_{j \text{ odd} \geq 1}$ with $\|c\|_{\mathcal{E}} = \sum_j j R^j |c_j| < \infty$. For $c \in \mathcal{E}$ let $\psi_c(w) = \sum_j c_j w^j$; the series for ψ_c and ψ'_c converge absolutely and uniformly on $|w| \leq R$. Put

$$p_c = \psi'_c(x_1 + iy), \quad q_c = \frac{\operatorname{Im} \psi_c(x_1 + iy)}{y} \quad (y \neq 0), \quad q_c(x_1, 0) = \operatorname{Re} \psi'_c(x_1).$$

Since $\operatorname{Im}(w^j) = r^j \sin j\theta$ for $w = re^{i\theta}$, $y = r \sin \theta$ and $U_{j-1}(\cos \theta) = \sin j\theta / \sin \theta$,
(21.1)

$$q_c = \sum_j c_j r^{j-1} U_{j-1}(\xi), \quad \partial_{x_1} q_c = \sum_{j \geq 3} j c_j r^{j-2} U_{j-2}(\xi), \quad \frac{\partial_y q_c}{y} = -2 \sum_{j \geq 3} c_j r^{j-3} C_{j-3}^{(2)}(\xi).$$

The second identity follows from $\partial_{x_1} \operatorname{Im}(w^j) = \operatorname{Im}(jw^{j-1})$. For the third, a computation in polar coordinates gives

$$\frac{1}{y} \partial_y \frac{\operatorname{Im} w^j}{y} = r^{j-3} \frac{(j-1) \sin \theta \cos(j-1)\theta - \sin(j-1)\theta \cos \theta}{\sin^3 \theta} = -r^{j-3} U'_{j-2}(\cos \theta),$$

and $U'_{j-2} = 2C_{j-3}^{(2)}$. Moreover, with $p_a = (a+1)c_{a+1}$,

$$(21.2) \quad |p_c|^2 = \sum_{\alpha, \beta \text{ even}} p_\alpha p_\beta \operatorname{Re}(w^\alpha \bar{w}^\beta) = \sum_{\alpha, \beta \text{ even}} p_\alpha p_\beta r^{\alpha+\beta} T_{|\alpha-\beta|}(\xi).$$

Since ψ_c is odd with real coefficients, $\psi_c(\bar{w}) = \overline{\psi_c(w)}$ and $\psi_c(-\bar{w}) = -\overline{\psi_c(w)}$; hence q_c , $y^{-1} \partial_y q_c$ and $|p_c|^2$ are even functions of x_1 , and $\partial_{x_1} q_c$ is odd.

For $c \in \mathcal{E}$ define $\Theta_c: B_R \rightarrow \mathbb{R}^3$ by

$$(21.3) \quad \Theta_c(x) = (\operatorname{Re} \psi_c(x_1 + i|x'|), q_c(x) x').$$

Lemma 21.1. *Let $c \in \mathcal{E}$ and suppose that $\operatorname{Re} \psi'_c > 0$ on a disc $\{|w| < R_0\}$ with $1 < R_0 < R$. Then:*

- (a) q_c , $\partial_{x_1} q_c$, $y^{-1} \partial_y q_c$, $|p_c|^2$ and $\operatorname{Re} \psi_c(x_1 + iy)$, regarded as functions of x , are real analytic on B_{R_0} , and $q_c > 0$ there;
- (b) Θ_c is a real-analytic diffeomorphism of B_{R_0} onto an open subset of $\mathbb{R}^3$, with $\det D\Theta_c = q_c |p_c|^2$; it commutes with the action of $O(2)$ on x' , and $\Theta_c(-x_1, x') = (-X_1, X')$ if $\Theta_c(x_1, x') = (X_1, X')$;
- (c) for every axisymmetric C^1 function U on B_{R_0} , $(\det D\Theta_c)(D\Theta_c^T D\Theta_c)^{-1} \nabla U = q_c \nabla U$.

Proof. (a) The function $F(x_1, y) = \operatorname{Im} \psi_c(x_1 + iy)$ is real analytic on the disc $\{|w| < R_0\}$ and odd in y . By Lemma 10.2 there is a real-analytic G with $F = yG(x_1, y^2)$, and $x \mapsto G(x_1, |x'|^2)$ is real analytic on B_{R_0} ; it equals q_c off the axis, and on the axis $G(x_1, 0) = \partial_y F(x_1, 0) = \operatorname{Re} \psi'_c(x_1)$ by the Cauchy–Riemann equations. Hence q_c is real analytic, and so are $\partial_{x_1} q_c = (\partial_1 G)(x_1, y^2)$ and $y^{-1} \partial_y q_c = 2(\partial_2 G)(x_1, y^2)$. Functions of (x_1, y) that are real analytic and even in y , such as $\operatorname{Re} \psi_c$ and $|\psi'_c|^2$, are treated by applying Lemma 10.2 to their product with y . For $y \neq 0$, the Cauchy–Riemann equations give $q_c = y^{-1} \int_0^y \operatorname{Re} \psi'_c(x_1 + i\varsigma) d\varsigma > 0$, and $q_c = \operatorname{Re} \psi'_c > 0$ on the axis.

(b) Both components of Θ_c are real analytic by (a). For $w_1 \neq w_2$ in the convex disc $\{|w| < R_0\}$, $(\psi_c(w_2) - \psi_c(w_1))/(w_2 - w_1) = \int_0^1 \psi'_c(w_1 + \varsigma(w_2 - w_1)) d\varsigma$ has positive real part, so ψ_c is injective there. Since $q_c > 0$, the second component of $\Theta_c(x)$ has modulus $\text{Im } \psi_c(x_1 + i|x'|)$. Hence $\Theta_c(x) = \Theta_c(z)$ implies $\psi_c(x_1 + i|x'|) = \psi_c(z_1 + i|z'|)$, so $x_1 = z_1$ and $|x'| = |z'|$, and then $q_c(x)x' = q_c(z)z'$ gives $x' = z'$. At a point with $x' \neq 0$ let $e_\rho = x'/|x'|$ and let e_ϕ be the unit vector in the x' -plane orthogonal to e_ρ . On the half-plane spanned by e_1 and e_ρ , Θ_c acts as ψ_c in the coordinates (x_1, y) , and $D\Theta_c$ maps this plane to itself as $|p_c|$ times a rotation; it maps e_ϕ to $q_c e_\phi$, since Θ_c maps the circle of radius y about the axis onto the circle of radius $q_c y$. Hence $\det D\Theta_c = q_c |p_c|^2 > 0$ off the axis, and by continuity on B_{R_0} . By the inverse function theorem Θ_c is a real-analytic diffeomorphism onto its open image. The symmetries follow from those of ψ_c .

(c) In the frame (e_1, e_ρ, e_ϕ) of (b), $D\Theta_c^\top D\Theta_c = \text{diag}(|p_c|^2, |p_c|^2, q_c^2)$, and ∇U lies in the span of e_1 and e_ρ . Off the axis the identity follows, and on the axis by continuity. $\square$

Lemma 21.2. *Let c be as in Lemma 21.1, $\Omega = \Theta_c(B)$, and let $U \in C^1(\overline{B})$ be axisymmetric and satisfy*

$$(21.4) \quad \int_B (q_c \nabla U \cdot \nabla \varphi - q_c |p_c|^2 U \varphi) dx = 0 \quad (\varphi \in C_c^\infty(B)).$$

Then $u = U \circ \Theta_c^{-1} \in C^1(\overline{\Omega})$ satisfies $\int_\Omega (\nabla u \cdot \nabla \varphi - u \varphi) dX = 0$ for all $\varphi \in C_c^\infty(\Omega)$. If $U = 1$ and $\nabla U = 0$ on ∂B , then $u = 1$ and $\nabla u = 0$ on $\partial \Omega$.

Proof. Let $\varphi \in C_c^\infty(\Omega)$ and $\tilde{\varphi} = \varphi \circ \Theta_c \in C_c^\infty(B)$. Since $(\nabla u) \circ \Theta_c = D\Theta_c^{-\top} \nabla U$, the change of variables $X = \Theta_c(x)$ and Lemma 21.1(b),(c) give

$$\int_\Omega (\nabla u \cdot \nabla \varphi - u \varphi) dX = \int_B ((\det D\Theta_c)(D\Theta_c^\top D\Theta_c)^{-1} \nabla U \cdot \nabla \tilde{\varphi} - q_c |p_c|^2 U \tilde{\varphi}) dx,$$

which vanishes by Lemma 21.1(c) and (21.4). The boundary values follow from $(\nabla u) \circ \Theta_c = D\Theta_c^{-\top} \nabla U$. $\square$

21.2. Zonal harmonics, Jacobi polynomials and coefficient spaces. For $\ell \geq 0$ let $H_\ell(x) = r^\ell P_\ell(\xi)$, the axisymmetric harmonic polynomial on $\mathbb{R}^3$, homogeneous of degree ℓ , with $H_\ell(1, 0, 0) = 1$; axisymmetric harmonic homogeneous polynomials of degree ℓ are multiples of H_ℓ [64, Ch. IV]. We have $|H_\ell| \leq 1$ on $\overline{B}$, $H_\ell = P_\ell(\xi)$ on ∂B , and, with $H_{-1} = 0$,

$$(21.5) \quad x_1 H_\ell = \frac{(\ell+1)H_{\ell+1} + \ell t H_{\ell-1}}{2\ell+1}, \quad \partial_{x_1} H_\ell = \ell H_{\ell-1}, \quad (x \cdot \nabla) H_\ell = \ell H_\ell.$$

The first identity is Bonnet's recurrence for P_ℓ multiplied by $r^{\ell+1}$; for the second, $\partial_{x_1} H_\ell$ is an axisymmetric harmonic homogeneous polynomial of degree $\ell-1$, hence a multiple of $H_{\ell-1}$, and the multiple is ℓ because $H_\ell(x_1, 0, 0) = x_1^\ell$.

For real $\beta > -1$ and integers $s \geq 0$ let $J_s^\beta(t) = P_s^{(0,\beta)}(2t-1)$, so that

$$(21.6) \quad J_s^\beta(t) = \sum_{k=0}^s (-1)^{s-k} \frac{\Gamma(\beta+s+k+1)}{k!(s-k)!\Gamma(\beta+k+1)} t^k = \frac{t^{-\beta}}{s!} \frac{d^s}{dt^s} \{t^{\beta+s}(t-1)^s\}, \quad J_s^\beta(1) = 1,$$

and $J_{-1}^\beta = 0$. We write $[t^k]f$ for the coefficient of t^k in a polynomial f , and $\|f\|_{J^\beta,1} = \sum_s |f_s|$ if $f = \sum_s f_s J_s^\beta$. Define the axisymmetric polynomials

$$\Psi_{\ell,s} = H_\ell J_s^{\ell+1/2}(t) \quad (\ell, s \geq 0),$$

of degree $\ell+2s$; thus $\Psi_{\ell,0} = H_\ell$.

Lemma 21.3. *Let $\beta > 0$ and $s, s' \geq 0$.*

(a) *For every integer $m \geq 0$,*

$$\int_0^1 t^{\beta+m} J_s^\beta(t) dt = \frac{m(m-1) \cdots (m-s+1)}{(\beta+m+1)(\beta+m+2) \cdots (\beta+m+s+1)};$$

in particular J_s^β is orthogonal in $L^2([0, 1], t^\beta dt)$ to all polynomials of degree less than s , and $\int_0^1 t^\beta J_s^\beta J_{s'}^\beta dt = \delta_{ss'}/(\beta+2s+1)$.

(b) $J_s^\beta = \frac{\beta+s+1}{\beta+2s+1} J_s^{\beta+1} + \frac{s}{\beta+2s+1} J_{s-1}^{\beta+1}$, *which holds also for $-1 < \beta \leq 0$, and*
 $tJ_s^\beta = \frac{\beta+s}{\beta+2s+1} J_s^{\beta-1} + \frac{s+1}{\beta+2s+1} J_{s+1}^{\beta-1}$. *The coefficients are nonnegative and sum to one, and in the first identity the coefficient of the term with the lower index is at most $1/2$.*

(c) *The coefficient of J_s^β in the expansion of $(1-t)^m$ is $(-1)^s$ times a positive number for $0 \leq s \leq m$, and vanishes for $s > m$. Moreover $\|(1-t)^m\|_{J^\beta, 1} = m!/(\beta/2+1)_m$, which is decreasing in β .*

(d) *For $M \geq 0$, the coefficients of t^M in the basis $(J_s^\beta)_s$ are nonnegative and sum to one, the coefficient of J_0^β is $(\beta+1)/(\beta+M+1)$, and $\|t^M - 1\|_{J^\beta, 1} = 2M/(\beta+M+1)$.*

Proof. (a) Insert the Rodrigues formula (21.6) and integrate by parts s times. The boundary terms vanish: at $t = 1$ the derivatives of order less than s of $t^{\beta+s}(t-1)^s$ vanish, and at $t = 0$ they are $O(t^{\beta+1})$. This gives $\frac{m(m-1) \cdots (m-s+1)}{s!} \int_0^1 t^{\beta+m} (1-t)^s dt$, which is the stated value. For $m < s$ the product vanishes. The leading coefficient of J_s^β is $\Gamma(\beta+2s+1)/(s! \Gamma(\beta+s+1))$, and multiplying it by the value for $m = s$ gives $1/(\beta+2s+1)$.

(b) Divide the coefficient of t^k of each side by $[t^k]J_s^{\beta+1}$, respectively by $[t^k]J_s^{\beta-1}$, using (21.6). For $0 \leq k \leq s$ the first identity reduces to $(\beta+k+1)(\beta+2s+1) = (\beta+s+1)(\beta+s+k+1) - s(s-k)$, and for $1 \leq k \leq s$ the second to $-k(\beta+2s+1) = (\beta+s)(s-k+1) - (s+1)(\beta+s+k)$; both are verified by expansion. In the second identity the cases $k = 0$ and $k = s+1$ are checked directly. The computation for the first identity uses only $\beta > -1$. The remaining assertions are clear, and $s \leq (\beta+2s+1)/2$ for $\beta \geq -1$.

(c) By the orthogonality in (a), the coefficient is $(\beta+2s+1) \int_0^1 t^\beta (1-t)^m J_s^\beta dt$. Integrating by parts s times in the Rodrigues formula gives $\frac{(-1)^s}{s!} m(m-1) \cdots (m-s+1) \int_0^1 t^{\beta+s} (1-t)^m dt$ times $\beta+2s+1$, which proves the sign pattern. Let $\Lambda(f) = \frac{\beta}{2} \int_0^1 t^{\beta/2-1} f(t) dt$. The same integration by parts, now against $t^{-\beta/2-1}$, gives $\Lambda(J_s^\beta) = (-1)^s \frac{\beta}{2} (\frac{\beta}{2}+1)_s B(\frac{\beta}{2}, s+1)/s! = (-1)^s$. Hence $\|(1-t)^m\|_{J^\beta, 1} = \Lambda((1-t)^m) = \frac{\beta}{2} B(\frac{\beta}{2}, m+1) = m!/(\beta/2+1)_m$.

(d) By (b), multiplication by t is the lowering step followed by the raising step (with $\beta-1$ in place of β); both are given by nonnegative matrices with column sums one. Hence $t^M = t^M J_0^\beta$ has nonnegative coefficients summing to $t^M|_{t=1} = 1$. The coefficient of $J_0^\beta = 1$ is $(\beta+1) \int_0^1 t^{\beta+M} dt$. Therefore $\|t^M - 1\|_{J^\beta, 1}$ equals $(1 - \frac{\beta+1}{\beta+M+1})$ plus the sum of the other coefficients, which is the same number. $\square$

Chebyshev and Legendre coordinates. Let $a_j = \binom{2j}{j} 4^{-j}$. Then $a_0 = 1$, $a_{j+1}/a_j = (2j+1)/(2j+2)$, so (a_j) is decreasing, and $a_k \sqrt{k+1/4}$ is nondecreasing in k , because $(2k+1)^2(4k+5) - (2k+2)^2(4k+1) = 1$. Consequently

$$(21.7) \quad \frac{a_\ell}{a_m} \leq \left(\frac{m+1/4}{\ell+1/4} \right)^{1/2} \quad (\ell \leq m), \quad a_{\ell-j} \leq \sqrt{2} a_\ell \quad (0 \leq j \leq \ell/2).$$

Factoring the generating function, $(1 - 2\xi z + z^2)^{-1/2} = (1 - e^{i\theta} z)^{-1/2} (1 - e^{-i\theta} z)^{-1/2}$ for $\xi = \cos \theta$, gives $P_\ell(\cos \theta) = \sum_{j=0}^\ell a_j a_{\ell-j} e^{i(\ell-2j)\theta}$, that is,

$$P_\ell = \sum_n D_{n\ell} T_n, \quad D_{\ell-2j,\ell} = (2 - \delta_{\ell,2j}) a_j a_{\ell-j} \geq 0.$$

Conversely write $T_n = \sum_\ell C_{\ell n} P_\ell$ and $U_n = \sum_\ell v_{\ell n} P_\ell$. Let $\langle \cdot \rangle_R$ be the linear functional on polynomials in ξ with $\langle T_n \rangle_R = R^n$, and put

$$h_\ell = \langle P_\ell \rangle_R = \sum_n D_{n\ell} R^n, \quad \kappa_R = \frac{4}{\sqrt{1 - R^{-2}}} - 1, \quad B_0 = \frac{2\sqrt{2}}{\sqrt{1 - R^{-2}}}.$$

Numerically $\kappa_R \approx 12.118596$ and $B_0 \approx 9.276248$.

Lemma 21.4. (a) $C_{00} = 1$, $C_{nn} = 1/(2a_n)$ for $n \geq 1$, and for $n = \ell + 2j$ with $j \geq 1$,

$$C_{\ell n} = -\frac{n(2\ell + 1) a_j}{2(\ell + j)(2\ell + 2j + 1) a_{\ell+j} (2j - 1)} < 0.$$

(b) For $n = \ell + 2j$, $v_{\ell n} = (2\ell + 1) a_j / ((2\ell + 2j + 1) a_{\ell+j}) > 0$. The Legendre coefficients of $C_n^{(2)}$ are positive.

Proof. We use the connection formula for ultraspherical polynomials [22, 18.18.16],

$$C_n^{(\lambda)} = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(\lambda - \mu)_j (\lambda)_{n-j}}{(\mu + 1)_{n-j} j!} \frac{\mu + n - 2j}{\mu} C_{n-2j}^{(\mu)},$$

with $\mu = \frac{1}{2}$, so that $C_\ell^{(\mu)} = P_\ell$ and $(\mu + n - 2j)/\mu = 2\ell + 1$ for $\ell = n - 2j$. We use $(\frac{1}{2})_j = j! a_j$, $(-\frac{1}{2})_j = -j! a_j / (2j - 1)$ and $(\frac{3}{2})_m = (2m + 1) m! a_m$. For $\lambda = 1$, $C_n^{(1)} = U_n$, and the coefficient is $(\frac{1}{2})_j (n - j)! (2\ell + 1) / ((\frac{3}{2})_{n-j} j!)$, which is the stated $v_{\ell n}$. For $\lambda \rightarrow 0$, $C_n^{(\lambda)} / \lambda \rightarrow (2/n) T_n$ for $n \geq 1$ [22, §18.7(iii)] and $(\lambda)_{n-j} / \lambda \rightarrow (n - j - 1)!$, which gives (a). For $\lambda = 2$ all coefficients are positive, since $(\frac{3}{2})_j > 0$. $\square$

Lemma 21.5. For all $\ell, n \geq 0$:

- (a) $1 \leq h_\ell \leq B_0 a_\ell R^\ell$;
- (b) $h_{\ell+1} \leq R h_\ell$, and $h_{\ell-1} \leq R^{-1} \frac{\ell}{\ell-1} h_\ell$ for $\ell \geq 2$;
- (c) $\sum_\ell |C_{\ell n}| h_\ell \leq \kappa_R R^n$;
- (d) $C_{nn} h_n \leq \frac{B_0}{2} R^n$; for $j \geq 1$ and $\ell = n - 2j \geq 0$, $|C_{\ell n}| h_\ell \leq B_0 \frac{a_j R^{-2j}}{2j-1} R^n$ and $v_{\ell n} h_\ell \leq B_0 a_j R^{-2j} R^n$;
- (e) $D_{n\ell} R^n \leq \sqrt{2} R^{n-\ell} h_\ell$.

Proof. (a) $h_\ell \geq \sum_n D_{n\ell} = P_\ell(1) = 1$. By (21.7),

$$h_\ell \leq 2 \sum_{j \leq \ell/2} a_j a_{\ell-j} R^{\ell-2j} \leq 2\sqrt{2} a_\ell R^\ell \sum_j a_j R^{-2j},$$

and $\sum_j a_j z^j = (1 - z)^{-1/2}$.

(b) Write $R h_\ell - h_{\ell+1} = \sum_e \delta_e R^e$, where $e \equiv \ell + 1 \pmod{2}$ runs over $0 \leq e \leq \ell + 1$. Comparing coefficients, $\delta_e = 2a_i(a_{\ell-i} - a_{\ell+1-i}) \geq 0$ for $e \geq 2$, where $i = (\ell + 1 - e)/2$. The lowest exponent $e_0 \in \{0, 1\}$ has a negative coefficient: $\delta_1 = a_{\ell/2}(a_{\ell/2} - 2a_{\ell/2+1}) < 0$ if ℓ is even, and $\delta_0 = -a_{(\ell+1)/2}^2$ if ℓ is odd. For $R = 1$ all h_k equal one, so $\sum_e \delta_e = 0$. Hence $R h_\ell - h_{\ell+1} = \sum_{e > e_0} \delta_e (R^e - R^{e_0}) \geq 0$ for $R \geq 1$. For the second inequality we compare the coefficients of R^e in $R h_{\ell-1}$ and h_ℓ . For $e = \ell - 2i \geq 2$ they are $2a_i a_{\ell-1-i}$ and $2a_i a_{\ell-i}$, whose

ratio is $2m/(2m-1)$ with $m = \ell - i \geq (\ell + 2)/2$, hence at most $(\ell + 2)/(\ell + 1) \leq \ell/(\ell - 1)$. For $e = 1$ (ℓ odd, $i = (\ell - 1)/2$) they are a_i^2 and $2a_i a_{i+1}$, with ratio $(\ell + 1)/(2\ell) \leq \ell/(\ell - 1)$. For ℓ even, h_ℓ has an additional nonnegative coefficient at $e = 0$.

(c) For $n = 0$ the sum is $1 \leq \kappa_R$. For $n \geq 1$, since $h_\ell = \langle P_\ell \rangle_R$, we have $\sum_\ell C_{\ell n} h_\ell = \langle T_n \rangle_R = R^n$. By Lemma 21.4(a) all $C_{\ell n}$ with $\ell < n$ are negative, so $\sum_\ell |C_{\ell n}| h_\ell = 2C_{nn} h_n - R^n \leq (B_0 - 1)R^n \leq \kappa_R R^n$ by (a).

(d) The first bound follows from (a). For $j \geq 1$ put $m = \ell + j$. By (a) and Lemma 21.4, the second bound reduces to $n(2\ell + 1)a_\ell \leq 2m(2m + 1)a_m$. By (21.7) it suffices that $4m^2(2m + 1)^2(\ell + \frac{1}{4}) - n^2(2\ell + 1)^2(m + \frac{1}{4}) \geq 0$; with $m = \ell + j$, $n = \ell + 2j$ this is a polynomial in (ℓ, j) whose coefficients are all nonnegative. The third bound reduces to $(2\ell + 1)a_\ell \leq (2m + 1)a_m$, which follows from (21.7) and the monotonicity of $(2x + 1)^2/(x + \frac{1}{4})$ on $[0, \infty)$.

(e) For $\ell \geq 1$, $h_\ell \geq D_{\ell\ell} R^\ell = 2a_\ell R^\ell$, and $D_{n\ell} \leq 2a_{\ell-j} \leq 2\sqrt{2}a_\ell$ by (21.7). For $\ell = 0$ the claim is trivial. $\square$

Coefficient spaces. Let $\mathcal{L}$ be the Banach space of real families $f = (f_{\ell,s})_{\ell,s \geq 0}$ with $\|f\|_{\mathcal{L}} = \sum_{\ell,s} h_\ell |f_{\ell,s}| < \infty$, identified with the formal series $\sum f_{\ell,s} \Psi_{\ell,s}$, and let $\mathcal{L}^{\text{ev}}$ be its subspace with $f_{\ell,s} = 0$ for odd ℓ . The field space is

$$\mathcal{G} = \{g \in \mathcal{L}^{\text{ev}} : g_{\ell,0} = 0 \text{ for all } \ell\}.$$

For even $n \geq 0$ let $\mathcal{T}_n = \sum_\ell C_{\ell n} H_\ell$; this is the harmonic polynomial with boundary values $T_n(\xi)$ on ∂B , and $|\mathcal{T}_n| \leq 1$ on $\bar{B}$ by the maximum principle. The output space $\mathcal{Y}$ is the Banach space of families $f = ((f_n)_{n \text{ even}}, (f_{\ell,s})_{\ell \text{ even}, s \geq 1})$, identified with $\sum_n f_n \mathcal{T}_n + \sum_{s \geq 1} f_{\ell,s} \Psi_{\ell,s}$, with

$$\|f\|_{\mathcal{Y}} = \sum_n R^n |f_n| + \sum_{\ell,s \geq 1} h_\ell |f_{\ell,s}|.$$

Thus the harmonic layer of $\mathcal{Y}$ is measured through the Chebyshev coefficients of its boundary trace. Since $H_\ell = \sum_n D_{n\ell} \mathcal{T}_n$ with $D_{n\ell} \geq 0$, the inclusion $\mathcal{L}^{\text{ev}} \rightarrow \mathcal{Y}$ satisfies $\|f\|_{\mathcal{Y}} \leq \|f\|_{\mathcal{L}}$. The unknowns lie in $\mathcal{X} = \mathcal{G} \times \mathcal{E}$, with norm $\|(g, c)\|_{\mathcal{X}} = \|g\|_{\mathcal{L}} + \sigma \|c\|_{\mathcal{E}}$ for a constant $\sigma > 0$ fixed in Section 21.5. All three spaces are weighted ℓ^1 spaces, and Lemma 15.2 applies to operators between them.

Lemma 21.6. (a) *The functions $\Psi_{\ell,s}$ are pairwise orthogonal in $L^2(B)$, $\|\Psi_{\ell,s}\|_{L^2(B)}^2 = 4\pi/((2\ell + 1)(2\ell + 4s + 3))$, and $\Psi_{\ell,s}$ is orthogonal to all polynomials on $\mathbb{R}^3$ of degree less than $\ell + 2s$.*

(b) *The series defining the elements of $\mathcal{L}$ and $\mathcal{Y}$ converge in $L^2(B)$, and the resulting maps $\iota: \mathcal{L} \rightarrow L^2(B)$ and $\iota: \mathcal{Y} \rightarrow L^2(B)$ are continuous, injective and compatible with the inclusion $\mathcal{L}^{\text{ev}} \subset \mathcal{Y}$. The harmonic part $\sum_n f_n \mathcal{T}_n$ of $f \in \mathcal{Y}$ converges uniformly on $\bar{B}$.*

Proof. (a) In spherical coordinates,

$$\int_B \Psi_{\ell,s} \Psi_{\ell',s'} dx = 2\pi \int_{-1}^1 P_\ell P_{\ell'} d\xi \cdot \frac{1}{2} \int_0^1 t^{(\ell+\ell'+1)/2} J_s^{\ell+1/2} J_{s'}^{\ell'+1/2} dt,$$

and Lemma 21.3(a) gives orthogonality and the norm. A polynomial of degree less than $\ell + 2s$ is a sum of terms $|x|^{2m} Y_k$ with Y_k harmonic and homogeneous of degree k , $k + 2m < \ell + 2s$ [64, Ch. IV]. The angular integral of $\Psi_{\ell,s} |x|^{2m} Y_k$ vanishes unless $k = \ell$, and then $m < s$ and the radial integral $\frac{1}{2} \int_0^1 t^{\ell+1/2+m} J_s^{\ell+1/2} dt$ vanishes by Lemma 21.3(a).

(b) Continuity follows from $\|\Psi_{\ell,s}\|_{L^2} \leq (4\pi/3)^{1/2} \leq (4\pi/3)^{1/2} h_\ell$ and $|\mathcal{T}_n| \leq 1 \leq R^n$. On $\mathcal{L}$, injectivity follows from (a). Let $f \in \mathcal{Y}$ with $\iota f = 0$. The $\mathcal{T}_n$ are harmonic polynomials, hence orthogonal to all $\Psi_{\ell,s}$ with $s \geq 1$ by (a); pairing with $\Psi_{\ell,s}$ gives $f_{\ell,s} = 0$ for $s \geq 1$. The

harmonic part $h = \sum_n f_n \mathcal{T}_n$ converges uniformly on $\overline{B}$, so it is harmonic in B , continuous on $\overline{B}$, and vanishes in B ; hence its boundary values $\sum_n f_n \mathcal{T}_n(\cos \theta) = \sum_n f_n \cos n\theta$ vanish, and all $f_n = 0$. $\square$

21.3. Multiplication.

Lemma 21.7 (Adams). *For $a, \ell \geq 0$, $P_a P_\ell = \sum_k G_{a\ell k} P_k$, where $G_{a\ell k} \geq 0$ vanishes unless $|a - \ell| \leq k \leq a + \ell$ and $k \equiv a + \ell \pmod{2}$, and $\sum_k G_{a\ell k} = 1$. Consequently $H_a H_\ell = \sum_k G_{a\ell k} H_k t^{(a+\ell-k)/2}$ and*

$$\sum_k G_{a\ell k} h_k \leq h_a h_\ell.$$

Proof. Adams' formula [1] gives, with $v = (a + \ell + k)/2$ in the support,

$$G_{a\ell k} = (2k + 1) \left[\frac{v!}{(v-a)!(v-\ell)!(v-k)!} \right]^2 \frac{(2v-2a)!(2v-2\ell)!(2v-2k)!}{(2v+1)!} \geq 0;$$

the sum is one by evaluation at $\xi = 1$, and the identity for $H_a H_\ell$ follows after multiplication by $r^{a+\ell}$. Finally, $\sum_k G_{a\ell k} h_k = \langle P_a P_\ell \rangle_R$, and $P_a P_\ell = \sum_{m,n} D_{ma} D_{n\ell} \frac{1}{2} (T_{m+n} + T_{|m-n|})$ with $D \geq 0$, so $\langle P_a P_\ell \rangle_R \leq \sum_{m,n} D_{ma} D_{n\ell} R^{m+n} = h_a h_\ell$. $\square$

Every axisymmetric polynomial m has unique expansions

$$m = \sum_{a,k \geq 0} m_{a,k} H_a t^k = \sum_{n,k \geq 0} \tilde{m}_{n,k} r^{n+2k} T_n(\xi),$$

the first by the Fischer decomposition, the second because $r^n T_n(\xi) = \operatorname{Re}(x_1 + iy)^n$. The homogeneous part of degree d of m consists of the terms with $a+2k = d$, respectively $n+2k = d$; we call $H_a t^k$ a *solid monomial* of degree $a + 2k$. We put

$$\tau(m) = \sum_{a,k} h_a |m_{a,k}|, \quad \|m\|_{\text{ch}} = \sum_{n,k} R^n |\tilde{m}_{n,k}|,$$

and extend both to series with finite norm; such series converge uniformly on $\overline{B}$, since $|H_a t^k| \leq 1$ and $|r^{n+2k} T_n(\xi)| \leq 1$ there.

Lemma 21.8. (a) *Let $H_a t^k$ be a solid monomial and $\ell, s \geq 0$. Then*

$$(21.8) \quad H_a t^k \Psi_{\ell,s} = \sum_v G_{a\ell v} \sum_{\varsigma \geq 0} \pi_\varsigma^{(v)} \Psi_{v,\varsigma},$$

where, for each v , $(\pi_\varsigma^{(v)})_\varsigma$ is a probability vector. Consequently $\|mf\|_{\mathcal{L}} \leq \tau(m) \|f\|_{\mathcal{L}}$ for every $f \in \mathcal{L}$ and every m with $\tau(m) < \infty$.

(b) $\tau(m) \leq \kappa_R \|m\|_{\text{ch}}$. If $m = r^{n+2k} U_n(\xi)$ or $m = r^{n+2k} C_n^{(2)}(\xi)$, then $\tau(m) = \|m\|_{\text{ch}}$.

(c) $\|m_1 m_2\|_{\text{ch}} \leq \|m_1\|_{\text{ch}} \|m_2\|_{\text{ch}}$ and $\|\operatorname{Re}(w^\alpha \overline{w}^\beta)\|_{\text{ch}} \leq R^{\alpha+\beta}$.

(d) For m with $\tau(m) < \infty$ and $f \in \mathcal{L}$, $\iota(mf) = m \iota(f)$.

Proof. (a) By Lemma 21.7, $H_a t^k \Psi_{\ell,s} = \sum_v G_{a\ell v} H_v t^e J_s^{\ell+1/2}$ with $e = (a + \ell - v)/2 + k$. If $v \geq \ell$, we apply the first identity of Lemma 21.3(b) $v - \ell$ times, raising the parameter from $\ell + \frac{1}{2}$ to $v + \frac{1}{2}$, and then write the factor t^e as e multiplications by t , each a lowering step followed by a raising step. If $v < \ell$, we apply the second identity $\ell - v$ times, which uses $\ell - v$ factors t ; this is possible because $e \geq \ell - v$, as $\ell - v \leq a$. The remaining $e - (\ell - v)$ factors t are treated as before. Every step is given by a nonnegative matrix with column sums one, which proves (21.8). Since $\|\Psi_{v,\varsigma}\|_{\mathcal{L}} = h_v$, Lemma 21.7 gives $\|H_a t^k \Psi_{\ell,s}\|_{\mathcal{L}} \leq h_a h_\ell$, and the bound for mf follows by absolute summation.

(b) Since $r^{n+2k}T_n(\xi) = \sum_{\ell} C_{\ell n} H_{\ell} t^{(n-\ell)/2+k}$, Lemma 21.5(c) gives $\tau(r^{n+2k}T_n) \leq \kappa_R R^n$. For U_n the Legendre coefficients are nonnegative (Lemma 21.4(b)), so $\tau(r^{n+2k}U_n) = \sum_{\ell} v_{\ell n} h_{\ell} = \langle U_n \rangle_R$, which is the Chebyshev norm, because $U_n(\cos \theta) = \sum_{|\nu| \leq n, \nu \equiv n \pmod{2}} e^{i\nu\theta}$ has non-negative Chebyshev coefficients. For $C_n^{(2)}$ the same argument applies, since $C_n^{(2)}(\cos \theta) = \sum_{p+p'=n} (p+1)(p'+1)e^{i(p-p')\theta}$.

(c) $T_m T_n = \frac{1}{2}(T_{m+n} + T_{|m-n|})$ and $R^{|m-n|} \leq R^{m+n}$; and $\operatorname{Re}(w^{\alpha} \bar{w}^{\beta}) = r^{\alpha+\beta} T_{|\alpha-\beta|}(\xi)$.

(d) Both sides are continuous in $f \in \mathcal{L}$ with values in $L^2(B)$, by (a) and Lemma 21.6, and m is bounded; they agree when f is a finite combination of the $\Psi_{\ell,s}$ and m is a polynomial, and the general case follows by uniform convergence of m and density. $\square$

Lemma 21.9. *Let $H_a t^k$ be a solid monomial of degree $d = a + 2k$, let $\ell, s \geq 0$, and let $\pi^{(v)}$ be as in (21.8). Then:*

- (a) $\pi_{\zeta}^{(v)} = 0$ unless $s - d \leq \zeta \leq s + d$, and $G_{\alpha v} = 0$ unless $|\ell - a| \leq v \leq \ell + a$;
- (b) $\pi_0^{(v)} \leq \vartheta_d(\ell, s)$, where $\vartheta_d(\ell, s) = (d/(\ell + d + \frac{5}{2}))^s$ if $d \geq s$ (with $0^0 = 1$) and $\vartheta_d(\ell, s) = 0$ if $d < s$;
- (c) for every integer $j \geq 1$, $\sum_{\zeta \leq s-j} \pi_{\zeta}^{(v)} \leq 2^{-d} \sum_{i \geq j} \binom{d}{i}$.

Proof. In the proof of Lemma 21.8(a), only the raising steps can decrease the radial index, each by at most one, and only the lowering steps can increase it, each by at most one. The number of raising steps is $e + v - \ell = (a - \ell + v)/2 + k$ and the number of lowering steps is $e = (a + \ell - v)/2 + k$; since $|\ell - v| \leq a$, both are at most $a + k \leq d$. This proves (a), the statement about v being part of Lemma 21.7.

(b) Since $J_0^{v+1/2} = 1$, Lemma 21.3(a) gives $\pi_0^{(v)} = (v + \frac{3}{2}) \int_0^1 t^{v+1/2+e} J_s^{\ell+1/2} dt$, which is the value in Lemma 21.3(a) with $\beta = \ell + \frac{1}{2}$ and $m = e + v - \ell = (a - \ell + v)/2 + k$, multiplied by $v + \frac{3}{2}$. Here $0 \leq m \leq a + k \leq d$. If $m < s$, in particular if $d < s$, it vanishes. Otherwise we pair the factor $v + \frac{3}{2}$ with $\ell + m + \frac{3}{2}$, which is not smaller because $\ell + m = (a + \ell + v)/2 + k \geq v$, and each factor $m - i$ ($0 \leq i < s$) with $\ell + m + \frac{5}{2} + i$; since $x \mapsto x/(\ell + x + \frac{5}{2})$ is increasing, each quotient $(m - i)/(\ell + m + \frac{5}{2} + i)$ is at most $d/(\ell + d + \frac{5}{2})$.

(c) The expansion coefficients define a Markov chain on the radial index: a raising step at parameter β moves ζ to $\zeta - 1$ with probability $\zeta/(\beta + 2\zeta + 1) \leq \frac{1}{2}$ and otherwise leaves it unchanged, and lowering steps never decrease ζ . Let N_i be the number of decreases among the first i raising steps. Then $\Pr(N_{i+1} \geq j) \leq \Pr(N_i \geq j) + \frac{1}{2} \Pr(N_i = j - 1) = \frac{1}{2} (\Pr(N_i \geq j) + \Pr(N_i \geq j - 1))$, and by induction $\Pr(N_i \geq j) \leq \Pr(\operatorname{Bin}(i, \frac{1}{2}) \geq j)$. Since there are at most d raising steps and the final index is at least $s - N$, the claim follows. (The probabilistic language only organises sums of products of the nonnegative coefficients.) $\square$

21.4. The operator K_3 . For $\ell \geq 0$ and $s \geq 1$ put $\beta = \ell + \frac{1}{2}$, $d = \beta + 2s = \ell + 2s + \frac{1}{2}$, and define

$$(21.9) \quad K_3 \Psi_{\ell,s} = \frac{\Psi_{\ell,s-1}}{4d(d+1)} - \frac{\Psi_{\ell,s}}{2d(d+2)} + \frac{\Psi_{\ell,s+1}}{4(d+1)(d+2)} = H_{\ell} k_{\ell,s}(t),$$

and let $E = x \cdot \nabla = r \partial_r$ be the Euler operator.

Lemma 21.10. *For $\ell \geq 0$ and $s \geq 1$, with β, d as above:*

- (a) $k'_{\ell,s} = (J_s^{\beta+1} - J_{s-1}^{\beta+1})/(4(d+1))$;
- (b) $\Delta K_3 \Psi_{\ell,s} = \Psi_{\ell,s}$;
- (c) $K_3 \Psi_{\ell,s} = 0$ and $\nabla K_3 \Psi_{\ell,s} = 0$ on ∂B ;

(d) with $\Psi_{-1,\cdot} = 0$,

$$\partial_{x_1} K_3 \Psi_{\ell,s} = \frac{\ell+1}{(2\ell+1)2(d+1)} (\Psi_{\ell+1,s} - \Psi_{\ell+1,s-1}) + \frac{\ell}{(2\ell+1)2(d+1)} (\Psi_{\ell-1,s+1} - \Psi_{\ell-1,s}),$$

$$EK_3 \Psi_{\ell,s} = -\frac{d+\frac{1}{2}}{4d(d+1)} \Psi_{\ell,s-1} + \frac{\Psi_{\ell,s}}{4d(d+2)} + \frac{d+\frac{3}{2}}{4(d+1)(d+2)} \Psi_{\ell,s+1},$$

and $y\partial_y K_3 \Psi_{\ell,s} = EK_3 \Psi_{\ell,s} - x_1 \partial_{x_1} K_3 \Psi_{\ell,s}$;

(e) K_3 , $\partial_{x_1} K_3$, EK_3 and $y\partial_y K_3$ extend to bounded operators $\mathcal{G} \rightarrow \mathcal{L}$ with $\|K_3\| = 4/45$, $\|\partial_{x_1} K_3\| \leq 2R/7$, $\|EK_3\| \leq 6/35$ and $\|y\partial_y K_3\| \leq 6/35 + 2R^2/7$.

Proof. Write $A = \frac{1}{4d(d+1)}$, $B = -\frac{1}{2d(d+2)}$, $C = \frac{1}{4(d+1)(d+2)}$ for the coefficients in (21.9), and $J_\zeta = J_\zeta^\beta$. For $0 \leq k \leq s$ put $u = s - k$; then $\beta + s + k + 1 = d - u + 1$, and (21.6) gives

$$\begin{aligned} \frac{[t^{k+1}]J_{s-1}}{[t^k]J_s} &= \frac{u(u-1)}{(k+1)(\beta+k+1)}, & \frac{[t^{k+1}]J_s}{[t^k]J_s} &= -\frac{u(d-u+1)}{(k+1)(\beta+k+1)}, \\ \frac{[t^{k+1}]J_{s+1}}{[t^k]J_s} &= \frac{(d-u+1)(d-u+2)}{(k+1)(\beta+k+1)}. \end{aligned}$$

Since $u(u-1)(d+2) + 2u(d+1)(d-u+1) + d(d-u+1)(d-u+2) = d(d+1)(d+2)$ identically in u and d ,

$$(21.10) \quad [t^{k+1}]k_{\ell,s} = \frac{[t^k]J_s}{4(k+1)(\beta+k+1)} \quad (0 \leq k \leq s).$$

(a) The coefficient of t^k in $k'_{\ell,s}$ is $(k+1)[t^{k+1}]k_{\ell,s}$. On the other hand, by (21.6),

$$\frac{[t^k]J_s^{\beta+1}}{[t^k]J_s} = \frac{\beta+s+k+1}{\beta+k+1}, \quad \frac{[t^k]J_{s-1}^{\beta+1}}{[t^k]J_s} = -\frac{s-k}{\beta+k+1},$$

so the coefficient of t^k on the right-hand side is

$$[t^k]J_s \frac{\beta+2s+1}{4(d+1)(\beta+k+1)} = \frac{[t^k]J_s}{4(\beta+k+1)}.$$

Both sides have degree s .

(b) For a polynomial $v(t)$, $\Delta(H_\ell v) = v\Delta H_\ell + 2\nabla H_\ell \cdot \nabla v + H_\ell \Delta v = 4H_\ell(tv'' + (\ell + \frac{3}{2})v')$, by (21.5) and $\Delta v(|x|^2) = 4tv'' + 6v'$. The coefficient of t^k in $4(tk'' + (\beta+1)k')$ is $4(k+1)(\beta+k+1)[t^{k+1}]k_{\ell,s} = [t^k]J_s$ by (21.10).

(c) Since $J_\zeta(1) = 1$, $k_{\ell,s}(1) = A + B + C = 0$, and $k'_{\ell,s}(1) = 0$ by (a). Hence $K_3 \Psi_{\ell,s} = 0$ and $\nabla(H_\ell k_{\ell,s}) = k_{\ell,s} \nabla H_\ell + 2k'_{\ell,s} H_\ell x = 0$ on ∂B .

(d) By (21.5), $\partial_{x_1}(H_\ell k) = \frac{2(\ell+1)}{2\ell+1} H_{\ell+1} k' + \frac{2\ell}{2\ell+1} H_{\ell-1}(\beta k + tk')$ and $E(H_\ell k) = H_\ell(\ell k + 2tk')$, with $k = k_{\ell,s}$. The $H_{\ell+1}$ term is given by (a). For the others we expand in the basis $J_\zeta = J_\zeta^\beta$: by (a) and the second identity of Lemma 21.3(b) (with $\beta+1$ in place of β),

$$tk' = \frac{1}{4(d+1)} \left(-\frac{\beta+s}{d} J_{s-1} + \frac{\beta(d+1)}{d(d+2)} J_s + \frac{s+1}{d+2} J_{s+1} \right),$$

where we used $(\beta+s+1)d - s(d+2) = \beta(d+1)$. Adding βk , respectively ℓk , gives

$$\begin{aligned} \beta k + tk' &= -\frac{s}{4d(d+1)} J_{s-1} - \frac{\beta}{4d(d+2)} J_s + \frac{\beta+s+1}{4(d+1)(d+2)} J_{s+1}, \\ \ell k + 2tk' &= -\frac{d+\frac{1}{2}}{4d(d+1)} J_{s-1} + \frac{1}{4d(d+2)} J_s + \frac{d+\frac{3}{2}}{4(d+1)(d+2)} J_{s+1}. \end{aligned}$$

The second line is the formula for EK_3 . By the first identity of Lemma 21.3(b) (with $\beta - 1$ in place of β), $(J_{s+1}^{\beta-1} - J_s^{\beta-1})/(4(d+1))$ has the same expansion as the first line, which gives the formula for $\partial_{x_1}K_3$. The last identity holds because $E = x_1\partial_{x_1} + y\partial_y$ on axisymmetric functions.

(e) We use Lemma 15.2. The operator K_3 preserves ℓ , and its column sum is $|\mathbf{A}| + |\mathbf{B}| + |\mathbf{C}| = 1/(d(d+2))$, which is largest, $4/45$, for $\ell = 0$, $s = 1$. For EK_3 the column sum is $(2d+1)/(4d(d+1))$, decreasing in d , with value $6/35$ at $d = \frac{5}{2}$. For $\partial_{x_1}K_3$ and even ℓ , the normalised column norm is at most $((\ell+1)h_{\ell+1} + \ell h_{\ell-1})/((2\ell+1)(d+1)h_\ell)$. For $\ell = 0$ this is at most $R/(d+1) \leq 2R/7$. For $\ell \geq 2$, Lemma 21.5(b), $\ell^2/(\ell-1) \leq \ell+2$ and $d+1 \geq \ell + \frac{7}{2}$ bound it by $R(2\ell+3)/((2\ell+1)(\ell + \frac{7}{2})) \leq 7R/27.5 < 2R/7$. Finally $\|x_1f\|_{\mathcal{L}} \leq h_1\|f\|_{\mathcal{L}} = R\|f\|_{\mathcal{L}}$ by Lemma 21.8(a). $\square$

The following lemma shows that K_3g is a genuine C^1 function with vanishing Cauchy data. Let $\mathcal{P}_n$ be the space of real polynomials of degree at most n on $\mathbb{R}^3$, with the inner product of $L^2(B)$, and let V_n be the orthogonal complement of $\mathcal{P}_{n-1}$ in $\mathcal{P}_n$.

Lemma 21.11. (a) For $P \in V_n$ and $x \in \overline{B}$, $|P(x)|^2 \leq \frac{(n+1)(n+2)(2n+3)}{8\pi} \|P\|_{L^2(B)}^2$.
 (b) For $\ell \geq 0$ and $s \geq 1$, $\sup_{\overline{B}} |\nabla K_3 \Psi_{\ell,s}| \leq 1$ and $\sup_{\overline{B}} |K_3 \Psi_{\ell,s}| \leq 1$.
 (c) For $g \in \mathcal{G}$, the series $W = \sum g_{\ell,s} K_3 \Psi_{\ell,s}$ converges in $C^1(\overline{B})$; $W = 0$ and $\nabla W = 0$ on ∂B ; $\Delta W = \iota(g)$ in $\mathcal{D}'(B)$; and $\iota(K_3g) = W$, $\iota(\partial_{x_1}K_3g) = \partial_{x_1}W$, $\iota(y\partial_yK_3g) = y\partial_yW$.

Proof. (a) Let $S^4 \subset \mathbb{R}^5 = \mathbb{R}^3 \times \mathbb{R}^2$, with points (x, z) and surface measure $d\zeta$. Parametrising S^4 by $(x, \phi) \mapsto (x, \sqrt{1-|x|^2}(\cos \phi, \sin \phi))$ one finds $d\zeta = dx d\phi$, so $\int_{S^4} f(x) d\zeta = 2\pi \int_B f dx$ for integrable f . Let $Q(x, z) = P(x)$. Its restriction to S^4 is a sum $\sum_{k \leq n} Y_k$ of spherical harmonics of degree k [64, Ch. IV]. Since Q is invariant under rotations of z and the projections onto the spaces of spherical harmonics commute with rotations, each Y_k is the restriction of a harmonic homogeneous polynomial invariant under these rotations, that is, of a polynomial in x and $|z|^2$; as $|z|^2 = 1 - |x|^2$ on S^4 , $Y_k(x, z) = P_k(x)$ for some $P_k \in \mathcal{P}_k$. For $k < n$, $\|Y_k\|_{L^2(S^4)}^2 = \int_{S^4} QY_k d\zeta = 2\pi \int_B PP_k dx = 0$. Hence $Q|_{S^4} = Y_n$ is a spherical harmonic of degree n . For an orthonormal basis (E_i) of the spherical harmonics of degree n on S^4 , $\sum_i |E_i|^2$ is invariant under rotations, hence equal to $N_n/|S^4|$, where $N_n = \binom{n+4}{4} - \binom{n+2}{4} = (n+1)(n+2)(2n+3)/6$ and $|S^4| = 8\pi^2/3$. By the Cauchy–Schwarz inequality, $|P(x)|^2 = |Y_n(x, z)|^2 \leq \frac{N_n}{|S^4|} \|Y_n\|_{L^2(S^4)}^2 = \frac{2\pi N_n}{|S^4|} \|P\|_{L^2(B)}^2$.

(b) Let $n = \ell + 2s$ and $\Xi = K_3 \Psi_{\ell,s}$, a polynomial of degree $n+2$ whose homogeneous components have the parity of n . By Lemma 21.6(a), Ξ is orthogonal to $\mathcal{P}_{n-3}$. For $\varphi \in \mathcal{P}_{n-2}$ and $i = 1, 2, 3$, $\int_B \partial_i \Xi \varphi = - \int_B \Xi \partial_i \varphi = 0$, because $\Xi = 0$ on ∂B . Hence $\partial_i \Xi \in V_{n-1} \oplus V_n \oplus V_{n+1}$. The reflection $x \mapsto -x$ preserves each V_m , and an element of V_m of parity $(-1)^{m+1}$ has no homogeneous part of degree m , hence lies in $\mathcal{P}_{m-1} \cap V_m = 0$; since $\partial_i \Xi$ has parity $(-1)^{n+1}$ and the orthogonal projections onto the V_m commute with the reflection, the component of $\partial_i \Xi$ in V_n vanishes, and $\partial_i \Xi = P_i + Q_i$ with $P_i \in V_{n-1}$, $Q_i \in V_{n+1}$. By (a) and the Cauchy–Schwarz inequality, with $\gamma_m = (m+1)(m+2)(2m+3)/(8\pi)$,

$$|\nabla \Xi(x)|^2 \leq (\gamma_{n-1} + \gamma_{n+1}) \sum_i (\|P_i\|^2 + \|Q_i\|^2) = (\gamma_{n-1} + \gamma_{n+1}) \|\nabla \Xi\|_{L^2(B)}^2.$$

By (c) of Lemma 21.10, (b) of the same lemma and orthogonality, $\|\nabla \Xi\|_{L^2}^2 = - \int_B \Xi \Delta \Xi = - \int_B \Xi \Psi_{\ell,s} = \|\Psi_{\ell,s}\|_{L^2}^2/(2d(d+2))$ with $d = n + \frac{1}{2}$. Since $\gamma_{n-1} + \gamma_{n+1} = (2n+3)(n^2+3n+5)/(4\pi)$ and

$$\|\Psi_{\ell,s}\|_{L^2}^2 = 4\pi/((2\ell+1)(2n+3)),$$

$$|\nabla\Xi(x)|^2 \leq \frac{2(n^2+3n+5)}{(2\ell+1)(2n+1)(2n+5)} \leq 1,$$

because $(4n^2+12n+5) - (2n^2+6n+10) = 2n^2+6n-5 > 0$ for $n \geq 1$. As $\Xi = 0$ on ∂B , integration along a radius gives $|\Xi| \leq 1$ on $\bar{B}$.

(c) By (b), $\sum |g_{\ell,s}| \leq \|g\|_{\mathcal{L}}$ bounds the series in $C^1(\bar{B})$. The partial sums $W_N = K_3 g_N$ are polynomials with $\Delta W_N = g_N$ and vanishing Cauchy data (Lemma 21.10); since $g_N \rightarrow \iota(g)$ in $L^2(B)$ and $W_N \rightarrow W$ in $C^1(\bar{B})$, the Cauchy data and the distributional equation pass to the limit. The coefficient series $K_3 g$, $\partial_{x_1} K_3 g$ and $y \partial_y K_3 g$ converge in $\mathcal{L}$ (Lemma 21.10(e)), hence in $L^2(B)$, to the same limits. $\square$

21.5. The quartic equation. Let

$$b = 7626062690497999 \cdot 2^{-46}, \quad \sigma = 8b,$$

a fixed dyadic rational independent of the varying coefficient c_1 . For $x = (g, c) \in \mathcal{X}$ let $W = K_3 g$, $W_1 = \partial_{x_1} K_3 g$ and $W_2 = y \partial_y K_3 g$, elements of $\mathcal{L}$ by Lemma 21.10(e), and define

$$(21.11) \quad F(g, c) = q_c g + (\partial_{x_1} q_c) W_1 + \left(\frac{\partial_y q_c}{y} \right) W_2 + q_c |p_c|^2 (1 + W), \quad \mathcal{F} = F/b,$$

where the multipliers are given by (21.1), (21.2), the products are those of Lemma 21.8, and $1 = \Psi_{0,0}$.

Lemma 21.12. *$\mathcal{F}$ is a bounded polynomial map of degree four from $\mathcal{X}$ to $\mathcal{Y}$. If $\mathcal{F}(g, c) = 0$ and $\operatorname{Re} \psi'_c > 0$ on $\{|w| < R_0\}$ for some $1 < R_0 < R$, then $U = 1 + \iota(K_3 g) \in C^1(\bar{B})$ is axisymmetric, satisfies (21.4), and $U = 1$, $\nabla U = 0$ on ∂B .*

Proof. By (21.1), $\|r^{j-1} U_{j-1}\|_{\text{ch}} = \sum_{|\nu| \leq j-1, \nu \text{ even}} R^{|\nu|}$, which is at most $j R^{j-1}$ and at most $2R^{j+1}/(R^2 - 1)$; similarly $\|r^m C_m^{(2)}\|_{\text{ch}} = \sum_{p+p'=m} (p+1)(p'+1) R^{|p-p'|} \leq 2(m+1)R^{m+4}/(R^2 - 1)^2$. Hence

$$(21.12) \quad \begin{aligned} \|q_c\|_{\text{ch}} &\leq \frac{\|c\|_{\mathcal{E}}}{R}, & \|\partial_{x_1} q_c\|_{\text{ch}} &\leq \frac{2\|c\|_{\mathcal{E}}}{R^2 - 1}, \\ \left\| \frac{\partial_y q_c}{y} \right\|_{\text{ch}} &\leq \frac{4R\|c\|_{\mathcal{E}}}{(R^2 - 1)^2}, & \| |p_c|^2 \|_{\text{ch}} &\leq \frac{\|c\|_{\mathcal{E}}^2}{R^2}, \end{aligned}$$

the last by (21.2), Lemma 21.8(c) and the identity $\sum_a |p_a| R^a = R^{-1} \|c\|_{\mathcal{E}}$. Using parts (a) and (b) of Lemma 21.8 and Lemma 21.10(e), it follows that (21.11) is a sum of bounded multilinear maps of degree at most four. Their values are even in x_1 , by the parities recorded after (21.2), so they lie in $\mathcal{L}^{\text{ev}}$, which is contained in $\mathcal{Y}$.

Suppose $\mathcal{F}(g, c) = 0$. By Lemma 21.11(c), $W = \iota(K_3 g) \in C^1(\bar{B})$ has vanishing Cauchy data and $\Delta W = \iota(g)$, and by Lemma 21.8(d) and Lemma 21.6(b),

$$0 = b \iota(\mathcal{F}(g, c)) = q_c \iota(g) + \partial_{x_1} q_c \partial_{x_1} W + \frac{\partial_y q_c}{y} y \partial_y W + q_c |p_c|^2 U \quad \text{a.e. in } B.$$

Off the axis, $\nabla q_c \cdot \nabla U = \partial_{x_1} q_c \partial_{x_1} W + \partial_y q_c \partial_y W$. Let $\varphi \in C_c^\infty(B)$. As q_c is smooth on $\bar{B}$ (Lemma 21.1(a)), $q_c \varphi \in C_c^\infty(B)$ and

$$\begin{aligned} \int_B (q_c \nabla U \cdot \nabla \varphi - q_c |p_c|^2 U \varphi) &= \int_B (\nabla W \cdot \nabla (q_c \varphi) - \varphi \nabla q_c \cdot \nabla U - q_c |p_c|^2 U \varphi) \\ &= - \int_B \varphi b \iota(\mathcal{F}(g, c)) = 0, \end{aligned}$$

using $\int \nabla W \cdot \nabla (q_c \varphi) = - \int \iota(g) q_c \varphi$. $\square$

Nonlinear bounds. Let $x^\circ = (g^\circ, c^\circ) \in \mathcal{X}$ and let $G_0 \geq \|g^\circ\|_{\mathcal{L}}$, $C_0 \geq \|c^\circ\|_{\mathcal{E}}$. Put

$$\begin{aligned} L_2 &= \kappa_R \left[\frac{1}{R} + \frac{2}{R^2 - 1} \cdot \frac{2R}{7} + \frac{4R}{(R^2 - 1)^2} \left(\frac{6}{35} + \frac{2R^2}{7} \right) \right], & L_3 &= \frac{\kappa_R}{R^3}, & L_4 &= \frac{4}{45} \cdot \frac{\kappa_R}{R^3}, \\ c_2 &= \frac{1}{b} \left[\frac{L_2 + 3L_4 C_0^2}{\sigma} + \frac{3C_0(L_3 + L_4 G_0)}{\sigma^2} \right], & c_3 &= \frac{1}{b} \left[\frac{L_3 + L_4 G_0}{\sigma^3} + \frac{3L_4 C_0}{\sigma^2} \right], & c_4 &= \frac{L_4}{b\sigma^3}. \end{aligned}$$

Lemma 21.13. For $h \in \mathcal{X}$ with $\|h\|_{\mathcal{X}} \leq r$,

$$\begin{aligned} \|\mathcal{F}(x^\circ + h) - \mathcal{F}(x^\circ) - D\mathcal{F}(x^\circ)h\|_{\mathcal{Y}} &\leq c_2 r^2 + c_3 r^3 + c_4 r^4, \\ \|D\mathcal{F}(x^\circ + h) - D\mathcal{F}(x^\circ)\|_{\mathcal{X} \rightarrow \mathcal{Y}} &\leq 2c_2 r + 3c_3 r^2 + 4c_4 r^3. \end{aligned}$$

Proof. Let $\mathbf{b}(c, g) = q_c g + (\partial_{x_1} q_c) W_1(g) + (y^{-1} \partial_y q_c) W_2(g)$ and let $\mathbf{t}(c^1, c^2, c^3)$ be the symmetrisation of $q_{c^1} \operatorname{Re}(p_{c^2} \overline{p_{c^3}})$, so that $b\mathcal{F}(g, c) = \mathbf{b}(c, g) + \mathbf{t}(c, c, c)(1 + K_3 g)$. By Lemma 21.8, (21.12) and Lemma 21.10(e),

$$\|\mathbf{b}(c, g)\|_{\mathcal{Y}} \leq L_2 \|c\|_{\mathcal{E}} \|g\|_{\mathcal{L}}, \quad \|\mathbf{t}(c^1, c^2, c^3)f\|_{\mathcal{L}} \leq L_3 \prod_i \|c^i\|_{\mathcal{E}} \|f\|_{\mathcal{L}},$$

and $\|K_3 \gamma\|_{\mathcal{L}} \leq \frac{4}{45} \|\gamma\|_{\mathcal{L}}$. Write $h = (\gamma, \eta)$; then $\|\gamma\|_{\mathcal{L}} \leq r$ and $\|\eta\|_{\mathcal{E}} \leq r/\sigma$. The terms of degree at least two in h of $b\mathcal{F}(x^\circ + h)$ are $\mathbf{b}(\eta, \gamma)$, $3\mathbf{t}(c^\circ, \eta, \eta) + \mathbf{t}(\eta, \eta, \eta)$, and

$$3\mathbf{t}(c^\circ, c^\circ, \eta)K_3 \gamma + 3\mathbf{t}(c^\circ, \eta, \eta)K_3 g^\circ + 3\mathbf{t}(c^\circ, \eta, \eta)K_3 \gamma + \mathbf{t}(\eta, \eta, \eta)K_3 g^\circ + \mathbf{t}(\eta, \eta, \eta)K_3 \gamma.$$

Their norms are at most $L_2 \frac{r}{\sigma} r$, $3L_3 C_0 \frac{r^2}{\sigma^2} + L_3 \frac{r^3}{\sigma^3}$, and $3L_4 C_0^2 \frac{r}{\sigma} r + 3L_4 C_0 G_0 \frac{r^2}{\sigma^2} + 3L_4 C_0 \frac{r^2}{\sigma^2} r + L_4 G_0 \frac{r^3}{\sigma^3} + L_4 \frac{r^3}{\sigma^3} r$, and collecting powers of r gives the first bound. Each of these terms is the restriction to the diagonal of a multilinear map bounded by a product in which every factor η contributes r/σ and every factor γ contributes r . Differentiating in the direction k with $\|k\|_{\mathcal{X}} \leq 1$ replaces one factor by a component of k , of norm at most $1/\sigma$, respectively 1; hence the derivative of a term of degree m is bounded by m/r times the bound of the term, which gives the second estimate. $\square$

Proposition 21.14. Let $A: \mathcal{Y} \rightarrow \mathcal{X}$ be bounded, linear and injective, and suppose that $\|A\mathcal{F}(x^\circ)\|_{\mathcal{X}} \leq Y$, $\|I - AD\mathcal{F}(x^\circ)\|_{\mathcal{X} \rightarrow \mathcal{X}} \leq Z$, $\|A\| \leq \mathbf{a}$, and that $r > 0$ satisfies

$$(21.13) \quad Y + (Z - 1)r + \mathbf{a}(c_2 r^2 + c_3 r^3 + c_4 r^4) < 0, \quad Z + \mathbf{a}(2c_2 r + 3c_3 r^2 + 4c_4 r^3) < 1.$$

Then $\mathcal{F}$ has exactly one zero in the closed ball $\bar{B}_{\mathcal{X}}(x^\circ, r)$.

Proof. Let $\mathcal{N}(x) = x - A\mathcal{F}(x)$. For $\|h\|_{\mathcal{X}} \leq r$, $\mathcal{N}(x^\circ + h) - x^\circ = -A\mathcal{F}(x^\circ) + (I - AD\mathcal{F}(x^\circ))h - A(\mathcal{F}(x^\circ + h) - \mathcal{F}(x^\circ) - D\mathcal{F}(x^\circ)h)$, whose norm is less than r by Lemma 21.13 and the first inequality. On the convex ball, $D\mathcal{N}(x^\circ + h) = (I - AD\mathcal{F}(x^\circ)) - A(D\mathcal{F}(x^\circ + h) - D\mathcal{F}(x^\circ))$ has norm less than one by the second inequality, so $\mathcal{N}$ is a contraction of the ball into itself. Its unique fixed point is the unique zero of $\mathcal{F}$ in the ball, since A is injective. $\square$

21.6. The approximate inverse. From now on $M = 180$ and $S = 80$. The exact dyadic centre $x^\circ = (g^\circ, c^\circ)$ consists of 7280 field coefficients with even $\ell \leq 180$ and $1 \leq s \leq 80$, and 91 odd shape coefficients c_j° with $j \leq 181$. These 7371 coefficients are stored in `anc/r3_convex/candidate.json`; their SHA-256 is recorded in the ancillary README. The centre is not assumed to be an exact zero. Write $q^\circ = q_{c^\circ}$, $p^\circ = \sum_a p_a w^a$ with $p_a = (a+1)c_{a+1}^\circ$, and $W^\circ = K_3 g^\circ$.

Finite and tail coordinates. The *finite inputs* are the 7280 field coordinates (ℓ, s) with $\ell \leq M$ even and $1 \leq s \leq S$, and the 91 shape coordinates c_j with $j \leq M+1$. The *finite rows* are the same 7280 field rows (ℓ, s) and the 91 harmonic rows $\mathcal{T}_n$, $n \leq M$ even. All other inputs and rows are *tail* inputs and rows; tail rows are *harmonic* ($\mathcal{T}_n$, $n \geq 182$) or *nonharmonic* ((ℓ, s) with $s \geq 1$ and $\ell \geq 182$ or $s \geq 81$). Accordingly $\mathcal{X} = \mathcal{X}_f \oplus \mathcal{X}_t$ and $\mathcal{Y} = \mathcal{Y}_f \oplus \mathcal{Y}_t^H \oplus \mathcal{Y}_t^N$. On the shape tail and on the harmonic rows we use normalised coordinates

$$\hat{a}_n = b(n+1)R^{n+1}c_{n+1}, \quad \hat{y}_n = R^n y_n \quad (n \text{ even}),$$

so that the shape part of $\|\cdot\|_{\mathcal{X}}$ is $8 \sum |\hat{a}_n|$ and the harmonic part of $\|\cdot\|_{\mathcal{Y}}$ is $\sum |\hat{y}_n|$.

The principal part on high shape modes. For odd j , the derivative of $b^{-1}q_c|p_c|^2$ at c° in the direction e_j contains the *principal polynomial* $\Pi_j = \frac{2}{b} q^\circ \operatorname{Re}(\overline{p^\circ} j w^{j-1})$, in which q is frozen. On ∂B we have $w = e^{i\theta}$, $\xi = \cos \theta$, $q^\circ = \sum_k q_k e^{ik\theta}$ with $q_k = \sum_{i>|k|} c_i^\circ$ (k even, i odd), and therefore

$$\Pi_j|_{\partial B} = \frac{2j}{b} \sum_{a,k} p_a q_k T_{|j-1-a+k|}(\xi).$$

Let $\operatorname{Me}_j = \frac{2j}{b} \sum_{a,k} p_a q_k \mathcal{T}_{|j-1-a+k|} \in \mathcal{Y}$ be the harmonic element with these boundary values, and define the symbol and its defect

$$s_\delta = \frac{1}{b^2} \sum_{k-a=2\delta} p_a q_k R^{2\delta} \quad (\delta \in \mathbb{Z}), \quad \varepsilon = \sum_{\delta} |s_\delta - \delta_{\delta 0}|.$$

Lemma 21.15. *Let $n \geq 182$ be even. In normalised coordinates, $\operatorname{Me}_{n+1}/(b(n+1)R^{n+1})$ has the entry $\frac{2}{R} s_\delta R^{|n+2\delta|-(n+2\delta)}$ in the harmonic row $|n+2\delta|$, summed over δ . Only $-180 \leq \delta \leq 90$ occur, so the rows $|n+2\delta| \leq M$ occur only for $n \leq 540$, and $n+2\delta \geq -178$. Let T be the part of $\mathbf{M}$ from the shape tail to the harmonic tail rows and $\mathcal{B}$ the part from the shape tail to the finite harmonic rows. Then, on ℓ^1 of the normalised coordinates,*

$$T = \frac{2}{R}(I + E), \quad (E\hat{a})_m = \sum_{n \geq 182} (s_{(m-n)/2} - \delta_{mn}) \hat{a}_n \quad (m \geq 182), \quad \|E\| \leq \varepsilon.$$

If $\varepsilon < 1$, then T is invertible, $T^{-1} = \frac{R}{2} \sum_{k \geq 0} (-E)^k$, $\|T^{-1}\| \leq R/(2(1-\varepsilon))$, and $\|T^{-1}\hat{y} - \frac{R}{2} \sum_{k=0}^L (-E)^k \hat{y}\|_1 \leq \frac{R\varepsilon^{L+1}}{2(1-\varepsilon)} \|\hat{y}\|_1$ for $L \geq 0$.

Proof. The entry of Me_{n+1} in the row $\mathcal{T}_{|n+2\delta|}$ is $\frac{2(n+1)}{b} \sum_{k-a=2\delta} p_a q_k$; multiplying by $R^{|n+2\delta|}$ and dividing by $b(n+1)R^{n+1}$ gives the stated value, since $R^{2\delta} = R^{(n+2\delta)-n}$. As $0 \leq a \leq 180$ and $|k| \leq 180$, $-180 \leq \delta \leq 90$. Hence rows $m \geq 182$ are reached only with $m = n+2\delta$, and T is $\frac{2}{R}$ times the compression to indices ≥ 182 of the Laurent operator with symbol (s_δ) ; the compression of a convolution has ℓ^1 norm at most the ℓ^1 norm of its kernel. The remaining statements are the Neumann series. $\square$

The operator A . Let J^{fl} be the frozen 7371×7371 binary64 Galerkin matrix and let $\hat{A}$ be its frozen binary64 approximate inverse, regarded as an exact linear map $\mathcal{Y}_f \rightarrow \mathcal{X}_f$. They are reconstructed from a nearby numerical seed; the discrepancy from the true finite Jacobian at the full exact centre is enclosed separately. The boundary symbol of the tail inverse uses the full exact centre. Suppose $\varepsilon < 1$ and

$$(21.14) \quad \eta := \|I - \hat{A}J^{\text{fl}}\|_b < 1,$$

where $\|\cdot\|_b$ is the operator norm for the weights of $\mathcal{X}$ and $\mathcal{Y}$ with σ replaced by b . For $y = y_f + y_t^H + y_t^N \in \mathcal{Y}$ define

$$(21.15) \quad Ay = \hat{A}(y_f - \mathcal{B}T^{-1}y_t^H) + T^{-1}y_t^H + y_t^N,$$

where $T^{-1}y_t^H$ is a shape tail vector and y_t^N is read as a field tail vector with the same coordinates. For a finite row ι with weight w_ι (h_ℓ or R^n) let $\alpha_\iota = \|\hat{A}e_\iota\|_{\mathcal{X}}/w_\iota$, let $\alpha_n^H = \alpha_{\tau_n}$ for $n \leq M$, $\alpha_{\max} = \max_\iota \alpha_\iota$, and

$$\rho_{\mathcal{B}} = \max_{n \geq 182} \sum_{m \leq M} |\hat{\mathcal{B}}_{mn}| \alpha_m^H, \quad A_t = \max \left\{ 1, \frac{R(8 + \rho_{\mathcal{B}})}{2(1 - \varepsilon)} \right\},$$

where $\hat{\mathcal{B}}$ is the matrix of $\mathcal{B}$ in normalised coordinates.

Lemma 21.16. *$A: \mathcal{Y} \rightarrow \mathcal{X}$ is a bounded bijection with $\|A\| \leq \max\{\alpha_{\max}, A_t\}$. It maps every harmonic tail row, normalised, to a vector of norm at most A_t , and every nonharmonic tail row to the identical field tail input. Moreover $A(\text{Me}_j) = e_j$ for every odd $j \geq 183$, and $A\Psi_{\ell,s} = e_{(\ell,s)}$ for every field tail input (ℓ, s) .*

Proof. By (21.14), $\hat{A}J^{\text{fl}}$ and hence $\hat{A}$ are invertible, and T is invertible by Lemma 21.15. The map $(x_f, \hat{a}, x^N) \mapsto (\hat{A}^{-1}x_f + \mathcal{B}\hat{a}) + T\hat{a} + x^N$ is a two-sided inverse of A . We use Lemma 15.2: a finite row e_ι is mapped to $\hat{A}e_\iota$; a nonharmonic tail row to itself; the normalised harmonic tail row $\hat{y} = e_m$ to $T^{-1}e_m$, of shape norm $8\|T^{-1}e_m\|_1 \leq 8R/(2(1 - \varepsilon))$, plus $-\hat{A}\mathcal{B}T^{-1}e_m$, of norm at most $\rho_{\mathcal{B}}\|T^{-1}e_m\|_1$. Finally, $\text{Me}_j = \mathcal{B}e_j + Te_j$ in normalised coordinates, so $A\text{Me}_j = \hat{A}(\mathcal{B}e_j - \mathcal{B}T^{-1}Te_j) + T^{-1}Te_j = e_j$. $\square$

21.7. The defect bounds. We bound $Y = \|A\mathcal{F}(x^\circ)\|_{\mathcal{X}}$ and $Z = \|I - AD\mathcal{F}(x^\circ)\|$ by normalised column norms. The disjoint field partition is: 7280 finite inputs; 443 exceptional inputs $S < s < S_\ell$ for even $\ell \leq 180$; the 91 radial classes $s \geq S_\ell$; the 14 explicit inputs $(\ell, 1)$ and the 14 radial classes $s \geq 2$ for even $182 \leq \ell \leq 208$; and the uniform angular tail $\ell \geq 210$, $s \geq 1$. Thus there are $443 + 14 = 457$ explicitly checked exceptional field inputs in total. Here S_ℓ equals 93 for $\ell = 0, 2, \dots, 12$; 92 for $14, \dots, 30$; 91 for $32, \dots, 50$; 90 for $52, \dots, 62$; 89 for $64, \dots, 72$; 88 for $74, 76, 78$; 87 for $80, \dots, 90$; 86 for $\ell = 92$; 85 for $\ell = 94$; and 81 for $96, \dots, 180$. The shape inputs are the 310 explicit odd indices $j \leq 619$ and all odd indices $j \geq 621$. These sets exhaust all input columns without overlap.

The residual and the explicit columns. The residual $\mathcal{F}(x^\circ)$ is a polynomial. It is expanded in the basis $\Psi_{\ell,s}$ without truncation, in ball arithmetic, using (21.5) and Lemma 21.3(b) for the multiplications by x_1 and t (which give $x_1\Psi_{\ell,s}$ and $t\Psi_{\ell,s}$ as combinations of at most four, respectively three, Ψ 's), the recurrences $r^{n+1}Q_{n+1} = 2x_1r^nQ_n - tr^{n-1}Q_{n-1}$ for $Q = T, U$ and the three-term recurrence of $C_n^{(2)}$ for the series (21.1), (21.2), and (21.9) and Lemma 21.10(d). Its harmonic part is converted to the coordinates of $\mathcal{Y}$ by the matrix D . Since A has norm at most $\alpha_{\max}$ on $\mathcal{Y}_f$, one on $\mathcal{Y}_t^N$ and A_t on $\mathcal{Y}_t^H$ (Lemma 21.16),

$$Y \leq \alpha_{\max}\|\mathcal{F}(x^\circ)_f\|_{\mathcal{Y}} + \|\mathcal{F}(x^\circ)_t^N\|_{\mathcal{Y}} + A_t\|\mathcal{F}(x^\circ)_t^H\|_{\mathcal{Y}}.$$

For the 457 exceptional field inputs and 310 explicit shape inputs, the column $D\mathcal{F}(x^\circ)e$ is polynomial. The shape columns are expanded completely. For exceptional field columns alone, evaluation retains the shape coefficients only through $j = 121$ and adds the rigorous omitted-tail bound below. In both cases A uses the full centre and the full boundary symbol. It is applied in the form (21.15), with T^{-1} evaluated by 24 terms of the Neumann series of Lemma 21.15 plus the stated remainder bound, and e is subtracted. Let Z_{ex} and Z_{sh} be the resulting maxima.

Put $c^t = (c_j^\circ \mathbf{1}_{j \leq 121})_j$, $C = \|c^\circ\|_{\mathcal{E}}$, $\delta = \|c^\circ - c^t\|_{\mathcal{E}}$, $\kappa_R = 4(1 - R^{-2})^{-1/2} - 1$, and $z = \ell + 2s + \frac{1}{2}$. For a normalised field basis vector the exact three-term inverse and derivative formulas give

$$K_z = \frac{1}{z(z+2)}, \quad D_z = \frac{2R}{z+1}, \quad E_z = \frac{1}{z} + RD_z$$

as upper bounds for the norms of K_3e , $\partial_{x_1}K_3e$ and $y\partial_yK_3e$, respectively. The linear multiplier inequalities (21.12) and the telescoping expansion of the three factors of $q|p|^2$ then give

$$\|(D_g\mathcal{F}(c^\circ) - D_g\mathcal{F}(c^t))e\|_{\mathcal{Y}} \leq \frac{\kappa_R\delta}{b} \left[\frac{1}{R} + \frac{2D_z}{R^2 - 1} + \frac{4RE_z}{(R^2 - 1)^2} + \frac{3C^2K_z}{R^3} \right].$$

Multiplication by the certified full $\|A\|$ bounds the omitted contribution to the X -defect. For these inputs $z \geq 325/2$, giving a common error strictly below 0.007935; the verifier uses the individual z of each column. No truncation enters A or the residual.

Finite field columns. Let $e = e_{(\ell,s)}$ be a finite field input, and let Je and $(D\mathcal{F}(x^\circ)e)_t$ be the components of $D\mathcal{F}(x^\circ)e$ in the finite rows and in the tail rows. Then

$$(I - AD\mathcal{F}(x^\circ))e = (I - \hat{A}J^{\text{fl}})e + \hat{A}(J^{\text{fl}} - J)e - A(D\mathcal{F}(x^\circ)e)_t.$$

The entries of Je are integrals over B of polynomials, namely $\|\Psi_{\ell',s'}\|_{L^2}^{-2} \int_B (D\mathcal{F}(x^\circ)e) \Psi_{\ell',s'}$ for the field rows and, for the harmonic rows, $\sum_{\ell' \leq 540} D_{n\ell'} \|H_{\ell'}\|_{L^2}^{-2} \int_B (D\mathcal{F}(x^\circ)e) H_{\ell'}$; all angular degrees of the output (at most $3M = 540$, because $q^\circ|p^\circ|^2$ has degree at most $2M$ in ξ at fixed r) are integrated before the conversion to Chebyshev coordinates. In spherical coordinates the integrands are even polynomials in ξ of degree at most 1080 and in r of degree at most $(4M + 2S + 2) + 3M + 2 = 1424$. They are integrated exactly by Gauss–Legendre rules with 542 nodes in ξ (exact to degree 1083) and 713 nodes in r (applied to the even extension to $[-1, 1]$, exact to degree 1425), whose nodes and weights are enclosed in ball arithmetic: each nonnegative root is bracketed by a sign change of the Legendre polynomial, the brackets are disjoint, together with their reflections their number equals the degree, and the weights are computed from the enclosures. Let ξ_{fin} bound $\max_e \sum_{\iota} |J_{\iota e}^{\text{fl}} - J_{\iota e}| \alpha_{\iota}^b w_{\iota} / w_e$, where α_{ι}^b is the normalised column norm of $\hat{A}$ in the norm $\|\cdot\|_b$. Converting from $\|\cdot\|_b$ to $\|\cdot\|_{\mathcal{X}}$ costs at most a factor eight, so the first two terms contribute at most $Z_{\text{head}} = 8(\eta' + \xi_{\text{fin}})$, where η' is η increased by explicit allowances for the representation of the weights h_{ℓ} , $b_j R^j$, R^n by binary64 numbers (relative error below 10^{-12} , checked) and for the rounding in their application. The last term is bounded with the output-tail envelope below, whose maximum over the finite field inputs we call $Z_{\text{fin},t}$.

Envelopes for field columns. We define step functions of the angular index L and the radial index ς . For even $\ell \leq M$ and $1 \leq \varsigma \leq S$ let

$$A(\ell, \varsigma) = \max\{1, \alpha_{(\ell', \varsigma')} : \ell' \geq \ell, \varsigma' \geq \varsigma, (\ell', \varsigma') \text{ a finite field row}\},$$

let $A(\ell, \varsigma) = 1$ if $\ell > M$ or $\varsigma > S$, and $A(\ell, \varsigma) = A(\ell + 1, \varsigma)$ for odd ℓ . Let $\alpha_n^H = A_t$ for $n \geq 182$,

$$\beta_\ell = \frac{1}{h_\ell} \sum_n D_{n\ell} R^n \alpha_n^H, \quad \beta^\infty = A_t + \sqrt{2} \sum_{n \leq M} (\alpha_n^H - A_t)_+ R^{n-1200},$$

$B(L) = \max(\{\beta_\ell : \ell \text{ even}, L \leq \ell \leq 1200\} \cup \{\beta^\infty\})$, and $\Gamma(L) = \max\{1, A(L, 1), B(L)\}$. These functions are nonincreasing in L and ς .

Lemma 21.17. *For $\ell \geq L$ (with ℓ even) and $\varsigma \geq 1$, $\|AH_\ell\|_{\mathcal{X}} \leq B(L)h_\ell$ and $\|A\Psi_{\ell, \varsigma}\|_{\mathcal{X}} \leq A(L, \varsigma)h_\ell$.*

Proof. $H_\ell = \sum_n D_{n\ell} \mathcal{T}_n$, and $\|A\mathcal{T}_n\| \leq \alpha_n^H R^n$ by Lemma 21.16; hence $\|AH_\ell\| \leq \beta_\ell h_\ell$. For $\ell > 1200$, since $\sum_n D_{n\ell} R^n = h_\ell$ and by Lemma 21.5(e), $\beta_\ell \leq A_t + \sum_{n \leq M} \sqrt{2} R^{n-\ell} (\alpha_n^H - A_t)_+ \leq \beta^\infty$. The second bound holds by definition for finite rows and because A is the identity on nonharmonic tail rows. $\square$

For $d \geq 0$, $\ell, s \geq 0$ put $\ell_d = \max(0, \ell - d)$, $s_d = \max(1, s - d)$, let X_d be a $\text{Bin}(d, \frac{1}{2})$ random variable, and let $\phi_L(\varsigma) = A(L, \varsigma)$ for $\varsigma \geq 1$ and $\phi_L(\varsigma) = \max\{A(L, 1), B(L)\}$ for $\varsigma \leq 0$. Define

$$\begin{aligned} \mathfrak{B}_d(\ell, s) &= \min \left\{ A(\ell_d, s_d) + (B(\ell_d) - A(\ell_d, s_d))_+ \vartheta_d(\ell, s), \quad \mathbb{E} \phi_{\ell_d}(s - X_d) \right\}, \\ \mathfrak{T}_d(\ell, s) &= \begin{cases} 0, & \ell + d \leq M \text{ and } s + d \leq S, \\ 1 + (A_t - 1) \vartheta_d(\ell, s) \mathbf{1}_{\ell+d > M}, & \text{otherwise,} \end{cases} \end{aligned}$$

with ϑ_d as in Lemma 21.9(b). Both bounds in $\mathfrak{B}_d$ are nonincreasing in ℓ and in s : the first because A, B, ϑ_d are and the expression is nondecreasing in A and in ϑ_d ; the second because ϕ_L is nonincreasing in ς and in L .

Lemma 21.18. *Let $H_a t^k$ be a solid monomial of degree d and $\ell, s \geq 0$ with $a + \ell$ even. Then $\|A(H_a t^k \Psi_{\ell, s})\|_{\mathcal{X}} \leq h_a h_\ell \mathfrak{B}_d(\ell, s)$, and the component $(H_a t^k \Psi_{\ell, s})_t$ in the tail rows satisfies $\|A(H_a t^k \Psi_{\ell, s})_t\|_{\mathcal{X}} \leq h_a h_\ell \mathfrak{T}_d(\ell, s)$.*

Proof. By (21.8) and Lemma 21.7 it suffices to show, for every v with $G_{av} \neq 0$, that

$$\Sigma_v := \sum_{\varsigma} \pi_{\varsigma}^{(v)} \frac{\|A\Psi_{v, \varsigma}\|}{h_v} \leq \mathfrak{B}_d(\ell, s).$$

By Lemma 21.9(a), $v \geq \ell_d$ and $\varsigma \geq s - d$. By Lemma 21.17 and monotonicity, the quotient $\|A\Psi_{v, \varsigma}\|/h_v$ is at most $A(\ell_d, \varsigma) \leq A(\ell_d, s_d)$ for $\varsigma \geq 1$, and at most $B(\ell_d)$ for $\varsigma = 0$. With $\pi_0^{(v)} \leq \vartheta_d(\ell, s)$ this gives the first bound in $\mathfrak{B}_d$. For the second, the quotient is at most $\phi_{\ell_d}(\varsigma)$, the function ϕ_{ℓ_d} is nonincreasing, the index is at least $s - N$ with N as in the proof of Lemma 21.9, and N is stochastically dominated by X_d (Lemma 21.9(c)); hence $\Sigma_v \leq \mathbb{E} \phi_{\ell_d}(s - X_d)$.

For the tail component: by Lemma 21.9(a) the outputs have $v \leq \ell + d$ and $\varsigma \leq s + d$. If $\ell + d \leq M$ and $s + d \leq S$, all of them are finite rows, including the harmonic outputs $H_v = \sum_{n \leq v} D_{nv} \mathcal{T}_n$. Otherwise the tail component consists of nonharmonic tail rows, on which A is the identity, and, only if $\ell + d > M$, of the parts $\sum_{n \geq 182} D_{nv} \mathcal{T}_n$ of harmonic outputs, of norm at most h_v , on which A has norm at most A_t . Weighting with $\pi_0^{(v)} \leq \vartheta_d$ gives the bound. $\square$

At the centre, let $\chi_1 = q^\circ/b - 1$, $\chi_2 = \partial_{x_1} q^\circ/b$, $\chi_3 = y^{-1} \partial_y q^\circ/b$, $\chi_4 = x_1 \chi_3$ and $\chi_5 = q^\circ |p^\circ|^2/b$. For $1 \leq i \leq 5$ let $\tau_i(d)$ be the τ -norm of the homogeneous part of degree d of χ_i . By Lemma 21.8(b), τ_1, τ_2, τ_3 are the Chebyshev norms of the terms of (21.1), with $\tau_1(0) = |c_1^\circ/b - 1|$; $\tau_4(d+1) \leq R \tau_3(d)$; and τ_5 is computed from the expansion of χ_5 in solid monomials. For

$\mathfrak{X} \in \{\mathfrak{B}, \mathfrak{T}\}$ put $\mathfrak{X}^{(i)}(\ell, s) = \sum_d \tau_i(d) \mathfrak{X}_d(\ell, s)$ and, with $\bar{d} = \ell + 2s + \frac{1}{2}$ (the parameter d of Lemma 21.10) and ratios $\rho_{\pm}$,

$$\begin{aligned} \mathfrak{z}\mathfrak{X}(\ell, s; \rho_+, \rho_-, \lambda_+, \lambda_-) &= \mathfrak{X}^{(1)}(\ell, s) + \sum_{i=2,4} \left[\frac{\lambda_+ \rho_+}{2(\bar{d}+1)} (\mathfrak{X}^{(i)}(\ell+1, s) + \mathfrak{X}^{(i)}(\ell+1, s-1)) \right. \\ &\quad \left. + \frac{\lambda_- \rho_-}{2(\bar{d}+1)} (\mathfrak{X}^{(i)}(\ell-1, s+1) + \mathfrak{X}^{(i)}(\ell-1, s)) \right] \\ &\quad + \frac{(\bar{d} + \frac{1}{2}) \mathfrak{X}^{(3)}(\ell, s-1)}{4\bar{d}(\bar{d}+1)} + \frac{\mathfrak{X}^{(3)}(\ell, s)}{4\bar{d}(\bar{d}+2)} + \frac{(\bar{d} + \frac{3}{2}) \mathfrak{X}^{(3)}(\ell, s+1)}{4(\bar{d}+1)(\bar{d}+2)} \\ &\quad + \frac{\mathfrak{X}^{(5)}(\ell, s-1)}{4\bar{d}(\bar{d}+1)} + \frac{\mathfrak{X}^{(5)}(\ell, s)}{2\bar{d}(\bar{d}+2)} + \frac{\mathfrak{X}^{(5)}(\ell, s+1)}{4(\bar{d}+1)(\bar{d}+2)}, \end{aligned}$$

where terms with $\ell-1 < 0$ are omitted. The exact values are $\rho_{\pm} = h_{\ell \pm 1}/h_{\ell}$, $\lambda_+ = (\ell+1)/(2\ell+1)$, $\lambda_- = \ell/(2\ell+1)$.

- Proposition 21.19.** (a) *For a field tail input (ℓ, s) , $\|(I - AD\mathcal{F}(x^{\circ}))e_{(\ell,s)}\|_{\mathcal{X}}/h_{\ell} \leq \mathfrak{z}\mathfrak{B}(\ell, s)$ (exact parameters), and for fixed ℓ this bound is nonincreasing in s . Hence the radial class of ℓ is bounded by $Z_{\text{rad}}(\ell) = \mathfrak{z}\mathfrak{B}(\ell, S_{\ell})$.*
(b) *The uniform angular class is bounded by $Z_{\text{ang}} = \mathfrak{z}\mathfrak{B}(210, 1; R, \frac{210}{209R}, \frac{211}{421}, \frac{1}{2})$.*
(c) *For a finite field input (ℓ, s) , $\|A(D\mathcal{F}(x^{\circ})e_{(\ell,s)})_t\|_{\mathcal{X}}/h_{\ell} \leq \mathfrak{z}\mathfrak{T}(\ell, s)$ (exact parameters).*

Proof. For a field input, by (21.11) and Lemma 21.10(d),

$$D\mathcal{F}(x^{\circ})e_{(\ell,s)} = \Psi_{\ell,s} + \chi_1 \Psi_{\ell,s} + \chi_2 \partial_{x_1} K_3 \Psi_{\ell,s} + \chi_3 EK_3 \Psi_{\ell,s} - \chi_4 \partial_{x_1} K_3 \Psi_{\ell,s} + \chi_5 K_3 \Psi_{\ell,s}.$$

For a tail input, $A\Psi_{\ell,s} = e_{(\ell,s)}$ (Lemma 21.16), so $(I - AD\mathcal{F}(x^{\circ}))e_{(\ell,s)}$ is $-A$ applied to the sum of the five remaining terms. Expanding them with Lemma 21.10(d) and (21.9), and bounding each product of a homogeneous part of χ_i with a Ψ by Lemma 21.18, gives $\mathfrak{z}\mathfrak{B}(\ell, s)h_{\ell}$; the factors $\rho_{\pm}$ convert $h_{\ell \pm 1}$ to h_{ℓ} . Every quantity in $\mathfrak{z}\mathfrak{B}$ is nonincreasing in s (the coefficients decrease in $\bar{d}$, and each $\mathfrak{B}_d$ is nonincreasing), which proves (a). For (b), let $\ell \geq 210$ and $s \geq 1$. By Lemma 21.5(b), $\rho_+ \leq R$ and $\rho_- \leq \frac{210}{209R}$; moreover $\lambda_+ \leq \frac{211}{421}$, $\lambda_- \leq \frac{1}{2}$, $\bar{d} \geq 212.5$, and $\mathfrak{B}_d$ is nonincreasing in both arguments, so every term is bounded by the corresponding term at $\ell = 210$, $s = 1$. For (c), $\Psi_{\ell,s}$ is a finite row, so the tail component of $D\mathcal{F}(x^{\circ})e_{(\ell,s)}$ is that of the five remaining terms, and Lemma 21.18 applies with $\mathfrak{T}$. $\square$

For the fourteen near angular degrees $182 \leq \ell \leq 208$, the same bound with the exact ℓ and $s = 2$, monotone in s , covers each infinite radial class; the input $(\ell, 1)$ is included among the 457 explicit exceptional columns. We denote the maximum of these fourteen infinite-class bounds by $Z_{\text{ang, near}}$. Thus the finite field columns are bounded by $Z_{\text{fin}} = Z_{\text{head}} + Z_{\text{fin}, t}$, where $Z_{\text{fin}, t} = \max \mathfrak{z}\mathfrak{T}(\ell, s)$ over the 7280 finite field inputs.

Far shape columns. For a polynomial $f = \sum_a H_a f_a(t) \in \mathcal{L}$ write $f_a = \sum_s f_{a,s} J_s^{a+1/2} = \sum_m f_{a,m}^{\sharp} (1-t)^m$, and let

$$V_f(L) = \sum_a h_a \Gamma((L-a)_+) \min \left\{ \sum_s |f_{a,s}|, \sum_m |f_{a,m}^{\sharp}| \frac{m!}{\left(\frac{(L-a)_+ + 1/2}{2} + 1\right)_m} \right\},$$

which is nonincreasing in L . With $\zeta = R^{-2}$, $N \geq 0$ and $J = \lfloor N/2 \rfloor$ put

$$\begin{aligned} \mathsf{T}_f(N) &= \frac{B_0}{2} V_f(N) + \sum_{j=1}^J \frac{B_0 a_j \zeta^j}{2j-1} V_f(N-2j) + \frac{B_0 a_{J+1} \zeta^{J+1}}{(2J+1)(1-\zeta)} V_f(0), \\ \mathsf{U}_f(N) &= \sum_{j=0}^J B_0 a_j \zeta^j V_f(N-2j) + \frac{B_0 a_{J+1} \zeta^{J+1}}{1-\zeta} V_f(0). \end{aligned}$$

Lemma 21.20. *Let $f \in \mathcal{L}$ be a polynomial, $e \geq 0$, and let L, n have the parity for which the products below are even in x_1 . Then $\|A(H_L t^e f)\|_{\mathcal{X}} \leq h_L V_f(L)$. For $n \geq N$, $\|A(r^n T_n(\xi) t^e f)\|_{\mathcal{X}} \leq R^n \mathsf{T}_f(N)$ and $\|A(r^n U_n(\xi) t^e f)\|_{\mathcal{X}} \leq R^n \mathsf{U}_f(N)$. More generally, for every m with $\tau(m) < \infty$ whose homogeneous parts have degree at most d_m , $\|A(m r^n T_n(\xi) t^e f)\| \leq \tau(m) R^n \mathsf{T}_f(N - d_m)$ for $n \geq N \geq d_m$.*

Proof. By Lemma 21.7, $H_L t^e H_a f_a = \sum_k G_{Lak} H_k t^{e+(L+a-k)/2} f_a$ with $k \geq |L-a|$. The $\mathcal{L}$ -coefficients of $H_k t^e f_a$ lie in angular index k , and their ℓ^1 norm in the basis $J^{k+1/2}$ is at most $\sum_s |f_{a,s}|$, by the stochastic steps in the proof of Lemma 21.8(a), and at most $\sum_m |f_{a,m}^\natural| \|(1-t)^m\|_{J^{k+1/2},1}$, by Lemma 21.3(c),(d) (multiplication by t does not increase $\|\cdot\|_{J^{\beta},1}$; the latter norm is at most $m!/((L-a)_+ + \frac{1}{2})/2 + 1)_m$ since $k \geq (L-a)_+$. On outputs of angular index at least $(L-a)_+$, A has normalised norm at most $\Gamma((L-a)_+)$ by Lemma 21.17. Summing with $\sum_k G_{Lak} h_k \leq h_L h_a$ gives the first bound. Next, $r^n T_n(\xi) = \sum_j C_{n-2j,n} H_{n-2j} t^j$, so by the first bound, Lemma 21.5(d) and the monotonicity of V_f ,

$$\begin{aligned} \|A(r^n T_n t^e f)\| &\leq \sum_{j \leq n/2} |C_{n-2j,n}| h_{n-2j} V_f(n-2j) \\ &\leq R^n \left(\frac{B_0}{2} V_f(N) + \sum_{j \geq 1} \frac{B_0 a_j \zeta^j}{2j-1} V_f((N-2j)_+) \right), \end{aligned}$$

and the terms with $j > J$ are bounded using $\sum_{j > J} a_j \zeta^j / (2j-1) \leq a_{J+1} \zeta^{J+1} / ((2J+1)(1-\zeta))$. The bound for U_n is the same with v . For a multiplier m we write $m = \sum m_{i,k'} H_i t^{k'}$ with $i \leq d_m$ and use $H_i H_{n-2j} = \sum_k G_{i,n-2j,k} H_k t^{(\cdots)}$ with $k \geq n-2j-d_m$, together with $\sum_k G_{i,n-2j,k} h_k \leq h_i h_{n-2j}$. $\square$

Lemma 21.21. (a) $y^{-1} \partial_y H_\ell = -\sum_{k=\ell-2, \ell-4, \dots \geq 0} (2k+1) H_k t^{(\ell-2-k)/2}$. Hence, for a polynomial $W = \sum_\ell H_\ell w_\ell(t)$, $Z_W = y^{-1} \partial_y W = \sum_\ell ((y^{-1} \partial_y H_\ell) w_\ell + 2H_\ell w'_\ell)$ is a polynomial, and $y \partial_y W = t(1-\xi^2) Z_W$.

(b) $2(1-\xi^2) C_{n-2}^{(2)}(\xi) = \frac{1}{2} (U_n(\xi) + U_{n-2}(\xi)) - n T_n(\xi)$ for $n \geq 2$.

Proof. (a) We have

$$\partial_y (r^\ell P_\ell(x_1/r)) = y r^{\ell-2} (\ell P_\ell(\xi) - \xi P'_\ell(\xi)) = -y r^{\ell-2} P'_{\ell-1}(\xi),$$

and $P'_{\ell-1} = \sum_{k \equiv \ell-2} (2k+1) P_k$ by the classical identities $\xi P'_\ell - P'_{\ell-1} = \ell P_\ell$ and $P'_{k+1} - P'_{k-1} = (2k+1) P_k$. The rest follows from $\partial_y w(t) = 2y w'(t)$ and $y^2 = t(1-\xi^2)$. (b) We have $2C_{n-2}^{(2)} = U'_{n-1}$ and $2\xi U_{n-1} = U_n + U_{n-2}$; differentiating $U_{n-1}(\cos \theta) = \sin n\theta / \sin \theta$ gives $(1-\xi^2) U'_{n-1}(\xi) = \xi U_{n-1}(\xi) - n T_n(\xi)$. $\square$

Let $H^\circ = g^\circ + |p^\circ|^2(1+W^\circ)$, $W_1^\circ = \partial_{x_1} W^\circ$ and $Z^\circ = Z_{W^\circ}$. These are polynomials; their Jacobi coefficients and boundary Taylor coefficients enter V_f . By Lemma 21.10(c), every angular component f_a of W° has a double zero at $t=1$, and every angular component of W_1°

and of Z° vanishes at $t = 1$, so the corresponding Taylor coefficients are exactly zero. Let $q_{(d)}^\circ = c_{d+1}^\circ r^d U_d(\xi)$ be the homogeneous parts of q° , with $\tau(q_{(d)}^\circ) = |c_{d+1}^\circ| \|r^d U_d\|_{\text{ch}}$. Finally, for $N \geq 360$ let

$$P(N) = \frac{2}{b^2 R} \sum_{a,i,\varsigma} |p_a c_i^\circ| R^{\varsigma-a} \left[\sum_{\nu=0}^{J_K} w_\nu^T \Gamma(K-2\nu) \frac{2(m+\nu)}{K-\nu+m+\frac{3}{2}} + \frac{2B_0 a_{J_K+1} \zeta^{J_K+1}}{(2J_K+1)(1-\zeta)} \Gamma(0) \right],$$

where $a \in \{0, 2, \dots, 180\}$, $i \in \{1, 3, \dots, 181\}$, $\varsigma \in \{-(i-1), -(i-3), \dots, i-1\}$, $K = N - a + \varsigma$, $m = a + (i-1-\varsigma)/2$, $J_K = \lfloor K/2 \rfloor$, $w_0^T = B_0/2$ and $w_\nu^T = B_0 a_\nu \zeta^\nu / (2\nu-1)$ for $\nu \geq 1$.

Proposition 21.22. *Let $N = 620$. For every odd $j \geq N + 1$,*

$$\begin{aligned} \frac{\|(I - AD\mathcal{F}(x^\circ))e_j\|_{\mathcal{X}}}{\sigma j R^j} &\leq \frac{1}{8} \left[\frac{\mathsf{U}_{H^\circ}(N)}{b^2(N+1)R} + \frac{\mathsf{U}_{W_1^\circ}(N-1)}{b^2 R^2} \right. \\ &\quad + \frac{\mathsf{T}_{Z^\circ}(N) + (\mathsf{U}_{Z^\circ}(N) + R^{-2} \mathsf{U}_{Z^\circ}(N-2))/(2N+2)}{b^2 R} \\ &\quad \left. + \frac{2}{b^2 R} \sum_a |p_a| R^{-a} \sum_d \tau(q_{(d)}^\circ) \mathsf{T}_{W^\circ}(N-a-d) + P(N) \right] =: Z_{\text{far}}. \end{aligned}$$

Proof. Let $j = n + 1$ with $n \geq N$ even. Differentiating (21.11) in the direction e_j , with $\delta q = r^n U_n$, $\delta \partial_{x_1} q = j r^{n-1} U_{n-1}$, $\delta(y^{-1} \partial_y q) = -2r^{n-2} C_{n-2}^{(2)}$ and $\delta p = j w^n$ by (21.1), gives

$$b D\mathcal{F}(x^\circ) e_j = r^n U_n H^\circ + j r^{n-1} U_{n-1} W_1^\circ - 2r^{n-2} C_{n-2}^{(2)} W_2^\circ + 2q^\circ \text{Re}(\overline{p^\circ} j w^n) (1 + W^\circ),$$

with $W_2^\circ = y \partial_y W^\circ = r^2(1 - \xi^2) Z^\circ$ by Lemma 21.21(a). Since $A M e_j = e_j$ (Lemma 21.16), $(I - AD\mathcal{F}(x^\circ))e_j = -A(D\mathcal{F}(x^\circ)e_j - M e_j)$, and we bound five parts, dividing by the input weight $\sigma j R^j = 8b(n+1)R^{n+1}$.

The first two terms are bounded by Lemma 21.20, using $n+1 \geq N+1$. For the third, Lemma 21.21(b) gives

$$-2r^{n-2} C_{n-2}^{(2)} W_2^\circ = -r^n \left(\frac{U_n + U_{n-2}}{2} - n T_n \right) Z^\circ, \quad r^n U_{n-2} = t r^{n-2} U_{n-2},$$

and Lemma 21.20 applies, since $n/(n+1) < 1$. For the fourth, we write

$$2 \text{Re}(\overline{p^\circ} w^n) = 2 \sum_a p_a t^a r^{n-a} T_{n-a}(\xi), \quad q^\circ = \sum_d q_{(d)}^\circ,$$

and apply the last statement of Lemma 21.20 to each product $q_{(d)}^\circ r^{n-a} T_{n-a} t^a W^\circ$.

The last part is $\frac{2}{b} q^\circ \text{Re}(\overline{p^\circ} j w^n) - M e_j$. Since $T_{n-a} U_{i-1} = \sum_\varsigma T_{n-a+\varsigma}$, with ς as in the definition of P and all indices positive because $n-a-(i-1) \geq N-360 > 0$,

$$\frac{2}{b} q^\circ \text{Re}(\overline{p^\circ} j w^n) = \frac{2j}{b} \sum_{a,i,\varsigma} p_a c_i^\circ r^{K+2m} T_K(\xi), \quad K = n-a+\varsigma, \quad m = a + \frac{i-1-\varsigma}{2},$$

and $M e_j$ is obtained by replacing $r^{K+2m} T_K$ by $\mathcal{T}_K$, which has the same boundary values. Now $r^{K+2m} T_K - \mathcal{T}_K = \sum_\nu C_{K-2\nu,K} H_{K-2\nu}(t^{m+\nu} - 1)$, and by Lemma 21.3(d) the function $H_\ell(t^M - 1)$ has $\mathcal{L}$ -coefficients in angular index ℓ only, with ℓ^1 norm $2M/(\ell + M + \frac{3}{2})$ in the basis $J^{\ell+1/2}$. With Lemmas 21.17 and 21.5(d),

$$\|A(r^{K+2m} T_K - \mathcal{T}_K)\| \leq R^K \sum_{\nu \leq K/2} w_\nu^T \Gamma(K-2\nu) \frac{2(m+\nu)}{K-\nu+m+\frac{3}{2}}.$$

For fixed (a, i, ς) , $R^{K-n} = R^{\varsigma-a}$ and m do not depend on n , while increasing n increases K ; each summand is nonincreasing in K , and the new summands with $\nu > J_K$ that appear for larger K are bounded by $2w_\nu^T \Gamma(0)$, whose sum over $\nu > J_K$ is at most the last term of the bracket. Hence the bracket in $P(N)$, evaluated at $K = N - a + \varsigma$, bounds the corresponding sum for every $n \geq N$. Collecting the five parts proves the claim. $\square$

The program evaluates an upper bound for $P(N)$ by grouping the sum over ν into blocks and bounding each block by its mass times the values of the factors $\Gamma(K - 2\nu)$ and $2(m + \nu)/(K - \nu + m + \frac{3}{2})$, both nondecreasing in ν , at the end of the block.

Collecting the finite, exceptional, radial, near-angular, uniform-angular, explicit-shape and far-shape classes,

$$Z = \max \{ Z_{\text{fin}}, Z_{\text{ex}}, \max_{\ell} Z_{\text{rad}}(\ell), Z_{\text{ang}}, Z_{\text{ang,near}}, Z_{\text{sh}}, Z_{\text{far}} \}$$

bounds $\|I - ADF(x^\circ)\|_{\mathcal{X} \rightarrow \mathcal{X}}$.

21.8. Certified bounds and the exact zero. The bounds below are outward roundings of the interval enclosures in `anc/r3_convex/VERIFICATION_RECEIPT.json`. The residual uses the entire polynomial support and the fixed dyadic b , not a moving value of c_1 . The certified finite inverse is nonsingular: the exact dyadic product defect, including a rigorous binary64 dot-product error, is less than 0.000173566. The boundary-symbol defect is less than 0.015579, hence both blocks of A are invertible.

Quantity	Certified bound
$\ A\ $	< 103241.73
$Y = \ A\mathcal{F}(x^\circ)\ _{\mathcal{X}}$	$< 9.358 \cdot 10^{-9}$
Z_{fin} , all 457 exceptional field columns	$< 0.283115, < 0.669070$
$Z_{\text{ang,near}}$ (fourteen radial classes)	< 0.559208
$\max_{\ell} Z_{\text{rad}}(\ell), Z_{\text{ang}}$	$< 0.776701, < 0.767365$
$Z_{\text{sh}}, Z_{\text{far}}$	$< 0.547873, < 0.501461$
$Z = \ I - ADF(x^\circ)\ _{\mathcal{X} \rightarrow \mathcal{X}}$	< 0.776701
c_2, c_3, c_4	$< 3.163938, < 1.304336 \cdot 10^{-5}, < 1.317589 \cdot 10^{-11}$
$r = 10^{-7}$: radii polynomial	$< -9.705 \cdot 10^{-9}$
$r = 10^{-7}$: contraction bound	< 0.8421

TABLE 14. The convex three-dimensional certificate for $M = 180$, $S = 80$, $R = 21/20$ and $\sigma = 8b$. The final Z is the maximum over every field and shape class. The full interval values are in the receipt.

Theorem 21.23. *There is exactly one zero $x^* = (g^*, c^*)$ of $\mathcal{F}$ in $\overline{B}_{\mathcal{X}}(x^\circ, 10^{-7})$.*

Proof. The finite dyadic product and the Toeplitz symbol bounds give an injective A by Lemmas 21.15 and 21.16. The partition above covers all derivative columns and yields $Z < 0.776701$. The verifier evaluates both inequalities (21.13) with the full Arb enclosures of $Y, Z, \|A\|, c_2, c_3, c_4$ at the exact radius 10^{-7} . Their left sides are respectively below $-9.705 \cdot 10^{-9}$ and $0.8421 < 1$, so Proposition 21.14 applies. $\square$

Remark 21.24 (Shape scale). The exact b is held fixed while the shape varies. At the same centre, the scale $\sigma = 3b$ cannot certify this fixed-point operator: the normalised defect of shape column $j = 183$ is strictly greater than 1.15486483728, as recorded in `anc/r3_convex/SCALE3_OBSTRUCTION.json`. This excludes contraction for that operator in that norm; it says nothing about other operators or centres. The scale $8b$ used above gives $Z < 0.776701$ and $c_2 < 3.163938$.

21.9. Reconstruction and convexity. Let $\psi = \psi_{c^*}$, and put $\varepsilon = \|c^* - c^\circ\|_{\mathcal{E}} \leq r/\sigma$. The existence ball projects inside the shape ball of $\mathcal{E}$ -radius one; in fact the projected radius is $68719476736/595786147695156171875 < 1$. Write

$$D_1 = \sum_{j \geq 3} j|c_j^\circ|, \quad D_2 = \sum_{j \geq 3} j(j-1)|c_j^\circ|, \quad H = \sup_{j \geq 3, j \text{ odd}} (j-1)R^{-j}.$$

The last supremum is a finite maximum through $j = 21$, since the ratio of successive odd-index terms is at most one for $j \geq 21$.

Proposition 21.25. *For every c with $\|c - c^\circ\|_{\mathcal{E}} \leq 1$, the map ψ_c is univalent on $|w| \leq 101/100$, its image $D = \psi_c(\mathbb{D})$ is strictly convex and strictly star-shaped, and $c_{19} \neq 0$. More precisely, on the closed unit disc,*

$$(21.16) \quad \operatorname{Re} \left(1 + \frac{w\psi_c''(w)}{\psi_c'(w)} \right) > 0.7878,$$

and $\operatorname{Re} \psi_c' > 106.2027$ on $|w| \leq 101/100$, $\operatorname{Re}(w\psi_c'/\psi_c) > 0.9717$ on $|w| = 1$, and $|c_{19}| > 0.000665$.

Proof. For $|w| \leq 1$ the triangle inequality gives

$$\operatorname{Re} \left(1 + \frac{w\psi_c''}{\psi_c'} \right) \geq 1 - \frac{D_2 + \varepsilon H}{c_1^\circ - D_1 - \varepsilon/R}.$$

The denominator is positive, and the interval computation gives a lower bound > 0.7878989239 at the worst case $\varepsilon = 1$. The same coefficient inequalities on the larger disc give $\operatorname{Re} \psi_c' > 106.2027$ there. Integration along line segments proves univalence; vertical segments give $q_c = \operatorname{Im} \psi_c/y > 0$. The separate boundary estimate gives the stated strict star-shapedness, and the weighted coefficient bound gives $|c_{19}| \geq |c_{19}^\circ| - \varepsilon/(19R^{19}) > 0.000665$.

For $\gamma(\theta) = \psi_c(e^{i\theta})$, its tangent angle derivative is $\operatorname{Re}(1 + w\psi_c''/\psi_c')$, which is positive by (21.16). As the simple tangent winds once, every oriented tangent line supports the interior, so D is strictly convex. It is symmetric under reflection in both axes because ψ_c is odd with real coefficients. The lift $\Theta_c(B)$ is convex exactly when D is: one direction follows by intersecting with a meridian plane; conversely, the vertical sections of D are symmetric intervals, and for X, Y in the solid, convexity of D and $|tX' + (1-t)Y'| \leq t|X'| + (1-t)|Y'|$ show that $tX + (1-t)Y$ lies in the solid for $0 \leq t \leq 1$. $\square$

Lemma 21.26. *Let $n \geq 2$ and let $\Omega \subset \mathbb{R}^n$ be a bounded domain with real-analytic boundary, and let $u \in C^1(\overline{\Omega})$ satisfy $\int_{\Omega} (\nabla u \cdot \nabla \varphi - u\varphi) = 0$ for all $\varphi \in C_c^\infty(\Omega)$ and $u = 1$ on $\partial\Omega$. Then u extends to a real-analytic function on an open neighbourhood of $\overline{\Omega}$ which satisfies $\Delta u + u = 0$ there.*

Proof. Let $v = u - 1$. For $\varepsilon > 0$ the function $\max(|v| - \varepsilon, 0) \operatorname{sign} v$ is Lipschitz with compact support in Ω and converges to v in $W^{1,2}(\Omega)$ as $\varepsilon \rightarrow 0$, so $v \in W_0^{1,2}(\Omega)$; it is a weak solution of $\Delta v + v = -1$. By global regularity for the Dirichlet problem [35, Theorem 8.13], applied for every k , $v \in W^{k,2}(\Omega)$ for all k , hence $v \in C^\infty(\overline{\Omega})$. By the theorem of Morrey and Nirenberg on analyticity up to the boundary [51] (constant coefficients, analytic right-hand side, analytic boundary and zero Dirichlet data), every point of $\partial\Omega$ has a ball around it to which $v|_{\Omega}$ extends real-analytically; the extension solves $\Delta v + v = -1$ by the identity theorem. Interior analyticity is classical (see also Lemma 18.3). The local extensions are glued to an extension on a neighbourhood of $\overline{\Omega}$ exactly as in the last step of the proof of Lemma 18.4. $\square$

Proof of Theorem 1.4. The estimates of Proposition 21.25 hold for c^* because $r/\sigma < 1$. The proofs of Lemmas 21.1 and 21.2 apply on the smaller disc $|w| < 101/100$, a neighbourhood of the closed unit disc; they give an analytic diffeomorphism Θ_{c^*} near $\bar{B}$, with Jacobian $q_{c^*}|p_{c^*}|^2 > 0$. Let $\Omega = \Theta_{c^*}(B)$, $U = 1 + \iota(K_3 g^*)$, and $u = U \circ \Theta_{c^*}^{-1}$. Since $\mathcal{F}(x^*) = 0$, Lemma 21.12 yields the pulled-back weak Helmholtz equation and zero Cauchy traces of $U - 1$; Lemma 21.2 transfers these to u . The analytic lift across the rotation axis is valid because ψ has odd real coefficients and $q = \text{Im } \psi/y$ is analytic in (x_1, y^2) . Interior analyticity also follows by harmonic lifting or elliptic regularity, and Lemma 21.26 extends u analytically across the boundary.

The positive boundary bound makes Ω strictly star-shaped. The solid-of-revolution argument in Proposition 21.25 makes it convex. The lift of the sphere is a compact real-analytic surface diffeomorphic to S^2 and has the stated $O(2) \times \mathbf{Z}_2$ invariance. A constant solution of $\Delta u + u = 0$ would be zero, contradicting $u = 1$ on the boundary. If Ω were a ball, its symmetries would put its centre at zero, and D would be a disc centred at zero. Schwarz's lemma applied to the normalised conformal map and its inverse would force ψ to be linear, contrary to $c_{19} \neq 0$. This proves the theorem.

21.10. An additional nonconvex example. The earlier three-dimensional construction is independent of the convex one and remains useful as a second exact branch.

Proposition 21.27. *There is a bounded, strictly star-shaped, non-ball domain $\Omega_{\text{old}} \subset \mathbb{R}^3$ with real-analytic boundary diffeomorphic to S^2 and a nonconstant analytic solution of (1.1). It has the same $O(2) \times \mathbf{Z}_2$ symmetry, but is not convex: both principal curvatures at the two poles, with respect to the inner normal, are negative.*

Proof. Apply the preceding analytic lemmas and the contraction criterion with the independent centre of the nonconvex example and its parameters $M = 180$, $S = 64$, $R = 21/20$, $b_{\text{old}} = 0 \times 1.85067\text{a9afca2dp}+5$, $\sigma = 3b_{\text{old}}$ and $r = 3 \cdot 10^{-7}$. Its complete verifier and two certificate files are in `anc/r3/`; the certified bounds are displayed in Table 15. The corresponding radius polynomial is $< -4.81 \cdot 10^{-9}$ and contraction bound is < 0.959201 . Its geometry certificate gives $\text{Re}(w\psi'/\psi) > 0.83$ on the unit circle and a nonzero c_{13} , so the same reconstruction gives an analytic, strictly star-shaped non-ball domain and solution. Nonconvexity follows from Proposition 21.28. $\square$

(M, S)	σ	$\ A\ $	Y	Z	r	Geometry
$(180, 64)$	$3b_{\text{old}}$	< 5849.52	$< 2.10029 \cdot 10^{-8}$	< 0.858929	$3 \cdot 10^{-7}$	nonconvex at poles

TABLE 15. The independent nonconvex certificate; see `anc/r3/certificate_r3.json`.

Proposition 21.28. *For the centre of the nonconvex example $c^{\text{old}, o}$, exact rational arithmetic gives*

$$42.81903 < \sum_j j c_j^{\text{old}, o} < 42.81904, \quad -7.758015 < \sum_j j^2 c_j^{\text{old}, o} < -7.758014.$$

Throughout its certified existence ball, $1 + \psi''(1)/\psi'(1) < -0.181$; hence both principal curvatures at each pole are negative.

Proof. The shape perturbation has $\mathcal{E}$ norm at most $r/(3b_{\text{old}}) < 2.06 \cdot 10^{-9}$. Its effect on $\psi'(1)$ is at most e/R , and on $\psi'(1) + \psi''(1)$ at most $\sup_{t>0} tR^{-t}e < 7.55e < 1.6 \cdot 10^{-8}$. Thus $0 < \psi'(1) < 42.82$ and $\psi'(1) + \psi''(1) < -7.758$, giving the claimed ratio. The meridian

curvature at a pole, with inward normal, is $(1 + \psi''(1)/\psi'(1))/\psi'(1) < 0$; rotation symmetry makes both surface principal curvatures equal to it. The other pole follows by reflection. The exact dyadic check is `anc/r3/check_nonconvex_r3.py`. $\square$

22. DIMENSION FIVE: WEIGHTED SPACES

In this section we prove Theorem 1.1 for $n = 5$. The construction follows Section 21: the problem is pulled back to the unit ball $B \subset \mathbb{R}^5$ by the axisymmetric extension of a conformal map of the meridian section, the unknown field is written as $1 + K_5 g$, and the existence of a zero of a quartic polynomial map is proved by a contraction argument around an exact dyadic centre. The new ingredient is functional-analytic. For $n \geq 5$ the Jacobi coefficient norm of Section 21 does not control $K_n g$ in C^1 (Remark 22.2), so that a zero in that norm would not yield a function with the two boundary traces. We therefore grade all coefficient norms by the weight

$$\Upsilon(D) = 1 + \frac{D}{10^6}$$

of the polynomial degree D . The C^1 bound then holds with a large but finite constant, which enters only the reconstruction, while the multiplication constants are those of the unweighted norms and all other operator bounds acquire at most factors $\Upsilon(\Delta)$ with $\Delta \leq 1000$, that is, at most 1.001.

Notation. Points of the physical space are $X = (X_1, X') \in \mathbb{R} \times \mathbb{R}^4$ and points of the reference space are $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^4$. The meridian variables $y = |x'|$, $w = x_1 + iy$, $r = |w| = |x|$, $t = r^2$, $\xi = x_1/r$, the constant $R = 21/20$, the polynomials T_n , U_n , $C_n^{(2)}$ and J_s^β , the map $\psi_c(w) = \sum_{j \text{ odd}} c_j w^j$ and the functions $p_c = \psi'_c$ and $q_c = \text{Im } \psi_c/y$, with the expansions (21.1) and (21.2), are as in Sections 21.1 and 21.2; they involve only the meridian variables. Now B is the unit ball of $\mathbb{R}^5$, and $\zeta = R^{-2}$. The objects H_ℓ , $\Psi_{\ell,s}$, $D_{n\ell}$, h_ℓ , $C_{\ell n}$, $\mathcal{T}_n$, $\mathcal{L}$, $\mathcal{G}$, $\mathcal{Y}$, $\mathcal{E}$, $\mathcal{X}$, τ , $\|\cdot\|_{\text{ch}}$, $\mathcal{F}$ and A of Section 21 are redefined below for $n = 5$; we keep the letters.

22.1. The pull-back and the obstruction. For $c = (c_j)_{j \text{ odd}}$ with $\sum_j j R^j |c_j| < \infty$ let $\Theta_c(x) = (\text{Re } \psi_c(x_1 + i|x'|), q_c(x) x')$, now with $x' \in \mathbb{R}^4$.

Lemma 22.1. *Let c be as above with $\text{Re } \psi'_c > 0$ on $\{|w| < R_0\}$ for some $1 < R_0 < R$, and let $\rho > 0$.*

- (a) *The functions q_c , $\partial_{x_1} q_c$, $y^{-1} \partial_y q_c$, $|p_c|^2$ and $\text{Re } \psi_c(x_1 + iy)$ are real analytic on $B_{R_0} = \{x \in \mathbb{R}^5 : |x| < R_0\}$, and $q_c > 0$ there. The map Θ_c is a real-analytic diffeomorphism of B_{R_0} onto an open subset of $\mathbb{R}^5$, commuting with the action of $O(4)$ on x' and with $x_1 \mapsto -x_1$; $\det D\Theta_c = q_c^3 |p_c|^2$; and $(\det D\Theta_c)(D\Theta_c^\top D\Theta_c)^{-1} \nabla U = q_c^3 \nabla U$ for every axisymmetric C^1 function U .*
- (b) *Let $U \in C^1(\overline{B})$ be axisymmetric and satisfy*

$$(22.1) \quad \int_B q_c^3 (\nabla U \cdot \nabla \varphi - \rho^2 |p_c|^2 U \varphi) dx = 0 \quad (\varphi \in C_c^\infty(B)).$$

Then $u = U \circ \Theta_c^{-1}$ satisfies $\int_{\Theta_c(B)} (\nabla u \cdot \nabla \varphi - \rho^2 u \varphi) dX = 0$ for all $\varphi \in C_c^\infty(\Theta_c(B))$, and if $U = 1$ and $\nabla U = 0$ on ∂B , then $u = 1$ and $\nabla u = 0$ on $\partial \Theta_c(B)$.

Proof. The proofs of Lemmas 21.1 and 21.2 apply with $x' \in \mathbb{R}^4$. At a point with $x' \neq 0$ let $e_\varrho = x'/|x'|$ and let $e_{\phi,1}, e_{\phi,2}, e_{\phi,3}$ be an orthonormal basis of the orthogonal complement of e_ϱ in $\mathbb{R}^4$. On the half-plane spanned by e_1 and e_ϱ , Θ_c acts as ψ_c , and $D\Theta_c$ is $|p_c|$ times a rotation there; Θ_c maps the 3-sphere $\{(x_1, z') : |z'| = y\}$ onto the 3-sphere of radius $q_c y$ in the hyperplane $\{X_1 = \text{Re } \psi_c(x_1 + iy)\}$ by $z' \mapsto q_c z'$, so that $D\Theta_c e_{\phi,i} = q_c e_{\phi,i}$. Hence

$D\Theta_c^\top D\Theta_c = \text{diag}(|p_c|^2, |p_c|^2, q_c^2, q_c^2, q_c^2)$ in this frame, which gives the determinant, and, since ∇U lies in the span of e_1 and e_ρ , the identity for ∇U . This identity is pointwise, so in the change of variables of Lemma 21.2 the test functions $\varphi \circ \Theta_c$ need not be axisymmetric. $\square$

For an axisymmetric $U = 1 + W$ with W of class C^2 one has, off the axis, $\nabla q_c \cdot \nabla W = \partial_{x_1} q_c \partial_{x_1} W + (y^{-1} \partial_y q_c) y \partial_y W$ and hence

$$(22.2) \quad \nabla \cdot (q_c^3 \nabla U) + \rho^2 q_c^3 |p_c|^2 U = q_c^2 \left(q_c \Delta W + 3 \partial_{x_1} q_c \partial_{x_1} W + 3 \frac{\partial_y q_c}{y} y \partial_y W + \rho^2 q_c |p_c|^2 U \right).$$

The factor $q_c^{n-3} = q_c^2$ is positive, and the bracket is linear in q_c , $\partial_{x_1} q_c$, $y^{-1} \partial_y q_c$, $q_c |p_c|^2$ and in W . With $\Delta W = g$ it is a polynomial of degree four in the field and shape coefficients, and no division by the unknown q_c occurs. (For $n = 3$ the same computation gives (21.11); in $\mathbb{R}^n$ the exponents are $n - 2$ and $n - 3$.)

Remark 22.2 (Why a weight is needed). Let $n \geq 3$, $\lambda = (n-2)/2$, and let K_n act on $\Psi_{0,s} = J_s^\lambda(t)$ by the three-term formula (21.9) with $d = \lambda + 2s$ (for $n = 3$ this is K_3 , for $n = 5$ it is (22.4) below). By Lemma 21.10(a), which holds for every β , $\partial_r K_n \Psi_{0,s} = \frac{r}{2(d+1)} (J_s^{\lambda+1}(t) - J_{s-1}^{\lambda+1}(t))$, and $J_s^\beta(t) = (-1)^s P_s^{\beta,0}(1-2t)$. By the Mehler–Heine formula [22, 18.11.5], which holds uniformly for bounded arguments, for every fixed $z > 0$

$$\lim_{s \rightarrow \infty} s^{1-\lambda} \left| \partial_r K_n \Psi_{0,s} \right|_{r=z/(2s)} = \frac{1}{2} \left(\frac{z}{2} \right)^{-\lambda} |J_{\lambda+1}(z)|.$$

Thus $\sup |\nabla K_n \Psi_{0,s}|$ is at least of order $s^{(n-4)/2}$, while the norm of $\Psi_{0,s}$ in the space $\mathcal{L}$ of Section 21 (and in its analogue for $\mathbb{R}^n$) is $h_0 = 1$ for all s . For $n \geq 5$ there is therefore no estimate $\|K_n g\|_{C^1} \leq C \|g\|_{\mathcal{L}}$, and the reconstruction step of Section 21.9 fails in these norms. (For $n = 3$ the growth is $s^{-1/2}$, in accordance with Lemma 21.11.) The growth is polynomial in the degree, and the weight $(1 + D/c)^{(n-3)/2}$ suffices (Theorem 22.4 and the remark after it); for $n = 5$ this is $1 + D/c$, and we take $c = 10^6$.

22.2. Graded coefficient spaces. For $\ell \geq 0$ let $Y_\ell = C_\ell^{(3/2)}/C_\ell^{(3/2)}(1)$, where $C_\ell^{(3/2)}(1) = (\ell+1)(\ell+2)/2$, and let $H_\ell(x) = r^\ell Y_\ell(\xi)$, the axisymmetric harmonic polynomial on $\mathbb{R}^5$, homogeneous of degree ℓ , with $H_\ell(1,0) = 1$; $|H_\ell| \leq 1$ on $\overline{B}$. With $\lambda_\ell^+ = (\ell+3)/(2\ell+3)$, $\lambda_\ell^- = \ell/(2\ell+3)$ and $H_{-1} = 0$,

$$(22.3) \quad \begin{aligned} \xi Y_\ell &= \lambda_\ell^+ Y_{\ell+1} + \lambda_\ell^- Y_{\ell-1}, & x_1 H_\ell &= \lambda_\ell^+ H_{\ell+1} + \lambda_\ell^- t H_{\ell-1}, \\ \partial_{x_1} H_\ell &= \ell H_{\ell-1}, & (x \cdot \nabla) H_\ell &= \ell H_\ell, \end{aligned}$$

by the three-term recurrence of $C_\ell^{(3/2)}$ and the argument after (21.5). Let $\beta_\ell = \ell + \frac{3}{2}$ and $\Psi_{\ell,s} = H_\ell J_s^{\beta_\ell}(t)$, of degree $\ell+2s$. Lemma 21.3 holds for every $\beta > 0$ and is used without change, and $\Delta(H_\ell v(t)) = 4H_\ell(tv'' + (\beta_\ell+1)v')$ in $\mathbb{R}^5$, by (22.3) and $\Delta v(|x|^2) = 4tv'' + 10v'$. In spherical coordinates, Lemma 21.3(a) and the Fischer decomposition show as in Lemma 21.6 that the $\Psi_{\ell,s}$ are pairwise orthogonal in $L^2(B)$, that $\Psi_{\ell,s}$ is orthogonal to all polynomials of degree less than $\ell+2s$, and that $\|\Psi_{\ell,s}\|_{L^2(B)}^2 = |S^4|/(2\nu_\ell(\ell+2s+\frac{5}{2}))$, where $\nu_\ell = (2\ell+3)(\ell+1)(\ell+2)/6$ is the dimension of the spherical harmonics of degree ℓ on S^4 .

Let $\mathbf{a}_k = (\frac{3}{2})_k/k!$, a nondecreasing sequence with $\sum_k \mathbf{a}_k \zeta^k = (1-\zeta)^{-3/2}$. Factoring $(1-2\xi z + z^2)^{-3/2}$ as in Section 21.2 gives $Y_\ell = \sum_n D_{n\ell} T_n$ with

$$D_{\ell-2k,\ell} = (2 - \delta_{\ell,2k}) \mathbf{a}_k \mathbf{a}_{\ell-k} / C_\ell^{(3/2)}(1) \geq 0,$$

and we put $h_\ell = \sum_n D_{n\ell} R^n \geq Y_\ell(1) = 1$. We write $T_K = \sum_\ell C_{\ell K} Y_\ell$ and $\mathcal{T}_K = \sum_\ell C_{\ell K} H_\ell$, the harmonic polynomial with boundary values $T_K(\xi)$; $|\mathcal{T}_K| \leq 1$ on $\overline{B}$.

Graded norms. The weight $\Upsilon(D) = 1 + D/10^6$ is nondecreasing and submultiplicative, $\Upsilon(a + b) \leq \Upsilon(a)\Upsilon(b)$. The spaces $\mathcal{L}$, $\mathcal{G}$, $\mathcal{Y}$, $\mathcal{E}$ and $\mathcal{X} = \mathcal{G} \times \mathcal{E}$ are defined as in Section 21.2, with the norms

$$\begin{aligned} \|f\|_{\mathcal{L}} &= \sum_{\ell, s \geq 0} h_{\ell} \Upsilon(\ell + 2s) |f_{\ell, s}|, & \|f\|_{\mathcal{Y}} &= \sum_n R^n \Upsilon(n) |f_n| + \sum_{\ell, s \geq 1} h_{\ell} \Upsilon(\ell + 2s) |f_{\ell, s}|, \\ \|c\|_{\mathcal{E}} &= \sum_{j \text{ odd}} j R^j \Upsilon(j - 1) |c_j|, & \|(g, c)\|_{\mathcal{X}} &= \|g\|_{\mathcal{L}} + \sigma \|c\|_{\mathcal{E}}, \end{aligned}$$

where $f \in \mathcal{Y}$ is identified with $\sum_n f_n \mathcal{T}_n + \sum_{s \geq 1} f_{\ell, s} \Psi_{\ell, s}$ and σ is fixed in Section 22.3. Thus every basis element carries its weight from Section 21 times Υ of its polynomial degree: $\ell + 2s$ for $\Psi_{\ell, s}$, n for $\mathcal{T}_n$, and $j - 1 = \deg j w^{j-1}$ for the shape mode w^j . Since $\sum_{m \leq \ell} D_{m\ell} R^m \Upsilon(m) \leq h_{\ell} \Upsilon(\ell)$, the inclusion $\mathcal{L}^{\text{ev}} \subset \mathcal{Y}$ is norm-decreasing. For polynomials and series $m = \sum m_{a,k} H_a t^k = \sum \tilde{m}_{n,k} r^{n+2k} T_n(\xi)$ we put

$$\tau(m) = \sum_{a,k} h_a \Upsilon(a + 2k) |m_{a,k}|, \quad \|m\|_{\text{ch}} = \sum_{n,k} R^n \Upsilon(n + 2k) |\tilde{m}_{n,k}|.$$

All layers are graded by the same function of the degree. Every product occurring below raises the degree by at most the degree of the multiplier, and the graded shape norm pays exactly for the degree of the multipliers it produces; if only the field layer were graded, the columns of the linearisation at high shape modes would lose their decay.

Lemma 22.3. (a) $C_{00} = 1$, and for $K \geq 1$ and $0 \leq i \leq K/2$,

$$C_{K-2i,K} = \frac{K}{2} \cdot \frac{(-\frac{3}{2})_i (K - i - 1)!}{i! (\frac{5}{2})_{K-i}} \cdot \frac{K - 2i + \frac{3}{2}}{\frac{3}{2}} \cdot C_{K-2i}^{(3/2)}(1).$$

- (b) Let $e_i = |(-\frac{3}{2})_i|/i!$ and $\mathbf{w}_i = e_i \zeta^i (1 - \zeta)^{-3/2}$. Then $|C_{K-2i,K}| h_{K-2i} \leq \mathbf{w}_i R^K$, and $\sum_i \mathbf{w}_i = \kappa_5 := 1 + 3\zeta(1 - \zeta)^{-3/2} \approx 96.99$.
 (c) $\lambda_{\ell}^+ h_{\ell+1} + \lambda_{\ell}^- h_{\ell-1} \leq R h_{\ell}$, and $D_{\ell-2k,\ell} R^{\ell-2k} \leq \mathbf{a}_k \zeta^k h_{\ell}$.

Proof. (a) Divide the connection formula [22, 18.18.16] for $C_K^{(\mu)}$ in terms of the $C_{K-2i}^{(3/2)}$ by μ and let $\mu \rightarrow 0$, using $C_K^{(\mu)}/\mu \rightarrow (2/K)T_K$ [22, 18.7.25] and $(\mu)_{K-i}/\mu \rightarrow (K - i - 1)!$; this is the argument of Lemma 21.4(a). (Alternatively, both sides satisfy the recurrence in K that follows from $T_{K+1} = 2\zeta T_K - T_{K-1}$ and (22.3).)

(b) Let $\ell = K - 2i$. For $\ell \geq 1$, $D_{\ell\ell} = 2\mathbf{a}_{\ell}/C_{\ell}^{(3/2)}(1)$, and (a) gives $|C_{\ell K}| D_{\ell\ell} = e_i \varrho_i$ with $\varrho_0 = 1$ and, for $i \geq 1$,

$$\varrho_i = \frac{K}{K - i + \frac{3}{2}} \prod_{r=1}^{i-1} \frac{\ell + r}{\ell + r + \frac{3}{2}} \leq \frac{K(\ell + 1)}{(K - i + \frac{3}{2})(\ell + i)} \leq 1,$$

by telescoping and $(K - i + \frac{3}{2})(\ell + i) - K(\ell + 1) = i^2 - \frac{3}{2}i + \frac{K}{2} > 0$. For $\ell = 0$ the same holds with $D_{00} = 1$ and an extra factor $\frac{1}{2}$. Since $\mathbf{a}_k$ is nondecreasing, $h_{\ell} \leq D_{\ell\ell} R^{\ell} \sum_k \mathbf{a}_k \zeta^k = D_{\ell\ell} R^{\ell} (1 - \zeta)^{-3/2}$, hence $|C_{\ell K}| h_{\ell} \leq e_i R^{K-2i} (1 - \zeta)^{-3/2} = \mathbf{w}_i R^K$. The numbers $e_i \zeta^i$ are the moduli of the Taylor coefficients of $(1 - \zeta)^{3/2}$, of which only the coefficient $-\frac{3}{2}$ of ζ is negative; so $\sum_i e_i \zeta^i = (1 - \zeta)^{3/2} + 3\zeta$.

(c) Apply the linear functional $\langle \cdot \rangle_R$ of Section 21.2 to the first identity in (22.3). The right-hand side gives $\lambda_{\ell}^+ h_{\ell+1} + \lambda_{\ell}^- h_{\ell-1}$, and the left-hand side is $\sum_n D_{n\ell} (R^{n+1} + R^{|n-1|})/2 \leq R h_{\ell}$, because all $D_{n\ell}$ are nonnegative. The second bound follows from $D_{\ell-2k,\ell} \leq \mathbf{a}_k D_{\ell\ell}$ and $h_{\ell} \geq D_{\ell\ell} R^{\ell}$. $\square$

The three basic theorems. The following three results make the scheme of Section 21 work in dimension five: a C^1 bound for the compatible inverse of the Laplacian, a graded multiplier algebra, and the transfer of the operator bounds to the graded norms.

For $\ell \geq 0$ and $s \geq 1$ let $d = \beta_\ell + 2s = \ell + 2s + \frac{3}{2}$ and

$$(22.4) \quad K_5 \Psi_{\ell,s} = \frac{\Psi_{\ell,s-1}}{4d(d+1)} - \frac{\Psi_{\ell,s}}{2d(d+2)} + \frac{\Psi_{\ell,s+1}}{4(d+1)(d+2)} = H_\ell k_{\ell,s}(t).$$

Theorem 22.4 (C^1 bound). *For $\ell \geq 0$ and $s \geq 1$, $\Delta K_5 \Psi_{\ell,s} = \Psi_{\ell,s}$, the function $K_5 \Psi_{\ell,s}$ and its gradient vanish on ∂B , and, with $D = \ell + 2s$,*

$$\max \left(\sup_{\bar{B}} |K_5 \Psi_{\ell,s}|, \sup_{\bar{B}} |\nabla K_5 \Psi_{\ell,s}| \right) \leq \frac{D+5}{4} \leq 1.25 \cdot 10^6 h_\ell \Upsilon(D).$$

Consequently, for $g \in \mathcal{G}$ the series $W = \sum g_{\ell,s} K_5 \Psi_{\ell,s}$ converges in $C^1(\bar{B})$, the functions $|W|$ and $|\nabla W|$ are bounded on $\bar{B}$ by $1.25 \cdot 10^6 \|g\|_{\mathcal{L}}$, $W = 0$ and $\nabla W = 0$ on ∂B , and $\Delta W = \iota(g)$ in $\mathcal{D}'(B)$.

Proof. The first assertions are Lemma 21.10(b),(c); their proofs use only Lemma 21.3, which holds for every β , and the formula for $\Delta(H_\ell v(t))$ above. For the bound we follow the proof of Lemma 21.11, with $S^6 \subset \mathbb{R}^5 \times \mathbb{R}^2$ in place of $S^4 \subset \mathbb{R}^3 \times \mathbb{R}^2$. The parametrisation $(x, \phi) \mapsto (x, \sqrt{1-|x|^2}(\cos \phi, \sin \phi))$ again gives $\int_{S^6} f(x) d\zeta = 2\pi \int_B f dx$; the dimension of the spherical harmonics of degree m on S^6 is $N_m = (m+1)(m+2)(m+3)(m+4)(2m+5)/120$, and $|S^6| = \frac{2\pi}{5}|S^4|$. Part (a) of that lemma becomes $|P(x)|^2 \leq 5N_m |S^4|^{-1} \|P\|_{L^2(B)}^2$ for $P \in V_m$ and $x \in \bar{B}$. As there, $\partial_i K_5 \Psi_{\ell,s} \in V_{D-1} \oplus V_{D+1}$ and $\|\nabla K_5 \Psi_{\ell,s}\|_{L^2}^2 = \|\Psi_{\ell,s}\|_{L^2}^2 / (2d(d+2))$, and $D + \frac{5}{2} = d + 1$, so that

$$|\nabla K_5 \Psi_{\ell,s}(x)|^2 \leq \frac{5(N_{D-1} + N_{D+1})}{4\nu_\ell d(d+1)(d+2)} \leq \frac{(D+2)(D+3)(D+4)(D+5)}{24(D+\frac{3}{2})(D+\frac{5}{2})} \leq \frac{(D+5)^2}{16},$$

using $\nu_\ell \geq 1$, $N_{D-1} \leq N_{D+1}$, $2(d+2) = 2D+7$ and, in the last step, $3(D+\frac{3}{2})(D+\frac{5}{2})(D+5) - 2(D+2)(D+3)(D+4) = D^3 + 9D^2 + \frac{77}{4}D + \frac{33}{4} > 0$. Integration along a radius bounds $|K_5 \Psi_{\ell,s}|$ by the same number. Finally $(D+5)/4 \leq \frac{5}{4} \cdot 10^6 \Upsilon(D)$ and $h_\ell \geq 1$. The consequences follow as in Lemma 21.11(c), with the embedding ι of Lemma 21.6, which is continuous because $\|\Psi_{\ell,s}\|_{L^2} \leq |S^4|^{1/2}$. $\square$

Remark 22.5. The same argument with $S^{n+1} \subset \mathbb{R}^n \times \mathbb{R}^2$ bounds the C^1 norm of $K_n \Psi_{\ell,s}$ in $\mathbb{R}^n$ by a constant times $D^{(n-3)/2}$, so the weights $(1+D/c)^{(n-3)/2}$ give a C^1 bound in every dimension $n \geq 4$; by Remark 22.2 the exponent cannot be smaller than $(n-4)/2$. The constant in Theorem 22.4 is used only qualitatively, in the proof of Theorem 1.1 for $n=5$ at the end of this section; it never enters the contraction argument, which is why $c=10^6$ may be taken so large that the weights are nearly flat on all indices that enter the computation.

Theorem 22.6 (Graded multiplier algebra). (a) *For $a, \ell \geq 0$, $Y_a Y_\ell = \sum_v G_{a\ell v} Y_v$ with $G_{a\ell v} \geq 0$, supported on $|a-\ell| \leq v \leq a+\ell$, $v \equiv a+\ell \pmod{2}$, with $\sum_v G_{a\ell v} = 1$ and $\sum_v G_{a\ell v} h_v \leq h_a h_\ell$.*

(b) *For a solid monomial $H_a t^k$ of degree $d = a+2k$ and $\ell, s \geq 0$, the expansion (21.8) holds with probability vectors $\pi^{(v)}$, and Lemma 21.9 holds with $\vartheta_d(\ell, s) = (d/(\ell+d+\frac{7}{2}))^s$ for $d \geq s$ and $\vartheta_d(\ell, s) = 0$ for $d < s$.*

(c) *For all m, m_1, m_2 with finite norms and $f \in \mathcal{L}$: $\|mf\|_{\mathcal{L}} \leq \tau(m)\|f\|_{\mathcal{L}}$ and $\tau(m_1 m_2) \leq \tau(m_1)\tau(m_2)$; moreover $\tau(m) \leq \kappa_5 \|m\|_{\text{ch}}$ and $\|m_1 m_2\|_{\text{ch}} \leq \|m_1\|_{\text{ch}} \|m_2\|_{\text{ch}}$; and $\iota(mf) = m \iota(f)$.*

(d) For $c \in \mathcal{E}$ the bounds (21.12) hold for the graded norms of this section.

Proof. (a) Up to the positive factors $C_v^{(3/2)}(1)/(C_a^{(3/2)}(1)C_\ell^{(3/2)}(1))$, the G_{alv} are the coefficients of the linearisation formula [22, 18.18.22] for $C_a^{(3/2)}C_\ell^{(3/2)}$, all of whose factors are positive. Evaluation at $\xi = 1$ gives $\sum_v G_{alv} = 1$, and the last bound is proved as in Lemma 21.7, since all D_{nv} are nonnegative.

(b) The proofs of Lemmas 21.8(a) and 21.9 use only (a), the resulting identity $H_a H_\ell = \sum_v G_{alv} H_v t^{(a+\ell-v)/2}$ and Lemma 21.3; in part (b) of Lemma 21.9 the factors are now $\beta_v + 1 = v + \frac{5}{2} \leq \beta_\ell + m + 1$ and $(m - i)/(\beta_\ell + m + 2 + i) \leq d/(\ell + d + \frac{7}{2})$.

(c) Every $\Psi_{v,\varsigma}$ in the expansion of $H_a t^k \Psi_{\ell,s}$ has degree $v + 2\varsigma \leq (a + 2k) + (\ell + 2s)$, so $\Upsilon(v + 2\varsigma) \leq \Upsilon(a + 2k)\Upsilon(\ell + 2s)$ and, by (a) and (b), $\|H_a t^k \Psi_{\ell,s}\|_{\mathcal{L}} \leq h_a \Upsilon(a + 2k) h_\ell \Upsilon(\ell + 2s)$; absolute summation gives the first bound, and products of two solid monomials are treated in the same way. The expansion $r^{n+2k} T_n = \sum_i C_{n-2i,n} H_{n-2i} t^{i+k}$ preserves the degree $n + 2k$, so Lemma 22.3(b) gives $\tau(r^{n+2k} T_n) \leq \kappa_5 R^n \Upsilon(n + 2k)$. For Chebyshev products, $T_m T_n = \frac{1}{2}(T_{m+n} + T_{|m-n|})$ adds the degrees and $R^{|m-n|} \leq R^{m+n}$. The last identity is proved as Lemma 21.8(d).

(d) By (21.1), the mode w^j contributes to q_c , $\partial_{x_1} q_c$ and $y^{-1} \partial_y q_c$ homogeneous polynomials of degrees $j - 1$, $j - 2$ and $j - 3$, and to p_c the monomial jw^{j-1} . Their graded norms are at most $\Upsilon(j - 1)$ times the ungraded ones, which is matched by the factor $\Upsilon(j - 1)$ in $\|c\|_{\mathcal{E}}$, and the ungraded bounds are those of the proof of Lemma 21.12. For $|p_c|^2$ use (21.2) and $R^{\alpha+\beta} \Upsilon(\alpha + \beta) \leq R^\alpha \Upsilon(\alpha) R^\beta \Upsilon(\beta)$. $\square$

Lemma 22.7. For $\ell \geq 0$, $s \geq 1$ and d as in (22.4), with $\Psi_{-1,\cdot} = 0$,

$$\begin{aligned} \partial_{x_1} K_5 \Psi_{\ell,s} &= \frac{\lambda_\ell^+}{2(d+1)} (\Psi_{\ell+1,s} - \Psi_{\ell+1,s-1}) + \frac{\lambda_\ell^-}{2(d+1)} (\Psi_{\ell-1,s+1} - \Psi_{\ell-1,s}), \\ EK_5 \Psi_{\ell,s} &= -\frac{d + \frac{3}{2}}{4d(d+1)} \Psi_{\ell,s-1} + \frac{3 \Psi_{\ell,s}}{4d(d+2)} + \frac{d + \frac{1}{2}}{4(d+1)(d+2)} \Psi_{\ell,s+1}, \end{aligned}$$

where $E = x \cdot \nabla$, and $y \partial_y K_5 = EK_5 - x_1 \partial_{x_1} K_5$. The operators K_5 , $\partial_{x_1} K_5$, EK_5 and $y \partial_y K_5$ are bounded from $\mathcal{G}$ to $\mathcal{L}$, with norms at most, for $d_0 = \frac{7}{2}$,

$$\begin{aligned} k_K &= \frac{1}{d_0(d_0+2)} + \frac{\Upsilon(2) - 1}{4(d_0+1)(d_0+2)}, & k_D &= \frac{R(1 + \Upsilon(1))}{2(d_0+1)}, \\ k_E &= \frac{d_0 + \frac{3}{2}}{2d_0(d_0+1)} + \frac{(\Upsilon(2) - 1)(d_0 + \frac{1}{2})}{4(d_0+1)(d_0+2)}, & k_Y &= k_E + R \Upsilon(1) k_D. \end{aligned}$$

Proof. By (22.3) and $2\lambda_\ell^- \beta_\ell = \ell$, $\partial_{x_1} (H_\ell k) = 2\lambda_\ell^+ H_{\ell+1} k' + 2\lambda_\ell^- H_{\ell-1} (\beta_\ell k + tk')$ and $E(H_\ell k) = H_\ell (2(\beta_\ell k + tk') - (\beta_\ell + \frac{3}{2})k)$. The expansions of $k'_{\ell,s}$ and of $\beta k_{\ell,s} + tk'_{\ell,s}$ in the proof of Lemma 21.10(a),(d) hold for every β and give the two formulas. For the norms we use Lemma 15.2. The stencil of K_5 changes the degree $D = \ell + 2s$ by $-2, 0, 2$, has column sum zero and absolute column sum $1/(d(d+2))$; with $\Upsilon(D-2) \leq \Upsilon(D)$ and $\Upsilon(D+2) \leq \Upsilon(2)\Upsilon(D)$ its weighted column sum is at most $1/(d(d+2)) + (\Upsilon(2) - 1)/(4(d+1)(d+2))$, which decreases in $d \geq d_0$. The stencil of EK_5 also has column sum zero, with absolute column sum $(d + \frac{3}{2})/(2d(d+1))$, which gives k_E in the same way. For $\partial_{x_1} K_5$ the degrees change by ± 1 , $\Upsilon(D+1) + \Upsilon(D-1) \leq (1 + \Upsilon(1))\Upsilon(D)$, and $\lambda_\ell^+ h_{\ell+1} + \lambda_\ell^- h_{\ell-1} \leq R h_\ell$ by Lemma 22.3(c). Finally $\tau(x_1) = h_1 \Upsilon(1) = R \Upsilon(1)$. $\square$

Theorem 22.8 (Transfer to graded norms). (a) Let M be a linear map between weighted ℓ^1 spaces whose weights are base weights times Υ of the degree of the basis elements,

and suppose that $M_{\kappa\iota} \neq 0$ implies $\deg \kappa \leq \deg \iota + \Delta$. Then $\|M\| \leq \Upsilon(\max(\Delta, 0))\|M\|_0$, where $\|\cdot\|_0$ is the operator norm for the base weights. The same holds for multilinear maps, with $\deg \iota$ replaced by the sum of the degrees of the inputs.

(b) Let $(s_\delta)_{\delta \in \mathbb{Z}}$ be finitely supported with $s_0 > 0$, and let

$$\varepsilon_\Upsilon = \frac{1}{s_0} \sum_{\delta \neq 0} |s_\delta| R^{2\delta} \Upsilon(\max(2\delta, 0)) < 1.$$

For $N_0 \geq 0$ let E act on sequences $(\hat{a}_N)_{N \geq N_0}$, N even by

$$(E\hat{a})_m = \frac{1}{s_0} \sum_{N \neq m} s_{(m-N)/2} R^{m-N} \hat{a}_N \quad (m \geq N_0 \text{ even}).$$

On the ℓ^1 space with weights $\Upsilon(N)$, $\|E\| \leq \varepsilon_\Upsilon$; hence $I + E$ is invertible, $\|(I + E)^{-1}\| \leq (1 - \varepsilon_\Upsilon)^{-1}$, and the remainder of the Neumann series after the terms $k \leq L$ has norm at most $\varepsilon_\Upsilon^{L+1}/(1 - \varepsilon_\Upsilon)$.

(c) In particular, the outputs of a product of a solid monomial of degree d with $\Psi_{\ell,s}$ carry at most the factor $\Upsilon(d)$ relative to the weight of $\Psi_{\ell,s}$, and a polynomial output of degree at most $N + \Delta$ produced from a shape input of degree N carries at most the factor $\Upsilon(\Delta)$.

Proof. (a) By Lemma 15.2 it suffices to consider single columns (tuples of basis vectors for multilinear maps). The weighted column norm is a sum of base-weighted entries multiplied by $\Upsilon(\deg \kappa)/\Upsilon(\deg \iota)$, which is at most $\Upsilon(\deg \kappa - \deg \iota) \leq \Upsilon(\Delta)$ if $\deg \kappa \geq \deg \iota$, by submultiplicativity, and at most one otherwise. (b) Column N of E has entries only in the rows $m = N + 2\delta \geq N_0$, and $\Upsilon(N + 2\delta)/\Upsilon(N) \leq \Upsilon(\max(2\delta, 0))$; so its weighted column sum is at most $\varepsilon_\Upsilon \Upsilon(N)$. The rest is the Neumann series. (c) is (a) applied to the degree bound of Theorem 22.6(c). $\square$

22.3. The quartic equation and the approximate inverse. From now on $M = 160$, $S = 57$ and $\rho = 7951044098899425 \cdot 2^{-47} \approx 56.4955663$. The exact dyadic centre $x^\circ = (g^\circ, c^\circ)$ consists of field coefficients $g_{\ell,s}^\circ$ with even $\ell \leq M$ and $1 \leq s \leq S$ and odd shape coefficients c_j° with $j \leq M + 1$; it is embedded in the verifier of Section 34.9 (SHA-256 recorded there). We put

$$b = c_1^\circ \approx 1.0002252, \quad \sigma = 30\rho^2,$$

a fixed dyadic number and a fixed scale, independent of the varying coefficient c_1 . For $x = (g, c) \in \mathcal{X}$ let $W = K_5 g$, $W_1 = \partial_{x_1} K_5 g$ and $W_2 = y \partial_y K_5 g$, elements of $\mathcal{L}$ by Lemma 22.7, and

$$(22.5) \quad \mathcal{F}(g, c) = \frac{1}{b} \left[q_c g + 3(\partial_{x_1} q_c) W_1 + 3 \left(\frac{\partial_y q_c}{y} \right) W_2 + \rho^2 q_c |p_c|^2 (1 + W) \right].$$

Lemma 22.9. $\mathcal{F}$ is a bounded polynomial map of degree four from $\mathcal{X}$ to $\mathcal{Y}$. If $\mathcal{F}(g, c) = 0$ and $\operatorname{Re} \psi'_c > 0$ on $\{|w| < R_0\}$ for some $1 < R_0 < R$, then $U = 1 + \iota(K_5 g) \in C^1(\overline{B})$ is axisymmetric, satisfies (22.1) for all $\varphi \in C_c^\infty(B)$, and $U = 1$, $\nabla U = 0$ on ∂B .

Proof. Boundedness follows from Theorem 22.6(c),(d) and Lemma 22.7, and the values are even in x_1 as in Lemma 21.12. Suppose $\mathcal{F}(g, c) = 0$. By Theorem 22.4, $W = \iota(K_5 g) \in C^1(\overline{B})$ has vanishing Cauchy data and $\Delta W = \iota(g)$. Let $\varphi \in C_c^\infty(B)$; then $q_c^3 \varphi \in C_c^\infty(B)$ by

Lemma 22.1(a), and, as in the proof of Lemma 21.12,

$$\begin{aligned} \int_B q_c^3 (\nabla U \cdot \nabla \varphi - \rho^2 |p_c|^2 U \varphi) &= \int_B (\nabla W \cdot \nabla (q_c^3 \varphi) - 3q_c^2 \varphi \nabla q_c \cdot \nabla W - \rho^2 q_c^3 |p_c|^2 U \varphi) \\ &= - \int_B q_c^2 \varphi b \iota(\mathcal{F}(g, c)) = 0, \end{aligned}$$

using $\int \nabla W \cdot \nabla (q_c^3 \varphi) = - \int \iota(g) q_c^3 \varphi$ and the identity for $\nabla q_c \cdot \nabla W$ before (22.2). No averaging over $O(4)$ is needed: the test function φ is arbitrary. $\square$

Nonlinear bounds. Let $G_0 \geq \|g^\circ\|_{\mathcal{L}}$ and $C_0 \geq \|c^\circ\|_{\mathcal{E}}$, let k_K, k_D, k_Y be as in Lemma 22.7, and put

$$\begin{aligned} B' &= \frac{1}{R} + 3 \left(\frac{2k_D}{R^2 - 1} + \frac{4R k_Y}{(R^2 - 1)^2} \right), \quad c_2 = \frac{\kappa_5}{b} \left[\frac{B'}{\sigma} + \frac{3\rho^2 k_K C_0^2}{R^3 \sigma} + \frac{3\rho^2 C_0 (1 + k_K G_0)}{R^3 \sigma^2} \right], \\ c_3 &= \frac{\kappa_5 \rho^2}{b R^3} \left[\frac{1 + k_K G_0}{\sigma^3} + \frac{3k_K C_0}{\sigma^2} \right], \quad c_4 = \frac{\kappa_5 \rho^2 k_K}{b R^3 \sigma^3}. \end{aligned}$$

Then the two estimates of Lemma 21.13 hold for (22.5). The proof is that of Lemma 21.13, with $L_2 = \kappa_5 B'$, $L_3 = \kappa_5 \rho^2 R^{-3}$ and $L_4 = k_K \kappa_5 \rho^2 R^{-3}$ (Theorem 22.6 and Lemma 22.7; each multiplier is converted from Chebyshev to solid form once). Proposition 21.14 applies verbatim to $\mathcal{F}: \mathcal{X} \rightarrow \mathcal{Y}$.

Finite and tail coordinates. The *finite inputs* are the 4617 field coordinates (ℓ, s) with $\ell \leq M$ even and $1 \leq s \leq S$, and the 81 shape coordinates c_j with $j \leq M + 1$; the *finite rows* are the same 4617 field rows and the 81 harmonic rows $\mathcal{T}_n$, $n \leq M$ even. All other inputs and rows are tail inputs and rows, and tail rows are harmonic or nonharmonic, as in Section 21.6. On the shape tail and on the harmonic rows we use the coordinates $\hat{a}_N = (N + 1)R^{N+1}c_{N+1}$ and $\hat{y}_n = R^n y_n$ (N, n even), in which the corresponding parts of the norms are $\sigma \sum_N \Upsilon(N) |\hat{a}_N|$ and $\sum_n \Upsilon(n) |\hat{y}_n|$.

The principal part on high shape modes. Let $q^\circ = q_{c^\circ}$ and $p^\circ = p_{c^\circ} = \sum_a p_a w^a$, $p_a = (a+1)c_{a+1}^\circ$. For $j = N + 1$ the derivative of $b^{-1} \rho^2 q_c |p_c|^2$ at c° in the direction e_j contains the principal polynomial $\Pi_j = \frac{2(N+1)\rho^2}{b} q^\circ \operatorname{Re}(\overline{p^\circ} w^N)$. On ∂B , $q^\circ = \sum_k q_k e^{ik\theta}$ with $q_k = \sum_{i>|k|} c_i^\circ$ (k even, i odd), and

$$\Pi_j|_{\partial B} = \frac{2(N+1)\rho^2}{b} \sum_{\delta} s_{\delta} T_{|N+2\delta|}(\xi), \quad s_{\delta} = \sum_{k-a=2\delta} p_a q_k \quad (-160 \leq \delta \leq 80).$$

Let $Me_j \in \mathcal{Y}$ be the harmonic element with these boundary values, and define ε_{Υ} from (s_{δ}) as in Theorem 22.8(b). As in Lemma 21.15, for $N \geq M + 2$ the part of $\mathbf{M}$ from the shape tail to the harmonic tail rows is, in the coordinates $\hat{a}, \hat{y}$,

$$T = \frac{2\rho^2 s_0}{bR} (I + E),$$

with E as in Theorem 22.8(b) for $N_0 = M + 2$, and the part $\mathcal{B}$ from the shape tail to the finite harmonic rows collects the rows $|N + 2\delta| \leq M$. If $\varepsilon_{\Upsilon} < 1$, then T is invertible on the graded tail, with $\|T^{-1}\| \leq \tau_T := bR/(2\rho^2 s_0(1 - \varepsilon_{\Upsilon}))$, and T^{-1} is evaluated by the Neumann series with the remainder bound of Theorem 22.8(b).

The operator A. Let $\tilde{J}$ be the finite Jacobian $D\mathcal{F}(x^\circ)$, restricted to finite inputs and rows, in scaled coordinates (inputs multiplied by their $\mathcal{X}$ -weights and rows by their $\mathcal{Y}$ -weights, so that both norms become plain ℓ^1 norms). It is computed at the exact centre with interval entries. Let $\hat{A}$ be a binary64 approximation of the inverse of its midpoint matrix, regarded as an exact dyadic matrix, and define, for $y = y_f + y_t^H + y_t^N \in \mathcal{Y}$,

$$(22.6) \quad Ay = \hat{A}(y_f - \mathcal{B}T^{-1}y_t^H) + T^{-1}y_t^H + y_t^N,$$

as in (21.15). For a finite row ι let α_ι be an upper bound for the normalised norm of $\hat{A}e_\iota$, let $\alpha_n^H = \alpha_{\tau_n}$ for $n \leq M$, and put

$$\rho_{\mathcal{B}} = \sup_{N > M} \frac{1}{\Upsilon(N)} \sum_{n \leq M} \alpha_n^H \Upsilon(n) |\hat{\mathcal{B}}_{nN}|, \quad A_t = \tau_T(\sigma + \rho_{\mathcal{B}}),$$

where $\hat{\mathcal{B}}$ is the matrix of $\mathcal{B}$ in the coordinates $\hat{a}, \hat{y}$; we set $\alpha_n^H = A_t$ for $n > M$.

Lemma 22.10. *Suppose that $\varepsilon_\Upsilon < 1$ and that the column sums of $|I - \hat{A}\tilde{J}|$ are less than one. Then $A: \mathcal{Y} \rightarrow \mathcal{X}$ is a bounded bijection with $\|A\| \leq \max\{\max_\iota \alpha_\iota, A_t, 1\}$. It maps every normalised harmonic tail row to a vector of norm at most A_t and every nonharmonic tail row to the identical field tail input; $Ame_j = e_j$ for every odd $j \geq M + 3$, and $A\Psi_{\ell,s} = e_{(\ell,s)}$ for every field tail input (ℓ, s) .*

Proof. As for Lemma 21.16. For a normalised harmonic tail row $\hat{y} = e_m$, the shape part $T^{-1}e_m$ has norm at most $\sigma\tau_T\Upsilon(m)$, and $\hat{A}\mathcal{B}T^{-1}e_m$ has norm at most $\rho_{\mathcal{B}}\tau_T\Upsilon(m)$. $\square$

22.4. The defect bounds. We bound $Y = \|A\mathcal{F}(x^\circ)\|_{\mathcal{X}}$ and $Z = \|I - A\mathcal{F}(x^\circ)\|_{\mathcal{X} \rightarrow \mathcal{X}}$ by normalised column norms (Lemma 15.2). The input columns are partitioned into disjoint classes that exhaust all inputs; the class parameters are frozen in the file `params.json` and rechecked by the verifier.

- Field inputs: the 4617 finite inputs; for even $\ell \leq 160$, the radial class $s \geq S_\ell$, with $S_\ell = 59$ for $\ell \leq 10$ and $S_\ell = 58$ for $12 \leq \ell \leq 160$, and the six exceptional inputs $(\ell, 58)$, $\ell \leq 10$; for even $162 \leq \ell \leq 254$, the angular gap class $s \geq S_\ell$, with $S_\ell = 3$ for $\ell \leq 178$, $S_\ell = 2$ for $180 \leq \ell \leq 248$ and $S_\ell = 1$ for $250 \leq \ell \leq 254$, and the 53 exceptional inputs (ℓ, s) with $s < S_\ell$; and the uniform angular class $\ell \geq 256$, $s \geq 1$.
- Shape inputs: the 81 finite inputs $j \leq 161$, the 219 explicit inputs $163 \leq j \leq 599$, and the far class $j \geq 601$.

The residual and the explicit columns. The residual $\mathcal{F}(x^\circ)$, which has 70299 nonzero coefficients, and the columns $D\mathcal{F}(x^\circ)e$ of all 4698 finite inputs, of the 59 exceptional field inputs and of the 219 explicit shape inputs are polynomials. They are expanded with their complete support in ball arithmetic at the final centre. For the field columns, the products of the four multipliers $q^\circ, \partial_{x_1}q^\circ, y^{-1}\partial_yq^\circ, q^\circ|p^\circ|^2$ with $\Psi_{\ell,s}$ are obtained for all s from the exact three-term expression of $t\Psi_{\ell,s}$ in $\Psi_{\ell,s-1}, \Psi_{\ell,s}, \Psi_{\ell,s+1}$ (Lemma 21.3(b)), at 1024 bits; the shape columns are obtained from exact recurrences in m for $r^mU_m, r^mC_m^{(2)}$ and $\text{Re}(\overline{p^\circ}w^m)$, at 1536 bits. Every column is recorded by enclosures of its finite rows and of all its harmonic tail coordinates, together with an upper bound for the norm of its nonharmonic tail part; a relative-width test rejects any loss of precision. Let $y = y_f + y_t^H + y_t^N$ be such a column and w_e the weight of the input e . By (22.6):

- for a finite input, $\|(I - A\mathcal{F}(x^\circ))e\|_{\mathcal{X}}/w_e \leq \delta_e + (A_t\|y_t^H\|_{\mathcal{Y}} + \|y_t^N\|_{\mathcal{Y}})/w_e$, where δ_e bounds the column sum of $|I - \hat{A}\tilde{J}|$ and is certified from the interval columns by a long-double matrix product with rigorous rounding-error bounds (unit roundoff 2^{-64});

- for an exceptional field input, whose image under A is the identical field tail input, the bound is $(\|\hat{A}y_f\| + A_t\|y_t^H\| + \|y_t^N - e\|)/w_e$;
- for an explicit shape input e_j , with a the Neumann approximation of $T^{-1}y_t^H$ and rem its remainder bound, the bound is $(\|\hat{A}(y_f - \mathcal{B}a)\| + \sigma\|a - e_j\| + \|y_t^N\| + (\sigma + \rho_{\mathcal{B}})\text{rem})/w_j$; the same expression without e_j and w_j , applied to $y = \mathcal{F}(x^\circ)$, bounds Y .

We denote the maxima over the three groups of columns by Z_{fin} , Z_{ex} and Z_{sh} .

Field tail classes. For a field tail input, $A\Psi_{\ell,s} = e_{(\ell,s)}$, so that $(I - AD\mathcal{F}(x^\circ))e_{(\ell,s)} = -Av$ with

$$v = \chi_1\Psi_{\ell,s} + 3\chi_2\partial_{x_1}K_5\Psi_{\ell,s} + 3\chi_3EK_5\Psi_{\ell,s} - 3\chi_4\partial_{x_1}K_5\Psi_{\ell,s} + \rho^2\chi_5K_5\Psi_{\ell,s},$$

where $\chi_1 = q^\circ/b - 1$, $\chi_2 = \partial_{x_1}q^\circ/b$, $\chi_3 = y^{-1}\partial_y q^\circ/b$, $\chi_4 = x_1\chi_3$ and $\chi_5 = q^\circ|p^\circ|^2/b$. For $1 \leq i \leq 5$ let $\tau_i(d)$ be the graded τ -norm of the homogeneous part of degree d of χ_i , multiplied by 3 for $i = 2, 3, 4$ and by ρ^2 for $i = 5$; it is computed exactly from the solid-monomial expansion. Define $\mathbf{A}(\ell, \varsigma)$, $\mathbf{B}(L)$ and $\mathfrak{B}_d(\ell, s)$ as before Lemma 21.18, now with the column norms α_i of this section, with ϑ_d as in Theorem 22.6(b), and with

$$\beta_v = \frac{1}{h_v\Upsilon(v)} \sum_n D_{nv} R^n \Upsilon(n) \alpha_n^H \quad (v \leq 1400),$$

$$\beta^\infty = A_t + \sum_{n \leq M} (\alpha_n^H - A_t)_+ \mathbf{a}_{k_n} \zeta^{k_n}, \quad k_n = \frac{1402 - n}{2},$$

in place of β_ℓ and β^∞ (and 1400 in place of 1200 in the definition of $\mathbf{B}$). Let $\mathfrak{B}^{(i)}(\ell, s) = \sum_d \tau_i(d) \mathfrak{B}_d(\ell, s)$, $\bar{d} = \ell + 2s + \frac{3}{2}$, and

$$\begin{aligned} \mathfrak{z}(\ell, s) = & \mathfrak{B}^{(1)}(\ell, s) + \frac{\Upsilon(1)}{2(\bar{d} + 1)} \sum_{i=2,4} \left[\lambda_\ell^+ \frac{h_{\ell+1}}{h_\ell} (\mathfrak{B}^{(i)}(\ell + 1, s) + \mathfrak{B}^{(i)}(\ell + 1, s - 1)) \right. \\ & \left. + \lambda_\ell^- \frac{h_{\ell-1}}{h_\ell} (\mathfrak{B}^{(i)}(\ell - 1, s + 1) + \mathfrak{B}^{(i)}(\ell - 1, s)) \right] \\ & + \frac{(\bar{d} + \frac{3}{2})\mathfrak{B}^{(3)}(\ell, s - 1)}{4\bar{d}(\bar{d} + 1)} + \frac{3\mathfrak{B}^{(3)}(\ell, s)}{4\bar{d}(\bar{d} + 2)} + \frac{\Upsilon(2)(\bar{d} + \frac{1}{2})\mathfrak{B}^{(3)}(\ell, s + 1)}{4(\bar{d} + 1)(\bar{d} + 2)} \\ & + \frac{\mathfrak{B}^{(5)}(\ell, s - 1)}{4\bar{d}(\bar{d} + 1)} + \frac{\mathfrak{B}^{(5)}(\ell, s)}{2\bar{d}(\bar{d} + 2)} + \frac{\Upsilon(2)\mathfrak{B}^{(5)}(\ell, s + 1)}{4(\bar{d} + 1)(\bar{d} + 2)}, \end{aligned}$$

where terms with $\ell - 1 < 0$ are omitted. Let $\mathfrak{z}_u(\ell, s)$ be the same expression with the bracket replaced by $R \max\{\mathfrak{B}^{(i)}(\ell + 1, s) + \mathfrak{B}^{(i)}(\ell + 1, s - 1), \mathfrak{B}^{(i)}(\ell - 1, s + 1) + \mathfrak{B}^{(i)}(\ell - 1, s)\}$.

Proposition 22.11. (a) *For every field tail input, $\|(I - AD\mathcal{F}(x^\circ))e_{(\ell,s)}\|_{\mathcal{X}}/(h_\ell\Upsilon(\ell + 2s)) \leq \mathfrak{z}(\ell, s) \leq \mathfrak{z}_u(\ell, s)$.*
 (b) *For fixed ℓ , $\mathfrak{z}(\ell, s)$ is nonincreasing in s , and $\mathfrak{z}_u(\ell, s)$ is nonincreasing in ℓ and in s . Hence the radial class and the angular gap class of ℓ are bounded by $\mathfrak{z}(\ell, S_\ell)$, and the uniform angular class by $\mathfrak{z}_u(256, 1)$.*

Proof. (a) As in Proposition 21.19(a): expand v with Lemma 22.7 and bound each product of a homogeneous part of χ_i with a Ψ by Lemma 21.18, which holds in the graded norms with $h_a h_\ell$ replaced by $h_a \Upsilon(a + 2k) h_\ell \Upsilon(\ell + 2s)$, by Theorems 22.6(b) and 22.8(c). In its proof, Lemma 21.17 is used; the bound for $v > 1400$ now follows from $\sum_n D_{nv} R^n \Upsilon(n) \leq h_v \Upsilon(v)$, Lemma 22.3(c) and the monotonicity of $\mathbf{a}_k \zeta^k$ for $k \geq 4$. The factors $\Upsilon(1)$ and $\Upsilon(2)$ bound the ratios of the weights of the outputs of the stencils to $\Upsilon(\ell + 2s)$. The second inequality follows from $\lambda_\ell^+ h_{\ell+1} + \lambda_\ell^- h_{\ell-1} \leq R h_\ell$ (Lemma 22.3(c)).

(b) Each $\mathfrak{B}_d$ is nonincreasing in both arguments: A , B and ϑ_d are nonincreasing (the step functions are suffix maxima), the first bound in $\mathfrak{B}_d$ is nondecreasing in A , B and ϑ_d , and the second is an expectation of a function that is nonincreasing in both arguments. The stencil coefficients decrease in $\bar{d}$, and in $\mathfrak{z}_u$ no other dependence on ℓ remains. $\square$

We write $\max_\ell Z_{\text{rad}}(\ell)$ and $\max_\ell Z_{\text{gap}}(\ell)$ for the maxima of $\mathfrak{z}(\ell, S_\ell)$ over the radial and angular gap classes, and $Z_{\text{ang}} = \mathfrak{z}_u(256, 1)$.

Far shape columns. Let $j = N + 1$ with $N \geq N_* = 600$ even. Let $W^\circ = K_5 g^\circ$, $W_1^\circ = \partial_{x_1} W^\circ$, $Z^\circ = y^{-1} \partial_y W^\circ$ (a polynomial, since W° is a polynomial in x_1 and y^2 ; $y \partial_y W^\circ = t(1 - \xi^2) Z^\circ$), and $H^\circ = g^\circ + \rho^2 |p^\circ|^2 (1 + W^\circ)$. With $\delta q = r^N U_N$, $\delta \partial_{x_1} q = (N + 1) r^{N-1} U_{N-1}$, $\delta(y^{-1} \partial_y q) = -2r^{N-2} C_{N-2}^{(2)}$ and $\delta p = (N + 1) w^N$, Lemma 21.21(b) gives

(22.7)

$$\begin{aligned} b D\mathcal{F}(x^\circ) e_j &= r^N U_N H^\circ + 3(N + 1) r^{N-1} U_{N-1} W_1^\circ + 3 \left(N r^N T_N - \frac{r^N U_N + t r^{N-2} U_{N-2}}{2} \right) Z^\circ \\ &\quad + 2(N + 1) \rho^2 q^\circ \operatorname{Re}(\bar{p}^\circ w^N) W^\circ + b \Pi_j. \end{aligned}$$

Since $A M e_j = e_j$, $(I - A D\mathcal{F}(x^\circ)) e_j = -A(D\mathcal{F}(x^\circ) e_j - M e_j)$. From $\operatorname{Re}(\bar{w}^a w^N) = r^{N+a} T_{N-a}(\xi)$, $r^{i-1} U_{i-1} = \sum_k (2 - \delta_{2k, i-1}) t^k r^{i-1-2k} T_{i-1-2k}$ and $T_K T_n = \frac{1}{2} (T_{K+n} + T_{K-n})$ we obtain

$$q^\circ \operatorname{Re}(\bar{p}^\circ w^N) = \sum_{(\eta, e)} \gamma_{\eta, e} t^e r^{N+\eta} T_{N+\eta}(\xi),$$

a finite sum over $-320 \leq \eta \leq 160$ and $e \geq 0$ with coefficients independent of N ; let $\gamma_{\eta, e} \geq 0$ be the sum of the moduli of the individual contributions to the coefficient of $t^e r^{N+\eta} T_{N+\eta}$. The element $M e_j$ is obtained from Π_j by replacing each $t^e r^K T_K$ with $\mathcal{T}_K$.

Column profiles of A . For $v \geq 0$ let $\Gamma_F(v) \geq 1$ and $\Gamma_H(v) \geq A_t$ be nonincreasing bounds, computed from the α_t , for the normalised norms $\|A \Psi_{v', \varsigma}\|_{\mathcal{X}} / (h_{v'} \Upsilon(v' + 2\varsigma))$ ($\varsigma \geq 1$) and for $\beta_{v'}$, over all $v' \geq v$; let $\Gamma_* = \max(\Gamma_F, \Gamma_H)$, $\Gamma_{\max} = \Gamma_*(0)$, $\alpha^H(K) = \alpha_K^H$, and let $\mathcal{H}_v = \sum_n D_{nv} \mathcal{T}_n$ be the harmonic element with boundary values Y_v , so that $\|A \mathcal{H}_v\| \leq \Gamma_H(v) h_v \Upsilon(v)$. Let $I = 140$, $S_p(n) = \sum_{k \geq 0} \zeta^k (n + k)^p$ and, for $p \in \{0, 1, 2\}$ and $i_1 = \lfloor (K - M)/2 \rfloor + 1$,

$$G_p(K) = \begin{cases} \mathbf{w}_{I+1} S_p(I + 1) \Gamma_{\max}, & i_1 \leq I + 1, \\ \mathbf{w}_{I+1} (S_p(I + 1) \Gamma_*(M + 1) + \zeta^{i_1 - I - 1} S_p(i_1) \Gamma_{\max}), & i_1 > I + 1. \end{cases}$$

Since e_i is nonincreasing for $i \geq 1$, $\mathbf{w}_i \leq \mathbf{w}_{I+1} \zeta^{i-I-1}$ for $i > I$, and $\sum_{I < i \leq K/2} \mathbf{w}_i i^p \Gamma_*(K - 2i) \leq G_p(K)$, because $K - 2i \geq M$ for $i < i_1$ and $\Gamma_*(M) = \Gamma_*(M + 1)$ for the certified profiles.

Lemma 22.12. For $K \geq 2$, $\sum_i i (K - i + \frac{3}{2}) C_{K-2i, K} Y_{K-2i} = -\frac{3K}{4} U_{K-2}$.

Proof. In $\mathbb{R}^5$, $\Delta(t^i H_v) = 4i(i + v + \frac{3}{2}) t^{i-1} H_v$. The function $r^K T_K(\xi) = \operatorname{Re} w^K$ is harmonic in the meridian variables, so $\Delta \operatorname{Re} w^K = \frac{3}{y} \partial_y \operatorname{Re} w^K = -\frac{3K}{y} \operatorname{Im} w^{K-1} = -3K r^{K-2} U_{K-2}(\xi)$. Apply Δ to $r^K T_K = \sum_i C_{K-2i, K} H_{K-2i} t^i$ and restrict to $t = 1$. $\square$

Lemma 22.13. Let $K \geq 0$, $e \geq 0$, $m \geq 0$, $\beta = K + \frac{3}{2}$, $c_m = \max(2, m + 1)$, $F_m(\beta') = m! / (\beta' / 2 + 1)_m$, and

$$f(x) = \frac{\beta - 2x + 1}{\prod_{k=1}^{m+1} (\beta - x + e + k)}, \quad \varrho_2(i) = f(i) - f(0) - f'(0) i.$$

Let $u = t^e r^K T_K(\xi)(1-t)^m$ if $m \geq 1$, and $u = t^e r^K T_K(\xi) - T_K$ if $m = 0$ and $K > M$. Then

$$\begin{aligned} \frac{\|Au\|_{\mathcal{X}}}{R^K \Upsilon(K+2e+2m)} &\leq \sum_i \mathbf{w}_i \mathbf{g}_i \Gamma_F(K-2i) \\ &\quad + m! \left(\mathbf{f}_0 \alpha^H(K) + \frac{3}{4} |\mathbf{f}'(0)| \mathbf{u}(K) + \sum_{i \geq 1} \mathbf{w}_i \left(\frac{|\mathbf{f}'(0)| i^2}{\beta} + |\varrho_2(i)| \right) \Gamma_H(K-2i) \right), \end{aligned}$$

where $\mathbf{g}_i = \mathbf{F}_m(\beta-2i)$ and $\mathbf{f}_0 = \mathbf{f}(0)$ if $m \geq 1$, $\mathbf{g}_i = (e+i)/(\beta-i+e+1)$ and $\mathbf{f}_0 = 1 - \mathbf{f}(0) = e/(\beta+e+1)$ if $m = 0$, and $\mathbf{u}(K) = 2A_t \zeta/(1-\zeta) + 2 \sum_{n \leq \min(M, K-2), n \equiv K} R^{n-K} (\alpha_n^H - A_t)_+$. Moreover, for $i \leq K/2$,

$$|\mathbf{f}'(0)| \leq \frac{c_m \mathbf{f}(0)}{\beta+1}, \quad |\varrho_2(i)| \leq \frac{i^2}{2} \cdot \frac{(\beta+1)(c_m^2 + m + 5)}{(\beta-2i+1)^2 \prod_{k=1}^{m+1} (\beta-i+e+k)}.$$

Proof. Let $v_i = K - 2i$. By definition of $C_{\ell K}$, $u = \sum_i C_{v_i, K} H_{v_i} t^{e+i} (1-t)^m$, minus $T_K = \sum_i C_{v_i, K} H_{v_i}$ if $m = 0$, and all outputs have degree at most $K + 2e + 2m$; so it suffices to bound the norms with $\Upsilon \equiv 1$ (Theorem 22.8(a)). *Field layer.* By Lemma 21.3(c),(d), the $J^{\beta_{v_i}}$ -coefficients of $t^{e+i}(1-t)^m$ have ℓ^1 norm at most $\mathbf{F}_m(\beta_{v_i}) = \mathbf{F}_m(\beta-2i)$, and those of $t^{e+i} - 1$ with $s \geq 1$ have sum $(e+i)/(\beta_{v_i} + e + i + 1)$; with $|C_{v_i, K}| h_{v_i} \leq \mathbf{w}_i R^K$ this gives the first sum. *Harmonic layer.* By Lemma 21.3(a), the coefficient of $J_0^{\beta_v}$ in $t^\mu (1-t)^m$ is $(\beta_v + 1) \int_0^1 t^{\beta_v + \mu} (1-t)^m dt = m! (\beta_v + 1) / \prod_{k=1}^{m+1} (\beta_v + \mu + k)$, which for $v = v_i$ and $\mu = e + i$ equals $m! \mathbf{f}(i)$. Hence the boundary trace of the harmonic layer of u is $m! \sum_i C_{v_i, K} \mathbf{f}(i) Y_{v_i}$, minus T_K if $m = 0$. By $\sum_i C_{v_i, K} Y_{v_i} = T_K$ and Lemma 22.12, $\sum_i i C_{v_i, K} Y_{v_i} = \beta^{-1} (-\frac{3K}{4} U_{K-2} + \sum_i i^2 C_{v_i, K} Y_{v_i})$, and the trace equals

$$m! \left(\mathbf{f}(0) T_K - \frac{3K \mathbf{f}'(0)}{4\beta} U_{K-2} + \sum_i C_{v_i, K} \left(\frac{\mathbf{f}'(0) i^2}{\beta} + \varrho_2(i) \right) Y_{v_i} \right),$$

with $\mathbf{f}(0)-1$ in place of $\mathbf{f}(0)$ if $m = 0$. We apply A to the three parts: $\|A T_K\| \leq \alpha^H(K) R^K \Upsilon(K)$; the Chebyshev coefficients of U_{K-2} are 2 (and 1 for T_0), which gives $\mathbf{u}(K)$, since $K < \beta$; and Y_v is the trace of $\mathcal{H}_v$. (Thus the harmonic layer is, up to terms of relative size $O(i^2/\beta)$, a single Chebyshev mode; the triangle inequality applied directly to $\sum_i C_{v_i, K} \mathbf{f}(i) Y_{v_i}$ would lose a factor close to κ_5 .) For the bounds on $\mathbf{f}$ write $\mathbf{f} = g/P$ with $g(x) = \beta - 2x + 1$ and $P(x) = \prod_k (\beta - x + e + k)$, and let $S_1 = \sum_k (\beta - x + e + k)^{-1}$, $S_2 = \sum_k (\beta - x + e + k)^{-2}$. Then $\mathbf{f}' = \mathbf{f}(S_1 - 2/g)$ and $\mathbf{f}'' = P^{-1}(g(S_1^2 + S_2) - 4S_1)$. At $x = 0$, $0 < S_1 \leq (m+1)/(\beta+1)$ and $2/g = 2/(\beta+1)$, which gives the bound for $\mathbf{f}'(0)$. For $0 \leq x \leq i \leq K/2$ the two terms of $\mathbf{f}''$ have opposite signs, so $|\mathbf{f}''(x)| \leq P(x)^{-1} \max\{g(S_1^2 + S_2), 4S_1\}$; with $P(x) \geq P(i)$, $g \leq \beta + 1$, $S_1 \leq (m+1)/(\beta-i+1)$, $S_2 \leq (m+1)/(\beta-i+1)^2$ and $(\beta-2i+1)^2 \leq (\beta+1)(\beta-i+1)$ this is at most $(\beta+1) \max\{(m+1)(m+2), 4(m+1)\}/((\beta-2i+1)^2 P(i))$, and $\max\{(m+1)(m+2), 4(m+1)\} \leq c_m^2 + m + 5$. Taylor's formula gives the bound for ϱ_2 . $\square$

Let $\Phi_m(K)$ ($m \geq 1$) and $\Phi_P(K, e)$ ($m = 0$) be the right-hand side of Lemma 22.13, evaluated as follows. For $m \geq 1$ the bounds are made independent of e by putting $e = 0$ in $\mathbf{f}(0)$ and in the bound for ϱ_2 . The sums over i are evaluated exactly for $i \leq 24$; for $24 < i \leq \min(I, K/2)$ the factors $\mathbf{g}_i$ and $|\varrho_2(i)|/i^2$, which increase with i , are replaced by their values at the largest index; and the terms with $i > I$ are bounded by means of G_0, G_1, G_2 , using $\mathbf{g}_i \leq 1$, $|\mathbf{f}(i)| \leq \mathbf{f}_{\max}$ and $|\varrho_2(i)| \leq 2\mathbf{f}_{\max} + |\mathbf{f}'(0)|i$, where $\mathbf{f}_{\max} = (\beta+1)/(K/2 + \frac{5}{2})^{m+1}$ for $m \geq 1$ and $\mathbf{f}_{\max} = 1$ for

$m = 0$. Let $\varphi_m(K) = \sup_{K' \geq K} \Phi_m(K')$, and

$$\bar{\Gamma}(K) = \sup_{K' \geq K} \sum_{i \leq I} w_i \Gamma_*(\max(K' - 2i, 0)) + w_{I+1} S_0(I+1) \Gamma_{\max}.$$

For f one of the polynomials $H^\circ, W_1^\circ, Z^\circ, W^\circ$, write $f = \sum_a H_a f_a(t)$, $f_a = \sum_s f_{a,s} J_s^{\beta_a} = \sum_m f_{a,m}^\natural (1-t)^m$ and $\|f_a\|_J = \sum_s |f_{a,s}|$. By Theorem 22.4, every f_a has a double zero at $t = 1$ for $f = W^\circ$ and a zero for $f = W_1^\circ, Z^\circ$, so the corresponding $f_{a,m}^\natural$ vanish exactly. For $K' \geq 0$ let

$$\text{HH}_a(K') = \sup_{K'' \geq K'} \sum_{i \leq \min(100, K''/2)} w_i \Gamma_*(K'' - 2i) \sum_{m \geq 3} |f_{a,m}^\natural| F_m(\beta_{K''-2i}),$$

let $\text{JT}(K')$ be the supremum over $K'' \geq K'$ of $\sum_{100 < i \leq \min(I, K''/2)} w_i \Gamma_*(K'' - 2i)$ plus $G_0(K'')$ if $K'' > 2I$, and put, for $K \geq a$,

$$\begin{aligned} \text{Mix}_a(K) &= |f_{a,0}^\natural| \bar{\Gamma}(K - a) + \sum_{m=1}^3 |f_{a,m}^\natural| \varphi_m(K - a) + \text{HH}_a(K - a) \\ &\quad + \left(\|f_a\|_J + \sum_{m \leq 3} |f_{a,m}^\natural| F_m(\beta_a) \right) \text{JT}(K - a), \end{aligned}$$

$\text{Jac}_a(K) = \|f_a\|_J \bar{\Gamma}(\max(K - a, 0))$, and finally

$$V_f(K) = \sup_{K' \geq K} \sum_a h_a \min\{\text{Jac}_a(K'), \text{Mix}_a(K')\},$$

where $\text{Mix}_a(K')$ is omitted for $K' < a$. All these tables are nonincreasing by construction.

Lemma 22.14. For $f \in \{H^\circ, W_1^\circ, Z^\circ, W^\circ\}$, $K \geq 0$ and $e \geq 0$, $\|A(t^e r^K T_K(\xi)f)\|_{\mathcal{X}} \leq R^K \Upsilon(K + 2e + \deg f) V_f(K)$.

Proof. It suffices to treat one angular component $H_a f_a$, with $\Upsilon \equiv 1$, and to show that its contribution is at most $R^K h_a \text{Jac}_a(K)$ and, for $K \geq a$, at most $R^K h_a \text{Mix}_a(K)$. *Jacobi route.* Expand $r^K T_K = \sum_i C_{K-2i, K} H_{K-2i} t^i$ and multiply by $H_a f_a$ with Theorem 22.6(a),(b): the outputs have angular index at least $K - 2i - a$, the lowering steps of the Jacobi parameter that are needed come with enough factors t (as in the proof of Lemma 21.8(a)), all steps are stochastic, and $\sum_v G_{v'av} h_v \leq h_v h_a$; with Lemma 22.3(b) this gives Jac_a . *Mixed route,* for $K \geq a$. By $H_a = \sum_n D_{na} t^{(a-n)/2} r^n T_n$ and $T_K T_n = \frac{1}{2}(T_{K+n} + T_{K-n})$, $t^e r^K T_K H_a$ is a nonnegative combination of raw monomials $t^{e'} r^{K'} T_{K'}$ with $K' \geq K - a$ and R -weighted mass at most $h_a R^K$. Split $f_a = P_a + (f_a - P_a)$ with $P_a = \sum_{m \leq 3} f_{a,m}^\natural (1-t)^m$. The part P_a is bounded by $\bar{\Gamma}$ for $m = 0$ and by Lemma 22.13 for $1 \leq m \leq 3$. For $f_a - P_a$, in the connection expansion of $t^{e'} r^{K'} T_{K'}$ the components with index $i \leq 100$ are bounded with Lemma 21.3(c),(d) term by term in $m > 3$, which gives HH_a , and those with $i > 100$ by the Jacobi norm, which gives JT : here

$$\|t^\mu (f_a - P_a)\|_{J^{\beta_v}, 1} \leq \|f_a - P_a\|_{J^{\beta_a}, 1} \leq \|f_a\|_J + \sum_{m \leq 3} |f_{a,m}^\natural| F_m(\beta_a),$$

because every lowering of the Jacobi parameter from β_a to $\beta_v < \beta_a$ comes with a factor t (one checks $\mu \geq a - v$ whenever $K \geq a$), and the raising and lowering stencils of Lemma 21.3(b) are nonnegative with column sums one. Finally, $\bar{\Gamma}$, φ_m , HH_a and JT are nonincreasing and $K' \geq K - a$. $\square$

Proposition 22.15. *Let $\mathbf{p} = \Upsilon(1000)/(b\sigma R)$, $N_2 = 1000$, $k_0 = N_2/2$, and for even $N \geq N_*$*

$$\begin{aligned} \mathfrak{F}(N) = & \mathbf{p} \left[\frac{1}{N+1} \sum_{k=0}^{N/2} (2 - \delta_{2k,N}) \zeta^k V_{H^\circ}(N-2k) + \frac{6}{R} \sum_{k=0}^{N/2-1} \zeta^k V_{W_1^\circ}(N-1-2k) \right. \\ & + 3V_{Z^\circ}(N) + \frac{3}{2(N+1)} \left(\sum_{k=0}^{N/2} (2 - \delta_{2k,N}) \zeta^k V_{Z^\circ}(N-2k) \right. \\ & \quad \left. \left. + \sum_{k=0}^{N/2-1} (2 - \delta_{2k,N-2}) \zeta^{k+1} V_{Z^\circ}(N-2-2k) \right) \right. \\ & \left. + 2\rho^2 \sum_{(\eta,e)} \gamma_{\eta,e} R^\eta (\Phi_P(N_* + \eta, e) + V_{W^\circ}(N_* + \eta)) \right], \\ \mathfrak{F}_\infty = & \mathfrak{F}(N_2) + \mathbf{p} \frac{\zeta^{k_0}}{1-\zeta} \left(\frac{2V_{H^\circ}(0)}{N_2+1} + \frac{6V_{W_1^\circ}(0)}{R} + \frac{6V_{Z^\circ}(0)}{N_2+1} \right). \end{aligned}$$

Then for every odd $j \geq N_* + 1$,

$$\frac{\|(I - \text{ADF}(x^\circ))e_j\|_{\mathcal{X}}}{\sigma j R^j \Upsilon(j-1)} \leq Z_{\text{far}} := \max(\{\mathfrak{F}(N) : N_* \leq N \leq N_2 \text{ even}\} \cup \{\mathfrak{F}_\infty\}) + 10^{-17}.$$

Proof. Let $j = N + 1$. We divide the five parts of (22.7), with Me_j subtracted from the last, by b and by the input weight $\sigma(N+1)R^{N+1}\Upsilon(N)$. All polynomials involved have degree at most $N + 596$ (the degree of H° is at most 596), so the factor $\Upsilon(1000)$ in $\mathbf{p}$ covers the grading (Theorem 22.8(c)). By $r^N U_N = \sum_k (2 - \delta_{2k,N}) t^k r^{N-2k} T_{N-2k}$ and Lemma 22.14, the first part contributes the first sum of $\mathfrak{F}(N)$; the second, where all Chebyshev coefficients of U_{N-1} equal two, contributes the second sum; for the third, $3N/(N+1) \leq 3$ and $r^N U_{N-2} = t r^{N-2} U_{N-2}$. For the last two parts, Lemmas 22.13 and 22.14 give the bound with $\Phi_P(N+\eta, e) + V_{W^\circ}(N+\eta)$ in place of the values at $N_* + \eta$.

Monotonicity in N . The tables V_f are nonincreasing. For $K \geq M + 2I + 2 = 442$ the function $K \mapsto \Phi_P(K, e)$ is nonincreasing: the index ranges $i \leq 24$, $24 < i \leq I$, $i > I$ no longer depend on K ; Γ_F and Γ_H are nonincreasing; $\alpha^H(K) = A_t$; the terms of $\mathbf{u}(K)$ are multiplied by ζ when K increases by two; the factors $\mathbf{g}_i = (e+i)/(\beta-i+e+1)$, $\mathbf{f}_0 = e/(\beta+e+1)$, the bound $2/(\beta+e+1)$ for $|\mathbf{f}'(0)|$, $1/\beta$ and the bound for $|\varrho_2(i)|/i^2$ decrease in $\beta = K + \frac{3}{2}$; and $G_p(K)$ is nonincreasing, because i_1 increases with K and $S_p(i) - \zeta S_p(i+1) = i^p \geq 0$. Since $N_* + \eta \geq 442$ for $\eta \geq -158$, these terms of $\mathfrak{F}(N)$ bound the corresponding terms at every $N \geq N_*$. For the monomials with $\eta \leq -160$ and $N > N_*$ we use instead $\|A(t^e r^K T_K - \mathcal{T}_K)\|_{\mathcal{X}} \leq \|A\| \|t^e r^K T_K - \mathcal{T}_K\|_{\mathcal{Y}} \leq 2\kappa_5 \|A\| R^K \Upsilon(K+2e)$, which follows from $t^e r^K T_K - \mathcal{T}_K = \sum_i C_{v_i, K} H_{v_i}(t^{e+i} - 1)$ and $\|H_v(t^\mu - 1)\|_{\mathcal{Y}} \leq 2h_v \Upsilon(v+2\mu)$; the exact total weight $\mathbf{p} 2\rho^2 \sum_{\eta \leq -160} \gamma_{\eta,e} R^\eta$ of these monomials is below $1.2 \cdot 10^{-25}$, so their contribution is below 10^{-17} for every N .

The k -sums. For $N_* \leq N \leq N_2$ the sums are evaluated exactly. Let $N > N_2$. In the first sum and in the first Z -sum the terms with $k < k_0$ have the same weight 2 as at N_2 ; at $k = k_0$ the weight is 2 instead of 1; in the second sum and in the second Z -sum, which at N_2 end at $k = k_0 - 1$, the terms with $k \geq k_0$ are new, and in the second Z -sum the weight at $k = k_0 - 1$ is 2 instead of 1. All values $V_f(N-2k)$ are at most $V_f(N_2-2k)$ for $k \leq k_0$ and at most $V_f(0)$ in general, and $1/(N+1) \leq 1/(N_2+1)$. Hence the excess over the sums evaluated at N_2 is at most $\zeta^{k_0} V_f(0) + 2 \sum_{k > k_0} \zeta^k V_f(0) \leq 2\zeta^{k_0} V_f(0)/(1-\zeta)$ for the first sum and for the first Z -sum, and at most $2\zeta^{k_0} V_f(0)/(1-\zeta)$ for the second sum and for the second Z -sum (where the powers are ζ^{k+1} with $k \geq k_0 - 1$). With the prefactors $1/(N_2+1)$, $3/R$ and $3/(2(N_2+1))$ this gives

$\mathfrak{F}_\infty$. Thus every $N > N_2$ is covered by $\mathfrak{F}_\infty + 10^{-17}$. (The bound for Z_{far} in Table 16 includes the term 10^{-17} .) $\square$

Collecting all classes,

$$Z = \max \{ Z_{\text{fin}}, Z_{\text{ex}}, \max_{\ell} Z_{\text{rad}}(\ell), \max_{\ell} Z_{\text{gap}}(\ell), Z_{\text{ang}}, Z_{\text{sh}}, Z_{\text{far}} \}$$

bounds $\|I - \text{ADF}(x^\circ)\|_{\mathcal{X} \rightarrow \mathcal{X}}$.

22.5. Certified bounds, geometry and reconstruction. The bounds in Table 16 are outward roundings of the interval enclosures in the receipt of the verifier (Section 34.9). The residual uses the entire polynomial support and the fixed dyadic b , and all finite columns are computed at the final centre, without transfer from another matrix. The symbol defect is $\varepsilon_\Upsilon < 0.025209$, the finite column sums δ_e are bounded by $Z_{\text{fin}} < 1$, and $\rho_{\mathcal{B}} < 2896.07$.

Quantity	Certified bound
$\ A\ , A_t$	$< 104568.72, < 16.60796$
$Y = \ \text{AF}(x^\circ)\ _{\mathcal{X}}$	$< 3.19608 \cdot 10^{-12}$
Z_{fin} (4698 columns), Z_{ex} (59 columns)	$< 0.0585324, < 0.496612$
field classes: 81 radial, 47 angular gap, Z_{ang} ($\ell \geq 256$)	all < 0.887235 ; the maximum is Z_{ang}
Z_{sh} ($163 \leq j \leq 599$), Z_{far} ($j \geq 601$)	$< 0.753050, < 0.605269$
$Z = \ I - \text{ADF}(x^\circ)\ _{\mathcal{X} \rightarrow \mathcal{X}}$	< 0.887235
G_0, C_0	$< 107535.85, < 1.079742$
c_2, c_3, c_4	$< 1.540189, < 6.608341 \cdot 10^{-6}, < 1.582007 \cdot 10^{-11}$
$r = 10^{-10}$: radii polynomial	$< -8.0788 \cdot 10^{-12}$
$r = 10^{-10}$: contraction bound	< 0.887267

TABLE 16. The five-dimensional certificate for $M = 160$, $S = 57$, $R = 21/20$, $\Upsilon(D) = 1 + D/10^6$ and $\sigma = 30\rho^2$. The final Z is the maximum over every field and shape class; the full interval values are in the receipt.

Theorem 22.16. *There is exactly one zero $x^* = (g^*, c^*)$ of $\mathcal{F}$ in $\overline{B}_{\mathcal{X}}(x^\circ, 10^{-10})$.*

Proof. By Lemma 22.10 and the bounds $\varepsilon_\Upsilon < 1$, $\max_e \delta_e < 1$, the operator A is injective. The classes of Section 22.4 cover all input columns, and $Z < 0.887235$. The verifier evaluates both inequalities (21.13) with the full enclosures of Y , Z , $\|A\|$, c_2 , c_3 , c_4 at the exact radius $r = 10^{-10}$; their left-hand sides are below $-8.0788 \cdot 10^{-12}$ and 0.887267 , so Proposition 21.14 applies. $\square$

The existence ball projects onto the shape ball $\|c - c^\circ\|_{\mathcal{E}} \leq \varepsilon_0$ with $\varepsilon_0 = 1044359479949477 \cdot 10^{-30} \geq r/\sigma$. Let

$$S_1 = \sum_{j \geq 3} j |c_j^\circ|, \quad S_2 = \sum_{j \geq 3} j(j-1) |c_j^\circ|, \quad H_{\mathcal{E}} = \max_{j \geq 3 \text{ odd}} \frac{j-1}{R^j \Upsilon(j-1)};$$

the maximum is attained for $j \leq 21$, since the ratio of consecutive odd terms is at most $(j+1)/((j-1)R^2) \leq 1$ for $j \geq 21$.

Proposition 22.17. *For every c with $\|c - c^\circ\|_{\mathcal{E}} \leq \varepsilon_0$: $\text{Re } \psi'_c > 0.981143$ on $\{|w| \leq \frac{101}{100}\}$; $\text{Re}(1 + w\psi''_c/\psi'_c) > 0.839460$ on $\overline{\mathbb{D}}$; $\text{Re}(w\psi'_c/\psi_c) > 0.984794$ on $|w| = 1$; and $|c_3| > 4.45696 \cdot 10^{-4}$.*

Proof. Let $\delta = c - c^\circ$. For $|w| \leq R$, $|\sum_j j \delta_j w^{j-1}| \leq \|\delta\|_\mathcal{E}/R$, $|\sum_j \delta_j w^j| \leq \|\delta\|_\mathcal{E}/R$ for $|w| \leq 1$, and $\sum_j j(j-1)|\delta_j| \leq H_\mathcal{E}\|\delta\|_\mathcal{E}$. Hence $\operatorname{Re} \psi'_c \geq c_1^\circ - \sum_{j \geq 3} j|c_j^\circ|(\frac{101}{100})^{j-1} - \varepsilon_0/R$ on the disc of radius $\frac{101}{100}$, and on $\mathbb{D}$

$$\operatorname{Re} \left(1 + \frac{w\psi''_c}{\psi'_c}\right) \geq 1 - \frac{S_2 + \varepsilon_0 H_\mathcal{E}}{c_1^\circ - S_1 - \varepsilon_0/R}, \quad \operatorname{Re} \frac{w\psi'_c}{\psi_c} \geq 1 - \frac{\sum_{j \geq 3} (j-1)|c_j^\circ| + \varepsilon_0/R}{c_1^\circ - \sum_{j \geq 3} |c_j^\circ| - \varepsilon_0/R} \quad (|w| = 1),$$

and $|c_3| \geq |c_3^\circ| - \varepsilon_0/(3R^3\Upsilon(2))$. The right-hand sides are evaluated in exact rational arithmetic. $\square$

Proof of Theorem 1.1 for $n = 5$. Let $x^* = (g^*, c^*)$ be the zero of Theorem 22.16, and $\psi = \psi_{c^*}$. By Proposition 22.17, $\operatorname{Re} \psi' > 0$ on the disc of radius $\frac{101}{100}$, so by Lemma 22.1, with $R_0 = \frac{101}{100}$, $\Theta = \Theta_{c^*}$ is a real-analytic diffeomorphism of a neighbourhood of $\bar{B}$ onto its image, with Jacobian $q_{c^*}^3 |p_{c^*}|^2 > 0$. Let $\Omega_0 = \Theta(B)$, $U = 1 + \iota(K_5 g^*)$ and $u_0 = U \circ \Theta^{-1}$. By Lemma 22.9 and Lemma 22.1(b), $u_0 \in C^1(\bar{\Omega}_0)$ is a weak solution of $\Delta u_0 + \rho^2 u_0 = 0$ with $u_0 = 1$ and $\nabla u_0 = 0$ on $\partial\Omega_0$. Let $\Omega = \rho\Omega_0$ and $u(X) = u_0(X/\rho)$. Then u is a weak solution of $\Delta u + u = 0$ with the same boundary data, and Lemma 21.26 extends u to a real-analytic solution on a neighbourhood of $\bar{\Omega}$. The boundary $\partial\Omega = \rho\Theta(\partial B)$ is a compact real-analytic hypersurface diffeomorphic to S^4 , and Ω is invariant under $O(4)$ acting on X' and under $X_1 \mapsto -X_1$. The lower bound for $\operatorname{Re}(w\psi'/\psi)$ makes the meridian domain $\psi(\mathbb{D})$, and hence Ω , strictly star-shaped with respect to the origin. The lower bound for $\operatorname{Re}(1 + w\psi''/\psi')$ makes $\psi(\mathbb{D})$ strictly convex, and the solid-of-revolution argument in the proof of Proposition 21.25, with $X' \in \mathbb{R}^4$, shows that Ω is convex. A constant solution of $\Delta u + u = 0$ vanishes, but $u = 1$ on $\partial\Omega$. If Ω were a ball, its symmetries would put its centre at the origin, $\psi(\mathbb{D})$ would be a disc centred at the origin, and Schwarz's lemma would force ψ to be linear, contrary to $|c_3^*| > 4.45696 \cdot 10^{-4}$. This proves Theorem 1.1 for $n = 5$.

Remark 22.18. The number ρ agrees with $j_{5/2,17}$ to within $3 \cdot 10^{-15}$ and $c_1^\circ \approx 1.000225$, so Ω is close to the Schiffer ball of radius $j_{5/2,17}$, which corresponds to $\psi(w) = (j_{5/2,17}/\rho)w$. That ball is far from the existence ball: its coefficient c_3 vanishes, so its distance from x° in $\mathcal{X}$ is at least $3\sigma R^3\Upsilon(2)|c_3^\circ| > 148$, while the existence ball has radius 10^{-10} . The domain Ω lies at the near-resonance $(j_{5/2,17}, 12)$ of Table 2, and its dominant nonconstant zonal harmonic has degree twelve (Remark 1.5).

23. DIMENSION SEVEN

We prove Theorem 1.1 for $n = 7$ by the axisymmetric formulation of Section 22, with the weight $\Upsilon(D) = (1 + D/10^6)^2$, the square of the weight used for $n = 5$.

23.1. Coefficient spaces and reconstruction. Let $B \subset \mathbb{R}^7$ be the unit ball, $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^6$, $y = |x'|$, $w = x_1 + iy$, $r = |x|$, $t = r^2$ and $\xi = x_1/r$. Throughout this section

$$\lambda = \frac{5}{2}, \quad R = \frac{6}{5}, \quad \zeta = R^{-2} = \frac{25}{36}, \quad \Upsilon(D) = \left(1 + \frac{D}{10^6}\right)^2.$$

Put $Y_\ell = C_\ell^{(\lambda)}/C_\ell^{(\lambda)}(1)$, $H_\ell = r^\ell Y_\ell(\xi)$, $\beta_\ell = \ell + \lambda$ and $\Psi_{\ell,s} = H_\ell J_s^{\beta_\ell}(t)$. Since $\lambda = (n-2)/2$, H_ℓ is the axisymmetric harmonic polynomial on $\mathbb{R}^7$, homogeneous of degree ℓ , with $H_\ell(1, 0) = 1$ and $|H_\ell| \leq 1$ on $\bar{B}$, and $\Delta(H_\ell v(t)) = 4H_\ell(tv'' + (\beta_\ell + 1)v')$. Here

$$C_\ell^{(\lambda)}(1) = \frac{(\ell+1)(\ell+2)(\ell+3)(\ell+4)}{24}, \quad \lambda_\ell^+ = \frac{\ell+5}{2\ell+5}, \quad \lambda_\ell^- = \frac{\ell}{2\ell+5}.$$

The identities $x_1 H_\ell = \lambda_\ell^+ H_{\ell+1} + \lambda_\ell^- t H_{\ell-1}$ and $(x \cdot \nabla) H_\ell = \ell H_\ell$ follow from the Gegenbauer recurrence and homogeneity. For $\ell \geq 1$, $\partial_{x_1} H_\ell = \ell H_{\ell-1}$: the left side is axisymmetric, harmonic and homogeneous of degree $\ell - 1$, hence a multiple of $H_{\ell-1}$, and the multiple is ℓ because $H_\ell(x_1, 0) = x_1^\ell$. For $\ell = 0$ the derivative is zero. Let $\mathbf{a}_k = (\lambda)_k/k!$ and

$$Y_\ell = \sum_m D_{m\ell} T_m, \quad D_{\ell-2k,\ell} = \frac{(2 - \delta_{\ell,2k}) \mathbf{a}_k \mathbf{a}_{\ell-k}}{C_\ell^{(\lambda)}(1)}, \quad h_\ell = \sum_m D_{m\ell} R^m.$$

$D_{m\ell} \geq 0$ and $\sum_m D_{m\ell} = 1$. Write $T_K = \sum_\ell C_{\ell K} Y_\ell$ and $\mathcal{T}_K = \sum_\ell C_{\ell K} H_\ell$. All field and harmonic indices in $\mathcal{X}, \mathcal{Y}$ below are even, and all shape indices are odd. Define

$$\begin{aligned} \|f\|_{\mathcal{L}} &= \sum_{\ell, s \geq 0} h_\ell \Upsilon(\ell + 2s) |f_{\ell, s}|, & \mathcal{G} &= \{g \in \mathcal{L}^{\text{ev}} : g_{\ell, 0} = 0\}, \\ \|f\|_{\mathcal{Y}} &= \sum_m R^m \Upsilon(m) |f_m| + \sum_{\ell, s \geq 1} h_\ell \Upsilon(\ell + 2s) |f_{\ell, s}|, \\ \|c\|_{\mathcal{E}} &= \sum_{j \text{ odd}} j R^j \Upsilon(j - 1) |c_j|, & \|(g, c)\|_{\mathcal{X}} &= \|g\|_{\mathcal{L}} + \sigma \|c\|_{\mathcal{E}}. \end{aligned}$$

Here $f \in \mathcal{Y}$ represents $\sum_m f_m \mathcal{T}_m + \sum_{s \geq 1} f_{\ell, s} \Psi_{\ell, s}$. For multipliers put $\tau(\sum m_{a,k} H_a t^k) = \sum h_a \Upsilon(a + 2k) |m_{a,k}|$ and $\|\sum \tilde{m}_{n,k} r^{n+2k} T_n\|_{\text{ch}} = \sum R^n \Upsilon(n + 2k) |\tilde{m}_{n,k}|$. $\mathcal{L}^{\text{ev}} \hookrightarrow \mathcal{Y}$ has norm at most one.

Lemma 23.1 (Connection coefficients). $C_{00} = 1$, and for $K \geq 1$, $0 \leq i \leq K/2$ and $\ell = K - 2i$,

$$C_{\ell K} = \frac{K(-\lambda)_i (K - i - 1)! (\ell + \lambda) C_\ell^{(\lambda)}(1)}{2i! (\lambda)_{K-i+1}}.$$

Let $e_i = |(-\lambda)_i|/i!$ and $\mathbf{w}_i = e_i \zeta^i (1 - \zeta)^{-\lambda}$. Then

$$\begin{aligned} |C_{K-2i,K}| h_{K-2i} &\leq \mathbf{w}_i R^K, & \kappa_7 &:= \sum_i \mathbf{w}_i = (1 - \zeta)^{-5/2} \left(2 + \frac{15}{4} \zeta^2\right) - 1, \\ \lambda_\ell^+ h_{\ell+1} + \lambda_\ell^- h_{\ell-1} &\leq R h_\ell, & D_{\ell-2k,\ell} R^{\ell-2k} &\leq \mathbf{a}_k \zeta^k h_\ell. \end{aligned}$$

Proof. Take the connection formula [22, 18.18.16] for $C_K^{(\mu)}$ in terms of the $C_{K-2i}^{(\lambda)}$ with $\lambda = \frac{5}{2}$, divide by μ and let $\mu \rightarrow 0$; by [22, 18.7.25], $C_K^{(\mu)}/\mu \rightarrow (2/K) T_K$ for $K \geq 1$, and $(\mu)_{K-i}/\mu \rightarrow (K - i - 1)!$ since $K - i \geq 1$. As $\lambda(\lambda + 1)_{K-i} = (\lambda)_{K-i+1}$, this gives the stated $C_{\ell K}$. For $\ell > 0$ it gives $|C_{\ell K}| D_{\ell\ell} = e_i \varrho_i$, where $\varrho_0 = 1$ and

$$\varrho_i = \frac{K}{K - i + \lambda} \prod_{q=1}^{i-1} \frac{\ell + q}{\ell + q + \lambda} \leq \frac{K(\ell + 1)}{(K - i + \lambda)(\ell + i)} \leq 1 \quad (i \geq 1).$$

Telescoping, using $(\ell + q)/(\ell + q + \lambda) \leq (\ell + q)/(\ell + q + 1)$ since $\lambda \geq 1$, gives the first inequality; the second uses

$$(K - i + \lambda)(K - i) - K(K - 2i + 1) = i^2 - \lambda i + (\lambda - 1)K \geq i^2 + (\lambda - 2)i \geq 0.$$

This uses $K \geq 2i$ and $\lambda \geq 2$. For $\ell = 0$ one has $D_{00} = 1$, hence $|C_{0K}| D_{00} = \frac{1}{2} e_i \varrho_i \leq e_i$. Since $\mathbf{a}_k$ is nondecreasing $((k + \lambda)/(k + 1) \geq 1)$, $h_\ell \leq D_{\ell\ell} R^\ell \sum_k \mathbf{a}_k \zeta^k = D_{\ell\ell} R^\ell (1 - \zeta)^{-\lambda}$, proving the bound. In $(1 - \zeta)^{5/2}$ only the coefficients of degrees 0 and 2 are positive; taking absolute values gives $\sum_i e_i \zeta^i = 2 + 15\zeta^2/4 - (1 - \zeta)^{5/2}$. The last two inequalities follow by applying the positive functional $T_m \mapsto R^m$ to $\xi Y_\ell = \lambda_\ell^+ Y_{\ell+1} + \lambda_\ell^- Y_{\ell-1}$, and by $D_{\ell-2k,\ell} \leq \mathbf{a}_k D_{\ell\ell}$. $\square$

Lemma 23.2 (Graded multiplication). *Theorem 22.6 holds in the spaces above, with κ_7 in place of κ_5 and harmonic fraction $\vartheta_{d_a}(\ell, s) = \mathbf{1}_{d_a \geq s}(d_a/(\ell + d_a + 9/2))^s$. The expansion of $H_a t^k \Psi_{\ell, s}$ is nonnegative, has angular index $v \geq |\ell - a|$, radial index $s' \geq s - d_a$, degree at most $\ell + 2s + d_a$, and its radial lowering is dominated by $\text{Binomial}(d_a, 1/2)$, where $d_a = a + 2k$. Theorem 22.8 holds with the present Υ .*

Proof. The linearisation formula [22, 18.18.22], at $\lambda = 5/2 > 0$, gives nonnegative coefficients G_{alv} of $Y_a Y_\ell$, with sum one and $\sum_v G_{alv} h_v \leq h_a h_\ell$. In the proof of Theorem 22.6(b), every Jacobi raising or lowering step remains a probability vector because $\beta_\ell > 0$. The factors in its harmonic estimate become $\beta_v + 1 = v + 7/2$ and $(d_a - i)/(\ell + d_a + 9/2 + i)$; the binomial domination is the same sequence of lowering steps of probability at most $1/2$ as in Lemma 21.9. The degree bound, the submultiplicativity of Υ , and Lemma 23.1 prove the norm bounds in part (c). Part (d) involves only the meridian expansions, whose degrees are unchanged. Finally, the proof of Theorem 22.8 uses only that Υ is nondecreasing and submultiplicative; these properties hold for its square as well. $\square$

Lemma 23.3 (Compatible inverse and stencils). *For $s \geq 1$, let $d = \ell + 2s + \lambda$ and $E = x \cdot \nabla$. Then*

$$\begin{aligned} K_7 \Psi_{\ell, s} &= \frac{\Psi_{\ell, s-1}}{4d(d+1)} - \frac{\Psi_{\ell, s}}{2d(d+2)} + \frac{\Psi_{\ell, s+1}}{4(d+1)(d+2)}, \\ \partial_{x_1} K_7 \Psi_{\ell, s} &= \frac{\lambda_\ell^+(\Psi_{\ell+1, s} - \Psi_{\ell+1, s-1}) + \lambda_\ell^-(\Psi_{\ell-1, s+1} - \Psi_{\ell-1, s})}{2(d+1)}, \\ EK_7 \Psi_{\ell, s} &= -\frac{d+\lambda}{4d(d+1)} \Psi_{\ell, s-1} + \frac{\lambda}{2d(d+2)} \Psi_{\ell, s} + \frac{d+2-\lambda}{4(d+1)(d+2)} \Psi_{\ell, s+1}. \end{aligned}$$

One has $\Delta K_7 \Psi_{\ell, s} = \Psi_{\ell, s}$ and zero value and gradient on ∂B . The norms of K_7 , $\partial_{x_1} K_7$, EK_7 and $y \partial_y K_7 = EK_7 - x_1 \partial_{x_1} K_7$, from $\mathcal{G}$ to $\mathcal{L}$, are bounded by

$$\begin{aligned} k_K &= \frac{1}{d_0(d_0+2)} + \frac{\Upsilon(2)-1}{4(d_0+1)(d_0+2)}, & k_D &= \frac{R(1+\Upsilon(1))}{2(d_0+1)}, \\ k_E &= \frac{d_0+\lambda}{2d_0(d_0+1)} + \frac{(\Upsilon(2)-1)(d_0+2-\lambda)}{4(d_0+1)(d_0+2)}, & k_Y &= k_E + R\Upsilon(1)k_D, \quad d_0 = \frac{9}{2}. \end{aligned}$$

Proof. Write $K_7 \Psi_{\ell, s} = H_\ell k(t)$ and $\beta = \beta_\ell$. Then $\Delta K_7 \Psi_{\ell, s} = \Psi_{\ell, s}$ reads $(t^{\beta+1} k')' = \frac{1}{4} t^\beta J_s^\beta$. By the Rodrigues formula the polynomial solution is

$$t^{\beta+1} k' = \frac{1}{4s!} \frac{d^{s-1}}{dt^{s-1}} [t^{\beta+s} (t-1)^s].$$

Thus $\deg k = s+1$, $k'(1) = \frac{1}{4} \int_0^1 t^\beta J_s^\beta dt = 0$ by Lemma 21.3(a), and we normalise $k(1) = 0$. For a polynomial φ of degree at most $s-1$, two integrations by parts with $k(1) = k'(1) = 0$ give

$$\int_0^1 t^\beta k(t\varphi'' + (\beta+1)\varphi') dt = \frac{1}{4} \int_0^1 t^\beta J_s^\beta \varphi dt = 0.$$

The map $\varphi \mapsto t\varphi'' + (\beta+1)\varphi'$ sends the polynomials of degree at most $s-1$ onto those of degree at most $s-2$. Hence k is orthogonal for the weight t^β to the polynomials of degree at most $s-2$, so it lies in the span of $J_{s-1}^\beta, J_s^\beta, J_{s+1}^\beta$. Its leading coefficient and the two boundary conditions give the first stencil. The Jacobi identities give

$$k' = \frac{J_s^{\beta_\ell+1} - J_{s-1}^{\beta_\ell+1}}{4(d+1)}, \quad \beta_\ell k + tk' = \frac{J_{s+1}^{\beta_\ell-1} - J_s^{\beta_\ell-1}}{4(d+1)}.$$

Substitute these into $\partial_{x_1}(H_\ell k) = 2\lambda_\ell^+ H_{\ell+1} k' + 2\lambda_\ell^- H_{\ell-1}(\beta_\ell k + tk')$. Substitution into $E(H_\ell k) = H_\ell(2(\beta_\ell k + tk') - (\beta_\ell + \lambda)k)$ gives the third stencil. All formulas include $\ell = 0$, with $\lambda_0^- = 0$. For the norm bounds, the absolute column sums of the first and third stencils are $1/(d(d+2))$ and $(d+\lambda)/(2d(d+1))$. The degree increases are at most two; only the last entry requires $\Upsilon(2)$ instead of one. The derivative stencil has degree shifts ± 1 and uses $\lambda_\ell^+ h_{\ell+1} + \lambda_\ell^- h_{\ell-1} \leq Rh_\ell$. These bounds decrease for $d \geq d_0$; multiplication by x_1 costs $R\Upsilon(1)$. $\square$

Proposition 23.4 (C^1 reconstruction). *The series $K_7 g$ converges in $C^1(\overline{B})$ for $g \in \mathcal{G}$, has zero value and gradient on ∂B , satisfies $\Delta K_7 g = \iota(g)$ in distributions, and*

$$\|K_7 g\|_{C^1(\overline{B})} \leq 3.73 \cdot 10^{10} \|g\|_{\mathcal{L}},$$

where the C^1 norm is the maximum of the two suprema of value and Euclidean gradient.

Proof. We use the correspondence between polynomials on the ball and $O(2)$ -invariant spherical harmonics on S^8 (cf. [27, Ch. 4]); its measure and degree argument is as follows. Let V_m be the orthogonal complement of polynomials of degree at most $m-1$ in those of degree at most m , in $L^2(B)$. An $O(2)$ -invariant homogeneous harmonic polynomial of degree m on $\mathbb{R}^7 \times \mathbb{R}^2$ restricts, by $|z|^2 = 1 - |x|^2$, to V_m . Indeed, the parametrisation $(x, \theta) \mapsto (x, \sqrt{1 - |x|^2}(\cos \theta, \sin \theta))$ has surface element $dx d\theta$, and $\int_{S^8} FG = 2\pi \int_B fg$. Different harmonic degrees are orthogonal. Conversely, homogenise each polynomial of degree at most m to degree m or $m-1$, according to parity, using $|x|^2 + |z|^2$, and apply the equivariant Fischer decomposition. The restrictions therefore exhaust $V_0 \oplus \dots \oplus V_m$, and the map is injective by the displayed norm identity.

As in Lemma 21.6(a), by Fischer decomposition and Lemma 21.3(a), the $\Psi_{\ell,s}$ are pairwise orthogonal in $L^2(B)$ and $\Psi_{\ell,s}$ is orthogonal to all polynomials of degree less than $\ell + 2s$. The map $\iota: \mathcal{L} \rightarrow L^2(B)$ is continuous, since $\|\Psi_{\ell,s}\|_{L^2(B)}^2 \leq |S^6|/7$ by the formula below. Let $N_m^{(q)}$ denote the dimension of degree- m harmonics on S^{q-1} . The spherical addition theorem and $|S^8| = (2\pi/7)|S^6|$ give $|f(x)|^2 \leq 7N_m^{(9)}\|f\|_{L^2(B)}^2/|S^6|$ for $f \in V_m$. Jacobi orthogonality gives, with $D = \ell + 2s$ and $d = D + 5/2$,

$$\|\Psi_{\ell,s}\|_{L^2(B)}^2 = \frac{|S^6|}{2N_\ell^{(7)}(D + 7/2)}, \quad \|\nabla K_7 \Psi_{\ell,s}\|_{L^2(B)}^2 = \frac{\|\Psi_{\ell,s}\|_{L^2(B)}^2}{2d(d+2)}.$$

Integration by parts and parity put every Cartesian derivative of $K_7 \Psi_{\ell,s}$ in $V_{D-1} \oplus V_{D+1}$. Cauchy–Schwarz hence bounds its squared gradient supremum by

$$\Gamma_7(\ell, s)^2 = \frac{7(N_{D-1}^{(9)} + N_{D+1}^{(9)})}{4N_\ell^{(7)}(D + 5/2)(D + 7/2)(D + 9/2)} \leq B_7(s) := \frac{\prod_{i=2}^7 (2s + i)}{720(2s + 5/2)(2s + 7/2)}.$$

For the last inequality use $N_{D-1}^{(9)} \leq N_{D+1}^{(9)}$ and the harmonic dimension formula: with $D = \ell + 2s$ the resulting factors $(D + 1 + i)/(\ell + i)$ ($1 \leq i \leq 4$), $(D + 6)/(D + 5/2)$, $(D + 7)/(D + 7/2)$ and $1/(2\ell + 5)$ all decrease in ℓ ; their value at zero is $B_7(s)$. Integrating along a radius gives the same bound for the value. For $c = 10^6$ and $x = 2s$, $(x+2)(x+3) \leq (x+5/2)(x+7/2)$ and $x+i \leq x+c$ for $4 \leq i \leq 7$, so $B_7(s)/\Upsilon(2s)^2 \leq c^4/720$. Since $373^2 \cdot 720 > 10^8$, $c^2/\sqrt{720} < 3.73 \cdot 10^{10}$. Since $h_\ell \geq 1$ and $\Upsilon(D) \geq \Upsilon(2s)$, absolute summation proves convergence and the traces; L^2 convergence of g passes the Laplacian identity to distributions. This constant is used only in reconstruction. $\square$

23.2. The equation, Taylor bounds and inverse. The exact dyadic centre $x^\circ = (g^\circ, c^\circ)$ has $(M, S) = (210, 84)$, $\rho = 731659610854495 \cdot 2^{-43}$, $b = c_1^\circ \approx 1.0000971073$ and $\sigma = 25\rho^2$. Its

8904 field coefficients and 106 shape coefficients are fixed exact data. Put $\psi_c(w) = \sum_{j \text{ odd}} c_j w^j$, $p_c = \psi'_c$, $q_c = \text{Im } \psi_c/y$, $W = K_7 g$, $W_1 = \partial_{x_1} W$ and $W_2 = y \partial_y W$. Define

$$(23.1) \quad \mathcal{F}(g, c) = b^{-1} [q_c g + 5((q_c)_{x_1} W_1 + ((q_c)_y/y) W_2) + \rho^2 q_c |p_c|^2 (1 + W)].$$

Lemma 23.5. $\mathcal{F}: \mathcal{X} \rightarrow \mathcal{Y}$ is a bounded quartic polynomial. If $\mathcal{F}(g, c) = 0$ and $\text{Re } \psi'_c > 0$ on a disc of radius R_0 , $1 < R_0 < R$, then $U = 1 + \iota(K_7 g) \in C^1(\overline{B})$ has $U = 1$, $\nabla U = 0$ on ∂B and satisfies

$$\int_B q_c^5 (\nabla U \cdot \nabla \varphi - \rho^2 |p_c|^2 U \varphi) = 0 \quad (\varphi \in C_c^\infty(B)).$$

Proof. Lemma 23.2 and the stencils prove boundedness, and the values are even in x_1 , because q_c , $|p_c|^2$, W and $y \partial_y W$ are even in x_1 while $\partial_{x_1} q_c \partial_{x_1} W$ is a product of two odd functions. The meridian series make q_c analytic in $(x_1, |x'|^2)$, with $q_c = \int_0^1 \text{Re } \psi'_c(x_1 + iuy) du > 0$. Proposition 23.4 gives $\Delta W = \iota(g)$ and the boundary traces. For any test function, integrate the distributional identity against $q_c^5 \varphi$ to obtain

$$\int_B q_c^5 (\nabla U \cdot \nabla \varphi - \rho^2 |p_c|^2 U \varphi) = - \int_B b q_c^4 \varphi \iota(\mathcal{F}(g, c)) = 0.$$

The factor five comes from $\nabla q_c^5 = 5q_c^4 \nabla q_c$; the test function need not be axisymmetric. $\square$

Let $G_0 \geq \|g^\circ\|_{\mathcal{L}}$, $C_0 \geq \|c^\circ\|_{\mathcal{E}}$ and

$$\begin{aligned} B' &= \frac{1}{R} + 5 \left(\frac{2k_D}{R^2 - 1} + \frac{4Rk_Y}{(R^2 - 1)^2} \right), \\ c_2 &= \frac{\kappa_7}{b} \left[\frac{B'}{\sigma} + \frac{3\rho^2 k_K C_0^2}{R^3 \sigma} + \frac{3\rho^2 C_0 (1 + k_K G_0)}{R^3 \sigma^2} \right], \\ c_3 &= \frac{\kappa_7 \rho^2}{b R^3} \left[\frac{1 + k_K G_0}{\sigma^3} + \frac{3C_0 k_K}{\sigma^2} \right], \quad c_4 = \frac{\kappa_7 \rho^2 k_K}{b R^3 \sigma^3}. \end{aligned}$$

For $\|h\|_{\mathcal{X}} \leq r$ the Taylor remainder and derivative remainder of $\mathcal{F}$ are bounded by $\sum_{k=2}^4 c_k r^k$ and $\sum_{k=2}^4 k c_k r^{k-1}$. To prove this, write $h = (\delta g, \delta c)$ with norms at most $r, r/\sigma$. The raw multiplier bounds (Theorem 22.6(d), which does not depend on n) are $\|\delta q\|_{\text{ch}} \leq \|\delta c\|_{\mathcal{E}}/R$, $\|\delta q_{x_1}\|_{\text{ch}} \leq 2\|\delta c\|_{\mathcal{E}}/(R^2 - 1)$ and $\|\delta q_y/y\|_{\text{ch}} \leq 4R\|\delta c\|_{\mathcal{E}}/(R^2 - 1)^2$. The first two terms of (23.1) give $\kappa_7 B' r^2/(b\sigma)$. In the potential term q° has raw norm and p° analytic coefficient norm at most $a = C_0/R$, and their increments at most $e = r/(\sigma R)$. Expand $(a + \eta e)^3(1 + k_K(G_0 + \eta r))$ in η ; the coefficients of orders two, three and four give the remaining terms. Convert each raw product to solid multipliers once, costing κ_7 . The same multilinear bounds, with one increment left free, prove the derivative estimate.

The finite inputs are (ℓ, s) with even $\ell \leq 210$, $1 \leq s \leq 84$, and odd $j \leq 211$. The finite rows are the same field rows and the harmonic rows $\overline{T}_m$, $m \leq 210$ even. Thus the scaled finite Jacobian $\tilde{J}$ has size 9010. Let $\hat{A}$ be an exact dyadic approximate inverse. In tail coordinates $\hat{a}_N = (N + 1)R^{N+1}c_{N+1}$, $\hat{y}_m = R^m y_m$ (N, m even), the principal part of the shape column is as in Lemma 21.15 and Section 22.3. The input e_{N+1} has $\hat{a}_N = (N + 1)R^{N+1}$, and its principal polynomial $\frac{2(N+1)\rho^2}{b} q^\circ \text{Re}(\overline{p^\circ} w^N)$ has boundary values $\frac{2(N+1)\rho^2}{b} \sum_{\delta} s_{\delta} T_{[N+2\delta]}$; let Me_{N+1} be the harmonic element with these boundary values. For $N, m \geq M + 2$, the output coordinate $\hat{y}_m$ in row $m = N + 2\delta$, divided by the input coordinate $\hat{a}_N$, is $\frac{2\rho^2}{bR} s_{\delta} R^{2\delta}$. Thus the part T of M from the shape tail to the harmonic tail rows is

$$T = \frac{2\rho^2 s_0}{bR} (I + E), \quad (E\hat{a})_m = \sum_{N \neq m} \frac{s_{(m-N)/2}}{s_0} R^{m-N} \hat{a}_N, \quad m, N \geq 212 \text{ even},$$

where $s_\delta = \sum_{k-a=2\delta} p_a q_k$, $p_a = (a+1)c_{a+1}^\circ$, $q_k = \sum_{j>|k|} c_j^\circ$, $s_0 > 0$ and $-210 \leq \delta \leq 105$. Since $N + 2\delta \geq -208$ for $N \geq 212$, negative frequencies fold only into finite rows. The finite fold $\mathcal{B}$ collects the modes $|N + 2\delta| \leq 210$. The bound

$$\varepsilon_\Upsilon = \sum_{\delta \neq 0} |s_\delta/s_0| R^{2\delta} \Upsilon((2\delta)_+) < 1$$

gives $\|T^{-1}\| \leq \tau_T = bR/(2\rho^2 s_0(1 - \varepsilon_\Upsilon))$, including a Neumann remainder of norm at most $\tau_T \varepsilon_\Upsilon^{L+1}$ after powers through L . Define

$$(23.2) \quad Ay = \hat{A}(y_f - \mathcal{B}T^{-1}y_t^H) + T^{-1}y_t^H + y_t^N.$$

Let α_ℓ bound the normalised finite-row column norms of $\hat{A}$, set $\alpha_m^H = \alpha_{\tau_m}$ for $m \leq M$, and put

$$\rho_{\mathcal{B}} = \sup_{N>M} \frac{\sum_{m \leq M} \alpha_m^H \Upsilon(m) |\hat{\mathcal{B}}_{mN}|}{\Upsilon(N)}, \quad A_t = \tau_T(\sigma + \rho_{\mathcal{B}}), \quad \alpha_m^H = A_t \ (m > M).$$

Then $\|A\| \leq \max(\max_\ell \alpha_\ell, A_t, 1)$, $A\Psi_{\ell,s} = e_{(\ell,s)}$ for field tail inputs, and $A\text{Me}_j = e_j$ for shape tail inputs, where Me_j is the harmonic extension of the principal boundary polynomial. If $\|I - \hat{A}\tilde{J}\| < 1$, $\hat{A}$ is invertible; the upper triangular form (23.2), with invertible T and the identity on field tails, makes A a bounded bijection.

23.3. All field columns. There are 106 radial classes $s \geq S_\ell$ for even $\ell \leq 210$, where $S_\ell = 90, 89, 88, 87, 86, 85$ on the ranges 0–2, 4–10, 12–18, 20–26, 28–36, 38–210, respectively. The intervening 51 field inputs are explicit. For even $212 \leq \ell \leq 386$, the 88 gap classes have $S_\ell = 3$ on 212–218, 2 on 220–366 and 1 on 368–386, with 82 explicit inputs below these thresholds. The uniform angular class is $\ell \geq 388$, $s \geq 1$.

For each finite input e , the complete column $y = D\mathcal{F}(x^\circ)e$ gives the normalised defect bound $\delta_e + (A_t\|y_t^H\| + \|y_t^N\|)/w_e$, where δ_e bounds its finite inverse defect. For an exceptional field input the bound is $(\|\hat{A}y_f\| + A_t\|y_t^H\| + \|y_t^N - e\|)/w_e$. For a shape input e_j with $213 \leq j \leq 701$ odd, use

$$\frac{\|\hat{A}(y_f - \mathcal{B}a)\| + \sigma\|a - e_j\| + \|y_t^N\| + (\sigma + \rho_{\mathcal{B}})\text{rem}}{w_j},$$

where a approximates $T^{-1}y_t^H$ and rem bounds its remainder. The same numerator without e_j , applied to $y = \mathcal{F}(x^\circ)$, bounds $Y_0 = \|A\mathcal{F}(x^\circ)\|$.

For the infinite field classes define $\chi_1 = q^\circ/b - 1$, $\chi_2 = 5q_{x_1}^\circ/b$, $\chi_3 = 5(q_y^\circ/y)/b$, $\chi_4 = x_1\chi_3$, $\chi_5 = \rho^2 q^\circ |p^\circ|^2/b$. Let $\tau_i(d)$ be the graded solid norm of the degree- d part of χ_i . For each angular grid row ν let $g_\nu(t)$ be the maximum of one and the normalised finite field-row column norms of A over $\ell' \geq 2\nu$, $s' \geq t+1$, with $g_\nu(t) = 1$ outside the finite grid, including $t \geq S$. Let $\beta^H(L)$ be a nonincreasing upper bound for

$$\beta_v^H = \frac{\sum_m D_{mv} R^m \Upsilon(m) \alpha_m^H}{h_v \Upsilon(v)} \quad (v \geq L \text{ even}).$$

Odd profile indices carry the maximum of their even neighbours. For even $v > 1400$ use $\beta_v^H \leq A_t + \sum_{m \leq M} (\alpha_m^H - A_t) + a_{k_m} \zeta^{k_m}$, $k_m = (1402 - m)/2$. This follows from Lemma 23.1; $a_k \zeta^k$ decreases for $k \geq k_m$, since $(k + 5/2) \leq (k + 1)R^2$ at these starting indices.

For $\nu = \lceil (\ell - d)_+/2 \rceil$, $a = g_\nu(\max(1, s - d) - 1)$, $b_d = \beta^H((\ell - d)_+)$, $\theta = \vartheta_d(\ell, s)$ and $X_d \sim \text{Binomial}(d, 1/2)$, put

$$\mathfrak{B}_d(\ell, s) = \min \left\{ a + (b_d - a)_+ \theta, \right. \\ \left. 1 + \sum_{t=0}^{S-1} (g_\nu(t) - g_\nu(t+1)) \Pr(X_d \geq s - t - 1) + (b_d - g_\nu(0))_+ \Pr(X_d \geq s) \right\}.$$

Set $\mathfrak{B}^{(i)}(\ell, s) = \sum_d \tau_i(d) \mathfrak{B}_d(\ell, s)$. With $d_\ell = \ell + 2s + \lambda$, define

$$\mathfrak{z}(\ell, s) = \mathfrak{B}^{(1)}(\ell, s) + \frac{\Upsilon(1)}{2(d_\ell + 1)} \sum_{i=2,4} \left[\lambda_\ell^+ \frac{h_{\ell+1}}{h_\ell} (\mathfrak{B}^{(i)}(\ell + 1, s) + \mathfrak{B}^{(i)}(\ell + 1, s - 1)) \right. \\ \left. + \lambda_\ell^- \frac{h_{\ell-1}}{h_\ell} (\mathfrak{B}^{(i)}(\ell - 1, s + 1) + \mathfrak{B}^{(i)}(\ell - 1, s)) \right] \\ + \frac{(d_\ell + \lambda) \mathfrak{B}^{(3)}(\ell, s - 1)}{4d_\ell(d_\ell + 1)} + \frac{\lambda \mathfrak{B}^{(3)}(\ell, s)}{2d_\ell(d_\ell + 2)} + \frac{\Upsilon(2)(d_\ell + 2 - \lambda) \mathfrak{B}^{(3)}(\ell, s + 1)}{4(d_\ell + 1)(d_\ell + 2)} \\ + \frac{\mathfrak{B}^{(5)}(\ell, s - 1)}{4d_\ell(d_\ell + 1)} + \frac{\mathfrak{B}^{(5)}(\ell, s)}{2d_\ell(d_\ell + 2)} + \frac{\Upsilon(2) \mathfrak{B}^{(5)}(\ell, s + 1)}{4(d_\ell + 1)(d_\ell + 2)}.$$

Omit the $\ell - 1$ terms when $\ell = 0$. In $\mathfrak{z}_u$ replace each bracket by R times the maximum of its two parenthesised sums.

Lemma 23.6 (Field envelope). *For every field tail input, its normalised defect is at most $\mathfrak{z}(\ell, s) \leq \mathfrak{z}_u(\ell, s)$. For fixed ℓ , $\mathfrak{z}$ decreases in s ; $\mathfrak{z}_u$ decreases in both ℓ and s . Thus the infinite classes are bounded at (ℓ, S_ℓ) and (388, 1).*

Proof. The column defect is $-Av$, with $v = \chi_1 \Psi + \chi_2 \partial_{x_1} K_7 \Psi + \chi_3 EK_7 \Psi - \chi_4 \partial_{x_1} K_7 \Psi + \chi_5 K_7 \Psi$. Lemma 23.2 bounds each multiplier by its degree norm. Its support and harmonic fraction give the first branch of $\mathfrak{B}_d$; binomial domination of the radial lowering gives the second, by summation of the layers $g_\nu(t) - g_\nu(t+1)$. The index $\max(1, s - d) - 1$ selects precisely $s' \geq \max(1, s - d)$. Applying the stencils proves the formula, and Lemma 23.1 proves the uniform replacement.

The first branch is nondecreasing in a, b_d, θ , each of which decreases with ℓ, s . For the second, put $p_{-1} = \Pr(X_d \geq s)$ and $p_t = \Pr(X_d \geq s - t - 1)$. Summation by parts rewrites it as

$$1 - p_{S-1} + \max(b_d, g_\nu(0)) p_{-1} + \sum_{t=0}^{S-1} g_\nu(t) (p_t - p_{t-1}).$$

All coefficients of the grid values are nonnegative, proving decrease in ℓ ; its original expression proves decrease in s . The stencil coefficients decrease for $d_\ell \geq 9/2$. For the only less immediate numerator $d_\ell + 2 - \lambda$, the derivative has numerator $-d_\ell^2 + d_\ell + 7/2 < 0$; for $d_\ell + \lambda$ it is $-d_\ell^2 - 2\lambda d_\ell - \lambda < 0$. The nonuniform h -ratios are fixed in a radial class; the uniform replacement removes them. This proves the stated monotonicity. $\square$

23.4. Far shape columns. Let $W^\circ = K_7 g^\circ$, $W_1^\circ = \partial_{x_1} W^\circ$, $Z^\circ = y^{-1} \partial_y W^\circ$ and $H^\circ = g^\circ + \rho^2 |p^\circ|^2 (1 + W^\circ)$. These are polynomials, with W° vanishing to order two and W_1°, Z° to order one at $t = 1$. The axis identity

$$\frac{\partial_y H_\ell}{y} = - \sum_{k=\ell-2, \ell-4, \dots \geq 0} \frac{2(k + \lambda) C_k^{(\lambda)}(1)}{C_\ell^{(\lambda)}(1)} H_k t^{(\ell-2-k)/2}$$

follows from $\xi C'_\ell = \ell C'_\ell + C'_{\ell-1}$ and $C'_{\ell-1} = \sum_k 2(k+\lambda)C_k^{(\lambda)}$, where $0 \leq k \leq \ell-2$ and $k \equiv \ell-2 \pmod{2}$ (see [22, 18.9]). Differentiating (23.1) in c_j , $j = N+1$, gives

$$(23.3) \quad \begin{aligned} bD\mathcal{F}(x^\circ)e_j &= r^N U_N H^\circ + 5(N+1)r^{N-1}U_{N-1}W_1^\circ \\ &+ 5\left(Nr^N T_N - \frac{r^N U_N + tr^{N-2}U_{N-2}}{2}\right)Z^\circ \\ &+ 2(N+1)\rho^2 q^\circ \operatorname{Re}(\bar{p}^\circ w^N)W^\circ + b\Pi_j, \end{aligned}$$

where $b\Pi_j = 2(N+1)\rho^2 q^\circ \operatorname{Re}(\bar{p}^\circ w^N)$. Indeed $\delta q = r^N U_N$, $\delta q_{x_1} = (N+1)r^{N-1}U_{N-1}$ and $\delta(q_y/y) = -2r^{N-2}C_{N-2}^{(2)}$; use $2(1-\xi^2)C_{N-2}^{(2)} = (U_N + U_{N-2})/2 - NT_N$ for the third term.

Expand the principal polynomial as $q^\circ \operatorname{Re}(\bar{p}^\circ w^N) = \sum_{(\eta,e)} \gamma_{\eta,e} t^e r^{N+\eta} T_{N+\eta}$, with shifts $-420 \leq \eta \leq 210$. In estimates $\gamma_{\eta,e}$ denotes the sum of the moduli of its individual contributions. Its harmonic model replaces $t^e r^K T_K$ by $\mathcal{T}_K$.

Let $\Gamma_F(v) \geq 1$ and $\Gamma_H(v) \geq A_t$ be nonincreasing bounds for all normalised field-row columns and for $\beta_{v'}^H$, respectively, with $v' \geq v$. Put $\Gamma_* = \max(\Gamma_F, \Gamma_H)$ and $\Gamma_{\max} = \Gamma_*(0)$. Fix $I = 140$, $I_1 = 24$, $i_c = 100$, $I_+ = I+1$, $i_1 = \lceil (K-M)/2 \rceil$ and $S_p(j) = \sum_{k \geq 0} \zeta^k(j+k)^p$. Define

$$G_p(K) = \mathbf{w}_{I_+} \begin{cases} \Gamma_{\max} S_p(I_+), & i_1 \leq I_+, \\ \Gamma_*(M+1)S_p(I_+) + \Gamma_{\max} \zeta^{i_1-I_+} S_p(i_1), & i_1 > I_+, \end{cases} \quad p = 0, 1, 2.$$

These bound $\sum_{I < i \leq K/2} \mathbf{w}_i i^p \Gamma_*(K-2i)$: e_i decreases for $i \geq I_+$, and $K-2i > M$ for $i < i_1$.

Lemma 23.7 (Geometric switch). *If $\Gamma_*(M+1)S_p(I_+) \leq \Gamma_{\max} I_+^p$ for $p = 0, 1, 2$, each G_p is nonincreasing in i_1 , including its first switch.*

Proof. The identity $S_p(I_+) - \zeta S_p(I_++1) = I_+^p$ gives the first switch inequality. In the second branch increasing i_1 drops a nonnegative summand from $\sum_{i \geq i_1} \zeta^{i-I_+} i^p$. For the present profiles the three assumptions hold with $\Gamma_*(M+1)S_p(I_+)/(\Gamma_{\max} I_+^p) \approx 7.9 \cdot 10^{-4}$, $8.0 \cdot 10^{-4}$, $8.1 \cdot 10^{-4}$ for $p = 0, 1, 2$. $\square$

Lemma 23.8. *For $K \geq 2$, $\sum_i i(K-i+\frac{5}{2})C_{K-2i,K}Y_{K-2i} = -\frac{5K}{4}U_{K-2}$.*

Proof. In $\mathbb{R}^7$, $\Delta(t^i H_v) = 4i(i+v+\frac{5}{2})t^{i-1}H_v$. The function $r^K T_K(\xi) = \operatorname{Re} w^K$ is harmonic in the meridian variables, so $\Delta \operatorname{Re} w^K = \frac{5}{y} \partial_y \operatorname{Re} w^K = -\frac{5K}{y} \operatorname{Im} w^{K-1} = -5K r^{K-2} U_{K-2}(\xi)$. Apply Δ to $r^K T_K = \sum_i C_{K-2i,K} H_{K-2i} t^i$ and restrict to $t = 1$. $\square$

Lemma 23.9. *Let K, e, m be nonnegative integers with K even, $\beta = K + \frac{5}{2}$, $c_m = \max(2, m+1)$, $F_m(\beta') = m! / (\beta'/2 + 1)_m$, and*

$$f(x) = \frac{\beta - 2x + 1}{\prod_{k=1}^{m+1} (\beta - x + e + k)}, \quad \varrho_2(i) = f(i) - f(0) - f'(0)i.$$

All connection sums below have $i \leq K/2$. Let $u = t^e r^K T_K(\xi)(1-t)^m$ if $m \geq 1$, and $u = t^e r^K T_K(\xi) - \mathcal{T}_K$ if $m = 0$ and $K > M$. Then

$$\begin{aligned} \frac{\|Au\|_{\mathcal{X}}}{R^K \Upsilon(K+2e+2m)} &\leq \sum_i \mathbf{w}_i g_i \Gamma_F(K-2i) \\ &+ m! \left(f_0 \alpha^H(K) + \frac{5}{4} |f'(0)| u(K) + \sum_{i \geq 1} \mathbf{w}_i \left(\frac{|f'(0)| i^2}{\beta} + |\varrho_2(i)| \right) \Gamma_H(K-2i) \right), \end{aligned}$$

where $\mathbf{g}_i = \mathbf{F}_m(\beta - 2i)$ and $\mathbf{f}_0 = \mathbf{f}(0)$ if $m \geq 1$, $\mathbf{g}_i = (e + i)/(\beta - i + e + 1)$ and $\mathbf{f}_0 = 1 - \mathbf{f}(0) = e/(\beta + e + 1)$ if $m = 0$, and $\mathbf{u}(K) = 2A_t\zeta/(1 - \zeta) + 2\sum_{n \leq \min(M, K-2), n \equiv K} R^{n-K}(\alpha_n^H - A_t)_+$. Here $\alpha^H(K) = \max(A_t, \alpha_K^H)$ for $K \leq M$ and $\alpha^H(K) = A_t$ for $K > M$; $\mathcal{H}_v = \sum_n D_{nv}\mathcal{T}_n$ denotes the harmonic element with trace Y_v . Moreover, for $i \leq K/2$,

$$|\mathbf{f}'(0)| \leq \frac{c_m \mathbf{f}(0)}{\beta + 1}, \quad |\varrho_2(i)| \leq \frac{i^2}{2} \cdot \frac{(\beta + 1)(c_m^2 + m + 5)}{(\beta - 2i + 1)^2 \prod_{k=1}^{m+1} (\beta - i + e + k)}.$$

Proof. Let $v_i = K - 2i$, with every connection sum restricted to $0 \leq i \leq \lfloor K/2 \rfloor$. By definition of $C_{\ell K}$, $u = \sum_i C_{v_i, K} H_{v_i} t^{e+i} (1-t)^m$, minus $\mathcal{T}_K = \sum_i C_{v_i, K} H_{v_i}$ if $m = 0$, and all outputs have degree at most $K + 2e + 2m$; so it suffices to bound the norms with $\Upsilon \equiv 1$ (Lemma 23.2). *Field layer.* By Lemma 21.3(c),(d), the $J^{\beta_{v_i}}$ -coefficients of $t^{e+i}(1-t)^m$ have ℓ^1 norm at most $\mathbf{F}_m(\beta_{v_i}) = \mathbf{F}_m(\beta - 2i)$, and those of $t^{e+i} - 1$ with $s \geq 1$ have sum $(e + i)/(\beta_{v_i} + e + i + 1)$; with $|C_{v_i, K}| h_{v_i} \leq w_i R^K$ this gives the first sum. *Harmonic layer.* By Lemma 21.3(a), the coefficient of $J_0^{\beta_v}$ in $t^\mu (1-t)^m$ is $(\beta_v + 1) \int_0^1 t^{\beta_v + \mu} (1-t)^m dt = m!(\beta_v + 1)/\prod_{k=1}^{m+1} (\beta_v + \mu + k)$, which for $v = v_i$ and $\mu = e + i$ equals $m! \mathbf{f}(i)$. Hence the boundary trace of the harmonic layer of u is $m! \sum_i C_{v_i, K} \mathbf{f}(i) Y_{v_i}$, minus T_K if $m = 0$. By $\sum_i C_{v_i, K} Y_{v_i} = T_K$ and Lemma 23.8, $\sum_i i C_{v_i, K} Y_{v_i} = \beta^{-1}(-\frac{5K}{4} U_{K-2} + \sum_i i^2 C_{v_i, K} Y_{v_i})$, and the trace equals

$$m! \left(\mathbf{f}(0) T_K - \frac{5K \mathbf{f}'(0)}{4\beta} U_{K-2} + \sum_i C_{v_i, K} \left(\frac{\mathbf{f}'(0) i^2}{\beta} + \varrho_2(i) \right) Y_{v_i} \right),$$

with $\mathbf{f}(0) - 1$ in place of $\mathbf{f}(0)$ if $m = 0$. We apply A to the three parts: $\|A\mathcal{T}_K\| \leq \alpha^H(K) R^K \Upsilon(K)$; the Chebyshev coefficients of U_{K-2} are 2 (and 1 for T_0), which gives $\mathbf{u}(K)$, since $K < \beta$; and Y_v is the trace of $\mathcal{H}_v$. For the bounds on $\mathbf{f}$ write $\mathbf{f} = g/P$ with $g(x) = \beta - 2x + 1$ and $P(x) = \prod_k (\beta - x + e + k)$, and let $S_1 = \sum_k (\beta - x + e + k)^{-1}$, $S_2 = \sum_k (\beta - x + e + k)^{-2}$. Then $\mathbf{f}' = \mathbf{f}(S_1 - 2/g)$ and $\mathbf{f}'' = P^{-1}(g(S_1^2 + S_2) - 4S_1)$. At $x = 0$, $0 < S_1 \leq (m+1)/(\beta+1)$ and $2/g = 2/(\beta+1)$, which gives the bound for $\mathbf{f}'(0)$. For $0 \leq x \leq i \leq K/2$ the two terms of $\mathbf{f}''$ have opposite signs, so $|\mathbf{f}''(x)| \leq P(x)^{-1} \max\{g(S_1^2 + S_2), 4S_1\}$; with $P(x) \geq P(i)$, $g \leq \beta + 1$, $S_1 \leq (m+1)/(\beta - i + 1)$, $S_2 \leq (m+1)/(\beta - i + 1)^2$ and $(\beta - 2i + 1)^2 \leq (\beta + 1)(\beta - i + 1)$ this is at most $(\beta + 1) \max\{(m+1)(m+2), 4(m+1)\}/((\beta - 2i + 1)^2 P(i))$, and $\max\{(m+1)(m+2), 4(m+1)\} \leq c_m^2 + m + 5$. Taylor's formula gives the bound for ϱ_2 . $\square$

Let $\Phi_m(K)$ ($m \geq 1$) and $\Phi_P(K, e)$ ($m = 0$) be the right-hand side of Lemma 23.9, evaluated as follows. For $m \geq 1$ the bounds are made independent of e by putting $e = 0$ in $\mathbf{f}(0)$ and in the bound for ϱ_2 . For $m = 0$ use $|\mathbf{f}'(0)| \leq 2/(\beta + e + 1)$, $\mathbf{f}_{\max} = 1$ and the second-derivative constant 9. The sums over i are evaluated exactly for $i \leq I_1$; for $I_1 < i \leq \min(I, K/2)$ the factors $\mathbf{g}_i$ and $|\varrho_2(i)|/i^2$, which increase with i , are replaced by their values at the largest index; and the terms with $i > I$ are bounded by means of G_0, G_1, G_2 , using $\mathbf{g}_i \leq 1$, $|\mathbf{f}(i)| \leq \mathbf{f}_{\max}$ and $|\varrho_2(i)| \leq 2\mathbf{f}_{\max} + |\mathbf{f}'(0)|i$, where $\mathbf{f}_{\max} = (\beta + 1)/(K/2 + \frac{7}{2})^{m+1}$ for $m \geq 1$ and $\mathbf{f}_{\max} = 1$ for $m = 0$. The same formulas define placeholder values at odd indices for suffix maximisation. For odd $K \leq M$, which enters only through suffix maxima, $\alpha^H(K)$ is the maximum of A_t and of the two even neighbours, and the sum in $\mathbf{u}(K)$ is empty. Let $\varphi_m(K) = \sup_{K' \geq K} \Phi_m(K')$, and

$$\bar{\Gamma}(K) = \sup_{K' \geq K} \sum_{i \leq I} w_i \Gamma_*(\max(K' - 2i, 0)) + w_{I_+} S_0(I_+) \Gamma_{\max}.$$

For f one of the polynomials $H^\circ, W_1^\circ, Z^\circ, W^\circ$, write $f = \sum_a H_a f_a(t)$, $f_a = \sum_s f_{a,s} J_s^{\beta_a} = \sum_m f_{a,m}^\natural (1-t)^m$ and $\|f_a\|_J = \sum_s |f_{a,s}|$. By Proposition 23.4, every f_a has a double zero at $t = 1$ for $f = W^\circ$ and a zero for $f = W_1^\circ, Z^\circ$, so the corresponding $f_{a,m}^\natural$ vanish exactly. For

$K' \geq 0$ let

$$\text{HH}_a(K') = \sup_{K'' \geq K'} \sum_{i \leq \min(100, K''/2)} \mathbf{w}_i \Gamma_*(K'' - 2i) \sum_{m \geq 3} |f_{a,m}^\natural| \mathbf{F}_m(\beta_{K''-2i}),$$

let $\text{JT}(K')$ be the supremum over $K'' \geq K'$ of $\sum_{100 < i \leq \min(I, K''/2)} \mathbf{w}_i \Gamma_*(K'' - 2i)$ plus $G_0(K'')$ if $K'' > 2I$, and put, for $K \geq a$,

$$\begin{aligned} \text{Mix}_a(K) &= |f_{a,0}^\natural| \bar{\Gamma}(K - a) + \sum_{m=1}^3 |f_{a,m}^\natural| \varphi_m(K - a) + \text{HH}_a(K - a) \\ &\quad + \left(\|f_a\|_J + \sum_{m \leq 3} |f_{a,m}^\natural| \mathbf{F}_m(\beta_a) \right) \text{JT}(K - a), \end{aligned}$$

$\text{Jac}_a(K) = \|f_a\|_J \bar{\Gamma}(\max(K - a, 0))$, and finally

$$V_f(K) = \sup_{K' \geq K} \sum_a h_a \min\{\text{Jac}_a(K'), \text{Mix}_a(K')\},$$

where $\text{Mix}_a(K')$ is omitted for $K' < a$. For HH_a , replacing $\mathbf{F}_m(\beta_v)$ by $\mathbf{F}_m(\beta_{8\lfloor v/8 \rfloor})$ is also permitted: this is a nonincreasing upper step bound and is the bound evaluated in Section 34.10.

Lemma 23.10. *For $f \in \{H^\circ, W_1^\circ, Z^\circ, W^\circ\}$ and integers $K, e \geq 0$, with K odd for W_1° and even otherwise, $\|A(t^{eR^K} T_K(\xi)f)\|_{\mathcal{X}} \leq R^K \Upsilon(K + 2e + \deg f) V_f(K)$.*

Proof. It suffices to treat one angular component $H_a f_a$, with $\Upsilon \equiv 1$, and to show that its contribution is at most $R^K h_a \text{Jac}_a(K)$ and, for $K \geq a$, at most $R^K h_a \text{Mix}_a(K)$. *Jacobi route.* Expand $r^K T_K = \sum_i C_{K-2i, K} H_{K-2i} t^i$ and multiply by $H_a f_a$ with Lemma 23.2: the outputs have angular index at least $K - 2i - a$, the lowering steps of the Jacobi parameter that are needed come with enough factors t (as in the proof of Lemma 21.8(a)), all steps are stochastic, and $\sum_v G_{v' av} h_v \leq h_v h_a$; with Lemma 23.1 this gives Jac_a . *Mixed route.* for $K \geq a$. By $H_a = \sum_n D_{na} t^{(a-n)/2} r^n T_n$ and $T_K T_n = \frac{1}{2}(T_{K+n} + T_{K-n})$, $t^e r^K T_K H_a$ is a nonnegative combination of raw monomials $t^{e'} r^{K'} T_{K'}$ with $K' \geq K - a$ and R -weighted mass at most $h_a R^K$. Split $f_a = P_a + (f_a - P_a)$ with $P_a = \sum_{m \leq 3} f_{a,m}^\natural (1 - t)^m$. The part P_a is bounded by $\bar{\Gamma}$ for $m = 0$ and by Lemma 23.9 for $1 \leq m \leq 3$. For $f_a - P_a$, in the connection expansion of $t^{e'} r^{K'} T_{K'}$ the components with index $i \leq 100$ are bounded with Lemma 21.3(c),(d) term by term in $m > 3$, which gives HH_a , and those with $i > 100$ by the Jacobi norm, which gives JT : here

$$\|t^\mu (f_a - P_a)\|_{J\beta_v, 1} \leq \|f_a - P_a\|_{J\beta_a, 1} \leq \|f_a\|_J + \sum_{m \leq 3} |f_{a,m}^\natural| \mathbf{F}_m(\beta_a),$$

because every lowering of the Jacobi parameter from β_a to $\beta_v < \beta_a$ comes with a factor t (for the monomials $t^{e'} r^{K'} T_{K'}$ above one has $e' + K'/2 = e + (a + K)/2 \geq a$ when $K \geq a$, so that $\mu = e' + i \geq a - (K' - 2i) = a - v$ for every $i \leq K'/2$), and the raising and lowering stencils of Lemma 21.3(b) are nonnegative with column sums one. Finally, $\bar{\Gamma}$, φ_m , HH_a and JT are nonincreasing and $K' \geq K - a$. $\square$

Lemma 23.11 (Principal difference and infinite suffix tables). *Assume the three inequalities of Lemma 23.7. Then $\Phi_P(K, e)$ is nonincreasing on the even integers $K \geq \max(M + 2, 2I + 2)$. Let $K_{\max}$ be an even integer with*

$$K_{\max} \geq \max(2I + 2, M + 2, 2i_c).$$

Then the suffix maxima of the raw tables defining $\bar{\Gamma}$, φ_m , HH_a and JT , computed through $K_{\max}$ and extended constantly afterwards, bound the suprema of these tables over all integers $K' \geq K$. The same construction gives nonincreasing upper bounds V_f at all required indices.

Proof. For $K \geq 2I + 2$ all i -ranges in Φ_P are fixed, and the geometric tail is present. The weights w_i are independent of K , the profiles decrease, and $\alpha^H(K) = A_t$. The factors $(e + i)/(\beta - i + e + 1)$, $e/(\beta + e + 1)$, $2/(\beta + e + 1)$ and $1/\beta$ decrease with $\beta = K + 5/2$. So does the second-derivative majorant $9(\beta + 1)/((\beta - 2i + 1)^2(\beta - i + e + 1))$, since $(\beta + 1)/(\beta - 2i + 1)^2$ has negative logarithmic derivative. On even indices the finite part of $u(K)$ is multiplied by ζ at each step once $K > M$; on odd indices it is absent. Lemma 23.7 controls every branch change of G_p .

For the other raw tables the terms of $\bar{\Gamma}$ decrease without a threshold. In Φ_m , once $K \geq \max(2I + 2, M + 2)$, $F_m(\beta - 2i)$, $f(0)$, its derivative and second-derivative bounds, and $f_{\max}$ all decrease. For the last factor the logarithmic derivative is $1/(K + 7/2) - (m + 1)/(K + 7) < 0$ for $m \geq 1$, $K > 0$. In JT the range $(i_c, I]$ is fixed for $K \geq 2I + 2$; in HH_a the range $[0, i_c]$ is fixed for $K \geq 2i_c$. The remaining factors are nonincreasing suffix profiles. Odd indices of u contribute only $2A_t\zeta/(1 - \zeta)$ and are bounded by the preceding even value. Thus every raw value beyond the even $K_{\max}$ is at most its value there, for either parity.

Suffix maximisation now bounds the whole infinite sequence. For V_f both routes are non-negative combinations of these tables at $\max(K - a, 0)$ or $K - a$; when $K = a$ the additional mixed route enters a minimum and can only lower it. Hence their summed minimum is non-increasing, and its constant extension after $K_{\max}$ is an upper bound. Tables are formed for both parities. Only odd K is used for $f = W_1^\circ$, and only even K for $H^\circ, Z^\circ, W^\circ$; entries of the other parity enter solely through suffix maxima and can only enlarge the bounds.

For the frozen parameters $K_{\max} = N_2 + 2 = 1202$ is even and $\max(2I + 2, M + 2, 2i_c) = 282$. The extension condition involves no Taylor angular degrees. $\square$

Proposition 23.12 (All far shape columns). *Let $N_* = 702$, $N_2 = 1200$, $k_0 = N_2/2$, $p = \Upsilon(1000)/(b\sigma R)$, and for even $N \geq N_*$ put*

$$\begin{aligned} \mathfrak{F}(N) = & p \left[\frac{1}{N+1} \sum_{k=0}^{N/2} (2 - \delta_{2k,N}) \zeta^k V_{H^\circ}(N - 2k) + \frac{10}{R} \sum_{k=0}^{N/2-1} \zeta^k V_{W_1^\circ}(N - 1 - 2k) \right. \\ & + 5V_{Z^\circ}(N) + \frac{5}{2(N+1)} \left(\sum_{k=0}^{N/2} (2 - \delta_{2k,N}) \zeta^k V_{Z^\circ}(N - 2k) \right. \\ & \quad \left. + \sum_{k=0}^{N/2-1} (2 - \delta_{2k,N-2}) \zeta^{k+1} V_{Z^\circ}(N - 2 - 2k) \right) \\ & \left. + 2\rho^2 \sum_{(\eta,e)} \gamma_{\eta,e} R^\eta (\Phi_P(N_* + \eta, e) + V_{W^\circ}(N_* + \eta)) \right], \\ \mathfrak{F}_\infty = & \mathfrak{F}(N_2) + \frac{p\zeta^{k_0}}{1 - \zeta} \left(\frac{2V_{H^\circ}(0)}{N_2 + 1} + \frac{10V_{W_1^\circ}(0)}{R} + \frac{10V_{Z^\circ}(0)}{N_2 + 1} \right). \end{aligned}$$

Every odd $j \geq 703$ has normalised defect at most $Z_{\text{far}} = \max(\{\mathfrak{F}(N) : 702 \leq N \leq 1200 \text{ even}\} \cup \{\mathfrak{F}_\infty\})$.

Proof. Divide (23.3), with Me_j subtracted, by $b\sigma(N + 1)R^{N+1}\Upsilon(N)$. All output degrees are at most $N + 800$, so $\Upsilon(1000)$ covers the grading. The expansions of U_N, U_{N-1}, U_{N-2} and Lemma 23.10 give the first four terms. Lemma 23.9 bounds the principal difference and

Lemma 23.10 the W° term. The smallest principal frequency is $N_* - 420 = 282 = 2I + 2 > M$, so Lemma 23.11 bounds these terms at N_* for every later N .

For $N \leq N_2$ the finite k -sums are evaluated as written. For $N > N_2$, every old term decreases because V_f does. In the H° sum and the first Z° sum, the endpoint $k = k_0$ has weight two instead of one. The W_1° sum has new terms from $k = k_0$. In the second Z° sum the old endpoint $k = k_0 - 1$ changes from weight one to two, with power ζ^{k_0} , and new terms follow. Write the W_1° sum as $\frac{5}{R} \sum_k 2\zeta^k V_{W_1^\circ}(N - 1 - 2k)$, all Chebyshev coefficients of U_{N-1} being two. Then each sum, with its weights, exceeds its value at N_2 by at most $2\zeta^{k_0} V_f(0)/(1 - \zeta)$, using $1/(N + 1) \leq 1/(N_2 + 1)$. The prefactors are $1/(N_2 + 1)$, $5/R$ and $5/(2(N_2 + 1))$, and the two Z° sums together carry the weight four; this gives exactly $\mathfrak{F}_\infty$. Thus all infinitely many far columns are covered. $\square$

23.5. Certified zero and geometry. Table 17 bounds all input columns. The finite inverse defect is below $5.94 \cdot 10^{-7}$, $\varepsilon_\Upsilon < 0.010791473$, $\rho_B < 846.350140$ and $\tau_T < 8.758304 \cdot 10^{-5}$. The value $Z_0 \approx 0.899$ is the bound for the uniform angular class $\ell \geq 388$, $s \geq 1$, given by the envelope $\mathfrak{J}_u(388, 1)$; every explicitly computed column has certified defect below 0.68.

Quantity	Certified upper bound
$\ A\ , A_t$	342234.383037, 15.223634
Y_0, Z_0	$5.950563 \cdot 10^{-11}$, 0.898934481
finite columns (9010)	0.058622927
exceptional field columns (133)	0.676824352
field envelopes (106 radial, 88 gap, angular)	0.898934481
explicit shape tail (245 columns), far shape tail	0.651602561, 0.305605856
G_0, C_0	366106.776644, 1.243142911
c_2, c_3, c_4	0.745260675, $1.946871609 \cdot 10^{-6}$, $1.925257056 \cdot 10^{-12}$
$p(10^{-9})$, contraction bound	$-4.13048 \cdot 10^{-11}$, 0.899444589

TABLE 17. Outward roundings of the seven-dimensional certificate: $(M, S) = (210, 84)$, $R = 6/5$, $\sigma = 25\rho^2$.

Theorem 23.13 (Seven-dimensional certificate). *For $(M, S) = (210, 84)$, $R = 6/5$ and $\sigma = 25\rho^2$, equation (23.1) has exactly one zero $x^* = (g^*, c^*)$ in $\overline{B}_X(x^\circ, 10^{-9})$. Its residual bound is $Y_0 < 5.950563 \cdot 10^{-11}$, its linear defect bound is $Z_0 < 0.898934481$, and the Newton map has contraction bound below 0.899444589. For every shape in this existence ball,*

$$\operatorname{Re} \psi' > 0.993557571646 \quad (|w| \leq 101/100),$$

$$\begin{aligned} \operatorname{Re}(1 + w\psi''/\psi') &> 0.951916263887 \quad (|w| \leq 1), \quad \operatorname{Re}(w\psi'/\psi) > 0.994796343179 \quad (|w| = 1), \\ |c_3| &> 0.000189913656777, \quad \kappa_\perp > 0.934750157145 \quad \text{on the unscaled revolved boundary.} \end{aligned}$$

In particular, the domain reconstructed from x^ is a strictly convex non-ball domain in $\mathbb{R}^7$ satisfying Theorem 1.1.*

Proof. The block inverse is injective. The finite, exceptional and envelope field classes are disjoint and exhaustive, as are the finite shape inputs $j \leq 211$, explicit tail $213 \leq j \leq 701$ and far tail $j \geq 703$. The preceding bounds therefore give Y_0 and Z_0 for the full operator by Lemma 15.2. At $r = 10^{-9}$,

$$p(r) = Y_0 + (Z_0 - 1)r + \|A\| \sum_{k=2}^4 c_k r^k < -4.13048 \cdot 10^{-11}, \quad Z_0 + \|A\| \sum_{k=2}^4 k c_k r^{k-1} < 0.899444589.$$

Proposition 21.14 gives existence and uniqueness. The shape projection has radius at most $\varepsilon_0 = 2890623035776027/(5 \cdot 10^{29}) \geq r/\sigma$.

Put $S_1 = \sum_{j \geq 3} j|c_j^\circ|$, $S_2 = \sum_{j \geq 3} j(j-1)|c_j^\circ|$, $S_{\text{rad}} = \sum_{j \geq 3} j^2|c_j^\circ|$ and $H_{\mathcal{E}} = \max_{j \geq 3 \text{ odd}} (j-1)/(R^j \Upsilon(j-1))$. Its consecutive odd ratio is at most $(j+1)/((j-1)R^2) \leq 1$ for $j \geq 7$; thus the maximum uses only $j \leq 7$. The triangle estimates of Proposition 22.17, with these weights and ε_0 , prove the three real-part bounds and $|c_3| \geq |c_3^\circ| - \varepsilon_0/(3R^3 \Upsilon(2))$.

For $w = e^{i\theta}$ on the upper semicircle the outward meridian normal is $w\psi'/|\psi'|$. Each of the five transverse principal curvatures equals

$$\kappa_\perp = \frac{\text{Im}(w\psi')}{|\psi'| \text{Im} \psi} \geq \frac{c_1^\circ - S_{\text{rad}} - \varepsilon_0(H_{\mathcal{E}} + 1/R)}{(c_1^\circ + S_1 + \varepsilon_0/R)^2} > 0.934750157145.$$

Indeed $|\sin(j\theta)/\sin \theta| \leq j$, and $\sum_j j^2|\delta c_j| \leq \varepsilon_0(H_{\mathcal{E}} + 1/R)$. Both factors of the denominator, after division of $\text{Im} \psi$ by $\sin \theta$, are positive and at most $c_1^\circ + S_1 + \varepsilon_0/R$. The formula extends continuously to the poles by real analyticity. The meridian curvature is $\text{Re}(1 + w\psi''/\psi')/|\psi'| > 0$; all six principal curvatures are positive throughout the shape ball. These are triangle-inequality lower bounds uniform over the shape ball; at the centre itself one finds numerically $\min_{|w|=1} \text{Re}(1 + w\psi''/\psi') \approx 0.9712$ and $\min \kappa_\perp \approx 0.9988$.

Let $\Theta(x_1, x') = (\text{Re} \psi(x_1 + i|x'|), q_c x')$. The proof of Lemma 22.1 applies in $\mathbb{R}^7$: the meridian differential is $|p|$ times a rotation and the five transverse differentials equal q_c , so $\det D\Theta = q_c^5 |p|^2 > 0$. Univalence and $q_c > 0$ make Θ injective near $\overline{B}$; its meridian series give analyticity across the axis. Lemma 23.5 and change of variables give $u_0 = (1 + K_7 g^*) \circ \Theta^{-1} \in C^1(\Theta(B))$, solving $\Delta u_0 + \rho^2 u_0 = 0$ weakly with the prescribed traces. Set $\Omega = \rho\Theta(B)$ and $u(X) = u_0(X/\rho)$. Lemma 21.26 gives analyticity near the closed domain. Its boundary is diffeomorphic to S^6 , with $O(6) \times \mathbb{Z}_2$ symmetry. The symmetric strictly convex meridian section gives a convex solid of revolution, and the curvature bounds give strict convexity (physical curvatures are divided by ρ). The lower bound for $\text{Re}(w\psi'/\psi)$ on $|w| = 1$ makes $\psi(\mathbb{D})$, and hence the solid of revolution Ω , strictly star-shaped with respect to the origin. A ball would be centred at zero by symmetry and would force ψ linear by Schwarz's lemma, contradicting $c_3 \neq 0$. A constant Helmholtz solution is zero, whereas $u = 1$ on the boundary. This proves Theorem 1.1 for $n = 7$. $\square$

Part 5. The exact inverse in general dimension

24. THE GENERAL-DIMENSIONAL CERTIFICATE

This part proves the cases $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\} \setminus \{5\}$ of Theorem 1.1 and gives a second proof of the case $n = 5$.

Theorem 24.1. *For every $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\}$ there is a bounded strictly convex domain $\Omega \subset \mathbb{R}^n$, different from a ball and strictly star-shaped with respect to the origin, with real-analytic boundary diffeomorphic to S^{n-1} , invariant under $O(n-1)$ on the last $n-1$ coordinates and under reflection of the first coordinate. It carries a nonconstant real-valued solution, real analytic on a neighbourhood of $\overline{\Omega}$, of*

$$(24.1) \quad (\Delta + 1)u = 0 \quad \text{in } \Omega, \quad u = 1, \quad \partial_{\mathbf{n}} u = 0 \quad \text{on } \partial\Omega.$$

Here $\mathbf{n}$ is the outward unit normal. In particular, Ω fails the Pompeiu property.

For $n = 5$ this is a second proof of the five-dimensional case of Theorem 1.1; it uses none of the bounds of Section 22. The argument below is parametrised by an integer $n \geq 5$. The angular conversion uses $\lambda \geq 1$, and root coverage uses $\nu = l + \lambda \geq 3/2$. Its exact parameters

and constants are in Tables 18–23; the finite centres are the compressed files `anc/gd/frozen_centers/nN.json.gz` described in Subsection 34.11. The scale ρ , analytic weight $R > 1$, shape weight $\sigma > 0$ and existence radius $r_* > 0$ are exact, as are λ , L_0 and the mode and state counts. Entries carrying an inequality sign are strict one-sided bounds.

n	stored modes		ρ	R	σ	r_*
5	481	56.49556626181198	1.05	10^{21}	10^{-30}	
9	385	169.58703412984173	1.02	10^{34}	10^{-43}	
11	321	57.86006297284511	1.1	10^{32}	10^{-38}	
12	257	42.77848161319951	1.1	10^{31}	10^{-50}	
13	481	84.57459591630585	1.02	10^{44}	10^{-50}	
15	417	32.1111962396826	1.03	10^{40}	10^{-49}	
16	641	369.83634515753704	1.02	10^{58}	10^{-72}	
17	257	74.91722119388515	1.03	10^{48}	10^{-55}	
18	321	20.807047789264114	1.05	10^{40}	10^{-52}	
20	513	83.43918979610575	1.03	10^{58}	10^{-71}	
21	641	112.60858355309749	1.03	10^{64}	10^{-82}	

TABLE 18. Exact inputs. The stored-mode count is the length of each finite shape and field array, not a count of converged coefficients; the clamping lift and the solution have infinite expansions.

24.1. Meridian coordinates and the equation. Write reference points as $x = (x_1, x') \in \mathbb{R} \times \mathbb{R}^{n-1}$ and set

$$y = |x'|, \quad w = x_1 + iy, \quad r = |x|, \quad \xi = x_1/r, \quad a = n - 2, \quad \lambda = a/2, \quad \mu = \rho^2.$$

As in Sections 22 and 23, ξ is the angular variable. The radial variable here is r ; below t denotes a time or a parameter. All scalar unknowns are $O(n - 1)$ -invariant and even in x_1 . The regular lift to the full ball interprets the meridian operator

$$(24.2) \quad \Delta_n = \partial_{x_1}^2 + \partial_y^2 + \frac{a}{y} \partial_y.$$

There is no boundary condition on the axis.

Let ψ be odd holomorphic with real coefficients, $p = \psi'$ and $q = \text{Im } \psi/y$. As in Lemma 22.1, put

$$(24.3) \quad \Theta_\psi(x_1, x') = (\text{Re } \psi(x_1 + i|x'|), q(x_1, |x'|)x').$$

Hereafter b is the positive coefficient of w in the fixed centre ψ_0 ; it is held constant throughout the equation. The unscaled domain is $\Theta_\psi(B^n)$, with u_0 of frequency ρ ; the domain of the theorem is $\Omega = \rho \Theta_\psi(B^n)$. On the meridian plane the differential is $|p|$ times a rotation; on each of the $n - 2$ transverse directions it is multiplication by q . Consequently

$$\det D\Theta_\psi = |p|^2 q^{n-2}.$$

The proof of Lemma 22.1, with $n - 2$ transverse vectors, gives the change of variables in every dimension.

For $U = u_0 \circ \Theta_\psi$, with frequency ρ , the pull-back equation is

$$\text{div}(q^a \nabla U) + \mu q^a |p|^2 U = 0.$$

Divide by the positive factor bq^{a-1} and define

$$(24.4) \quad \begin{aligned} BU &= \frac{1}{b} \left\{ q\Delta_n U + a \left(q_{x_1} U_{x_1} + \frac{q_y}{y} y U_y \right) + \mu q |p|^2 U \right\}, \\ \mathcal{L}U &= \Delta_n U + \frac{a}{q} \nabla q \cdot \nabla U + \mu |p|^2 U, \quad B = (q/b)\mathcal{L}. \end{aligned}$$

In particular the multiplier relating B to the operator $\Delta + \mu$ is

$$(24.5) \quad B(u_0 \circ \Theta_\psi) = \frac{q|p|^2}{b} ((\Delta + \mu)u_0) \circ \Theta_\psi.$$

The quotients in (24.4) are regular analytic scalars: $q = \int_0^1 \operatorname{Re} p(x_1 + isy) ds$ is even in y , and so is q_y/y . This is the quartic normalisation of (22.2), with the fixed scalar factor $1/b$.

25. ANGULAR COEFFICIENT SPACES AND THE COMPATIBLE POISSON INVERSE

25.1. **The angular algebra.** Set

$$G_l(\xi) = \frac{C_l^\lambda(\xi)}{C_l^\lambda(1)}, \quad H_l(x) = r^l G_l(\xi).$$

For $n = 5, 7$ these are the polynomials denoted Y_ℓ, H_ℓ in Sections 22.2 and 23.1. The generating function of the Gegenbauer polynomials gives

$$(25.1) \quad \begin{aligned} G_l(\cos \theta) &= \sum_m D_{ml} \cos(m\theta), \\ D_{l-2k,l} &= \frac{(2 - \delta_{l,2k})(\lambda)_k(\lambda)_{l-k}}{k!(l-k)!C_l^\lambda(1)} \quad (0 \leq k \leq l/2), \quad h_l = \sum_m D_{ml} R^m. \end{aligned}$$

In particular $D_{ml} \geq 0$, $\sum_m D_{ml} = 1$ and $h_l \geq \sum_m D_{ml} = 1$.

Let $\mathcal{Y}_n$ consist of expansions $f = \sum_{l \geq 0} f_l(r) G_l(\xi)$, with $f_l \in C([0, 1])$, $f_l(0) = 0$ for $l > 0$, and norm

$$(25.2) \quad \|f\|_{\mathcal{Y}_n} = \sum_{l \geq 0} h_l \|f_l\|_\infty.$$

Let Y be its real even sector, so only even l occur. The radial norm is a supremum over all radii. It replaces the sum of Jacobi radial coefficients, graded by polynomial degree, in Sections 22.2 and 23.1. In particular a continuous radial profile is admitted without a Jacobi expansion. The Poisson estimates below provide its C^1 reconstruction.

The raw cosine norm is $\|f\|_{\text{raw}} = \sum_m R^m \|\tilde{f}_m\|_\infty$ for the corresponding cosine expansion. The boundary norms W_0, W_1 have exactly the Fourier convention of (2.7), now with weight $R^{|m|}$. For a real cosine series they are

$$\|h\|_{W_0} = \sum_{m \geq 0} R^m |h_m|, \quad \|h\|_{W_1} = \sum_{m \geq 0} (1 + m) R^m |h_m|.$$

Thus $\|f\|_{\text{raw}} \leq \|f\|_{\mathcal{Y}_n}$ and boundary evaluation has norm at most one into W_0 .

Lemma 25.1. *The spaces $\mathcal{Y}_n$ and Y are Banach algebras with $\|fg\|_{\mathcal{Y}_n} \leq \|f\|_{\mathcal{Y}_n} \|g\|_{\mathcal{Y}_n}$.*

Proof. Completeness is the weighted $C([0, 1])$ argument of Lemma 2.1; the origin conditions define a closed subspace. Since $|G_l| \leq 1$, convergence in the norm gives uniform convergence on the full closed ball. For the product, write $G_l G_k = \sum_d L_{lk}^d G_d$. The addition theorem represents G_l as a positive semidefinite zonal kernel on S^{n-1} . The product kernel is positive semidefinite by tensoring its feature vectors. Its scalar eigenvalues on spherical harmonics are

nonnegative, and orthogonality identifies them, up to positive normalisations, with L_{lk}^d . Hence $L_{lk}^d \geq 0$; evaluation at $\xi = 1$ gives $\sum_d L_{lk}^d = 1$. This is the positivity used in Theorem 22.6(a), also given by the Gegenbauer linearisation formula [22, 18.18.22]. The nonnegative cosine expansions and $T_m T_j = (T_{m+j} + T_{|m-j|})/2$ give

$$\sum_d L_{lk}^d h_d \leq h_l h_k.$$

Taking radial suprema proves the product estimate for finite sums. Absolute convergence proves it for infinite sums. At the origin only constant angular terms remain, so products preserve origin compatibility. $\square$

Lemma 25.2 (Conversion of the full cosine expansion). *Put $z = R^{-2}$ and $e_i = |(-\lambda)_i|/i!$. Then*

$$(25.3) \quad \|f\|_{\mathcal{Y}_n} \leq \kappa \|f\|_{\text{raw}}, \quad \kappa = (1-z)^{-\lambda} \sum_{i \geq 0} e_i z^i.$$

This bound applies to arbitrary continuous radial profiles and includes all angular degrees.

Proof. The connection formula of Lemma 23.1, whose derivation of the coefficient formula holds for every $\lambda > 0$, is

$$(25.4) \quad T_N = \sum_{i=0}^{\lfloor N/2 \rfloor} C_{N-2i,N} G_{N-2i}, \quad C_{l,N} = \frac{N(-\lambda)_i (N-i-1)! (l+\lambda) C_l^\lambda(1)}{2i! (\lambda)_{N-i+1}}, \quad l = N-2i,$$

for $N > 0$; $T_0 = G_0 = 1$. It follows by the same limit of the Gegenbauer connection formula [22, 18.18.16, 18.7.25]. For $l > 0$ the top cosine coefficient is $2(\lambda)_l/(l! C_l^\lambda(1))$. Since $\lambda \geq 1$, the remaining positive coefficients in (25.1) give

$$h_l \leq \frac{2(\lambda)_l R^l}{l! C_l^\lambda(1)} (1-z)^{-\lambda}.$$

For $l = 0$ use the top cosine coefficient one and $h_0 \leq (1-z)^{-\lambda}$. For $i \geq 1$, multiplication by $|C_{l,N}|$ leaves $e_i R^N z^i (1-z)^{-\lambda}$ times

$$\frac{N}{N-i+\lambda} \prod_{s=1}^{i-1} \frac{l+s}{l+s+\lambda} \leq 1.$$

For $i = 0$, the product of $C_{N,N}$ and the top cosine coefficient is exactly one. For $i \geq 1$ the displayed product is at most $(l+1)/(l+i)$, and the desired inequality reduces to $i^2 - \lambda i + N(\lambda-1) \geq i(i+\lambda-2) \geq 0$, using $N \geq 2i$. Summation proves (25.3). Conversion lowers degree and preserves the origin condition.

For integer λ , the infinite series is $(1+z)^\lambda$. For $\lambda = m+1/2$, the Taylor coefficients $(-\lambda)_i/i!$ have sign $(-1)^i$ for $i \leq m+1$ and fixed sign $\epsilon_\lambda = (-1)^{m+1}$ for $i \geq m+1$. Hence

$$\sum_{i \geq 0} e_i z^i = \epsilon_\lambda (1-z)^\lambda + 2 \sum_{\substack{0 \leq i \leq m \\ i \equiv m \pmod{2}}} e_i z^i.$$

For $\lambda = 3/2, 5/2$ this is the expression in Lemmas 22.3(b) and 23.1. Thus (25.3) has a finite closed expression in both cases. $\square$

Lemma 25.3 (Weights and polynomial moments). *One has $1 = h_0 \leq h_1 \leq h_2 \leq \dots$ and $h_l \leq R^l$. For a representation $f = \sum_{l,s} c_{ls} r^{2s} H_l$ define $\mathbf{m}_k(f) = \sum_{l,s} |c_{ls}| h_l (l+2s)^k$. Then*

$$(25.5) \quad \|\partial_{x_1} f\|_{\mathcal{Y}_n} \leq C_x^{\text{pol}} \mathbf{m}_1(f), \quad \|y \partial_y f\|_{\mathcal{Y}_n} \leq C_y^{\text{pol}} \mathbf{m}_1(f), \quad C_x^{\text{pol}} = 1 + R, \quad C_y^{\text{pol}} = 1 + R C_x^{\text{pol}}.$$

Proof. Choose a random ζ with distribution $\text{Beta}(\lambda, \lambda)$, then independent signs of mean $2\zeta - 1$, and let S_l be their sum. Integration of the binomial law gives $\binom{l}{k}(\lambda)_k(\lambda)_{l-k}/(2\lambda)_l$. Pairing k and $l-k$ identifies (25.1) with $h_l = \mathbb{E} R^{|S_l|}$. The conditional expectation is $\mathbb{E}(S_{l+1} \mid S_l) = S_l(1 + (2\lambda + l)^{-1})$. Jensen's inequality for the even convex function $s \mapsto R^{|s|}$ proves monotonicity; $|S_l| \leq l$ proves the upper bound. The harmonic identity $\partial_{x_1} H_l = l H_{l-1}$ is proved in Section 23.1 by an argument valid in every dimension. Differentiating $r^{2s} H_l$ and using $\|x_1\|_{\mathcal{Y}_n} = R$ gives the first estimate. The Euler derivative multiplies that monomial by $l+2s$; $y \partial_y = r \partial_r - x_1 \partial_{x_1}$ gives the second. Absolute convergence with the indicated moments justifies all derivatives. $\square$

25.2. The zero-Dirichlet inverse and its trace. Let K_n^D be the regular zero-Dirichlet inverse of Δ_n . For angular degree l its radial equation is

$$\mathcal{D}_l v = v'' + \frac{n-1}{r} v' - \frac{l(l+n-2)}{r^2} v = f_l, \quad v(1) = 0.$$

Regularity at the origin is that of the solution on B^n .

Lemma 25.4. *For $f \in Y$ the following bounds hold:*

$$(25.6) \quad \begin{aligned} \|K_n^D f\|_Y &\leq C_K \|f\|_Y, & C_K &= \frac{1}{2n}, \\ \|\partial_{x_1} K_n^D f\|_{\mathcal{Y}_n} &\leq C_x^K \|f\|_Y, & \|y \partial_y K_n^D f\|_Y &\leq C_y^K \|f\|_Y, \\ C_x^K &= \frac{R}{n} + \frac{R}{n+2} + \frac{2(2R+1)}{n+1}, \\ C_E^K &= \frac{1}{n} + \frac{1}{n+2} + \frac{2}{n+1}, & C_y^K &= C_E^K + R C_x^K. \end{aligned}$$

Moreover,

$$(25.7) \quad (K_n^D f)'_l(1) = \int_0^1 r^{l+n-1} f_l(r) dr, \quad \|\partial_r K_n^D f\|_{W_1} \leq \|f\|_Y.$$

Proof. For $-\mathcal{D}_l$ the function $(1-r^2)/(2n)$ is a supersolution with right side at least one. The radial maximum principle therefore gives $\|v\|_\infty \leq \|f_l\|_\infty/(2n)$. For $l \geq 2$, $-\mathcal{D}_l r = (l-1)(l+n-1)/r$; comparison with $r\|f_l\|_\infty/d_l$, where $d_l = (l-1)(l+n-1)$, gives $|v(r)|/r \leq \|f_l\|_\infty/d_l$. For $l = 0$, apply the maximum principle on B^n to the radial functions $(1-r^2)\|f_0\|_\infty/(2n) \pm v$. For $l \geq 1$, apply it on $[\epsilon, 1]$ and let $\epsilon \rightarrow 0$. Continuity gives $v(\epsilon) = o(1)$ for $l \geq 1$. Full-ball Dirichlet regularity for continuous data [35] gives a C^1 solution; its constant and linear Taylor terms have angular degrees zero and one. Thus $v(\epsilon) = o(\epsilon)$ for $l \geq 2$, as required for the second barrier. Writing $v = r^l \phi$ and integrating its equation gives

$$(25.8) \quad v' = \frac{lv}{r} + \mathcal{I}_l(r), \quad \mathcal{I}_l(r) = r^{-l-n+1} \int_0^r s^{l+n-1} f_l(s) ds, \quad |\mathcal{I}_l(r)| \leq \frac{r}{l+n} \|f_l\|_\infty.$$

For $l = 0$ this gives $|v'| \leq \|f_0\|_\infty/n$.

The Gegenbauer identity $(1-\xi^2)G'_l = l(G_{l-1} - \xi G_l)$, obtained by differentiating H_l , gives the cancellation

$$\partial_{x_1}(v G_l) = \xi \mathcal{I}_l G_l + (lv/r) G_{l-1}.$$

Multiplication by $\xi = G_1$ costs R . For even $l \geq 2$, $l/d_l \leq 2/(n+1)$, and $h_{l-1} \leq h_l$. Thus the coefficient bound for $\partial_{x_1} K_n^D$ is at most $\max\{R/n, R/(n+2) + 2/(n+1)\}$. For the Euler derivative $r\partial_r$ the corresponding bound is $\max\{1/n, 1/(n+2) + 2/(n+1)\}$. The enlarged constants in (25.6) dominate these bounds, and $y\partial_y = r\partial_r - x_1\partial_{x_1}$ gives C_y^K . Inputs have even degree, so no $l = 1$ radial estimate is required; ∂_{x_1} may have odd outputs in $\mathcal{Y}_n$.

At $r = 1$, (25.8) and $v(1) = 0$ give the moment formula. The boundary cosine frequencies of G_l are at most l , hence

$$\|\partial_r K_n^D f\|_{W_1} \leq \sum_l (l+1) h_l |v'_l(1)| \leq \sum_l \frac{l+1}{l+n} h_l \|f\|_\infty \leq \|f\|_Y.$$

For continuous data the full-ball Poisson solution belongs to $W^{2,p}(B^n)$ for every finite p and to $C^1(\overline{B^n})$ for $p > n$, by Dirichlet regularity [35]. Approximation of the data proves the moment formula for its classical normal trace. $\square$

Define the closed compatible subspace and product space

$$(25.9) \quad \begin{aligned} \mathcal{G} &= \left\{ g \in Y : \int_0^1 r^{l+n-1} g_l(r) dr = 0 \text{ for every even } l \right\}, \\ K_n &= K_n^D|_{\mathcal{G}}, \quad W = \left\{ p = \sum_{j \geq 0} p_{2j} w^{2j} : p_{2j} \in \mathbb{R}, \|p\|_W = \sum_{j \geq 0} R^{2j} |p_{2j}| < \infty \right\}, \\ X &= \mathcal{G} \times W, \quad \|(g, p)\|_X = \|g\|_Y + \sigma \|p\|_W. \end{aligned}$$

The moments are bounded functionals, so $\mathcal{G}$ is complete. Lemma 25.4 gives both zero traces of $K_n g$. The normalised primitive is $\psi(w) = \int_0^w p(z) dz$. The nonlinear equation is

$$(25.10) \quad F(g, p) = B(1 + K_n g) = 0.$$

It is affine in g and cubic in p , thus a polynomial of degree four in (g, p) . The constant σ changes the product norm, not this equation.

n	λ	κ	C_x^K	C_y^K	L_0
5	$\frac{3}{2}$	$< 9.69898 \cdot 10^1$	$< 1.39334 \cdot 10^0$	$< 2.1392 \cdot 10^0$	96
9	$\frac{7}{2}$	$< 9.1983 \cdot 10^5$	$< 8.14061 \cdot 10^{-1}$	$< 1.23237 \cdot 10^0$	204
11	$\frac{9}{2}$	$< 3.98269 \cdot 10^4$	$< 7.17949 \cdot 10^{-1}$	$< 1.12425 \cdot 10^0$	110
12	$\frac{5}{2}$	$< 1.29082 \cdot 10^5$	$< 6.62546 \cdot 10^{-1}$	$< 1.03741 \cdot 10^0$	50
13	$\frac{11}{2}$	$< 2.33676 \cdot 10^9$	$< 5.80748 \cdot 10^{-1}$	$< 8.7881 \cdot 10^{-1}$	168
15	$\frac{13}{2}$	$< 8.73889 \cdot 10^9$	$< 5.11755 \cdot 10^{-1}$	$< 7.77598 \cdot 10^{-1}$	114
16	$\frac{7}{2}$	$< 8.38181 \cdot 10^{11}$	$< 4.78064 \cdot 10^{-1}$	$< 7.23328 \cdot 10^{-1}$	622
17	$\frac{15}{2}$	$< 2.95691 \cdot 10^{11}$	$< 4.54799 \cdot 10^{-1}$	$< 6.91009 \cdot 10^{-1}$	78
18	$\frac{8}{2}$	$< 3.13399 \cdot 10^{10}$	$< 4.3715 \cdot 10^{-1}$	$< 6.69826 \cdot 10^{-1}$	40
20	$\frac{9}{2}$	$< 5.82046 \cdot 10^{13}$	$< 3.89747 \cdot 10^{-1}$	$< 5.92132 \cdot 10^{-1}$	168
21	$\frac{19}{2}$	$< 3.38598 \cdot 10^{14}$	$< 3.72013 \cdot 10^{-1}$	$< 5.65179 \cdot 10^{-1}$	436

TABLE 19. Angular conversion, Poisson bounds, and the starting degree L_0 of the high angular complement. Here $C_K = 1/(2n)$ exactly.

26. THE SYMMETRIC-SECTOR DIRICHLET INVERSE

All operators in this section are evaluated at the finite shape ψ_0 . Set $e = a/2 = \lambda$. Direct differentiation of q^{-e} yields

$$(26.1) \quad q^e \mathcal{L}(q^{-e}Z) = (\Delta_n + \mathcal{V})Z, \quad \mathcal{V} = \mu|p|^2 - \frac{a(a-2)}{2} \frac{q_y}{yq} - \frac{a(a-2)}{4} \frac{|\nabla q|^2}{q^2}.$$

Here $\Delta_2 = \partial_{x_1}^2 + \partial_y^2$. The identity $\Delta_2(yq) = 0$ implies $\Delta_n q = (a-2)q_y/y$; substitution cancels the first derivatives of Z . Let $L = \Delta_n + \mathcal{V}$, $c = \rho b$, and $v_0 = c^2$.

26.1. The complete window. With reduced measure $r^{n-1} dr (1 - \xi^2)^{(n-3)/2} d\xi$, an orthonormal Dirichlet basis in the invariant even sector is

$$(26.2) \quad b_{l,\alpha}(r, \xi) = \frac{\sqrt{2} r^{-\lambda} J_{l+\lambda}(\alpha r) C_l^\lambda(\xi)}{|J_{l+\lambda+1}(\alpha)| \sqrt{h_l^{(2)}}}, \quad h_l^{(2)} = \frac{\pi 2^{1-2\lambda} \Gamma(l+2\lambda)}{l!(l+\lambda)\Gamma(\lambda)^2},$$

where l is even and α runs through all positive zeros of $J_{l+\lambda}$. The Bessel equation and differentiation in α give the radial norm, as in (2.12). Angular orthogonality and the self-adjoint ball Dirichlet spectral decomposition give completeness. The analytic weight h_l in (25.2) is distinct from the L^2 normalisation $h_l^{(2)}$.

Let P include every state with

$$(26.3) \quad |\alpha - c| < 12, \quad Q = I - P.$$

There are two complete coverage arguments for this window.

Lemma 26.1. *Opposite-sign root brackets satisfying the spacing and guard conditions below enumerate all states of (26.3).*

Proof. Put $\nu = l + \lambda$. The leading series of J_ν makes $J_\nu, J'_\nu > 0$ near zero. At a first stationary point $x < \nu$ with $J_\nu > 0$, the Bessel equation would give $J''_\nu = (\nu^2/x^2 - 1)J_\nu > 0$, contrary to a first loss of positive derivative. A first zero of J_ν is also impossible while its derivative is positive. Thus both remain positive for $x < \nu$. If $J'_\nu(\nu) = 0$, its equation gives $J''_\nu(\nu) = 0$ and $J'''_\nu(\nu) = -2J_\nu(\nu)/\nu < 0$; Taylor expansion would make J'_ν negative immediately to the left. Hence $J_\nu, J'_\nu > 0$ also at ν .

For $z_\nu(x) = \sqrt{x}J_\nu(x)$ one has

$$z''_\nu + \left(1 - \frac{\nu^2 - 1/4}{x^2}\right) z_\nu = 0.$$

On an interval ending at $B > \nu$, its coefficient is at most $q_B = 1 - (\nu^2 - 1/4)/B^2$. The integration-by-parts and interval Poincaré argument of Lemma 2.3 gives spacing at least $\pi/\sqrt{q_B}$. Starting from $z_\nu(\nu), z'_\nu(\nu) > 0$, comparison with the constant-coefficient equation gives first-zero distance at least $\pi/(2\sqrt{q_B})$. Consequently the first bracket ending strictly before $\nu + 3\pi/(2\sqrt{q_B})$ cannot miss an earlier zero. If the outer endpoints of successive brackets have separation less than $2\pi/\sqrt{q_B}$, no zero is missed between them. Each bracket has opposite nonzero signs and width below the spacing; it therefore contains exactly one simple zero. Enumeration from ν through a guard root above $c + 12$ covers the window in that angular sector. Positivity up to ν excludes all orders $\nu \geq c + 12$.

For the second argument evaluate J_ν on the exact grid $c - 12 + k/4$, $0 \leq k \leq 96$, in each remaining sector. Here $\nu \geq 3/2$, so the differential-equation coefficient is at most one on the positive axis. Distinct positive roots are separated by at least π . A grid cell contains at most one root. Simplicity, and nonzero endpoint signs, make equal signs equivalent to no root and

opposite signs equivalent to one root. Exclude equality at the two window edges. The same positivity bound excludes all larger angular orders. $\square$

26.2. The polynomial potential and its full error. Index shape coefficients throughout this part by their actual odd degree: $\psi_0(w) = \sum_{j \text{ odd}} c_j w^j$, $c_1 = b$. For a full, retained or discarded list define

$$s_1 = \sum_{j>1} j|c_j|, \quad s_2 = \sum_{j>1} j(j-1)|c_j|, \quad s_3 = \frac{1}{3} \sum_{j>1} j(j-1)(j-2)|c_j|.$$

The integral formula for q gives $|p-b|, |q-b| \leq s_1$, $|q_{x_1}|, |q_y| \leq s_2$ and $|q_y/y| \leq s_3$. For the last bound, $q_y(x_1, 0) = 0$ because q is even in y , and $q_{yy} = -\int_0^1 s^2 \operatorname{Re} p''(x_1 + isy) ds$ gives $|q_{yy}| \leq s_3$, hence $|q_y| \leq s_3|y|$. Let the subscripts f, t, d denote full, retained and discarded data. Retain degrees at most 61 in the polynomial potential only. Put $q_* = b - s_{1,f} > 0$ and $A_n = a(a-2)$. Then

$$(26.4) \quad \|\mathcal{V} - v_0\|_\infty \leq v_f := \mu(2bs_{1,f} + s_{1,f}^2) + \frac{A_n s_{3,f}}{2q_*} + \frac{A_n s_{2,f}^2}{2q_*^2}.$$

The difference caused by discarding shape degrees is at most

$$(26.5) \quad \begin{aligned} \delta_{\text{shape}} = & \mu(2b + s_{1,f} + s_{1,t})s_{1,d} \\ & + \frac{A_n}{2} \left(\frac{s_{3,d}}{q_*} + \frac{s_{3,t}s_{1,d}}{q_*^2} \right) \\ & + \frac{A_n}{4} \left\{ \frac{2(s_{2,f} + s_{2,t})s_{2,d}}{q_*^2} + \frac{2s_{2,t}^2(2b + s_{1,f} + s_{1,t})s_{1,d}}{q_*^4} \right\}. \end{aligned}$$

The first line bounds the difference of $|p|^2$. The second uses $|q_f^{-1} - q_t^{-1}| \leq s_{1,d}/q_*^2$. For the last line split the gradient square and its squared reciprocal; $|\nabla q|^2 \leq 2s_2^2$ and $|q_f^{-2} - q_t^{-2}| \leq (2b + s_{1,f} + s_{1,t})s_{1,d}/q_*^4$ give the displayed bound.

Replace q_t^{-1} by the polynomial $S_4 = b^{-1} \sum_{k=0}^4 [-(q_t - b)/b]^k$ and q_t^{-2} by S_4^2 . Since $s_{1,t} \leq s_{1,f} < b$, the geometric remainder is at most $\delta_S = (s_{1,t}/b)^5/(b - s_{1,t})$. Also $|S_4| \leq (b - s_{1,t})^{-1}$. The resulting polynomial potential $\mathcal{V}_t$ satisfies

$$(26.6) \quad \|\mathcal{V} - \mathcal{V}_t\|_\infty \leq \delta_V := \delta_{\text{shape}} + \frac{A_n}{2} s_{3,t} \delta_S + \frac{A_n s_{2,t}^2 \delta_S}{b - s_{1,t}}.$$

This includes the squared-reciprocal error $2\delta_S/(b - s_{1,t})$. As a polynomial in (x_1, y) , $\mathcal{V}_t$ has degree at most 598: the term $|\nabla q_t|^2 S_4^2$ has degree $118 + 480$. Set $V_t = \mathcal{V}_t - v_0$ and $v_* = v_f + \delta_V$.

26.3. The Gram identity and Schur complement. Form the exact finite matrices

$$(26.7) \quad \mathbf{B} = \mathbf{P}V_t\mathbf{P}, \quad \mathbf{B}_2 = \mathbf{P}V_t^2\mathbf{P}, \quad H_P = \operatorname{diag}(\alpha^2 - v_0) - \mathbf{B}, \quad \mathbf{G} = \mathbf{B}_2 - \mathbf{B}^2.$$

The sans-serif symbols distinguish these matrices from the differential operator B in (24.4). The Gram identity is

$$(26.8) \quad \mathbf{G} = (\mathbf{Q}V_t\mathbf{P})^*(\mathbf{Q}V_t\mathbf{P}) \succeq 0.$$

It is the identity (2.22) on the entire orthogonal complement; in particular it includes every omitted radial and angular mode.

On the complement the ball operator $-\Delta_n - v_0$ has absolute spectral gap at least $12(2c-12)$, with $c > 12$. Therefore

$$M_Q = \mathbf{Q}(-\Delta_n - v_0 - V_t)\mathbf{Q}, \quad \|M_Q^{-1}\| \leq \gamma^{-1}, \quad \gamma = 12(2c-12) - v_* > 0.$$

This is the Neumann factorisation (2.20); the operator is allowed to be indefinite.

Proposition 26.2. *Let C be a square real matrix of the window size, D a real diagonal matrix with nonzero diagonal, and suppose*

$$(26.9) \quad \begin{aligned} \|C^T C - I\|_2 &\leq \epsilon_C < 1, \\ \| |D|^{-1/2} (C^T H_P C - D) |D|^{-1/2} \|_2 &\leq \epsilon_S, \\ \gamma^{-1} \operatorname{tr}(|D|^{-1} C^T G C) &\leq \eta_S, \quad \epsilon_S + \eta_S < 1. \end{aligned}$$

Set

$$(26.10) \quad \begin{aligned} S_* &= \frac{1 + \epsilon_C}{\min_i |D_i| (1 - \epsilon_S - \eta_S)}, \\ B_{\text{pol}} &= (1 + (v_*/\gamma)^2) S_* + \gamma^{-1}, \\ B_{L^2} &= \frac{B_{\text{pol}}}{1 - B_{\text{pol}} \delta_V} \quad (B_{\text{pol}} \delta_V < 1). \end{aligned}$$

Then L has a symmetric-sector Dirichlet inverse with $\|L_D^{-1}\|_{L^2 \rightarrow L^2} \leq B_{L^2}$.

Proof. The proof of Proposition 2.4 applies with this ball basis and V_t , the two defects being bounded directly in the ℓ^2 operator norm. It uses only the orthogonal splitting of the Dirichlet operator domain, the complement gap and (26.8). In particular, for $T = QV_t P C |D|^{-1/2}$,

$$\|T^* M_Q^{-1} T\| \leq \gamma^{-1} \|T^* T\| \leq \gamma^{-1} \operatorname{tr}(T^* T) \leq \eta_S.$$

After congruence the Schur matrix is $\operatorname{sgn}(D)$ plus a perturbation of norm at most $\epsilon_S + \eta_S$, so its inverse is bounded by S_* . The block inverse has norm at most $(1 + (v_*/\gamma)^2) S_* + \gamma^{-1}$, exactly as in (2.26). Finally the bounded perturbation $\mathcal{V} - \mathcal{V}_t$ is transferred by the factorisation of (2.30), giving B_{L^2} . $\square$

The reduced measure in (26.2) differs from Lebesgue measure on invariant functions on B^n by the factor $|S^{n-2}|$. On the even invariant sector, the map $f(r, \xi) \mapsto |S^{n-2}|^{-1/2} f(|x|, x_1/|x|)$ is a unitary identification of the reduced space with the invariant subspace of $L^2(B^n, dx)$. It intertwines the Dirichlet operators, so B_{L^2} is also the inverse bound on this subspace with Lebesgue measure.

26.4. Quadrature enclosures. Here are the finite expressions defining the matrix enclosures. For odd n , the angular rule is Gauss–Legendre with polynomial weight $(1 - \xi^2)^{(n-3)/2}$. For even n , it is Gauss–Chebyshev of the second kind, with

$$\xi_k = \cos \frac{k\pi}{N_\theta + 1}, \quad \omega_k = \frac{\pi}{N_\theta + 1} \sin^2 \frac{k\pi}{N_\theta + 1} \quad (1 \leq k \leq N_\theta),$$

and remaining polynomial weight $(1 - \xi^2)^{(n-4)/2}$. Choose the angular order so that the entire polynomial degree of each integrand is at most $2N_\theta - 1$; this makes angular integration exact. This covers both integral and half-integral λ .

For radial Gauss–Legendre order N_r , the Bernstein ellipse of parameter two in $2r - 1$ gives $|r| \leq 9/8$ and $|\operatorname{Im} r| \leq 3/8$. Poisson’s Bessel integral [22, 10.9.4] gives

$$|z^{-\nu} J_\nu(z)| \leq \frac{e^{|\operatorname{Im} z|}}{2^\nu \Gamma(\nu + 1)}.$$

For each window basis vector its radial integrand factor, including $r^{(n-1)/2}$, is bounded by

$$(26.11) \quad \hat{e}_i = \frac{\sqrt{2}C_l^\lambda(1)}{\sqrt{h_l^{(2)}|J_{\nu+1}(\alpha)|}} \frac{\alpha^\nu(9/8)^{l+(n-1)/2}e^{3\alpha/8}}{2^\nu\Gamma(\nu+1)}, \quad \nu = l + \lambda.$$

The products are entire: after removing the Bessel leading powers, the radial power is $l + l' + n - 1$, an integer. Let $\hat{e}_* = \max_i \hat{e}_i$ and let V_e bound $|V_t(r, \xi)|$ for r on the ellipse and $-1 \leq \xi \leq 1$, using the s_i sums at $9/8$ and the finite polynomial S_4 . The entrywise radial errors can be taken as

$$(26.12) \quad q_1 = 32\hat{e}_*^2(1 + V_e)2^{-2N_r}, \quad q_2 = 32\hat{e}_*^2(1 + V_e^2)2^{-2N_r}.$$

Indeed the Chebyshev coefficients of an analytic integrand bounded by M are at most $2M2^{-k}$. The radial Gaussian rule is exact through degree $2N_r - 1$; bounding both its positive mass and the integration mass and summing the remaining geometric series gives the factor 16. The angular integration has mass at most two, giving 32.

Let $u_{\text{fl}} = 2^{-52}$, $\gamma_k = ku_{\text{fl}}/(1 - ku_{\text{fl}})$, and $\tau = 10^{-280}$. Use these bounds only when $ku_{\text{fl}} < 1$, with finite intermediate values. Regard every binary64 export as its exact dyadic rational. At a tensor node the radial and angular factors are

$$\frac{\sqrt{2\omega_r r} J_\nu(\alpha r)}{|J_{\nu+1}(\alpha)|}, \quad C_l^\lambda(\xi) \sqrt{\omega_\theta w_{\text{poly}}(\xi)/h_l^{(2)}}.$$

Here ω_r is the radial Gauss–Legendre weight on $[0, 1]$, ω_θ is the angular weight, and $w_{\text{poly}}(\xi) = (1 - \xi^2)^{(n-3)/2}$ for odd n , $(1 - \xi^2)^{(n-4)/2}$ for even n . Let their exported vectors be r_i, a_i , with enclosed norm errors $e_{r,i}, e_{a,i}$. The column error is bounded by

$$(26.13) \quad e_i^{\text{col}} = e_{r,i}\|a_i\|_2 + \|r_i\|_2 e_{a,i} + e_{r,i}e_{a,i} + u_{\text{fl}}\|r_i\|_2\|a_i\|_2 + \tau.$$

Thus $e_{\text{bas}} = (\sum_i (e_i^{\text{col}})^2)^{1/2}$ bounds the basis matrix Frobenius error. Bessel argument uncertainty is bounded by differentiating the Poisson integral:

$$\sup_{0 \leq x \leq X} |J'_\nu(x)| \leq \frac{\nu X^{\nu-1} + X^\nu}{2^\nu \Gamma(\nu+1)} \quad (\nu \geq 1).$$

The potential arrays are given by the exact identities

$$(26.14) \quad \begin{aligned} q - b &= \sum_{j>1} c_j r^{j-1} U_{j-1}(\xi), & \text{Re } p - b &= \sum_{j>1} j c_j r^{j-1} T_{j-1}(\xi), \\ q_{x1} &= \sum_{j>1} j c_j r^{j-2} U_{j-2}(\xi), & q_y/y &= -2 \sum_{j>1} c_j r^{j-3} C_{j-3}^{(2)}(\xi). \end{aligned}$$

Each array is expanded in $r^i T_m(\xi)$ using

$$\begin{aligned} U_m &= \sum_{k=0}^{\lfloor m/2 \rfloor} (2 - \delta_{m,2k}) T_{m-2k}, \\ C_N^{(2)} &= \sum_{k=0}^{\lfloor N/2 \rfloor} (2 - \delta_{N,2k}) (k+1)(N-k+1) T_{N-2k}. \end{aligned}$$

Here S_{pol} is the absolute sum of the resulting coefficients. These coefficients are exported, and only the T_m are evaluated by the recurrence below. For degree at most d , radial power error costs γ_{d+1} . The recurrence at an exported node $|\xi| < 1$ is $\hat{T}_{j+1} = 2\xi\hat{T}_j - \hat{T}_{j-1}$, $\hat{T}_0 = 1$, $\hat{T}_1 = \xi$.

Suppose $10u_{\text{fl}}(d+1)^3 < 10^{-3}$. If $|\widehat{T}_i| \leq 1.001$ for $i \leq j$, the product and subtraction have total rounding error $|\epsilon_j| \leq 6u_{\text{fl}}$, apart from the separate underflow allowance. Since $|U_m| \leq m+1$,

$$\left| \widehat{T}_{j+1} - T_{j+1}(\xi) \right| = \left| \sum_{i=1}^j \epsilon_i U_{j-i}(\xi) \right| \leq 3u_{\text{fl}}j(j+1) < 10^{-3}.$$

Thus $|\widehat{T}_{j+1}| \leq 1.001$ by induction, and the error at every degree $k \leq d$ is at most $3u_{\text{fl}}d(d-1) \leq 10u_{\text{fl}}(d+1)^3$, the recurrence allowance of the array error. Node-export uncertainty is separate: $|T_m| \leq 1$ and $|T'_m| \leq m^2$ imply the array error $S_{\text{pol}}(d\delta r + d^2\delta\xi)$. Add coefficient export errors and both dense-product allowances. Propagate them through the seven-variable polynomial potential by interval differentiation on a box containing both the exact and exported inputs, so the mean value theorem applies; an additional $2\gamma_{400}$ times the absolute arithmetic majorant covers its fewer than 100 scalar operations. Let δ_{pt} be the resulting pointwise potential error, and v_{exp} the exported absolute bound.

For an exported basis matrix E_F with $\|E_F\|_F \leq K_F$, window size m and total node count N_q , valid matrix errors are

$$(26.15) \quad \begin{aligned} e_B &= v_*(2K_F e_{\text{bas}} + e_{\text{bas}}^2) + \delta_{\text{pt}} K_F^2 + \gamma_{N_q+4} v_{\text{exp}} K_F^2 + mq_1 + \tau, \\ e_{B_2} &= v_*^2(2K_F e_{\text{bas}} + e_{\text{bas}}^2) + \delta_{\text{pt}}(2v_* + \delta_{\text{pt}}) K_F^2 + \gamma_{N_q+6} v_{\text{exp}}^2 K_F^2 + mq_2 + \tau. \end{aligned}$$

These follow by adding and subtracting the exact factors in $E_F^T \text{diag}(V_F) E_F$. Include root-diagonal export, subtraction and symmetrisation errors in e_H . Bound ϵ_C by the Frobenius norm of the computed $C^T C - I$ plus $\gamma_{m+3} \|C\|_F^2 + \tau$. Let H_F be the exported symmetrised H_{P} . Bound the numerator of ϵ_S by the Frobenius norm of the computed $C^T H_F C - D$ plus

$$\gamma_{2m+8} (\|C\|_F^2 \|H_F\|_F + \|D\|_F) + \tau + (1 + \epsilon_C) e_H.$$

Division by $\min |D_i|$ bounds the normalised defect in (26.9). For the exported Gram difference $G_F = B_{2,\text{exp}} - B_{\text{exp}}^2$ add

$$(26.16) \quad e_G = e_{B_2} + e_B(2v_* + e_B) + \gamma_{m+4} \|B_{\text{exp}}\|_F^2 + u_{\text{fl}} (\|B_{2,\text{exp}}\|_F + \|B_{\text{exp}}^2\|_F) + \tau.$$

The weighted trace allowance is

$$\frac{\gamma_{4m+20} \|C\|_F^2 \|G_F\|_F + m(1 + \epsilon_C) e_G}{\min |D_i|} + \tau.$$

Divide the upper trace bound by γ to obtain η_S . For an exported array A_F with N entries, let s_F be its computed sum of squares. The Frobenius reduction uses

$$\|A_F\|_F \leq \sqrt{\frac{s_F}{1 - \gamma_{2N+2}}} + \tau.$$

27. TRANSFER OF THE DIRICHLET INVERSE TO THE WORKING NORM

27.1. Low angular degrees by heat smoothing. Let $M \geq \|\mathcal{V}\|_Y$, so $|\mathcal{V}| \leq M$, and let $Lv = f$ with zero Dirichlet data in the invariant sector. For the Dirichlet heat semigroup with this bounded real potential,

$$(27.1) \quad \|e^{tL}\|_{2 \rightarrow \infty} \leq e^{Mt} (8\pi t)^{-n/4}, \quad \|e^{tL}\|_{\infty \rightarrow \infty} \leq e^{Mt} \quad (t > 0).$$

To obtain these bounds, the maximum principle first dominates the Dirichlet heat kernel by the positive free Gaussian kernel, after extension of data by zero. Iterating Duhamel's formula for multiplication by $\mathcal{V}$, taking absolute values, and using $|\mathcal{V}| \leq M$ sums to the factor e^{Mt} . The L^2 norm of $(4\pi t)^{-n/2} \exp(-|x|^2/(4t))$ is $(8\pi t)^{-n/4}$. Its L^1 norm is one. This proves both estimates, also on invariant inputs with the full Lebesgue measure.

The spectral inverse places v in the Dirichlet operator domain. Integrating the semigroup derivative gives

$$v = e^{tL}v - \int_0^t e^{sL}f ds.$$

Since $\|f\|_2 \leq \sqrt{|B^n|}\|f\|_Y$ and $\|v\|_2 \leq B_{L^2}\|f\|_2$, it follows that

$$(27.2) \quad \|v\|_\infty \leq B_\infty\|f\|_Y, \quad B_\infty = e^{Mt}(8\pi t)^{-n/4} \sqrt{|B^n|}B_{L^2} + \frac{e^{Mt} - 1}{M}, \quad |B^n| = \frac{\pi^{n/2}}{\Gamma(n/2 + 1)}.$$

We take $t = 1/M$. In particular v is bounded before any analytic coefficient norm of v is used. The equation $\Delta_n v = f - \mathcal{V}v$ now has bounded data; full-ball elliptic regularity gives $v \in W^{2,p}$ for every finite p , and hence a continuous radial expansion [35].

The dimension of degree- l spherical harmonics on S^{n-1} is

$$d_l^{\text{sph}} = \frac{2l + n - 2}{n - 2} \binom{l + n - 3}{l}.$$

The addition theorem gives $\int_{S^{n-1}} G_l^2 d\mathcal{H}^{n-1}/|S^{n-1}| = 1/d_l^{\text{sph}}$. At each r , angular projection and Cauchy–Schwarz imply

$$(27.3) \quad |v_l(r)| \leq \sqrt{d_l^{\text{sph}}} \|v(r, \cdot)\|_\infty \leq \sqrt{d_l^{\text{sph}}} B_\infty \|f\|_Y.$$

Only finitely many angular degrees will use this estimate.

27.2. The entire high angular complement. Let $v_Y \geq \|\mathcal{V} - v_0\|_Y$. The tabulated L_0 is the least even degree with $d_h > 2v_Y$. For the argument, choose any even L_0 with

$$(27.4) \quad d_h = L_0(L_0 + n - 2) - v_0 > v_Y, \quad \beta = v_Y/d_h < 1.$$

For every $l \geq L_0$, the positive zero-order part of $-\mathcal{D}_l - v_0$ is at least d_h . The radial maximum principle therefore bounds its zero-Dirichlet inverse in supremum norm by $1/d_h$. First define this high-mode inverse in L^2 by coercivity. For a source in Y , its finite angular truncations converge in Y ; the radial bound makes their inverse images converge in Y and in L^2 . Closedness of the Dirichlet operator places their limit in $H^2 \cap H_0^1$ and identifies it with the same inverse. Let Π_H be angular projection to these degrees, and let $v_{\text{low}} = (I - \Pi_H)v$. By (27.3) this low part belongs to Y . Solve

$$(\Delta_n + v_0)\tilde{v}_H = \Pi_H(f - (\mathcal{V} - v_0)(v_{\text{low}} + \tilde{v}_H))$$

with zero Dirichlet data by contraction on the high part of Y . The algebra bound gives contraction factor at most β .

We next identify $\tilde{v}_H$ with $\Pi_H v$, as in the proof of Proposition 2.6. Its Poisson right side, including $-v_0\tilde{v}_H$, lies in Y . Thus $\tilde{v}_H$ equals its zero-Dirichlet Poisson potential, and belongs to $H^2 \cap H_0^1$. The original high projection belongs to the same operator domain. Their difference z has only degrees $l \geq L_0$ and satisfies

$$(\Delta_n + v_0)z = -\Pi_H((\mathcal{V} - v_0)z).$$

Angular orthogonality and $r \leq 1$ give $\int |\nabla z|^2 \geq L_0(L_0 + n - 2) \int |z|^2$. Testing the equation with $-z$, using self-adjointness of Π_H , gives

$$d_h \int |z|^2 \leq \int |\nabla z|^2 - v_0 \int |z|^2 = \int (\mathcal{V} - v_0)|z|^2 \leq v_Y \int |z|^2.$$

Hence $z = 0$. This proves membership of the original solution in Y before rearranging any inequality involving its Y norm.

Add the low norm to the high contraction inequality, use $h_l \leq R^l$, and obtain

$$(27.5) \quad \|L_D^{-1}f\|_Y \leq B_Y \|f\|_Y, \quad B_Y = \frac{B_\infty \sum_{\substack{0 \leq l < L_0 \\ l \text{ even}}} R^l \sqrt{d_l^{\text{sph}} + d_h^{-1}}}{1 - \beta}.$$

The Poisson trace formula applied to $\Delta_n v = f - \mathcal{V}v$ also gives

$$(27.6) \quad \|\partial_r v\|_{W_1} \leq (1 + MB_Y) \|f\|_Y.$$

27.3. Undoing the conjugation. Let $\|q - b\|_Y \leq q_b < b$ and define

$$(27.7) \quad \mathfrak{I} = (b - q_b)^{-1}, \quad S_q = b^e (1 - q_b/b)^{-|e-1|}, \quad M_0 = b^{-e} (1 - q_b/b)^{-e}.$$

The binomial series gives $\|bq^{e-1}\|_Y \leq S_q$ and $\|q^{-e}\|_Y \leq M_0$. For a positive exponent, absolute falling-factorial coefficients are bounded by the corresponding rising-factorial coefficients. This proves the stated enlarged bound for either integral or half-integral e .

For the original Dirichlet equation $B\Phi = f$, set $Z = q^e \Phi$. Then $LZ = bq^{e-1}f$. Write

$$X_* \geq \|q_{x_1}\|_{\mathcal{Y}_n}, \quad Y_* \geq \|q_y/y\|_Y, \quad Y_2 \geq \|y^2\|_Y, \quad Q^{[1]} \geq \mathfrak{m}_1(q - b), \\ B_Z = S_q B_Y, \quad J_Z = S_q + MB_Z.$$

Consequently $\|Z\|_Y \leq B_Z \|f\|_Y$ and $\|\Delta_n Z\|_Y \leq J_Z \|f\|_Y$. Define

$$(27.8) \quad J_W = M_0 [J_Z + 2e\mathfrak{I}(X_* C_x^K + Y_* C_y^K) J_Z \\ + \{e(n-4)\mathfrak{I}Y_* + e(e+1)\mathfrak{I}^2(X_*^2 + Y_2 Y_*^2)\} B_Z], \\ N_W = M_0 (1 + e\mathfrak{I}Q^{[1]}) J_Z.$$

Proposition 27.1. *The original symmetric-sector zero-Dirichlet inverse satisfies*

$$\|\Delta_n B_D^{-1}f\|_Y \leq J_W \|f\|_Y, \quad \|\partial_r B_D^{-1}f\|_{W_1} \leq N_W \|f\|_Y.$$

Proof. Expand

$$\Delta_n(q^{-e}Z) = q^{-e}[\Delta_n Z - 2eq^{-1}\nabla q \cdot \nabla Z + \{-eq^{-1}\Delta_n q + e(e+1)q^{-2}|\nabla q|^2\}Z].$$

The zero-Dirichlet identity $Z = K_n^D \Delta_n Z$ and Lemma 25.4 control both gradient terms. Use $\Delta_n q = (n-4)q_y/y$ and $|\nabla q|^2 = q_{x_1}^2 + y^2(q_y/y)^2$ to obtain J_W . At the boundary $Z = 0$, so $\partial_r \Phi = q^{-e} \partial_r Z$. Lemma 25.4 bounds $\partial_r Z$ by $J_Z \|f\|_Y$ in W_1 . Moreover

$$\|q^{-e}|\partial|\|_{W_1} \leq M_0 + eM_0\mathfrak{I}\|\partial_\theta q|_\partial\|_{W_0} \leq M_0(1 + e\mathfrak{I}Q^{[1]}).$$

The boundary algebra of (2.7) gives N_W . □

28. THE EXACTLY CLAMPED CENTRE AND ITS RESIDUAL

28.1. Finite field and infinite clamping lift. The finite input arrays are exact decimals c_{2j+1}, a_j, s_j , $0 \leq j \leq J$, with $s_j > 0$: these are the keys **c**, **a**, **scales** of the frozen centre, and $J+1$ is the mode count of Table 18. Their shape convention is $\psi_0(w) = \sum_{j \text{ odd}} c_j w^j$, $b = c_1$ and $p_0 = \psi'_0$. Put, in unscaled meridian coordinates (X_1, Y_ψ) ,

$$(28.1) \quad u_* = \sum_{j=0}^J \frac{a_j}{s_j} \mathcal{R}^{-\lambda} J_{2j+\lambda}(\rho \mathcal{R}) G_{2j}(X_1/\mathcal{R}), \quad \mathcal{R}^2 = X_1^2 + Y_\psi^2, \quad U_* = u_* \circ \Theta_{\psi_0}.$$

n	states	B_{L^2}	B_Y	J_W	N_W
5	165	$< 1.57384 \cdot 10^0$	$< 7.64322 \cdot 10^9$	$< 4.45681 \cdot 10^{14}$	$< 9.58269 \cdot 10^{13}$
9	496	$< 9.55464 \cdot 10^0$	$< 2.10086 \cdot 10^{18}$	$< 2.12299 \cdot 10^{23}$	$< 1.20829 \cdot 10^{23}$
11	159	$< 1.40122 \cdot 10^0$	$< 3.93706 \cdot 10^{19}$	$< 9.78801 \cdot 10^{24}$	$< 4.76229 \cdot 10^{24}$
12	111	$< 5.99436 \cdot 10^0$	$< 2.81456 \cdot 10^{15}$	$< 2.60965 \cdot 10^{19}$	$< 1.75305 \cdot 10^{19}$
13	236	$< 1.58733 \cdot 10^0$	$< 1.76013 \cdot 10^{21}$	$< 1.24785 \cdot 10^{27}$	$< 7.23999 \cdot 10^{26}$
15	74	$< 1.09998 \cdot 10^0$	$< 6.99886 \cdot 10^{20}$	$< 4.52519 \cdot 10^{28}$	$< 2.82705 \cdot 10^{28}$
16	1078	$< 2.9373 \cdot 10^1$	$< 6.16265 \cdot 10^{38}$	$< 2.23123 \cdot 10^{45}$	$< 1.28403 \cdot 10^{45}$
17	198	$< 2.11596 \cdot 10^1$	$< 7.69351 \cdot 10^{21}$	$< 9.94338 \cdot 10^{25}$	$< 9.00711 \cdot 10^{25}$
18	33	$< 1.45968 \cdot 10^0$	$< 8.91113 \cdot 10^{15}$	$< 3.33056 \cdot 10^{21}$	$< 3.03179 \cdot 10^{21}$
20	217	$< 3.13955 \cdot 10^0$	$< 1.49392 \cdot 10^{30}$	$< 2.26518 \cdot 10^{36}$	$< 1.92607 \cdot 10^{36}$
21	300	$< 2.30008 \cdot 10^0$	$< 3.26875 \cdot 10^{42}$	$< 3.05724 \cdot 10^{52}$	$< 1.59458 \cdot 10^{52}$

TABLE 20. Complete Dirichlet inverse bounds. B_Y is for the conjugated operator; J_W, N_W are for the original operator B .

The Bessel power series expresses each summand as a solid harmonic times an entire series in $\mathcal{R}^2$, and separation gives $(\Delta_n + \mu)u_* = 0$. The s_j are fixed divisors; no identity between them and Bessel values is required. By (24.5), $BU_* = 0$ exactly.

Let D_l, N_l be the full Gegenbauer coefficients of $D = U_*|_{r=1} - 1$ and $N = \partial_r U_*|_{r=1}$. The double-trace lifts are

$$(28.2) \quad h_l^D = r^l[1 - l(r^2 - 1)/2]G_l, \quad h_l^N = \frac{1}{2}r^l(r^2 - 1)G_l.$$

Their value and radial derivative traces are respectively $(G_l, 0)$ and $(0, G_l)$. Define the exact centre by

$$(28.3) \quad \ell = \sum_l (D_l h_l^D + N_l h_l^N), \quad U_0 = U_* - \ell, \quad g_0 = \Delta_n U_0, \quad x_0 = (g_0, p_0).$$

The boundary decay proved below gives analytic convergence on a neighbourhood of the closed ball. Thus $U_0 = 1$ and $\partial_r U_0 = 0$ on the sphere, $g_0 \in \mathcal{G}$, and $U_0 = 1 + K_n g_0$ by Dirichlet uniqueness. This is the clamping construction of (3.4) with solid zonal harmonics in place of planar harmonics.

28.2. Complex majorants for the finite field. Choose $S > R > 1$, $\delta > 0$, and $r_0 = 1/\rho < 1$. For $0 \leq r \leq 1$, $|\operatorname{Im} \theta| \leq \log S$, allow independent perturbations of size δ in $w = re^{i\theta}$ and $\bar{w} = re^{-i\theta}$. Define

$$(28.4) \quad \begin{aligned} s &= b\delta S/r_0 + S \sum_{j>1} |c_j|(S + \delta/r_0)^j, & d_s &= 2s + s^2/b, \\ z_* &= \sum_{j \text{ odd}} |c_j|(r_0 S + \delta)^j, & H_s &= S(b + s), & A_s &= H_s/(b - d_s). \end{aligned}$$

Assume $d_s < b$. For $r \geq r_0$ write the two perturbed coordinates as $X_1 \pm iY_\psi = re^{\pm i\theta} A_\pm$. Then $|A_\pm - b| \leq s$. To verify the nonlinear part of this estimate, set $\varsigma = |re^{\pm i\theta}| \in [r_0/S, S]$. For odd $m \geq 3$, $f(\varsigma) = (\varsigma + \delta)^m/\varsigma$ has derivative $f'(\varsigma) = (\varsigma + \delta)^{m-1}((m-1)\varsigma - \delta)/\varsigma^2$, so f decreases for $\varsigma < \delta/(m-1)$ and increases afterwards; hence its supremum on $[r_0/S, S]$ is attained at an endpoint. Since $r_0 < 1 < S$,

$$\begin{aligned} f(S) &= (S + \delta)^m/S \leq S(S + \delta/r_0)^m, \\ f(r_0/S) &= Sr_0^{m-1}(1/S + \delta/r_0)^m \leq S(S + \delta/r_0)^m. \end{aligned}$$

The linear perturbation is at most $b\delta S/r_0$.

The product A_+A_- lies in the disc of radius $2bs + s^2 = bd_s < b^2$ about b^2 , so it admits a square root B_s with positive real part. It satisfies $|B_s - b| \leq d_s$ because $|B_s + b| \geq b$. The complex radius is rB_s , with positive real part and $|\operatorname{Im}(rB_s)| \leq d_s$. The positive cosine expansion of the solid harmonic gives $|H_l(X_1, Y_\psi)| \leq (rH_s)^l$ and $|G_l(X_1/\mathcal{R})| \leq A_s^l$.

For $\nu > 0$, $\operatorname{Re} z > 0$, Schläfli's integral [22, 10.9.6] gives

$$|J_\nu(z)| \leq e^{|\operatorname{Im} z|} + \frac{1}{\pi\nu}.$$

Indeed the finite cosine integral has modulus at most $e^{|\operatorname{Im} z|}$, and the remaining integral is bounded by $\pi^{-1} \int_0^\infty e^{-\nu t} dt$. The normalised Poisson integral also applies at the removable origin. Writing $l = 2j$, $\nu = l + \lambda$ and $P_\nu = (\rho/2)^\nu / \Gamma(\nu + 1)$, a uniform field majorant is

$$(28.5) \quad M_* = \sum_j \frac{|a_j|}{s_j} \left[P_\nu z_*^l e^{\rho z_*} + \min \left\{ (r_0(b - d_s))^{-\lambda} A_s^l \left(e^{\rho d_s} + \frac{1}{\pi\nu} \right), P_\nu H_s^l e^{\rho d_s} \right\} \right].$$

The first term covers $r \leq r_0$ by the entire solid-harmonic representation; the minimum covers $r \geq r_0$ by the two Bessel bounds. Adding these nonnegative bounds enlarges their maximum.

Angular Cauchy estimates and Lemma 25.2 give, with $C = \kappa(S + R)/(S - R)$,

$$(28.6) \quad \begin{aligned} U_b &= CM_*, & X_b &= 2U_b/\delta, & Y_b &= RX_b, \\ XX_b &= 6U_b/\delta^2, & YX_b &= RXX_b, & YY_b &= R^2XX_b. \end{aligned}$$

These bound U_* , $(U_*)_{x_1}$, $y(U_*)_y$, $(U_*)_{x_1x_1}$, $y(U_*)_{x_1y}$, $y^2(U_*)_{yy}$ in Y or $\mathcal{Y}_n$ according to parity. For the first derivative the two independent complex coordinates give the factor two; their second derivatives and twice the mixed derivative give six by the polydisc Cauchy estimates. The factors y, y^2 are multiplied in the raw Fourier algebra before conversion of the resulting regular scalar. Thus y itself is not treated as an analytic scalar on B^n . For these fixed-field estimates use $S = (102/100)R$ and $\delta = (1000\rho)^{-1}$. The boundary moment estimates use their own larger strip radius S .

28.3. Boundary coefficients and their entire tails. Let K and $J_b < K$ be the boundary sample and retained coefficient parameters. Reflection of the values at $\theta_k = (k + 1/2)\pi/(2K)$, $0 \leq k < K$, gives $2K$ equispaced samples in $\phi = 2\theta$. For D and N the complex strip majorants are $M_D = M_* + 1$, $M_N = 2SM_*/\delta$. The corresponding continuum cosine coefficient of frequency $2j$ has modulus at most $2M_{D,N}S^{-2j}$. The exact discrete Fourier transform, with its half-grid phase, differs at each retained index $0 \leq j \leq J_b$ by at most

$$(28.7) \quad e_Q = \frac{4M_{D,N}S^{-2(2K-J_b)}}{1 - S^{-4K}}.$$

This is the alias sum at indices $j + 2Kq$, as in (3.14); the constant mode is included with the enlarged same bound. Above J_b use the full continuum coefficient bound.

Define the actual Gegenbauer moments

$$(28.8) \quad D^{[k]} = \sum_l h_l(l+1)^k |D_l|, \quad N^{[k]} = \sum_l h_l(l+1)^k |N_l|, \quad 0 \leq k \leq 3.$$

The connection lowers degree and hence the moment factor. Below, $D^{[k]}, N^{[k]}$ denote the upper bounds obtained by multiplying the retained cosine moments $\sum_{j=0}^{J_b} R^{2j}(2j+1)^k(|d_j| + e_Q)$ and the full tail of (28.9) by κ , where d_j denotes the corresponding cosine coefficient of D or N . All

later constants are monotone in these bounds. For $q_{\text{tail}} = (R/S)^2$, expand $(2(J_b + 1 + t) + 1)^k$ and use

$$(28.9) \quad \begin{aligned} \sum_{t \geq 0} q_{\text{tail}}^t &= \frac{1}{1 - q_{\text{tail}}}, & \sum_{t \geq 0} t q_{\text{tail}}^t &= \frac{q_{\text{tail}}}{(1 - q_{\text{tail}})^2}, \\ \sum_{t \geq 0} t^2 q_{\text{tail}}^t &= \frac{q_{\text{tail}}(1 + q_{\text{tail}})}{(1 - q_{\text{tail}})^3}, & \sum_{t \geq 0} t^3 q_{\text{tail}}^t &= \frac{q_{\text{tail}}(1 + 4q_{\text{tail}} + q_{\text{tail}}^2)}{(1 - q_{\text{tail}})^4}. \end{aligned}$$

The geometric coefficient decay also shows that the lift in (28.3) converges with all its derivatives on a neighbourhood of the closed ball.

Direct differentiation of (28.2) gives

$$\Delta_n h_l^D = -l(2l + n)H_l, \quad \Delta_n h_l^N = (2l + n)H_l.$$

Consequently set

$$(28.10) \quad \Lambda_k = 2^k(D^{[k+1]} + N^{[k]}) \quad (0 \leq k \leq 2), \quad \Lambda_k^\Delta = (n + 2)(D^{[k+2]} + N^{[k+1]}) \quad (0 \leq k \leq 1).$$

These bound the degree moments $\mathbf{m}_k(\ell)$ and $\mathbf{m}_k(\Delta_n \ell)$. For the first use that a lift monomial has degree at most $l + 2 \leq 2(l + 1)$, with total absolute coefficient at most $l + 1$ for h_l^D and one for h_l^N . For the second use $2l + n \leq (n + 2)(l + 1)$. The radial factors themselves also give

$$\|\ell\|_Y \leq D^{[0]} + N^{[0]}/2 :$$

the Dirichlet lift increases to one on $[0, 1]$, and the Neumann lift has modulus at most $1/2$.

28.4. Residual and centre derivatives. Let Q_*, X_*, Y_*, P_* bound $q, q_{x_1}, q_y/y, |p_0|^2$ in $\mathcal{Y}_n$, and let $Q^{[1]}, X^{[1]}, Y^{[1]}, P^{[1]}$ bound their total-degree coefficient moments. Here X_*, Y_* are the bounds of Section 27.3, and $Q_* = b + q_b$. They arise from the finite shape identities (26.14) and the full product

$$|p_0|^2 = \sum_{i,j} (2i + 1)(2j + 1) c_{2i+1} c_{2j+1} r^{2(i+j)} \cos(2(i - j)\theta).$$

Retained connection coefficients can be converted directly; Lemma 25.2 bounds every higher angular degree. Products add degree moments with nonnegative majorants.

Since $BU_* = 0$, the actual residual is $E = F(x_0) = -B\ell$. The product and derivative bounds give

$$(28.11) \quad E_0 = \frac{Q_* \Lambda_0^\Delta + a(X_* C_x^{\text{pol}} + Y_* C_y^{\text{pol}}) \Lambda_1 + \mu Q_* P_* \Lambda_0}{b} \geq \|E\|_Y$$

and the coefficient degree moment bound

$$(28.12) \quad \begin{aligned} E_d &= \frac{1}{b} [Q^{[1]} \Lambda_0^\Delta + Q_* \Lambda_1^\Delta \\ &\quad + a\{(X^{[1]} C_x^{\text{pol}} + Y^{[1]} C_y^{\text{pol}}) \Lambda_1 + (X_* C_x^{\text{pol}} + Y_* C_y^{\text{pol}}) \Lambda_2\} \\ &\quad + \mu\{(Q^{[1]} P_* + Q_* P^{[1]}) \Lambda_0 + Q_* P_* \Lambda_1\}]. \end{aligned}$$

Thus

$$\|E_{x_1}\|_{\mathcal{Y}_n} \leq C_x^{\text{pol}} E_d, \quad \|y E_y\|_Y \leq C_y^{\text{pol}} E_d, \quad \|\partial_\theta E|_\partial\|_{W_0} \leq E_d.$$

The inverse below uses derivatives of the fixed centre only. Define $A^q = q_{x_1}/q$, $B^q = (q_y/y)/q$, $C^q = \mu|p_0|^2$. With $\mathfrak{J}$ of (27.7), upper bounds for these coefficients and their derivatives are

$$(28.13) \quad \begin{aligned} A_*^q &= X_* \mathfrak{J}, & B_*^q &= Y_* \mathfrak{J}, & C_*^q &= \mu P_*, \\ A_x^q &= C_x^{\text{pol}} X^{[1]} \mathfrak{J} + X_*^2 \mathfrak{J}^2, & A_y^q &= C_y^{\text{pol}} X^{[1]} \mathfrak{J} + X_* Y_2 Y_* \mathfrak{J}^2, \\ B_x^q &= C_x^{\text{pol}} Y^{[1]} \mathfrak{J} + Y_* X_* \mathfrak{J}^2, & B_y^q &= C_y^{\text{pol}} Y^{[1]} \mathfrak{J} + Y_*^2 Y_2 \mathfrak{J}^2, \\ C_x^q &= \mu C_x^{\text{pol}} P^{[1]}, & C_y^q &= \mu C_y^{\text{pol}} P^{[1]}. \end{aligned}$$

Here subscripts x and y indicate ∂_{x_1} and $y\partial_y$ respectively. Use $\Delta_n U_* = -a(A^q(U_*)_{x_1} + B^q y(U_*)_y) - C^q U_*$ and (28.6) to define

$$(28.14) \quad \begin{aligned} \Gamma_b &= a(A_*^q X_b + B_*^q Y_b) + C_*^q U_b, \\ \Gamma_x &= a(A_x^q X_b + A_*^q X X_b + B_x^q Y_b + B_*^q Y X_b) + C_x^q U_b + C_*^q X_b, \\ \Gamma_y &= a(A_y^q X_b + A_*^q Y X_b + B_y^q Y_b + B_*^q (Y_b + Y Y_b)) + C_y^q U_b + C_*^q Y_b. \end{aligned}$$

The term $Y_b + Y Y_b$ is the derivative of $y(U_*)_y$ by $y\partial_y$. Finally the bounds for the clamped centre are

$$(28.15) \quad \begin{aligned} U_c &= U_b + D^{[0]} + N^{[0]}/2, & U_x &= X_b + C_x^{\text{pol}} \Lambda_1, & U_y &= Y_b + C_y^{\text{pol}} \Lambda_1, \\ \Gamma_0 &= \Gamma_b + \Lambda_0^\Delta, & \Gamma_{0x} &= \Gamma_x + C_x^{\text{pol}} \Lambda_1^\Delta, & \Gamma_{0y} &= \Gamma_y + C_y^{\text{pol}} \Lambda_1^\Delta. \end{aligned}$$

They bound $U_0, (U_0)_{x_1}, y(U_0)_y, g_0, (g_0)_{x_1}, y(g_0)_y$ in their respective parity spaces.

29. THE EXACT INVERSE OF THE CORRECTED LINEARISATION

29.1. The material identity, including the axis term. All coefficients here are evaluated at x_0 . For $\delta p \in W$ define the odd real-coefficient holomorphic functions

$$\eta(w) = \int_0^w \delta p(z) dz, \quad v = \eta/p_0.$$

The associated full-ball vector field is $(\text{Re } v, (\text{Im } v/y)x')$, regular on the axis. Its action on U_0 is $T = \text{Re } v (U_0)_{x_1} + \text{Im } v (U_0)_y$.

To derive the material identity, let Θ_t vary the shape, $m_t = q_t |p_t|^2/b$, and keep the function $u_0 = U_0 \circ \Theta_0^{-1}$ fixed. The centre is analytic on a neighbourhood of the closed ball, so this function is defined on a neighbourhood as well. Equivalently the following differentiation is a local differential identity. Differentiating $B_t(u_0 \circ \Theta_t) = m_t((\Delta + \mu)u_0) \circ \Theta_t$ gives

$$(29.1) \quad BT + (\delta B)U_0 = v \cdot \nabla E + (\delta \log m - v \cdot \nabla \log m)E.$$

For example the last transport term before expansion is $m v \cdot \nabla(E/m)$, which explains the minus sign.

The cancellation for $|p|^2$ uses $\eta' = p'v + pv'$:

$$\delta \log |p|^2 - v \cdot \nabla \log |p|^2 = 2 \text{Re } v'.$$

For q one has, with $Y_\psi = yq = \text{Im } \psi$, $v \cdot \nabla Y_\psi = \text{Im}(pv) = \text{Im } \eta$. Hence $v \cdot \nabla q = \text{Im } \eta/y - q \text{Im } v/y$ and

$$\delta \log q - v \cdot \nabla \log q = \text{Im } v/y.$$

Define

$$(29.2) \quad \mathcal{E}(\delta p) = \text{Re } v E_{x_1} + \text{Im } v E_y + (2 \text{Re } v' + \text{Im } v/y)E, \quad \tilde{D} = DF(x_0) - \mathcal{E}.$$

The coefficient of $\text{Im } v/y$ is one, independent of n , because the multiplier in (24.5) contains one factor q . The dimension enters the operator B through $a = n - 2$. For the divergence

multiplier $|p|^2 q^a$, the same calculation gives coefficient $a = n - 2$ instead. Equations (25.10) and (29.1) now imply

$$(29.3) \quad \tilde{D}(\delta g, \delta p) = B(K_n \delta g - T).$$

The material mechanism is the Hadamard shape derivative used in Proposition 3.1; see, for example, [40, 65].

29.2. Boundary reconstruction and both inverse identities. Exact clamping gives $\nabla U_0 = 0$ on the sphere. Differentiating this identity in each tangential direction shows that the Hessian annihilates every tangent vector. By symmetry of the Hessian,

$$D^2 U_0 = (\Delta_n U_0) \mathbf{n} \otimes \mathbf{n} \quad \text{on } \partial B^n.$$

Thus, with $w = e^{i\theta}$ on the meridian circle,

$$(29.4) \quad \begin{aligned} T &= 0, & \partial_r T &= -d_\partial \operatorname{Re}(v\bar{w}), \\ d_\partial &= -\Delta_n U_0|_\partial = \mu|p_0|^2 - bE/q. \end{aligned}$$

The term from differentiating v multiplies ∇U_0 and vanishes. At the poles these identities are full-ball identities, hence also hold by continuity.

Assume that B_D^{-1} exists as proved above, p_0^{-1} belongs to the holomorphic weighted algebra, and $d_\partial^{-1} \in W_1$. For $f \in Y$ solve

$$(29.5) \quad B\Phi = f, \quad \Phi|_\partial = 0, \quad N_\partial = \partial_r \Phi, \quad H_\partial = N_\partial/d_\partial.$$

Write its cosine coefficients as $H_\partial = H_{\partial,0} + \sum_{j>0} H_{\partial,2j} \cos(2j\theta)$ and set

$$(29.6) \quad \mathbf{a}(w) = H_{\partial,0} + \sum_{j>0} H_{\partial,2j} w^{2j}, \quad v = w\mathbf{a}.$$

Then $\operatorname{Re}(v\bar{w}) = H_\partial$ on the circle and $\|v'\|_W = \|\mathbf{a}\|_{W_1} = \|H_\partial\|_{W_1}$. The holomorphic real-part problem is the one of (3.35), with cosine coefficients in place of paired complex Fourier coefficients. Its real-coefficient normalisation excludes the imaginary constant; the real constant $H_{\partial,0}$ determines the dilation variation.

Proposition 29.1 (Two-sided inverse). *Under these hypotheses the corrected derivative $\tilde{D} : X \rightarrow Y$ has a bounded two-sided inverse $A : Y \rightarrow X$. It is given by*

$$(29.7) \quad \delta\psi = p_0 v, \quad \delta p = (p_0 v)', \quad \delta U = \Phi + T, \quad \delta g = \Delta_n \delta U, \quad Af = (\delta g, \delta p).$$

Proof. Equation (29.4) makes both traces of δU zero, since $\partial_r \delta U = N_\partial - d_\partial H_\partial = 0$. The Laplacian estimate in the next subsection proves $\delta g \in Y$; the moment formula therefore gives $\delta g \in \mathcal{G}$ and Dirichlet uniqueness gives $K_n \delta g = \delta U$. The shape variation is even holomorphic with real coefficients. Then (29.3) gives $\tilde{D}Af = f$.

For the other identity, take any $(\delta g, \delta p) \in X$, form its η, v, T , and put $\Phi = K_n \delta g - T$. It has zero Dirichlet trace and satisfies $B\Phi = \tilde{D}(\delta g, \delta p)$. Dirichlet uniqueness recovers precisely this Φ when the recipe is applied to that right side. Its normal trace is $N_\partial = d_\partial \operatorname{Re}(v\bar{w})$. The function $\mathbf{a} = v/w = \eta/(wp_0)$ has a removable singularity at zero. It belongs to W_1 , since $\|\eta/w\|_{W_1} = \|\delta p\|_W$ and $p_0^{-1} \in W_1$ by differentiation of its reciprocal. The unique real-coefficient real-part solution therefore recovers $\mathbf{a}$ and then $v, \delta p, \delta U, \delta g$. Hence $A\tilde{D} = I_X$ as well as $\tilde{D}A = I_Y$. The bounds below prove boundedness on the entire spaces. $\square$

The sector is essential to the Dirichlet inverse. For a Schiffer solution, physical translation derivatives solve the homogeneous Dirichlet equation. The axial derivative is odd in the first coordinate and the transverse derivatives are not invariant under $O(n-1)$, so they lie outside this sector.

29.3. Cancellation of higher shape derivatives. Cauchy–Riemann cancellation gives

$$(29.8) \quad \begin{aligned} \Delta_n T &= v \cdot \nabla g_0 + 2 \operatorname{Re} v' g_0 - a \frac{\operatorname{Im} v'}{y} (U_0)_{x_1} \\ &\quad + a \frac{\operatorname{Im} v/y - \operatorname{Re} v'}{y^2} y (U_0)_y. \end{aligned}$$

To see the axis terms directly, write $A_v = \operatorname{Re} v$, $C_v = \operatorname{Im} v$. The planar Laplacian gives $\Delta_2(v \cdot \nabla U_0) = v \cdot \nabla \Delta_2 U_0 + 2 \operatorname{Re} v' \Delta_2 U_0$. For the remaining operator $ay^{-1}\partial_y$, subtracting its transport derivative and $2 \operatorname{Re} v'$ times that operator leaves

$$a \frac{(A_v)_y}{y} (U_0)_{x_1} + a \left(\frac{(C_v)_y - 2 \operatorname{Re} v'}{y} + \frac{C_v}{y^2} \right) (U_0)_y.$$

Use $(A_v)_y = -\operatorname{Im} v'$ and $(C_v)_y = \operatorname{Re} v'$ to obtain (29.8). Thus no second derivative of the unknown holomorphic variation appears.

For the odd monomial $v = w^j$, $j \geq 3$, the regular quotient is

$$\frac{\operatorname{Im} v/y - \operatorname{Re} v'}{y^2} = 2r^{j-3} C_{j-3}^{(2)}(\xi);$$

for $j = 1$, both $\operatorname{Im} v'/y$ and $(\operatorname{Im} v/y - \operatorname{Re} v')/y^2$ vanish. The positive cosine expansions of U_m and $C_m^{(2)}$, summed with weight R , give

$$(29.9) \quad \begin{aligned} c_a &= \frac{2}{R(1-R^{-2})}, & c_v &= \frac{4}{R^2(1-R^{-2})^2}, \\ \|\operatorname{Im} v'/y\|_{\text{raw}} &\leq c_a \|v'\|_W, & \|(\operatorname{Im} v/y - \operatorname{Re} v')/y^2\|_{\text{raw}} &\leq c_v \|v'\|_W, \\ \|\operatorname{Im} v/y\|_{\text{raw}} &\leq \|v'\|_W, & \|\operatorname{Re} v\|_{\text{raw}} &\leq R \|v'\|_W. \end{aligned}$$

For the first inequality use the sum of the coefficients of U_m with decreasing powers R^{-2k} , bounded by $2/(1-R^{-2})$ after the leading power. For the second use the Gegenbauer generating coefficients $(2)_k/k! = k+1$; their weighted sum is $(1-R^{-2})^{-2}$. The leading powers and division by the derivative coefficient j give the stated enlarged c_a, c_v . For $\operatorname{Im} v/y$ the unweighted sum of the coefficients of U_{j-1} is j , and every frequency is at most $j-1$. The last inequality follows coefficientwise from $R^j/j \leq RR^{j-1}$. Lemma 25.2 converts each regular scalar once. Consequently

$$(29.10) \quad M_T = \kappa [R\Gamma_{0x} + \Gamma_{0y} + 2\Gamma_0 + ac_a U_x + ac_v U_y]$$

bounds $\|\Delta_n T\|_Y$ per unit $\|v'\|_W$. This proves the field membership required in Proposition 29.1 without changing the analytic radius.

29.4. Reciprocals, inverse norm and left Newton defect. Define the finite shape quantities

$$P = \|p_0\|_W, \quad P' = \sum_{m>0} m |(p_0)_m| R^{m-1}, \quad q_0 = 2b - P > 0.$$

The holomorphic reciprocal of p_0 and the raw scalar reciprocal of q have norm at most $1/q_0$, by Neumann series about b . Set

$$(29.11) \quad \begin{aligned} d_* &= \mu(2b^2 - P^2) - bE_0/q_0 > 0, \\ D_\theta &= 2\mu PRP' + b(E_d/q_0 + E_0RP'/q_0^2), \\ I_0 &= d_*^{-1}, \quad I_1 = I_0 + I_0^2 D_\theta. \end{aligned}$$

Indeed $\|d_\partial - \mu b^2\|_{W_0} \leq \mu(P^2 - b^2) + bE_0/q_0 < \mu b^2$, so $\|d_\partial^{-1}\|_{W_0} \leq I_0$. Since $\|\partial_\theta q\|_{W_0} \leq RP'$, differentiating d_∂ gives the bound D_θ . Differentiating $d_\partial d_\partial^{-1} = 1$ then gives $\|d_\partial^{-1}\|_{W_1} \leq I_1$. This is the weaker-norm reciprocal argument in Proposition 3.1; smallness in W_1 is unnecessary.

By (29.5) and (27.8), $\|\mathbf{a}\|_{W_1} \leq I_1 N_W \|f\|_Y$. Equation (29.7) therefore gives

$$(29.12) \quad B_g = J_W + M_T I_1 N_W, \quad B_p = (P + RP') I_1 N_W, \quad A_* = B_g + \sigma B_p, \quad \|A\|_{Y \rightarrow X} \leq A_*.$$

Here $\delta p = (wp_0 \mathbf{a})'$ has the norm of $p_0 \mathbf{a}$ in W_1 . For an arbitrary input variation, $v = \eta/p_0$ gives

$$(29.13) \quad \|v'\|_W \leq J_v \|\delta p\|_W, \quad J_v = q_0^{-1} + RP' q_0^{-2},$$

using $\|\eta\|_W \leq R \|\delta p\|_W$. The material correction has the bound

$$(29.14) \quad \|\mathcal{E}\|_{X \rightarrow Y} \leq \varepsilon_{\mathcal{E}} = \frac{\kappa}{\sigma} \left[(RC_x^{\text{pol}} + C_y^{\text{pol}}) E_d + 3E_0 \right] J_v.$$

The factor three is $2 + 1$ from the last two terms of (29.2). The transport terms use $\text{Im } v E_y = (\text{Im } v/y) y E_y$. Since $A\tilde{D} = I_X$, the left Newton defect is

$$(29.15) \quad \|AF(x_0)\|_X \leq Y_0 = A_* E_0, \quad \|I_X - ADF(x_0)\|_{X \rightarrow X} \leq Z = A_* \varepsilon_{\mathcal{E}}.$$

The field right sides need only be continuous. Their Dirichlet potentials have C^1 traces by the preceding Poisson and elliptic bounds, whereas T is analytic near the closed ball. Thus the identities apply to every element of X and Y .

n	A_*	Y_0	Z	$4C_2 Y_0$	$\mathbf{c}(r_*)$
5	$< 1.54023 \cdot 10^{32}$	$< 5.5497 \cdot 10^{-40}$	$< 6.72503 \cdot 10^{-55}$	$< 1.35527 \cdot 10^{-22}$	$< 1.22103 \cdot 10^{-13}$
9	$< 6.28334 \cdot 10^{52}$	$< 2.43517 \cdot 10^{-59}$	$< 1.64019 \cdot 10^{-83}$	$< 1.54869 \cdot 10^{-29}$	$< 3.17983 \cdot 10^{-14}$
11	$< 5.78969 \cdot 10^{53}$	$< 2.16886 \cdot 10^{-46}$	$< 6.87444 \cdot 10^{-70}$	$< 3.78775 \cdot 10^{-16}$	$< 8.73215 \cdot 10^{-9}$
12	$< 1.93402 \cdot 10^{47}$	$< 7.70138 \cdot 10^{-76}$	$< 4.65561 \cdot 10^{-98}$	$< 1.12753 \cdot 10^{-50}$	$< 7.32031 \cdot 10^{-26}$
13	$< 3.97845 \cdot 10^{67}$	$< 1.87769 \cdot 10^{-67}$	$< 4.3153 \cdot 10^{-98}$	$< 1.8351 \cdot 10^{-29}$	$< 4.88659 \cdot 10^{-13}$
15	$< 4.68743 \cdot 10^{66}$	$< 5.77465 \cdot 10^{-54}$	$< 5.15043 \cdot 10^{-80}$	$< 1.14824 \cdot 10^{-12}$	$< 9.94207 \cdot 10^{-9}$
16	$< 1.10148 \cdot 10^{98}$	$< 2.71173 \cdot 10^{-87}$	$< 2.91643 \cdot 10^{-129}$	$< 4.3328 \cdot 10^{-30}$	$< 7.98901 \cdot 10^{-16}$
17	$< 4.89228 \cdot 10^{70}$	$< 5.21583 \cdot 10^{-71}$	$< 6.23668 \cdot 10^{-104}$	$< 3.8058 \cdot 10^{-32}$	$< 3.64832 \cdot 10^{-17}$
18	$< 1.45567 \cdot 10^{59}$	$< 1.45364 \cdot 10^{-70}$	$< 3.45744 \cdot 10^{-96}$	$< 1.25466 \cdot 10^{-36}$	$< 4.31556 \cdot 10^{-19}$
20	$< 6.37761 \cdot 10^{90}$	$< 1.05248 \cdot 10^{-91}$	$< 6.35578 \cdot 10^{-132}$	$< 2.07659 \cdot 10^{-40}$	$< 9.86528 \cdot 10^{-21}$
21	$< 7.92495 \cdot 10^{112}$	$< 1.6709 \cdot 10^{-96}$	$< 8.05097 \cdot 10^{-142}$	$< 2.77832 \cdot 10^{-28}$	$< 8.31387 \cdot 10^{-15}$

TABLE 21. Corrected inverse and contraction bounds. $\mathbf{c}$ is defined in (30.3).
The quantity $4C_2 Y_0$ is defined by (29.15) and (30.3).

30. QUARTIC BOUNDS AND THE EXACT ZERO

For a variation $(h, k) \in X$ write $u_h = K_n h$ and $q_k = \operatorname{Im} \int_0^w k(z) dz/y$. Lemma 25.2 and the axis multiplier estimates give

$$(30.1) \quad \begin{aligned} \|q_k\|_Y &\leq \kappa \|k\|_W, \quad \|(q_k)_{x_1}\|_{Y_n} \leq \kappa c_a \|k\|_W, \quad \|(q_k)_y/y\|_Y \leq \kappa c_v \|k\|_W, \\ \|p_0 \bar{k} + \bar{p}_0 k\|_Y &\leq 2\kappa P \|k\|_W, \quad \| |k|^2 \|_Y \leq \kappa \|k\|_W^2. \end{aligned}$$

The last two estimates multiply in the raw angular algebra before conversion. These bounded linear operators, together with Lemma 25.4, make $F : X \rightarrow Y$ a continuous quartic polynomial on this fixed Banach space.

To bound its remainder, put $P_k = p_0 \bar{k} + \bar{p}_0 k$. The terms of degree two, after removal of $E + DF(x_0)(h, k)$, are

$$\begin{aligned} b\mathcal{N}_2 &= q_k h + a\{(q_k)_{x_1}(u_h)_{x_1} + ((q_k)_y/y)y(u_h)_y\} \\ &\quad + \mu\{(qP_k + q_k|p_0|^2)u_h + (q|k|^2 + q_k P_k)U_0\}. \end{aligned}$$

The terms of degrees three and four are

$$b\mathcal{N}_3 = \mu\{(q|k|^2 + q_k P_k)u_h + q_k |k|^2 U_0\}, \quad b\mathcal{N}_4 = \mu q_k |k|^2 u_h.$$

Projection to the field has norm at most one and projection to the shape has norm at most $1/\sigma$ in X . The algebra, Poisson, and centre estimates therefore give

$$(30.2) \quad \begin{aligned} k_2 &= \frac{1}{b} \left[\frac{\kappa}{\sigma} + \frac{a\kappa(c_a C_x^K + c_v C_y^K)}{\sigma} + \frac{\mu(2\kappa Q_* P + \kappa P_*)C_K}{\sigma} \right. \\ &\quad \left. + \frac{\mu(\kappa Q_* + 2\kappa^2 P)U_c}{\sigma^2} \right], \\ k_3 &= \frac{\mu}{b} \left[\frac{(\kappa Q_* + 2\kappa^2 P)C_K}{\sigma^2} + \frac{\kappa^2 U_c}{\sigma^3} \right], \\ k_4 &= \frac{\mu \kappa^2 C_K}{b \sigma^3}. \end{aligned}$$

In particular $\|\mathcal{N}_j(h, k)\|_Y \leq k_j \|(h, k)\|_X^j$. Each displayed product also defines a multilinear form with this same bound on the product of input norms. Symmetrisation averages over the positions of the field and shape inputs, and introduces no larger constant. Differentiating the diagonal form consequently bounds its derivative by $jk_j s^{j-1}$ on $\|(h, k)\|_X \leq s$, as in the polarisation argument of Section 4.

Set $C_j = A_* k_j$ and define

$$(30.3) \quad \begin{aligned} \mathfrak{p}(s) &= Y_0 + (Z - 1)s + C_2 s^2 + C_3 s^3 + C_4 s^4, \\ \mathfrak{c}(s) &= Z + 2C_2 s + 3C_3 s^2 + 4C_4 s^3. \end{aligned}$$

Proposition 30.1. *If $\mathfrak{p}(r_*) < 0$ and $\mathfrak{c}(r_*) < 1$, then F has exactly one zero in $\overline{B}_X(x_0, r_*)$.*

Proof. The proof of Lemma 4.1 applies to the additional quartic term. For $\mathcal{T}(x) = x - AF(x)$, the estimates give

$$\|\mathcal{T}(x_0 + h) - x_0\|_X \leq Y_0 + Z\|h\|_X + \sum_{j=2}^4 C_j \|h\|_X^j.$$

The first inequality makes the closed ball invariant. Its derivative is bounded there by $\mathfrak{c}(r_*) < 1$, so completeness gives a unique fixed point. The two-sided inverse A is injective, hence $AF(x) = 0$ implies $F(x) = 0$. Every zero is a fixed point. $\square$

n	E_0	C_2	C_3	C_4	$\mathbf{p}(r_*)$
5	$< 3.60318 \cdot 10^{-72}$	$< 6.10514 \cdot 10^{16}$	$< 9.55874 \cdot 10^{-4}$	$< 4.62346 \cdot 10^{-25}$	$< -9.99999 \cdot 10^{-31}$
9	$< 3.8756 \cdot 10^{-112}$	$< 1.58992 \cdot 10^{29}$	$< 1.70017 \cdot 10^0$	$< 8.49417 \cdot 10^{-35}$	$< -9.99999 \cdot 10^{-44}$
11	$< 3.74606 \cdot 10^{-100}$	$< 4.36608 \cdot 10^{29}$	$< 2.86311 \cdot 10^1$	$< 1.3967 \cdot 10^{-31}$	$< -9.99999 \cdot 10^{-39}$
12	$< 3.98207 \cdot 10^{-123}$	$< 3.66016 \cdot 10^{24}$	$< 4.94961 \cdot 10^{-3}$	$< 2.45652 \cdot 10^{-34}$	$< -9.9999 \cdot 10^{-51}$
13	$< 4.71966 \cdot 10^{-135}$	$< 2.4433 \cdot 10^{37}$	$< 1.20781 \cdot 10^1$	$< 5.97467 \cdot 10^{-44}$	$< -9.99999 \cdot 10^{-51}$
15	$< 1.23195 \cdot 10^{-120}$	$< 4.97104 \cdot 10^{40}$	$< 2.54826 \cdot 10^8$	$< 1.23371 \cdot 10^{-32}$	$< -9.99942 \cdot 10^{-50}$
16	$< 2.46191 \cdot 10^{-185}$	$< 3.99451 \cdot 10^{56}$	$< 6.61634 \cdot 10^9$	$< 3.30765 \cdot 10^{-49}$	$< -9.99999 \cdot 10^{-73}$
17	$< 1.06614 \cdot 10^{-141}$	$< 1.82416 \cdot 10^{38}$	$< 1.41374 \cdot 10^0$	$< 7.06073 \cdot 10^{-49}$	$< -9.99999 \cdot 10^{-56}$
18	$< 9.98606 \cdot 10^{-130}$	$< 2.15778 \cdot 10^{33}$	$< 3.60003 \cdot 10^1$	$< 1.72899 \cdot 10^{-39}$	$< -9.99999 \cdot 10^{-53}$
20	$< 1.65027 \cdot 10^{-182}$	$< 4.93264 \cdot 10^{50}$	$< 7.57525 \cdot 10^4$	$< 3.76177 \cdot 10^{-54}$	$< -9.99999 \cdot 10^{-72}$
21	$< 2.1084 \cdot 10^{-209}$	$< 4.15694 \cdot 10^{67}$	$< 5.53845 \cdot 10^{16}$	$< 2.74234 \cdot 10^{-48}$	$< -9.99999 \cdot 10^{-83}$

TABLE 22. The full residual, homogeneous remainder constants and negative radii polynomial, at the exact radius of Table 18.

Tables 21 and 22 give $\mathbf{p}(r_*) < 0$ and $\mathbf{c}(r_*) < 10^{-5}$, so Proposition 30.1 applies.

31. GEOMETRY ON THE WHOLE EXISTENCE BALL

Write $\psi_0 = \sum_{j \text{ odd}} c_j w^j$ and let $\psi = \psi_0 + \delta\psi$ be any shape in the existence ball. Then $\|\delta p\|_W \leq s_* = r_*/\sigma$. Define the unweighted centre sums

$$\begin{aligned}
 S_1 &= \sum_{j>1} j|c_j|, & S_2 &= \sum_{j>1} j(j-1)|c_j|, \\
 S_T &= \sum_{j>1} j^2|c_j| = S_1 + S_2, & H_R &= \frac{R}{(R-1)^2}, \\
 \beta_0 &= 2b - P - s_*, & L_* &= b - S_1 - s_*, & \chi_* &= b - S_1 - S_2 - (1 + H_R)s_*, \\
 \vartheta_3 &= |c_3| - \frac{s_*}{3R^2}.
 \end{aligned}
 \tag{31.1}$$

Table 23 gives strict positive lower bounds. Since $P - b \geq S_1$, $\beta_0 > 0$ also implies $L_* > 0$. The two sums S_T and $S_1 + S_2$ are identical, so the meridian and transverse estimates below use one common numerator χ_* .

31.1. Univalence and analytic lift. On $|w| \leq R$,

$$\operatorname{Re} p(w) \geq b - (P - b) - s_* = \beta_0 > 0.$$

On the closed unit disc the sharper coefficient estimate is

$$\operatorname{Re} \psi'(w) \geq b - S_1 - s_* = L_* > 0 \quad (|w| \leq 1).$$

The line-segment proof in Section 4 applies verbatim: for distinct points w_1, w_2 of the disc $|w| < R$,

$$\frac{\psi(w_2) - \psi(w_1)}{w_2 - w_1} = \int_0^1 p(w_1 + t(w_2 - w_1)) dt$$

has positive real part. Thus ψ is univalent on $|w| < R$. Also $q = \int_0^1 \operatorname{Re} p(x_1 + isy) ds > 0$ there, so ψ preserves the upper half-disc. The real-coefficient identities express $\operatorname{Re} \psi$ and q as convergent power series in x_1, y^2 . Replacing y^2 by $|x'|^2$ makes Θ_ψ analytic across the axis.

Off the axis the meridian differential and the $n - 2$ transverse differentials give $\det D\Theta_\psi = |p|^2 q^{n-2} > 0$. On the axis its differential is $\operatorname{diag}(p(x_1), p(x_1)I_{n-1})$, also nonsingular. If two

n	β_0	χ_*	ϑ_3
5	$> 9.72125 \cdot 10^{-1}$	$> 8.25031 \cdot 10^{-1}$	$> 4.45696 \cdot 10^{-4}$
9	$> 9.99203 \cdot 10^{-1}$	$> 9.93044 \cdot 10^{-1}$	$> 1.45326 \cdot 10^{-5}$
11	$> 9.76159 \cdot 10^{-1}$	$> 9.10902 \cdot 10^{-1}$	$> 1.02429 \cdot 10^{-3}$
12	$> 9.93062 \cdot 10^{-1}$	$> 9.75281 \cdot 10^{-1}$	$> 4.53906 \cdot 10^{-4}$
13	$> 9.8983 \cdot 10^{-1}$	$> 9.37264 \cdot 10^{-1}$	$> 5.67768 \cdot 10^{-4}$
15	$> 9.61839 \cdot 10^{-1}$	$> 8.32915 \cdot 10^{-1}$	$> 4.37252 \cdot 10^{-3}$
16	$> 9.99839 \cdot 10^{-1}$	$> 9.98507 \cdot 10^{-1}$	$> 2.66517 \cdot 10^{-6}$
17	$> 9.98964 \cdot 10^{-1}$	$> 9.94623 \cdot 10^{-1}$	$> 7.9917 \cdot 10^{-5}$
18	$> 9.4782 \cdot 10^{-1}$	$> 8.25454 \cdot 10^{-1}$	$> 7.7897 \cdot 10^{-3}$
20	$> 9.92479 \cdot 10^{-1}$	$> 9.64444 \cdot 10^{-1}$	$> 5.79555 \cdot 10^{-4}$
21	$> 9.90823 \cdot 10^{-1}$	$> 9.483 \cdot 10^{-1}$	$> 5.76413 \cdot 10^{-4}$

TABLE 23. Whole-ball geometry: analytic univalence, the common curvature numerator and the absolute w^3 coefficient. The numerator χ_* is the common numerator of (31.3) and (31.4).

lifted points have the same image, their meridian coordinates have the same image under ψ ; univalence gives equality of x_1, y , and positivity of q then gives equality of x' . Thus the lift is an analytic diffeomorphism on a neighbourhood of $\bar{B}^n$. Its restriction to the sphere gives a real-analytic boundary diffeomorphic to S^{n-1} . Oddness and reality give the required $O(n-1) \times \mathbb{Z}_2$ symmetry.

31.2. Meridian and transverse curvatures. For the perturbation,

$$\sup_{|w| \leq 1} |\delta p'| \leq \sup_{m \geq 1} m R^{-m} \|\delta p\|_W \leq H_R s_*.$$

Therefore $|p| \geq L_*$ and $|wp'| \leq S_2 + H_R s_*$ on the closed disc. The curvature formula (4.7) gives

$$(31.3) \quad \operatorname{Re}(1 + wp'/p) \geq \frac{\chi_*}{L_*} > 0, \quad \kappa_{\text{mer}} = \frac{\operatorname{Re}(1 + wp'/p)}{|p|} > 0.$$

The simple meridian boundary consequently encloses a strictly convex planar domain symmetric in both axes.

For the transverse curvatures and the graph property, write $\delta\psi = \sum_{j \text{ odd}} \delta c_j w^j$ and parameterise the upper arc as $\psi(e^{i\theta}) = (X(\theta), Y_\psi(\theta))$, $0 < \theta < \pi$. Its outward normal is $(Y'_\psi, -X')/|p|$ and speed is $|p|$. Using $U_m(\cos \theta) = \sin((m+1)\theta)/\sin \theta$ gives

$$(31.4) \quad -\frac{X'}{\sin \theta} = \sum_{j \text{ odd}} j(c_j + \delta c_j) U_{j-1}(\cos \theta) \geq b - S_T - (1 + H_R) s_* = \chi_*.$$

Indeed $|U_{j-1}| \leq j$, and

$$\sum_{j \text{ odd}} j^2 |\delta c_j| = \sum_{m \text{ even}} (m+1) |\delta p_m| \leq (1 + H_R) \|\delta p\|_W.$$

Since $Y_\psi/\sin \theta = q > 0$, each of the $n-2$ transverse principal curvatures of the unscaled body is

$$(31.5) \quad \kappa_\perp = \frac{-X'}{|p| Y_\psi} = \frac{-X'/\sin \theta}{|p|(Y_\psi/\sin \theta)} \geq \frac{\chi_*}{(b + S_1 + s_*)^2} > 0.$$

Both denominators have the displayed upper bound by the coefficient and integral estimates for p, q . The quotients extend to the poles analytically. At $\theta = 0$, $Y'_\psi = p(1) > 0$ and $-X'/\sin\theta \rightarrow -X''(0) = p(1)+p'(1)$, so $\kappa_\perp$ tends to $(p(1)+p'(1))/p(1)^2$, the meridian curvature there. The other pole has the same conclusion by symmetry. Thus every principal curvature is positive on the whole boundary. Scaling by ρ divides all principal curvatures by ρ .

31.3. Global convexity and the non-ball coefficient. By (31.4), $X'(\theta) < 0$ on the open upper arc. It is therefore a graph $Y_\psi = f(X)$ between its two poles. The curvature numerator of this oriented graph is $X'Y''_\psi - Y'_\psi X'' = (X')^3 f''$; positive meridian curvature and $X' < 0$ give $f'' < 0$ in the interior. Continuity extends its concavity to the closed interval, where f vanishes at the endpoints and is positive inside. The lifted domain is exactly

$$\Theta_\psi(B^n) = \{(X_1, X_\perp) : X_1 \text{ between the poles, } X_\perp \in \mathbb{R}^{n-1}, |X_\perp| < f(X_1)\}.$$

For two interior points $\mathbf{X} = (X_1, X_\perp)$ and $\tilde{\mathbf{X}} = (\tilde{X}_1, \tilde{X}_\perp)$ and $0 < t < 1$, the triangle inequality and concavity give

$$\begin{aligned} |(1-t)X_\perp + t\tilde{X}_\perp| &\leq (1-t)|X_\perp| + t|\tilde{X}_\perp| \\ &< (1-t)f(X_1) + tf(\tilde{X}_1) \\ &\leq f((1-t)X_1 + t\tilde{X}_1). \end{aligned}$$

Hence the body is convex globally. Its smooth boundary has positive principal curvatures by (31.3) and (31.5), so it contains no boundary line segment and is strictly convex. This is the solid-of-revolution argument used in Sections 22.5 and 23.5.

Finally the coefficient estimate gives $|c_3 + \delta c_3| \geq \vartheta_3 > 0$. A ball invariant under $O(n-1)$ and first-coordinate reflection has centre zero. Its meridian section would be a centred disc. The odd conformal map fixing zero with positive derivative there would then be linear, by uniqueness of the normalised conformal map, contradicting its nonzero w^3 coefficient. This excludes balls throughout the shape projection of the existence ball. The origin $\Theta_\psi(0)$ is an interior point of the convex domain. Every tangent hyperplane supports the domain and therefore does not contain the origin. Every ray from the origin meets the boundary exactly once and transversally, so Ω is strictly star-shaped with respect to the origin.

32. THE SCHIFFER FIELD AND THE POMPEIU CONCLUSION

Proof of Theorem 24.1. For each $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\}$, the centre satisfies $c = \rho b > 20.69$, $q_* > 0.9548$ and $\gamma > 299.4$. The window of Lemma 26.1 has the state count of Table 20. The hypotheses (26.9) hold with $\epsilon_C < 2.7 \cdot 10^{-10}$, $\epsilon_S + \eta_S < 0.0242$ and $B_{\text{pol}}\delta_V < 1.1 \cdot 10^{-4}$, so Proposition 26.2 gives B_{L^2} . With the tabulated L_0 , one has $\beta < 0.4987$ and $q_b < 0.3412 < b$. Thus (27.5) and Proposition 27.1 give B_Y, J_W, N_W . For both strip radii, $d_s < 0.13b$ in (28.4); also $q_0 > 0.9478$ and $d_* > 387.0$ in (29.11). Proposition 29.1 therefore applies with the bound A_* of Table 21. The strict radii inequalities in Tables 21 and 22 give the unique zero (g, p) in $\overline{B}_X(x_0, r_*)$ by Proposition 30.1. Put $U = 1 + K_n g$. Lemma 25.4 gives $U \in W^{2,p}(B^n)$ for every finite p , with $U = 1$ and $\partial_r U = 0$ on the sphere. Its tangential derivatives also vanish there, so $\nabla U = 0$. Since $U \in W^{2,p}$, $F(g, p) = 0$ says $BU = 0$ almost everywhere. Multiplication by bq^{a-1} and integration by parts give

$$\int_{B^n} q^a (\nabla U \cdot \nabla \varphi - \mu |p|^2 U \varphi) dx = 0 \quad (\varphi \in C_c^\infty(B^n)).$$

The analytic diffeomorphism of Section 31.1 and the change of variables of Lemma 22.1(b), with $n-2$ transverse directions, give $u_0 = U \circ \Theta_\psi^{-1} \in C^1(\overline{\Theta_\psi(B^n)})$, a weak solution of $(\Delta + \mu)u_0 = 0$

with $u_0 = 1$ and $\nabla u_0 = 0$ on the boundary. Set

$$\Omega = \rho \Theta_\psi(B^n), \quad u(X) = u_0(X/\rho).$$

Since $\mu = \rho^2$, $u \in C^1(\overline{\Omega})$ solves $\Delta u + u = 0$ weakly with $u = 1$, $\nabla u = 0$ on the real-analytic boundary. Lemma 21.26 gives a real-analytic extension to a neighbourhood of $\overline{\Omega}$, solving (24.1). It cannot be constant, because its boundary value would make it the constant one, which does not solve $(\Delta + 1)u = 0$. Sections 31.1–31.3 give all geometric assertions and the symmetry.

The Green identity of Section 33 applies in every dimension. For a unit vector $\mathbf{k}$ put $v(X) = e^{i\mathbf{k} \cdot X}$. Then

$$0 = \int_{\Omega} (u \Delta v - v \Delta u) = \int_{\partial\Omega} \partial_{\mathbf{n}} v = - \int_{\Omega} e^{i\mathbf{k} \cdot X} dX.$$

Thus $\widehat{\mathbf{1}}_{\Omega}$ vanishes on the unit sphere. For a rigid motion $X \mapsto QX + A$,

$$\int_{Q\Omega+A} e^{iX_1} dX = e^{iA_1} \int_{\Omega} e^{i(Q^T e_1) \cdot X} dX = 0.$$

Taking real parts gives zero integrals of the nonzero continuous function $\cos X_1$ over every rigid motion of Ω . Hence Ω fails the Pompeiu property. $\square$

Part 6. Consequences, computations and remarks

33. THE POMPEIU PROPERTY

Proof of Corollary 1.2. Let n satisfy $2 \leq n \leq 18$ or $n \in \{20, 21\}$, let Ω and u be as in Theorem 1.4 or Theorem 1.1 (proved in Sections 4, 21, 22, 23, 18, 19, 20 and 32), let $\xi \in \mathbb{R}^n$ with $|\xi| = 1$, and let $v(x) = e^{i\xi \cdot x}$, so that $\Delta v + v = 0$. Since u and v are smooth up to the real-analytic boundary and solve the same equation, Green's identity and (1.1) give

$$0 = \int_{\Omega} (u \Delta v - v \Delta u) dx = \int_{\partial\Omega} (u \partial_{\nu} v - v \partial_{\nu} u) dS = \int_{\partial\Omega} \partial_{\nu} v dS = \int_{\Omega} \Delta v dx = - \int_{\Omega} e^{i\xi \cdot x} dx.$$

Thus $\widehat{\mathbf{1}}_{\Omega}$ vanishes on the unit sphere. If $\sigma(x) = Qx + a$ is a rigid motion, then

$$\int_{\sigma(\Omega)} e^{ix_1} dx = e^{ia_1} \int_{\Omega} e^{i(Q^T e_1) \cdot x} dx = 0,$$

and taking real parts, $\int_{\sigma(\Omega)} \cos x_1 dx = 0$. Since $\cos x_1$ is continuous and not identically zero, Ω fails the Pompeiu property. By Theorems 1.4 and 1.1, Ω is a bounded domain, not a ball, whose boundary is real analytic and diffeomorphic to S^{n-1} , so the Pompeiu conjecture fails in $\mathbb{R}^n$; the domain is convex for every $2 \leq n \leq 18$ and for $n = 20, 21$, by Theorems 1.4 and 1.1. $\square$

34. COMPUTER-ASSISTED VERIFICATION

This section describes, for each dimension, what the scripts check and how the computations are reproduced; Section 34.12 describes the ancillary files. In all dimensions the inequalities are checked in the ball arithmetic of Arb [43], through `python-flint` 0.9.0 [30]: an order relation between balls is true only if it holds for all points of the balls, every check raises an exception when it fails, and NaN or infinite enclosures are rejected. The analytic content of the proofs, namely all lemmas and all tail estimates, is contained in Sections 2–22, Section 23 and Parts 5 and 2; the scripts only evaluate the finitely many explicit expressions defined there.

34.1. The plane.

34.1.1. *Inputs and arithmetic.* The ancillary file `verify_planar_convex_v2.py` rebuilds the numerical hypotheses used in the planar proof sections. Its inputs are two finite sets of exact decimal strings: the eight-mode reference shape and the forty-mode centre, the latter consisting of the arrays `p`, `a` and `scales`. Every decimal string is interpreted as an exact rational and enclosed without a binary64 conversion. In particular the centre’s `p` array contains the coefficients c_j of ψ_0 ; differentiation to obtain p_0 is exact. The field divisors `scales` are positive exact decimals.

The single-file verifier embeds the stage sources, the reference and the centre in a compressed bundle. Hashes identify both the bundle and each embedded file. It embeds no computed Bessel roots, spectral matrices or boundary outputs. Stage functions construct floating-point proposals where useful, then validate them with enclosures. Opposite endpoint signs enclose Bessel roots; the spacing and sign-grid arguments certify coverage; enclosed Gauss rules and basis errors certify matrix entries; the congruence defects certify the inverse. The boundary computation encloses the values and adds the analytic aliases and infinite Fourier tail before forming the residual.

Special functions, quadrature data, matrix errors and the boundary traces are enclosed with FLINT/Arb through `python-flint` [30, 43]. Root-sign evaluations increase through 256, 1024 and 4096 bits when necessary; spectral Bessel values use 2048 bits and boundary values use 4096 bits. Binary64 matrix proposals are exact dyadics with separately bounded rounding errors. A second implementation in `mpmath.iv`, at 400 bits, recombines the scalar chain from the spectral inverse bound and the eight boundary moment bounds. It independently reconstructs the lift and coupled constants, but does not independently certify the special functions, spectral matrices or boundary traces.

The acceptance checks use certain one-sided interval comparisons. Nonfinite values and comparisons that an interval cannot resolve are rejected. The final check recomputes the coupled constants in the driver process rather than accepting constants supplied by a stage-output file. It also compares the two spectral enumerations at the level of individual roots. A successful run writes the receipt and prints `NUMERICAL_GATES_PASS`, certifying the numerical hypotheses of the analytic lemmas. The receipt field `full_proof=false` records that the analytic lemmas of this paper are not machine-checked. The proof consists of those lemmas together with the enclosed inequalities. Symbolic polynomial control examples test the material identity, while its general proof is Proposition 3.1.

34.1.2. *Stages and recorded timings.* Table 24 gives the stage timings in the recorded receipt. The leaf groups report sums of individual stage elapsed times, so they are not wall times for the parallel groups. The recorded run used five single-threaded workers and had wall time 242.21 seconds. The number of workers affects only the scheduling of the stages: the two-worker command below reproduces the same numerical hypotheses with lower concurrency.

The built-in failure controls reject 144 deliberate perturbations, including nonfinite decimal inputs, altered coupled bounds, missing or duplicated spectral states, changed grid size and violated geometric thresholds. A nonzero stage exit or a timeout terminates the run without a receipt. The final artifact hash map is read back before the status line is printed. These controls exercise the numerical acceptance path; they do not replace the analytic estimates.

34.1.3. *Ancillary files and replay.* The directory `anc/r2/` contains the verifier `verify_planar_convex_v2.py`, the recorded receipt `VERIFICATION_RECEIPT.json`, the frozen centre `center_j40_frozen.json`, the eight-mode reference `reference.json` and the nine stage sources that the verifier embeds. It also contains the two files from which the single-file verifier is assembled, the list of software versions, the scalar robustness calculation `ROBUSTNESS_CHECK.json`, the

Stage group	Stages	Recorded seconds
Independent sign-grid coverage	1	30.42
Spectral preparation	1	11.95
Spectral basis leaves	17	455.06
Spectral assembly and inverse	1	13.68
Boundary trace leaves	8	219.60
Boundary merge	1	0.48
Coupled constants and radii	1	0.28
Symbolic identities	1	3.99
Second interval scalar chain	1	0.33

TABLE 24. The 32 stages of the recorded run. Each leaf is single-threaded. Group sums are displayed to two decimal places and may differ slightly from a sum of rounded individual times.

script `exact_constants.py`, a description of the files, and the checksum file `SHA256SUMS`. The verifier needs only its embedded inputs to run; the separate centre and sources expose the exact input and implementation for inspection. The receipt records the enclosures in Table 4, the complete stage list, software versions, the rejected perturbations and hashes of replay artifacts. The script `exact_constants.py` evaluates in exact rational arithmetic the algebraic constants of the two inputs that are quoted in Subsection 2.2 through Section 4. The directory `anc/r2/diagnostics/` contains the scripts and output logs of the non-interval computations quoted in Sections 1 and 36.

The verifier’s SHA-256 is

`ed4f65a5d5a31ee86bfe6c8c742ea86ba76f4cdb888b3c8fb70b53b8334e7226`

The recorded environment was Python 3.10.19, `python-flint` 0.9.0, NumPy 1.26.4, SciPy 1.15.2, SymPy 1.14.0 and mpmath 1.3.0.

From the directory containing `anc/r2/`, with those packages available, choose a new or empty replay directory and an output receipt path that does not exist. A command with at most two single-threaded stage workers is

```
nice -n 19 env OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
  MKL_NUM_THREADS=1 NUMEXPR_NUM_THREADS=1 \
  PYTHONDONTWRITEBYTECODE=1 PYTHONPATH= \
  python -O anc/r2/verify_planar_convex_v2.py \
  --workdir ./replay --jobs 2 \
  --output ./REPLAY_RECEIPT.json
```

The driver refuses an occupied working directory and a pre-existing output receipt. The default timeout is 7200 seconds for each stage and can be set with `--stage-timeout`. The files of `anc/r2/` can be checked by running `sha256sum -c SHA256SUMS` inside that directory.

The computational trust base consists of the Python runtime, the enclosure libraries and their arithmetic, and the correspondence between the verifier and the formulas of Sections 2–4; the analytic arguments of those sections are not machine-checked. The frozen centre is a finite proposal with a nonzero residual. All existence and geometry statements concern the exact zero of F in the certified infinite-dimensional ball.

34.2. The convex planar Berenstein domain. The files in `anc/berenstein/` reproduce the numerical hypotheses of Part 2. The single-file verifier `verify_berenstein_planar_`

`convex_v3.py` embeds the exact-decimal centre and ten stage sources. It contains no computed roots, spectral matrices, boundary enclosures or existence receipts. The two coverage arguments reconstruct the complete Robin window; enclosed radial and boundary integration gives the form matrices and their complementary Gram estimates. Boundary interval sums include the aliases and infinite tails of Subsection 8.1. The driver merges those sums and recomputes the coupled, nonlinear and geometric quantities in its own process.

The parent process rechecks the quoted upper and lower bounds listed in the receipt field `parent_gates`, complete coverage, actual matrix and basis dimensions, and the scalar-chain consistency before acceptance. An interval identity stage tests Cartesian translations at the frozen centre and the residual-corrected material and boundary identities on finite polynomial examples with nonzero residual. It also tests both clamping traces and rejects wrong-sign and omitted-term variants. The boundary coefficient $d = \operatorname{Re}(wp'_0(w)/p_0(w))$ is independently recovered from the exact centre by Taylor division in $z = w^m$. The eleven Fourier coefficients of d that enter the boundary form are compared with the Taylor-division values, and all 90000 entries of the form with Laurent products of these values and the radial traces of the basis functions; the assembled operator is checked against that form.

The spectral Bessel values use 2048 bits and the boundary evaluations use 4096 bits in Arb through `python-flint` [30,43]. Exact decimal input is enclosed without a binary64 conversion; floating-point proposals are read as exact dyadics with explicit errors. The 403 semantic controls and 31 identity negative controls test input syntax, coverage, centre binding, matrix signs, published bounds and the radii gates. Nonfinite values, unresolved strict comparisons, a failed subprocess or a timeout prevent a success receipt. Before finalisation the driver reads back all artifact hashes. The displayed bounds are rounded outward from the recorded enclosures. The bound $E_0 < 9.699 \cdot 10^{-67}$ is the all-mode, weight-amplified clamped residual.

The recorded isolated run took 173.042 seconds with at most 4 single-threaded workers. Its environment was Python 3.10.19, `python-flint` 0.9.0, NumPy 1.26.4, SciPy 1.15.2 and `mpmath` 1.3.0. A second implementation recomputes 56 scalar quantities in 400-bit `mpmath.iv` arithmetic from six Arb spectral primitives and eight boundary moments, including the complement and Schur estimates, norm transfer, lifts, corrected inverse, radii and geometry. It does not reconstruct the special functions, spectral matrices or boundary moments. The receipt field `full_proof=false` records that the analytic lemmas are not machine-checked. They are proved in the paper; the verifier certifies their numerical hypotheses.

With these dependencies installed in an environment visible to isolated Python, run

```
nice -n 19 env OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
  MKL_NUM_THREADS=1 NUMEXPR_NUM_THREADS=1 \
  PYTHONDONTWRITEBYTECODE=1 \
  python -I -O verify_berenstein_planar_convex_v3.py \
  --workdir replay --jobs 4
```

The working directory `replay` must not exist. The verifier installs no packages. A successful run exits with status zero, writes `replay/VERIFICATION_RECEIPT.json` with status `NUMERICAL_GATES_PASS`, and ends with a line beginning `NUMERICAL_GATES_PASS` followed by the receipt path and a line beginning `RADIUS` that also prints `CONTRACTION`. The recorded verifier SHA-256 is

```
375ea2e68225d3419309d5359c7d5d2a589c07bb0985e71f2cd0ac9bc3fc95fb
```

The frozen centre's SHA-256 is

```
272abaf4b002d8b63cf68344616ca501eb1a3c1f2636b27280af4960713c4472
```

The ancillary directory contains the verifier, separate frozen centre, recorded isolated receipt, requirements, `README.md` and `SHA256SUMS`. Only the machine-specific Python executable path in the receipt is replaced by `python`. A fresh run regenerates its stage sources and outputs. The scripts and records in `anc/diagnostics/berenstein/` describe the numerical curvature, eigenvalue-index and stationary-phase comparisons of Subsection 9.5. The script `csb_nonconvex.py` encloses in Arb the triangle witness for the non-convexity of the domain of [15]. These records are not inputs to the contraction.

34.3. Dimension four: what the script checks. All finite computations for $n = 4$ are carried out by a single script, `verify_r4.py` (569 lines), which uses only the Python standard library and `python-flint` 0.9.0 [30], the Python interface to the ball arithmetic of FLINT/Arb [43]. Every real number is represented by a ball $[m \pm \varepsilon]$ whose arithmetic is rigorous, including all rounding errors. The 525 centre coefficients are embedded in the script as hexadecimal binary64 strings, which are converted to exact dyadic rationals (exactness is checked); $\rho = 21/20$, $r = 1/50000$ and the rational majorants of Theorem 17.1 are exact. The script performs the following steps.

- (1) It expands $\mathcal{F}(x^\circ)$ and the 525 finite columns of $D\mathcal{F}(x^\circ)$ in ball arithmetic, using (12.3), (13.1) and (14.3); no coefficient is discarded.
- (2) It assembles M_f , computes a ball enclosure of M_f^{-1} , and defines $\hat{A}$ as the matrix of the (exact, dyadic) midpoints of this enclosure. It then computes $I - \hat{A}M_f$ in ball arithmetic and verifies (15.5) for the weighted column norm. The matrix $\hat{A}$ is never treated as an interval-valued operator.
- (3) It computes $\|R\|$, H , the numbers h_n for $M + 2 \leq n \leq N_H = 443$ and the remainder bound of Lemma 15.6, hence A_t and a bound for $\|A\|$.
- (4) It computes Y and Z_f by applying A , in the form (15.6), (15.3), (15.4), to the residual and to the finite columns.
- (5) It computes the 3650 tail g -columns with $n \leq 123$, $s \leq 67$ with the short polynomial $p_{\leq 16}$ and adds the error term of (16.1); it evaluates $Z_{g,\infty}$.
- (6) It computes the 480 shape columns $43 \leq j \leq 1001$ without truncation, checks the hypotheses of Proposition 16.4, and evaluates $U_1, \dots, U_5, U_W$ and $Z_{\text{sh},\infty}$.
- (7) It evaluates Z , c_2 , c_3 , C_2 , C_3 , checks $Y < 7 \cdot 10^{-6}$, $Z < 37/100$, $C_2 < 4$, $C_3 < 10^{-3}$, and then the two inequalities (15.1) at $r = 1/50000$.
- (8) It checks the geometric inequalities (18.2) and (18.3).

In `python-flint`, an order relation between balls evaluates to true only if it holds for all points of the balls. Every check is an explicit test that raises an exception when it fails (also under `python -O`); maxima over columns are taken over rigorous upper endpoints, and nonfinite enclosures are rejected. After step (4) the script writes an intermediate checkpoint file (`*.base.json`), which carries no status field; only when all checks succeed does it write the certificate file (JSON) with the enclosures of all quantities of Table 8 and the status `PROVED`, and print `PROVED`. A diagnostic mode (`--stage base`) stops after step (4) and never prints `PROVED`.

A second script, `verify_convex.py`, checks that the centre data file coincides with the data embedded in `verify_r4.py`, that the main certificate has status `PROVED` for the radius $r = 1/50000$, and then verifies the bounds of Proposition 18.2, which involve only the finite sums A_1, A_2 and the explicit tail constants E_1, E_2 .

34.4. Dimension four: reproduction. From a directory containing the files `verify_r4.py`, `verify_convex.py` and `data/centre.json`, the certificate is reproduced by

```

mkdir -p logs
nice -n 19 env OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
  MKL_NUM_THREADS=1 python3 verify_r4.py --stage all --bits 128 --jstar 1001 \
  --output logs/certificate_r4_128.json
python3 verify_convex.py --bits 128 \
  --main-certificate logs/certificate_r4_128.json \
  --output logs/convex_128.json

```

The second command prints `PROVED` as its last line, the third prints `PROVED_CONVEX`. The runs use one thread. The distributed certificates are in the subdirectory `certificates/` of `anc/` (Section 34.12); the files written to `logs/` are byte-identical to them. On a single core of a Linux server the 128-bit run took between 163 and 167 seconds of wall-clock time in several independent runs, and the same run at 192 bits (`--bits 192`) took about 198 seconds; the convexity check takes a fraction of a second. Runs under Python 3.10.19 and Python 3.12.3, both with `python-flint` 0.9.0, produced byte-identical certificate files. At 192 bits all inequalities hold with the same displayed bounds, and the finite inverse defect (15.5) is below $7.5 \cdot 10^{-54}$.

34.5. Dimension six. For $n = 6$ the finite computations are split into five stage scripts, which evaluate the quantities of Section 19 in ball arithmetic at 128 bits, in batches, and write interval receipts (JSON files), and a final script that combines the receipts. All scripts use Python, `python-flint` 0.9.0 and, only to read the matrix $\hat{A}$, NumPy.

- (1) `finite_interval_r6.py` expands $\mathcal{F}_6(x^\circ)$ without truncation and computes $A\mathcal{F}_6(x^\circ)$ and Y (receipt `residual_bound_r6.json`); in eight batches covering all 465 finite columns it computes the inverse defect (19.12) and Z_f (`finite_batches_r6.json`). The matrix $\hat{A}$ is read from `inverse_r6_M58_S30.npy`, and its binary64 entries are converted to exact dyadic rationals.
- (2) `tail_inverse_r6.py` computes H , the numbers h_n for $M + 4 \leq n \leq N_H = 458$ and the remainder bound of Lemma 19.12, hence A_t (`tail_inverse_bound_r6.json`).
- (3) `g_tail_interval_r6.py` computes the 705 near tail g -columns in seven batches with the short polynomial $p_{\leq 24}$ and adds the error term of (19.15) (`g_tail_batches_r6.json`); it also evaluates the two far bounds of Lemma 19.13 (`g_far_bound_r6.json`).
- (4) `shape_tail_interval_r6.py` computes the 85 near shape columns $61 \leq j \leq 397$ without truncation, in five batches (`shape_tail_batches_r6.json`).
- (5) `far_shape_r6.py` computes Γ_1, Γ_0 , the expansion of Q° by the connection recurrences, the coefficients $\gamma_{m,t}^W$, U_Γ , U_W and $Z_{\text{sh},\infty}$ for $j_* = 401$, and checks the support condition of Proposition 19.14 (`far_shape_bound_r6.json`).
- (6) `verify_r6.py --stage final` checks the SHA-256 checksums of the centre, of $\hat{A}$, of the seven receipts and of the five stage scripts against values frozen in its source; checks that the batches of each receipt are contiguous and cover all columns of their class; recomputes $\|R\|$ from $\hat{A}$; forms Z and the constants c_2, c_3, c_4 of Lemma 19.7 with $\|A\| \leq 1402.314$; checks the rational majorants of Theorem 19.16 and the inequalities (19.6) at $r = 1/5000$; and checks the geometric inequalities (19.18). Only then does it write the certificate `certificate_r6.json`, with the status `PROVED`, all enclosures, the frozen checksums and its own checksum, and print `PROVED`.
- (7) `verify_convex_r6.py` checks the checksums of `verify_r6.py` and of the centre, and the status, radius, weight and checksums recorded in the certificate, and then verifies the bounds of Proposition 19.19 (`convex_r6.json`, status `PROVED_CONVEX`).

As for $n = 4$, an order relation between balls is true only if it holds for all points of the balls, every check raises an exception when it fails, and maxima are taken over rigorous upper endpoints; a NaN or infinite enclosure, whether computed or read back from a receipt, stops the verification. The script `test_fail_closed_r6.py` tests this behaviour on artificial inputs. A further script, `check_identities_r6.py`, checks in exact rational arithmetic the identities of Lemma 13.1(a)–(d) for all $n \equiv 2 \pmod{4}$, $n \leq 202$, and $1 \leq s \leq 80$ (4080 pairs), reruns the connection recurrences used for Q° in rational arithmetic (450 pairs, all coefficients positive), checks $\|Kg^\circ - (1 - |w|^2)^2 Q^\circ\|_\rho \leq 2.02 \cdot 10^{-31}$ in ball arithmetic, and compares (19.8) with the column algebra of the verifier, exactly for $5 \leq j \leq 2001$ and in ball arithmetic for $5 \leq j \leq 1001$. These identities are proved in Section 19; the script is a consistency check.

The complete six-dimensional verification is reproduced, from the directory `anc/r6/` of the ancillary files, by

```
sha256sum -c SHA256SUMS
bash reproduce_r6.sh
```

The script `reproduce_r6.sh` uses the interpreter given by the environment variable `PYTHON`, if it is set, checks the `python-flint` version, runs `test_fail_closed_r6.py`, regenerates all seven receipts from the centre and $\hat{A}$ with the stage scripts, each batch in a separate single-core process under `nice -n 19` with a time limit of 600 seconds, and compares them byte for byte with the distributed receipts. It then reruns steps (6) and (7) and `check_identities_r6.py` in a scratch directory, compares the resulting `certificate_r6.json`, `convex_r6.json` and `identities_r6.json` byte for byte with the distributed files, and prints `REPRODUCTION COMPLETE`; any mismatch stops it with a nonzero exit code. On one core of a Linux server, with Python 3.10.19 and `python-flint` 0.9.0, the complete reproduction took about three minutes of wall-clock time (3 min 2 s and 3 min 5 s in two independent runs). The individual final steps are

```
python3 verify_r6.py --stage final \
  --output certificate_r6.json
python3 verify_convex_r6.py
python3 check_identities_r6.py --output identities_r6.json
```

which print `PROVED`, `PROVED_CONVEX` and `IDENTITIES VERIFIED` and rewrite the three files byte-identically; they take about two seconds, less than a second and about fifteen seconds, respectively.

34.6. Consistency checks for $n = 4$ and $n = 6$. The following checks are not part of the proof. The algebraic identities of Sections 12, 13 and Lemma 15.3 were verified symbolically for small indices. An independently written floating-point implementation, which computes projections by quadrature instead of the recurrences (12.3) and uses its own approximate inverse, reproduced $\|R\|$, H , Z_f and the largest shape column of Table 8 to the displayed digits. The function u was also evaluated directly in $\mathbb{R}^4$ from a Fourier–Bessel representation $F = \sum_{m \text{ odd}} a_m J_m(r) \sin m\phi$ fitted to the boundary data of the centre: at random points of $\partial\Omega$ the defects $|u - 1|$ and $|\nabla u|$ were below 10^{-10} , and the Fourier transform of $\mathbf{1}_\Omega$, computed by quadrature, was below 10^{-10} in modulus on the unit sphere and between 18 and 28 in modulus on the sphere of radius 0.999.

For $n = 6$, Lemma 19.1 and the averages $\langle Y_\ell^3 \rangle / \langle Y_\ell^2 \rangle$ of Section 35.2, for $\ell = 4, 8, 12, 16$, were verified symbolically, and seven exact finite columns of $D\mathcal{F}_6(x^\circ)$ agreed with an independently written quadrature implementation to relative accuracy $1.7 \cdot 10^{-13}$. The Fourier–Bessel solution from which the centre was obtained (Remark 19.17) has Cauchy defects $8.2 \cdot 10^{-14}$ and $2.2 \cdot 10^{-13}$ at 2001 boundary points off the collocation grid, with 20 modes, and a solution with 28 modes

agrees with it. Using only its boundary, the first 14 volume integrals over Ω of $O(3) \times O(3)$ -invariant regular solutions of $\Delta v + v = 0$, which vanish when $\widehat{\mathbf{1}}_\Omega$ vanishes on the unit sphere, were computed by quadrature; relative to the corresponding integrals in which the angular integrand is replaced by its modulus, they were between 0 and $3.3 \cdot 10^{-13}$, compared with $4.7 \cdot 10^{-3}$ for a comparison domain that retains only the dominant degree-eight term of the boundary.

34.7. Dimensions eight, ten and fourteen. The computations for $n = 8, 10, 14$ are distributed in the directories `anc/r8`, `anc/r10` and `anc/r14`. They use Python, `python-flint` 0.9.0 [30, 43] and, only to read $\widehat{A}$, NumPy. Each dimension has stage scripts that evaluate the quantities of Sections 20.5–20.7 in ball arithmetic at 128 bits and write interval receipts (JSON files), a final script that combines the receipts, and a separate convexity script.

- (1) For $n = 8$ the algebra of disk polynomials is contained in `verify_r8.py`; the stage scripts are `finite_interval_r8.py` ($Y, Z_f, (20.15), \|R\|$), `tail_inverse_r8.py` (H, h_n for $M + 2m \leq n \leq N_H, A_t$), `g_tail_r8.py` (the 396 near tail g -columns in 20 batches, and the far bounds of Lemma 20.17), `shape_tail_r8.py` (the 29 near shape columns in two batches) and `far_shape_r8.py` ($\Gamma_1, \Gamma_0, Q^\circ, \gamma_{k,t}^W$ and Proposition 20.18).
- (2) For $n = 10, 14$ the algebra is in `rank2_cap_core.py`, shared by the stage scripts `finite_rank2.py` (for $n = 14$: `finite_batch_rank2.py`, in 26 batches, and `aggregate_finite_r14.py`), `tail_inverse_rank2.py`, `g_tail_rank2.py` (358 columns in 18 batches, resp. 1215 columns in 13 batches), `shape_tail_rank2.py` (31 columns in 4 batches, resp. 10 columns in 8 batches), and `far_shape_rank2.py` for $n = 10$, `far_shape_split_r14.py` for $n = 14$ (the first bound of Proposition 20.18).
- (3) The final scripts `verify_r8.py` (with the option `--stage final`), `verify_r10.py` and `verify_r14.py` check a SHA-256 digest, frozen in their source, of a bundle consisting of the centre, $\widehat{A}$, all stage scripts and all receipts; check that the batches of each receipt class are contiguous and cover all columns of the class; check the support conditions of Lemma 20.17 and Proposition 20.18; form Z and $c_2, \dots, c_{m+2}$ by (20.9) with the rational ceiling for $\|A\|$; check the rational majorants and (20.10); and check the bounds of Proposition 20.21. Only then do they write a certificate with status `PROVED` and print `PROVED`.
- (4) `verify_convex_r8.py`, `verify_convex_r10.py` and `verify_convex_r14.py` check the status of the main certificate and the checksum of the main verifier, and recompute the bounds of Table 13 (status `PROVED_CONVEX`).
- (5) `check_identities_r8.py` and `check_identities_rank2.py` verify the identities of Lemma 13.1(a),(b) in exact symbolic arithmetic for 12 index pairs, compare (20.12) with the column algebra at 256 bits for three shape indices, check positivity and total mass one of $q_s(n)$, and check in ball arithmetic that all coefficients of $Kg^\circ - (1 - |w|^2)^2 Q^\circ$ are below 10^{-69} in absolute value. These identities are proved in Section 20; the scripts are consistency checks.

As for $n = 4, 6$, the centre coefficients, B and the entries of $\widehat{A}$ are converted exactly to dyadic rationals (exactness is checked), and ρ, κ_m, r and all thresholds are exact rationals. An order relation between balls is true only if it holds for all points of the balls, maxima are taken over rigorous upper endpoints, and a NaN or infinite enclosure, computed or read from a receipt, stops the verification. (In a preliminary version of the scripts for $n = 10, 14$, κ_m was a binary64 number, which for $m = 4$ lies $1.2 \cdot 10^{-18}$ below $1/48$; the distributed scripts use κ_m exactly, and the displayed bounds are unaffected.)

In each directory, `bash reproduce_r8.sh` (resp. `reproduce_r10.sh`, `reproduce_r14.sh`) copies the scripts, the centre and $\hat{A}$ into a scratch directory, recomputes every receipt there, each job in a separate single-core process under `nice -n 19`, reruns the final and convexity scripts, and compares every distributed JSON file byte for byte with its recomputed counterpart. The final steps alone are

```
python3 verify_r8.py --stage final \
  --centre center_r8_M45_S24.json --output certificate_r8.json
python3 verify_convex_r8.py
python3 verify_r10.py && python3 verify_convex_r10.py
python3 verify_r14.py && python3 verify_convex_r14.py
```

which print `PROVED` and `PROVED_CONVEX`; each takes less than four seconds on one core. In an independent rerun of all stage scripts, the 106 interval receipts of the three dimensions were reproduced byte for byte, with a total of about 930 seconds of single-core CPU time.

Consistency checks. The following checks are not part of the proof. The script `check_radial_rank2.py` verifies Proposition 20.2 for $\mathfrak{su}(3)$ in exact symbolic arithmetic in explicit matrix coordinates, and for the root data of B_2 and G_2 at 100 random regular points each. An independently written check with explicit matrix realisations of $\mathfrak{so}(5)$ and of $\mathfrak{g}_2 \subset \mathfrak{so}(7)$, as the stabiliser of a generic 3-form, confirmed (20.1) and the harmonicity of ϖ at random regular points to relative accuracy $5.1 \cdot 10^{-13}$. A pointwise evaluation of $\mathcal{H}_m(x^\circ)$ confirmed the sign conventions (relative residuals between 10^{-15} and 10^{-12} , against 2 with the opposite sign). Centred differences at 256 bits reproduced the shape column $j = 2m + 1$ of $D\mathcal{H}_m(x^\circ)$ to weighted accuracy $3.4 \cdot 10^{-12}$, $7.4 \cdot 10^{-16}$ and $2.9 \cdot 10^{-15}$ for $n = 8, 10, 14$; the tail inverse (20.14) was checked against T on four tail shape columns in each dimension (errors below 10^{-62}); the minimal output frequencies and radial indices of the first far tail g -columns were computed directly (for $n = 14$: $126 > M$ and $61 > S$); and 13 artificially corrupted receipts (NaN or infinite values, coverage gaps, bounds above the thresholds) were all rejected by the final scripts.

34.8. Dimension three. The convex $n = 3$ files are in `anc/r3_convex/`. The single-file verifier `verify_r3_convex.py` is frozen; its full SHA-256 is recorded in the ancillary README. It embeds the exact dyadic centre, the stage programs and the C ball-arithmetic accelerator. A fresh isolated replay completed all 17 stages, printed `PROVED`, and wrote the full receipt after 6515.9 seconds with five single-thread workers. A second from-scratch run of the same unmodified verifier, in a separate fresh directory, also printed `PROVED`; its receipt agrees with the first in every mathematical field. The distributed receipts record the centre and matrix hashes, the complete inverse and derivative bounds, the radii inequalities, the shape-ball geometry and the exact coverage counts.

The stages construct the 7371-dimensional finite matrix and certify its dyadic product defect with a fixed-order dot-product error bound; certify the full-centre boundary-symbol inverse; compute the full-support residual; enclose all 7280 finite field columns; compute 457 exceptional field columns with the $j \leq 121$ evaluation truncation and rigorous omitted-tail error; certify 91 radial classes, the fourteen near angular classes and $\ell \geq 210$; check 310 explicit shape columns and every $j \geq 621$; and assemble the nonlinear, geometric and coverage gates. Every strict inequality is checked with Arb balls. Missing columns, stale inputs, nonfinite enclosures or a failed gate prevent `PROVED`.

From an empty scratch directory, reproduction is requested by

```
python3 -O verify_r3_convex.py --workdir <empty dir> \
  --output receipt.json --jobs 5
```

with Python 3, NumPy, SciPy, `python-flint` 0.9.0, `gcc` and 64-bit Linux. The run takes about 1.8 hours on five cores. The numbers reported here are those of the frozen receipt; the SHA-256 of the verifier is recorded in the ancillary README.

The independent nonconvex example has its own verifier, two completed certificates and a separate exact rational nonconvexity check in `anc/r3/`. Its computation uses $M = 180$, $S = 64$ and the different centre and shape scale of Proposition 21.27. Its previously certified values are Table 15. They do not enter the convex certificate.

34.9. Dimension five. The five-dimensional files are in `anc/r5/`. The frozen single-file verifier is `verify_r5.py`; its SHA-256 begins with `b49dc914` and is recorded in full in the files `README.md` and `SHA256SUMS` of that directory. It embeds, in a compressed payload that is checked against its own SHA-256, the exact dyadic centre (SHA-256 `c672b3fb...`), the frozen class parameters of Section 22.4 (also distributed as `params.json`), the stage programs and a C library of ball-arithmetic and long-double kernels; nothing computed is read from disk. In a new empty work directory it compiles the kernels and runs a self-test, which includes the identities $\sum_n D_{nl} = 1$ and $\sum_\ell C_{\ell N} D_{m\ell} = \delta_{mN}$ for $N \leq 40$ and Lemma 22.12 for $K \leq 30$ in exact arithmetic. In 23 stages, run by at most six single-threaded worker processes, it then computes the boundary Taylor and Jacobi data of H° , W_1° , Z° and W° ; the residual; all 4698 finite columns and the 59 exceptional and 219 explicit shape columns; the matrix $\hat{A}$, its certified defect and its column norms; ε_T , τ_T , $\mathcal{B}$, $\rho_{\mathcal{B}}$, A_t and the profiles Γ_F , Γ_H ; the defects of all explicit columns and Y ; the field envelopes for the frozen classes; the far shape bound of Proposition 22.15; and the constants c_2, c_3, c_4 and the bounds of Proposition 22.17. Finally it checks that the classes cover every input exactly once and that all gates of Theorem 22.16 hold. Only then does it print `PROVED` and write the receipt; otherwise it prints `NOT_CERTIFIED` and exits with status 2. Every inequality is checked with Arb balls; binary64 and long-double numbers enter the bounds only through exact dyadic conversion or with rigorous rounding-error terms, maxima are taken over rigorous upper endpoints, and nonfinite enclosures are rejected.

A fresh run of the distributed verifier in an empty directory printed `PROVED` after 2364 seconds of wall-clock time with six single-threaded workers. Its receipt is distributed as `VERIFICATION_RECEIPT.json`, in which only the machine-specific work-directory string was replaced, and Table 16 is taken from it. A second from-scratch run of the same verifier, in a separate empty directory, also printed `PROVED` (after 2355 seconds); its receipt `VERIFICATION_RECEIPT_replay2.json` agrees with the first in every mathematical field. The computation is reproduced, from the directory `anc/r5/` and with an empty scratch directory, by

```
python3 -O verify_r5.py --workdir <empty dir> \
--output receipt.json --jobs 6
```

with 64-bit Linux (the long-double kernels check the x87 format), Python 3, NumPy, `python-flint` 0.9.0 and `gcc`; it takes about 40 minutes on six cores. The uniform bound for $N > N_2$ in Proposition 22.15 includes the edge terms at $k = k_0$ described in its proof; their total size is below $2 \cdot 10^{-16}$.

Consistency checks. The following checks are not part of the proof. An independently written floating-point implementation reproduced the uniform angular bound Z_{ang} and its five parts to ten digits and Y and the column defects to eight digits. Eight field columns computed by direct evaluation, and seven shape columns computed from a formula without recurrences, agreed with those of the verifier. An independent multiprecision evaluation of the approximate solution defined by the centre gave a Helmholtz residual below $7 \cdot 10^{-19}$ at 250 sample points, compared with values of $|u|$ up to about 10^3 , and $|u - 1| < 3 \cdot 10^{-65}$, $|\nabla u| < 3 \cdot 10^{-33}$ on

the boundary; for the centre, the principal curvatures of $\partial\Theta_{c^0}(B^5)$ are at least 0.912 in the meridian direction and at least 0.993 in the three transverse directions.

34.10. Dimension seven. The files in `anc/r7/` contain the frozen single-file verifier `verify_r7.py`, its two full-run receipts and checksums. Its SHA-256 is

`f8fd0ae17f4de35e7031b17ae61bde49e7d6c1e98705456e82cd1cca070989ed`

The compressed payload has SHA-256 `a9335ae7...`; it embeds the exact dyadic centre (SHA-256 `31de5cfd...`), the class parameters, all stage sources and the C kernels. The field and shape numbers are the frozen centre data; no development file is used as a proof input.

In 28 stages the script rebuilds the kernels and runs exact self-tests, computes the boundary Taylor data and the full-support residual, encloses all 9010 finite columns, 133 exceptional field columns and 245 explicit shape tail columns, constructs the scaled finite inverse and certifies its defect, computes the boundary Toeplitz inverse and its Neumann remainders, the column profiles, every infinite field envelope and the all-index shape bound, and checks the quartic Taylor, radii, contraction and whole-ball geometry inequalities of Theorem 23.13. It also checks that the input classes cover every coordinate exactly once. Arb balls supply the enclosures; dyadic conversions and explicit dot-product rounding bounds cover the floating-point kernels. Missing data, nonfinite enclosures, a failed worker or a failed inequality prevent **PROVED**. The integer gates of the far-shape stage (N_* and N_2 even with $N_2 \geq N_* \geq 500$, and $N_* + \eta > M$, $N_* + \eta \geq 2I + 2$ for every monomial (η, e) of the principal polynomial) imply the hypotheses of Lemma 23.11: the monomial $(0, 0)$ has the single contribution $(c_1^0)^2 > 0$, so $N_* \geq 2I + 2$ and $K_{\max} = N_2 + 2 > N_*$, and $2i_c = 200 \leq 500$. The inequalities of Lemma 23.7 are checked in ball arithmetic.

From `anc/r7/`, with a new or empty scratch directory, run

```
python3 -0 verify_r7.py --workdir <empty dir> \
--output receipt.json --jobs 2
```

It requires 64-bit Linux with x87 extended precision, Python 3, NumPy, version 0.9.0 of `python-flint`, and `gcc`. The recorded complete run printed **PROVED** after 6175 seconds (about 103 minutes) with eight single-threaded workers; `--jobs` accepts 1 through 8, and the two-worker command uses less concurrency. On success the receipt is written; a failure prints **NOT_CERTIFIED** and exits with status 2. In the distributed receipt only the machine-specific work-directory string was replaced. A second from-scratch run of the same unmodified verifier, in a separate empty directory, also printed **PROVED** (after 10629 seconds with six single-threaded workers); its receipt `VERIFICATION_RECEIPT_replay2.json` agrees with the first in every mathematical field. Its mathematical fields supply Table 17; the full hashes and the integrity command are in the README.

34.11. The exact inverse in general dimension. The directory `anc/gd/` contains the single-file verifier `verify_axisym_exact_v2.py` for Theorem 24.1. It embeds the finite exact-decimal centres, the dimension parameters and the stage sources in a hash-checked compressed bundle. It embeds no computed spectral matrix, boundary coefficient or existence receipt. A fresh run extracts the sources and inputs, checks their hashes and the module bindings, and recomputes the numerical chain in one process for each dimension. Relative to the single-file verifier that was first run for $n \in \{5, 9, 11, 12, 13\}$, this file changes the dimension lists, the embedded centres and parameters of the six further dimensions, and the two spectral settings for $n = 16$ stated below. For $n \in \{5, 9, 11, 12, 13\}$ it reproduces every certified bound of the first verifier.

The spectral stage encloses all roots in (26.3), compares the two enumerations root by root, encloses the tensor Gaussian data and polynomial potential, adds the errors of Section 26.4, and evaluates the full Schur bound. The spectral arithmetic uses 768 bits and 384 radial Gauss nodes, except for $n = 16$, where it uses 1536 bits and 640 radial nodes. The boundary stage encloses the trigonometric samples and the Acb Fourier transform, including the half-grid phase, and adds every retained alias allowance and the entire analytic tail. The angular conversion, heat and maximum-principle bounds supply the norm transfer. The coupled stage forms the residual of the exactly clamped centre, the corrected inverse bound, the left Newton defect, the quartic remainder and the whole-ball geometry. Every retained decimal is an exact rational input. Special functions and enclosure arithmetic use Arb/Acb through `python-flint` [30,43]; binary64 matrix proposals use exact dyadic interpretation and the explicit error budget.

The acceptance path accepts a strict interval comparison only when it holds for every pair of points of the two enclosures, rejects nonfinite or unresolved bounds, and is run under `python -E -O`. A successful run performs 125 exact symbolic connection-formula and lift examples, 25 in each of dimensions 5, 9, 11, 12, 13, as well as the symbolic- n material identity, and rejects 52 semantic failure controls. The connection examples test the closed formula; they do not test the implementation's connection recurrence. After the controls, the final scalar chain is recomputed and all receipt artifacts are read back against their hashes.

A second implementation in `mpmath.iv`, at 120 decimal digits, recombines the final residual, coupled, quartic, radii and geometric expressions from the Arb primitives. It compares overlap of the nine final scalar enclosures. It does not independently enclose the special functions, spectral matrices, boundary samples, shape-norm constants, norm-transfer constants or fixed-field primitives. The scalar gates also include $4C_2Y_0 < 10^{-10}$; existence uses the invariant-ball and contraction inequalities of (30.3). Every printed inequality bound is rounded outward from its own enclosure; products of rounded entries need not recover the sharper product bounds. The analytic lemmas of this paper and the stated numerical arithmetic together give the certificate.

From `anc/gd/`, with an absent or empty output directory, run

```
nice -n 19 env OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
  MKL_NUM_THREADS=1 NUMEXPR_NUM_THREADS=1 \
  python -E -O verify_axisym_exact_v2.py \
  --dimension 12 --workdir replay12
```

The successful final line is `PROVED dimensions=12`, with exit status zero. The overall and per-dimension receipts must both have status `PROVED`. A search for the substring `PROVED` is insufficient: it occurs in the failure status `NOT_PROVED`, the overall failure receipt status `NOT_ALL_PROVED`, the spectral stage status `PROVED_L2_DIRICHLET_INVERSE_ONLY` and the per-dimension lines. The option `--dimension all` executes all supported dimensions sequentially; its final line lists those dimensions. The spectral module pins to the lowest allowed CPU; an explicit `taskset` restriction to one allowed CPU fixes that choice. Table 25 records, for each dimension, the strip radius S and the parameters K , J_b of Subsection 28.3, the boundary precision in bits, and the times of the final isolated runs.

The recorded environment is Python 3.10.19, `python-flint` 0.9.0, NumPy 1.26.4, SciPy 1.15.2, SymPy 1.14.0 and `mpmath` 1.3.0. The verifier's SHA-256 is

b33fd6c7e9d72a283c9f752ec55b8e1d4dec48b132872759ac4a32553d1e3940

The separate frozen centres `frozen_centers/nN.json.gz` are the gzip-compressed exact-decimal inputs (keys `n`, `rho`, `c`, `a`, `scales`, `status`); `gzip -dc` returns the JSON inputs of the replays byte for byte. The ancillary directory retains the overall `receipts/nN/VERIFICATION_RECEIPT.json` of the final isolated run for each dimension; for dimension 21 this is the only

n	boundary S	K	J_b	bits	chain (s)	process (s)
5	1.22	2048	1800	2048	528.104	535.312
9	1.2	2048	1800	2048	637.539	645.653
11	1.25	2048	1800	2048	370.256	378.099
12	1.25	2048	1800	2048	286.531	294.107
13	1.2	2048	1800	2048	581.188	588.951
15	1.35	1024	900	2048	209.550	217.013
16	1.45	1024	900	2048	1243.897	1252.022
17	1.45	1024	900	2048	170.385	177.179
18	1.4	1024	900	2048	150.032	157.956
20	1.45	1024	900	2048	320.510	328.003
21	1.5	1024	900	2048	426.939	434.442

TABLE 25. Boundary-stage parameters and times of the final isolated runs of the verifier, rounded to three decimals. The process time also includes extraction and symbolic checks. Each used one allowed CPU, one Arb/BLAS thread and priority `nice 19`.

completed isolated replay of the verifier. These receipts bind the verifier, payload and dimension and record the wall time. A fresh run regenerates the extracted sources, stage outputs, identity receipt, both final receipts and artifact-hash readbacks in its output directory; compare the numerical enclosures with the tables in this paper. The checksum command is `sha256sum -c SHA256SUMS`, run inside `anc/gd/`. The ancillary requirements record the versions actually used by these receipts.

34.12. Availability. All scripts, data and certificates are distributed as ancillary files with the arXiv version of this paper, in the directory `anc/`. The file `anc/README.md` describes the layout, the requirements, the commands and the running times for all dimensions. The planar verifier, recorded receipt, frozen forty-mode centre, reference shape, stage sources and diagnostic scripts are in `r2/`; the four-dimensional files are `verify_r4.py`, `verify_convex.py`, `data/centre.json` and the four certificates in `certificates/`; the six-dimensional files, with instructions in `RELEASE_README.md`, are in `r6/`; the files for $n = 8, 10, 14$, each directory with its own `README.md`, are in `r8/`, `r10/` and `r14/`; the convex three-dimensional verifier and its receipts are in `r3_convex/`, and the independent nonconvex example with its two certificates and exact nonconvexity check is in `r3/`; the five-dimensional verifier, its receipt and the frozen class parameters are in `r5/`; the seven-dimensional verifier and its two full-run receipts are in `r7/`; the general-dimension verifier, frozen centres and overall receipts are in `gd/`; the convex planar Berenstein files are in `berenstein/`; their diagnostic comparisons and the interval triangle witness are in `diagnostics/berenstein/`; and `check_radial_rank2.py` is the consistency check of Proposition 20.2 described in Section 34.7. The file `anc/SHA256SUMS` lists the SHA-256 checksums of all other files under `anc/`, and each of the subdirectories `r2/`, `r3/`, `r3_convex/`, `r5/`, `r6/`, `r7/`, `r8/`, `r10/`, `r14/`, `gd/`, `berenstein/`, `diagnostics/berenstein/` carries its own file `SHA256SUMS`; the integrity of the files is checked by `sha256sum -c SHA256SUMS`, run in `anc/` and in each of these subdirectories. The files for $n = 4$ and $n = 6$ are also available at <https://github.com/aster2024/schiffer-pompeiu-r4> (in `verification/` and `verification/r6/`) and archived on Zenodo in [36].

35. REMARKS ON THE MECHANISM, OTHER DIMENSIONS AND CURVED SPACES

The statements in this section are not used in the proofs of Theorems 1.4 and 1.1. The uncertified branches listed in Section 35.3 are based on numerical computations only.

35.1. Symmetry classes for which the reduction is exact. Remark 10.5 shows that, among splittings $\mathbb{R}^n = \mathbb{R}^{d_1} \times \mathbb{R}^{d_2}$ with symmetry group $O(d_1) \times O(d_2)$, a reduction to the planar Laplacian without potential exists only for $n \in \{2, 4, 6\}$. The classes used for $n = 4$ and $n = 6$ are the adjoint classes of $\mathfrak{u}(2)$ and $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ (Remark 20.3), and Section 20 carries out the reduction for all nonabelian compact Lie algebras of rank two; by Remark 20.22, the dimensions 4, 6, 8, 10 and 14 are the only ones reached by exact planar reductions of adjoint type. The three-, five- and seven-dimensional problems admit no such reduction; Sections 21, 22 and 23 use fixed-domain formulations in the corresponding unit balls. The axisymmetric exact-inverse construction of Part 5 uses the full ball Laplacian and has no rank-two restriction.

35.2. A transcritical mechanism. Let $n \geq 3$ and let ρ be a positive zero of $J_{n/2}$, so that the unit ball of $\mathbb{R}^n$ carries the radial solution of (1.6) with $\mu = \rho^2$. Consider domains $\{x : |x| < 1 + sY_\ell(x_1/|x|) + O(s^2)\}$ invariant under $O(n-1) \times \mathbb{Z}_2$, where Y_ℓ is the zonal spherical harmonic of even degree ℓ on S^{n-1} , normalised by $Y_\ell(1) = 1$. For $n = 6$ we use instead the class of Section 19, of domains invariant under $O(3) \times O(3)$ and the exchange of the factors; there the invariant spherical harmonics of degree $\ell = 2k$ are the functions $Y_\ell(\omega) = U_k(t)/(k+1)$ of $t = |\omega'|^2 - |\omega''|^2$, and the exchange symmetry requires k to be even. The linearisation of the Schiffer problem at the ball in the direction Y_ℓ is degenerate exactly when $J_{\ell+n/2-1}(\rho) = 0$, and this never happens for integer n and $\ell \geq 2$: for even n this is Bourget's hypothesis, proved by Siegel [62], that Bessel functions of distinct nonnegative integer orders have no common positive zero, and for odd n it follows from a similar transcendence argument based on the Lindemann–Weierstrass theorem. So there is no bifurcation from balls in this class. However, $J_{\ell+n/2-1}(\rho)$ can be small. A formal Lyapunov–Schmidt expansion in the direction Y_ℓ then gives, at second order, a reduced equation of the form

$$-\sigma s + \frac{\rho^2}{2} \frac{\langle Y_\ell^3 \rangle}{\langle Y_\ell^2 \rangle} s^2 + O(s^3, \sigma s^2) = 0,$$

where $\langle \cdot \rangle$ is the average over S^{n-1} and σ is a small quantity proportional to $J_{\ell+n/2-1}(\rho)$. This suggests a non-ball solution with boundary amplitude $s_* \approx 2\sigma \langle Y_\ell^2 \rangle / (\rho^2 \langle Y_\ell^3 \rangle)$.

In the plane the corresponding cubic average $\langle \cos^3 N\phi \rangle$ vanishes and the reduced equation is odd in s . The cubic amplitude balance is described in Section 1; the exact planar solution is proved in Section 4. For $n \geq 3$ and even ℓ the cubic average does not vanish: for $n = 3$, $\langle P_\ell^3 \rangle / \langle P_\ell^2 \rangle = (2\ell + 1) \begin{pmatrix} \ell & \ell & \ell \\ 0 & 0 & 0 \end{pmatrix}^2 > 0$; for $n = 4$ the zonal harmonics are $U_\ell(\cos \phi)/(\ell + 1)$, where $U_\ell(\cos \phi)$ is the character of the irreducible representation of $SU(2) \cong S^3$ of dimension $\ell + 1$, and the Clebsch–Gordan rule gives $\langle U_\ell^3 \rangle = \langle U_\ell^2 \rangle = 1$ for even ℓ ; for $n = 5$ and $n = 7$, $\langle Y_\ell^3 \rangle / \langle Y_\ell^2 \rangle$ is the linearisation coefficient $G_{\ell\ell\ell}$ of Theorem 22.6(a), respectively Lemma 23.2, which is positive for even ℓ by [22, 18.18.22]; for $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\} \setminus \{5\}$ it is the coefficient $L_{\ell\ell}^\ell$ in the proof of Lemma 25.1, positive for even ℓ by the same formula. For $n = 6$ in the $O(3) \times O(3)$ -invariant class, the invariant measure of S^5 is carried by $t = |\omega'|^2 - |\omega''|^2$ to a multiple of $\sqrt{1-t^2} dt$, for which the U_k are orthonormal after normalisation, and $U_k^2 = \sum_{i=0}^k U_{2i}$ gives $\langle U_k^3 \rangle = \langle U_k^2 \rangle = 1$ for even k , hence $\langle Y_\ell^3 \rangle / \langle Y_\ell^2 \rangle = 1/(k+1)$. That quadratic terms make axisymmetric bifurcations in the even-degree representations of $O(3)$ generically transcritical is a classical fact of equivariant bifurcation theory [34, 42]. This classical

mechanism applies to the Schiffer problem near a near-resonance, with the quadratic coefficient given by the projection of the squared harmonic onto the original harmonic.

For $(n, \rho, \ell) = (4, j_{2,4}, 6)$, the zero $j_{2,4} \approx 14.79595$ of J_2 is close to the zero $j_{7,2} \approx 14.82127$ of $J_7 = J_{\ell+n/2-1}$, and the formal prediction for the relative Y_6 -amplitude is about $2.4 \cdot 10^{-2}$. The domain of Theorem 1.1 has Y_6 -amplitude about $2.2 \cdot 10^{-2}$ (Remark 1.5); in the meridian plane the Y_6 direction corresponds to the mode $J_7(r) \sin 7\phi$ (Section 10). This is how the example was located. Numerically, Newton iterations started from the ball perturbed by a small multiple of Y_6 converge either back to the ball or to our domain, depending on the size of the perturbation, as expected from a transcritical picture.

For $(n, \rho, \ell) = (6, j_{3,8}, 8)$, the zero $j_{3,8} \approx 28.90835$ of J_3 is close to the zero $j_{10,5} \approx 28.88738$ of $J_{10} = J_{\ell+n/2-1}$, and $J_{10}(j_{3,8}) \approx -3.0 \cdot 10^{-3}$. The formal prediction for the relative amplitude of the degree-eight harmonic is about $7.3 \cdot 10^{-3}$, and the six-dimensional domain of Theorem 1.1 has amplitude about $7.2 \cdot 10^{-3}$ (Remark 1.5). In the meridian plane this direction corresponds to the mode $J_{10}(r) \sin 10\phi$ (Section 19). The centre of Section 19 was located in this way (Remark 19.17).

For $(n, \rho, \ell) = (3, j_{3/2,15}, 12)$ one has $J_{25/2}(j_{3/2,15}) \approx 2.3 \cdot 10^{-3}$, and the additional nonconvex domain of Proposition 21.27 has a dominant degree-twelve harmonic of relative amplitude about $1.5 \cdot 10^{-2}$. The convex domain of Theorem 1.4 instead lies near $(j_{3/2,34}, 18)$, as recorded in Table 2. For $(n, \rho, \ell) = (5, j_{5/2,17}, 12)$ one has $J_{27/2}(j_{5/2,17}) \approx -6.1 \cdot 10^{-4}$, and the five-dimensional domain of Section 22 has a dominant degree-twelve zonal harmonic of relative amplitude about $2.5 \cdot 10^{-3}$ (Remark 1.5). For $(n, \rho, \ell) = (7, j_{7/2,25}, 14)$ one has $J_{33/2}(\rho) \approx 2.3 \cdot 10^{-4}$, and Section 23 gives the corresponding convex domain. Numerically, the centres of Part 5 for $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\} \setminus \{5\}$ lie at the near-resonances listed in Table 2; their dominant zonal harmonics have relative polar amplitudes between about $1.6 \cdot 10^{-5}$ (for $n = 16$) and $2.0 \cdot 10^{-2}$ (for $n = 18$) in absolute value (Remark 1.5). For the adjoint classes of Section 20 the invariant harmonics and the near-resonances are described in Section 20.4, and the centres were located at the near-resonances of Table 2 (Remark 20.20). In these classes the cubic average does not vanish either: by the proof of Proposition 20.2, the average over S^{n-1} of an $\text{Ad}(G)$ -invariant function is proportional to the integral of its restriction to the unit circle of $\mathfrak{t}$ against $\sin^2(m\theta) d\theta$; the invariant harmonic of degree ℓ restricts to a multiple of $U_{\ell/m}(\cos m\theta)$, and in the variable $m\theta$ this is again the character computation of $SU(2)$, which gives $\langle U_k^3 \rangle = \langle U_k^2 \rangle = 1$ for even $k = \ell/m$, as the enlarged symmetry requires.

35.3. Further solutions in dimensions three and five. With a spectral collocation solver (expansions in exact Helmholtz solutions $r^{1-n/2} J_{\ell+n/2-1}(\sqrt{\mu} r) Y_\ell$, boundary given as a radial graph, Gauss–Newton iteration), we found numerical solutions of (1.6) on non-ball, star-shaped, $O(n-1) \times \mathbb{Z}_2$ -invariant domains near the near-resonances predicted in Section 35.2. For $n = 3$, $\sqrt{\mu} = j_{3/2,15} \approx 48.6741$ (unit ball normalisation) and $\ell = 12$, with amplitude about $1.5 \cdot 10^{-2}$ and off-grid residual about $2 \cdot 10^{-11}$ with 150 modes, such a solution was the starting point for the additional nonconvex example of Proposition 21.27, which certifies an exact solution nearby. The following further numerical branches have not been certified:

- $n = 3$, $\sqrt{\mu} = j_{3/2,42} \approx 133.5102$, $\ell = 20$, boundary amplitude about $9.6 \cdot 10^{-4}$. In multiprecision the collocation residual is about 10^{-36} , and the residual off the collocation grid decreases spectrally with the truncation (about 10^{-11} , 10^{-14} , 10^{-17} and 10^{-20} for 120, 160, 200 and 240 modes). A test of the Pompeiu condition that uses only the computed boundary gives $2 \cdot 10^{-12}$, compared with $1.4 \cdot 10^{-2}$ for a comparison domain.

- $n = 5$, $\sqrt{\mu} = j_{5/2,30} \approx 97.3586$, $\ell = 16$, amplitude about $7.1 \cdot 10^{-3}$, and $\sqrt{\mu} = j_{5/2,24} \approx 78.5016$, $\ell = 14$, amplitude about $2.1 \cdot 10^{-2}$; residuals about 10^{-11} , stable under changes of the truncation.

By Remark 10.5, a reduction to the planar Laplacian of the type used in Sections 10 and 19 is not available for $n = 3$ and $n = 5$; the formulations of Sections 21 and 22 give the corresponding equations for these numerical branches. The certified five-dimensional domain of Section 22 lies at a different near-resonance, $(j_{5/2,17}, 12)$. Part 5 gives a second proof of the five-dimensional case, using different spaces and a different inverse. The seven-dimensional branch at $(j_{7/2,25}, 14)$ is certified in Section 23; its dominant zonal harmonic has relative polar amplitude about $9.5 \cdot 10^{-4}$.

35.4. Curved spaces. In spheres and hyperbolic spaces the curvature provides a genuine one-parameter family in which radial Neumann eigenvalues and non-radial Dirichlet eigenvalues of geodesic balls can cross, which allows local bifurcation from geodesic balls. Cao-Labora and Fernández [13] used this to construct contractible Schiffer domains on the half-sphere S^2 , and they pointed out [13, Remark 1.2] that their method generalises to other non-flat spaces, with numerically observed crossings for S^3 , S^4 and H^2 . Earlier examples in spheres are bounded by isoparametric hypersurfaces [56, 61], whose boundaries are not topological spheres in general. The mechanism of Section 35.2 is also relevant in spheres and hyperbolic spaces of dimension $n \geq 3$.

36. RELATION WITH RIGIDITY RESULTS AND OTHER CLAIMS

The planar geometric theorems below concern the domain of Theorem 1.3; the spectral-sequence theorem concerns all the domains of Theorems 1.1 and 5.1, and the higher-dimensional spectral and perturbation comparisons concern the corresponding domains of Theorem 1.1. Apart from the certified bounds already stated, the planar numerical values are binary64 evaluations for the forty-mode centre ψ_0 , from `anc/r2/diagnostics/`, and are not interval enclosures. The planar domain is uniformly convex and centrally symmetric, with real-analytic boundary.

The Berenstein domain of Theorem 5.1 is treated separately in Subsection 9.5: its boundary arc-length measure replaces the indicator measure, its eigenfunction has zero Dirichlet data and constant nonzero flux, and the comparison uses the Dirichlet version of the rigidity results.

36.1. Dai–Sun–Wei–Zhang: high frequency in the plane. Dai, Sun, Wei and Zhang [25, Theorem 1.2, version 2] prove that a bounded uniformly convex planar domain with Lipschitz connected boundary carrying a Schiffer solution at an eigenvalue $\alpha > \alpha_0(\Omega)$ must be a disc. The threshold depends on the domain. Their quantitative argument uses support coordinates in Steiner position. Write $h(\theta)$ for the support function and $\varrho(\theta) = h + h''$ for the radius of curvature. The relevant amplitude defect is

$$(36.1) \quad \mathcal{B}(\Omega) = \max_{\theta, t \in \{-1, 1\}} \left| \sqrt{\varrho(\theta)} e^{th'(\theta)} - \sqrt{\varrho(\theta + \pi)} e^{-th'(\theta + \pi)} \right|.$$

For a controlled C^5 class with size, curvature and support-function bounds, their two-point stationary-phase expansion has remainders satisfying

$$(36.2) \quad |R_D| + |R_N| \leq C_* \lambda^{-3/2}, \quad \lambda \geq 3, \quad |t| \leq 1.$$

The constant C_* depends on the class bounds and is not supplied numerically. Their exclusion estimate, for $\mathcal{B}(\Omega) \geq \delta > 0$, is

$$(36.3) \quad \alpha > \left[\max \left\{ 3, \frac{C_*}{\sqrt{2\pi}\delta} \right\} \right]^2 - 1.$$

The proof of their qualitative theorem chooses $\delta = \mathcal{B}(\Omega)/2$.

Remark 36.1 (The remainder for this domain). Let $\Omega_1 = \rho^{-1}\Omega$ be the domain rescaled to conformal radius one; its eigenvalue is $\alpha = \rho^2 \approx 6.6334 \cdot 10^5$. In this remark h , ϱ and $\mathcal{B}$ refer to Ω_1 , and their values are evaluated for the normalised centre ψ_0/ρ . By central symmetry h and ϱ have period π , so $\mathcal{B} = 2 \max_{\theta} \sqrt{\varrho(\theta)} \sinh |h'(\theta)|$; its value is $\mathcal{B} \approx 0.0078698$. With the most favourable choice $\delta = \mathcal{B}$, the estimate (36.3) excludes the eigenvalue α only if

$$C_* < \sqrt{2\pi(\alpha + 1)} \mathcal{B} \approx 16.07.$$

The first remainder in (36.2) belongs to the boundary integral

$$I_D(\lambda, t, \theta) = \int_{\partial\Omega} e^{t\eta \cdot x} e^{i\lambda \xi \cdot x} ds, \quad \xi = (\cos \theta, \sin \theta), \quad \eta \perp \xi,$$

whose leading two-point expression is

$$\sqrt{\frac{2\pi}{\lambda}} \left[\sqrt{\varrho(\theta)} e^{th'(\theta)} e^{i\lambda h(\theta) - i\pi/4} + \sqrt{\varrho(\theta + \pi)} e^{-th'(\theta + \pi)} e^{-i\lambda h(\theta + \pi) + i\pi/4} \right];$$

R_D is the difference. A floating-point computation (trapezoidal quadrature with 2^{17} and 2^{21} nodes, without error enclosures) for Ω_1 at $t = 0$ gives $\max_{\theta} |R_D| \approx 0.118$ and $\lambda^{3/2} \max_{\theta} |R_D| \approx 2.75 \cdot 10^3$ at $\lambda = \rho \approx 814.4574$, and $\lambda^{3/2} \max_{\theta} |R_D| \approx 1.28 \cdot 10^5$ at $\lambda = 1.6 \cdot 10^5$. These computed values indicate lower bounds for any constant C_* valid in (36.2) for a class containing the domain; they exceed 16.07 by factors larger than 170 and 7900, respectively. The support function in the normal angle has $\sup |h^{(5)}| \approx 1.8 \cdot 10^9$, so the C^5 bound of such a class is large although the radial displacement is small.

The threshold (36.3) is not invariant under dilation, and the theorem applies to every dilate $s\Omega_1$, $s > 0$, whose eigenvalue is α/s^2 . The remainder and the defect scale as

$$R_D^{s\Omega_1}(\lambda, 0, \theta) = s R_D^{\Omega_1}(s\lambda, 0, \theta), \quad \mathcal{B}(s\Omega_1) = 2\sqrt{s} \max_{\theta} \sqrt{\varrho(\theta)} \sinh(s|h'(\theta)|).$$

Hence (36.3), with the most favourable choice $\delta = \mathcal{B}(s\Omega_1)$, excludes the eigenvalue of $s\Omega_1$ only if, for every $\lambda' \geq 3s$,

$$\lambda'^{3/2} \max_{\theta} |R_D^{\Omega_1}(\lambda', 0, \theta)| < T(s) := \sqrt{2\pi(\alpha + s^2)} \max_{\theta} 2\sqrt{\varrho(\theta)} \sinh(s|h'(\theta)|).$$

At the sampled values of $s \leq \sqrt{\alpha/8} \approx 288$, $T(s)$ increases and remains below $5.8 \cdot 10^3$, with $T(1) \approx 16.07$; for larger s the eigenvalue α/s^2 is less than 8 and does not exceed the right side of (36.3) for any C_* . Since the left side is approximately $1.28 \cdot 10^5$ at $\lambda' = 1.6 \cdot 10^5$, the computation indicates that no dilate is excluded. This agrees with the remark of the authors of [25] that their threshold depends on the full geometry of the domain and does not contradict the planar counterexamples [12, 14].

Dependence of a high-frequency cutoff on the domain is necessary over the full class of uniformly convex planar domains: the dilates $s\Omega$ of the domain of Theorem 1.3 are uniformly convex, are not discs, and carry solutions with eigenvalue s^{-2} , which is arbitrarily large as $s \downarrow 0$.

36.2. Planar geometric criteria.

36.2.1. *Brown–Kahane: width.* Brown and Kahane [6] prove that a convex planar region has the Pompeiu property if its minimum width is at most half its maximum width. This hypothesis is not satisfied by Ω . Indeed, for $|w| = 1$,

$$808 < \operatorname{Re} \int_0^1 p(tw) dt \leq |\psi(w)| \leq \int_0^1 |p(tw)| dt \leq P + r_* < 821,$$

by (4.4) and the bound $P < 820.7003$ of Subsection 3.2. The support function of the convex domain Ω about the origin lies between the minimum and the maximum of $|\psi|$ on the unit circle, so every width of Ω lies between 1616 and 1642. For the centre, the ratio of minimum to maximum width is approximately 0.99996.

36.2.2. *Mondal: normals through the centre.* Since $\psi(-w) = -\psi(w)$, the domain is centrally symmetric about the origin. Mondal [52] proves that a bounded centrally symmetric planar domain with analytic boundary and centre c satisfies Schiffer's conjecture if the number τ of boundary points whose normal line passes through c is less than 8. Each of the 182 reflection axes of D_{182} meets $\partial\Omega$ in two points. At such a point the tangent line is invariant under the reflection and, being a supporting line of Ω , does not pass through the origin; it is therefore perpendicular to the axis, and the normal line is the axis. Hence $\tau \geq 364$, and the hypothesis is not satisfied.

36.2.3. *Dalmasso: a support-function integral.* Dalmasso [16, Theorem 4] considers bounded convex planar domains with C^2 boundary of positive curvature. Such a domain has the Pompeiu property if its support function h satisfies

$$\int_0^{2\pi} h(\theta)(h(\theta) + h''(\theta))e^{2i\theta} d\theta \neq 0.$$

For Ω , with the origin at its centre, h and $h + h''$ have period $2\pi/182$, so the integral vanishes. It also vanishes for every translate of Ω , because a translation adds a combination of $\cos \theta$ and $\sin \theta$ to h and leaves $h + h''$ unchanged. The hypothesis is therefore not satisfied by Ω in any position.

36.3. **Deng, Aviles and Dai–Liu–Sun: low spectral index.** Deng [20] proves disc rigidity for a bounded planar domain with connected smooth boundary when the eigenvalue is below the eighth Neumann eigenvalue, and for a strictly convex centrally symmetric such domain when it is below the thirteenth; the Neumann eigenvalues are numbered $0 = \mu_1 < \mu_2 \leq \dots$ with multiplicity. The eigenvalue 1 of Theorem 1.3 lies outside both ranges. To see this we use a min–max comparison that requires only the bound $|\psi'| > 811$ on $\mathbb{D}$ from Theorem 1.3. Since ψ is a diffeomorphism of $\mathbb{D}$ onto $\overline{\Omega}$, the pull-back $v \mapsto v \circ \psi$ is an isomorphism of $H^1(\Omega)$ onto $H^1(\mathbb{D})$ and of $H_0^1(\Omega)$ onto $H_0^1(\mathbb{D})$. Under it, the Dirichlet energy is unchanged and the mass becomes $\int_{\mathbb{D}} |v \circ \psi|^2 |\psi'|^2$. Every Rayleigh quotient on Ω is therefore at most 811^{-2} times the corresponding quotient on $\mathbb{D}$, and the min–max principle gives

$$(36.4) \quad \mu_{13}(\Omega) \leq \frac{\mu_{13}(\mathbb{D})}{811^2} \leq \frac{84}{811^2} < 1.$$

For the second inequality use the 13-dimensional real harmonic-polynomial trial space

$$1, \quad r^n \cos n\theta, \quad r^n \sin n\theta \quad (1 \leq n \leq 6).$$

Its energy and mass forms are diagonal by angular orthogonality. For either degree- n function their ratio is $2n(n+1)$, whose maximum is 84; the constant has ratio zero. This proves (36.4).

For the Dirichlet eigenvalues $\lambda_1 < \lambda_2 \leq \dots$ of Ω , use the 13-dimensional zero-Dirichlet trial space

$$1 - r^2, \quad r^n(1 - r^2) \cos n\theta, \quad r^n(1 - r^2) \sin n\theta \quad (1 \leq n \leq 6).$$

For $n \geq 0$, the radial mass factor is

$$\int_0^1 r^{2n+1}(1 - r^2)^2 dr = \frac{1}{(n+1)(n+2)(n+3)}.$$

The identity $-\Delta(r^n(1 - r^2) \cos n\theta) = 4(n+1)r^n \cos n\theta$ gives the Rayleigh quotient $2(n+1)(n+3)$, which is at most 126. The same conformal comparison proves

$$(36.5) \quad \lambda_{13}(\Omega) \leq 126/811^2 < 1.$$

The estimates (36.4) and (36.5) also cover the other low-index results. Aviles [4] proved rigidity for convex planar domains when the eigenvalue does not exceed the seventh eigenvalue, in the Dirichlet numbering as quoted in [9, §0] and in the Neumann numbering as quoted in [20, §1] and [46, (1.4.2)]; and Dai, Liu and Sun [23], as quoted in [25, §1], treat the case in which the eigenvalue equals the second Dirichlet eigenvalue. Since $\mu_7(\Omega) \leq \mu_{13}(\Omega) < 1$ and $\lambda_2(\Omega) \leq \lambda_7(\Omega) \leq \lambda_{13}(\Omega) < 1$, the eigenvalue 1 lies above all these ranges.

36.3.1. The higher-dimensional examples. The three-dimensional example has the radial scale $j_{3/2,34} \approx 108.37572$, and the certified inscribed ball of radius 106 puts $\lambda_2(\Omega) < 0.002$; hence its eigenvalue 1 is far above the second-Dirichlet threshold quoted for [23] in [25, §1]. The certified bounds of Sections 18, 19, 20, 21, 22 and 23 show that Ω contains the ball of radius r_0 about the origin, with $r_0 = 106, 14.54, 55.4, 28.69, 82.6, 20.0, 37.9$ and 87.1 for $n = 3, 4, 5, 6, 7, 8, 10, 14$ (for $n = 3, 8, 10, 14$, the lower bound for $\operatorname{Re} \psi^{*'} on the closed unit disc gives $|\psi^*(w)| \geq \operatorname{Re}(\psi^*(w)/w) \geq r_0$ for $|w| = 1$, and for $n = 5$ the same argument applied to $\rho\psi$, with $\operatorname{Re} \psi' > 0.9811$ and $\rho > 56.49$, gives $r_0 > 55.4$; for $n = 7, \rho > 83.18$ and $\operatorname{Re} \psi' > 0.9935575$ give $r_0 > 82.6$). By domain monotonicity, $\lambda_2(\Omega) \leq \dots \leq \lambda_{n+1}(\Omega) \leq (j_{n/2,1}/r_0)^2 < 0.15$ in every case; in fact, 1 exceeds more than five hundred Dirichlet eigenvalues of Ω , counted with multiplicity, and hence more than five hundred Neumann eigenvalues, since $\mu_k(\Omega) \leq \lambda_k(\Omega)$ for every k .$

36.3.2. The exact-inverse axisymmetric examples. Let $\mathcal{N}_{\text{ax}} = \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\}$ be the set of dimensions in which Theorem 24.1 gives a domain $\Omega = \rho \Theta_\psi(B^n)$ by the exact-inverse method, where $\psi(w) = \sum_{j \text{ odd}} c_j w^j$ has real coefficients and $\Theta_\psi(x_1, x') = (\operatorname{Re} \psi(x_1 + i|x'|), \operatorname{Im} \psi(x_1 + i|x'|) x'/|x'|)$. These domains are strictly convex and invariant under $x \mapsto -x$. In Tables 26 and 27, ρ is rounded; T is the bound L_* of (31.2), evaluated from (31.1) for the centres of Table 18, and $r_0 = \rho T$ is rounded down. The Bessel zeros, the quantities $(j_{n/2,1}/r_0)^2$ and U , the count N_n and d are fifty-digit numerical evaluations; the meridian map ψ of Ω differs from that of the centre, at which d is evaluated, by less than 10^{-50} on $\overline{\mathbb{D}}$.

The bound $T = L_*$ of (31.2) gives

$$|\psi(w)| \geq \operatorname{Re}(\psi(w)/w) = \operatorname{Re} \int_0^1 \psi'(tw) dt \geq T \quad (|w| = 1).$$

Thus Ω contains the origin-centred ball of radius $r_0 = \rho T$. Domain monotonicity gives

$$\lambda_2(\Omega) \leq \dots \leq \lambda_{n+1}(\Omega) \leq (j_{n/2,1}/r_0)^2, \quad \lambda_k(\Omega) \leq \lambda_k(B_{r_0}) \quad (k \geq 1).$$

The numerical count N_n estimates the number of Dirichlet eigenvalues of B_{r_0} below 1, counted with multiplicity; the same comparison applies to Neumann eigenvalues because $\mu_k(\Omega) \leq \lambda_k(\Omega)$. Table 26 gives these evaluations for $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\}$, where

$(j_{n/2,1}/r_0)^2 < 0.4522$. The trial space spanned by $1 - r^2$ and $x_i(1 - r^2)$, $1 \leq i \leq n$, on B^n gives by min-max

$$\lambda_{n+1}(\Omega) \leq \frac{(n+2)(n+6)}{2r_0^2} < 0.609 < 1 \quad (n \in \mathcal{N}_{\text{ax}}).$$

The energy and mass forms are diagonal by spherical orthogonality; their quotients are $n(n+4)/2$ and $(n+2)(n+6)/2$, respectively. The eigenvalue 1 therefore lies above the second-Dirichlet threshold quoted for [23] in [25, §1], which is stated there for every dimension. The results of Deng [20] and Aviles [4] stated above concern the plane and give no comparison for $n \in \mathcal{N}_{\text{ax}}$.

n	ρ	$j_{n/2,k} \approx \rho$	T	r_0	$(j_{n/2,1}/r_0)^2$	N_n
5	56.4956	$j_{5/2,17}$	> 0.98281	55.52	< 0.0108	$> 1.28 \cdot 10^6$
9	169.587	$j_{9/2,52}$	> 0.99934	169.47	$< 2.34 \cdot 10^{-3}$	$> 7.41 \cdot 10^{13}$
11	57.8601	$j_{11/2,16}$	> 0.98610	57.05	< 0.0269	$> 8.12 \cdot 10^{10}$
12	42.7785	$j_{6,11}$	> 0.99561	42.59	< 0.0545	$> 9.32 \cdot 10^9$
13	84.5746	$j_{13/2,24}$	> 0.99097	83.81	< 0.0158	$> 2.51 \cdot 10^{14}$
15	32.1112	$j_{15/2,7}$	> 0.96616	31.02	< 0.142	$> 7.99 \cdot 10^8$
16	369.836	$j_{8,114}$	> 0.99986	369.78	$< 1.10 \cdot 10^{-3}$	$> 1.02 \cdot 10^{27}$
17	74.9172	$j_{17/2,20}$	> 0.999109	74.85	< 0.0293	$> 2.01 \cdot 10^{16}$
18	20.8070	$j_{9,3}$	> 0.95484	19.86	< 0.4522	$> 1.28 \cdot 10^5$
20	83.4392	$j_{10,22}$	> 0.99341	82.88	< 0.0306	$> 7.42 \cdot 10^{18}$
21	112.609	$j_{21/2,31}$	> 0.99226	111.73	< 0.0182	$> 1.97 \cdot 10^{22}$

TABLE 26. Spectral comparison for $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\}$. In fifty-digit numerical evaluation, ρ differs from the zero $j_{n/2,k}$ of $J_{n/2}$ nearest to it by less than $2 \cdot 10^{-14}$. Here T is the certified lower bound for $\text{Re } \psi'$ on $\mathbb{D}$, $r_0 = \rho T$ is rounded down, and N_n is the numerical count of Dirichlet eigenvalues of B_{r_0} below 1, counted with multiplicity.

36.4. Berenstein and Berenstein–Yang: spectral sequences. Berenstein and Yang [9] prove that a bounded Lipschitz domain with connected boundary in any dimension carrying overdetermined eigenfunctions for infinitely many eigenvalues is a ball; for simply connected planar domains this is attributed to Berenstein [5] in [9, §0]. Theorems 1.1 and 5.1 provide one eigenvalue for each domain. By [9], each of these domains carries overdetermined eigenfunctions for at most finitely many eigenvalues.

36.5. Kobayashi: perturbations of balls. Kobayashi [46] considers a $C^{1,\beta}$ family of star-shaped domains $\Omega_t = \{r\omega : 0 \leq r < g(t, \omega)\}$ with $g(0, \cdot) = 1$. He proves that Ω_{t_0} is a disc if the real zero set of the Fourier transform of $\mathbf{1}_{\Omega_{t_0}}$ contains a circle inside the open disc of radius R about the origin and t_0 satisfies a smallness condition [46, Proposition 4.4]. This condition involves the frequency radius R and the size and regularity of the family, not the radial displacement alone. Its first part reads $t_0 B(g)e^{\|g\|_\infty} < \varepsilon(2, R)\delta(2, R)$. By the definitions [46, (3.3.1), (3.3.2), (4.2.1)], the left side is at least $2\pi e \sup_\omega |g(t_0, \omega) - 1|$, and by [46, (2.3.3), (2.4.1)] the right side is at most $2\pi |J_0(j)|/j$ for every positive zero $j \leq R$ of J_1 . Consider the rescaled domain $\Omega_1 = \rho^{-1}\Omega$ of Remark 36.1, for which the zero circle has radius ρ , so that $R > \rho$. For every family with $\Omega_{t_0} = \Omega_1$ the left side is then at least $3.45 \cdot 10^{-4}$, while the right side, evaluated at the zero $j \approx 811.3158$ of J_1 , is at most $2.17 \cdot 10^{-4}$. The smallness condition is therefore not satisfied.

For the higher-dimensional examples, the same smallness condition gives the following comparison. Its smallness condition implies $t_0 B(g) < \varepsilon(n, R)\delta(n, R)$ [46, (4.2.2b), (4.4.1b) and Lemma 4.5], where $B(g)$ [46, (4.2.1)] is at least $\omega_{n-1} \sup_{\omega} |g(t_0, \omega) - 1|/t_0$, with $\omega_{n-1} = |S^{n-1}|$, while by [46, (2.3.2), (2.3.4)] the product $\varepsilon(n, R)\delta(n, R)$ is smaller than the modulus of the Fourier transform of the indicator function of the unit ball at a point between two consecutive zeros of $J_{n/2}$ below R , which is of order $R^{-(n+1)/2}$. Hence the condition forces the relative deviation of $\partial\Omega_{t_0}$ from the unit sphere to be at most of order $R^{-(n+1)/2}$. Kobayashi's theorem concerns a controlled perturbation family with a smallness condition; neither is part of Theorem 1.1.

For $n \in \mathcal{N}_{\text{ax}}$ the comparison is explicit, as in the plane. Let $\Omega_1 = \rho^{-1}\Omega = \Theta_{\psi}(B^n)$; the Fourier transform of $\mathbf{1}_{\Omega_1}$ vanishes on the sphere of radius ρ , so $R > \rho$. Let h be the support function of Ω_1 about its centre and $d = (\max h - \min h)/(\max h + \min h)$. Since Ω_1 is centrally symmetric, its widths are $2h$. If a translate of Ω_1 has a radial function g about some point with $\sup |g - 1| \leq s$, it lies between the balls of radii $1 - s$ and $1 + s$ about that point, so all widths $2h$ of Ω_1 lie in $[2 - 2s, 2 + 2s]$; then $(1 + s)/(1 - s) \geq \max h / \min h$, that is, $s \geq d$. By the definitions [46, (3.3.1), (3.3.2), (4.2.1)] and $\|g\|_{\infty} \geq 1$, the left side of the first part $t_0 B(g)e^{\|g\|_{\infty}} < \varepsilon(n, R)\delta(n, R)$ of the smallness condition is therefore at least $e\omega_{n-1}d$ for every family with $\Omega_{t_0} = \Omega_1 + v$, $v \in \mathbb{R}^n$. By [46, (2.3.1), (2.3.3), (2.4.1)],

$$\varepsilon(n, R)\delta(n, R) \leq (2\pi)^{n/2} j^{-n/2} |J_{n/2+1}(j)|$$

for every positive zero $j \leq R$ of $J_{n/2}$; for $n = 2$ the right side is $2\pi|J_0(j)|/j$. We take for j the largest zero not exceeding ρ , listed for each tabulated dimension in Table 27. For $n = 13, 15, 17, 20, 21$, ρ lies below its nearest zero by less than 10^{-14} , and j is the preceding zero. The centre evaluations in Table 27 give the numerical factors $e\omega_{n-1}d/U$ by which the left side exceeds this upper estimate for the right side.

n	j	ω_{n-1}	d	$e\omega_{n-1}d$	U	$e\omega_{n-1}d/U$
5	$j_{5/2,17}$	26.32	$> 1.30 \cdot 10^{-3}$	> 0.0934	$< 4.38 \cdot 10^{-4}$	> 213
9	$j_{9/2,52}$	29.69	$> 4.36 \cdot 10^{-5}$	$> 3.52 \cdot 10^{-3}$	$< 2.23 \cdot 10^{-8}$	$> 1.58 \cdot 10^5$
11	$j_{11/2,16}$	20.73	$> 1.68 \cdot 10^{-3}$	> 0.0948	$< 5.21 \cdot 10^{-7}$	$> 1.82 \cdot 10^5$
12	$j_{6,11}$	16.02	$> 6.31 \cdot 10^{-4}$	> 0.0275	$< 1.22 \cdot 10^{-6}$	$> 2.25 \cdot 10^4$
13	$j_{13/2,23}$	11.84	$> 9.95 \cdot 10^{-4}$	> 0.0320	$< 5.18 \cdot 10^{-9}$	$> 6.18 \cdot 10^6$
15	$j_{15/2,6}$	5.722	$> 4.45 \cdot 10^{-3}$	> 0.0693	$< 1.58 \cdot 10^{-6}$	$> 4.40 \cdot 10^4$
16	$j_{8,114}$	3.765	$> 8.00 \cdot 10^{-6}$	$> 8.19 \cdot 10^{-5}$	$< 2.88 \cdot 10^{-16}$	$> 2.84 \cdot 10^{11}$
17	$j_{17/2,19}$	2.397	$> 1.17 \cdot 10^{-4}$	$> 7.66 \cdot 10^{-4}$	$< 9.61 \cdot 10^{-11}$	$> 7.98 \cdot 10^6$
18	$j_{9,3}$	1.479	$> 7.33 \cdot 10^{-3}$	> 0.0294	$< 3.47 \cdot 10^{-6}$	$> 8.49 \cdot 10^3$
20	$j_{10,21}$	0.5161	$> 8.15 \cdot 10^{-4}$	$> 1.14 \cdot 10^{-3}$	$< 7.66 \cdot 10^{-13}$	$> 1.49 \cdot 10^9$
21	$j_{21/2,30}$	0.2929	$> 9.15 \cdot 10^{-4}$	$> 7.28 \cdot 10^{-4}$	$< 7.09 \cdot 10^{-15}$	$> 1.02 \cdot 10^{11}$

TABLE 27. Kobayashi comparison for $n \in \{5, 9, 11, 12, 13, 15, 16, 17, 18, 20, 21\}$: $\omega_{n-1} = |S^{n-1}|$, d the relative Hausdorff distance of $\Omega_1 = \rho^{-1}\Omega$ to centred balls, and $U = (2\pi)^{n/2} j^{-n/2} |J_{n/2+1}(j)|$, the upper-bound expression for $\varepsilon(n, R)\delta(n, R)$. Entries other than n and the zero index are fifty-digit numerical evaluations; inequality signs record their rounding direction.

Agranovsky [2] and Canuto [10] give further perturbation results near balls; the quantitative comparison here uses Kobayashi's stated smallness condition.

36.6. Consequences for the conformal map. Theorems restricting polynomial and rational conformal maps have a different role: they give consequences for the exact map once

existence is established. Garofalo and Segàla [32] prove that a simply connected bounded domain whose simple closed boundary is parametrised by a polynomial in $e^{i\theta}$ has the Pompeiu property unless it is a disc. Their later result for convex domains [33, Corollary 3] gives the Pompeiu property for a convex domain $h(\mathbb{D})$ when the conformal map h has a pole on the circle of convergence of its Taylor series at zero and is not a Möbius transformation. Ebenfelt [28, Theorem 2.1] proves that the only bounded quadrature domains failing the Pompeiu property are discs. By the theorem of Aharonov and Shapiro [3], a simply connected bounded domain is a quadrature domain if and only if its conformal map from $\mathbb{D}$ is rational; compare the remark following Corollary 3 in [33].

Corollary 36.2. *The normalised conformal map ψ of Theorem 1.3 is not rational; in particular it is not a polynomial. If the Taylor series of ψ at zero has a finite radius of convergence, then ψ has no pole on its circle of convergence.*

Proof. If ψ were rational, $\Omega = \psi(\mathbb{D})$ would be a bounded quadrature domain [3]. Ebenfelt's theorem, together with the failure of the Pompeiu property, would make Ω a disc, contrary to Theorem 1.3. If ψ had a pole on the circle of convergence of its Taylor series, then, Ω being convex, Corollary 3 of [33] would give the Pompeiu property unless ψ were a Möbius transformation. A Möbius transformation that maps $\mathbb{D}$ onto a bounded domain maps it onto a disc, which is again excluded. $\square$

36.7. Spheres and isoparametric hypersurfaces. The principal orbits of the adjoint representations of $\mathfrak{su}(3)$, $\mathfrak{so}(5)$ and $\mathfrak{g}_2$ on their unit spheres are homogeneous isoparametric hypersurfaces of S^7 , S^9 and S^{13} , with 3, 4 and 6 distinct principal curvatures [39], so domains of these spheres bounded by adjoint orbits are among the isoparametric tubes that fail the Pompeiu property [56, Corollary 7]; see also [61]. These are problems of cohomogeneity one in spheres, reducing to ordinary differential equations, and the boundaries of the domains are not topological spheres. Our domains are Euclidean, of cohomogeneity two, and bounded by topological spheres, and their construction requires the solution of a two-dimensional free boundary problem.

36.8. Claims in dimension three and in general dimension. Several published or posted statements are contradicted by Theorem 1.4 and Corollary 1.2. Ramm [57, Theorems 1 and 2] states that a bounded domain in $\mathbb{R}^3$ with C^1 boundary on which (1.6) has a solution with constant 1, or which fails the Pompeiu property, is a ball. Ramm [59, Theorem 1] states that a bounded connected domain in $\mathbb{R}^m$, $m \geq 2$, with smooth boundary, on which an overdetermined problem that includes (1.6) as a special case is solvable, is a ball; [59, Theorem A(a)] states that a bounded connected domain in $\mathbb{R}^3$ with Lipschitz boundary whose indicator function has Fourier transform vanishing on a sphere is a ball; and [59, Theorem C] states the corresponding answer to the Pompeiu problem in $\mathbb{R}^m$. Ramm [58, Theorem B] states the same Fourier transform statement for connected bounded domains in $\mathbb{R}^3$ with C^2 boundary. Chen [11, Theorem 1.1] states, in a scattering formulation of the problem with the Sommerfeld radiation condition, that a star-shaped domain in $\mathbb{R}^3$ containing the origin, with Lipschitz boundary, on which (1.6) has a nonconstant solution with $\mu \geq 1$, is a ball centred at the origin. The domain of Theorem 1.4 satisfies the hypotheses of each of these statements, with $\mu = 1$, and is not a ball; [59, Theorems 1 and C] are also contradicted by the planar counterexamples [12, 14] and by Theorem 1.1. Hence these statements are not correct as stated. Finally, Disser [21, Theorem 11] showed that a bounded $C^{2,1}$ domain of $\mathbb{R}^3$ is a “bad domain” for the fluid-elastic semigroup if and only if it carries a nonconstant solution of (1.6), and conjectured that balls are the only bad domains [21, Conjecture 6]; by Theorem 1.4, this conjecture does not hold.

37. OPEN PROBLEMS

We list some questions left open by our results. Throughout, the domains are bounded, their boundaries are homeomorphic to a sphere, and solutions of (1.6) are nonconstant.

1. *Other dimensions.* The results above give convex counterexamples in every dimension $2 \leq n \leq 18$ and in dimensions 20 and 21. The Cartan reduction of Section 1.6 is tied to compact Lie algebras of rank two, whereas the axisymmetric pull-back and the exact-inverse argument of Part 5 are parametrised by the dimension. The analytic estimates of Part 5 are stated for every integer $n \geq 5$; an existence statement in a further dimension requires a finite centre satisfying the spectral, nonlinear and geometric inequalities. The dimension enters through the drift coefficient $n - 2$, the Chebyshev–Gegenbauer conversion, the heat-kernel smoothing factor, the spherical-harmonic dimensions and the high angular threshold. Dimension nineteen and every dimension above twenty-one remain open. The methods of this paper certify one dimension at a time. The existence of convex counterexamples in infinitely many dimensions is not addressed here.

Question 1. Does the Schiffer conjecture fail within the class of convex domains in every remaining dimension?

2. *Berenstein’s conjecture.* The overdetermined Dirichlet problem

$$\Delta u + \mu u = 0 \text{ in } \Omega, \quad u = 0 \text{ and } \partial_\nu u = \text{const} \neq 0 \text{ on } \partial\Omega$$

is conjectured to be solvable only on balls. This fails in the plane [15], and Theorem 5.1 gives a strictly convex non-disc planar domain on which it is solvable. In every dimension a bounded Lipschitz domain with connected boundary on which the problem is solvable for infinitely many eigenvalues μ is a ball [9, Theorem 1]. For an Ad-invariant domain Ω in a compact Lie algebra of rank two and $\mu = 1$, formula (1.7) turns it into

$$\Delta F + F = 0 \text{ in } D, \quad F = 0 \text{ and } \partial_\nu F = c \varpi \text{ on } \partial D,$$

for the W -anti-invariant function $F = \varpi u|_{\mathfrak{t}}$ on the Cartan section $D = \Omega \cap \mathfrak{t}$, where $c \neq 0$ is the constant Neumann datum.

Question 2. Does Berenstein’s conjecture fail in every dimension $n \geq 3$, and within the convex class in those dimensions?

USE OF AI

The author conceived and directed the project and takes full responsibility for the content of this paper. The constructions, proofs, verification code and text were produced with AI models (Claude Opus 5.5, Claude Sonnet 5.5, GPT-6 Astra, GPT-6 Sol and GPT-6.1 Sol) under the author’s direction, and were audited with AI assistance.

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